REMONLOGIC PRIVATE LIMITED

RemonLogic Ecosystem Master Blueprint

Products, Technology, Infrastructure and Business Plan

Learn → Create → Build → Earn

NexaTute · PlayBoom · Live Study Hall · Remon Motion Studio · NexaLogicDev · Reseller Program · Security & Control Centre

Edition	Corporate Edition v32.10 | 29 October 2026
Status	Final planning baseline for board and investor review
Classification	Strictly Confidential — owners, board and invited investors only
Company	RemonLogic Private Limited, Dhaka, Bangladesh
Owners	Md Samsuzzaman · Mst Sonia Sabuz · Md Lablu Mia

All capacity, cost and revenue figures are planning estimates based on decisions approved up to 29 October 2026. They are replaced by measured results (benchmarks, Free Beta and first-cohort data) before any external claim is made. This document does not establish that any feature is built or production-ready.

Document Control and Approvals

Document control	Value
Company	RemonLogic Private Limited
Ecosystem	RemonLogic Ecosystem — Learn → Create → Build → Earn: NexaTute (with PlayBoom and Live Study Hall), Remon Motion Studio and NexaLogicDev on one shared platform; the Reseller Program and the Security & Control Centre cover all products (Section 00)
Corporate structure	RemonLogic Private Limited (Bangladesh) owns all code and brands and pays the shared costs; RemonLogic LLC, a Delaware LLC with the same owners, sells internationally under licence and pays RemonLogic a monthly service fee (0.2)
Funding	Funding A ৳5.00 crore (final): ecosystem salaries and marketing, NexaTute and Motion Studio infrastructure; Funding B ৳1.50 crore: NexaLogicDev infrastructure, raised separately (0.5)
Canonical architecture baseline	16-Oct-26_Final_Hybrid_Architecture_v24.0.docx
Consolidated from	Project Proposal v24.1; Investor Project Proposal v22.3; NexaTute Architecture v19 reconciled MD; final Hybrid Architecture v24.0
Status	Final planning baseline for board and investor review; not a substitute for implementation, security review or load testing
Capacity statement	16,000+ combined CCU is a design target, not a benchmark-verified result
Launch programme	Free Beta with an admission cap, daily limits and no cloud fallback (11.11): Remon Motion Studio 30 days, NexaTute 20 days, PlayBoom 20 days
Launch dates	Servers rented from 5 November 2026; NexaTute Free Beta 20 November–9 December 2026, paid plans from 1 January 2027; Remon Motion Studio Free Beta from 20 December 2026 (30 days), paid packages from a date to be set (to confirm); PlayBoom Free Beta 1–20 January 2027, paid from 21 January 2027 after its readiness checklist; NexaLogicDev development from March 2027 and launch in September 2027 (0.4)
Fleet statement	16 self-hosted launch servers (14 core + 2 dedicated recording nodes), App-3 from December 2026 and the recovery replica; NexaLogicDev adds 3 Heavy GPU servers (June 2027), 4 Fast RTX PRO 6000 GPUs and 2 sandbox servers by September 2027 (27 servers) and a third sandbox server (SBX-3) at its trigger (28) — all in one Kubernetes cluster, sharing idle capacity owner first (0.6, 8.17)
Commercial authority	NexaTute Section 11 (monthly Batch, Combo and Personal plans, 11.1, with wallet top-ups for extra minutes; all prices include VAT); Motion Studio Appendix G.3 (entitlements), G.4 (remakes and Motion Studio credits), G.8 (packages, per-minute rates, delivery-speed tiers G.8.3 and USD prices G.8.4, recording and metering), G.9 (Islamic-content discount); live package limits, tiers and rates are managed in the admin panel (4.9).

Authority rule: if any earlier document conflicts with the final Hybrid Architecture v24.0, the v24.0 server inventory, node assignments, client-side rendering rule, priority policy, failover design and scratch-file safety policy take precedence.

Approval

Name	Role	Signature	Date
Md Samsuzzaman	Owner — accountable service owner		
Mst Sonia Sabuz	Owner — security owner		
Md Lablu Mia	Owner — operations / SRE owner		

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Edition history

Edition	Date	Summary
v24.0	16 October 2026	Final Hybrid Architecture — infrastructure baseline (still authoritative for locked infrastructure decisions)
v30.0	19 October 2026	Final Corporate Edition for NexaTute and Remon Motion Studio
v31.0–v31.8	21–22 October 2026	RemonLogic Ecosystem: PlayBoom, Live Study Hall, NexaLogicDev, Reseller Program, Security & Control Centre, ecosystem funding, revenue model, 28-server architecture, resource sharing and AI lane design
v32.0	22 October 2026	Corporate Edition: the single master blueprint for board and investor review
v32.1	22 October 2026	Browser-first NexaLogicDev IDE (32.14)
v32.2	22 October 2026	Browser-first execution for all products (8.18): device tiers, Device Check, Data Saver and measured browser offload
v32.3	22 October 2026	Domains, official channels and platform approvals (0.7); Instagram Reels in Motion Studio direct publishing
v32.4	23 October 2026	Product names — Nexa Teacher, Nexa Co-Teacher, Nexa Study Hall Moderator; Co-Teacher voice input and read-aloud; NexaTute hi-fi mockup; admin-managed helplines, sponsors and Eid dates (0.8)
v32.5	24 October 2026	MindForge (37); platform admin panel, Institute Admin and company accounting (38); Study Hall Q&A (31.3); PlayBoom additions, curricula and NCTB books (30.8–30.9); RemonLogic Growth Platform (3.26); clickable prototypes X1–X19 (0.8)
v32.6	25 October 2026	Admin notifications and staff chat (38.15); Avatar & Voice Studio (38.16); Nexa Notes with videos and landing-page additions (3.25); Marketing Hub channels, import and export (3.26); software register and placement updated, duplicates merged (18.1, 8.17)
v32.7	25 October 2026	Tests, written-answer checking and exam integrity (3.27); error notebook in the 1:1 class (3.18); tests and marking in the admin panel (38.2, 38.3, 38.5)

Edition	Date	Summary
v32.8	29 October 2026	A-to-Z edition: classroom voice rules and weak networks (3.28); recordings, class reports, dashboard and All Exams (3.29); Exam & Admission Hub (3.30); sign language and accessibility (3.31); 2027 engagement features with free and paid limits (3.32); Free-Beta safety (3.33); storage by plan (G.3); roadmap 2027–2028 (17.6); legal opinion (36.6); every admin value and software updates (38.17, 38.18); new dates
v32.9	29 October 2026	Admission limits, waiting list and capacity guard (38.19); Nexa Co-Admin in the admin panel and on WhatsApp, Messenger and imo (38.20); prototype structure (0.8)
v32.10	29 October 2026	USD rounding final: an exact .5 rounds down with no exception (৳500 = $7); USD figures corrected (11.2, 3.22, 11.14)

Every decision behind these editions is recorded with its date in Appendix D (Change Log).

Executive Dashboard — The Ecosystem at a Glance

RemonLogic Private Limited is building one shared AI platform for learning, content creation and software building, priced for Bangladesh and sold internationally. This dashboard summarises the plan; each line points to the section that holds the detail.

Products and launch

Product	What it does	Launch	Section
NexaTute	AI-run learning from Play class to professional exams: Nexa Teacher, Nexa Co-Teacher, batch classes, labs, mock tests, recordings	Free Beta 20 Nov–9 Dec 2026; paid 1 Jan 2027	03, 11
PlayBoom (part of NexaTute)	Play class–Class III: games, AI Play Buddy, handwriting practice, parent dashboard	Free Beta 1–20 Jan 2027; paid 21 Jan 2027	30
Live Study Hall (part of NexaTute)	Free silent study hall and Q&A boards; paid class-wise groups with the Nexa Study Hall Moderator	With NexaTute; groups as they pass their tests	31
MindForge (part of NexaTute)	Thinking-skills practice for Class 4 to adults — free daily puzzles and the MindForge Pro add-on; practice, not an IQ test	Pro open to all in the NexaTute Free Beta 20 Nov–9 Dec 2026; then a 7-day trial	37
Remon Motion Studio	AI video studio: 3D animation, Cinematic and Islamic content, series and promos; Express, Normal and Slow delivery	Free Beta from 20 Dec 2026 (30 days); paid date to confirm	04, G, 35
NexaLogicDev	Browser-first AI cloud IDE, website builders and the Auto-Architect app factory	Development Mar 2027; launch Sep 2027	32
Reseller Program; Security & Control Centre	Partner sales for every product; company-wide control baseline and go-live gates	With each launch	33, 34

Financial plan (base case)

Item	Figure
Funding A — ecosystem, Oct 2026–Mar 2028	৳5.00 crore: salaries ৳2.30 cr, infrastructure ৳1.65 cr, marketing ৳55 lakh, content ৳15 lakh, legal ৳10 lakh, R&D ৳5 lakh, working capital ৳10 lakh, contingency ৳10 lakh (15.2)
Funding B — NexaLogicDev infrastructure, Mar 2027–Aug 2028	৳1.50 crore; 18-month cost ≈৳1.44 crore (32.13)
Net revenue, 18 months, all products	≈৳6.68 crore incl. MindForge Pro (15.8)
Break-even	Monthly from July 2027; cumulative from February 2028
March 2028 run rate	≈৳83 lakh net revenue a month against ≈৳38 lakh spend
Month 12 (September 2027)	≈15,500 users; ≈3,270 paying NexaTute learners incl. PlayBoom; NexaTute MRR ≈৳21.7 lakh; Motion Studio MRR ≈৳28.8 lakh (22)

Item	Figure
NexaLogicDev break-even	≈570 paid users (32.13)

Infrastructure and capacity

Item	Figure
Server fleet	28 rented servers in the USA at full build-out: 16 at launch (from 5 November 2026), App-3 and a recovery replica, and for NexaLogicDev 3 Heavy GPU servers, 4 Fast GPUs and 3 sandbox servers — one Kubernetes cluster (7.2)
GPUs	6 launch GPUs (RTX PRO 6000 / 5000 / 4000), 4 × RTX PRO 6000 (Fast Lane), 12 × RTX A6000 (Heavy Lane); every product may use idle capacity, owner first (8.17)
NexaTute + Motion Studio	≈7,000–9,000 concurrent users on the launch fleet with App-3 (≈87,000–112,000 active users); 16,000 concurrent users by scale-out (8.5, 8.6)
NexaLogicDev	≈750–1,800 paid users if 28% are online at peak, ≈2,100–4,500 at a typical 10%; the browser-first IDE keeps the sandbox servers for AI loops and tests (32.13, 32.14)
Browser-first execution	All three products run in the user's browser whatever a normal device can run; servers keep large AI models, secrets, trusted results and the fallback; Device Check sets each user's tier and Data Saver protects mobile data (8.18)
Domains	remonlogic.com (company and one Nexa ID), nexatute.com, remonlogicstudio.com and nexalogicdev.com — all owned; customer sites on nexalogicdev.app (0.7)
New modules	MindForge, Study Hall Q&A, the PlayBoom additions, the platform admin panel, company accounting, the Growth Platform, admin notifications and staff chat, the Avatar & Voice Studio, Nexa Notes videos and written-answer checking run on the existing servers — no new servers or GPUs (7.1, 37.11, 38.14); so do the additions of 26–29 October 2026 — classroom voice rules, recordings by choice, the Exam & Admission Hub, the sign-language avatar, the 2027 engagement features and the admin controls (7.1)
Video rendering	Normal and Slow delivery and night lending of the Heavy servers spread renders across the day; a new video GPU is ordered at 60% peak load (8.17)

AI models

Use	Model
NexaTute tutoring, Motion Studio scripts, safety	Qwen3-30B-A3B and Qwen3Guard on our own GPUs
Motion Studio video and images	Wan 2.2 (TI2V-5B, A14B) and FLUX.1-schnell on our own GPUs; Quick mode renders in the creator's browser
NexaLogicDev Heavy Lane	MiniMax-M2.7 on our own servers (GLM-5.2 if the commercial licence is refused)
NexaLogicDev Fast Lane	Qwen3-Coder-Next on our own GPUs

Use	Model
External APIs (fallback only, monthly caps)	MiniMax M3 for the Fast Lane; Claude Sonnet 5.5 for Heavy-lane plan review and rescue (Pro, Team, Auto-Architect); capped cloud fallback for NexaTute overload (8.9)

Top risks and responses

Risk	Response
Launch scope too large for the launch dates	Day-1 must-have list, “coming soon” switches and go-live gates (17.5, 34)
Revenue below the base case	Monthly re-run from actual data, staged spending, marketing tied to measured CAC (15.8)
Child safety or privacy failure in PlayBoom	Guardian accounts, no face templates, local AI only, readiness checklist and lawyer review (30.5, 36)
MiniMax-M2.7 commercial licence refused	GLM-5.2 fallback on the same three Heavy servers (32.1)
Single GPU site or supplier change	Cloud failover within a spend cap; month-to-month terms and alternative suppliers (8.11, 32.13)

Next 60 days

When	Action
By 5 November 2026	Privacy Policy, Terms and Data Deletion pages live on every product domain; support email addresses set up (0.7)
5 November 2026	Launch fleet rented; GPU, CPU and storage benchmarks start
Before 20 November 2026	Load test on the staging copy (38.18); lawyer review of Section 36 and the draft legal opinion (36.6), including the MindForge questions; MindForge DPIA and partner-university ethics approval for the Free Beta pilot (37.9); the Free-Beta safety items ready (3.33)
20 November–9 December 2026	NexaTute Free Beta (20 days): latency, edge-cache, capacity and conversion measurements (11.11, 9.5)
1–15 December 2026	App-3 ordered, load-tested and in service
From 20 December 2026	Remon Motion Studio Free Beta (30 days); the video-optimisation benchmark before its paid packages (date to confirm)
Before 1 January 2027	External penetration test with high findings fixed and retested; the legal go-live checklist and the lawyer’s signed review (36.6); a second staging load test (38.18)
Now	YouTube and TikTok API applications; Meta Business Verification and App Review; NCMEC hash-sharing request; reserve official handles (0.7); MiniMax-M2.7 licence request; USA LLC through doola and Paddle; StackBlitz quote

Document Map

Sections keep their numbers from the master so that every cross-reference stays valid. Start with the rows that match your role.

Reader	Start with
Board and investors	Executive Dashboard; 00 Ecosystem Overview; 01 Executive Summary; 15 Funding and 15.8 Revenue Model; 22 OKRs; 23 Risk Register
NexaTute and PlayBoom	03 Feature Catalogue (3.25 landing page and Nexa Notes; 3.27 tests and written-answer checking); 11 Commercial Model; 17.5 Day-1 list; 30 PlayBoom (30.8 additions, 30.9 curricula and NCTB books); 31 Live Study Hall and Q&A; 37 MindForge; 3.28 classroom voice and weak networks; 3.29 recordings, reports and dashboard; 3.30 Exam & Admission Hub; 3.31 accessibility and sign language; 3.32 engagement and 2027 features; 3.33 Free-Beta safety; 17.6 roadmap 2027–2028
Remon Motion Studio	04 Feature Catalogue; 11.10 Pricing; 35 Rights, Safety and Pricing; Appendices F and G
NexaLogicDev	32 AI Cloud IDE, Builders and App Factory (lanes, API fallback, credits, Funding B, 32.14 browser-first IDE)
Architecture and operations	05–09 Architecture, Hardware (7.2 server register), Scheduling (8.17 resource sharing, 8.18 browser-first execution), Data; 18 Software Register
Security, privacy and legal	10 and 19 Security; 34 Security & Control Centre; 36 Legal and Compliance Drafts; 36.6 draft legal opinion and legal go-live checklist
Partners, sales and launch operations	0.7 Domains, channels and platform approvals; 3.26 AI Marketing Hub and RemonLogic Growth Platform; 26 Partnerships; 33 Reseller Program; 35.5 Launch actions
Platform administration and accounting	38 Platform admin panel, Institute Admin, company accounting, access and approvals, cloud export; 0.8 prototypes X10–X23; 38.15 notifications and staff chat; 38.16 Avatar & Voice Studio; tests and marking in 38.2, 38.3 and 38.5; 38.17 every user-visible value; 38.18 software updates, load tests and delivery monitoring; 38.19 admission limits and capacity guard; 38.20 Nexa Co-Admin
Product names, hi-fi mockup and admin settings	0.8 Nexa Teacher, Nexa Co-Teacher, Nexa Study Hall Moderator, Nexa Guard; NexaTute hi-fi mockup with clickable prototypes X1–X23; helplines, sponsors and Eid dates in the admin panel
Organisation	14 Organisation and Hiring; 25 Advisory Board; 28 Customer Support

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Terminology: “locked” means a user-approved design reference, not proof that the configuration has already been deployed or load-tested. “Target”, “proposed”, and “requires validation” are used where the source material does not establish production evidence.

00. RemonLogic Ecosystem Overview

This blueprint is the single master document for the whole RemonLogic Ecosystem. Section 00 sets out how the products fit together, the corporate structure, the shared platform, the launch timeline, the unified funding, salary and marketing plan and the server architecture (27 servers from September 2027, 28 once the third sandbox server is added at its trigger). Sections 01–29 and Appendices A–I keep the detailed NexaTute and Motion Studio design (updated in place where 21 October decisions changed them); Sections 30–35 add PlayBoom, the Study Hall groups and moderator, NexaLogicDev, the Reseller Program, the Security & Control Centre and the new Motion Studio rights, safety and international-pricing rules; Section 36 holds the legal drafts, Section 37 MindForge and Section 38 the platform administration, Institute Admin and company accounting.

Status: planning master. Nothing here proves implementation, load-test results, legal approval or certification; figures marked as estimates must be confirmed by quotes, benchmarks and the accountant or lawyer named in each section.

0.1 Vision and product map — Learn → Create → Build → Earn

One account, one wallet and one AI platform carry a learner from studying, to making media, to building software, to earning from it.

Stage	Product	What it does	Launch	Section
Learn	NexaTute (with PlayBoom and Live Study Hall)	AI-run tutoring from Play class to professional exams: Nexa Teacher, Nexa Co-Teacher, batch classes, labs, recordings; PlayBoom for Play class–Class III; free Study Hall with paid groups	NexaTute Free Beta 20 Nov–9 Dec 2026, paid 1 Jan 2027; PlayBoom Free Beta 1–20 Jan 2027, paid 21 Jan 2027	03, 11, 30, 31
Create	Remon Motion Studio	AI video studio: 3D animation, Cinematic and Islamic content, series, promos, direct publishing	Free Beta from 20 Dec 2026 (30 days); paid date to confirm	04, F, G, 35
Build	NexaLogicDev (formerly NexaPrompt)	AI Cloud IDE, Website/WordPress/Wireframe builders and Auto-Architect app factory	Development from March 2027; launch September 2027	32
Earn	Reseller Program, NexaLogicDev Marketplace, Expert Help, creator monetisation, job pathways	Resellers sell every product from one wallet; builders sell templates; certified builders take paid help jobs	Reseller Program with paid launches; Marketplace with NexaLogicDev	32, 33
Company-wide	RemonLogic Security & Control Centre	18 control domains, AI read-only boundary, go-live gates	Before each paid launch	34

Cross-product flows

- NexaTute coding learners start projects in NexaLogicDev (Learn → Build) with Explain mode and a Student Plan; course + real project completion earns the "NexaLogicDev Certified Builder" certificate and priority on the Job Board.
- Websites and apps built in NexaLogicDev get promo videos from Motion Studio in one click; NexaLogicDev's client-handover voice-over video uses the Motion Studio voice pipeline; NexaLogicDev credits are deducted at the Motion Studio price (৳5 = 1 credit).
- NexaTute topic Shorts and the NexaTute Wrapped recap are made with Motion Studio (existing, 3.20, 3.26).
- One Nexa ID and one Nexa Wallet across products; NexaCoins (NexaLogicDev) and NexaTute XP become one reward system.
- One reseller sells NexaTute, PlayBoom, Motion Studio and NexaLogicDev from the same reseller wallet (Section 33).

0.2 Corporate structure

	RemonLogic Private Limited (Bangladesh)	RemonLogic LLC (Delaware)
Owners	Founders	The same owners
Role	Owns the code and brands of NexaTute (with PlayBoom), Remon Motion Studio and NexaLogicDev; runs the shared platform	Sells internationally under a licence agreement from RemonLogic
Income	Bangladesh customers; the monthly service fee from the LLC	International customers, collected through Paddle (Merchant of Record: sales tax and VAT)
Costs	All shared costs — servers, Cloudflare, platform — and the Bangladesh team	Service fee to RemonLogic, domain purchases (Name.com), Paddle fees, US compliance
Money flow	Receives the service fee as software/IT export income (Bangladesh export incentive as applicable)	Funded and paid from the USA; no money is sent from Bangladesh to the LLC
Compliance	Trade licence, TIN, VAT (BIN), aamarPay KYC, data-law registration (Appendix G checklist)	Formed through doola.com: registered agent, EIN, bookkeeping, annual Form 5472 + pro-forma 1120 (15 April; 15 October with Form 7004); Delaware franchise tax US$300 by 1 June

- Service fee: the LLC pays RemonLogic each month for international customers' share of servers and services, so costs and income sit in the right company; the fee and the licence are written in one intercompany agreement.
- Every related-party transaction (service fee, licence fee, owner contributions) is recorded with date, amount, currency and exchange rate for Form 5472.
- To confirm with a Bangladeshi chartered accountant before the LLC opens: Bangladesh Bank rules on foreign ownership, transfer pricing of the service fee, and how the owners' LLC income is taxed in Bangladesh. This document is not legal or tax advice.

0.3 Shared platform

Layer	Shared by all products
Identity — Nexa ID	One account per person across products (Keycloak); NexaTute learners add face verification; PlayBoom uses guardian accounts with child profiles, guardian PIN and OTP and no face templates

Layer	Shared by all products
Wallet and billing — Nexa Wallet	One wallet and ledger; aamarPay (bKash, Nagad, cards) in Bangladesh, Paddle internationally; guardian approval for under-18 purchases; reseller wallets on the same ledger
Rewards	NexaTute XP and NexaLogicDev NexaCoins in one reward system; no money-bought rewards for children
AI gateway and GPU pool	One Kubernetes cluster and one priority-scheduled GPU pool (Portkey + vLLM); Qwen3 family for NexaTute and the Fast Lane (Qwen3-Coder-Next), MiniMax-M2.7 for the NexaLogicDev Heavy Lane (GLM-5.2 fallback; the MiniMax M3 API backs the Fast Lane and Claude Sonnet 5.5 reviews and rescues Heavy-lane plans, 32.1); overflow lending between pools
Browser-first execution	Every product runs in the user's browser whatever it can; our servers keep large AI models, secrets, trusted results and the fallback for light or weak devices; Device Check sets each user's tier and Data Saver protects mobile data (8.18)
Data	PostgreSQL (Patroni), Valkey, Cloudflare R2, ClickHouse, TimescaleDB — separate schemas and row-level security per product
Services	Document Intake Service, notification service (push, email, SMS, WhatsApp), Centrifugo, Chatwoot support, platform admin panel and company accounting (38), RemonLogic Growth Platform (3.26), admin notifications and staff chat (38.15), Avatar & Voice Studio (38.16)
Safety and governance	Qwen3Guard, NSFW and child-safety hash matching, content policies, human controllers, Security & Control Centre (34)
Support	AI first in Bangla and English, then a person; SLAs per product

0.4 Launch timeline

Date	Milestone
5 November 2026	Launch fleet rented (16 servers)
20 November–9 December 2026	NexaTute Free Beta (20 days), after the staging load test (38.18)
1 / 15 December 2026	App-3 ordered / in service after its load test
From 20 December 2026	Remon Motion Studio Free Beta (30 days)
Before 1 January 2027	External penetration test, legal go-live checklist and second staging load test (36.6, 38.18)
1 January 2027	NexaTute paid plans; the 2027 engagement features switch on (3.32, 17.6)
1–20 January 2027	PlayBoom Free Beta (20 days)
21 January 2027	PlayBoom paid launch — only after its readiness checklist passes (30.8)
After the Motion Studio Free Beta (date to confirm)	Motion Studio paid packages; delivery-speed tiers (G.8.3) start
March 2027 (Month 6)	NexaLogicDev development starts; Funding B's 18 months start; no GPU servers yet — AI features are tested on the NexaTute pool off-peak

Date	Milestone
June 2027	3 Heavy-lane GPU servers rented; MiniMax-M2.7 (and GLM-5.2 fallback) benchmark and integration
September 2027 (Month 12)	NexaLogicDev launch: IDE, Website Builder, Wireframe Generator, Auto-Architect Basic and five engagement features; 4 Fast-lane RTX PRO 6000 GPUs added
≈March–June 2028	SBX-3 sandbox server at its trigger (7.2); more Fast GPUs one at a time at >70% load
Later updates	WordPress Mode, Complex Module, the nine approved growth features (32.8), managed hosting when demand triggers it

0.5 Unified funding, salaries and marketing

Funding A (৳5.00 crore, final) pays the ecosystem's salaries and marketing and the NexaTute and Motion Studio infrastructure. Funding B (৳1.50 crore) is raised separately for NexaLogicDev infrastructure. All three SaaS products' salaries and marketing are shown together here.

Funding A — ৳5.00 crore over 18 months (15.2)

Category	Amount	Share	Change from the earlier plan
Salaries (all three products)	৳2.30 crore	46%	+৳30 lakh: NexaLogicDev AI/ML + Full-stack from Month 6; one B2B Sales hire deferred; MindForge part-time psychometrician and item reviewer (+৳10 lakh, 14.4)
Infrastructure (NexaTute + Motion Studio)	৳1.65 crore	33%	−৳15 lakh: 17-month rent window (15.6)
Marketing	৳55 lakh	11%	NexaTute + Motion Studio ৳52 lakh; NexaLogicDev launch ৳3 lakh
Content	৳15 lakh	3%	−৳10 lakh: moved to Salaries for the MindForge part-time roles (to be confirmed)
Legal	৳10 lakh	2%	—
R&D	৳5 lakh	1%	−৳5 lakh: NexaLogicDev benchmarks run in Funding B
Working capital	৳10 lakh	2%	—
Contingency	৳10 lakh	2%	—
Total	৳5.00 crore	100%	Unchanged

Combined salaries — all SaaS products

Team	Staff	Monthly base (full team)	18 months incl. 10% on-cost
NexaTute + Motion Studio + shared roles (CTO, DevOps ×2, UX, Marketing Manager, Partnership Manager, Backend ×2, AI/ML, Full-stack, Content Lead, Writers ×2, Support ×2, B2B Sales ×1 from Month 13)	15 → 16	৳9.9 → ৳10.5 lakh	≈৳1.94 crore
NexaLogicDev core (AI/ML Engineer for Heavy and Fast model serving, Full-stack Engineer) from Month 6	+2	+৳1.8 lakh	≈৳25.7 lakh
MindForge part-time psychometrician (Month 2) and item reviewer (Month 3), contract	+2 part-time	+≈৳55,000	≈৳9.2 lakh (estimate, to be confirmed)
Total in Funding A	17 → 18 (+2 part-time)	≈৳12.9 lakh	৳2.30 crore
NexaLogicDev revenue-gated (2nd Full-stack at MRR ≥ ৳2 lakh; Support at ≥150 paid users; Developer Advocate at MRR ≥ ৳4 lakh)	+3	+৳1.65 lakh	From NexaLogicDev revenue
NexaLogicDev pre-launch QA contract (Months 9–12)	1	৳50,000	≈৳2 lakh — Funding B

Combined marketing

Product	Budget	Notes
NexaTute + Motion Studio	৳52 lakh (Funding A)	Plan in 3.26 and 15.7; AI Marketing Hub
NexaLogicDev launch campaign	৳3 lakh (Funding A)	September 2027 launch
NexaLogicDev performance ads	20% of NexaLogicDev revenue	Revenue-funded
NexaLogicDev hackathon prizes	≈৳2.2 lakh (Funding B)	Monthly cash hackathon from launch

Funding B — NexaLogicDev infrastructure, ৳1.50 crore over 18 months from March 2027 (32.13)

Item	18 months
Heavy servers (3 × 4× A6000 at US$1,599, Months 4–18, 45 server-months) and Fast GPUs (4 × RTX PRO 6000 at US$749, then US$648, Months 7–18, 48 GPU-months)	≈৳1.27 crore
Sandbox CPU servers (1, then 2; SBX-3 at its trigger) incl. installation	≈৳9.4 lakh
Cloudflare, domains, email, monitoring	≈৳3.6 lakh
Hackathon prizes	≈৳2.2 lakh
Pre-launch QA contract	≈৳2 lakh
Total	≈৳1.44 crore

Item	18 months
Contingency	≈৳5.6 lakh

Risks: Funding A contingency is 2% (Infrastructure headroom is now small), and NexaLogicDev reaches its ≈570-user break-even after the Funding A window — inside Funding A's window it brings about ৳15 lakh of net revenue.

0.6 Unified server architecture — 24 → 28 servers

Pool	Servers	From	Funding	Role
Launch fleet (Section 07)	16 — 6 GPU (Dallas), Core-1/Core-2, App-1/2, DB-1/2, Redis1/2, LB-1/2	Nov 2026	A	NexaTute, PlayBoom, Study Hall, Motion Studio, shared platform
App-3	1 (OVHcloud ADVANCE-2)	Dec 2026	A	App/API pool
Recovery replica	1 (OVHcloud Hillsboro)	Nov 2026	A	Second region
Heavy GPU pool	3 × (4× RTX A6000 48 GB, 512 GB RAM)	Jun 2027	B	MiniMax-M2.7 4-bit, one copy per server (GLM-5.2 TP=4 × PP=3 as fallback)
Fast GPU pool	4 × RTX PRO 6000 96 GB (Database Mart Dallas)	Sep 2027	B	Qwen3-Coder-Next, one copy per GPU; more GPUs one at a time
Sandbox pool	1 → 2 → 3 (OVHcloud ADVANCE-2)	Mar / Sep 2027; SBX-3 at its trigger (≈Mar–Jun 2028)	B	AI write → run → fix loops, tests, scans and database backends (gVisor/Kata, strict network policy); previews run in the browser (32.14)

- One Kubernetes cluster: zone labels and taints keep App pods in Virginia, GPU work in Dallas and user code only on the sandbox pool; untrusted code can never reach children's data (PlayBoom) or payment ledgers.
- Priority: NexaTute live classes and recording first; Motion Studio Express takes free GPU capacity at high priority; Normal/Slow renders, batch jobs and NexaLogicDev overflow fill quieter hours. Every product may use any idle GPU or CPU capacity, owner first, in the placement order of 8.17.
- Pool lending (8.17): the Heavy servers are lent one at a time in the 1–7 am window for Motion Studio Normal/Slow renders and batch work; idle Fast GPUs take NexaTute LLM and Motion Studio overflow; idle NexaTute GPUs take paid NexaLogicDev Fast-lane overflow; Heavy-server CPUs take trusted CPU work. Sandbox, data, edge and recovery servers are never lent. The Fast GPUs are the same RTX PRO 6000 as the launch pool, so models move between them easily.
- Free NexaLogicDev sites are served from one wildcard Cloudflare Worker + R2, so they add no server load.
- Every new server enters the Security & Control Centre asset inventory (D01) before it takes traffic.

0.7 Domains, brands, official channels and platform approvals

Each product has its own domain and remonlogic.com is the company umbrella. Separate domains keep a problem in one product — for example a phishing site built by a NexaLogicDev customer — from getting the others blocked, above all NexaTute, which serves children; they also give each brand its own trust, search ranking, platform app reviews and email reputation.

Domain	Use	Status
remonlogic.com	Company site, investors, careers and legal pages; one account for every product (Nexa ID at accounts.remonlogic.com), billing and the status page (status.remonlogic.com)	Owned
nexatute.com	NexaTute, PlayBoom and the Live Study Hall	Owned
remonlogicstudio.com	Remon Motion Studio	Owned
nexalogicdev.com	NexaLogicDev IDE, builders and website	Owned
nexalogicdev.app	Customers' free sites only (username-site.nexalogicdev.app), never our own pages; listed on the Public Suffix List so one customer's site cannot read another's cookies	To buy now (needed by September 2027)

- Every product site carries "by RemonLogic" in its footer, and one Nexa ID signs in to all products.
- DNS, SSL and email routing run on Cloudflare; each product sends email from its own domain (support@ and no-reply@), with SPF, DKIM and DMARC on every domain, so one product's email reputation never affects another.
- Every product domain publishes its Privacy Policy, Terms and Data Deletion page (36); the platform approvals below require them.

Official channels

Brand	Handle	Platforms
RemonLogic	@remonlogic	Facebook Page, Instagram (Business), LinkedIn Company Page, YouTube channel, TikTok (Business)
NexaTute	@nexatute	Same five platforms
Remon Motion Studio	@remonlogicstudio	Same five platforms
NexaLogicDev	@nexalogicdev	Same five platforms

- All pages sit in the RemonLogic Meta Business Manager with at least two owner admins — never a single personal account — and use the same logo, cover image, one-line description and website link everywhere.

Platform approvals

Platform	Used for	Approval needed	When
YouTube	Motion Studio direct publishing, with the AI-content disclosure on every upload	YouTube API Services audit and quota extension; Google OAuth verification for youtube.upload	Apply now; creators download and upload until approved

Platform	Used for	Approval needed	When
TikTok	Motion Studio direct posting	Content Posting API review (Login Kit + Direct Post); API posts stay private until audited	Apply now; if TikTok's branding rules exclude watermarked videos, direct posting is for paid packages only
Facebook Pages and Instagram Reels	Motion Studio direct publishing (Instagram Business or Creator accounts only); our own marketing pages need no review	Meta Business Verification; one Meta app per product in one Business Manager; App Review for Page and Instagram publishing permissions	Business Verification now; App Review as soon as the demo is ready
Facebook Pages (read)	NexaLogicDev "Facebook Page → Website"	App Review for Page-reading permissions in the NexaLogicDev Meta app	2–3 months before the September 2027 launch
WhatsApp Business Platform	Opt-in lead, order and build alerts, guardian reports and security messages	Business Verification, a dedicated number per sender and approved message templates	Before each product starts using it
LinkedIn	Motion Studio publishing to profiles and Company Pages (after launch); sign-in with LinkedIn	Share on LinkedIn and OpenID sign-in are self-serve; the Community Management API needs an application	After launch
NCMEC (child safety)	Hash lists for Hasher-Matcher-Actioner and CyberTipline reporting (4.15)	Electronic Service Provider registration and hash-sharing agreement through the USA LLC; the IWF list is a paid alternative	Request now

- Every application needs the product domain's public Privacy Policy, Terms and Data Deletion page and a short demo screencast of the real flow; application texts and demo scripts are prepared for YouTube, TikTok, Meta (Motion Studio and NexaLogicDev), WhatsApp, LinkedIn and NCMEC.
- "Add to LinkedIn Profile": NexaTute certificates and the NexaLogicDev Certified Builder certificate carry a button (on 18+ accounts) that pre-fills the certification on the learner's LinkedIn profile — a plain link with no API or review.

0.8 Product names, NexaTute hi-fi mockup and admin-managed settings

NexaTute's AI roles now carry product names. The names are used in the app, the website, every document and this blueprint. Wherever a learner meets the Nexa Teacher, a small "AI" label sits next to the name, so no learner or guardian takes it for a human teacher (AI transparency, 19.7).

Name	Earlier name	What it does	Rules
Nexa Teacher	AI Teacher (until v32.3)	Teaches live by voice, one-to-one and in batch classes, with the 3D avatar, the Smart Whiteboard (stops at the first wrong step) and the talk-ratio target (3.16, 3.17)	Always shown with an "AI" label; voice, avatar, whiteboard and labs run in the learner's browser, the server keeps the language model (8.18)

Name	Earlier name	What it does	Rules
Nexa Co-Teacher	AI Co-teacher (until v32.3)	Teaches in text: text lessons, step-by-step answers, links to the learner's own recordings and summaries (3.16). The learner can also speak a question (turned into text in the browser) and listen to any answer (browser voice)	No continuous voice conversation, avatar, whiteboard step-watching or talk ratio — these stay with the Nexa Teacher, so 1:1 minutes keep their value; 25 topics a day unchanged; the Co-Teacher Pass keeps its price
Nexa Study Hall Moderator	Study Hall AI moderator (until v32.3)	Gives short help (hints, the wrong step, 2–3 minutes) in Study Hall groups by text and voice; answers a shared question once for the group; hands a full topic to the Nexa Co-Teacher or the Nexa Teacher with a handover note (31)	Learners of any age may ask the Moderator by voice; group chat stays text-only for minors; bad-language filter and private report button unchanged
Nexa Guard	Batch-class "AI moderator" (until v32.4)	The automated safety system in every class, group, Q&A board, upload and link: bad-language filter, pornography and child-safety blocking, harmful-link blocking, the report button and the behaviour-change ladder (10.2)	People who act on serious alerts are the Human Controllers (19.3); Nexa Guard never judges mood or attention

Other decisions of 23 October 2026

- Free Beta: 30 minutes a day of 1:1 Nexa Teacher time, as in the Free Beta allowances (11.11).
- Certificates: the "Add to LinkedIn Profile" button appears only on 18+ accounts; younger learners keep PDF download and link sharing.
- Standing rule: every subject follows each learner's own country curriculum — in NexaTute, PlayBoom, MindForge and every future feature (29.2, 30.9, 37.3).

Settings managed in the admin panel

These change often or differ case by case, so the admin enters them in the NexaTute admin panel and the app shows the new values as soon as they are saved. Every change is logged.

Setting	What the admin does	Effect in the app
Helpline numbers	Helplines are set country-wise: the numbers for each learner's country are filled in automatically and the admin can change them. The admin adds each helpline with its name, number, hours and where it is shown, and switches it on or off: national emergency 999, a mental-health helpline and a child helpline	Shown at once in the learner wellbeing screen and in guardian messages; while a number is empty, only 999 is shown
Sponsor-a-Learner	Adds each sponsor with their own amount, period, target group and plan; sets the allocation rules — rural girls first, guardian approval, a per-learner monthly maximum (starting value ৳600) and a reminder before a sponsor's balance runs out	Sponsored learners study free within the rules; sponsors never see learner names and get an optional anonymous monthly report (numbers and progress only)
Eid holidays	Enters the start and end dates of Eid-ul-Fitr and Eid-ul-Adha after the moon is sighted	Batch classes stop on those days and learners and guardians are notified; 1:1 teaching and the Nexa Co-Teacher continue

NexaTute hi-fi mockup

A complete hi-fi mockup of NexaTute (Bangla interface, mobile first at 390 × 844 with key desktop screens at 1440) is kept on a private design canvas for the UX and development teams: https://claude.ai/artifact/M3ZGBvHBRzmKsf9dC216Ri. It follows every decision in this blueprint and is the reference for the wireframe-to-build hand-off.

Area	Boards	Screens
Learner app — onboarding, home and plans, Nexa Teacher 1:1, batch class, Nexa Co-Teacher, tests, progress, Study Hall and labs, Moderator, more learning, beyond the syllabus, care and links, free tier and ways to buy	L1–L15	59 mobile
Guardian app — home, weekly report, purchase approval, consent and data, screen time, alerts, PlayBoom progress	G1–G2	7 mobile
PlayBoom — who is playing (guardian PIN), worlds, counting game, AI Play Buddy, Handwriting Studio, break	P1–P2	6 mobile
Teacher app — home, content check, class report	T1	3 mobile
Desktop — learner home, classroom, content studio, institution, admin overview, plans and prices, safety, quality gate, schedule, helplines and sponsors	D1–D10	10 desktop
Clickable prototypes — learner, guardian and teacher: new learner; class and Nexa Co-Teacher; purchase with guardian approval on two phones; Study Hall and Moderator; PlayBoom; batch class; mock test, OMR and exam eve; guardian (weekly report with the expected syllabus finish date and MindForge, approvals, consent and data, messages); teacher app; calculator and dictionary; avatar and voice for learners, guardians and teachers; the error notebook in the 1:1 class; test types, written checking and board-style tests	X1–X9, X11, X23	Clickable
Clickable prototypes — administration: platform admin (capacity, learners and usage, plans and modules with MindForge, institutions, events, certificates and jobs, accounting and money, roles and access, tests and marking, Nexa Guard, holidays, country helplines and sponsors); Institute Admin; Landing Page AI	X10, X25, X26, X12, X13	Clickable
Clickable prototypes — platform operations (with the written-check GPU queue); accounts, support and legal; content quality and money (with marking schemes and rechecks); growth; MindForge; AI Marketing Hub — Digital Marketing Automation (3.26, both products, linked from the platform admin sidebar); notifications and staff chat; Avatar & Voice Studio	X14–X19, X21, X22	Clickable
Clickable prototype — the NexaTute landing page: country and syllabus picker, plan finder, University, Polytechnic and Vocational courses, sign-up, the six legal pages and Nexa Notes with videos	X20, X24 (legal pages)	Clickable
Clickable prototype — Nexa Co-Admin: chat workspace with a role switcher, demo requests, the activity log and settings (38.20)	X27	Clickable
Total	29 boards + 27 clickable prototypes	75 mobile + 10 desktop screens; every prototype has tooltips

Design rules used throughout: the “AI” label next to the Nexa Teacher; touch targets of at least 44 px; text contrast of at least 4.5:1 (WCAG 2.2 AA); “understood / not understood” told apart by words as well as colour; no ads or tracking on any screen. Twenty-seven clickable prototypes (X1–X27) sit beside the boards. Every prototype has tooltips: hovering over an important icon, button, status or chart shows a short explanation, and on mobile a tap shows it.

Prototype rules (24–26 October 2026): every prototype page (X01–X23, the website preview W00 and Main) has mobile, tablet and desktop versions — one responsive page per board with a Mobile / Tablet / Desktop switch, never duplicate boards; the static wireframes (L, G, P, T, D) and W00/Main are in all six languages (bn, en, hi, ur, ar, fr); the platform admin sidebar menus expand and collapse. The 26–29 October 2026 decisions are built into the existing boards as tabs and sections (3.28–3.33, 38.17, 38.18). Structure from 29 October 2026: the landing page’s legal pages moved to X24; the platform admin board was split into X10 (overview, institutions, events, learners, accounts and finance, roles and access), X25 (plans and modules, platform switches, plans and prices, admission limits) and X26 (rules and settings, Free Beta safety); X27 is Nexa Co-Admin. Every board is kept at or below 1.8 MB because the design viewer drops files over 2 MiB.

01. Executive Summary and Project Mandate

RemonLogic Private Limited is building the RemonLogic Ecosystem — Learn → Create → Build → Earn — on one shared infrastructure: NexaTute (AI-run learning from Play class to professional exams, including the PlayBoom module and the Live Study Hall), Remon Motion Studio (AI video production) and NexaLogicDev (AI cloud IDE, builders and app factory, from September 2027), with a Reseller Program, a USA LLC for international sales and a company-wide Security & Control Centre (Section 00). NexaTute is an AI-assisted personalized learning platform with live learning, browser-rendered 3D teaching experiences, collaborative whiteboard and assessment; Remon Motion Studio automates the creator workflow from brief to delivered video; NexaLogicDev turns prompts and wireframes into working websites and apps priced for Bangladesh.

The products remain logically separate even when physical GPU capacity is shared. Each product requires its own API/runtime context, queue and cache namespaces, authorization boundary, job accounting and product-level usage records. The Hybrid Shared Resource Pool dynamically routes eligible workloads based on availability and priority; it must not erase product isolation.

Strategic objectives

- Provide localized learning and BCS preparation workflows, with multilingual and mobile-first learning experiences where model and curriculum quality support them.
- Automate a creator workflow from brief → script → storyboard → voice/audio → scene assets → assembly/encoding → validation → secure delivery.
- Run NexaTute’s voice, Nexa Teacher video, whiteboard, virtual lab, 3D classroom and AI-tutor recording in the learner’s browser to reduce server load, with a server speech fallback for weak devices (3.2).
- Sell NexaTute as simple monthly plans — Batch (AI group classes, one Nexa Teacher for up to 50 learners), Combo (group classes plus 30 minutes a day of 1:1) and Personal (1:1) — with a text-only Nexa Co-Teacher that keeps learning going when live teaching is busy (3.16, 3.17, 11.1).
- Use dedicated recording nodes for LiveKit recording/egress so recording and compilation workloads do not consume the central real-time media layer unnecessarily.
- Protect real-time classroom and speech workloads from background video backlog using the X-Priority policy and bounded queues.
- Keep local scratch storage bounded and delete task-owned files only after final upload and integrity validation.

- Validate costs, model licensing, feature maturity, throughput and concurrency through measured acceptance tests before making public or investor claims.

Locked baseline at a glance

Item	Baseline
Products	NexaTute (with PlayBoom and Live Study Hall); Remon Motion Studio; NexaLogicDev (from September 2027)
Physical fleet	16 rented servers at launch (14 core + 2 recording), the recovery replica and App-3 from December 2026; 28 servers once NexaLogicDev is fully built out (7.2), sharing idle capacity owner first (8.17); scales out to ≈37–39 servers plus hourly GPU burst for the 16,000 CCU target (8.5)
Combined capacity target	16,000+ CCU target: 8,000 NexaTute + 8,000 Motion Studio; capacity model in 8.6; must be load-tested
Hybrid GPU policy	Dynamic shared resource pool with priority-aware routing and workload isolation
Highest workload priority	NexaTute live-class speech / 3D virtual-lab related real-time workload, X-Priority: 100
Paid production priority	Motion Studio paid remake/render jobs, X-Priority: 50
Free/trial batch priority	Motion Studio free/trial generation, X-Priority: 10
Browser rendering	Voice, Nexa Teacher video, whiteboard, virtual lab, 3D classroom (3.15), AI-tutor recording and recording-download MP4s in the learner's browser; server speech fallback for weak devices (3.2); Motion Studio Quick-mode animation, 3D avatars and standard voices in the creator's browser, with a server takeover after 30 minutes without a connection (4.8); in batch classes the Nexa Teacher's voice is made in each learner's browser from the AI's text (3.17)
Recording nodes	Rec-1 and Rec-2 (RTX PRO 4000): lightweight track recording; download MP4s are made in the user's browser (9.5)
Object storage	Cloudflare R2 for validated final media and permitted archives; signed access and integrity checks required

02. Product Portfolio and Business Proposal

2.1 NexaTute

Product concept: an AI-Powered Learning Operating System (3.11) — an AI learning companion and education platform for individual learners, BCS candidates, tutors, coaching centres and institutions. It combines conversational teaching, guided practice, assessment, AI-assisted educational content creation, live classes, recordings and optional premium interactive experiences. The Bangladesh-first entry hypothesis uses local curriculum and BCS preparation as an initial segment; regional expansion depends on curriculum fit, language quality, willingness-to-pay and operating economics.

2.2 Remon Motion Studio

Product concept: a workflow-oriented AI video production SaaS for creators, educators, agencies and marketing teams. It automates text/script creation, scene-by-scene storyboard generation,

multilingual speech/voice workflows, rendering orchestration, editing/assembly, output validation and secure delivery. General cinematic generation and avatar animation are separate engines connected by a controlled orchestration pipeline.

NexaLogicDev (Section 32) is the third product: an AI cloud IDE, Website/WordPress/Wireframe builders and the Auto-Architect app factory for freelancers, agencies, small businesses and development teams, launching in September 2027. PlayBoom (Section 30) is part of NexaTute, teaching Play class to Class III.

2.3 Portfolio rationale

- Share physical compute only where compatible and schedulable; preserve separate product data and service boundaries.
- Use browser-side interactive rendering for NexaTute rather than generating every avatar/lab frame on the server.
- Use asynchronous queues and durable job state for long-running Motion Studio work rather than tying rendering to synchronous web requests.
- Offer separate product billing and usage measurement. Never let a shared GPU pool imply shared customer records or cross-product permissions.
- Validate each product's own unit economics, retention, quality and support load; a shared infrastructure design alone does not prove commercial viability.

2.4 Target customer groups

Product	Initial customer groups	Needs to validate
NexaTute	BCS candidates; school/college/university learners; young children through PlayBoom (Play class–Class III) and their guardians; self-directed learners; tutors and coaching centres; institutions; multilingual learners	Curriculum accuracy, learning outcomes, accessibility, model cost per minute, guardian/teacher workflows, willingness to pay
Remon Motion Studio	Bangladeshi content creators; local marketing agencies; educators; small businesses; production teams	Output quality, time-to-completion, render cost, licensing/consent, remake demand, retention and willingness to pay
NexaLogicDev	Freelancers; digital agencies; small businesses and F-commerce sellers; startup development teams; NexaTute coding learners	Build success rate, AI quality on Bangla prompts, abuse control, GPU cost per credit, free → paid conversion and retention

2.5 Strategic Growth Initiatives

These initiatives turn the two products into learning and creative infrastructure for Bangladesh that is hard to copy. They build on the features in Sections 03, 04 and 3.11; partners named are targets, not confirmed agreements.

Initiative	What it is	What it needs
Learn → Create → Earn	Learners gain skills on NexaTute (design, video, digital marketing, English), make content with Motion Studio and earn through its marketplace and freelancing; learner-passport credentials prove their skills — one ecosystem instead of two separate products	Single sign-on across both products with consent, while each product keeps its own data (6.4); credential links to creator profiles

Initiative	What it is	What it needs
NexaTute-Bangla model	An in-house Bangla education model trained on verified answers, teacher corrections and rubric data, aiming to beat general AI models on Bangladeshi syllabi; a small version may be released as open source	Consented and anonymised data only (minors only with guardian consent); fine-tuning and evaluation pipeline; GPU time (8.6)
Telco partnerships	Zero-rated NexaTute data or study data bundles with Grameenphone, Robi or Banglalink, removing the biggest cost barrier for rural learners	A dedicated learning domain and, if the telco rates by IP address, dedicated Cloudflare IPs (8.16); a data-light mode
NexaTute Box for rural schools	A low-cost mini PC in a school serves saved lessons, quizzes and flashcards over local Wi-Fi without internet, answers simple questions with a small local model, and syncs at night	Edge software stack (5.3); hardware bought per school under government or NGO projects
Government and NGO (B2G)	Nexa Co-Teacher for government school teachers and participation in a2i, Ministry of Education, UNICEF and World Bank education programmes	Institution workspaces and Teacher Assistant agent (3.11); possible in-country hosting for government data (8.16)
Proven learning outcomes	Randomised studies with universities measure score gains from NexaTute and publish the results	Anonymised research exports from the learning record store with ethics approval (9.1)
Diaspora market	Bangla language and Bangladeshi curriculum for children of Bangladeshi families in the Middle East, UK and USA	USD plan prices (11.2) and a Bangla-for-children course track
Learning guarantee	Learners who study regularly but do not improve get their minutes back as wallet credit	Learning Insights evidence (3.4); terms in 11.4

03. NexaTute — Detailed Feature Catalogue

3.1 AI Tutor / Personal Nexa Teacher

- Conversational explanations and follow-up questions.
- Step-by-step help, worked examples and guided practice.
- Explanations adapted to learner question, level, observed performance and learning context.
- Structured output validation for lab diagnostics, MCQs, true/false items and assistant responses using explicit schemas.
- Learning paths organized by topics, lessons and checkpoints; use uncertainty labels where learner evidence is insufficient.
- Two AI roles (3.16): the Nexa Teacher teaches live by voice, one-to-one or in a batch class (3.17); the Nexa Co-Teacher supports everything else in text — text lessons, subject answers, and the learner's own recordings and class summaries.

3.2 Voice, video and live learning

- Voice and video learning sessions, including spoken Q&A where enabled.
- LiveKit-based live classroom and group learning experiences.
- Mobile-first sessions with audio/text fallback when device, bandwidth or rendering support is limited.
- Low-internet mode: on a weak connection the avatar and video switch off and the class continues with voice and the whiteboard only (audio-only, about 30 kbps); learners can also switch it on themselves — important for rural learners. The microphone rules, the LiveKit connection features and the weak-network and weak-phone help outside class are in 3.28.
- Voice (speech-to-text and text-to-speech), Nexa Teacher video/3D avatar with lip-sync, Smart Whiteboard, Smart Virtual Laboratory, the 3D classroom (3.15), AI-tutor session recording and the MP4 of a downloaded class recording (9.5) run in the learner's browser (WebGPU/WebAssembly speech models, MediaRecorder); devices that fail the browser capability check fall back automatically to server-side speech under admission limits: speech-to-text on AI-2 Node A and the same in-house VITS voices on CPU (8.12). The 3D classroom never falls back to the server: weak devices get its Light or 2D version in the browser (3.15). This is the canonical browser-rendering rule; other sections refer to it.
- Live and group video: the speaker (teacher or active presenter) is sent and recorded at 720p and listeners at 160p using simulcast layers; on weak devices, low client bandwidth or high server load the speaker drops to 480p while listeners stay at 160p, switching automatically between the two profiles based on live server load and each client's measured internet speed (LiveKit adaptive streaming/dynacast).
- Recording is free and by choice (3.29, 11.3): a 1:1 class is recorded only when the learner chooses it, and a batch class once for the batch when at least one member asks for it, with a visible recording indicator, consent, access controls and the 12-month retention rule; replay is unlimited while the recording is kept.
- Multilingual experience only for languages supported and tested by the selected models and content pipeline.

3.3 AI-assisted learning content

- Generate course outlines, lessons, tutorials, assignments and learning videos from prompts or learning objectives.
- Create narrated educational content and AI-avatar teacher experiences where enabled.
- Generate reusable learning resources for educators, tutors and training providers.
- Produce personalized worksheets, assignments and evaluation reports; PDF output can be generated through WeasyPrint.

3.4 Personalization, assessment and practice

- Quizzes, question sets, assignments, knowledge checks and feedback loops.
- Identify likely knowledge gaps and misconceptions; these feed the Learning Insights and next-step plan below.
- How topic progress is measured: answer accuracy is the base, scored with interpretable models — Bayesian Knowledge Tracing (pyBKT) for topic mastery and Item Response Theory (girth) for question difficulty and learner ability; reasoning steps and written answers are graded against a rubric written for each question type, so correct steps earn partial credit and a right answer reached by a wrong method counts less; answers that needed hints carry less weight than unaided ones; each mistake is tagged by error type, and the same misconception three times marks the topic "misconception present" and triggers targeted practice; lab choices (step order and quality of the hypothesis) give a process-skill score; a "sure but wrong" confidence check lowers the score's certainty and raises revision priority; teacher feedback can confirm or adjust

a score and outweighs the AI estimate. Learners and teachers see which signals moved each score.

- Model policy: only rubric-based and interpretable models (rubrics, BKT, IRT) are used at launch, so every score can be explained to learners, guardians and teachers. Neural knowledge-tracing models are tested later, only once enough representative learner data has been collected, and are adopted only if they beat BKT/IRT on held-out data for every major learner group and their results can still be explained.
- Learning Insights: every topic gets a 0–100 mastery score, updated after each answer from the signals above, and is classified as strong, developing, weak or insufficient evidence with a confidence level and a plain-language explanation; topic scores roll up to each chapter and subject, so the learner sees which subjects are strong and which are weak.
- Skill map: each syllabus (NCTB, Cambridge, Edexcel, IB, BCS and others) is mapped as subject → chapter → topic by the AI content pipeline (3.13), with exam weight per topic, so every answer is counted against the right topic.
- Improvement trend: weekly change in mastery per subject and topic, shown as charts (for example “Mathematics +12% this month”), with the evidence behind each change.
- Where to improve first: weak subjects and topics are ranked by mastery gap, exam weight and days left to the exam, so the learner sees what needs work most.
- Next-step plan: for the three highest-priority weak topics the system prepares a targeted practice set, a short AI-tutor lesson and flashcards (3.10), and links them to the AI daily study plan (3.16; release timing in 3.10).
- Views: learner dashboard, guardian dashboard (3.10) and, for teachers and coaching centres, a class heatmap of weak topics across all their learners.
- Adapt lesson sequence, practice difficulty, examples and revision intervals after meaningful checkpoints.
- Present course-completion estimates as ranges with assumptions—not guarantees—and allow authorized learners/teachers to review or override recommendations.

Question types, written-answer checking, adaptive placement and practice tests, exam recovery and integrity, coding tests, the paper builder, certificates and proctoring are in 3.27; mistakes from every test go to the error notebook (3.18).

3.5 Premium 3D Nexa Teacher and browser rendering

- Render the 3D teacher/avatar in supported browsers using Three.js / React Three Fiber with WebGL/WebGPU where available.
- The 3D classroom (3.15) uses the same renderer, asset rules and device check as the 3D teacher.
- Synchronize voice playback and mouth movement through phoneme/viseme timing; expressions, blinking and gestures depend on the implemented avatar pipeline.
- Optimize GLB/glTF geometry, textures and assets; use adaptive visual quality and fallback modes for lower-end devices.
- Keep voice (STT/TTS), avatar display, lip-sync presentation, whiteboard rendering, supported interactive lab visuals and AI-tutor session recording client-side. Backend services provide AI inference, data, synchronization, media distribution and explicitly assigned processing.
- Download browser speech models once and cache them on the device to limit mobile-data use; detect WebGPU/WebAssembly capability and memory before enabling in-browser speech, otherwise use the server fallback.
- Do not promise a fixed frame rate on every device; measure performance across representative devices and browsers.

3.6 Interactive whiteboard and Smart Virtual Laboratory

- Collaborative canvas for writing, drawing, formulas, diagrams, highlighting, erasing and guided problem-solving.
- Math tools for equations, graphs, geometry and worked examples.
- Supported physics, chemistry and biology simulations, including a guided lab flow: objective → prerequisites/safety context → hypothesis → simulation → observation → explanation → knowledge check.
- Three.js / React Three Fiber renders 3D lab rooms, instruments and scenes locally; Ammo.js/WebAssembly can provide client-side rigid-body physics where supported.
- Yjs-based CRDT synchronization for whiteboard strokes and shared lab-object/state changes; define authorization, conflict behavior and persistence.
- Label simulations clearly; do not present simulated outputs as real measurements or substitutes for required supervision.
- Practice labs for technical and IT courses (11.2 catalogue, vocational and diploma): every lab is a clearly labelled practice environment with automatic checking of each task.
- Data Science and AI/ML: JupyterLite notebooks with Python data and machine-learning packages running in the browser.
- System administration, Cloud and DevOps: a small Linux system running inside the browser (v86) for command-line practice; Docker and Kubernetes exercises run in short-lived, network-less sandbox containers on the server (gVisor), capped in number and closed after 30 minutes.
- Cybersecurity: capture-the-flag style challenges against an intentionally vulnerable practice app (OWASP Juice Shop) inside the same isolated sandboxes; no real systems or internet targets are ever used.
- Robotics and Industrial Automation: robot and conveyor simulations in the 3D lab engine (Three.js and Ammo.js), programmed with Blockly blocks or Python, plus ladder-logic exercises in a browser simulator.
- Game Development: small game projects with Phaser in the coding sandbox.
- Electrical and electronics (diploma and engineering): circuit diagrams and ngspice circuit simulation (3.13).
- IFC/BIM asset preparation via IfcOpenShell/Bonsai may support selected asset conversion, but does not alone provide a complete browser-native CAD application.

NexaTute Laboratory — highlighted feature set. The laboratory is shown as its own feature group across every plan:

Laboratory feature	What the learner gets
Virtual Lab in every plan	Physics, Chemistry and Biology experiments in the learner's browser, free in the Batch, Combo and Personal plans (11.1); no server rendering
Lab in the group class	The Nexa Teacher demonstrates an experiment; learners first predict the result in a poll, then every learner runs the same experiment in their own browser and compares (3.17)
Lab in 1:1	The learner changes the variables freely; the AI watches each step and stops at the first wrong step to explain it
SSC/HSC practical exam preparation	Board practicals done virtually — procedure, observation and calculation — with help for the practical notebook and an AI practical viva mock
Coding and cybersecurity labs	Server sandboxes for skill courses (3.12) up to 1 hour a day per learner; each session closes after 30 minutes; more time with a Daily Pass (11.12)
Lab KPIs	Choices made in experiments (changing the right variable, step order, quality of the hypothesis) count in Learning Insights (3.4) and the Thinking Profile (3.18)

Laboratory feature	What the learner gets
From virtual to real	After a virtual lab the learner can test the same idea with a home experiment using household items, with photo or video upload and AI feedback (3.10)

Release timing of these laboratory items is set in the release plan after the wireframes; the interactive science simulations keep their timing in 3.10. The laboratory adds no server rendering: experiments run in the browser, and only the coding and cybersecurity sandboxes use spare CPU (8.12).

3.7 Class summary, learner profile and adaptive content

- Class summaries covering topics, learner questions, exercises, unresolved points and recommended revision, made automatically in the learner's language and delivered in the app and as a PDF: AI-tutor sessions within 1–2 minutes from the session text; live and group classes within 10–15 minutes from each speaker's audio track and the whiteboard event log; saved lessons once, when the lesson is first created.
- Topic-level evidence tracking and trend analytics such as D1/D7/D30 retention where appropriately measured.
- Personalized reports and progress indicators with transparent confidence and evidence.
- Recommendations can be challenged, annotated, reviewed, overridden or reverted by authorized learners/teachers.
- Use scikit-learn for exploratory analytics only where appropriate; validate predictions against observed outcomes before using them as product claims.

3.8 Institution, account and platform operations

- Learner, educator, guardian and institution roles where included in the release.
- Plan entitlements (Batch, Combo and Personal, 11.1), daily 1:1 minutes, the wallet for extra minutes and add-ons, recording allowances and account controls; one face-verified learner per account (3.16).
- Privacy-aware recording management, particularly for minors, including consent, access, retention and deletion workflows.
- Authenticated access, least privilege, auditable actions, reporting and operational monitoring.
- The full platform admin panel, the Institute Admin and the company accounting are in Section 38.

3.9 NexaTute software-to-feature map

Software / component	Primary responsibility	Important implementation boundary
FastAPI — NexaTute API (App node cluster)	Learning APIs, auth, lab coordination, evaluation submission	Validate identity, entitlements, rate limits and request schemas
vLLM Router / vLLM	Local LLM serving and inference routing	Use benchmarked model/runtime settings; isolate workload budgets
Qwen3-30B-A3B (MoE) with a per-request thinking switch (Apache-2.0)	Learning tutor inference	Shared LLM pool at priority 100; benchmark quality and Bangla accuracy
Instructor + Pydantic	Structured AI output validation	Reject/repair invalid output; validate safety-sensitive results

Software / component	Primary responsibility	Important implementation boundary
PostgreSQL	Durable accounts, assessments, billing and learning records	Backups, replication, recovery and schema migration required
TimescaleDB	Time-series lab events and assessment/response timing	Collect meaningful events and aggregates; avoid unnecessary surveillance
Redis1 / port 6379	Short-lived session, cache, presence and transient lab state	Isolate from Celery broker; durable records remain in database
LiveKit SFU	Live audio/video distribution	Capacity depends on tracks, layers, bitrate, topology and network
coturn	STUN/TURN connectivity fallback	TURN-relayed traffic and network path require real load testing
Yjs + Excalidraw	Collaborative whiteboard and state synchronization	MIT licences; verify authorization and persistence
Three.js / React Three Fiber	Browser-side 3D lab and avatar rendering	Adaptive quality and browser/device compatibility
Classroom builder (Three.js, React Three Fiber, CC0 asset libraries)	Build and redesign virtual classrooms; AI room designer (3.15)	Browser rendering only, also on weak devices (Light or 2D version); designs in PostgreSQL; assets from the CDN
Ammo.js + WebAssembly	Client-side physics/collision where supported	Test supported browser/device matrix
Browser STT/TTS (Whisper via transformers.js; in-house VITS voices via ONNX Runtime Web)	In-browser speech recognition and voice for the AI Tutor in all six teaching languages (4.2)	Device capability check; server fallback to Whisper large-v3-turbo on AI-2 Node A and the same VITS voices on CPU (ONNX Runtime); a language whose browser model has not passed its benchmark uses the server path (Chatterbox or Indic Parler-TTS on GPU for its voice)
Event-log recorder (in-house) + MediaRecorder (Opus) + signed R2 upload	Recording of an AI-tutor class the learner chooses to record, and one shared batch recording when a member asks (3.29), as an event log (AI text, whiteboard events, timing) with only the learner's own Opus audio (11.3)	Server-side consent, size and checksum verification; ≈5 MB per 75-minute class; the AI voice is re-created on replay by the in-browser VITS voice with the pinned voice version (server CPU voices on weak devices)
Classroom voice, reports and hub (LiveKit Agents, Centrifugo, ClickHouse, ts-fsrs, pyBKT/girth, BGE-M3)	Mic rules and turn detection (3.28); recording library, class reports, Mastery Map and All Exams (3.29); the Exam & Admission Hub (3.30)	No new servers; signed 15-minute links for private recordings (9.4); hub circulars read by the local Qwen3 and approved by a reviewer
Engagement and access (MindAR, model-viewer, Three.js, Motion Studio Quick, LiveKit Egress, WhatsApp/Messenger/SMS)	Team Quest, Quiz Battle, Live Game Show, AR Magic Book, Be the Teacher, study bots (3.32); the sign-language avatar and Voice mode (3.31)	Mostly in the browser; bots answer from the verified cache; each channel has its own budget (38.18)

Software / component	Primary responsibility	Important implementation boundary
Recording download builder (WebCodecs, Mediabunny, Web Audio, OffscreenCanvas)	Makes the MP4 of a recorded live or group class in the user's browser when it is downloaded (9.5)	Tracks and the whiteboard event log come from R2 through the CDN; 1-minute parts with resume; no compositing or encoding on the servers
IfcOpenShell / Bonsai	Selected IFC/BIM asset parsing and conversion	Not a complete browser-native CAD experience by themselves
scikit-learn	Learning trend/retention analysis experiments	Predictions are estimates; validate against real outcomes
WeasyPrint	PDF worksheets, reports, summaries and certificates in all six languages (HTML/CSS templates)	Template, accessibility and localization tests
Qwen3-VL-8B + PP-FormulaNet + Tesseract	Photo questions, formula reading and answer-script checking (3.10)	Qwen3-VL on the GPU pool at P100; OCR on spare CPU; numeric and formula answers checked with SymPy
wav2vec2 phoneme model + g2p_en + CMU dictionary	Pronunciation scoring for English speaking practice	Runs with the speech fallback on AI-2 Node A; scores validated against human raters
ts-fsrs	Flashcard spaced repetition	Runs in the browser; review state synced to PostgreSQL
Pyodide + CodeMirror 6	In-browser coding sandbox	Code never runs on the servers; runtime cached on the device
Skill-course file checker (openpyxl, python-docx, python-pptx, LibreOffice headless)	Automatic checking of uploaded Word, Excel and PowerPoint practice files (3.12)	Runs in an isolated worker with no network access; files deleted after grading unless the learner keeps them in the portfolio
Penpot (self-hosted)	In-browser design editor for the graphic design course (3.12)	Own database in the PostgreSQL cluster and files in R2; per-learner projects
AI content pipeline (Scrapy, Docling, Mistral Small 3.2, SymPy, Ragas, datasketch, promptfoo)	Collects, builds and checks the category knowledge bases and question banks without manual work (3.13)	Approved source list and licence check; every item versioned with its sources and check results
Medical and engineering tools (schemdraw, ngspice, JupyterLite, Z-Anatomy models)	Circuit diagrams and simulation, browser notebooks and 3D anatomy for medical and engineering subjects (3.13)	schemdraw and ngspice run in the isolated, network-less sandbox; notebooks and 3D anatomy run in the browser
Practice labs (v86, gVisor sandboxes, OWASP Juice Shop, Blockly)	Browser Linux, Docker/Kubernetes and cybersecurity practice, robot programming (3.6)	Sandboxes have no network access, are capped in number and close after 30 minutes
Language-learning tools (Chatterbox voices, hanzi-writer, KanjiVG, Noto Sans CJK)	Target-language voices, character stroke order and CJK text for foreign-language courses (11.2)	Lesson audio made once and cached; live practice voices from the speech pool

Software / component	Primary responsibility	Important implementation boundary
LiveKit SIP + LiveKit Agents	Phone tutor (IVR): connects phone calls from a telco SIP trunk to the AI Tutor voice pipeline (3.10)	Runs on the LiveKit cluster (Core-1/Core-2); speech on the server speech pool
Centrifugo + assessment engine + Web Crypto	National live model tests: encrypted paper released at start time, answers saved on the device and submitted, merit list in PostgreSQL (3.10)	Paper served from the CDN; key released by Centrifugo at the start time
Avatar and voice tools (three-vrm, glTF-Validator, glTF-Transform, Chatterbox, in-house VITS, SpeechBrain, AudioSeal, C2PA)	Own 3D avatars and cloned voices for verified teachers, guardians (PlayBoom family voices) and learners aged 18 or over; the Avatar & Voice Studio library and marketplace (3.10, 38.16, G.3)	Shared engine with Motion Studio; consent sentence, speaker verification, watermark, C2PA and owner-only deletion; no clones or photo avatars under 18
Markmap + Apache ECharts	Mind maps and guardian-dashboard charts	Browser rendering
In-house notification service + WhatsApp Business / SMS gateway	Guardian weekly reports, study reminders and human-controller safety alerts (push and email; SMS as well for serious cases, 10.2); admin push notifications in the platform and Institute admin panels (38.15)	Guardian consent, opt-out and message-template approval
Capacitor	Android app from the same web code	Feature parity with the browser; Play Store review; in-app purchases through the store's billing, with the app-store charge as its own line (11.7)
pypdf + python-docx + python-pptx + BGE-M3	Study from own notes or PDF	Uploads private to the learner; size limits; scanned pages go through OCR first
LanguageTool	English grammar checks for writing feedback	Separate internal service; Bangla feedback comes from the LLM
Avro phonetic rules (from avro-py)	Bangla phonetic typing in the browser	Runs on the device; no keyboard install
Class summary pipeline (faster-whisper → Qwen3 → WeasyPrint)	AI-tutor and live/group class summaries (3.7)	Per-speaker audio tracks, no diarization; 15-minute target; summary-only audio deleted after use (9.5)
pyBKT + girth (IRT) + rubrics + Qwen3-30B-A3B + SymPy + TimescaleDB + ECharts	Learning Insights: combined topic progress from accuracy, reasoning steps, hints, misconceptions, lab choices, confidence and teacher feedback (3.4)	Score recomputed per answer on App CPU; step grading and mistake tagging on the LLM pool; weekly aggregation in the batch window; teacher adjustments outweigh the AI estimate
Indic Parler-TTS / Chatterbox + FFmpeg (Opus)	Audio lessons and Bangla voice reports for guardians	Made once per lesson or per weekly report in the batch window; served from the CDN
LiveKit 1:1 room + in-app booking	Guardian–counsellor calls	Booking, payment and consent checks in the App; no recording without consent

Software / component	Primary responsibility	Important implementation boundary
Keycloak	Multi-tenant identity and single sign-on for institution workspaces (3.11)	One realm or organisation per institution; guardian and minor roles enforced
LangGraph	AI agent layer: routes each request to the right agent with shared learner context (3.11)	Agents call the existing caches, Learning Insights and LLM pool; memory only with consent
Apache AGE on PostgreSQL	Curriculum knowledge graph (3.11)	Runs inside the existing PostgreSQL cluster; stable outcome IDs
Ralph LRS + PyLTI1p3 + scorm-again	xAPI learning records, LTI 1.3 and SCORM import (3.11)	Conformance tests with partner LMSs
Apache APISIX + Redoc	Public API gateway and developer portal (3.11)	API keys, rate limits and permission scopes per app
Centrifugo	Real-time notices and messaging (3.11)	Minor-safety rules enforced in the App before delivery
@digitalbazaar/vc	Verifiable certificates and learner passport (3.11)	Signing keys kept in OpenBao
Phaser + Mermaid + WebXR + Three.js	Mini-games, diagrams in answers, AR 3D models and science simulations (3.10)	All in the browser; AR only on supported phones, normal 3D view otherwise
Motion Studio Quick pipeline (internal API)	Explainer videos for hard NexaTute topics (3.10)	Called through the Motion Studio API with a service account at P10 in the batch window; output kept in the lesson cache
Centrifugo + LiveKit Egress (HLS) + Cloudflare CDN	Weekly national live quiz (3.10)	Questions and answers over Centrifugo; host video as HLS through the CDN, not through the SFU
Nexa Co-Teacher (LangGraph agent, Qwen3-30B-A3B, BGE-M3 retrieval, PostgreSQL row-level security)	Text lessons, subject answers and search of the learner's own recordings and class summaries, 25 topics a day (3.16)	Retrieval filtered by learner ID with row-level security, so one learner's data never reaches another; on overload the cloud API in the order Mistral Large 3 → GPT-6 Luna (8.9)
Batch class service (FastAPI, Centrifugo, LiveKit, Yjs)	AI group classes for up to 50 learners: AI text stream voiced in every browser, polls, confusion meter, question wall, teams, pair rooms and event-log recording (3.17)	One Nexa Teacher LLM session per batch at P100; one server voice stream per batch only for weak devices; poll bursts count toward App/API capacity (8.6)
Load-aware router (in-house; Prometheus metrics, Valkey queue)	Starts live Nexa Teacher classes at once when the GPU pool is free; when busy offers recordings, mock tests and summaries first, with an ETA and a kept queue place (3.16)	Thresholds set in the admin panel; minutes lost when live classes stop are credited back automatically

Software / component	Primary responsibility	Important implementation boundary
Face verification (MediaPipe Face Landmarker, SFace through ONNX Runtime Web)	One account = one learner: face match with a liveness check, random re-checks and OTP fallback (3.16)	Runs in the browser; the server keeps only an encrypted face template, deleted with the account; guardian consent for minors
aamarPay payment gateway	Plan purchases and renewals (auto or manual), wallet top-ups and add-ons (11.7)	Callbacks verified on the server before a plan is activated or the wallet credited (20.4)
Shared learner record (PostgreSQL, pgvector, Ralph LRS, Valkey, LangGraph)	One record read and written by the Nexa Teacher and the Nexa Co-Teacher: handover notes, combined evaluation and Nexa Co-Teacher class summaries (3.16)	Row-level security per learner; summaries made on the local LLM pool, later during overload
Engagement models (scikit-learn, LightGBM, Vowpal Wabbit; Unleash with TimescaleDB for A/B tests)	Learner state, next activity, reminder timing, drop-out prediction, interest map and A/B tests (3.19)	App-tier CPU on the event data in ClickHouse; each feature has its own consent switch, and the guardian decides for under-18 learners (19.4)
Live Study Hall (Centrifugo presence, Yjs groups, Nexa Study Hall Moderator)	Silent hall with 25-minute rounds and a question-bank check (free); paid class-wise groups of 2–10 with a shared Lab and Whiteboard and a Help button to the Nexa Study Hall Moderator (3.20, 31)	Groups and boards in the browser; the moderator uses the existing Nexa Co-Teacher LLM and Qwen3Guard on reports, no new GPU; nicknames and text-only chat for minors
Motion Studio Quick mode in the learner's browser	NexaTute Wrapped recap videos and personal mistake videos (3.20)	Rendered in the browser; only the script uses the LLM
AI Parent–Teacher Meeting (Nexa Teacher voice pipeline)	Monthly 10-minute meeting of the Nexa Teacher with the guardian (3.20)	Held off-peak; about 10 GPU-minutes per learner a month
Circular tracker (Scrapy in the AI content pipeline)	Job circulars, admission notices, application deadlines and exam dates with daily alerts (3.20)	A reviewer verifies every circular before the alert goes out
Public Q&A pages (static pages on the CDN) and Nexa Teacher Live (LiveKit Egress)	Verified answers published for Google search with an "Ask the Nexa Teacher" button, and a weekly public live session on Facebook and YouTube (3.21)	Verified answers only, with no learner names or data; one live session a week
Script Checker (Qwen3-VL overnight batch)	Marks schools' and coaching centres' scanned answer scripts with the teacher's rubric (3.21)	Runs in the quietest GPU hours (for example 2–6 am); the teacher approves the final marks
Vouchers, affiliate codes and marketplace payouts (in-house; aamarPay; Motion Studio payout system)	Shikkha Salami gift cards, affiliate commission and teacher-marketplace payouts (3.21, 11.14)	Commission only after the 30-day refund window; tax withheld; a unique code per gift card

Software / component	Primary responsibility	Important implementation boundary
Guardian purchase approval (notification service and aamarPay)	Purchases by under-18 learners need the guardian's approval within a monthly spending limit (11.9)	Payment starts only after the guardian approves on their phone
Link step of the Document Intake Service (URLhaus and StevenBlack hosts blocklists, gVisor, Docling, Qwen3Guard, NSFW classifier, YouTube Data API)	Safe handling of links shared by learners, guardians and teachers (3.23)	Blocklist first; sandbox without login, cookies or scripts; link content used only to explain
Landing Page AI and AI CMS (in-house on Next.js and PostgreSQL; Qwen3, FLUX.1-schnell, axe-core, Lighthouse CI, sharp)	Builds and runs the landing page and the CMS of both products with admin approval (3.25)	Pages only from the component library; prices read live from the price table; every change approved and reversible
Drop-off analysis (TimescaleDB product events, Grafana funnels, web-vitals, Qwen3 analyst)	Admin page showing why users drop and where, with a weekly AI report (3.25)	Aggregated and anonymous; no session recording of children
Sales tools (Chatwoot channels, notification service, in-house CRM)	Pre-sale AI on Messenger, WhatsApp and website chat; plan finder; abandoned-checkout recovery; B2B CRM; review collection (3.25)	Purchases by minors go to the guardian; reminders only with consent
AI Marketing Hub (Qwen3, FLUX.1-schnell, Motion Studio API, platform APIs, outreach mailbox, CRM)	Articles, videos, Reels, publishing, SEO, query answers, B2B outreach and reports (3.26); Channels by OAuth, bulk import, export and the target-clients list and map (3.26)	Approval before anything new goes public; cold email only to organisations; channel tokens in OpenBao; imports and exports approved, exports watermarked
MindForge (OR-Tools, girth, Qwen3, Mistral Small 3.2, PostgreSQL, Workbox)	Thinking-skills practice, calibrated measurement and expected syllabus completion (37)	Code-made puzzles must have exactly one solution; measurement uses only calibrated items; no IQ-style score; no new GPU
Study Hall Q&A (Valkey, PostgreSQL with pgvector, jsQR, Tesseract, OpenCV, Nexa Guard)	Free question-and-answer boards with verification, rewards and Moderator answers (31.3)	Images yes, links never; nicknames only; a verified answer enters the question bank only after a reviewer approves it
Calculator and dictionary (math.js, Open English WordNet, CMU dictionary, in-house VITS)	Free calculator and dictionary for everyone (3.20)	Mock tests follow the real exam's calculator rule

Software / component	Primary responsibility	Important implementation boundary
Platform admin panel and company accounting (Casbin, pgaudit, rclone, Keycloak, OpenBao)	Admin panel, Institute Admin and accounting (38)	The maker cannot approve own work; every action logged; nothing counts in reports until approved
RemonLogic Growth Platform (in-house CRM and journey engine, Scrapy, Qwen3, pgvector, Valkey streams, ClickHouse)	CRM, event-driven journeys and the target-client database for all three products (3.26)	Organisations only, never individuals' personal data; consent and opt-out per channel; no marketing to minors
Admin notifications and staff chat (notification service with pywebpush, FCM, Centrifugo, PostgreSQL, LiveKit, Qwen3Guard, Qwen3, ClamAV)	The bell, push and the notification control panel; internal staff chat with calls in the platform admin panel (38.15)	Push text never shows secrets; compliance check on send; calls not recorded; summaries encrypted; staff only
Nexa Notes (AI CMS, FFmpeg HLS, hls.js, faster-whisper, Qwen3, ClamAV, R2 + CDN)	The NexaTute landing-page blog with articles and videos (3.25)	Approved by the marketing admin or super admin, never the maker; no autoplay; YouTube click-to-load; no ads or tracking
Tests and written-answer checking (Qwen3-VL-8B, Qwen3, math.js, SymPy, girth, CodeMirror 6, gVisor, headless Chromium, datasketch)	Question types, written checks with part-wise marking, adaptive placement and practice, coding tests and the paper builder (3.27)	Scripts private to the learner and guardian; nothing marked before the learner confirms unsure parts; never sent to an outside AI; similarity flags only for human review

3.10 Engagement and interactive learning

These features make NexaTute more interactive and bring learners back every day. Features marked Launch are part of the 1 January 2027 launch; the others are new offerings on the roadmap (17.1). Rows marked "Release plan" were approved on 18 October 2026; their timing is set in the release plan after the wireframes.

Feature	What the learner gets	Release
Photo questions and answer-script checking	Photograph a textbook or homework question to get a step-by-step explanation; photograph a handwritten answer to get marks and corrections (Qwen3-VL); written tests with part-wise board-scheme marking, plan limits and human recheck (3.27)	Launch
Streaks, XP, badges and leaderboards	Daily study streaks, XP points and badges; class, school and district leaderboards; weekly challenges; leaderboards off by default, nicknames only under 18, and the streak and freeze rules of 3.31	Launch

Feature	What the learner gets	Release
In-lesson questions	The Nexa Teacher pauses for quick checks during a lesson, and the learner can stop it at any moment to say “I didn’t understand this”	Launch
Flashcards with spaced repetition	Flashcards generated from each lesson and from the learner’s own mistakes, brought back for review at the right time (FSRS)	Launch
Mobile app	Install NexaTute on the phone from the website — the installable web app is the main mobile channel and also works on iPhone — or as an Android app built with Capacitor, with the same features as the browser version; apps are also on Google Play, the Apple App Store and the Microsoft Store, where purchases use the store’s billing with the app-store charge shown as its own line (11.7)	Launch
Certificates	A course-completion certificate after passing the final assessment, with a QR code anyone can scan to verify it online and an “Add to LinkedIn Profile” button (18+ accounts) that pre-fills the certification on the learner’s profile (0.7); competency levels and the retake rule (3.27)	Launch
Referral programme	Invite a friend; both receive bonus learning minutes in their wallets after the friend’s first payment (plan or wallet)	Launch
Study Hall Q&A (community forum)	Question-and-answer boards by subject and course, free for everyone; answers are verified by community signals and the AI against the verified knowledge base, with human review; the Nexa Study Hall Moderator answers when nobody has within 30 minutes; images yes, links never (31.3)	Launch
Bangla phonetic typing	Type “ami” to get “আমি” in chat and answers, Avro-style, without installing a keyboard	Launch

Feature	What the learner gets	Release
Placement test	A 10–15 minute diagnostic test at sign-up sets the learner's level and first learning path and seeds the mastery scores (3.4); adaptive (CAT) from IRT-calibrated items, with the level shown as a range (3.27)	Launch
Report a wrong answer	One tap reports a wrong AI answer; confirmed errors refund the minutes used and the corrected answer goes to the verified-answer cache. A paid panel of subject experts also reviews a weekly sample of AI answers (Content budget, 15.2)	Launch
Exam countdown and revision mode	SSC, HSC, BCS and admission exam dates with a countdown; a focused revision mode switches on as the exam gets close	Launch
Re-engagement messages	Push or SMS when a streak is at risk after 3 days without study, 5 days before a plan renews or ends or the wallet expires, and before exams with the learner's weak topics	Launch
Hint ladder	A small hint first, then a bigger hint, then the full solution; the number of hints used feeds the Learning Insights (3.4)	Launch
Confidence check	Before answering, the learner says how sure they are; "sure but wrong" answers feed the Learning Insights (3.4) and raise revision priority	Launch
Focus timer	25 minutes of study and a 5-minute break, with short rest tips and exam-stress advice	Launch
Study-time controls for guardians	Guardians set a study schedule, block the app after a chosen hour at night and cap daily use	Launch
Examples from the learner's interests	The learner picks interests (cricket, cooking, games, football ...) and the AI builds examples around them, for example percentages from a batter's strike rate	Launch
Puzzle of the day	One daily puzzle for everyone, Wordle-style, with a shareable result	Launch

Feature	What the learner gets	Release
Mixed practice	Practice sets that mix questions from different topics, so learners learn to recognise which method each problem needs	Launch
Diagrams with explanations	AI answers come with a diagram, flowchart or picture (Mermaid diagrams and the FLUX image cache)	Launch
English speaking practice	Conversation and role-play with the AI, pronunciation scoring and IELTS speaking mocks with an estimated band; marked on clarity, fluency, vocabulary and grammar, never on accent (3.27)	Launch
Writing feedback	Feedback on Bangla and English essays, letters and IELTS writing: grammar, structure and an estimated band	Launch
PlayBoom (Play class–Class III)	Voice-first lessons with pictures and animation, big buttons, slower speech, letter and number tracing on the whiteboard, star rewards and short sessions (25 minutes for Play/Nursery/KG); started by a guardian, with no forum, leaderboard or chat; replaces PlayBoom — full module in Section 30	Launch (paid from 21 January 2027)
OMR practice (on screen)	Board-style OMR sheets on screen for SSC, HSC, Admission and BCS MCQ practice, with exam timing and no changing a filled bubble; instant score, explanation of each mistake and weak topics; no printing needed	Launch
Board-book solutions	Step-by-step solutions to every exercise in the NCTB textbooks, chapter by chapter for Class 1–12, each linked to a short lesson and practice questions; each exercise is named by chapter and number and restated in the AI's own words; made and checked by the AI content pipeline (3.13)	Launch

Feature	What the learner gets	Release
National live model tests	SSC, HSC, admission and BCS model tests at a fixed time for the whole country, with the same paper for everyone, instant MCQ marks, written answers marked by answer-script checking, and a national and district merit list with percentile; shuffled question and option order and an answer-similarity check for human review, without proctoring, as in every practice test (3.27)	Launch
Own avatar and voice (from Motion Studio)	The same character-creation and voice-cloning tools as Motion Studio, inside NexaTute: teachers can let the AI teach their class with their own 3D avatar and voice; parents can record encouragement and story messages for PlayBoom in their own voice; students design their own 3D avatar for their profile, Quiz Battle and study buddy (voice cloning only for learners aged 18 or over); free and paid allowances as in G.3 The Avatar & Voice Studio (38.16) manages the libraries, clone requests and the marketplace: one of each free, more bought; guardians clone only their own voice (never a child's), and verified teachers may also make an avatar from their own face photo after a face check.	Launch
Quiz Battle	Live quiz contests in Group Study and 1-vs-1 quiz duels with friends	Launch
Socratic mode and difficulty control	The AI guides the learner with questions instead of giving the answer; "explain more simply" and "make it harder" controls	Launch
Guardian dashboard and weekly report	Progress, study time and weak topics for guardians, with a weekly WhatsApp or SMS summary and a 1-minute Bangla voice summary for guardians who prefer listening; for batch learners the weekly report also shows attendance, best and weak topics and next week's goal (3.17)	Launch
Exam simulator	Full-length, timed board, BCS and admission mock exams in a real-exam interface, with a predicted rank; fixed papers, never adaptive, with the practice-exam note (3.27)	Launch

Feature	What the learner gets	Release
Study from own notes or PDF	Upload class notes or a teacher's sheet (PDF, DOCX, PowerPoint or photo); the AI teaches from it and makes practice questions (3.14)	Launch
Family account and gifting	One wallet shared by siblings under a guardian; relatives in Bangladesh or abroad can top up or gift wallet balance	Launch
Share achievements	Shareable cards for badges, certificates and results on Facebook and WhatsApp, carrying the learner's referral code	Launch
Weekly live revision class	A weekly live revision class in the main subjects by expert human teachers, alongside the AI Tutor (teachers paid per class from the Content budget, 15.2); held inside the learner's Virtual Batch (3.17)	Launch
Mastery-based progression	The next chapter fully unlocks after about 80% mastery of the current one; until then the AI strengthens the weak parts first	Launch
Teach-back mode	The learner explains a topic to the AI in their own words, by voice or text, and the AI points out gaps in understanding	Launch
Past-paper bank	Previous SSC, HSC, BCS and admission questions by year and board, each with an AI explanation — Bangladesh board questions used verbatim with the source year and board, never as whole papers; Cambridge, Pearson and IB as original syllabus-matched questions with links to the official sites (3.30)	Launch
Audio lessons	5–10 minute audio summaries of each chapter for listening while travelling, using little mobile data — Nexa Radio (3.32)	Launch
Story missions	Chapters become story missions (for example a "Save the river" mission for the water cycle and pollution); the story moves on as steps are completed	Launch
Explainer videos from Motion Studio	Hard topics get a 1–2 minute animated explainer made once with the Motion Studio Quick pipeline and kept in the lesson cache	Launch

Feature	What the learner gets	Release
Topic mini-games	Browser mini-games per topic, such as a maths race, word games or matching chemical compounds (Phaser)	Launch
Interactive science simulations	Physics and chemistry simulations with sliders and live graphs (for example how mass and force change motion), built in-house with Three.js	Launch
Weekly national live quiz	A weekly live quiz show for learners across the country — the Live Game Show (3.32) — with bonus minutes and team results, never cash prizes for children	Launch
Sponsor-a-Learner	Companies, NGOs and people in Bangladesh or abroad buy learning minutes for a named learner or for learners chosen by partner schools and NGOs; sponsors see consented, anonymised progress and certificates and get a CSR receipt (11.9); sponsors can also pay a monthly plan (11.1), and rural girls are given priority	Launch
Coding sandbox	Run Python in the browser for ICT and programming courses (Pyodide); graded coding tests run on the server in the gVisor sandbox (3.27)	Launch
AI daily study plan and reminders	After the learner or guardian confirms the syllabus (and uploads the school routine if they wish, 3.14), the AI sets the daily study time, re-plans every week and sends push reminders (3.16)	Launch
Mind maps	Automatic concept maps for any chapter (Markmap)	Launch
Accessibility options	Dyslexia-friendly font, read-aloud of all text, captions on every lesson, slower speech, shorter segments, more breaks and less time pressure for learners with dyslexia, ADHD or slower learning, set in the guardian's settings; the whole platform is built to WCAG 2.2 AA (19.7); also the extra-time settings, Voice mode, the 20-20-20 eye break, the sign-language avatar and the Sensory-friendly mode (3.31)	Launch
My Notebook	Save explanations, highlight, and search all past lessons and chats	Launch

Feature	What the learner gets	Release
Career guidance	AI guidance on subject choice after SSC and HSC, universities and careers	Launch
Study buddy	Pairs two learners with a similar level and goal for Group Study and Quiz Battle; guardian approval for minors	Launch
Guardian goals and rewards	Guardians set a goal (for example 70% mastery in Mathematics this month); when it is reached the guardian gives their own reward and the app adds bonus minutes	Launch
Guardian–counsellor call	Each month the AI prepares a detailed report, and the guardian can book a 15-minute call with a human counsellor for ৳100 (11.4)	Launch
AR 3D models	View molecules, the solar system or a human heart in 3D on the table through the phone camera (WebXR); phones without AR get a normal 3D view	Launch
Phone tutor (IVR)	Learners without a smartphone call the licensed IPTSP hotline number (3.28) and learn by voice in Bangla: ask questions, hear audio lessons and answer quizzes with the keypad; billed from the wallet linked to the phone number or through the telco bill under a telco partnership (2.5)	Launch
Nexa Teacher remembers you	The Nexa Teacher remembers each learner's interests, weak points and last class, and opens the next class with them	Release plan
Thana leaderboards	Leaderboards by thana (sub-district) beside the class, school and district boards; nicknames and opt-in for minors as on the other leaderboards	Release plan
Spend XP in the Classroom Builder	XP buys posters, themes and avatar outfits for the learner's own room (3.15) — earned by studying, never bought with money	Release plan
Chapter Boss battle	A hard challenge at the end of each chapter; passing it unlocks the next chapter with a badge	Release plan

Feature	What the learner gets	Release
Shorts feed	60-second topic Shorts made once with the Motion Studio Quick pipeline and scrolled like a social feed, but only for learning; served from the CDN	Release plan
National-occasion events	21 February language challenge, 16 December quiz and a Ramadan after-Fajr study streak	Release plan
Guardian and child quiz	A weekly quiz played together by a guardian and child, especially in PlayBoom	Release plan
Exam-eve mode	On the day before an exam: no new topics, only the learner's own error notebook (3.18), the formula sheet and 10 MCQs, ending with a "go to sleep" message	Release plan
Daily reading habit	10 minutes a day of Bangla or English stories, news and science articles, with three comprehension questions	Release plan
Monthly mini-project	A small real project each month — plan, measure, result (for example cutting the home electricity bill) — presented as a 1-minute Motion Studio Quick-mode video	Release plan
Home science experiments	Simple experiments with household items after a virtual lab; photo or video upload with AI feedback (3.6)	Release plan
2-minute "explain it"	The learner records a 2-minute explanation of a topic and the AI gives feedback on clarity, steps and pace (extends Teach-back mode)	Release plan
Curiosity corner	15 minutes a week, text only, for any "why?" question outside the syllabus	Release plan
Guardian coaching tips	One short tip a week for guardians, for example how to respond to mistakes without pressure	Release plan
Life-skills sessions	One 20-minute session a month on money basics, digital safety and time management	Release plan
BCS Written and Viva mock board	Three AI examiners with different voices and personas ask Written and Viva questions and give feedback at the end	Release plan

- Server load: most of these features run in the browser or on the App and database tier. Photo questions and pronunciation scoring use the shared GPU pool at P100 and pass the cache layers and safety checks (8.14); written checks are a queued job within a daily GPU cap (3.27, 8.17); guardian reports use the WhatsApp Business Platform (a paid managed service) with SMS as fallback; weekly live revision classes use the existing LiveKit and recording capacity. The 18 October 2026 additions are text, cached content or browser work: Shorts are made once in the batch window and served from the CDN, and the Viva board uses the existing voices.
- Privacy: leaderboards show a nickname and are opt-in for minors; photographed answer scripts are seen only by the learner and the guardian (3.27); voice recordings and uploaded notes stay private to the learner; forum posts by minors are pre-moderated; a family account, study-time controls and study-buddy matching for minors are controlled by the guardian. All follow the learner-privacy rules in 10.2.

3.11 AI-Powered Learning Operating System

NexaTute is built as a learning operating system, not only a tutoring app: learners, guardians, teachers, schools, coaching centres and outside developers work on one platform that shares the same curriculum graph, learner record and AI services. The platform layers below build on the features in 3.1–3.10.

Platform layer	What it provides	Builds on
Institution workspaces (LMS)	Separate workspaces for schools and coaching centres: admins, teachers, classes, timetable, attendance, homework, gradebook and notices, with each institution's data kept apart	B2B (11.8), class heatmap (3.4)
Teacher content studio	Teachers create lessons, quizzes and worksheets with AI help, or upload their own lesson scripts and question banks (3.14), keep versions and share them with their class, their institution or everyone	AI-assisted content (3.3)
AI agent layer	Specialised AI agents — Tutor, Examiner (question setting and script marking), Planner, Counsellor, Content Creator and Teacher Assistant — sharing one learner context, with long-term learner memory kept only with consent	AI Tutor (3.1), Learning Insights (3.4)
Learner passport	A lifelong learning record that moves with the learner between schools; verifiable certificates in Open Badges 3.0 / W3C Verifiable Credentials format (the 3.10 certificates use it); learners own their data and can export it	Certificates (3.10)
Open API and app store	Developer portal, REST API and webhooks; outside teachers and companies add simulations, quizzes or games as sandboxed plug-in apps with permission scopes and revenue share	Virtual Lab (3.6)

Platform layer	What it provides	Builds on
Interoperability	LTI 1.3 to plug into other LMSs, xAPI learning records in a learning record store, SCORM course import and CSV roster import	Institution workspaces
Curriculum knowledge graph	NCTB and other boards' learning outcomes with stable IDs; every lesson, question, mastery score and plug-in app links to the same graph	Skill map (3.4)
Course marketplace	Teachers and coaching centres sell courses paid from the learner wallet, with revenue share (timing in 17.1)	Wallet (11.1)
Communication hub	Notices and messaging between teachers, learners and guardians; no private adult–minor chat without guardian visibility	Guardian dashboard (3.10)
Assessment engine	Secure online exams for institutions: question banks and the paper builder, randomised papers and sets A/B/C, limited proctoring with consent under the 3.27 rules, and result publishing; the roster and results API (18.3)	Exam simulator (3.10)
Admin and trust centre	Role-based access, audit logs, content moderation, child-safety controls, consent records and per-institution data-retention settings	Security (10.1, 10.2)

3.12 Skill Courses

Besides school, admission, BCS and international curricula, NexaTute teaches job and income skills. Every skill course has a level-by-level syllabus, short AI-tutor lessons, hands-on practice in the browser, graded projects, a final assessment and a verifiable certificate in the learner passport (3.11). The AI checks every practice task against a rubric first (3.4); a human mentor reviews only the final project, so the certificate carries weight with employers and clients while costs stay low.

Course	How it is taught, practised and assessed	Release
Spoken English	Levelled syllabus from beginner to job-interview level, built on English speaking practice and pronunciation scoring (3.10); role-plays for interviews, viva, customer calls and presentations	Launch
MS Office (Word, Excel, PowerPoint)	Each task comes with a practice file; the learner does the work in their own Office or LibreOffice and uploads it; the server opens the file and checks formatting, formulas, charts and slides automatically, then the AI explains every mistake	Launch
Video editing	Practice projects inside Motion Studio's editor (cuts, captions, sound, colour, export settings), reviewed by the AI against a rubric; finished work goes to the learner's portfolio	Launch

Course	How it is taught, practised and assessed	Release
Graphic design (catalogue: UX/UI & Product Design, 11.2)	Design tasks in a self-hosted Penpot editor in the browser (posters, social posts, logos, layouts); the AI reviews the exported design for layout, colour, type and brief fit	Launch
Digital and Facebook marketing (catalogue: Digital Marketing & Content Creation, 11.2)	Case studies and practice on audience targeting, ad copy, content calendars and budgets; a campaign simulator shows modelled results that are clearly labelled as practice, since no real ads are run	Launch
Freelancing	Building a Fiverr/Upwork-style profile, writing proposals and pricing; role-play with an AI client; portfolio from the other skill courses; links to the Motion Studio marketplace (2.5)	Launch
Coding	Project-based courses in the coding sandbox (3.10), with automatic test cases and AI code review (coding tests, 3.27)	Launch
Quran learning	Taught live by qualified human teachers over LiveKit; recitation and tajweed are corrected only by the teacher, not by AI; Quran text and translations follow the Islamic verification rules (4.5); the AI only helps with schedules, revision reminders and Arabic letter practice	Launch

3.13 Course Category Readiness

The AI Tutor can explain any subject, but a course category from 11.2 is sold only when it has its own knowledge base, its own question bank and a passed quality gate; until then it is shown as “coming soon”. All of this is built and checked by an automatic AI content pipeline — nobody writes questions or notes by hand. The AI always writes in its own words: sources such as the NCTB syllabus and textbooks are used only for the syllabus, learning outcomes and facts, and no book text or images are copied. People are involved only in the existing subject-expert panel that reviews answers learners report as wrong (3.10), and in licences for exam-board past papers if they are ever used. Only sources that RemonLogic has permission to use, or that carry an open licence allowing commercial use (for example OpenStax, CC BY 4.0), are collected.

Courses open in the launch order — BCS first, then English medium, then NCTB — and each one only after it passes its quality gate; part-time subject reviewers on contract support the Content Lead and writers (14.2).

Pipeline step	What the AI does
1. Collect	A crawler visits only an approved source list (official government and exam-body sites, open-licence repositories and files RemonLogic has a licence for), reads each source’s licence and rejects any that is unknown, non-commercial or no-derivatives; source, licence and date are stored with every document
2. Parse and map	Docling turns each document into clean text and tables; Qwen3 maps every passage to subject → chapter → topic in the curriculum knowledge graph (3.11); BGE-M3 embeds it into the category’s knowledge base
3. Generate	Qwen3-30B-A3B writes chapter notes, questions, answers, step-by-step explanations and mark schemes from the knowledge base only, with a set mix of easy, medium and hard items
4. Solve independently	A second model from a different family (Mistral Small 3.2) solves every question without seeing the first answer; if the two answers differ, the item goes back to step 3

Pipeline step	What the AI does
5. Rule checks	Maths and numeric answers are re-calculated with SymPy or code; Ragas checks that each answer is supported by the cited passage; a copy check (MinHash) makes sure no source text is reproduced word for word; Qwen3Guard checks safety; a language check scores grammar and clarity
6. Publish or discard	Items that pass every check are published automatically with a version number; failing items are regenerated up to three times, then dropped and logged
7. Weekly current affairs	Every week the crawler collects new official notices; a fact enters the BCS knowledge base only when two official sources agree, and each fact keeps its source and date
8. Learn from use	After launch, items that learners report as wrong, rate badly or that show poor IRT statistics (girth) are retired and regenerated automatically; confirmed errors from the expert panel go back into the pipeline
9. Quality gate	Before a category goes on sale, an automatic test set (promptfoo and lm-evaluation-harness) checks answer accuracy, language quality and, where official sample answers exist, score or band agreement; the gate re-runs after every model or knowledge-base change
10. Other teaching languages	For every course and language pair offered (table below), the AI translates notes, questions and explanations, keeps exam terms in the exam's language, translates them back and compares meaning with the original (BGE-M3 similarity and chrF score); items that drift are re-translated or dropped, and the pair then runs its own quality gate

What each category needs before it goes on sale:

Category (11.2)	Knowledge base	Questions and practice	Quality gate before sale
Bangla Medium and English Version (NCTB)	NCTB syllabus and textbooks (published free online by NCTB), used only for chapter structure, learning outcomes and facts; every note, explanation, question and solution is written by the AI in its own words, and answers name the book chapter	Chapter-wise AI question bank (creative (srijonshil) questions and MCQs) and board-book solutions (3.10)	Answer accuracy per subject and a Bangla language-quality benchmark; the NexaTute-Bangla model (2.5) when it beats the base model
Early classes — Play class to Class III (PlayBoom, Section 30)	Play/Nursery/KG: letters, numbers, rhymes, phonics, colours and shapes; Class I–III: the child's own country curriculum — in Bangladesh the NCTB books chapter by chapter (Bangla, English, Mathematics; Science, Bangladesh and Global Studies and religion in Class III, 30.9); all lessons written by the AI in its own words	Picture and voice questions, tracing and short games	Age-appropriate language and child-safety checks; sessions only with guardian start
BCS + Job Test Preparation (BCS, Bank and Government jobs)	Bangladesh general knowledge plus the weekly current-affairs base (pipeline step 7); every fact answer shows its source and date	AI model tests; past questions only with permission	Fact accuracy on Bangladesh topics; the tutor says it does not know rather than guessing, and flags facts newer than the last update
Admission Test Preparation (University, Medical and Engineering)	HSC syllabus content and each admission test's published syllabus	AI admission question bank calibrated with IRT (girth); past questions only with permission	Benchmark on admission-style questions per subject

Category (11.2)	Knowledge base	Questions and practice	Quality gate before sale
SSC/HSC Vocational and Diplomas (BTEB)	BTEB syllabi for each trade and technology	AI question bank; practice labs for technical subjects (3.6); workshop skills stay practical	Per-trade benchmark
Cambridge IGCSE, A-Level and Edexcel	Published syllabus learning outcomes mapped in the curriculum knowledge graph (3.11)	AI exam-style questions and mark schemes; board past papers only under licence	Per-subject benchmark against published specimen answers where licensed
IB	Subject-guide learning outcomes	AI practice questions; for the Internal Assessment, Extended Essay and Theory of Knowledge the AI only guides, questions and gives feedback — it never writes the work	Per-subject benchmark and academic-integrity checks
University subjects	Openly licensed textbooks that allow commercial use	AI practice sets	Per-subject benchmark; medical subjects are for study only, with no diagnosis or treatment advice for real cases; medical and engineering limits in the table below
IELTS, TOEFL and PTE Academic	Public test formats and band descriptors (PTE Academic scoring guide included), including the TOEFL iBT format used since January 2026 (1–6 score scale)	AI reading, listening, writing and speaking mocks; listening audio made with the server voices in several accents; speaking and writing scored with an estimated band from launch (3.10)	Band estimates agree with public sample answers that carry official band scores
Foreign languages, JLPT, TOPIK and EPS-TOPIK	Levelled courses from beginner upward; public test formats for JLPT, TOPIK and EPS-TOPIK	AI conversation partner and lessons in the target language with server voices; pronunciation feedback by comparing the learner's speech transcript with the target sentence; kana, kanji, Hangul and Chinese character stroke-order practice; computer-based mock tests in each test's format	Per-language benchmark and mock-score agreement with public sample tests
GRE, GMAT and SAT	Public test formats (GMAT Focus Edition, digital adaptive SAT)	AI adaptive question bank calibrated with IRT, with computer-adaptive mock tests	Score estimates checked against learners' reported real scores; quality routing if the gate is not met
ACCA, CA and CMA	Each exam year's syllabus and exam rules; Bangladesh and UK tax and law tables refreshed every exam year from the official Finance Act, NBR and HMRC publications	AI questions with worked solutions	Per-paper benchmark; tax and law answers state the year they apply to; quality routing if the gate is not met
PLAB, USMLE and OET	Public exam blueprints and content outlines; openly licensed medical sources and licensed texts only	AI single-best-answer and clinical-vignette questions; OET writing and speaking role-plays scored like IELTS; PLAB 2 clinical skills only as spoken practice (practical skills, 3.13 medical table)	Per-exam benchmark; medical safeguards as in the table below; quality routing if the gate is not met

Category (11.2)	Knowledge base	Questions and practice	Quality gate before sale
Other Popular Courses and any course	AI-built syllabus for every catalogue course; for "any course" the AI first builds an outline and labels it as AI-generated	Practice environments as in 3.12 and the practice labs in 3.6	Medical, legal and financial topics are for learning only; harmful topics are refused by Qwen3Guard

Quality routing: if Qwen3 does not pass a premium category's gate (GRE, GMAT, SAT, ACCA, CA, CMA, PLAB, USMLE), only the hard questions in that category go to the stronger cloud model (8.9) until the local model passes; the higher per-minute rate of these categories covers the cost.

NexaTute Verified badge: a category (or a subject inside it) gets the "NexaTute Verified" badge only when it passes its quality gate with at least 95% answer accuracy on its test set (proposed threshold). Next to the badge the learner sees the measured result — for example "96% accurate on 2,000 test questions, October 2026" — and a public page lists every verified category with its accuracy, test-set size, test date and model version.

The badge is kept up to date automatically: it is removed when the gate fails after a model or knowledge-base change, or when confirmed wrong-answer reports (3.10) stay above 1 per 1,000 answers for two weeks (proposed), and it returns once the category passes again. Marketing may use only these measured results — never a general claim such as "expert in every subject".

Course and language readiness: a course is offered in a teaching language only when that course-and-language pair passes its own quality gate; learners can pick only the languages that have passed for their course, and the NexaTute Verified badge is shown per language (for example "ACCA: English ✓, Bangla ✓"). The table shows which languages each category is offered in.

Category (11.2)	Teaching languages offered	Notes
Early classes (PlayBoom), Bangla Medium, English Version, BCS + Job Test Preparation, Admission Test Preparation, vocational and diplomas	Bangla, English	Bangladesh-specific courses; other languages add no value
IELTS, TOEFL, PTE Academic, SAT	All six for explanations	Practice, mocks and answers stay in English; rules and mistakes can be explained in the learner's language
Foreign languages, JLPT, TOPIK, EPS-TOPIK	All six for explanations	The language being learned is taught in that language; grammar and meaning explained in the learner's language
IGCSE, A-Level, Edexcel, IB, university subjects, ACCA/CA/CMA, GRE/GMAT, PLAB/USMLE/OET	All six	Exam terms kept in English as used in the exam; full curricula for other countries follow market entry (29.4)
Skill courses, Other Popular Courses and any course	All six	Bangla and English first; other languages as their gates pass
Quran learning	The human teacher's languages	Taught by qualified teachers (3.12)

Medical and engineering subjects: the AI is a strong study partner for theory, numericals, MCQs and viva practice, but it does not replace laboratory, ward or clinical training. The table shows what it teaches, where its limits are and which tools support it.

Area	What the AI teaches	Limits and safeguards	Tools
Engineering — theory and numericals	Concepts and derivations in mathematics, circuits, mechanics, thermodynamics, structures and programming; step-by-step numerical solutions	Every numeric answer is re-checked with SymPy or code before it is shown; long design problems go to quality routing if the gate is not met	SymPy, coding sandbox (3.10), JupyterLite notebooks in the browser
Engineering — diagrams and simulation	Circuit diagrams drawn from the AI's own description, circuit simulation with plotted results, graphs and free-body diagrams	The AI does not grade hand-drawn engineering or CAD drawings; drawings are checked only where a rule-based check exists	schemdraw for circuit diagrams, ngspice for circuit simulation, ECharts and Mermaid (already listed)
Engineering — design codes and labs	The method behind design calculations and lab reports, viva questions	BNBC, ACI and similar codes go into the knowledge base only under licence; until then the learner uses the code book and the AI explains the method; lab work stays practical	Knowledge base (3.13)
Medical — basic sciences	Anatomy, physiology, biochemistry, microbiology and pathology concepts and mechanisms; MCQ, viva and spoken OSCE practice; mnemonics	Copyrighted textbooks (for example Guyton, Robbins) only under publisher licence; open sources first	3D anatomy models (Z-Anatomy) in the 3D viewer; OpenStax Anatomy & Physiology in the knowledge base
Medical — clinical reasoning	Virtual-patient cases: the AI plays the patient, the learner takes a history, orders tests and reaches a diagnosis, then gets feedback	Study only — no advice about real patients; drug doses and treatment guidelines always show their source and year with a "check the current national guideline" note	AI agent layer (3.11) with a fixed case script and marking rubric
Medical — images (X-ray, histology, ECG)	Image-based questions with explanations	The AI's reading of a medical image is never shown as the answer by itself; image questions come only from labelled datasets whose correct answer is known and whose licence allows commercial use (for example PTB-XL ECGs, CC BY 4.0)	Labelled open datasets checked by the content pipeline (3.13)
Medical — practical skills	Explains steps of examinations and procedures for revision	Physical examination, procedures and ward skills are not taught by the AI; they need supervised clinical training	—

3.14 Uploads by Learners, Guardians and Teachers

Learners, guardians and teachers can all upload files, including images, from launch. Every upload goes through the shared Document Intake Service (5.3), which reads the file, converts Bijoy text to Unicode, reads images and scanned pages, and runs the 4.15 safety checks. Links shared by learners, guardians and teachers are handled in 3.23.

Who	What they upload	What happens	Charge
Learner	Homework sheets, own notes or a teacher's sheet, own written answers (photo, PDF, DOCX, PowerPoint)	Homework Help lists the questions and teaches them one by one; own notes become lessons and practice questions; own answers are checked with marks and corrections (3.10)	Uploads free; tutoring billed as normal learning minutes at the course rate; 10 free images a day, each extra image costs 1 learning minute at the course rate
Guardian	School diary or homework list, exam routine and syllabus, school report card	Daily study list and reminders for the learner; exam countdown and revision plan (3.10); report-card weaknesses added to Learning Insights (3.4)	10 free images a day; each extra image costs 1 learning minute at the course rate from the learner's wallet
Teacher	Lesson scripts and notes; question banks (Excel, CSV or Word template)	The Nexa Teacher delivers the lesson in the teacher's own avatar and voice (3.10), either as written ("as-is") or improved by AI with every change shown for approval; a lesson can also be made into a Motion Studio explainer video; imported questions are verified by the 3.13 pipeline	Free, no limit

- Homework mode: the default is "teach me" — the AI explains the method with the hint ladder and shows the full solution only after the learner has tried; a guardian or teacher can switch to "show full solution" or "hints only".
- Learning from teachers' question banks: RemonLogic's servers may learn from imported questions only after the teacher gives licence consent at upload (also in the terms of service); questions that match copyrighted board papers or guide books (MinHash check) stay in the teacher's own class only; the rest are solved independently and checked by the 3.13 pipeline, rewritten in the AI's own words and only then added to the shared question bank and knowledge base; contributors get a Contributor badge; later, verified data may help train the NexaTute-Bangla model (2.5) after its own quality gate.
- Privacy: learner and guardian uploads are visible only to the learner and their guardian; a teacher sees them only when shared; homework images are deleted automatically after 90 days, or earlier by the guardian; teacher uploads are private by default, can be shared with a class or institution, and become public only after the 3.13 quality gate.
- Admin control: daily free-image limits, the extra-image charge, the review and approval queue for imported questions, subject-wise approval, blocking of low-quality contributors and removal of any question are admin-panel settings.
- Images: image files (JPG, PNG, WebP, HEIC) and images inside PDF, DOCX and PowerPoint files are extracted and understood in place — diagrams, graphs and handwriting by Qwen3-VL, printed text and formulas by Tesseract and PP-FormulaNet — so a question with a diagram is explained with the diagram.

3.15 Classroom Builder and Redesign

Learners, teachers and institutions can build their own virtual classroom and redesign it at any time. The classroom is always rendered in the user’s web browser — on weak devices too — and never on the server; the server only stores a small design file. Everything in this section is available from the NexaTute launch on 1 January 2027.

Who	What they can do
Institution admin	Branded rooms for the whole institution (logo, colours, name board) in its workspace (3.11); can lock the design for all classes and approves templates shared inside the institution
Teacher	Builds a room for each class or subject from a template and redesigns it: board type and position, desk layout (rows, U-shape or groups), subject props (globe, periodic table, lab bench, maps), posters and the seat plan for live classes; saves a room as a template for colleagues
Learner	Own study room for AI-tutor sessions: theme, colours, wallpaper and desk items
PlayBoom	Only child-friendly rooms and props; the guardian chooses the room

- Templates: school classroom, university lecture hall, science lab, computer lab, library and quiet study room, kids' room, IELTS and speaking room, and a simple, plain room for human-taught Quran lessons; seasonal themes such as Eid and Pohela Boishakh.
- Editor: drag, rotate and scale any object (Three.js transform controls), change colours, materials, lighting and time of day, undo and redo, and keep a version history with one-click restore; a calm low-motion mode and a high-contrast mode support accessibility.
- AI room designer: the user describes a room (for example "a science lab with a green board") and Qwen3-30B-A3B with grammar-constrained JSON output (xgrammar) builds the design from the approved asset library; no images are generated, so a design costs one short LLM call.
- Own posters and logos: uploaded through the Document Intake Service (5.3) and checked like every other image (NSFW classifier and child-safety hash matching, 4.15); they stay private until checked; anything shown to minors needs teacher or admin approval, and PlayBoom rooms allow no free text on the walls.
- Browser rendering: Three.js and React Three Fiber draw the room in the browser (WebGPU, otherwise WebGL2); a device check picks the Full version, a Light version (fewer objects, baked lighting, no real-time shadows) or a 2D version (a flat drawing of the same layout) — every version is drawn in the browser. Assets follow the 3.5 asset rules and load progressively, room shell first, so a lesson starts within seconds.
- Inside the room: the Nexa Teacher's 3D avatar stands at the teacher's place; the Smart Whiteboard runs in the user's browser (Excalidraw with Yjs, as everywhere in NexaTute) and is shown on the room's board; live-class video tiles from LiveKit sit at the seats. In a live class everyone sees the teacher's room, but every browser draws it itself; LiveKit carries only voice, video and whiteboard events, so the room adds no bandwidth or server work.
- Storage and delivery: a design is a small versioned JSON file (a few KB) kept in PostgreSQL through the NexaTute API on the App nodes; 3D models and textures come from R2 through the Cloudflare CDN and stay cached in the browser after the first download. No GPU and no server rendering are used.
- Assets: only CC0 or RemonLogic-owned 3D models and textures — Poly Haven, Kenney and Quaternius (all CC0-1.0) and commissioned models — each reviewed before it enters the library; Motion Studio's Quick-mode 3D scenes use the same library (18.1).
- Charge and load: free for every user, with no learning minutes used; no effect on server production (browser rendering and CDN delivery only).

3.16 Nexa Teacher, Nexa Co-Teacher and Load-Aware Learning

NexaTute uses two AI roles. The Nexa Teacher teaches live by voice — one-to-one, or a whole batch class (3.17). The Nexa Co-Teacher supports everything else in text. Live teaching is kept for the moments that need it, and learning continues when the servers are busy.

Voice for the Nexa Co-Teacher (0.8): the learner can speak a question, which is turned into text in the browser, and can listen to any answer in the browser voice. There is no continuous voice

conversation, avatar, whiteboard step-watching or talk ratio — these stay with the Nexa Teacher. The 25-topics-a-day limit is unchanged and no GPU is added.

Role	What it does	GPU pool free	GPU pool busy	Overloaded or down
Nexa Teacher	Live teaching with voice, 3D avatar and whiteboard: 1:1 within the plan's daily minutes, and batch classes	Starts the class at once	Suggests the learner's earlier recorded class first, then mock tests, true/false and class summaries; starts the live class only if the learner still has not understood; shows an ETA and keeps the learner's place in the queue	No new live classes start; the learner continues with the Nexa Co-Teacher in text, and the lost minutes are added back automatically (11.6)
Nexa Co-Teacher	Text only: full text lessons with step-by-step working, subject answers, and finding the learner's own recorded classes (with timestamps) and class summaries; no voice, video or new images	Local LLM pool	Local LLM pool, caches first	Cloud API in the order verified-answer cache → Mistral Large 3 → GPT-6 Luna, within the spend cap (8.9); Free Beta users wait in the queue (11.11); free-tier and Nexa Co-Teacher Pass users get cached answers first and then wait in the queue — the cloud API serves plan holders only (3.22)

- Nexa Co-Teacher limits: 25 topics a day per learner; follow-up questions on the same topic ("explain again", "next step") count as one topic. The Nexa Co-Teacher teaches in the Nexa Teacher's pattern, at the learner's level (shared learner record below).
- Own data only: the Nexa Co-Teacher answers from the learner's own recordings, class summaries and progress and from the shared course content; it never shows one learner's data to another (row-level security, 6.2). A batch-class recording belongs to every member of that batch (3.17).
- Still not understood: the learner calls the Nexa Teacher (1:1). Beyond the plan's daily minutes the wallet is charged at the course's per-minute rate (11.2), 20% less from 1 am to 7 am for adult courses (11.12).
- Syllabus scope: help is unlimited inside the purchased syllabus. For BCS + Job Test Preparation the scope is the PSC syllabus areas at MCQ depth plus about two years of current affairs (3.13); for open courses it is the course's learning goals (for IELTS, the four modules). A question outside the scope gets a one- or two-line pointer to the right course and the offer of that course's Daily Pass or an upgrade (11.12); when the scope is unclear the AI decides in the learner's favour.
- AI daily study plan: built from the confirmed syllabus and the school routine and re-planned every week (3.10).
- Face-verified single user: one account = one learner. The browser matches the face with a liveness check (blink or head turn) and repeats the check at random moments; a weak camera falls back to an OTP; minors need guardian consent. Only an encrypted face template is stored, and it is deleted with the account (10.2, 19.4).
- End of the Free Trial: the Nexa Teacher's honest plan recommendation is described in 11.5.
- Routing thresholds (GPU-pool load and the P100 queue wait) are admin-panel settings; the ETA comes from the live queue, and the learner's place is kept in Valkey (6.2, 8.9).

Shared learner record — the Nexa Teacher and the Nexa Co-Teacher work as one team:

Item	How it works
One record	Both roles read and write one shared learner record: level and topic mastery, the explanation type that works for this learner (picture, story or example, step-by-step), the error list, the last class, the exit ticket and the next step.
Nexa Teacher → Nexa Co-Teacher	After every class the Nexa Teacher records what it taught, with which example and method, and where the learner got stuck; the Nexa Co-Teacher then teaches in text at the same level, with the same example and notation.
Nexa Co-Teacher → Nexa Teacher	The Nexa Co-Teacher records the learner's questions, what was understood, the hints needed, drill scores and where the learner is stuck; the Nexa Teacher starts the next class from there and switches to a different method instead of repeating one that failed.
Handover note	When the learner still does not understand and calls the Nexa Teacher, a handover note makes the 1:1 class start at that exact step, so no minutes are spent repeating.
One evaluation	Learning Insights (3.4) is updated from both roles' answers, and help-assisted answers weigh less. New KPIs: share solved independently, solved by the Nexa Co-Teacher compared with needing the Nexa Teacher, days to mastery and the method that works; the depth of questions asked to the Nexa Co-Teacher counts in the Thinking Profile (3.18).
Nexa Co-Teacher class summary	Every Nexa Co-Teacher topic gets a class summary — what was learned, mistakes, drill score and next step — shown in "My summaries" next to the Nexa Teacher's class summaries and in the guardian's weekly report.
Batches	The topics that batch members asked the Nexa Co-Teacher about most after a class (counts only, no names) feed the next class's 5-minute recap (3.17).
Privacy and systems	Only the learner's own data is used, as before. No new software or servers: PostgreSQL with row-level security holds the record, pgvector the long-term memory, Ralph LRS the xAPI statements from both roles, Valkey a short context card for active learners, and LangGraph gives both roles the same memory. Summaries are made on the local LLM pool; during overload, when the Nexa Co-Teacher runs on the cloud API, they are made later on the local pool so API cost does not rise.

3.17 Batch Classes, Combo and Virtual Batch

Batch classes are AI-taught group classes — one Nexa Teacher for up to 50 learners. They are the heart of the Batch and Combo plans (11.1).

Item	Rule
Class size and start	Up to 50 learners per batch — the "1:50 AI class" (3.28); a batch starts on its scheduled date whether it has 1 learner or 50. Bangla Medium and English Version have separate batches, as do the other boards, and batches are formed by placement-test level so the pace fits. Batches are also separated by teaching language (for example Bangla batch B-07, English batch EN-03); a learner who changes the teaching language moves to a batch in that language.
Timetable	School courses 4:00–6:10 pm; Admission and BCS + Job Test Preparation 7:00–9:00 am; 40-minute classes (30 minutes for IELTS-type modules) with 5-minute breaks, five days a week; subjects per course are in 11.1. No batch classes on government holidays and Eid (1:1 time and the Nexa Co-Teacher continue); in Ramadan, school batches move to 2:00–4:10 pm and adult batches to 10:00 am–12:00 pm to avoid the Iftar clash (admin-editable).

Item	Rule
Class format	8 minutes of teaching, then a 2-minute poll or MCQ — four cycles in a 40-minute class. Adaptive pace: if fewer than 60% answer correctly the AI re-explains with a new example; above 85% it moves on.
Engagement	Always-on “Bujhlam / Bujhini” (understood / not understood) confusion meter; random cold-call with a text answer; five teams of ten with team scores; an anonymous question wall with upvotes, where the AI answers the most-voted questions first. Microphones start off and learners speak by raising a hand (3.28).
Peer Instruction	When 30–70% answer a poll correctly, learners discuss in pairs for 2 minutes in small rooms and then answer again; pair rooms use text chat for minors and may use voice in adult batches (10.2).
Questions and recording	Every question asked in class is visible to the whole batch, as in a real classroom; personal questions go to the learner’s 1:1 time. When at least one learner wants the recording, the class is recorded once for the batch as an event log — the AI’s teaching text, whiteboard events, polls and the question wall — plus the audio of raised-hand questions, and the approved members can replay it (11.3, 3.29).
Make-up class	If a server problem stops a batch class, a make-up class runs the same day or the next day, and the recording follows (11.6).
Free guests	Free-tier learners can take one guest seat a week in a batch with empty seats — listening and answering polls only, with no 1:1 time (3.22).
After class	Individual homework for each learner from their own poll mistakes; a weak topic automatically becomes that day’s 1:1 time (5 minutes in Batch, 30 in Combo) with the recording timestamp and a Nexa Co-Teacher drill.
Small batches	With only a few learners, team scores, Peer Instruction and the leaderboard switch off and the AI takes more questions.
Rollout	Batch classes open first for BCS + Job Test Preparation and SSC; other courses follow once the Free Beta has measured class completion and the Batch → Combo/Personal upgrade rate (11.11).

Combo: the learner joins the group classes and also gets 30 minutes a day of 1:1 with the Personal Nexa Teacher at any time; the 1:1 session starts from the topics the learner got wrong in that day’s polls, then moves on to homework.

Virtual Batch: learners with the same exam date share a batch rank, an exam countdown and the weekly top 3. Personal-plan learners also join a Virtual Batch, with one free weekly batch live class (45–60 minutes, outside the evening peak) and a monthly batch mock test with live results and rank; the weekly human-teacher revision class (3.10) runs inside it. Only rank and name or nickname are shown to the batch.

Quality, KPIs and reports:

- Per-learner KPIs, collected automatically: attendance and time in class (join and leave, tab focus), poll accuracy and answer speed, confusion taps, questions asked (raised hand, Nexa Co-Teacher), the after-class quiz and the weekly improvement in Learning Insights (3.4).
- Weakness and strength: topic mastery compared with the batch average; the error type is read from the wrong MCQ option chosen (concept, calculation or careless); strong topics get challenge problems and a “helper” badge, with XP for answering other learners’ questions.
- AI teaching quality: a segment report after each class shows which 8-minute part caused the most confusion, and the next class opens with a 5-minute recap of it; lessons are pre-built and

verified by the 3.13 pipeline; two explanations are A/B-tested and the one with better poll results is kept; people review the worst-rated segments every week.
- Reports: a weekly guardian report (attendance, best topics, weak topics, next week's goal) and batch KPIs (attendance, completion, average poll accuracy, confusion rate, upgrade rate). Face verification is used only for presence, never to judge mood or attention.
- Safety: Nexa Guard (bad-language filter, report button and the behaviour-change ladder) and nicknames protect every batch (10.2).

3.18 Learning Science, 1:1 Teaching and Brain Grooming

Goal: world-class learning rather than conventional coaching — learners who truly understand, think faster, ask deeper questions and lose their fear of any subject ("brain grooming"). These methods apply in every plan. Release timing, except for mastery-based progression, is set in the release plan after the wireframes.

Learning-science methods:

Method	How NexaTute applies it
Pre-test	Three short questions before each new topic; wrong answers are expected and prepare the learner for the lesson
Retrieval warm-up	Every class opens with a 3-minute warm-up: one question each from 1 day, 1 week and 1 month ago
Worked → faded → independent	In Mathematics and Science the AI shows a full solution, then a half-finished one for the learner to complete, then a problem to solve alone
Prerequisite repair	The curriculum knowledge graph (3.11) finds the earlier topic behind a difficulty; a 5-minute mini-lesson repairs it before the class returns to the current topic
Error notebook	Every mistake is saved automatically and re-tested after 3 and 7 days; it leaves the notebook only after two correct answers in a row; in the 1:1 class it opens from the toolbar (below)
Visible progress	A chapter pre-test and post-test show the gain (for example "35% → 80%") to the learner and guardian
Weekly goal and reflection	A goal on Saturday and a 2-minute reflection on Friday: what was learned and what was hard
Mastery-based progression	About 80% mastery before the next topic; in every plan from launch (3.10)

1:1 teaching methods:

Method	How it works
Watching each step	The AI follows the learner's working on the whiteboard and stops at the first wrong step
The learner talks more	Target: the learner explains or answers for about 40% of the class; the AI asks more than it tells (talk ratio on the dashboard; Socratic mode, 3.10)
The explanation that works	The AI tracks which kind of explanation — picture, story or example, or step-by-step — gives this learner better quiz results, and uses it first
Exit ticket	One question and the learner's own one-line summary at the end; the next class starts from any gap

Error notebook in the 1:1 class (decided 25 October 2026; prototypes X2 and X8): an “Error notebook” button in the class toolbar shows today’s count and opens a side panel without leaving the class; after the exit ticket the learner sees “N mistakes went to the notebook” with a “See” button. The full notebook page filters by subject, topic and mistake type (concept, calculation, careless); each entry shows the question, the learner’s step, the correct step, the Nexa Teacher’s short explanation, where it came from (1:1, batch class, mock test, Nexa Co-Teacher or a written check, 3.27) and its retest dates. Buttons: “Practise now”, “Ask Co-Teacher” (inside the plan) and “Ask Nexa Teacher again”, which uses 1:1 minutes and says so on the button; each mistake can be practised once a day, so it is not cleared in one sitting (default, editable, 38.5). The home screen shows “N mistakes are due for review today”, and the guardian’s weekly report shows only mistake counts by topic — new and mastered — never the questions or answers.

Brain grooming — a thinking layer across every subject:

Method	How it works
Problem first	Every topic starts with a real puzzle (for example “why does a cricket ball swing?”); the learner tries before the AI explains
Why → How → What if	Each concept is taught as why it happens, how it works and what changes in another situation
The learner’s own question	In each class the learner writes one question; the AI scores its depth on Bloom’s levels (remember → create) and tracks the trend
No fear	Easy wins first; the AI says “not yet” instead of “wrong”; a “best mistake of the week”; quiet foundation repair for learners anxious about Mathematics or English
Argue both sides	In BCS, Bangla and Social Science the learner debates with the AI, giving claim and evidence for both sides before their own view
Subject links	Physics in cricket, mathematics in market sums, history in today’s news
Thinking speed	Mental mathematics, estimation (“how many rickshaws are there in Dhaka?”) and timed puzzles

Thinking Profile (KPIs): performance on unseen problems, the level of the learner’s own questions, transfer of ideas between subjects, confidence compared with actual accuracy, and persistence on hard problems; stored with Learning Insights in TimescaleDB. Claims to guardians and investors use these measured gains only — never general “brain training” claims (16.3).

3.19 Engagement Intelligence — Learning-First Personalisation

Like a social-media feed, NexaTute predicts from behaviour what each learner needs next — but the aim is more learning, not more time in the app: success is measured by mastery gain and regular study. Release timing is set in the release plan after the wireframes.

Capability	What it does	Software
Learner state	From interaction logs only — answer time, pauses, repeated mistakes, recording rewinds, hints and skips — the AI sees whether the learner understands, is stuck, bored or tired, and changes method or pace, gives a break or an easy win	scikit-learn, LightGBM
Next activity	Ranks the next lesson, quiz, Short or challenge by expected learning and interest, and learns every day from what the learner does	Vowpal Wabbit (contextual bandit)

Capability	What it does	Software
Smart reminder timing	Learns when each learner studies and reminds them then — at most one or two reminders a day, and none to children after 9 pm	Same models; notification service
Drop-out prediction	Falling logins, a broken streak or falling accuracy → 7 days ahead the Nexa Teacher sends a personal message and an easy win, and the guardian gets a tip; supports the target of at most 10% monthly churn (11.12)	LightGBM
Interest map	Interests from the learner's questions and choices (cricket, games, cooking …), so that examples use them	pgvector
Personal Shorts feed	Shorts ranked by weak topics and interests; a quiz after every 10 Shorts; no infinite scroll	Next-activity model
A/B tests	Every change — buttons, reminder text, explanation type — is tested on small groups, and the version that teaches best is kept	Unleash; results in TimescaleDB and Grafana

- No camera or voice mood or emotion reading: the face is used only for presence (3.16), and inferring emotions in education is banned under the EU AI Act; the log-based "understanding / stuck" signals get a legal review before launch.
- Consent: every feature in the table has its own on/off switch, and the consent rules — including the guardian's decision for under-18 learners — are in 19.4.
- For children: no infinite scroll, autoplay or "stay a bit longer" reward loops; guardians set a daily time limit and quiet hours; every suggestion shows "why am I seeing this".
- Systems: no GPU or new servers — the models run on App-tier CPU on the event data already in ClickHouse; the register adds Vowpal Wabbit (BSD-3-Clause) and LightGBM (MIT) (18.1). Success is tracked with the North Star metric (13.1).

3.20 Learning Programmes and Study Tools

Prices are in 11.14. Release timing is set in the release plan after the wireframes.

Programme	What the learner or guardian gets
AI Parent–Teacher Meeting	Every month the Nexa Teacher (3D avatar) talks with the guardian for 10 minutes in Bangla about progress, the Thinking Profile and next month's plan, and answers questions; free in all plans and held off-peak (about 10 GPU-minutes per learner a month).
Live Study Hall	A virtual hall where learners see others studying (avatar or nickname): 25-minute study and 5-minute break rounds, with one question-bank question after each round; the silent hall, XP and streak credit are free for everyone, and when someone gets stuck the hall offers the Nexa Co-Teacher, a Daily Pass or a plan. Paid users can form class-wise groups of 2–10, use the Laboratory and Whiteboard together and press Help to call the Nexa Study Hall Moderator, which answers by text and voice with short help and hands full topics to the Nexa Co-Teacher or Nexa Teacher (Section 31). Groups and shared boards run in the browser (Yjs); minors need guardian approval, appear by nickname and use text chat only.
NexaTute Wrapped	A monthly and a year-end 30–60-second personal recap video made in the learner's browser with Motion Studio Quick mode (minutes, chapters mastered, best streak, "best mistake"), shareable on Facebook and WhatsApp with the referral code; minors share only with guardian permission.

Programme	What the learner or guardian gets
Personal mistake video	When a learner makes the same mistake twice, the AI writes a 30-second script about that exact mistake using the learner's interests, and Motion Studio Quick mode renders it in the learner's browser (the explainer videos in 3.10 stay per topic); one a day in Combo and Personal, one a week on the free tier; only the script uses the LLM.
Choose your Nexa Teacher	The learner picks the Nexa Teacher's look, voice (male or female), style (calm, energetic or fun) and Bangla–English mix; cartoon teachers in PlayBoom; new looks unlock with XP, never with money for children; no extra cost (browser avatar, existing voices).
Head Start camps	In December, when schools are closed and the new class starts in January, a 30-day camp teaches the first chapters of next year's books and fills this year's gaps; an HSC Head Start runs after the SSC exam (March–June).
Olympiad track	Mathematics, Science and ICT for strong students, linked to brain grooming (3.18) and the Bangladesh Olympiad calendar.
Smart Suggestion	Before exams, practice sets built from how often each topic appeared in the last 10 years' board questions, clearly labelled "for practice, not a guarantee of what will come". The topic frequency is shown (for example "came 7 times in 10 years"); never "100%", "common" or "leak" (decided 29 October 2026).
Post-exam solution sprint	Full solutions to every board, admission and BCS paper within 2–3 hours of the exam on the public pages and Nexa Teacher Live (3.21), published only after a content reviewer verifies them; then Daily Pass or Crash Pack offers for the next paper, and Head Start or Second Chance after the last one.
Second Chance	For students who failed SSC or HSC, admission re-takers and BCS repeaters: a free result analysis on result day, a plan built around the failed subjects and a fresh 7-day Free Trial (11.5), with extra fear-removal and wellbeing care (10.2), gentle language and guidance for the guardian.
Circular and deadline tracker	For BCS + Job and Admission learners: daily alerts for new government and bank job circulars, university admission notices, application deadlines and exam dates; learners follow the jobs or universities they want. Free for everyone, with a "start preparing for this exam" button on every circular; collected from public circulars (BPSC, banks, university websites) by the AI content pipeline's Scrapy and verified by a reviewer before alerts go out.
Study Abroad Application Coach	For IELTS, TOEFL, PTE, SAT, GRE, GMAT and JLPT learners: SOP and essay feedback (the AI never writes them), a university and scholarship shortlist from scores and budget, an application timeline and an AI visa-interview mock.
Guardian learning	Guardians of active learners take Spoken English, computer, foreign-language and skill courses at 50% off.
Calculator	Free for everyone, including the free tier: scientific, graphing and unit-conversion modes. In mock tests the calculator follows the real exam's rule — off for the BCS Preliminary, scientific for SSC and HSC, graphing for the SAT; the rules per exam are kept in the admin panel (38.5)
Dictionary	Free for everyone: tap any word for its meaning and pronunciation, plus synonyms and antonyms, idioms and phrases (BCS, IELTS), Bangla ↔ English subject glossaries, "My words" with revision after 1, 3 and 7 days, and a picture dictionary with sound for PlayBoom

Season calendar (admin panel): January — Academic Year Pass; 30 days before exams — Crash Pack and Smart Suggestion; after SSC — HSC Head Start; Eid — Shikkha Salami gift cards; December — Head Start; dates are updated from the board announcements.

3.21 Growth, Partnerships and B2B Services

Prices and commissions are in 11.14; partnerships are led by the partnership manager who joins in Month 3–4 (14.2). Release timing is set in the release plan after the wireframes.

Programme	How it works
Free content funnel	Questions and answers from the verified-answer cache are published as public Bangla and English web pages for Google search, each with an “Ask the Nexa Teacher” button (verified answers only, no learner names or data); Shorts are auto-published to YouTube, Facebook and TikTok through Motion Studio’s direct publishing (4.6); a weekly “Nexa Teacher Live” on Facebook and YouTube solves the week’s most-asked questions (LiveKit Egress, one session a week).
Affiliate program	Coaching centres, campus ambassadors and YouTubers earn commission on plans sold with their code. School teachers get a free teacher tool — create a class, assign practice and see reports, with their students joining the free tier (3.22) — but no commission, so they never pressure their own students.
Teacher marketplace	Verified teachers and coaching centres sell their own mock tests, question banks and notes, which the AI explains to learners; sold only after the 3.13 quality check and the teacher-content check (10.2), through Motion Studio’s creator-marketplace payment and payout system.
Script Checker (B2B)	Schools and coaching centres upload scanned class-test and model-test answer scripts; the AI marks them overnight with the teacher’s rubric in the quietest GPU hours (for example 2–6 am, Qwen3-VL), and the teacher reviews and approves the final marks; students get detailed feedback and their weak topics the next day.
Job pathway	For EPS-TOPIK, JLPT, skill-course and Other Popular course learners: with consent, a verified job-ready profile (certificates, skill passport, AI mock-interview score) goes to employers and licensed agencies, who pay per hire or a monthly fee; 18+ only; overseas jobs only through the government process or licensed agencies — NexaTute is not a recruiting agent. Employer and organisation skill and recruitment tests for adults follow 3.27. Verified employer job posts and freelance gigs are on the job board (3.32).
Shikkha Salami gift cards	Daily Pass or 1- or 3-month plans as gifts for Eid, birthdays and result day, sold online and through bKash agents, mobile-recharge shops and bookshops, reaching cash buyers in villages; in-house voucher system with a unique code per card.

- Systems: no new servers — public pages on the CDN, Shorts through Motion Studio publishing, Nexa Teacher Live through LiveKit Egress, the Script Checker on the GPU pool at night, and vouchers, affiliate codes and payouts in-house with aamarPay (18.1).

3.22 Lifetime Free Tier, Help Button and Nexa Co-Teacher Pass

Learners who never pay can use NexaTute free for life: everything for self-study is free and help is paid, so they are most likely to buy when they need help. The 7-day Free Trial (11.5) stays as it is. Release timing is set in the release plan after the wireframes.

Free for life	Details
Study material	Chapter summaries (text notes) and all Shorts; the first chapter’s AI lesson in every subject; question-bank practice of 20 MCQs a day with auto-marking and written solutions
Studying together	Live Study Hall (3.20); one guest seat a week in a batch class with empty seats — listening and answering polls only, no 1:1 (3.17)

Free for life	Details
Play and labs	Mini-games, AR models and one Virtual Lab experiment a week; the free 2027 features and their limits are in 3.32
Progress	Weekly national quiz marks and rank; XP, streak, leaderboards and a simple progress chart; exam countdown and a fixed study plan (no AI re-planning)
Guardians	Weekly push or email summary and the Wrapped video

The Help button — the moment learners are most likely to buy — appears in practice, lessons and the Study Hall:

- First, free help: the written solution, a related Short and Nexa Co-Teacher lite (3 topics a day, from the verified cache only).
- Then the first 5 minutes of Nexa Teacher time each month free, starting from that exact question; after that, a one-tap Daily Pass (৳20 for school courses, 11.12).
- Offers at help moments: three Daily Passes in a month → the monthly plan's savings; a poor mock result → Combo or a Crash Pack; 30 days before the exam → Crash Pack; a topic weak for three weeks in the weekly summary → the cheapest plan (Batch), shown to the guardian. Every purchase by an under-18 learner needs the guardian's approval (11.9).
- Nexa Co-Teacher Pass: ৳100 a month per course ($1) for the full Nexa Co-Teacher — text lessons, step-by-step answers and topic summaries — for 25 topics a day, the same limit as the plans; during overload Pass users get cached answers first and then wait in the queue, and the cloud API serves plan holders only. Upgrade ladder: free (Nexa Co-Teacher lite) → Nexa Co-Teacher Pass → Daily Pass when voice is needed → Batch → Combo → Personal.
- Free users' practice, mistakes and progress go into the same learner record (3.16), so after paying, the Nexa Teacher knows their weak points from day one.
- Not free: 1:1 Nexa Teacher time beyond the help minutes, regular batch classes, the full Nexa Co-Teacher, recordings, full mock tests, downloads, the AI study plan, the Parent–Teacher Meeting and full Learning Insights.
- Cost and limits: GPU is used only for the 5 free help minutes — about 12 concurrent sessions even if all 50,000 free users used them over 12 hours, about 1% of the fleet; everything else runs on the CDN, in the browser and from the question bank. Phone OTP and device checks stop abuse; free users are last in the queue during overload; the Free Beta measures help-button → purchase % and free → paid % (13.1).

3.23 Links Shared by Learners, Guardians and Teachers

Links go through a link step added to the Document Intake Service (5.3). Safety comes first: a URL blocklist check (malware, phishing, pornography, gambling and piracy — the URLhaus and StevenBlack hosts lists, 18.1); then the page is opened in a gVisor sandbox with no login, no cookies, no scripts and size and time limits; text is checked by Qwen3Guard, and images by the NSFW classifier and child-safety hash matching (4.15).

Link type	What happens
Web page or article	Docling extracts the text; the AI explains it, makes notes and quizzes, and corrects mistakes against the verified knowledge base
Public PDF, Google Drive or Docs file	Handled by the Document Intake Service like an upload (3.14)

Link type	What happens
YouTube	Only the title, description and chapters through the official YouTube Data API — no download or copy; the AI teaches the topic with NexaTute's own lessons and recordings
Social-media post	Public text only; otherwise the learner pastes the text
Link that needs a login (private Drive, school portal)	Not opened; the user downloads the file and uploads it

- Students: "explain this", "make notes", "quiz me" or "is this correct?" — the AI maps the link to the syllabus topic, explains it at the learner's level and links NexaTute's own lessons.
- Guardians: links to school notices, routines or syllabuses update the AI study plan and the exam countdown (3.14).
- Teachers: lesson or question-bank links are imported into the teacher content studio with a copyright confirmation and the teacher-content check before publishing (10.2).
- Harmful links are blocked at once with a polite reason: pornography, child abuse or threats → the behaviour ladder and a human-controller alert (10.2); phishing or malware → a warning and a block; pirated books and leaked exam-question links are refused, and question leaks are reported to the human controller; minors cannot post links in batch chat or on the question wall except to approved sites (for example NCTB and the education boards).
- Copyright, privacy and cost: link content is used only to explain — never stored as a copy or published; a learner's links are their own private data; each link counts as one Nexa Co-Teacher topic (within the 3 a day on the free tier); the size limit is about 30 pages or 20 MB; fetching and text extraction run on CPU and only the explanation uses the LLM; the same link shared by many learners is processed once.

3.24 Learning Quality Proof and International Frameworks

NexaTute proves international-level learning with evidence, not only with features. Release timing is set in the release plan after the wireframes.

Item	How it works
Learning efficacy	A term-end "learning gain" report for every course (chapter pre-test compared with post-test, 3.18); after the Free Beta, a comparison study with a partner university — NexaTute learners against similar learners — to measure the effect size; a yearly "Learning Outcomes Report" is published.
Prediction accuracy	Every season the predicted rank, estimated IELTS band and GPA predictions are compared with the real results; the accuracy is published, and the models are recalibrated when it is off.
CEFR levels	English and foreign-language progress is reported on the CEFR scale (A1–C2), and certificates show the CEFR level.
Universal Design for Learning	Inclusion features — dyslexia-friendly font, read-aloud, captions, shorter segments, more breaks and less time pressure (3.10) — are organised under the three UDL principles: multiple means of engagement, of representation, and of action and expression.
External audits	A third-party WCAG 2.2 AA accessibility audit with an Accessibility Conformance Report (VPAT); after launch, a certification path for ISO/IEC 27001 and ISO/IEC 42001, which are "aligned" today (19.7); the yearly external penetration test stays.

Item	How it works
AI lesson quality rubric	Every AI lesson is scored on a rubric — correct, clear, checks understanding, right level — and subject experts review a random sample of lessons every week, not only the worst-rated segments (3.17).

- Only these measured results may be used in public learning claims (16.3).

3.25 Landing Page AI, AI CMS and Sales Tools

Approved 18–19 October 2026. One Landing Page AI and AI CMS engine serves both products, with separate sites, content, admins, analytics and data (6.4); Motion Studio's specifics are in 4.19. The AI builds and runs the landing page and the CMS, and an admin approves every change. Release timing is set in the release plan after the wireframes.

How the AI knows what to change and how a change goes live:

- Signals: the season calendar (3.20); funnel and conversion data (TimescaleDB product events, Grafana funnels, Unleash A/B tests); support and pre-sale questions; new verified results, consented success stories and the Learning Outcomes Report (3.24); new courses that pass the quality gate; Google Search Console queries; and site health — broken links, expired offers, slow pages and accessibility issues.
- Change requests: the AI sends the admin what changes and why, a before/after preview and the expected effect; the admin approves, edits, rejects or schedules it. Plain-language instructions — for example "make the December Head Start campaign page" — give the AI a draft to approve.
- Publishing: offer end dates set at approval remove offers automatically; every version is kept with one-click rollback, and the AI suggests a rollback on errors or falling sales; approved variants run as A/B tests, and the admin approves the switch to the winner.
- Roles per product: the AI drafts; editors and the marketing admin approve pages; the lawyer or DPO approves legal pages, which the AI only drafts; the superadmin handles site settings; every approval is in the audit log.
- Guardrails: prices are read live from the price table and are never written by the AI; no claim without proof (3.24, 16.3); scarcity only when true; only real, consented testimonials; no tracking-based personalisation for children — only season and language; every change passes axe-core and Qwen3Guard, in Bangla and English; pages are built only from a fixed component library — no free-form code — and a new component type needs a developer review.

Professional landing-page standard (checked before every publish):

Rule	What it means
5-second clarity	The headline says what it is, for whom and the benefit; one main button ("Start free") and one second button ("Watch the demo")
Demo first	A 20-second Motion Studio video of the Nexa Teacher solving a problem on the whiteboard in Bangla, and an "ask a question" box answered at once from the verified cache
Section order	Demo → how it works (3 steps) → plans with the exact price after choosing a course (Combo "Most popular") → proof (verified numbers, consented stories) → for guardians (safety, guardian controls, no ads) → FAQ → final button
Plan finder	Three questions — class or course, goal, daily time — show the right plan, its price and a "Start free" button

Rule	What it means
Mobile-first and fast	Designed for phones first; Core Web Vitals in the “good” range — about 2.5 seconds to load on a 4G phone — and a light version for slow internet
Language and design	Bangla first with an English toggle and simple words; a brand book for each product (logo, colours, fonts, icon and image style, Bangla and English voice and tone), which the component library and the AI’s copy follow; real product screenshots, no stock photos; WCAG 2.2 AA
Honest attention	Season banners and live numbers such as learners studying today — only real numbers
Tested with people	A UX designers’ review and a usability test with five guardians and five students before launch; A/B tests of every headline after launch

CMS options (both products):

Area	Options
Pages and content	Drag-and-drop page builder from the component library, templates, page copy and reusable blocks; pages, course or package pages built from the price table (G.8 for Motion Studio), blog (Nexa Notes on NexaTute) and news, FAQ, success stories and testimonials, campaign and season pages, announcement bar, forms (contact, B2B inquiry, lead) and legal pages; menu, header, footer and URL manager
Media library	Images, videos and PDFs; automatic resizing and WebP/AVIF (sharp); AI alt text; where each asset is used; a copyright and consent record per asset; banners by FLUX.1-schnell and short videos by Motion Studio Quick, with the usual safety checks (4.15); videos as HLS in several qualities with reviewed subtitles in six languages (Nexa Notes)
Languages	The NexaTute landing page in all six languages at launch (AI translation, native-speaker review afterwards); other pages Bangla and English first, later Hindi, Urdu, Arabic and French; AI translation with human review; hreflang (next-intl)
SEO	Meta, social preview, schema.org (Course, FAQPage, Article, VideoObject), sitemap, robots, canonical, 301 redirects, broken-link checker, keyword and internal-link suggestions; Google Search Console and Bing Webmaster
Workflow	Draft → review → approve → schedule → publish → expire; comments, side-by-side diff, version history and rollback; mobile, tablet and desktop preview and a staging link
AI features	Plain-language instructions; drafts of pages, blocks and copy; alt text, captions and translation; SEO suggestions; A/B variants; refresh proposals for outdated facts; fact check against the price table and the verified knowledge base; accessibility fixes
Growth and analytics	A/B tests; season and language pages; page-level visits, scroll, clicks and conversions; UTM, campaign and affiliate landing pages (3.21); leads to the admin panel and the B2B CRM; newsletter and push only with consent
Technical and safety	Static pages on the CDN, image optimisation and cache; real-user Core Web Vitals (web-vitals) and a page-speed and SEO check before publishing (Lighthouse CI); Cloudflare WAF and Turnstile on forms; a cookie consent manager (vanilla-cookieconsent); Terms & Conditions and Privacy Policy versions with an acceptance log; backup and restore; JSON export and import; headless API; maintenance mode and busy page
NexaTute only	PlayBoom pages get a child-safety check; course pages show the NexaTute Verified badge; a section for guardians; the public Q&A pages from the verified-answer cache (3.21)

Admin page “why users drop” (NexaTute; Motion Studio’s funnel is in 4.19):

- Funnel: visit → sign-up → placement test → first lesson → return on day 3 → trial end → purchase → first renewal, with the drop percentage of every step and weekly trends.
- Where: section scroll depth and button clicks as anonymous heatmaps, page speed by phone and network, form errors, payment-failure reasons by method, guardian approvals waiting or declined, and the lesson minute or topic where learners quit. Who: aggregated segments — device, internet speed, district, course, traffic source, language, new or returning. Why: a one-question exit survey at trial end or cancellation, and support-chat themes.
- Weekly AI analyst: the top three causes with evidence and fixes, sent as change requests to the Landing Page AI or the product team — for example “38% drop at the placement test: 15 minutes is long on mobile and slow on 3G → cut it to 8 minutes and switch on low-internet mode automatically”; alerts when a step’s drop rises suddenly; the before and after of every change.
- Privacy: aggregated and anonymous; no session recording of children; the consent rules of 19.4 apply.

Sales tools (both products):

Tool	How it works
Pre-sale AI	On Facebook Messenger, WhatsApp and the website chat through Chatwoot: answers price, plan and course questions 24/7, creates leads, hands B2B or complex cases to a person, and moves purchases for minors to the guardian (11.9)
Abandoned-checkout recovery	Reminders 1 hour and 24 hours after an unfinished payment — push, email or SMS, by consent — with the chosen plan saved; for minors the link goes to the guardian; help by failure reason (for example a low bKash balance or a card problem)
B2B CRM	An in-house pipeline in the admin panel for the partnership manager (14.2): lead → meeting → proposal → contract → onboarding for schools, coaching centres, agencies, employers and retailers, with reminders, contract documents and revenue per partner; leads arrive automatically from site forms and chat; built in-house because most open-source CRMs are AGPL
Reviews and ratings	Asked for at happy moments — a good mock result, a chapter mastered, after 30 days; consented reviews become testimonials; Play Store rating prompt and Google Business Profile reviews; the AI drafts replies and a person approves them; no fake reviews, and negative reviews are never hidden

- Systems: no new servers — one in-house engine on the existing stack: page blocks with versions in PostgreSQL, Next.js rebuilds and Cloudflare cache purge, copy and the weekly analyst on local Qwen3, Chatwoot channels, the notification service and the in-house CRM. New software: sharp, vanilla-cookieconsent, web-vitals and Lighthouse CI; new free services: Cloudflare Turnstile, Google Search Console, Bing Webmaster Tools and Google Business Profile (18.1). The sites need only a few LLM calls a week, off-peak; pre-sale chats use the cache first.

Landing Page AI additions

- The Landing Page AI lives in the platform admin panel (38.6; prototype X13): the marketing admin approves page changes and the superadmin handles site settings. The admin can also give the AI advice, ideas and orders in plain language.
- Keyword-research module: sources are Google Search Console, Bing Webmaster, Google Keyword Planner, competitors’ public pages and seasons (results, admission, BCS circulars); it covers Bangla, English and Banglish searches, maps each keyword to one page so that two pages never compete, tracks ranks weekly and sends proposals to the same approval queue.

Free tools are used by default; paid tools such as Ahrefs or Semrush are an optional connection (18.1).

- Check cadence: daily (broken links, page speed, expired offers, wrong prices), weekly (keywords, conversion, new FAQ from people's questions) and at once on events (a new course, result day, the start of a season).
- Version control: every publish is a full snapshot (copy, images, SEO meta, A/B settings, author, approver and reason); a rollback is itself a new version; any two versions can be compared; a rollback never brings back old prices (always from the price table) or expired offers, and redirects stay for changed URLs.
- Automatic rollback proposal when, 24–48 hours after publishing, conversion is more than 20% below the previous 7 days or errors or page speed get worse — only once there are enough visitors.
- Preview: mobile and desktop, a staging link, a red/green diff, the expected effect and a low, medium or high risk label.
- The NexaTute landing page is live in all six languages at launch with AI translation, followed by native-speaker review.
- Small technical fixes (broken links, image alt text, image compression) publish automatically after 60 days without a mistake (3.26); copy, images, offers and prices always need approval.

Nexa Notes and landing-page additions

- Nexa Notes (name decided 25 October 2026) is the blog of the NexaTute landing page, built on the same AI CMS: posts follow the CMS workflow and approvals, carry schema.org Article markup (VideoObject for videos); the address nexatute.com/notes, with a language prefix for the other languages, is to be confirmed. Our websites show no advertisements and no tracking.
- Nexa Notes videos (decided 25 October 2026): added in the AI CMS like posts — an MP4 upload, a video made with Motion Studio, or one of our own YouTube videos embedded click-to-load (youtube-nocookie; nothing loads from YouTube before the click, so no new tracking). Uploads go straight to R2 (Uppy), are virus-scanned (ClamAV) and transcoded by FFmpeg into HLS at 720p, 480p and 240p on the CPU worker pool at P10 in the night window; AI subtitles in six languages (Whisper transcript, Qwen3 translation) are checked by a reviewer; a transcript under the video, chapters and a thumbnail; no autoplay; played with hls.js from R2 through the CDN. Nexa Guard safety and copyright checks (NSFW frames, TMK+PDQF video hashes, Chromaprint audio fingerprints) and the same approval flow apply; the marketing admin or the super admin approves, never the person who made the video.
- Country and syllabus picker (decided 24 October 2026): a "Country & syllabus" button in the header and the first question of the hero — syllabus country → board or medium → class or course — with one URL per country and syllabus for SEO (for example nexatute.com/in/cbse, next-intl). The syllabus is chosen freely, while price and payment follow where the learner lives (Bangladesh → Taka through aamarPay; abroad → USD through Paddle); syllabi not yet through the quality check show "coming soon" with a waiting list; the choice is pre-filled into sign-up and the placement test for the learner to confirm.
- Also on the landing page (24–25 October 2026): six legal pages (privacy, terms, refund, data deletion, cookies, accessibility) in six languages as CMS legal pages drafted for lawyer or DPO approval, the English version prevailing, with versions in the acceptance log; University, Polytechnic and Vocational courses; a clearly visible sign-up button (Keycloak sign-up behind Turnstile); US learners under 13 go to a waiting list until verifiable parental consent under COPPA is in place.
- Recruitment tests on the landing page (decided 25 October 2026): the offer — free with the job-test courses, otherwise the Recruitment tests add-on for adults at ৳399 a month ($6) for up to 3 tests a day (11.14) — appears among the plan add-ons, on the BCS + Job Test Preparation course card, in an employer block for organisations (3.27) and in the FAQ.

- Paid campaigns on other platforms (adults 18+ only; every budget change approved) are measured only through the cookie banner's Marketing switch (vanilla-cookieconsent), which is off by default, opt-in and never offered to learners under 18.
- Contact and countries (24 October 2026): contact through chat, WhatsApp and Messenger (no email or phone number shown); learners outside Bangladesh join a waiting list until their country opens.
- Positioning and guarantee (26 October 2026): the landing page shows "AI-Powered Learning Operating System" and the learning guarantee for every paid plan (11.4), with the lawyer note; after launch the legal pages show "In force from 1 January 2027".
- Autism support (27 October 2026): highlighted prominently — a link in the desktop header menu in every language (and near the top of the phone menu), a hero chip, a footer link and its own section with the Sensory-friendly mode (3.31).
- Redesign (28 October 2026): a clean, standard EdTech landing page without empty areas — a clear hero with four path chips (SSC · HSC · English Medium · Autism-friendly), a "Choose your path" band of four cards, a dedicated English Medium section (Class 4–8, Checkpoint and O/A Level, with the past-question rule of 3.30), a combined SSC/HSC band with the hub countdown, Crash Packs and board questions, a sticky section nav, equal-height card grids, long details in tabs and expanders, original SVG illustrations and a slim cookie bar that never covers the hero; no filler text, fake numbers, fake reviews, partner or board logos or scarcity wording.
- Crash Pack menu (28 October 2026): Crash Pack has its own top menu item and highlighted section — every crash program with 1, 3 and 6 months, Taka and computed dollar prices, weekly milestones, never sold longer than the time left before the exam, no seat caps and no grade guarantee (11.12).
- Waiting list and admissions message (29 October 2026): when a class or course is full, or admissions are paused, course and price buttons show "Join the waiting list" with a one-line explanation and an FAQ entry "What is the waiting list?"; the admin's admissions message appears in all six languages (38.19). No seat counts, "only N left" or countdown pressure.
- Also added (26–29 October 2026): the Exam & Admission Hub teaser with the "not an official service" note (3.30); "Every class in Bangla Sign Language" (3.31); recording choice in the FAQ (3.29); a Learning Atlas teaser; weak-internet FAQ lines (3.28); the store-purchase lines in the subscription terms (11.7); the Free Beta list and hidden cross-links, the status-page link and the "Report a security problem" page (3.33, 36.6); the public "Coming in 2027" roadmap page (17.6); and the legal pages of 36.6.
- Systems: no new servers — the same AI CMS engine and Next.js pages on the CDN; Nexa Notes videos and subtitles in R2 (≈0.2–0.4 GB per 10-minute video with every rendition and the master, about 20–40 GB a year at two videos a week — estimate); transcoding on the CPU worker pool and subtitles on AI-2 Node A off-peak (8.6, 8.17); new software hls.js and ClamAV (18.1).

3.26 AI Marketing Hub and RemonLogic Growth Platform

The Landing Page AI (3.25) grows into one AI Marketing Hub — Digital Marketing Automation — for NexaTute and Motion Studio, and with the CRM, journeys and target-client database below it forms the RemonLogic Growth Platform shared by all three products, with separate brands, social accounts, mailboxes, data and approvers per product (6.4). The AI plans, makes, publishes, answers, researches and reports; a person approves before anything new goes public and before any email that needs approval is sent. Release timing is set in the release plan after the wireframes.

Area	What the AI does	Approval
Blog articles	Bangla and English articles from the keyword plan, with FLUX.1-schnell images, FAQ schema and internal links, on the NexaTute and Motion Studio blogs (CMS, 3.25); facts from the verified knowledge base, prices from the price table or G.8	Marketing admin; a subject reviewer when the article teaches
Videos, Reels and Shorts	Promo, explainer and demo videos from the Motion Studio Quick or Enhanced pipeline; 9:16 Reels and Shorts with Bangla captions, music, thumbnails, the platform's AI label and Content Credentials	Marketing admin
Publishing	Blog pages, YouTube, Facebook, Instagram, TikTok and LinkedIn, with a title, description and hashtags written for each platform and posting at the best time for each audience	Automatic once the content is approved
SEO	Keyword research (Search Console, Bing keyword data, Google Trends and Keyword Planner when access is granted), page and YouTube SEO, ranking from Search Console and refresh proposals for older pages	New pages and large changes
Demos	The live demos on both landing pages (3.25, 4.19), demo videos for each audience (student, guardian, school, coaching centre, creator, agency), B2B demo booking in the CRM and a 7-day institution sandbox with sample data	—
Query answers	Comments, DMs, chat and email answered 24/7 from verified content; spam and abuse hidden; complaints handed to a person; no direct messages to minors	Prices outside the table, complaints or low AI confidence
Market research and outreach	Competitor and trend reports; lists of target organisations from the target-client database (below); personalised B2B emails with up to two follow-ups; every reply read and answered	Each campaign before it starts; replies that need approval
Customer data	Every lead in the CRM (3.25) with its source, lawful basis or consent, opt-out status and conversation history; leads from forms, chat, comments, email and events	—
Reports	Daily report to the marketing admin; weekly and monthly reports to the founder; the B2B pipeline to the partnership manager — by email, a WhatsApp summary and the admin dashboard, each with three suggestions	—
Paid ads (switched on in the admin panel)	Ad copy, creatives and audiences for Facebook, Google, YouTube and TikTok; optimisation inside the approved budget; adults and guardians (18+) only	Every budget change

Email replies that need approval:

- The AI sorts every reply: interested, question, not interested (follow-ups stop), unsubscribe (added to the suppression list) or out of office; it answers by itself only from verified content — prices, features, the demo link.
- Approval is needed for a discount outside the B2B table (11.8), custom prices or contract terms, partnerships, legal or data-protection questions, complaints, government or tender requests, or when the AI is unsure. The email and the AI's draft go to the concerned person by topic — partnership manager, finance, lawyer or DPO, founder — who approves, edits or rejects it in the admin panel or on the phone; the AI sends it only after approval.
- The sender gets a short holding reply at once; a reminder goes after 4 working hours without a decision, and after 24 hours the request moves to the next approver. Routing (X19 defaults): a

discount, custom price or contract → finance lead; a partnership, complaint or uncertain AI → partnership manager; legal or data questions → lawyer or DPO; government or tender → founder; unanswered after 24 hours → founder; working hours Saturday–Thursday, 9 am–6 pm.

Rules:

- Cold email only to organisations — schools, colleges, coaching centres, agencies, companies and creators — at their official or public business addresses; never to students' or guardians' personal addresses, who are contacted only after they opt in through sign-up, a form or chat. The data law requires explicit consent and bans profiling and targeted advertising aimed at minors (19.4); the lawyer confirms the outreach rules.
- Every email names the sender and has a one-click unsubscribe honoured within 48 hours; sending starts at 30 emails a day per mailbox from a separate outreach domain (teamnexatute.com, teamremonstudio.com and teamnexalogicdev.com, to be bought — reminder set for 1 November 2027; cold email stays off in any country whose rules are not yet written, for example India) with SPF, DKIM and DMARC, keeping spam reports under 0.1%. Amazon SES stays for transactional email and opt-in newsletters, because its policy forbids unsolicited mass email.
- Data only from official platform APIs and public business information — no scraping of LinkedIn, Facebook or other sites against their terms; platform data is refreshed or deleted as each API policy requires.
- Platform approvals: TikTok keeps API posts private until its audit, LinkedIn page posting needs the Community Management API, Instagram needs a professional account and Meta app review, and YouTube needs its compliance audit (4.14). Until each approval arrives, the AI prepares a ready-to-post pack and a person posts it with one click.
- Approval level: at launch every new item needs approval; after 60 days without a mistake, low-risk items (for example Reels cut from an approved video) can publish automatically, with weekly spot checks.
- Starting rhythm per product: 3 articles and 2 videos a week and 1–2 Reels a day, changeable in the admin panel; prices only from the price table, claims only with proof (16.3), and AI-made media labelled as the platforms require.
- Systems: no new servers — the Landing Page AI and CMS (3.25), the Motion Studio pipeline through its API at P10 in the batch window and its direct publishing (4.6), Chatwoot, the notification service, the in-house CRM, local Qwen3 and FLUX.1-schnell, and TimescaleDB with Grafana for attribution and reports. New services: the LinkedIn Community Management API, Google Trends and Google Ads Keyword Planner APIs, and an outreach mailbox (18.1); open-source schedulers and email tools such as Postiz and listmonk are AGPL, so this part is built in-house like the CRM.

RemonLogic Growth Platform (all products)

The AI Marketing Hub, the CRM, event-driven journeys and the target-client database form one shared RemonLogic Growth Platform for NexaTute, Remon Motion Studio and NexaLogicDev (5.3). Each product keeps its own brand, accounts, mailboxes, data, pipeline, owners and approvers. The Marketing Hub appears in the platform admin panel (38.6; prototype X19).

- One CRM record per organisation across the three products; the Nexa ID shows which products it uses. Another product is offered to an organisation only with its consent, and one unsubscribe stops marketing from all three products.

Event-driven journeys: NexaTute sends events — demo requested, application started or abandoned, enrolment done, institute onboarding step stuck, webinar registered or attended, inactive days and placement confirmed. Each journey is trigger → wait → condition → segment → channel (push, email, SMS or WhatsApp) → stop at the goal. Reports show entered → messaged → goal (application or enrolment), with a 10% holdout group to measure the real lift.

Journey	What happens
Demo requested, application unfinished	Reminders at 24 and 72 hours with chat help; stops when the application is completed or the person opts out
Institute onboarding	Nudges the institution admin for each stuck step; after 7 days the partnership manager is alerted
Webinar follow-up	A no-show gets the recording; an attendee gets thanks and a demo link; attendance → demo → enrolment is reported
Course recommendations	From the learner's own goals, syllabus and progress; at most one a week; only with the "personalised offers" consent switch (3.19)
Inactive re-engagement	Learning-first messages at 3, 7 and 14 days, with no guilt or fear wording; stops after three
Placement success stories	From the placement module (18+ only) with written consent; AI draft → approval → publish; views → leads are reported

- Journey rules: no marketing message goes to an under-18 learner directly — only to the guardian, with the guardian's consent; no behaviour-based targeting of minors; consent and opt-out per channel; at most one marketing message a day and three a week; quiet hours 9 pm–8 am; each journey and its message templates are approved once, then sends run automatically; no grades or health details in any message.
- Target-client database: a worldwide database of organisations, not of individuals' personal data. Sources are government and open data (for example Bangladesh school and education-board lists, OpenStreetMap, Wikidata), organisations' own public contact pages (respecting robots.txt and site terms), official APIs, public fair and exhibitor lists, and our own form, chat and webinar leads. Every field records its source and lawful basis; prospects with no response for 12 months are deleted.
- AI target finding: ideal customer profiles per product — NexaTute: schools, colleges, coaching centres, universities, employers and sponsors; Motion Studio: agencies, small businesses, education and Islamic content organisations; NexaLogicDev: software agencies, freelancers, small businesses with outdated sites and startups. A fit score (size, type, area, language) and intent signals from public information (a new branch, hiring, exam results, an outdated site) produce a weekly top-target list; a person approves each campaign before it starts.
- Country rules engine: consent-first countries (for example Canada) get no cold email — only content, ads and inbound; the EU, UK and US follow their own rules, to official addresses with an unsubscribe; Bangladesh rules are confirmed by the lawyer; never personal addresses or bought email lists; individuals such as creators and developers are reached only through inbound, content or ads.
- Systems: no new servers — websites are read in the night batch on CPU with the existing Scrapy, details are extracted by local Qwen3, PostgreSQL with pgvector finds look-alike organisations, Valkey streams carry journey events, ClickHouse holds journey reports, and the in-house CRM and notification service send the messages; email addresses are checked only by format and domain (email-validator, dnspython), never by probing other mail servers; the WhatsApp Business Platform stays the paid exception (18.1).

Channels, import, export and target clients (decided 25 October 2026)

- Channels (a tab under Publishing and channels in X19): a Connect button per platform — YouTube (Google Cloud project and the YouTube Data API; audit before public API uploads), Facebook and Instagram (the Meta app, an Instagram Business account linked to a Page, App Review) and LinkedIn (Community Management API approval). Each signs in with the product's official account by OAuth, so we never see or keep a password; the chosen page or channel

and its permissions are shown; tokens are encrypted in OpenBao with automatic refresh and expiry alerts; the super admin connects and the security owner approves; NexaTute, Motion Studio and NexaLogicDev each keep their own accounts. X (Twitter) is left out for now because its API is paid; the owner decides after checking the price (18.1, blocked until approved).

- Import (a tab under Target clients and CRM): Excel, CSV or PDF lists (openpyxl, Docling) → AI column mapping (local Qwen3) → duplicate, format and country-rule checks (email-validator, dnspython) → source and lawful basis required for every import → preview → approval → a batch ID with one-click undo. Organisations only — no bought lists or personal email addresses.
- AI target finding uses the existing night batch (Scrapy on Core-1, local Qwen3, pgvector look-alikes), and every field shows its source. Export: Excel, CSV or PDF with chosen fields (openpyxl, WeasyPrint, pypdf), a fresh OTP, the log, the exporter's watermark and a link that expires in 24 hours, limited by role.
- Target clients list: filters, the weekly top list with "why this target", a map drawn with Apache ECharts from boundary shapes on our own CDN (no external map tiles), saved segments, add to a campaign with approval, an owner and the CRM record; unsubscribed organisations are never targeted.
- Import and export are approved by the marketing admin, plus the DPO when personal data is involved. The new parts are tabs inside the existing sections of X19, not new menu sections. No new servers.

3.27 Tests, Written-Answer Checking and Exam Integrity

Decided 25 October 2026. The learner side is in prototype X7 (choosing a test, written checking and board-style tests) and the admin side in X10, X14 and X16 (38.2, 38.3, 38.5). Mistakes from every test go to the error notebook (3.18). Release timing is set in the release plan after the wireframes.

Question types

Type	How it works and how it is checked
True/false	A statement to mark true or false; higher classes may add an optional one-line "why"; negative marking only where the real exam uses it
Fill in the blank	Checked automatically: a small spelling slip is accepted and the correct spelling is shown; Bangla and English digits count the same; units are checked; equivalent mathematical answers are accepted (math.js in the browser, re-checked with SymPy on the server). The rules are set in the admin panel (38.5)
Matching and ordering	Match pairs, or put steps, events or sentences in order; checked automatically
Short answers	Typed answers marked by the AI against the marking scheme, with the reason for each mark; a short answer counts as one written check
Written tests	SSC and HSC creative questions, essays and mathematics working, answered by photographing or scanning several pages or by typing (written-answer checking, below)
Board-style mixed tests	Mixed papers built like the real board exam of each country's curriculum, for example SSC Physics — MCQ 25 marks in 25 minutes, then creative questions 50 marks in 2 hours 35 minutes; the order of parts and the number of creative questions to answer are matched to each board's current rules, for example "answer 5 of 8". Questions stay in the exam's language whatever the app language, and explanations follow the app language (to confirm). Fixed papers, never adaptive

Written-answer checking

- Upload: every page is photographed or scanned (several pages per answer), or the answer is typed. Each image is checked by Nexa Guard (NSFW classifier and child-safety hash matching,

4.15), location data is removed, and the script is visible only to the learner and the guardian; for tests an institution sets, that class's teachers also see and review it (decided 26 October 2026).

- Reading: handwriting is read by Qwen3-VL-8B on our own GPUs. Parts the model is unsure of are shown to the learner to confirm or correct, and nothing is marked until they are confirmed.
- Marking: part-wise marks following the board's marking scheme — knowledge, comprehension, application and higher ability for creative questions, step marks in mathematics — by the local Qwen3 against the approved scheme (38.5); the result shows where and why marks were lost and a model full-mark answer.
- After marking: the mistakes go to the error notebook (3.18). "Request recheck" sends the script to a human reviewer — free once per test, then up to 2 more a day; the reviewer answers within 24 hours.
- Timing: results usually arrive in 10–15 minutes. At the evening peak (4–9 pm) checks wait in a queue and a notification says when the result is ready. Scripts are never sent to an outside AI service, even when the GPUs are busy — they wait in the queue (8.17).
- Limits: written checks per plan are set in the plans and modules table (38.2); true/false, fill-in-the-blank and matching are unlimited for everyone.

Exam recovery, integrity, adaptive tests and certificates

Item	Rule
Exam recovery	Every answer is saved on the device (IndexedDB) and synced to the server every 10 seconds (default, editable); after a connection loss the test resumes exactly where it stopped, until the test's end time. Practice tests: the timer pauses while offline. Timed ranked, board-style and certificate tests: the timer keeps running during the learner's own connection loss (the answers are kept), and time lost to a verified NexaTute server outage is added back
Randomisation and similarity review	In ranked, batch, live, certificate and institution tests the order of questions and of options is shuffled per learner, and institutions can use sets A, B and C of equal difficulty. After the test an answer-pattern similarity check across learners in the same session flags pairs for human review — never an automatic penalty; the learner is told and can appeal, and the decision is logged
Adaptive tests (CAT)	Only the placement test at sign-up (3.10) and "smart practice" are adaptive: the next question is chosen by difficulty from IRT-calibrated items — items with 200 or more answers, the MindForge rule (37.5) — and the test stops when the level estimate is precise enough, after about 10–20 questions (default, editable), showing the level as a range. Board-style mocks, ranks and certificates never use adaptive tests
Practice-exam note	Shown on every practice exam in every app language: "Practice exam by NexaTute — not recognised by any education board or exam authority. Real exams and their recognition belong to the concerned authority." The Terms and their plain-language summary carry the same line (19.4)
Speaking fairness	Speaking and oral tests are marked on clarity, fluency, vocabulary and grammar — never on having one particular accent; bias checks run regularly across regional Bangla accents and first languages, and a person reviews a weekly sample and every appeal
Coding tests	In skill courses (3.12): problems with visible and hidden test cases. The learner writes the code in the browser editor (CodeMirror 6); the graded run happens on the server in the existing gVisor sandbox with time and memory limits and no network — our own runner, not Judge0, which is GPL. Python, JavaScript and HTML/CSS at launch (JavaScript and HTML/CSS run in headless Chromium inside the sandbox). Score = tests passed, plus AI hints on style; a code-similarity check flags pairs for review, never an automatic penalty C++ tests can be switched on at any time from the admin panel (decided 26 October 2026).

Item	Rule
Paper and blueprint builder	For teachers and institutions: choose the syllabus, chapters and topics, question types, marks per section, difficulty mix and number of sets (A/B/C) → a draft paper from the question bank (approved questions only; AI-drafted questions only after validation) → the teacher edits → approval by the head teacher or coordinator (the maker cannot approve their own paper) → published to a class or batch online and/or printed as a PDF, with the answer key kept restricted
Certificates	A pass mark per course; competency levels Foundation, Proficient and Advanced (default, editable); a retake after 7 days and at most 3 attempts in 30 days (default, editable), the best result counting; certificate ID, QR verification and revocation as before (3.10); each certificate notes that NexaTute certificates are not accreditation
Live proctoring (limited)	Only certificate exams and an institution's own exams — never practice tests. Adults (18+) consent first; the options are a face check at the start and random presence checks (the existing face verification, 3.16), browser focus and tab events and — adults only, with consent — webcam snapshots every few minutes (not video), kept 6 months with the consent records (decided 25 October 2026) and then deleted; institutions may add a live human invigilator over LiveKit, not recorded. Under 18: no video or snapshots — only the face check, random presence checks and focus events, with guardian consent. No emotion or attention reading. Flags go to a human reviewer, who decides with a reason; the learner can appeal; AI flags alone never prove cheating
Employer and organisation assessments	Companies and agencies can run employee skill tests and recruitment tests for adults (18+) only; candidates consent before results are shared; the employer sees scores with explanations; the service follows ISO 10667 and every report states that results are never the only basis for hiring; it fits the job pathway (3.21). Free for learners on job-test courses; anyone else aged 18 or over buys the Recruitment tests add-on (price and limits in 11.14; per-plan setting in 38.2)
Institution roster and results API	For partner institutions (18.3): OAuth client-credential keys with scopes (roster read and write, assign tests, results read), webhooks for results, rate limits, approval by the institution head and the NexaTute security owner, an audit log, and key rotation and revocation; LTI 1.3 stays available (3.11)

Systems: no new servers and no new software components. Written checks are a queued job on the launch GPU pool — Qwen3-VL-8B on AI-2 Node A first — within a daily GPU-minute cap (8.6, 8.17); graded coding runs use the practice-lab gVisor sandboxes (8.12); autosave and the code editor run in the browser; the existing register rows are extended (18.1).

3.28 Classroom Voice Rules, Live Connection and Weak Networks

Decided 28 October 2026 (prototypes X02 1:1 class, X06 batch class and the X12 classroom view; admin X10 and X14). Every value below is an admin-panel setting with two-person approval and a change log (38.17); values marked (default, editable) are starting settings.

Voice in class

Rule	How it works
"1:50 AI class"	The AI batch class is labelled the "1:50 AI class": one Nexa Teacher and at most 50 learners per batch (3.17); the batch maximum is an admin setting (50, default, editable).

Rule	How it works
Microphone off by default	In the 1:1 class and the 1:50 AI class every learner's microphone starts off. To speak, the learner taps Raise hand; the Nexa Teacher opens the mic, explains naturally and may ask that same learner one short follow-up question (the mic reopens only for that answer); the mic turns off automatically at the end of every voice turn (turn detection, at most 60 seconds a turn). Hand-raises per class: unlimited (default, editable); follow-ups per hand-raise: 1 (default, editable); turn length: 60 seconds (default, editable).
Disturbance rule	Off-topic, noise or non-question turns detected by the AI: 3 within 10 minutes → the mic is paused for 2 minutes with a short, kind, age-appropriate motivational message from the Nexa Teacher (never shaming); typing questions stays allowed. The same rule applies in the 1:1 class and the 1:50 AI class (confirmed 28 October 2026); the rule can be switched off and its threshold and pause changed in the admin panel.
Interrupting the teacher	The Nexa Teacher runs as a LiveKit agent with turn detection, so a learner whose mic is open can interrupt it, as in a real class.
Batch speaking permissions	Raise hand → mic granted, mute all and cold-call; the institution's classroom view in the Institute Admin shows the same rules (38.7).

Live-class connection (LiveKit features, approved 28 October 2026)

Feature	How it works
Pre-class check	A device, microphone and network check before the class starts
Adaptive quality	Simulcast and dynacast with a connection indicator; automatic audio-only fallback at 15% packet loss (default, editable)
Reconnect	A "reconnecting…" banner; minutes lost to a dropped connection are given back (3.16)
Noise suppression	In the learner's browser, on by default (default, editable)
Weekly Nexa Teacher Live	LiveKit Egress streams it to YouTube and Facebook, scheduled and monitored in the admin panel (3.21)
Human-teacher revision class	LiveKit Ingress (OBS, RTMP or WHIP) brings the human teacher's stream into the batch (3.10)
Screen sharing	Only for learners aged 18 or over in skill and coding courses; locked for under-18 with the reason shown
Attendance	Join and leave webhooks feed the attendance KPIs (3.17)
Encryption and recording	No end-to-end encryption in AI classes (Nexa Guard needs the class transcript, 10.2); recordings stay browser-made (3.29)

The LiveKit infrastructure behind these features is in 9.5 and the X14 LiveKit panel in 38.18.

Weak network and weak phone (approved 28 October 2026)

Item	How it works
Offline or slow banner	Outside class (dashboard, hub, payments, uploads) a slim top banner: “No internet — we’ll try again automatically when you’re back online” or “Slow connection — some things may take longer”; automatic retry every 10 seconds (default, editable) and queued actions shown. On payment screens: “Your payment is safe — you won’t be charged twice. Don’t tap Pay again; we’ll confirm as soon as the connection is back.”
Lite mode	Suggested on slow or low-memory phones (3 GB or less, default, editable): 2D avatar, animations off, lower video; a setting and part of the pre-class check; a guardian can set it for a child under 13
Data saver	On mobile data an estimate of data per hour of class (audio and whiteboard ≈15 MB, light video ≈90 MB, full ≈250 MB — estimates, to be measured), video off, downloads on Wi-Fi only (on by default) and recordings streamed in low quality
Low battery	Below 20% in class (default, editable) the app offers audio only
Guardian notice	“Today’s class had a weak connection — N minutes were given back; the recording is ready” — a service notification, on by default, which the guardian can mute; sent when at least 5 minutes were lost (default, editable)
Area or operator notice	From the area and operator weak-network table the admin can notify users of an affected area or operator by in-app notice, SMS and banner, optionally suggesting another class time; two-person approval and a log (38.18)
Wi-Fi prefetch	The next day’s lessons, notes and practice download on Wi-Fi for offline use — on by default, Wi-Fi only, at most 200 MB (default, editable) — with a “Ready offline” chip on the dashboard
Image compression	Photos are compressed in the browser to WebP (AVIF where the browser supports it) before upload — at most 1,600 px, quality 75 (default, editable) — so uploads are small on mobile data (8.18)

Phone and voice channels (decided 28 October 2026): SIP calls through our own servers were not approved, because they need a Bangladesh government (BTRC) licence that is very hard to get. Instead a BTRC-licensed provider’s voice/IVR API makes outbound guardian voice reminders; smartphones use in-app internet calls, WhatsApp Business voice messages and push; feature phones get SMS through the licensed SMS provider; and the inbound phone-tutor line is a licensed IPTSP’s hotline number, available at launch with no licence needed by us. Provider, fallback, per-minute cost and each channel are switchable in the admin panel (38.17). SMS to guardians is always free for families and on by default; NexaTute pays the provider.

3.29 Recordings, Class Reports, Student Dashboard and All Exams

Decided 26 October 2026 from NexaTute_All_Reports_FRESH.docx, with the mistake timeline and per-class improvement added on 28 October 2026 (prototypes X01, X02, X06, X07, X08, X12, X14, X18 and D01). Every value below is an admin-panel setting with two-person approval and a change log (38.17); values marked (default, editable) are starting settings.

Recording by choice

Item	Rule
1:1 class	Recording never starts automatically. At the start the learner chooses “Record this class” or “Don’t record” and can start recording during the class; a recording indicator is always visible; the recording is available only in the learner’s own account (and to the guardian of a learner under 18).

Item	Rule
1:50 AI class	Each learner chooses “I want the recording” or “I don’t need it”. If at least one learner wants it, one shared full-class recording is made — not 50 copies — after the notice, consent and privacy checks. The voice and video of learners who do not want it are left out where possible, otherwise extra safeguards apply; only approved class members can watch it; under 18 the guardian’s consent is needed; the shared recording keeps only consenting learners’ audio and shows nicknames in chat and polls.
Workflow	Choice and consent check → processing queue → upload to R2 → integrity (SHA-256) and metadata check → the “Watch Recording” link is published only after verification → the class summary, evaluation and review history are attached.
Status labels	Recording: Not requested, Consent pending, Queued, Processing, Uploading, Verifying, Ready, Failed (with retry). Report: Generating, Ready, Updated, Not enough data, Unavailable.
Recording library	Named “Subject — Topic — Lesson/Unit — Class date” (for example Mathematics — Linear Equations — Lesson 03 — 26 Oct 2026); Subject → Unit → Topic → Class with filter and search; each card shows subject, topic, lesson, date, duration, AI tutor, status, a one-line summary, an improvement chip, the mistakes count and “Open report” and “Watch” buttons; repeat classes of a topic are separate entries linked to the review history.
Retention and access	Recordings are kept 12 months and then deleted automatically; the learner may keep them until the course ends. Guardians can see the recordings and reports of learners under 18 (locked on). No transcript of the learner’s own voice is stored — only the teacher’s text. Small recordings can be downloaded for offline use.
Protected links	Every private recording opens through a signed link valid 15 minutes (default, editable), made each time Watch is pressed; a copied link does not work for anyone else (9.4).
Priority	Live NexaTute classes have priority over recording processing and Motion Studio work on the shared infrastructure (8.17).

Recording player and class report

Part	What it does
Player	Ask-the-recording search (from the Nexa Teacher’s own text — no GPU transcript); auto-chapters and a 3-minute smart recap; “where I got stuck” bookmarks; private timestamped notes; one-tap clip → flashcard or error notebook; “skip what you know” replay that jumps only to unmastered parts; WebVTT captions and chapters and #t= timestamp links; red seek-bar markers at each mistake with “Next mistake”, a tap jumping 10 seconds before the moment (default, editable); the sign-language avatar (3.31)
Class report	Summary, topic coverage, performance, mistake analysis (shown as “possible cause” unless certain), knowledge gaps, review history (count, dates, before and after score), practice history, next steps and evidence links (recording timestamps, quiz results)
Mistake timeline	Each row: the class minute (mm:ss), what went wrong, the kind (careless, concept or calculation), the correct rule, 1–2 practice items (2, default, editable), “Watch in video” and a “Corrected” chip after 2 right answers in a row; an unrecorded class shows the minute with the whiteboard step and the Nexa Teacher’s words as text; in batch classes only the learner’s own mistakes
Per-class improvement	Topic mastery before → after the class (for example 42% → 61%); help-assisted answers weigh less; “Not enough data” below 5 answers (default, editable)

Part	What it does
For guardians	The weekly report shows the week's top 3 mistakes with minute links and improvement chips; "3 things to talk about with your child this week"; the report read aloud in Bangla; free 2-minute "how to help your child" tips made with Motion Studio
For teachers and institutions	An anonymised class-report view; a revision-class teacher brief with the batch's top 3 mistake topics and no names; scripts of tests an institution sets are seen by that class's teachers (3.27)

Main Student Dashboard and All Exams

Part	What it shows
Landing page	After login a learner lands on the Main Student Dashboard; "Dashboard" is the first item of the learner menu (decided 28 October 2026)
Overview	Progress and mastery as separate metrics, classes, time, scores, review count, repeated and corrected mistakes, weak and strong topics and overdue reviews; total time in the app and active learning minutes side by side; a timeline from the first session (weekly, monthly, term, all-time) with no causal claim without evidence; each KPI with its definition, formula and data freshness (ⓘ); "Not enough data" instead of a guessed score
Mastery Map	BKT and IRT mastery with a confidence range; cross-subject topic links; a target-grade tracker; the FSRS review scheduler with the overdue-review KPI; the Exam Readiness score with a what-if (3.32)
Advice	Explainable AI advice cards with reasons and evidence (a simpler version for guardians); encouraging, never labelling; a person reviews high-impact decisions
Review and mistake rules	Reviews are counted per topic as separate events; rewatching a video never counts as mastery (assessment or practice evidence is needed); each mistake keeps the question or skill, timestamp or assessment ID, category, correction status and later result; the history is append-only
All Exams	Every exam type — diagnostic, practice, topic, chapter, mock, term and final — from the first to the latest, with attempt history and statuses (unattempted, incomplete, submitted, evaluated, retake); question-wise and topic-wise analysis; a question heatmap, time per question, attempt comparison and answer-change analysis (wrong → right, right → wrong, time stuck); "where I lost marks" from part-wise board marking (missing steps, units, diagrams, presentation); handwriting and presentation progress from written-test scans (learner and guardian only, deleted after 90 days); a long-term trend only between comparable tests; no guessed final score before evaluation; data source, maximum marks and scoring rules shown; class participation, practice and formal exams kept as separate metrics
More	Self-assessment accuracy (confidence against correctness); chapter micro-badges (Open Badges 3.0) in the learner passport; a portfolio PDF and verifiable certificate export; one-click download of all the learner's own data (ready within a few minutes, link in the notifications, valid 7 days, a one-time code — default, editable); dyslexia-friendly and colour-blind-safe report modes; "Teach the AI", where the learner corrects a deliberately wrong AI student on the Nexa Co-Teacher and it counts as mastery evidence; a daily "question from my mistakes" widget at a time the learner picks (never AI timing for under-18 and nothing after 9 pm); exam-day mode (no new content, a 5-minute formula recap, a checklist with the admit card and calculator rule, a calm message and a 1-minute reflection after the paper)

Part	What it shows
Programmes	A monthly “Most Improved” scholarship: the top 3 improvers per course each month (default, editable) — biggest improvers, not top scorers — get a free month; the NexaTute Learning Atlas, an anonymised district × subject map of topic difficulty with cells under 20 learners hidden; a weekly “10 weakest AI explanations” quality loop from thumbs-up and thumbs-down ratings and wrong-answer reports into the review queue (38.5)

Standards: xAPI learning records (append-only, in ClickHouse through Ralph LRS), QTI 3.0 for questions and results, Open Badges 3.0 and W3C Verifiable Credentials, WebVTT captions and chapters, #t= timestamp links, a SHA-256 integrity check before Watch, the C2PA label, WCAG 2.2 AA with every chart also available as a table, the KPI dictionary, and a consent record and access log for every recording (19.4). Privacy: learners see only their own reports and permitted shared class reports — never another learner’s KPIs, mistakes, voice, video or scores. No new servers (9.5).

3.30 Exam & Admission Hub

Approved 27 October 2026 from the Academic Intelligence Hub prototype (exam registry by type, exam calendar, notification preferences and academic profile, made more futuristic), with the record rules of the draft legal opinion added on 28 October 2026 (36.6). Placement: an “Exam & Admission Hub” tab in the tests area (X07), a countdown card on the dashboard (X18), the guardian view (X08), institution schedules through the Institute Admin (X12), control rows in the plans table (38.2), the circular-verification queue in content quality (38.5) and the demand signal in growth (38.6) — no separate app. The hub is free for everyone; the paid features are marked below. Idea 51 (English-medium Class 4–8 courses) is in 11.1 and crash programs in 11.12.

Rule	How it works
Existing rules	Six languages and the NexaTute design system; all 11 education boards (Dhaka, Rajshahi, Cumilla, Jashore, Chattogram, Barishal, Sylhet, Dinajpur, Mymensingh, Madrasah, Technical); readiness from the KPI dictionary (3.29); no rank alerts in Calm mode; no reminders to children after 9 pm; advance notice of changes when the Sensory-friendly mode is on (3.31)
Personal numbers	Roll, registration and candidate numbers are optional, masked (for example ••••4821) and encrypted; the guardian consents for learners under 18; the hub never asks for passwords or OTPs of board, university or other systems
Record statuses	User-entered · Pending verification · Verified from official source · Updated by official notice · Unable to verify; every exam event keeps the source institution, the original notice link, the publication date, the last-verified time, the status and a history of corrected notices; sample dates are never shown as official
Sources	Public notices only, with a “source rights checked” tick per source; no logins to board or university accounts and no scraping of restricted systems without an approved API or written permission (36.6)
Not an official service	A one-line note in the hub and on the landing-page teaser: NexaTute is not an official service of any board, university, Cambridge or Pearson, and uses none of their logos
Notification categories	Exam and deadline · admit card and seat plan · result and shortlist · mock exam · marketing — kept separate; service categories say why they are sent; marketing is off by default with one-tap opt-out (19.4)

Rule	How it works
Past questions	Bangladesh board past questions (SSC, HSC, admission, BCS) are used verbatim in teaching and NexaTute's own exams with the source year and board; past papers are never shown as whole papers; Cambridge, Pearson and IB questions are not published verbatim — original syllabus-matched questions plus links to the official sites (decided 28 October 2026; IB treated the same as Cambridge and Pearson)
Centre uploads	A school that is an approved Cambridge or Pearson centre, or an IB-authorised school, may upload papers into its own private area of the Institute Admin for its own students only, free of charge: it confirms its centre number, private use with its own enrolled students, no charge, no sharing, removal of third-party material it may not share and deletion after the board's limit (Pearson: 5 years after the series); the decision and responsibility are the school's; NexaTute only stores them privately, keeps an upload log and removes a paper at once on a rights-holder request (36.6); never mixed into NexaTute's question bank
Order of work	The 54 ideas are built and reviewed before anything is added; later additions are decided from Free Beta data

#	Hub idea	Plan
1	AI circular reader — PDF → dates, units, eligibility, fees and seats; a reviewer approves; source and last-verified time on every event	Free
2	Eligibility checker from GPA, group and year, recalculated on result day	Free
3	Backward study plan from the exam date, readiness per exam with a what-if; Crash Pack 30 days before	Free
4	Date-clash and application-fee budget planner with fee-deadline reminders (NexaTute never takes the fee)	Free
5	Encrypted admit-card and document wallet with an offline copy, exam-day checklist, route and leave-by alert	Paid plans
6	Result day: alert, re-scrutiny deadline, Second Chance offer, eligibility recalculated	Free
7	Merit and waiting-list and migration tracker	Free
8	Exam-pattern mock tests (DU units, Medical, BUET, GST, Cambridge) with a predicted range only from comparable tests	Paid plans
9	Guardian copy of deadlines, in the guardian's own language	Free
10	Google and Apple calendar, ICS, WhatsApp and push sync with advance notice	Free
11	Institutions push their schedules and circulars; teachers see counts without names	Free
12	O/A Level entries and IELTS/SAT dates linked to the Study Abroad Coach (3.20)	Free
13	Voice: "when is my next deadline?" through the Nexa Co-Teacher	Free
14	Photo and signature resizer to each form's size and KB, in the browser	Free
15	SSC ↔ HSC name and date mismatch checker, with the exact certificate spelling to copy	Free
16	Step-by-step application guides per university, kept up to date by a reviewer	Free

#	Hub idea	Plan
17	Marksheet photo → GPA and subject marks read by our own AI and confirmed by the learner (never pulled from result sites)	Paid plans
18	Last 5 years' seats and merit positions where public, with the source	Free
19	Subject and unit choice simulator from past data (estimate only)	Paid plans
20	Link to career guidance	Paid plans
21	Scholarship and stipend finder with deadlines in the calendar	Free
22	One-tap "reached the centre" message to the guardian (no live location)	Free
23	Seat-plan finder in the official PDF on the phone (the roll number never goes to the server)	Free
24	Weekly one-question self-reported stress check-in → Calm or Sensory-friendly mode suggestion, helplines if serious; no camera or voice mood reading	Free
25	Guardian voice-call deadline reminders through the licensed voice provider (3.28), switchable in the admin panel	Free
26	Home-screen countdown widget	Free
27	Target-exam Study Hall groups (nicknames, under-18 rules)	Paid plans
28	"Date changed" alerts showing old → new	Free
29	Admin demand signal suggesting new batches or Crash Packs (counts only)	Admin only
30	Anonymised "where learners like you applied" (cells under 20 hidden)	Paid plans
31	School admission tracking — government school Class 1–9 lottery, cadet college Class 7	Free
32	Polytechnic (BTEB), National University and nursing admissions	Free
33	Private-university fee and GPA-waiver comparison from public sources	Free
34	Post-admission yearly cost estimate (hall or mess, transport, books) tied to the budget planner	Free
35	Viva and interview mocks like the BCS Viva board	Paid plans
36	Exam-day rehearsal mock at the real exam's start time and duration, with on-screen OMR	Paid plans
37	Disability accommodation guides (scribe, extra time where the board or university allows; reviewer-verified) with deadlines	Free
38	Lost admit or registration card guide with the wallet copy	Free
39	O/A Level entry manager — series, paper codes, entry and late-entry deadlines, school or private candidate through the British Council, statement-of-entry check	Free
40	Paper timetable clash and contingency-day reminder	Free
41	Syllabus-version tracker	Free
42	Correct grade scales everywhere (A*–G and 9–1 per subject)	Free

#	Hub idea	Plan
43	Estimated grade from mocks using past official grade thresholds (estimate only)	Paid plans
44	Post-results — enquiries about results and script access deadlines, resit planning, conditional-offer tracking	Free
45	O/A Level → Bangladesh GPA equivalence, the board's equivalence-certificate steps and DU/BUET/Medical English-medium rules (reviewer-verified), linked to the eligibility checker	Free
46	Predicted grades and UCAS / Common App deadlines linked to the Study Abroad Coach	Free
47	Command-word trainer marked against mark-scheme points (within the written-check limits, 38.2)	Paid plans
48	Topic-wise past-paper-style practice with examiner-report lessons (original questions, below)	Paid plans
49	Alternative to Practical with the Virtual Lab; A-Level practical endorsement and coursework deadlines	Paid plans
50	Exam data booklets and per-paper calculator rules in mocks	Paid plans
52	Bangla and Bangladesh Studies support for English-medium learners	Paid plans
53	IB EE/IA and A-Level project feedback (the AI never writes it)	Paid plans
54	English-medium schools push internal exam calendars through the Institute Admin; guardian reports use that school's grade scale	Free

3.31 Accessibility, Sign Language and Wellbeing

Decided 26–29 October 2026. These features sit beside the accessibility options of 3.10 and the standards of 19.7. Every value below is an admin-panel setting with two-person approval and a change log (38.17); values marked (default, editable) are starting settings.

Feature	Rule
Bangla Sign Language avatar	Built complete, not as a pilot, for deaf and hard-of-hearing learners in every class, subject and course, Bangla and English medium alike; the avatar always signs in Bangla Sign Language (BdSL) and captions follow the teaching language (English medium: BdSL with English captions); rendered in the user's browser; signs are checked by a BdSL expert and a Deaf reviewer (38.5), and a lesson not yet checked shows "sign check in progress". It costs nothing extra in any plan; with the avatar on for a whole 1:1 class, minutes are counted more slowly — 1.5 minutes used = 1 minute charged (default, editable). It is on and locked in every plan (only the platform switch and the defaults are editable). A Sign Language Fund (Sponsor-a-Learner and CSR, grants) and deaf schools and NGOs at normal seat prices (11.8) pay for it.
Sensory-friendly mode	The autism and speech-delay friendly mode (bn সংবেদন-বান্ধব মোড): soft colours, no sudden sounds, a picture routine and advance notice of changes; for everyone — every age, class and course — not only PlayBoom, whose calm mode is the same mode (30.8); on and locked in every plan; highlighted on the landing page (3.25)
Calm mode	Hides ranks and leaderboards before exams; the learner or guardian chooses; kept separate from the Sensory-friendly mode

Feature	Rule
Extra-time settings	The public name of the slow-learner settings of 3.10 (shorter segments, more breaks, less time pressure); the note says “your specialist” so that it also covers adults
Voice mode	App-wide for slow readers and rural learners: big icons, every button read aloud and voice commands such as “next lesson” and “say again”; set by the learner or guardian; always free
20-20-20 eye break	By time only, never by camera: after every 20 minutes of screen time a 20-second “look far away” nudge between polls or questions (quiet during polls), plus a night warm-screen mode (from 7 pm, default, editable); always free and on in the Free Beta
Under-18 study wellbeing	A break reminder after 45 minutes of continuous study (default, editable); the guardian may set a daily study-time limit; PlayBoom keeps its own limits (30.3)
Self check-in and breaks	An optional check-in chosen by the learner (happy, neutral or worried face) — nothing is inferred; time-based 2-minute breathing or stretch breaks; the guardian sees the check-in only if the learner shares it (sharing off by default, locked off under 13). This replaces AI mood detection, which is not used.
Streaks and leaderboards	A streak breaks after 2 missed days, with 1 free streak freeze a week (default, editable); leaderboards are off by default; learners under 18 are shown by nickname only, never by real name

No emotion recognition is used anywhere: faces are used only for presence and one account = one user (3.16, 19.7).

3.32 Engagement, Community and Future Learning Features

Approved 29 October 2026 (owner, with Claude’s changes; prototypes X04, X07, X08, X11, X12, X18 and X20). The features are built in full; those marked as 2027 features are switched off in the Free Beta with a “Coming soon — from 1 January 2027” teaser and switched on at the paid launch, while the source line, the 20-20-20 eye break (3.31) and the Beta poll are on in the Free Beta. Every value below is an admin-panel setting with two-person approval and a change log (38.17); values marked (default, editable) are starting settings. They run on the existing servers — Centrifugo, the question bank, the verified-answer cache, LiveKit Egress and Motion Studio Quick in the browser — and use the GPU only where the row says so (7.1).

Feature	What it does and its rules
Team Quest (Live Study Hall)	Teams of 3–5 from the same class (4, default, editable) that the AI mixes from stronger and weaker learners; one syllabus problem from the class’s weak topic split into 3–4 steps, each member solving a step and the team voting on each; one shared hint; only team progress is shown, never who made a mistake; 2–3 times a week (2, default, editable), 15 minutes, outside the study rounds; preset buttons and votes only, nicknames only; guardian approval under 18; XP and a team badge for members who contributed (30 XP a quest, weekly cap 120 — default, editable); teachers can launch a quest for their class and see the step where the class got stuck, without names
Quiz Battle	Team against team, 10 questions, a team answer and a team result only — never individual; under 13 quest only, no battle (default, editable)
Shorts question	An optional one-tap question under each Short with XP (5, default, editable); no infinite scroll and no coins for children; the quiz after every 10 Shorts stays (3.19)

Feature	What it does and its rules
AI Debate Chamber	A voice debate in which the Nexa Teacher takes the other side, scored on structure, rebuttal, evidence and fluency; for BCS, IELTS Speaking and Spoken English; topics from a vetted list; no political or religious debates; Nexa Guard on
Family quiz cross-rewards	In the weekly guardian + child quiz a guardian's right answer gives the child XP and the child's gives the guardian a "family badge"; Friday evening (default, editable)
Live Game Show	Weekly, with an AI-avatar host, timed questions and a homepage countdown; Saturday 7:00 pm (default, editable); nobody is knocked out — a wrong answer keeps playing as practice; no cash prizes for children; results by school or district team, not individual rank; LiveKit Egress HLS and Centrifugo
Be the Teacher	Learners make 1-minute explainer videos with Motion Studio Quick; under 18 no face or real voice (avatar, whiteboard, library voice) and the guardian's permission; visible to the learner's own class only; Nexa Guard and a subject expert review before publishing; no like counts, only "this helped"; XP and a Helper badge; the best videos of learners aged 18 or over may go to public Nexa Notes
AR Magic Book	Scanning an NCTB book page shows a 3D model on it, in the browser (MindAR image tracking and model-viewer; iPhone through Quick Look)
Time-Travel role-play	Only long-dead scientists and historical figures from a vetted list (for example Newton, Ibn Sina, al-Khwarizmi); no political or religious figures and no Bangladesh Liberation War or political figures; always labelled "AI role-play — not the real person"; answers from verified sources; a generic library voice, never voice imitation
Nexa Radio	The 5–10-minute audio summary of each chapter (the audio lessons of 3.10) for screen-off listening, downloaded by Wi-Fi prefetch (3.28)
Mystery missions	A clue for every correct answer; for children lost items, treasure or science mysteries — no crime or violence
Group mind maps	In Team Quest and paid Study Hall groups, with AI gap suggestions; every node checked by Nexa Guard
Learn → Earn	Under 18: simulated client briefs (portfolio, XP, certificate); 18 or over: small real projects for partner NGOs, schools or shops through NexaLogicDev or Motion Studio for learning credits and a verifiable certificate
Beta poll	An in-app "which feature do you want next?" poll — on in the Free Beta — with results in the admin Beta dashboard (3.33)
Teacher's AI assistant	In the Institute Admin and the teacher tools the AI drafts lesson plans, homework sheets and class quizzes from an NCTB chapter or the teacher's syllabus; the teacher edits and approves; answers from the verified question bank; off-peak GPU; 20 drafts a day per teacher (default, editable, by institution plan)
Helper rewards	Reviewed notes or videos that help others earn XP and a badge; learners aged 18 or over on paid plans also get learning credits (non-cash, non-transferable, for their own subscription only, with a monthly cap); this replaces a student-to-student note marketplace — the verified-teacher marketplace is unchanged (3.21)
Parent–Teacher Meeting by phone	The monthly AI Parent–Teacher Meeting summary (3.20) also as a phone call through the licensed voice provider or IPTSP line (3.28), with a keypad question menu for guardians without smartphones
Job board	Verified employer job posts that learners aged 18 or over can apply to, and freelance gigs; every post reviewed for fraud and overseas-job lures, under the job-pathway rules (3.21)

Feature	What it does and its rules
NexaTute for Partners	An API and LTI tool that embeds the AI tutor on other schools' or companies' sites; a DPA with each partner; per-seat or per-minute pricing (sample ৳150 a seat a month or ৳1.50 a minute — to confirm, 15.8); API keys and spend caps in the admin panel; phased in after the paid launch
Nexa Swap	A skill exchange by time-banking — 1 hour taught = 1 hour learned, never money; under 18 only within the learner's own school or class under teacher supervision (whiteboard and preset buttons, no private messages or contact sharing); 18 or over open matching after verification; Nexa Guard on
Public roadmap page	A "Coming in 2027" page on the landing page listing every approved future feature, edited in the Landing Page AI (17.6)
Smart Suggestion wording	The topic frequency is shown (for example "came 7 times in 10 years") and labelled practice, not a guarantee; never "100%", "common" or "leak" (3.20)
Study Buddy companion	A plant or small creature that grows with learning and is never sick or sad; it waits kindly and welcomes the learner back (it extends PlayBoom's garden); this replaces a virtual pet that gets sick or sad
White-label portal	For large coaching centres and schools: their own domain and brand with a small "powered by NexaTute", web and installable app (PWA); a monthly licence fee plus a per-seat price (the fee is a sample — to confirm, 15.8); their own store apps later
AI comic stories	A comic story per chapter, reviewed by a subject expert before publishing; no AI-drawn real people for sensitive history such as the Liberation War — symbolic images, facts checked against the book
Yearly sponsorship packages	For NRB and corporate sponsors, priced like the Academic Year Pass (for example SSC Batch ৳6,000 a year); a sponsor dashboard with progress summaries only — no child names or personal data — and an optional thank-you letter with the guardian's consent (11.9)
What-if simulator	Readiness and finish-date ranges labelled as estimates; no GPA-probability percentages (3.29)
Error Autopsy	Two more evidence-based causes beside concept, calculation and careless: "rushed" (very fast and wrong — under 20% of the expected time) and "didn't read the question" (reading time under 3 seconds) (default, editable); always shown as "possible cause"; the Nexa Teacher explains in 2–3 lines after a test
Source line	Under every AI answer, for example "NCTB Math 9–10, Ch 3, p. 54" or "Verified answer ✓", and "No source — AI explanation" when there is none; on in the Free Beta
Study bots	For registered users only (account-linked; guardian consent under 18) on WhatsApp, Messenger (13 or over; the bot is disclosed; reply within 30 seconds) and SMS short answers for feature phones; answers from the verified cache, one complete reply per question and an app link for long answers; Messenger preferred when the user is there (no per-message fee); WhatsApp service replies cost money after 1,000 free service messages a month per number (from 1 October 2026); each channel on or off with its own budget (38.18); Meta's 2026 AI Providers policy is checked by the lawyer (36.6)

Free and paid placement (approved 29 October 2026; Claude set the placement of the later items, which the owner adjusts in the admin panel)

Everything in free is also in paid. Free for everyone: Quiz Battle, the Shorts question with XP, the Live Game Show, family quiz cross-rewards, the AR Magic Book, Nexa Radio, the self check-in and

breaks, the Beta poll, 1 GB of storage (G.3), Helper rewards XP, the job board (18 or over), Nexa Swap, the Study Buddy companion and Error Autopsy cause labels; locked free: Voice mode, the 20-20-20 eye break with the warm screen, and the source line.

Feature	Free	Paid plans
Team Quest	1 a week	All scheduled sessions (2–3 a week)
Mystery missions	1 a week	Unlimited
AI Debate Chamber	1 a month, 5 minutes	Batch 1 a week; Combo 3 a week; Personal 1 a day
Time-Travel role-play	3 text questions a week	With voice — Batch 5 a day; Combo and Personal 15 a day
Be the Teacher videos	1 a month	Batch 2, Combo 4, Personal 8 a month
Group mind maps	Solo only	In groups
AI comic stories	The first chapter of every subject	All chapters
Smart Suggestion	Top 3 topics per chapter	Full practice sets
Error Autopsy	Cause labels on practice tests	Plus the Nexa Teacher's 2–3-line explanation and the All Exams trend
What-if simulator	Readiness range once there is enough data	Full what-if (study time, weeks, finish date)
Study bots	3 questions a day (WhatsApp, Messenger)	Within the Nexa Co-Teacher's 25 topics a day; the SMS bot for feature phones on paid plans only (5 a day)

Paid only: Learn → Earn (simulated briefs for under-18 on any paid plan; real projects for 18 or over on Combo and Personal); the monthly Parent–Teacher Meeting as a phone call; the SMS study bot for feature phones (5 a day); Helper learning credits (18 or over, monthly cap). Institution plans: the teacher's AI assistant (20 drafts a day per teacher), the white-label portal (separate licence) and NexaTute for Partners (contract). A Daily Pass gives Batch-level access while it is active; the AI Debate Chamber and Time-Travel do not use 1:1 minutes (they have their own caps); sponsored learners get the sponsored plan's full paid features. Every limit and placement is a (default, editable) row of the plans table (38.2).

3.33 Free Beta Safety, Trust Messages and Release Notes

Approved 28–29 October 2026 (prototypes X01–X08, X12, X14, X15, X18 and X20). Every value below is an admin-panel setting with two-person approval and a change log (38.17); values marked (default, editable) are starting settings.

Item	How it works
Report a problem / Give feedback	An app-wide button for learners, guardians and institutions, never covering content: a category (bug, wrong answer, payment, safety, idea), a text box and an automatic screenshot with the page, time and device (shown and removable) → a ticket number; reports reach the admin support queue tagged "Beta feedback" with an SLA timer (38.4)

Item	How it works
AI fallibility line	“AI can make mistakes. If something looks wrong, check with your teacher or report it.” under every Nexa Teacher and Nexa Co-Teacher answer, beside the rating buttons and “Report wrong answer”
Paused feature	When the admin turns a feature off for everyone, users see “This feature is paused for now — it will be back soon.” (reason-neutral, never blaming) instead of an error or a blank screen
Beta live dashboard	In platform operations: an hourly funnel sign-up → guardian consent → first class → day-2 return (counts only, cells under 20 hidden), drop-off points, errors and crashes, wrong-answer reports, feedback by category, busy-page hits and the Beta poll results, for the Free Beta of 20 November–9 December 2026 (38.18)
In the Beta / coming later	A list on the landing page of what is on in the Free Beta and what comes later; the “More from RemonLogic” links to Motion Studio and NexaLogicDev are hidden during the Beta and shown after launch (admin switch)
Beta community	An in-app moderated Beta channel — a learner room in the Study Hall and a guardian room — never a WhatsApp, Telegram or Facebook group; moderated by NexaTute staff with Nexa Guard; no phone numbers or social handles; learners under 18 can read and react, and posting needs the guardian’s approval
What’s new sheet	After an update a small sheet (a corner card on desktop) is shown once on the dashboard — never during a live class or test — with three short items, “See the roadmap” and “Got it”; users can turn it off in Settings (on by default); the notes are written per release in the software-updates panel in six languages for learners, guardians and institutions (38.18)
Maintenance notices	When an update needs it, customers are told by SMS, email or WhatsApp, with a maintenance banner on the landing page and a notice on learner, guardian and institution dashboards (38.18)
Public status page	status.nexatute.com, linked from the landing-page footer; incidents are published from platform operations with two-person approval of the public text (38.18)
Report a security problem	A responsible-disclosure page on the landing page (19.6)

04. Remon Motion Studio — Detailed Feature Catalogue

4.1 Script and pre-production

- Accept a topic, marketing objective or creative brief and generate a structured script, or take the creator's own uploaded script (4.17).
- Create scene-by-scene storyboard/visual staging and metadata to guide asset generation and assembly.
- Keep script, storyboard and job state durable so interrupted jobs can resume without losing the user’s work.
- Support a creator workflow for general cinematic video and a distinct avatar-animation path (Quick-mode Three.js characters with Rhubarb lip-sync, 4.10); do not collapse separate engines into one unobservable job.
- Provide an AI Director/orchestration capability to coordinate planning steps and production status; this is a workflow orchestrator, not a duplicate script or rendering engine.
- Support script and storyboard revision history with traceable versions and creator approval checkpoints before downstream generation.

- Maintain character/world continuity notes and approved project assets as reusable references; treat Character Bible and World Bible as project metadata modules, not separate generators.
- Define the project brief fields for audience, purpose, tone, target format and constraints; keep them linked to the approved script and storyboard.

4.2 Voice, speech and localization

- Text-to-speech and voice-cloning workflows where the selected model supports the target language.
- AI Tutor teaching languages at launch: Bangla, English, Hindi, Urdu, Arabic and French. Exact voice quality, availability and licensing must be validated per language; the Chatterbox Multilingual vendor language list does not include Bengali or Urdu, so Bengali and Urdu voices use Indic Parler-TTS (Apache-2.0), which must be validated before launch.
- In-browser speech for every teaching language: speech-to-text runs Whisper small/base in the browser through transformers.js, with in-house Bangla and Urdu fine-tunes trained on Common Voice (CC0) and consented recordings because the base models are weak in those languages. The AI Tutor voice in the browser is a small in-house VITS voice per language (Bangla, English, Hindi, Urdu, Arabic, French), trained on commissioned voice-artist recordings that RemonLogic owns and run through ONNX Runtime Web. The voices read characters directly, so no GPL phonemiser is needed; for Arabic and Urdu the LLM adds vowel marks to the text sent to the voice. A language whose browser model fails its benchmark uses the server voice until it passes.
- Scripts and text: Bengali, Devanagari, Urdu (Nastaliq), Arabic (Naskh) and Latin text use Noto fonts (SIL Open Font License); Urdu and Arabic screens are right-to-left, while formulas, code and equation numbers stay left-to-right; PDFs are made with WeasyPrint so joined letters and conjuncts render correctly; digits, dates and currency follow the learner's locale.
- Whisper large-v3-turbo running on faster-whisper (MIT) is used for local speech-to-text of voiceover files and the server speech fallback.
- Motion Studio voices in the browser (4.8): standard narration and character voices use the same in-house VITS voices in the creator's browser; cloned voices, Chatterbox, Indic Parler-TTS and auto-dubbing run on the server.
- Keep narration, voice, scene timing, subtitle timing and visual sequence synchronized, with review checkpoints before final render.
- Voice cloning must be permissioned: the speaker reads a randomly generated consent sentence aloud (as in NexaTute), and speaker consent, source-rights evidence, intended use and abuse-control records are captured before processing; route uncertain cases for review.
- Auto-dubbing: a finished video can be dubbed into other languages in one step — the AI transcribes the voice-over (Whisper), translates it (Qwen3), voices it in the target language (Chatterbox, Indic Parler-TTS or the in-house voices, or the creator's own cloned voice with consent), fits it to scene timing, redoes Quick-mode lip-sync and adds translated subtitles; the creator reviews each language before export.
- Provide voiceover transcription, translated/adapted script, subtitle generation and target-language review where the selected model and language are supported; preserve creator approval of meaning, names and factual claims.

4.3 Rendering and automated assembly

- Celery workers orchestrate long-running render/remake jobs asynchronously with bounded retries and idempotent job IDs.
- GStreamer and FFmpeg handle media assembly, stitching, transcoding and output generation in the assigned worker environment.
- Use NVIDIA NVENC / h264_nvenc only where the deployed driver, build and GPU encoder support have been confirmed.

- Heavy video processing and final MP4 compilation run on the local SSD of the GPU node executing the job (AI-2 Node B preferred; any unified-cluster GPU node may take it), following the scratch and cleanup rules in 9.3.
- After the final MP4 is compiled, validated and uploaded successfully, a background task may transfer the single finished artifact to Cloudflare R2.
- Maintain progress counters, queue status, retry state, error diagnostics and output validation results.
- Keep the general cinematic video path distinct from the avatar-animation path, with observable job state and separate processing stages where their engines differ.
- Support configured export profiles and output checks only for validated combinations of resolution, aspect ratio, frame rate, codec and duration; the initial 720p and 5-second-to-20-minute values remain design targets, not tested guarantees.
- Output tiers: 480p Native → 720p Escalating → 720p Native → 1080p Escalating → 1080p Native (highest tier, admin-controlled, 4.9). Tier definitions and rates are in G.8. Upscaling runs on the video generation pool after native generation and before final MP4 compilation and validation, following the quality pipeline and gate in 4.13.
- Support scene-level regeneration and targeted correction where the pipeline permits, reusing existing scripts, scene definitions and approved assets to avoid unnecessary full-project regeneration.

4.4 Content remake business policy

Hybrid freemium/paid remake model carried forward from the product decisions: free remake allowance is limited to text/script adjustments and scene/layout reordering that do not trigger heavy GPU rendering, new 3D asset generation or paid voice-cloning workloads. A single basic remake may be offered as a marketing trial if configured. Paid scope includes advanced voice cloning/tone changes, multilingual remakes, GPU/cinematic rendering and other compute-intensive transformations. Enforce entitlements before enqueueing a job; never start paid GPU work before billing/credit reservation succeeds.

- Avoid misleading religious clickbait and fabricated religious quotations; record unresolved source issues and reviewer decisions.
- Do not reserve video-generation GPU capacity while verification or human review is pending; enqueue render work only after required gates pass.
- For Islamic content, uncertain or disputed source cases remain pending human review rather than auto-approved on timeout (verification rules in 4.5).
- For every plan, apply common safety checks to the prompt and relevant downstream artifacts; run plan-specific checks on the generated script before storyboard/video generation and retain policy-versioned audit results.

4.5 Islamic and general content plans

- Client selects a content plan at the start because plan pricing and verification requirements differ: (1) Islamic Content Creation or (2) General Content Creation.
- Every prompt passes mandatory Common Safety Verification before generation; text, audio, image and video safety layers are defined in 4.15.
- Islamic plan adds religious verification for Quran/Hadith source and attribution, religious claims and sensitive depictions. Trace Quran ayah/translation and Hadith collection/reference/authenticity where applicable; do not label unverified quotations as verified.
- General plan uses standard safety verification plus relevant factual and rights checks.
- Content Plan Mismatch Guard: if a General plan prompt is clearly Islamic, explain that Islamic-specific verification will not apply and offer switching plans or proceeding; do not silently switch plan or prompt.

- Cache verification results using a key that includes prompt/content and policy version. Record plan/billing validation per job. Verify generated script before storyboard/video generation. Maintain a human-review queue for unresolved or high-risk cases; Islamic human review completes within 3–7 days, with a separate ETA shown and no GPU reserved or charge taken before approval (4.14).
- Use scoped client-review links and role-based collaborator access only if enabled; collaboration, simultaneous editing and fine-grained roles require separate acceptance tests before being advertised as available.
- Track per-job stage progress, queue wait, retries, error diagnostics, render utilization, upload outcomes and usage-versus-credit settlement for operational monitoring.
- Show storage usage and quota state, and apply configured retention/deletion rules consistently to generated media and project artifacts.
- Expose durable job states such as queued, processing, awaiting review, failed, completed and delivery pending where applicable; support safe cancellation and retry transitions.
- Provide a project workspace for managing multiple projects, their approved assets, scripts, storyboards, remake versions, feedback and approval history; version history is part of project management, not a duplicate media-generation feature.

4.6 Output delivery and account controls

- Final MP4 delivery follows 9.4: upload to Cloudflare R2 as soon as validation passes; clients without available storage receive a 72-hour emailed download link, after which the file is deleted.
- Validate final file existence, size and checksum/integrity before marking a job complete.
- Store durable job state, credit reservation/settlement/refund events and idempotency keys in PostgreSQL; Redis is for queueing/throttling, not the financial source of truth.
- Allow retry-safe uploads and recovery from upload failure; retain local file until durable upload verification succeeds.
- Use storage quotas, retention rules and per-job usage accounting.
- Do not claim that superficial remakes, voice/theme changes or animation changes evade copyright claims or platform reused-content restrictions.
- Avoid double charging for rendering, storage or delivery already included in an active plan; show any applicable add-on price before confirmation.
- Enforce free-allowance eligibility, one-time trial rules, rate/fair-use limits and duplicate-claim controls before job creation.
- Estimate paid compute/credits before enqueueing; show the operation scope, expected charge and storage impact, and require explicit confirmation before paid work begins.
- Direct publishing: creators connect their YouTube, Facebook, Instagram (Reels; Business or Creator accounts) and TikTok accounts (LinkedIn after launch; approvals in 0.7) and publish or schedule finished videos straight from Motion Studio, with title, description, hashtags, thumbnail and captions, and see views and engagement for each video; uploads use each platform's official API and follow its terms and reused-content rules.

4.7 Remon Motion Studio software-to-feature map

Software / component	Primary responsibility	Important implementation boundary
FastAPI — Motion Studio API (App node cluster)	Project creation, access control, paid remake validation, estimates and job status	Separate product runtime and API context from NexaTute
vLLM / local text engine	Script and storyboard text generation	Version, model/license and throughput to be validated

Software / component	Primary responsibility	Important implementation boundary
Celery workers	Asynchronous job orchestration, retries and priority queues	Idempotency, bounded retries, dead-letter/recovery process
Redis2 / port 6380	Celery broker, throttling and transient queue state	Isolated from NexaTute session cache; not durable job ledger
PostgreSQL	Jobs, idempotency, credit reservations, usage ledger and settlement/refund records	Durable source of truth
Chatterbox Multilingual V3	Voice generation/cloning — English, Hindi, Arabic, French, plus Japanese, Korean, German and Chinese for dubbing and language courses	MIT; consent process and quality per language
Indic Parler-TTS	Voice generation — Bengali and Urdu	Apache-2.0; validate voice quality, especially Urdu
Whisper large-v3-turbo	Speech-to-text for voiceover files	Benchmark memory/latency under concurrency
GStreamer	Media pipeline and compositing where configured	Verify plugins, muxing topology, codecs and local scratch path
FFmpeg + h264_nvenc	Final assembly and hardware-accelerated encoding	Confirm deployed driver/build/encoder support
Cloudflare R2 + S3-compatible SDK	Final MP4 and permitted media storage	Scoped credentials, checksum, retry-safe upload and signed access
LiveKit Egress on Rec-1/Rec-2	Lightweight track recording and whiteboard event log; download MP4s are made in the user's browser (9.5)	Do not overload Core-1 with avoidable recording work; benchmark track recordings per node
Portkey AI Gateway	Model routing/fallback and provider control	Provider identifiers, credentials, timeout/retry semantics need deployment review
Dubbing pipeline (Whisper → Qwen3 → TTS → FFmpeg)	Auto-dubbing into other languages with translated subtitles (4.2)	Creator review per language; cloned voices only with consent
Platform publishing (YouTube Data API, Meta Graph API for Facebook and Instagram, TikTok Content Posting API; LinkedIn Community Management API after launch)	Direct publish, scheduling and view statistics (4.6)	OAuth tokens encrypted in PostgreSQL; platform app review needed before launch

Software / component	Primary responsibility	Important implementation boundary
Creator tools (4.10–4.12)	Music (ACE-Step), sound-effect library, auto-reframe (MediaPipe), timeline editor (Revideo), captions (libass), Content Credentials (c2pa-rs), background removal (rembg + BiRefNet), Shorts cutting (PySceneDetect), music separation (Demucs), fingerprint check (Chromaprint)	Commercial licences verified before use (ACE-Step, Demucs weights); Chromaprint as a separate process; C2PA signing key in the secrets manager
Series and promo videos (Qwen3, faster-whisper, PySceneDetect, FFmpeg + NVENC, Revideo, ACE-Step, FLUX.1-schnell)	Intro, outro, recap, next-episode preview, thumbnails, teaser, trailer and highlights (4.18)	Cut from rendered episodes on the assembly node; new AI shots only on request; drafts free until export
Quality and safety (4.13, 4.15)	Video upscaling and quality gate (RealBasicVSR/BasicVSR++, GFPGAN, libvmaf, OpenCV); content safety for text, audio, images and video (Qwen3Guard, policy layer, NSFW classifier, Qwen3-VL) and child-safety hash matching (HMA + PDQ)	Model weights licences verified (VSR, GFPGAN face detector, NSFW classifier); thresholds set from benchmarks; unclear safety cases to human review
Script upload and pre-production fee (4.17)	Reads uploaded scripts and images through the Document Intake Service, splits scenes and speakers, and charges the pre-production fee in the usage ledger	Fee charged once per project before generation; edits free; refunds on technical failure
Shared Landing Page AI and AI CMS (3.25)	Motion Studio landing page, CMS and drop-off analysis (4.19)	Own site, content, admins and data; package and render rates read live from G.8
Shared AI Marketing Hub (3.26)	Motion Studio articles, showreels, Reels, creator and agency outreach, and reports	Own social accounts, mailbox, CRM data and approvers

- Performance and quality gates: validate per-language voice quality, subtitle timing, scene synchronization, export compatibility, GPU cost, retry cost, queue latency and upload recovery using reproducible tests before publishing SLAs.
- Implementation status and launch gates: each Motion Studio capability must have a canonical requirement, owner, dependency, model/license record, security review and acceptance test. A listed feature or software component does not imply production readiness.

Canonicalization / no-duplicate rule: script generation and AI Director orchestration are related but not counted as two script generators; voice generation, voice cloning and transcription remain distinct operations; storyboard, scene generation and final assembly are separate pipeline stages; basic editing and compute-heavy remake are access/billing classes over the same project workflow; Islamic/General plans share common safety gates but retain plan-specific verification; workspace/versioning, job telemetry, billing ledger and media delivery are supporting capabilities, not duplicate render engines.

4.8 Render Modes, Browser Rendering and Lip-Sync

Motion Studio offers three render modes. The AI Director chooses the cheapest suitable path for each scene, so only hero shots use full video generation. The modes differ in visual style, not in

delivery quality. Pricing: one subscription covers all modes; Quick and Enhanced use the G.8.1 per-minute rates; Cinematic uses the G.8.2 per-minute rates (+35%) within the same subscription; Islamic Content gets the G.9 discount in every mode.

Mode	How scenes are made	Look and resolution	GPU cost
Quick	Rendered in the creator's web browser: 2D/3D animated characters and motion graphics with Revideo (MIT) and Three.js; MP4 encoded by the browser with WebCodecs; reusable scenes from the scene library	Animation, cartoon, explainer and Reels motion graphics; true 1080p Native because frames are drawn, not generated	None
Enhanced	2.5D: a FLUX.1-schnell key image plus a depth map (Depth-Anything V2 Small) animated with camera moves and parallax; library and cached scenes where they fit	Cinematic-looking still or slow-moving scenes; up to 1080p Native (key image, depth map and parallax drawn at 1080p)	Low
Cinematic	Full Wan 2.2 generation for hero shots; other scenes from the cheaper paths above	Most realistic AI video; up to 720p Native and 1080p Escalating; 1080p Native only after its GPU benchmark and quality gate pass (4.9, G.7)	High (only for the scenes that need it)

- Scene library and generated-scene cache: popular scenes (mosques, cities, sunsets, offices, classrooms, nature) are generated off-peak and reused; any generated scene is cached by prompt, style and model version so an identical request reuses it instead of rendering again.
- Preview first: every order first produces a fast 480p, few-step preview; the full render starts only after the creator approves, which reduces wasted renders and remakes; script, storyboard, key images and this preview are covered by the pre-production fee (4.17).
- Delivery promise: each order shows an estimated completion time from the live queue, by video length and mode (typically 15–30 minutes for a 1-minute video; longer videos and Islamic human review show their own ETA, 4.14); the creator is notified by email or push when the video is ready, and delivery follows 9.4. Paid server renders use the delivery-speed tiers (G.8.3): Express (+20%, high priority, the system calculates and shows the expected delivery time), Normal (within 12 hours, usually much sooner) or Slow (within 24 hours, 10% off); Quick mode renders in the browser and has no delivery tier.
- Browser rendering is chunked: each scene (or 10-second segment) is rendered and uploaded to R2 as a resumable chunk immediately. Finished chunks wait in the browser's private storage (OPFS/IndexedDB) while offline and upload when the connection returns. If the PC is switched off, reopening the studio resumes from the next unfinished chunk because all project data is stored on the server. All files the job needs (characters, textures, the voice track, music and fonts) are downloaded before the first frame, so rendering continues offline; frames are drawn in a Web Worker with OffscreenCanvas, so rendering does not stop when the tab is in the background; chunks become MP4 files with Mediabunny (MPL-2.0) and upload straight to R2 with presigned multipart URLs (4.16).
- Server fallback: the studio sends a connection signal over the existing Centrifugo connection. If it stops for 30 minutes (admin setting, 4.9) — whether the internet dropped, the power failed or the creator closed the studio — the server renders the remaining chunks starting from the last one and working backwards, while a browser that comes back continues forwards; the first

finished copy of each chunk is kept and the other is discarded. Phones, tablets and devices that fail a 3-second test render when the order begins (estimated render time above the admin limit, by default 6× the video length) render on the server from the start. The server uses the same renderer (Revideo in headless Chromium through Puppeteer): 2D chunks on spare Core-1/Core-2 CPU, 3D avatar chunks with the GPU on Rec-1/Rec-2 when no recording is running (AI-2 Node B at night), or on CPU with a longer ETA if the GPU path fails the November benchmark; encoding uses Rec-node NVENC. Server-rendered chunks run at priority P10 and cost the creator the same price. A closing signal (Beacon API) records whether the creator closed the studio or lost the connection, for the KPI only.

- Keep-awake and warnings: before the render starts the studio says how long the PC will be busy and asks the creator to keep it on and plugged in; during the render a banner in Bangla or English shows progress, time left and "Please do not shut down the computer or close this tab"; the Screen Wake Lock API keeps the PC awake while the tab is visible and is requested again when the tab comes back into view; closing the tab shows the browser's own "Leave site?" prompt (browsers do not allow custom text in it). A website cannot detect or stop an operating-system shutdown, restart, sleep or power cut, so the saved chunks and the server fallback are the real protection.
- Voice in the browser: standard narration and character voices use the in-house VITS voices (six languages) in the browser through ONNX Runtime Web, the same voices NexaTute uses; the voice track is made at the preview stage, uploaded at once (≈1 MB per minute) and reused for the final render, so a server takeover keeps the same voice and lip-sync. Cloned voices (consent check, 4.15), Chatterbox and Indic Parler-TTS voices and auto-dubbing stay on the server GPU: their models are several gigabytes, and a cloned voice model never leaves the server.
- Cinematic look for 3D avatars: a Quick-mode style with film lighting, soft shadows, depth of field and camera moves, rendered by Three.js (WebGPU, with WebGL2 fallback) and billed at the Quick-mode rates (G.8.1). Realistic AI video in Cinematic mode (Wan 2.2, Wan2.2-S2V-14B lip-sync) needs data-centre GPUs and is always made on the server.
- Checks on browser-made chunks: they are treated like uploads. Before the Content Credentials (C2PA) are signed on the assembly node, the server runs the 4.15 frame checks, turns the audio into text with faster-whisper and compares it with the approved script, and renders one small frame per chunk itself to confirm (SSIM) that the chunk matches the approved storyboard; a chunk that fails is rendered again on the server.
- Quality rules: every chunk starts with a keyframe and uses the same codec, resolution, frame rate and bitrate, so chunks are joined without re-encoding; voice and music are kept as one separate audio track and muxed once at the end (no clicks at chunk boundaries); the browser renders at a moderate 8–10 Mbps to keep uploads short, and the server makes one final NVENC encode so browser-made and server-made parts look identical, then checks resolution, bitrate and quality automatically.
- Lip-sync: Quick-mode characters use the same browser viseme engine as the NexaTute 3D teacher, driven by mouth-shape timing from Rhubarb Lip Sync (MIT, audio-based, so it works for Bangla and Urdu); Enhanced mode mostly uses narration and needs no lip-sync; Cinematic mode lip-syncs realistic faces on the server GPU with Wan2.2-S2V-14B (licence check and quality gate first; it replaces MuseTalk/LatentSync, 8.15) for close-up dialogue shots only.

Step (1-minute 1080p video, Quick mode)	Where	Typical time (estimate)	Note
Draw and encode frames	Creator's PC (WebCodecs)	≈1–3 min (≈3–6 min on a weak PC)	Runs offline

Step (1-minute 1080p video, Quick mode)	Where	Typical time (estimate)	Note
Upload chunks (≈60–80 MB at 8–10 Mbps)	Creator's internet	≈1–2 min at 10 Mbps	Overlaps with rendering
Join chunks, add audio	Server CPU	Seconds	No re-encode
Final NVENC encode, validate, store in R2	Rec-node GPU	≈5–15 s (≈1.5–3 min for 20 minutes of video)	Same quality for every part
Total	—	≈3–6 min (Enhanced ≈4–8 min; Cinematic ≈12–15 min without a queue)	—

4.9 Admin-Controlled Package Settings

All Motion Studio package limits, output tiers, render modes and rates are managed from one Package Settings page in the admin panel, for the Free Trial, the Free Beta and every paid package, so the whole platform is controlled from one place. The G.8, G.11 and 11.11 values are the launch defaults. A change made in the admin panel takes effect without a code change or server restart; G.8 and F.2 are brought in line at the next document revision.

Setting	Free Trial	Free Beta	Paid packages
Daily limit — Quick / Enhanced	1 minute	1 minute	None by default (every minute billed at G.8 rates); optional cap per package
Daily limit — Cinematic	Not available (G.11)	1 minute, within the daily Cinematic pool (11.11)	None by default; optional cap per package
Maximum video length	15 seconds	15 seconds	Per package (G.8): 5 sec–45 sec (Freelancer Mini) up to 5 sec–20 min (Premium Agency)
Output tiers	480p Native and 720p Escalating	480p Native and 720p Escalating	Every tier marked with a rate in G.8, including 1080p Native in every paid package
Render modes	Quick and Enhanced	Quick, Enhanced and Cinematic	Quick, Enhanced and Cinematic; each mode on/off

Setting	Free Trial	Free Beta	Paid packages
Per-minute rates	Free	Free	G.8.1 base rate per tier is editable; Cinematic (+35%, G.8.2) and Islamic (−25%, G.9) rates recalculate automatically with G.8 rounding; any cell can be overridden
Delivery-speed tiers (G.8.3)	Not applicable (free)	Not applicable (free)	Express +20% (high priority, ETA shown, surcharge refunded if more than 20% late); Normal within 12 hours; Slow within 24 hours at 10% off; Quick mode exempt
Monthly fee	৳0	৳0	G.8.1 fee, editable
Validity	7 days	30 days from 20 December 2026; end date editable	Monthly billing cycle
Cloud fallback	Off	Off	On (8.4)
Admission	One trial per account (4.6)	Admission cap ≈2,000 combined CCU and waiting-room queue; cap editable (11.11)	No admission cap
Pre-production fee	Free (3 scripts a day)	Free (3 scripts a day)	৳20 per started 2 minutes; ৳15 for Islamic content (4.17)
Scene regeneration after edits	Up to 3 per scene per day	Up to 3 per scene per day	Up to 3 per scene per day
Upload size and length	Admin-set, below the paid limits	Admin-set, below the paid limits	Per package, for example 2 GB / 60 minutes (4.16)
Monthly spending cap	—	—	Set by the creator; alerts at 50%, 80% and 100% (4.16)
Quality gate and bitrates	Same thresholds	Same thresholds	Thresholds and minimum bitrates from the benchmark (4.13)
Content-safety policy	Same policy	Same policy	Categories, term lists and thresholds (4.15)

Setting	Free Trial	Free Beta	Paid packages
Browser rendering (4.8)	On; server takeover after 30 minutes without a connection; phones and devices that fail the test (default 6× the video length) render on the server	On; server takeover after 30 minutes without a connection; phones and devices that fail the test (default 6× the video length) render on the server	On; server takeover after 30 minutes without a connection; phones and devices that fail the test (default 6× the video length) render on the server
Browser voices (4.8)	On per language once its VITS voice passes the native-speaker test	On per language once its VITS voice passes the native-speaker test	On per language once its VITS voice passes the native-speaker test
Server takeovers per creator	No cap at launch; a monthly cap can be set	No cap at launch; a monthly cap can be set	No cap at launch; a monthly cap can be set
Series and promo videos (4.18)	Template intro/outro and frame thumbnails	Template intro/outro and frame thumbnails	Per F.2; lengths, autopilot and the daily AI-thumbnail limit editable

- 1080p Native controls: on/off for the whole platform, per package and per render mode. The base rate is ৳120 per minute (editable); the Cinematic rate (৳162) and the Islamic rates (৳90 and ৳122) are recalculated automatically by the G.8.2 and G.9 rules. Launch default: on in every paid package for Quick and Enhanced; Cinematic 1080p Native stays off until its GPU benchmark and quality gate pass (G.7), and 1080p Escalating is offered meanwhile.
- Load protection: when GPU cluster load passes an admin-set threshold (default 80%), 1080p Native is paused for new orders and 1080p Escalating is offered instead; it returns automatically when load falls. An optional daily or monthly 1080p Native minute cap can be set per user or per package.
- Changes apply to new jobs at once; jobs already confirmed, queued or rendering finish under the settings and price shown at confirmation. The usage ledger records the settings version used for every job.
- Daily limits reset at 00:00 Bangladesh time (UTC+6); unused allowance does not carry over.
- Every change is written to the audit log (admin, time, old value, new value, reason). Changes to paid fees, rates or limits need a second admin's approval (Admin and Superadmin roles, Section 19).
- When a paid rate rises or a paid limit falls, subscribers are told in the app and by email before it applies (default 7 days' notice); price cuts apply at once.
- Any package, tier or render mode can be paused with one click, for example during an incident, with a message shown to users.
- Implementation: settings are versioned rows in PostgreSQL (DB-1/DB-2 cluster), cached in Valkey and pushed to the API pods by Valkey pub/sub; on/off switches and gradual rollouts use Unleash feature flags (Apache-2.0, 18.1). No new software is needed.

4.10 Audio, Formats, Editing, Captions and AI Labels

These capabilities complete features already listed in 4.1–4.8 and Appendix F and are required before the Motion Studio Free Beta on 20 December 2026. They are included in every package fee; generated video is billed at the G.8 per-minute rates.

Capability	What the creator gets	Software (licence)	Where it runs
Music and sound effects	AI background music made for each video by mood, length and tempo; a sound-effect library; voice-over automatically lowers the music (auto-ducking); every export is loudness-normalised for YouTube, Facebook and TikTok	ACE-Step (Apache-2.0, licence to verify); RemonLogic CC0 sound-effect library; FFmpeg sidechaincompress and loudnorm. MusicGen and Stable Audio are excluded (non-commercial or restricted licences)	GPU cluster at P50 (paid) / P10 (free); popular music beds pre-made in the night batch window; library on R2 + CDN
Avatar-animation path	Talking 3D characters and presenters with lip-sync, kept as the separate avatar path named in 4.1 and 4.3	Three.js (MIT) characters from the G.3 character engine; Rhubarb Lip Sync (MIT); no new software	Creator browser (Quick mode); headless-Chromium server fallback (4.8)
Aspect ratios	16:9 (YouTube), 9:16 (Reels/Shorts/TikTok), 1:1 and 4:5 (Facebook/Instagram feed); the creator first picks a primary format, which is rendered natively; the other formats are made in one click by auto-reframe cropping plus light upscaling, keeping faces and the main subject in frame, and every extra format is billed at the normal per-minute rate	MediaPipe Face Detection (Apache-2.0) for auto-reframe; FFmpeg	Browser for Quick mode; spare Core-1/Core-2 cores for server renders (8.12)
Manual timeline editor	Drag-and-drop timeline to trim and split clips, move scenes, add text, stickers, music and transitions, alongside Natural-Language Editing; basic editor at launch, advanced tools added later	In-house editor on Revideo (MIT) and WebCodecs; no new library	Creator browser; project state saved on the server
AI content labels	Every finished video carries Content Credentials showing it was made with AI; direct publishing (4.6) ticks each platform's AI-disclosure option automatically for realistic AI content and cloned voices	c2pa-rs / c2patool (MIT/Apache-2.0); signing key in the secrets manager	Assembly node, after final validation and before R2 upload; the manifest hash is signed by a remote-signing service on the App cluster with the key held in OpenBao
Styled Bangla captions	Burned-in captions with word-by-word highlight, caption styles and correct Bangla conjuncts; also Urdu/Arabic right-to-left	libass (ISC) through FFmpeg; Noto Sans Bengali and other Noto fonts (OFL); faster-whisper word timestamps	Browser preview; burn-in during final assembly on the GPU node (AI-2 Node B CPU helpers)

4.11 Creator Growth Features

Charges: every feature below is included in the package monthly fee; projects that use AI pre-production carry the pre-production fee (4.17); videos made with them are billed at the normal G.8 per-minute rates for the chosen mode and tier; the two marketplaces take a 20% commission on each sale (G.8).

Feature	What it does	Software (licence)	Release
Template library	Ready templates (Islamic reminders, product ads, news, education, birthday and Eid greetings): pick one, change the text and media, render	Revideo scene JSON in PostgreSQL + R2; no new software	Launch
Product ad maker	Seller uploads product photos; background removed; ad video made with price, offer and order button for F-commerce	rembg (MIT) with BiRefNet (MIT)	Launch
Hook, title and thumbnail suggestions	First-3-second hooks, titles, hashtags and thumbnail options; results compared later in Content Performance Analytics	Qwen3 (existing); FLUX.1-schnell (existing)	Launch
Long video → Shorts	Creator's own long video (uploaded or imported) cut into several Reels/Shorts with captions, under the copyright controls in 4.12	faster-whisper + Qwen3 highlight selection (existing); PySceneDetect (BSD-3); MediaPipe reframe	Launch
Referral programme	Creators invite creators and earn account credit	Shared referral system with NexaTute (3.10)	Launch
Video translate	Creator's own uploaded video dubbed into other languages with subtitles	Existing auto-dubbing pipeline (4.2)	Q2 2027
AI presenter	The creator's own 3D avatar presents a script (news, education, courses)	Three.js + Rhubarb (existing)	Q2 2027
Podcast / audio → video	Audio file turned into a video with waveform, captions and images	FFmpeg showwaves + libass (existing)	Q2 2027
Team workspace	2–3 member teams with roles for Premium Starter and Premium Professional	Keycloak roles (existing)	Q2 2027
Mobile app	Order, preview, approve and publish from a phone; rendering stays on the server	Installable web app + Capacitor (MIT), shared with NexaTute	Q2 2027
Creator marketplace	Creators sell templates, characters and voices (with consent records); 20% commission	Existing payment gateway and payouts	Q4 2027
Client service marketplace	Businesses hire Motion Studio creators to make videos; 20% commission	Existing payment gateway and payouts	Q4 2027

4.12 Copyright Protection for Uploaded Videos

Making Reels/Shorts or translations from uploaded videos creates copyright risk for the creator and for RemonLogic. No system can remove that risk fully — Motion Studio has no Content ID-scale reference database — so these controls reduce it and are live at launch. Motion Studio never claims that a video is copyright-safe.

- Ownership declaration: before any upload is processed the creator confirms the video is their own or that they hold permission to use it; the declaration, time, account and file hash are stored with the project.
- Connected-channel import (default path): videos imported from the creator's own connected YouTube or Facebook channel prove ownership through the platform login; direct file upload is the secondary path.

- Music replacement: the background music of an uploaded video can be separated from the voice (Demucs, MIT; weights licence to verify) and replaced with music from the RemonLogic library or ACE-Step, removing the most common cause of copyright strikes.
- Audio fingerprint check at launch (moved forward from the earlier Q4 2027 plan): Chromaprint (LGPL-2.1, run as a separate process) checks uploads against the RemonLogic music library and a blocklist of known copyrighted sources (for example broadcast drama, film and sports clips reported to us); matches are blocked or flagged for review.
- Takedown and repeat-infringer policy: a copyright complaint removes the video from Motion Studio storage and publishing, the creator is notified, and repeated infringement suspends then closes the account; the rules are written into the terms of service.
- Pre-publish warning: when an uploaded or imported source is used, the creator sees a warning that third-party content may still be present and that platform claims remain possible before publishing.

4.13 Escalating-Tier Quality Assurance

Escalating tiers (720p Escalating, 1080p Escalating) must not lose sharpness, flicker or look blurred or waxy after upscaling. Real-ESRGAN is an image model that upscales each frame separately, so final escalated videos use a video upscaler and pass a measured quality gate before delivery. All software is free, open-source and commercially usable.

Step	What happens	Software (licence)	Where it runs
1. Video upscaling	Final escalated videos are upscaled with a video super-resolution model that uses neighbouring frames, so textures do not shimmer: the full model for 1080p Escalating and a lighter model (small BasicVSR++ variant, or Real-ESRGAN where it passes the quality gate) for 720p Escalating; FP16, torch.compile and CUDA graphs speed both up; Real-ESRGAN stays for previews and as the backup model	RealBasicVSR / BasicVSR++ (MMagic, Apache-2.0; weights to verify); SeedVR2 (Apache-2.0, to verify) benchmarked later for 1080p Escalating only	Unified GPU cluster (8.11) at P50 paid / P10 free
2. Exact sizing	Both steps are 1.5× (480p→720p, 720p→1080p): a 2× model is used and the result is scaled down to the exact size with Lanczos, avoiding invented detail	FFmpeg (Lanczos)	GPU node running the job
3. Text after upscaling	Captions, titles, logos and Brand Kit elements are added after upscaling at the final resolution, so text is always sharp	libass, FFmpeg overlay (existing)	Assembly on the GPU node
4. Face restoration	Light restoration of close-up faces only, at low strength so faces do not change; CodeFormer is excluded (non-commercial licence)	GFPGAN (Apache-2.0; face-detector weights licence to verify)	GPU cluster
5. Finishing	Light contrast-adaptive sharpening, debanding and fine grain so images do not look plastic	FFmpeg cas, deband and noise filters	Assembly
6. Measured quality gate	Every upscaled scene is checked (VMAF measured on one frame per second to save CPU): round-trip fidelity (output scaled back to source size and compared with VMAF/SSIM), sharpness (Laplacian variance) and flicker (frame-to-frame difference); thresholds are set from the launch benchmark and editable in the admin panel (4.9)	FFmpeg libvmaf (BSD-2-Clause-Patent); OpenCV (Apache-2.0)	CPU of the assembly node (8.12)

Step	What happens	Software (licence)	Where it runs
7. Failure handling	A failed scene is upscaled again with the alternative model at no charge; if it still fails, that scene is made at the next native tier and still charged at the escalating rate	Existing job pipeline	GPU cluster
8. Encoding floor	Minimum bitrates for escalated tiers: 720p 5–8 Mbps and 1080p 8–12 Mbps H.264 (final values from the benchmark, editable in 4.9)	FFmpeg / NVENC (existing)	GPU node
9. Side-by-side preview	Before ordering, the creator can compare native and escalated output side by side	Studio web app	Creator browser

- The video upscaler takes more GPU time than Real-ESRGAN; escalating-tier GPU-minutes per output minute are re-measured in the launch benchmark (13.2) and the 15.7 cost table and rates are checked against them.

4.14 Launch Readiness and Risk Controls

Twelve weak points found in the Motion Studio review and the controls approved for each:

Risk	Control	Where enforced	When
Short time to launch (Free Beta from 20 December 2026)	A minimum launch feature list is fixed; any feature not ready is shown as "coming soon" and switched on from the admin panel when it passes its tests	4.9, 17.4 release gates	Before launch
Free Beta Cinematic pool larger than capacity	The daily Free Beta Cinematic pool starts at 10–15% of the measured Cinematic render capacity and is raised only after the benchmark	11.11	Launch
Unproven render cost	Wan 2.2 GPU-minutes per output minute, including 1080p Native Cinematic, are benchmarked before paid packages start (date to confirm); rates are adjusted in the admin panel if needed	G.7, 15.7, 4.9	Before paid packages start
No realistic Cinematic lip-sync	Until Wan2.2-S2V-14B passes its licence check and quality gate (8.15), dialogue scenes use Quick or Enhanced mode; Cinematic uses narration and back or long shots	4.8, 18.2	Launch
Platform API approvals	YouTube and TikTok API audits are applied for immediately; until approved, creators download and upload themselves; LinkedIn Community Management API and Meta app review for Instagram are applied for at the same time (3.26)	4.6	Now
Deepfakes and real people	Motion Studio voice cloning uses the random-sentence spoken consent check; photo-real likeness of real public figures is blocked without proof of permission; separate political-content rules (4.15)	4.2, 10.3, 4.15	Launch
Bangla voice quality	Native-speaker listening test of Bangla voices; if Indic Parler-TTS fails, the in-house VITS Bangla voices are used	4.2	Before launch

Risk	Control	Where enforced	When
Long-video continuity	ETA shown by video length and mode (15–30 minutes applies to a 1-minute video); seams between ≈5-second clips smoothed by overlap blending; character LoRA for consistency	4.8	Launch
Monthly payment renewal	Renewal reminders, a 3-day grace period and a wallet top-up option where auto-debit is not available	G.8, G.7	Paid launch
Islamic human review delay	Human review completes within 3–7 days; while in review the user sees a separate "in review, up to 7 days" ETA; no GPU is reserved and no charge is taken before approval	4.5	Launch
Quality gap with closed models	Positioning is Bangla, Islamic verification, price, templates and product ads — not "most realistic video"	16, 26	Launch
Single GPU site (Dallas)	Cloud GPUs act as automatic failover when the Dallas GPU site is down, with a monthly spending cap	8.11, 15.2	Launch

4.15 Content Safety and Moderation

Motion Studio blocks adult content, religious hatred and communal incitement, anti-state violence, bad language and obscene gestures in text, audio, images and video. Every check uses free, open-source, self-hosted software; nothing is sent to a paid outside service.

Layer	What is checked	Software (licence)	Where it runs
Text	Prompts, AI and creator scripts, captions, titles and descriptions: sexual content, violence, hate, religious hatred, communal incitement, anti-state violence and bad language in Bangla, Banglish and English (including disguised spellings)	Qwen3Guard (existing) + Bangladesh policy layer (Qwen3 classifier and a maintained term list)	LLM pool, before generation
Images and video	Key images and sampled frames of every scene: nudity, sexual poses or dance, obscene or insulting gestures, gore and hate symbols; failed scenes are re-rendered safely or blocked and never reach the creator	Fast NSFW classifier (e.g. Falconsai ViT, Apache-2.0, licence to verify) on every sampled frame; Qwen3-VL (existing) policy check only when the classifier flags a frame or the scene contains people or characters; safety negative prompts always on in Wan and FLUX	GPU cluster (classifier also runs on CPU)
Audio	Voice-overs, cloned voices and the audio of uploaded videos are transcribed and checked with the text layer	faster-whisper (existing) + text layer	Speech pool
Uploads	Uploaded videos and photos pass the image/video and audio checks before any processing, alongside the copyright checks (4.12)	Same as above	Before processing

Layer	What is checked	Software (licence)	Where it runs
Child safety	Zero tolerance: any sexual content involving minors is blocked permanently, the account is closed and the case reported as the law requires; known illegal images are matched by hash	Meta Hasher-Matcher-Actioner with PDQ (images) and TMK+PDQF (video) hashing (BSD, free, self-hosted); hash lists obtained through free NCMEC-type access application	Spare Core-1/Core-2 cores (8.12)
Real people	Sexual or insulting content made from any real person's image is prohibited; photo-real public-figure likeness needs proof of permission	Policy layer + Qwen3-VL	Before generation and on frames

- Bad language: the AI never writes it; if a creator's own text contains it, rendering does not start, the words are highlighted and a clean alternative is suggested; uploaded audio with bad language is blocked.
- Obscene gestures: prompts asking for them are blocked; Quick-mode 3D characters use only an approved animation library, so such gestures cannot be made; generated frames of scenes that contain people or characters are checked separately for gestures.
- "Anti-state" is defined narrowly: incitement to violence or terrorism, calls for armed overthrow of the state and content clearly prohibited by law. Ordinary political criticism, news and satire are allowed; unclear cases go to human review, never automatic blocking. The definitions are finalised by a Bangladeshi lawyer under the current cyber-security law.
- Reporting and enforcement: every video has a Report button; enforcement steps are warning → temporary suspension → permanent ban, each decision is audit-logged and the creator can appeal.
- Admin control: policy categories, term lists and thresholds are admin-panel settings (4.9).
- Capacity: image/video safety checks add about 5–10% to safety GPU load, counted in the capacity model (8.6); no new servers are needed.

4.16 Creator Account, Spending and Rights

Account-level controls every Motion Studio creator gets. They use the existing platform (PostgreSQL usage ledger, R2, Grafana, next-intl); the only new software is Uppy, whose S3 multipart plugin uploads straight to R2.

Area	Rule	Software / placement	When
Monthly spending cap	The creator sets a monthly spending cap (for example ৳5,000); alerts at 50%, 80% and 100%; at the cap every new order needs fresh approval. Because packages have no included minutes, this prevents bill shock	PostgreSQL usage ledger; in-app, email and push alerts	Paid launch (date to confirm)
Ownership and commercial rights	Videos made on a paid package carry full commercial rights for the creator (ads, monetisation, client work); Free Trial and Free Beta videos are watermarked and for personal use only; marketplace templates and characters carry their own licence terms (4.11); all of this is written into the terms of service	Terms of service; licence record per asset	Launch

Area	Rule	Software / placement	When
Upload limits and resumable upload	Per-package maximum upload size and length (for example 2 GB / 60 minutes; admin-editable in 4.9); uploads resume from where they stopped after a connection drop; uploaded files count toward the G.3 storage allowance	Uppy (MIT) with its AWS S3 multipart plugin: the browser uploads parts straight to Cloudflare R2 with presigned URLs from the App, so files never pass through the load balancers or App nodes; an interrupted upload resumes from the last finished part	Launch
Interface language	Studio interface follows 29.2: Bangla and English at launch, Hindi and Urdu in Year 2, Arabic and French in Year 3; scripts, voices and subtitles work in all six languages from launch	next-intl (MIT), Noto fonts, right-to-left layout for Urdu and Arabic	Launch
Onboarding	Guided first-video tour, a sample video gallery and short Bangla tutorial videos	Studio web app; Unleash for onboarding tests	Launch
Project export and account deletion	The creator can download all projects, videos and scripts in one export; deleting the account removes all data, including cloned voices and characters, within the retention rules in 9.4 and G.3	App cluster worker; R2 lifecycle rules	Launch
VAT (Mushak) invoices	VAT invoices for business clients are handled with the deferred Bangladesh legal and tax items	—	After legal review

4.17 Script Upload and Pre-production Fee

Creators can bring their own script, and every project that uses AI pre-production carries a small pre-production fee. Motion Studio pricing is therefore: package fee + pre-production fee + per-minute video rate (G.8).

- Script Upload (every package, including the Free Trial): paste text or upload .txt, .docx, .pdf, .srt or .fountain files, with images; a sample script format can be downloaded in Bangla and English; Bijoy-encoded Bangla is converted to Unicode automatically (5.3).
- Two modes: "use as-is" — the AI does not change a single word, it only splits scenes, identifies speakers and builds the storyboard; "AI improve" — grammar, timing and scene descriptions are improved, every change is shown and applied only after approval.
- Automatic structure: narration and dialogue are separated and every speaker is mapped to a Character Bible character and voice; images in the script become reference or key images for their scenes (image-to-video).
- Before rendering: the expected video length, the pre-production fee and the estimated video charge are shown, the package's maximum video length is checked and the spending cap (4.16) is applied.
- Safety and rights: uploaded scripts and images pass the 4.15 checks; the creator confirms the script is their own or used with permission; on the Islamic plan, quotations in the script are verified (4.5).
- Document-to-Video stays separate: it writes a new script from a document and remains in Premium Professional and above; Script Upload uses the creator's own script in every package.

Pre-production fee	Rule
Amount	৳20 per started 2 minutes of planned video length (1 min = ৳20, 10 min = ৳100, 20 min = ৳200); Islamic content ৳15 per 2 minutes (−25%, G.9); same in every package, editable in the admin panel (4.9); Enterprise by contract
Covers	Script, storyboard, scene list, key images and the 480p preview
Charged for	AI-written scripts, uploaded scripts, Document-to-Video, the product ad maker and templates where the AI writes new text
Not charged for	Shorts from the creator's own video, promo cuts from the creator's own episodes (4.18), video translate or dubbing, podcast-to-video, remakes of finished videos (G.4) and templates filled only with the creator's own text
When charged	After the price is shown and confirmed, before generation starts; not charged if the safety check blocks the request at the start; refunded automatically on a technical failure (G.10); recorded in the usage ledger or wallet and counted toward the spending cap (4.16)
Final video	The fee is not credited back when the final video is ordered
Edits	The AI generates the pre-production once; the creator's own edits are free and unlimited; key images and previews of edited scenes regenerate free up to 3 times per scene per day (admin-editable); a completely new AI script needs a new fee
Free Trial / Free Beta	Free within the daily limits (3 scripts a day); script and storyboard download is off
Download	Paid creators can download the script and storyboard as PDF

4.18 Series and Promo Videos

Intro, outro, recap, next-episode preview, thumbnails, teaser, trailer and highlights are made by one system. Most of them are cut from episodes the creator has already rendered, so no new AI video is generated and they are cheap and fast; new AI shots are added only when the creator asks. Appendix F lists these features (F.3.3–F.3.6); this section is how they are made, where they run and how they are charged.

Item	How it is made	Length (admin-editable)	Charge
Intro	Brand Kit template — animated logo and title drawn with Revideo in the creator's browser, with an optional AI hero shot; made once, kept in the library and added to every episode automatically (can be switched off)	3–15 s	Billed once when made, at its mode and tier; free when reused
Outro / end screen	End screen with a subscribe or follow call to action, a "next episode" card and each platform's safe area for end-screen elements	5–20 s	As for the intro
Recap ("Previously on...")	Qwen3-30B-A3B picks 3–6 key moments of the previous episode or episodes from the transcript, storyboard and series bible; they are cut from the rendered episode and joined with a short voice-over	20–60 s	Quick-mode (G.8.1) rate of the chosen tier; no pre-production fee
Next-episode preview ("Coming up...")	Cut from the next episode when it is rendered, otherwise built from its approved key images with 2.5D camera moves; scenes the creator marks as spoilers are never used	10–20 s	Quick-mode (G.8.1) rate

Item	How it is made	Length (admin-editable)	Charge
Thumbnails	The best frames of the video (sharp, faces visible, close-ups) plus AI concept images from FLUX.1-schnell with the character LoRA; the title is overlaid separately in Brand Kit fonts, because AI images cannot write Bangla text reliably; exported at each platform's size and checked under 4.15	—	Included in the package fee; AI concept images within an admin-set daily limit
Teaser	A hook, 3–5 fast shots, the title and the release date; made in 9:16 first, other formats by auto-reframe (4.10)	15–30 s	Quick-mode (G.8.1) rate
Trailer	For a series or season: hook, story, a hint of the climax, title, release date and call to action, with trailer music (ACE-Step) and a voice-over	60–120 s	Quick-mode rate; any new AI shot at its own mode's rate
Highlights and best moments	The strongest moments of an episode or a season, chosen by the same selection	15–60 s	Quick-mode (G.8.1) rate

- Series autopilot: when an episode finishes rendering, drafts are made automatically at P10 — the recap for the next episode, a teaser and three thumbnail options — and the next-episode preview is drafted once the next episode is ready. The creator reviews and edits the drafts on the timeline; drafts are low-resolution previews, and nothing is charged until one is exported.
- Charges: as for Long video → Shorts (4.11), promos cut from the creator's own rendered episodes are billed at the Quick-mode (G.8.1) rate of the chosen tier with no pre-production fee (4.17) — for example a 30-second teaser at 720p Native on Premium Starter (৳45 a minute) costs ≈৳22.5. A new AI shot added to a promo is billed at its own mode's rate; extra aspect ratios follow 4.10.
- Packages (F.2): Freelancer Mini and Freelancer Starter — template intros and outros, thumbnails from video frames and a basic recap within the package's length limit, plus short promos through the Reels & Shorts Generator (F.3.4); Premium Starter and above — recap, next-episode preview, teaser, trailer, highlights, AI thumbnails and the series autopilot (F.2.3); Enterprise — by contract. Free Trial and Free Beta follow their own settings in 4.9.
- Where it runs: moment selection by Qwen3-30B-A3B with xgrammar JSON on the GPU pool, using faster-whisper word timestamps and the storyboard's scene list (PySceneDetect for uploaded videos); frame-accurate cuts and joins with FFmpeg and NVENC on the assembly node (AI-2 Node B, any GPU node as fallback); thumbnail frames scored on the same node — sharpness with OpenCV and faces with MediaPipe on its CPU, OpenCLIP through Triton — so no video crosses sites; title cards, end cards and thumbnail text drawn with Revideo templates in the creator's browser (headless Chromium fallback, 4.8); music from ACE-Step and the CC0 sound library with auto-ducking and EBU R128 loudness (4.10); voices under the 4.8 rules; captions with libass; every export signed with Content Credentials (4.10). No new software is needed.
- Safety and rights: cuts from the creator's own approved episodes reuse footage that already passed 4.15; new text, voice, music and thumbnails are checked again; promos made from uploaded or imported videos follow the copyright controls in 4.12.
- Production impact: ≈1–2% of Motion Studio GPU time, mainly for moment selection and AI thumbnails; cutting runs on CPU and NVENC. Admin settings (4.9): the length range of each item, autopilot on or off per package, the daily AI-thumbnail limit and whether intros and outros are attached automatically.

4.19 Motion Studio Landing Page, CMS and Drop-off Analysis

Motion Studio uses the shared Landing Page AI and AI CMS (3.25) with its own site, content, admins and data. Release timing is set in the release plan after the wireframes.

Item	Motion Studio rule
Landing page	A headline such as “from text to video, in Bangla, in minutes”; main button “Start Free Trial”; a 15–20-second showreel made with Motion Studio (cinematic, 3D, cartoon, Islamic); a “write an idea” box that makes a storyboard and thumbnail in the visitor’s browser with Quick mode (pre-made examples on weak phones)
Section order	Showreel → how it works (idea → script → video) → render modes and quality → live package prices from G.8 with the Express, Normal and Slow delivery tiers (G.8.3) → creator gallery with permission → rights and safety (voice-clone consent, C2PA, copyright check) → FAQ → final button
Honesty	No fake views or earnings promises; marketplace income shown only through real examples; the render ETA comes from the live queue (4.8)
CMS specifics	Showreel and creator gallery with consent; package and render rates from G.8; template and marketplace pages; Islamic content rules and discount (4.5, G.9)
Drop-off funnel	Visit → sign-up → first project → first preview → first full render → download or publish → Free Trial end → package purchase → second-month renewal
Where and who	Script-step abandonment, long render queues, render failures and browser-to-server takeovers, quality re-renders, payment failures, minutes running out, YouTube, Facebook or TikTok publish errors; by creator type (freelancer, agency, Islamic creator, business), device capability, country and traffic source
Why and fixes	A cancel survey, support themes and video quality ratings; a weekly AI analyst whose fixes become change requests for the landing page or the render settings (4.9)
Season calendar	Ramadan and Eid, Pohela Boishakh, 21 February, 16 December, the wedding season and the year-end recap
Marketing	Run by the AI Marketing Hub (3.26) with Motion Studio’s own accounts; outreach to agencies, businesses and creators at their public business addresses

05. Separate Software Inventories

5.1 NexaTute stack (canonical grouping)

NexaTute software is listed once, with licence and placement, in the Unified Software Register (18.1), rows marked N or B. Layer summary: browser apps (Next.js, Three.js for the 3D teacher and classrooms (3.15), Excalidraw and Yjs, browser Whisper and VITS voices, WebLLM); App-cluster services (FastAPI, Keycloak, LangGraph and the Learning OS components); AI on the GPU cluster (vLLM with Qwen3, Qwen3-VL, Qwen3Guard, Mistral Small 3.2 at night, speech models on AI-2 Node A); live classes (LiveKit, coturn, GStreamer recording on Rec-1/Rec-2); data (PostgreSQL, pgvector, Valkey); uploads through the Document Intake Service (5.3). Added 18 October 2026: the batch-class service, load-aware router and Nexa Co-Teacher on the App cluster (3.16, 3.17), face verification in the browser (MediaPipe, SFace) and payments through aamarPay (11.7). Engagement models (scikit-learn, LightGBM, Vowpal Wabbit) on App-tier CPU (3.19) and URL blocklists (URLhaus, StevenBlack hosts) for links shared by users (3.23). The shared Landing Page AI and AI CMS with sharp, vanilla-cookieconsent, web-vitals and Lighthouse CI (3.25); Revideo in the learner’s browser for Wrapped and mistake videos (3.20). The shared AI Marketing Hub (3.26).

5.2 Remon Motion Studio stack (canonical grouping)

Remon Motion Studio software is listed once, with licence and placement, in the Unified Software Register (18.1), rows marked M or B. Layer summary: browser studio (Revideo, Three.js, WebCodecs with Mediabunny, Rhubarb Lip Sync, in-house VITS voices, timeline editor; server fallback with Puppeteer and headless Chromium); App-cluster services (FastAPI, Celery, publishing, C2PA signing, usage ledger); video and image AI on the GPU cluster (Wan 2.2, FLUX.1-schnell, RIFE, RealBasicVSR/BasicVSR++, GFPGAN, OpenCLIP/DINOv2); creator tools (ACE-Step, BiRefNet, Demucs on Rec-1/Rec-2 when idle; PySceneDetect and Chromaprint on spare Core-1 cores; quality-gate measurements on the Dallas assembly node); safety (Qwen3Guard, NSFW classifier, Qwen3-VL, hash matching); assembly and promo cuts (4.18) with FFmpeg/GStreamer and NVENC on the GPU node; delivery through R2 and the CDN. The shared Landing Page AI and AI CMS (3.25, 4.19). The shared AI Marketing Hub (3.26).

5.3 Shared platform services — logical sharing only

- Shared physical GPU capacity is mediated by a scheduler/resource pool; do not share user-level authorization or product data.
- Document Intake Service (shared by both products): one service reads every upload — txt, docx, pdf, pptx, srt, fountain, csv, xlsx and images (JPG, PNG, WebP, HEIC) — converts Bijoy-encoded Bangla to Unicode, extracts embedded images, reads scanned pages and images (Tesseract, PP-FormulaNet, Qwen3-VL), runs the 4.15 safety checks and records the rights declaration; it feeds NexaTute uploads (3.14) and Motion Studio Script Upload (4.17). Files arrive straight in R2 from the browser (Uppy S3 multipart with presigned URLs, 4.16) and the service reads them from there, so large uploads never load the load balancers or App nodes. It runs on spare Core-2 cores and the App cluster; Qwen3-VL image reading uses the GPU pool (AI-2 Node A) within the daily image limits. Links (3.23): a link step checks the URL against the URLhaus and StevenBlack blocklists, opens the page in a gVisor sandbox without login, cookies or scripts, extracts the text with Docling (for YouTube only the title, description and chapters through the YouTube Data API) and runs the same safety checks; the result is kept only as a short summary and topic mapping by URL, so a link shared by many learners is processed once.
- PostgreSQL may use common physical nodes, but separate schemas/databases, credentials, migration boundaries and backup/restore tests must be explicit.
- Redis instances/ports/namespaces remain isolated: Redis1:6379 for NexaTute session/transient state; Redis2:6380 for Motion Studio queue/broker.
- Common ingress, observability, secrets management and object-storage infrastructure require scoped credentials, tenant/product labels and audit trails.
- Landing Page AI and AI CMS (shared by both products, 3.25): one engine with separate sites, content, admins, analytics and data per product; the same scoped credentials and audit trail as the other shared services.
- RemonLogic Growth Platform (shared by all three products, 3.26): one AI Marketing Hub, CRM, journey engine and target-client database, with separate brands, accounts, mailboxes, data, pipelines, owners and approvers per product; one CRM record per organisation and one unsubscribe for all products.
- Do not assume a software name in this plan means it is installed, licensed or production configured; the deployment inventory must track version, owner, license, configuration and validation status.
- Shared platform software (both products): Kubernetes with Kueue and the NVIDIA GPU Operator, Cilium (WireGuard between sites), HAProxy + keepalived, KEDA, Prometheus,

Grafana, Loki, Langfuse with ClickHouse, OpenBao, Harbor, and Cloudflare (WAF, CDN, R2, Workers). Licences and versions for every item in 5.1–5.3 are listed once in 18.1 and 18.2.

- Edge software for the NexaTute Box (2.5): k3s single-node Kubernetes, rclone for nightly lesson sync from R2, llama.cpp with a small Qwen3 model on CPU, and Nginx serving cached lessons over the school's local Wi-Fi.

06. Master Hybrid Software Architecture and Data Flow

6.1 Layer model

Layer	Canonical responsibility	Key boundaries
Client / experience	NexaTute browser UI, Three.js avatar/lab/3D classroom (3.15), Web Audio playback, Excalidraw whiteboard, small in-browser LLM on capable devices (WebLLM), offline saved lessons and lesson prefetch, flashcards, Quiz Battle, coding sandbox and mind maps (3.10); PlayBoom; browser speech (Whisper via transformers.js, in-house VITS voices); practice labs (v86, Blockly, JupyterLite, Phaser), 3D anatomy and the Penpot design editor; Motion Studio project editor, key-image approval and output review (AV1 or H.264 playback); batch-class client (polls, confusion meter, question wall, AI voice made from the AI's text) and face verification (3.16, 3.17); the public websites of both products with the plan finder, cookie consent manager and pre-sale chat (3.25, 4.19); test autosave in IndexedDB and the coding-test editor (3.27)	Client device renders supported interactive visuals; degrade gracefully on weaker devices
Ingress / routing	LB-1/LB-2 active-active HAProxy cluster with keepalived from launch (LB-3 at scale); product route to the NexaTute or Motion Studio API pool	Authentication, request limits, TLS, rate limits and request tracing

Layer	Canonical responsibility	Key boundaries
Application	App-1 + App-2 (+ App-3 from December 2026) Kubernetes app cluster running NexaTute and Motion Studio FastAPI replicas in separate namespaces (8.13); cache layers, request coalescing and rule-based answer checking before any LLM call; category knowledge-base retrieval with cited sources (3.13); skill-course file checks and practice-lab sandboxes in network-less gVisor containers; national live model tests and merit lists; Motion Studio publishing to YouTube, Facebook and TikTok; batch-class service, load-aware router and Nexa Co-Teacher (3.16, 3.17); plan billing and renewals through aamarPay (11.7); engagement models and the link step (3.19, 3.23); the shared Landing Page AI and AI CMS with page versions, approvals and site rebuilds, and the in-house B2B CRM (3.25); the shared AI Marketing Hub with approvals and outreach (3.26)	Separate API/runtime contexts, entitlements, data scopes and job lifecycle
Learning OS platform	Keycloak identity for institutions, LangGraph agent layer, curriculum graph (Apache AGE), Ralph LRS, LTI/SCORM, APISIX public API, Centrifugo messaging and verifiable credentials (3.11); the institution roster and results API with OAuth scopes and webhooks, and the paper builder (3.27)	Each institution’s data kept apart; outside apps only through the API gateway with scoped permissions
Administration and accounting	Platform admin panel and Institute Admin (38): plans and modules table and module switches (Unleash), operations pages (Langfuse prompt versions, Portkey caps, OpenCost, DCGM), accounts, support and legal, content quality and money, growth; company accounting (double-entry books, payroll, loans, shares) in its own PostgreSQL schema with Casbin permissions, pgaudit, an append-only action log and a nightly encrypted rclone export to Google Drive, OneDrive or Dropbox; admin notifications (notification service with pywebpush and FCM) and internal staff chat (Centrifugo, PostgreSQL, LiveKit calls, Qwen3Guard check on send, ClamAV, encrypted daily summaries, 38.15); the Avatar & Voice Studio (38.16)	Admin and finance roles only; 2-step login and a fresh OTP before money, role or export actions; the maker cannot approve own work; never reachable from NexaLogicDev sandboxes or learner apps; no new servers; push text never shows secrets; staff chat is monitored and not end-to-end encrypted; calls are not recorded

Layer	Canonical responsibility	Key boundaries
MindForge, Study Hall Q&A and Growth Platform	MindForge (37): puzzles, practice scoring and offline daily puzzles in the browser; generation with OR-Tools checks, IRT calibration (girth) and forecasts on CPU workers; verbal items on the shared GPU pool at P10. Study Hall Q&A (31.3): jsQR and Tesseract.js in the browser; server checks with OpenCV, Tesseract, the NSFW classifier, PDQ hashing, MediaPipe and Qwen3Guard; verified answers in Valkey with a PostgreSQL/pgvector copy. Growth Platform (3.26): Valkey streams, in-house CRM and notification service, ClickHouse reports, Scrapy and Qwen3 in the night batch; Marketing Hub Channels (OAuth, tokens in OpenBao), Import, Export and Target clients (3.26)	Measurement items, safety checks and reports decided on the server (trust rule, 8.18); no new GPU or servers
AI / compute	Unified 6-node GPU cluster at Database Mart, Dallas, joined by the NexaLogicDev Heavy pool (3 × 4× RTX A6000) from June 2027 and Fast pool (4 × RTX PRO 6000) from September 2027,: AI-1 Node A/B (LLM prefill/decode, safety), AI-2 Node A/B (speech, video) and Rec-1/Rec-2 (recording first, then post-processing work) under shared hybrid scheduling (8.11); late-night batch window for the AI content pipeline (Qwen3 + Mistral Small 3.2) and model training	Priority-aware GPU allocation; bounded concurrency; product-specific workload labels
NexaLogicDev build layer	Browser-first IDE (Monaco, ESLint/Prettier in Web Workers, Sandpack, WebContainer API, php-wasm, GrapesJS canvas, in-browser Whisper, offline PWA with Yjs and IndexedDB, 32.14); builders and Auto-Architect on the App cluster; Heavy pool (MiniMax-M2.7 on 3 × 4× RTX A6000) from June 2027 and Fast pool (Qwen3-Coder-Next on 4 × RTX PRO 6000) from September 2027; API fallback through Portkey with spend caps (MiniMax M3 for the Fast Lane; Claude Sonnet 5.5 plan review and rescue for the Heavy Lane); sandbox pool (gVisor/Kata) for user code; free sites on Cloudflare Workers + R2; domains through the Name.com API; deploy to the user's Vercel/Netlify (32)	User code runs only in the sandbox pool, with no route to NexaTute minors' data or payment ledgers; DNS changes only through the approved workflow after customer approval (34)

Layer	Canonical responsibility	Key boundaries
Ecosystem resource sharing	RemonLogic Resource Broker, Kueue cohorts with lending and borrowing limits, Inference Gateway (Envoy Gateway), vLLM priority scheduling, KEDA, DCGM Exporter and OpenCost across the launch GPU pool, the Heavy and Fast pools and the CPU workers (8.17)	Owner first; borrowed work only on idle capacity and returned within one scheduling cycle; sandbox, data and edge servers never lent; minors' raw media only on the launch pool
Browser execution layer	Whisper and VITS voices, Three.js, WebCodecs + Mediabunny, Web Audio, Tesseract.js, ONNX Runtime Web, WebLLM; Monaco, Sandpack, WebContainer, php-wasm, GrapesJS, isomorphic-git; Workbox, IndexedDB/OPFS and Yjs; Device Check and Data Saver; browser OpenTelemetry (8.18); hls.js, three-vrm avatars and the glTF-Validator pre-check	Browser first; the server keeps large AI models, secrets, trusted results (money, ranked grades, safety, certificates) and the fallback for light or weak devices
Real-time media	Core-1 + Core-2 LiveKit SFU cluster with embedded TURN; coturn fallback relay; also 1:1 guardian–counsellor calls; LiveKit SIP and LiveKit Agents for the phone tutor (IVR)	LiveKit capacity must be tested against tracks, subscribed layers, bitrate, topology and TURN share
Recording	Rec-1 and Rec-2: lightweight track recording with the whiteboard as an event log; the download MP4 is composited and encoded in the user's browser (9.5)	Local scratch and output compilation; do not put heavy scratch writes on Core-1
Data	Patroni PostgreSQL cluster DB-1/DB-2 with live pgBackRest backup to Cloudflare R2; Valkey cluster Redis1/Redis2 with Sentinel; PgBouncer as a sidecar on every App node (OVHcloud private network); per-category pgvector knowledge bases, quality-gate results and NexaTute Verified badge status (3.13)	Durable data in PostgreSQL; cache/queues not financial or job-state authority
Storage / delivery	Cloudflare R2 for validated final files and permitted archives; lesson cache, image cache and AV1 viewing copies served through the Cloudflare CDN; browser speech models, 3D anatomy models, lesson audio and the public verified-categories page; static website pages and CMS media in WebP/AVIF (3.25); Nexa Notes HLS videos with WebVTT subtitles and avatar GLB/VRM files (3.25, 38.16)	Scoped access, signed URLs, checksums and retry-safe uploads

Layer	Canonical responsibility	Key boundaries
Operations	Prometheus/Grafana/logging/secrets manager and admin capacity dashboard; Motion Studio Package Settings (4.9) and copyright takedown queue (4.12); Unleash experiments and Chatwoot support inbox; the drop-off analysis page with Grafana funnels and the weekly AI analyst (3.25)	Metrics, alerts, capacity and cost estimates are measured, not assumed

6.2 NexaTute request flow

- Browser submits a learning action or joins a live classroom.
- For a lesson in a 3D classroom, the browser loads the room design (a few KB) from the NexaTute API and its 3D assets from the CDN, and draws the room itself (3.15).
- Uploaded homework, notes, guardian documents and teacher scripts or question banks go through the Document Intake Service (5.3) and the safety checks before the AI uses them (3.14).
- LB-1/LB-2 route the request to the NexaTute API pods on the App cluster; identity, role, plan entitlement, rate limit and schema are checked.
- Before any LLM call the request passes the cheaper paths in order: the small in-browser LLM for light tasks on capable devices, the lesson cache, the verified-answer cache (exact and meaning match), request coalescing for identical questions, and rule-based checking for MCQ, numeric and formula answers (8.14, 8.15).
- Load-aware routing (3.16): before a live Nexa Teacher class starts, the router reads the GPU-pool load and the P100 queue. When the pool is free the class starts; when it is busy the learner is offered the recorded class, mock tests or summaries first, with an ETA and a kept queue place in Valkey; when it is overloaded or down, no new live class starts, the Nexa Co-Teacher continues in text and the lost minutes are credited back.
- Nexa Co-Teacher (3.16): text requests pass the same cache layers; retrieval covers only the learner's own recordings and summaries plus the category knowledge base, filtered by learner ID with PostgreSQL row-level security; the daily topic count is kept in Valkey; on overload the Portkey gateway uses Mistral Large 3, then GPT-6 Luna, with personal data removed (8.9).
- Shared learner record (3.16): before each turn the Nexa Teacher and the Nexa Co-Teacher read a short context card for the learner from Valkey, and afterwards write their notes, handover notes and summaries to PostgreSQL (row-level security), with every activity as an xAPI statement in Ralph LRS; LangGraph gives both roles the same memory.
- Engagement models (3.19): the learner-state, drop-out and reminder-timing models (scikit-learn, LightGBM) and the next-activity bandit (Vowpal Wabbit) run on App-tier CPU on the event data in ClickHouse and are retrained every night; they act only where the learner's or guardian's consent switch is on (19.4).
- Free tier and Nexa Co-Teacher Pass (3.22): free-tier requests are served from the CDN, the browser and the question bank; Nexa Co-Teacher lite answers only from the verified cache; the 5 free help minutes and Nexa Co-Teacher Pass requests join the local queue behind plan holders and are never sent to the cloud API.
- Batch classes (3.17): the batch-class service on the App cluster runs one Nexa Teacher session per batch; the AI's text and whiteboard steps go to every learner over Centrifugo and are voiced and lip-synced in each browser with the same in-house VITS voice version; weak devices receive one server CPU voice stream per batch through the batch's LiveKit room; raised-hand questions and adult pair rooms use LiveKit audio; polls, the confusion meter, the question wall

and team scores run on Centrifugo with results in PostgreSQL; the recording is written to R2 as an event log with the raised-hand audio tracks.

- Recording (11.3): every AI-tutor class is recorded by the learner's browser as an event log (AI text, whiteboard events and timing) with only the learner's own Opus audio, uploaded straight to R2; on replay the browser re-creates the AI voice with the pinned voice version.
- Face verification (3.16): the browser runs the liveness check (MediaPipe Face Landmarker) and the face match (SFace through ONNX Runtime Web); the server receives only the result and the encrypted template; a failed check falls back to an OTP.
- Payments (11.7): plan purchases, renewals and top-ups go through aamarPay; each callback is verified on the server before the plan is activated or the wallet credited (20.4).
- Purchases by minors (11.9): the purchase request goes to the guardian's phone through the notification service, and aamarPay is called only after the guardian approves, within the guardian's monthly limit; gift-card codes, affiliate commission and marketplace payouts are recorded in the usage ledger (11.14).
- Content safety (10.2): learners' spoken words in live classes are filtered in real time from the speech-to-text transcript, and Nexa Teacher and Nexa Co-Teacher answers are checked by Qwen3Guard before they are shown.
- Photo questions: the browser compresses the image; printed text and formulas are read by Tesseract and PP-FormulaNet on spare CPU, and handwriting by Qwen3-VL on the GPU pool. The extracted question then follows the same cache and LLM path, and numeric or formula answers are checked with SymPy. Written tests (3.27) take a queued path instead: pages go to R2 and are checked by Nexa Guard, Qwen3-VL reads them, the learner confirms unsure parts, the local Qwen3 marks against the approved scheme and a notification brings the result.
- Class summaries (3.7): after an AI-tutor session Qwen3-30B-A3B (thinking off) summarises the session text; after a live or group class faster-whisper (Whisper large-v3-turbo) transcribes each speaker's audio track — no speaker separation is needed because tracks are already separate — the whiteboard event log is added, Qwen3-30B-A3B writes the summary in a fixed format and WeasyPrint makes the PDF. Summary jobs start at P10 and are raised to P100 if they wait more than 10 minutes, so the 15-minute target holds; audio handling when recording is off follows 9.5.
- Learning Insights (3.4): every answer, reasoning step, hint, mistake type, lab action, confidence rating, teacher adjustment and time-on-task event is stored in TimescaleDB; written reasoning steps are graded against the question's rubric and mistakes tagged by Qwen3-30B-A3B (thinking off), with SymPy checking maths steps; the topic mastery score is then recomputed straight away on App CPU by combining pyBKT and IRT (girth) with these signals; weekly trends and the improvement ranking are computed in the late-night batch window; Qwen3-30B-A3B writes the plain-language explanation; ECharts draws the learner, guardian and teacher dashboards.
- Agent routing (3.11): LangGraph sends each AI request to the right agent — Tutor, Examiner, Planner, Counsellor, Content Creator or Teacher Assistant — which uses the same cache layers, Learning Insights, curriculum graph and, with consent, the learner's long-term memory in pgvector, before calling the LLM pool.
- Category knowledge bases (3.13): each course category has its own pgvector collection in PostgreSQL; the Tutor retrieves the best passages for the learner's category with BGE-M3 and the reranker and cites them in the answer. The AI content pipeline (collection, parsing, generation, independent solving and checks) runs as Celery jobs in the late-night batch window; Mistral Small 3.2 is woken with vLLM sleep mode only for that window, and only items that pass every check are published. Gate results and badge status are stored in PostgreSQL per course and teaching language, with the model and knowledge-base version; a nightly job re-checks live wrong-answer rates and updates the badges, and the public verified-categories page is served from the CDN.

- Medical and engineering tools (3.13): circuit diagrams (schemdraw) and circuit simulations (ngspice) run as short jobs in an isolated, network-less sandbox on spare Core-2 cores (the App cluster when Core-2 is busy), like the skill-course file checker on the App cluster; JupyterLite notebooks and Z-Anatomy 3D models run in the learner's browser and are served from the CDN; virtual-patient cases use the AI agent layer and the shared LLM pool.
- Practice labs and language courses (3.6, 11.2): v86 Linux, Blockly robots and Phaser games run in the learner's browser; Docker/Kubernetes and cybersecurity sandboxes run as gVisor-isolated containers with no network access on spare Core-1/Core-2 cores and App workers (8.12), at most 50 at once at launch (raised with App capacity) and closed after 30 minutes, with a queue when full. Foreign-language lesson audio is made once in the batch window and served from the CDN; live conversation practice uses Chatterbox voices from the speech pool.
- National live model tests (3.10): the paper is published to the CDN in encrypted form before the start time and the key is released over Centrifugo at the start, so thousands of learners can open it at once without loading the servers; answers are saved on the device and submitted in batches, MCQs are marked by rules, written answers by answer-script checking, and the merit list is calculated in PostgreSQL within minutes.
- Phone tutor (3.10): a Bangladesh telco or IPTSP SIP trunk connects to LiveKit SIP on the LiveKit cluster (Core-1/Core-2); LiveKit Agents runs the call — speech recognition and voice on the server speech pool, answers from the shared LLM pool — and keypad answers arrive as DTMF tones.
- Motion Studio dubbing and publishing (4.2, 4.6): dubbing jobs use the speech and LLM pools at P50 for paid and P10 for free work; scheduled posts and daily view statistics run as Celery beat jobs on the App cluster, uploading from R2 with encrypted OAuth tokens.
- Own avatar and voice (3.10): avatars are VRM files stored in R2 and rendered in the browser with the existing lip-sync; English, Hindi, Arabic and French voices are cloned by Chatterbox from a short sample on AI-2 Node A, while Bangla and Urdu voices are trained as small per-person VITS models from 30–60 minutes of recording in the late-night batch window and can then run in the teacher's learners' browsers; lessons in a teacher's voice are generated once and cached on the CDN. Consent records, voice samples and voice models are encrypted and deleted on request. The Avatar & Voice Studio (38.16) adds the library and approvals: uploads are checked with glTF-Validator in the browser and again on Core-2, approved avatars get lighter copies from glTF-Transform on the CPU worker pool, and NexaTute clone jobs run off-peak at P10 within a daily GPU-minute cap.
- Cross-product and live events: explainer videos are requested from the Motion Studio Quick pipeline through its API with a service account, rendered at P10 in the batch window and stored in the lesson cache, so product isolation (6.4) holds; the weekly live quiz sends questions and answers over Centrifugo and the host's video as an HLS stream through the Cloudflare CDN, so thousands of players do not load the LiveKit SFU.
- Programmes (3.20, 3.21): the AI Parent–Teacher Meeting uses the Nexa Teacher voice pipeline off-peak; Nexa Teacher Live streams one session a week through LiveKit Egress; the Script Checker runs Qwen3-VL as an overnight batch in the quietest GPU hours; public Q&A pages are static pages on the CDN; Wrapped and personal mistake videos are rendered in the learner's browser; the Live Study Hall uses Centrifugo presence only.
- Website and CMS (3.25): the AI drafts a change as page blocks in PostgreSQL; after the admin approves, Next.js rebuilds the page and the Cloudflare cache is purged; prices are read live from the price-table API; pre-sale chats arrive through Chatwoot and use the same caches and the local LLM; funnel events go to TimescaleDB and Grafana, and real-user page speed comes from web-vitals.
- Marketing (3.26): articles become CMS drafts; videos are requested from the Motion Studio Quick pipeline through its API at P10 in the batch window; approved posts go out through Motion Studio's direct publishing and the LinkedIn API; outreach email is sent and read over

SMTP/IMAP from the outreach mailbox; local Qwen3 sorts replies, and approvals wait in the admin panel; post and lead events go to TimescaleDB.
- Only the remaining requests go to the local LLM with X-Priority: 100 for the protected real-time class/speech/lab workload; AI-1 Node A processes the prompt and AI-1 Node B generates the answer (prefill/decode disaggregation), and the answer streams sentence by sentence to the browser voice and avatar.
- Durable assessment and learning records are written to PostgreSQL/TimescaleDB; the NexaTute Valkey instance is used only for short-lived cache/session/presence/transient state.
- Voice, avatar, whiteboard, lab, the 3D classroom and AI-tutor recording run in the browser (3.2, 3.15), so AI-tutor sessions exchange only text/audio data with the server. Human-led live and group classes use LiveKit with the video profiles in 3.2.
- Monitoring records latency, errors, queue depth, media quality and resource use.

6.3 Motion Studio request flow

- The creator creates a project and submits a brief or uploads a script; the Motion Studio API pods on the App cluster validate identity, content plan, rights/safety policy and billing entitlements.
- Create a durable job record and idempotency key first, so every later charge, retry and refund is tied to one job.
- An uploaded script or source video goes through the Document Intake Service (5.3); for a source video the ownership declaration, connected-channel check and Chromaprint fingerprint check (4.12) run before any processing, and blocked or flagged files never reach the GPU queue.
- Every prompt, script and upload passes the content-safety layers (4.15) before anything is generated.
- The API reads the live package settings (4.9) — allowed render modes and output tiers (including 1080p Native and its load pause), daily limit, maximum video length and per-minute rate — and checks the monthly spending cap (4.16); it shows the pre-production fee and the estimated video charge and asks for confirmation; the settings version is stored with the job.
- After confirmation, charge the pre-production fee (4.17) and reserve the estimated video credits in the PostgreSQL usage ledger before paid work begins.
- Generate script and storyboard with grammar-constrained JSON output (xgrammar) and run script-level safety/plan verification.
- Generate and lock the voice-over first, then set each scene's length from the audio timing (audio-first); queue asset and rendering tasks with X-Priority: 50 for paid work and 10 for free/trial batch work.
- For Cinematic hero shots, show a key image for creator approval; Wan 2.2 A14B animates the approved image (with the creator's character LoRA for series), hero shots of the same size render together in batches at 16 fps and RIFE interpolates them to 24/30 fps; other scenes use TI2V-5B, which renders natively at 24 fps (8.15). Previews use the compact upscaler and an AV1 viewing copy.
- The GPU node executing the job (AI-2 Node B preferred) keeps all heavy processing and intermediate files on its own local SSD; nothing is written over the network to Core-1.
- Every scene passes the automatic quality and frame-safety checks; only a failed scene is rendered again.
- For escalated tiers, upscale with the video upscaler, size with Lanczos, restore close-up faces, apply finishing filters and pass the measured quality gate on the same node (4.13).
- Mix music and sound effects with auto-ducking and loudness normalisation, add titles, logos and the chosen caption style at the final resolution, and make each requested aspect ratio (4.10).
- Compile and validate the final MP4, then embed the Content Credentials manifest on the assembly node; the manifest hash is signed by the remote-signing service on the App cluster, whose key never leaves OpenBao (4.10).

- Upload the signed final file to R2 and verify integrity and durable acknowledgement; direct publishing sets each platform's AI-disclosure option automatically.
- Only then purge the exact task-owned local files; keep them if upload or validation fails.
- Settle the reserved credits against the accepted output duration, update the durable job state and provide secure download/replay access.
- When an episode is finished, a P10 job drafts its promo set (4.18) — the next episode's recap, a teaser and three thumbnails; drafts are free, and an exported promo is cut from the rendered episode on the assembly node, signed and uploaded like any other video and billed at the Quick-mode rate.
- Website and CMS (4.19): the same flow as in 6.2 on Motion Studio's own site, with package and render rates read live from G.8.

6.4 Logical separation with shared compute

The resource pool can allocate physical compute dynamically, but a request must retain product identity, customer/tenant scope, priority, billing context and audit trace from ingress through completion. GPU sharing is not database sharing. Each product must have its own queue policy and failure behavior; a backlog in one product must not exhaust the other product's critical resources.

07. Hardware Inventory — 16-Server Launch Baseline and Ecosystem Additions

All 16 servers are rented as dedicated/bare-metal servers rather than purchased — all in the USA: the six GPU servers from Database Mart (Dallas, Texas), six servers from OVHcloud US (Vint Hill, Virginia) and four from RedSwitches (Washington DC) (8.16) — (rental budget in 15.2). This is the launch fleet; the 16,000 CCU scale-out fleet is defined in 8.5. The following inventory reproduces the final v24.0-aligned proposal values. These are architecture inputs, not independently verified procurement quotes or measured capacity results. Hardware changes require explicit architecture approval and updated testing.

#	Node	CPU / RAM	Storage	Network	GPU	Role
1	AI-1 Node A	32 cores / 84 GB RAM	400 GB SSD	1 Gbps unmetered	NVIDIA RTX PRO 6000, 96 GB	Unified GPU cluster node — NexaTute LLM prompt processing (prefill, P100) and safety model; can run Motion Studio LLM/video — Database Mart, Dallas (8.16)
2	AI-1 Node B	32 cores / 84 GB RAM	400 GB SSD	1 Gbps unmetered	NVIDIA RTX PRO 6000, 96 GB	Unified GPU cluster node — NexaTute LLM answer generation (decode) and hot standby; Motion Studio script LLM; Wan 2.2 A14B video in the daytime — Database Mart, Dallas (8.16)
3	AI-2 Node A	24 cores / 56 GB RAM	320 GB SSD	500 Mbps unmetered	NVIDIA RTX PRO 5000, 48 GB	Unified GPU cluster node — prefers speech (server STT/TTS fallback, Motion Studio voice-over), Qwen3-VL photo questions, class-summary transcription and pronunciation scoring; can run LLM overflow and upscaling — Database Mart, Dallas (8.16)

#	Node	CPU / RAM	Storage	Network	GPU	Role
4	AI-2 Node B	24 cores / 56 GB RAM	400 GB HP SSD	500 Mbps unmetered	NVIDIA RTX PRO 5000, 48 GB	Unified GPU cluster node — prefers video (batched Wan 2.2 TI2V-5B, RIFE), scene quality checks, upscaling and final MP4 compile on local SSD — Database Mart, Dallas (8.16)
5	Rec-1 Node A	24 cores / 56 GB RAM	320 GB SSD	500 Mbps unmetered	NVIDIA RTX PRO 4000, 24 GB	GPU cluster node — lightweight track recording unit 1 (top priority); download MP4s are made in the browser (9.5); when idle, preview upscaling, RIFE, AV1 viewing copies, scene quality checks and LLM/speech overflow — Database Mart, Dallas (8.16)
6	Rec-2 Node B	24 cores / 56 GB RAM	320 GB SSD	500 Mbps unmetered	NVIDIA RTX PRO 4000, 24 GB	GPU cluster node — lightweight track recording unit 2 (top priority); download MP4s are made in the browser (9.5); when idle, preview upscaling, RIFE, AV1 viewing copies, scene quality checks and LLM/speech overflow — Database Mart, Dallas (8.16)
7	Core-1	48 cores / 256 GB RAM	2 × 4 TB NVMe RAID 1	10 Gbps unmetered	None	LiveKit SFU node 1 (cluster with Core-2) with embedded TURN — the only 10 Gbps server; its 4 TB NVMe also hosts a low-priority local mirror (Harbor container registry + AI model cache) — RedSwitches, Washington DC (8.16)
8	Core-2	32 cores / 128 GB RAM	2 × 960 GB NVMe RAID 1	1 Gbps unmetered	None	LiveKit SFU node 2 (cluster with Core-1) with embedded TURN; coturn fallback; ClickHouse for Langfuse traces and Ralph LRS in a capped pod (8.12) — RedSwitches, Washington DC (8.16)
9	App-1	16 cores / 64 GB RAM	2 × 960 GB NVMe	1 Gbps unmetered	None	App cluster node 1 — NexaTute and Motion Studio API replicas (separate namespaces) with a PgBouncer sidecar; third member of the Kubernetes control plane/etcd, OpenBao Raft and Valkey Sentinel — OVHcloud US, Vint Hill, Virginia (8.16)
10	App-2	16 cores / 64 GB RAM	2 × 960 GB NVMe	1 Gbps unmetered	None	App cluster node 2 — NexaTute and Motion Studio API replicas (separate namespaces) with a PgBouncer sidecar — OVHcloud US, Vint Hill, Virginia (8.16)
11	DB-1 Primary	16 cores / 64 GB RAM	2 × 960 GB NVMe	1 Gbps unmetered	None	PostgreSQL Patroni cluster node 1 (primary) — live pgBackRest backup to Cloudflare R2 — OVHcloud US, Vint Hill, Virginia (8.16)

#	Node	CPU / RAM	Storage	Network	GPU	Role
12	DB-2 Replica	16 cores / 64 GB RAM	2 × 960 GB NVMe	1 Gbps unmetered	None	PostgreSQL Patroni cluster node 2 (synchronous standby, reporting reads) — takes over automatically — OVHcloud US, Vint Hill, Virginia (8.16)
13	Redis1	16 cores / 64 GB RAM	At least 960 GB NVMe (Prometheus 15-day local retention; logs in R2)	1 Gbps unmetered	None	Valkey cluster node 1 — NexaTute primary (6379), Motion Studio replica (6380), Sentinel; Kubernetes control-plane/etcd member; OpenBao Raft member; monitoring (Prometheus replica, Grafana, Alertmanager) — OVHcloud US, Vint Hill, Virginia (8.16)
14	Redis2	16 cores / 64 GB RAM	At least 960 GB NVMe (Prometheus 15-day local retention; logs in R2)	1 Gbps unmetered	None	Valkey cluster node 2 — Motion Studio primary (6380), NexaTute replica (6379), Sentinel; Kubernetes control-plane/etcd member; OpenBao Raft member; monitoring (Prometheus replica, Loki with R2 log storage) — OVHcloud US, Vint Hill, Virginia (8.16)
15	LB-1	8 cores / 16 GB RAM	As per final procurement record	500 Mbps unmetered	None	Edge cluster node 1 — HAProxy + keepalived — RedSwitches, Washington DC (8.16)
16	LB-2	8 cores / 16 GB RAM	As per final procurement record	500 Mbps unmetered	None	Edge cluster node 2 — HAProxy + keepalived — RedSwitches, Washington DC (8.16)

Inventory note: the source hardware table does not specify storage values for LB-1 and LB-2; Redis1 and Redis2 need at least 960 GB NVMe each for etcd, Prometheus (15-day local retention) and Loki buffers, with long-term logs in R2. “As per final procurement record” is intentionally used rather than inventing a capacity. Confirm the exact storage SKU, RAID, filesystem and mount configuration before procurement/deployment.

Approved addition: App-3 — OVHcloud US, Vint Hill, Virginia, ADVANCE-2 like App-1/App-2 (AMD EPYC 4344P, 8 cores/16 threads, 64 GB DDR5, 2 × 960 GB NVMe RAID 1, 1 Gbps unmetered), ≈US$198 a month plus a one-time installation of ≈US$198 (15.6). It is ordered on 1 December 2026 and put into service by 15 December 2026 after its load test, before the NexaTute launch, because the App/API tier is the first capacity limit when both products run together (8.6). App-3 is an App-pool worker only: NexaTute and Motion Studio API replicas, platform services and a PgBouncer sidecar, with pool sizes re-divided so the total PostgreSQL connections stay within the same cap. The Kubernetes control plane/etcd, OpenBao and Valkey Sentinel keep three members on Redis1, Redis2 and App-1, because a fourth member would weaken quorum. App-3 holds no data (no backup needed; it is rebuilt from the configuration in Forgejo), is added to the LB-1/LB-2 HAProxy backends with health checks, and runs the same agents as every node (Cilium with WireGuard, SPIRE, OpenTelemetry Collector, Falco). From December 2026 the fleet is 17 servers plus the recovery replica; the 16-row inventory above stays the v24.0 launch baseline.

Approved additions — NexaLogicDev GPU pools (Funding B, Section 32): a Heavy pool of three servers, each with 4× NVIDIA RTX A6000 48 GB (two NVLink pairs), 2× Intel Xeon E5-2699 v4 (44 cores), 512 GB RAM, 240 GB SSD + 4 TB NVMe + 16 TB SATA and 1 Gbps unmetered, USA, US$1,599 per server per month on every term, from June 2027 (MiniMax-M2.7, one copy per server); and a Fast pool of four RTX PRO 6000 96 GB GPUs at Database Mart, Dallas (US$749 each

on the 3-month term, US$648 on the 24-month term) from September 2027 (Qwen3-Coder-Next), with more GPUs added one at a time at >70% load; plus OVHcloud ADVANCE-2 sandbox servers for user code (one from March 2027, two from September 2027, SBX-3 at its trigger). They join the same Kubernetes cluster (0.6). From September 2027 the ecosystem runs 27 servers — the 16-server launch baseline, App-3, the recovery replica, three Heavy servers, four Fast GPUs and two sandbox servers — and 28 once SBX-3 is added. All pools share idle capacity by the placement order in 8.17.

7.1 Fleet totals and interpretation

- GPU-equipped nodes in the listed fleet: 6 (AI-1 A/B, AI-2 A/B, Rec-1/Rec-2).
- GPU allocations listed: 2 × RTX PRO 6000 96 GB; 2 × RTX PRO 5000 48 GB; 2 × RTX PRO 4000 24 GB. These model names and exact variants must be verified against supplier inventory and installed drivers.
- Outside the server fleet: NexaTute Box devices (2.5) are low-cost mini PCs bought per school under government or NGO projects; they need no GPU and are managed remotely.
- The fleet has dedicated application, database, cache/queue, media relay, load-balancer and recording roles. A role assignment does not imply high availability unless redundant paths and failover are configured and tested.
- The plan model of 18 October 2026 (Batch, Combo and Personal, 11.1), the Nexa Co-Teacher and batch classes need no new servers at launch: a batch class needs one LLM session for up to 50 learners, the Nexa Co-Teacher teaches in text (spoken questions and read-aloud answers run in the browser), and class recordings are small event logs (8.6).
- The learning-proof, website and sales-tool additions of 18–19 October 2026 (3.24, 3.25, 4.19) need no new servers either: the websites are static pages on the Cloudflare CDN, site rebuilds and checks run in low-priority pods on the App nodes (8.12), the CMS uses the local LLM pool off-peak (8.6) and media is stored in R2 (9.4). The AI Marketing Hub (3.26) needs none either: its videos render at P10 in the batch window.
- The additions of 24–25 October 2026 need no new servers either: admin push and staff chat run on the App cluster (notification service, Centrifugo, PostgreSQL) with staff calls on LiveKit (Core-1/Core-2); the Avatar & Voice Studio renders avatars in the browser, checks uploads on Core-2 and makes NexaTute voice clones on AI-2 Node A off-peak within a daily GPU-minute cap; Nexa Notes videos are transcoded to HLS on the CPU worker pool (App-1–App-3) at P10 in the night window — never on Core-1 (9.3) — subtitled on AI-2 Node A off-peak and stored in R2 behind the CDN (≈20–40 GB a year at two videos a week, estimate); the Marketing Hub channels, import, export and target-client map run on the App cluster (8.6, 8.17).
- The tests and written-answer checking decided later on 25 October 2026 (3.27) need no new servers either: written checks are a queued job on the launch GPU pool (AI-2 Node A first) within a daily cap of 1,500 GPU-minutes (default, editable); graded coding tests run in the existing practice-lab gVisor sandboxes; autosave and the code editor run in the browser (8.6, 8.17).
- The decisions of 26–29 October 2026 need no new servers either. LiveKit already runs on two nodes (Core-1 and Core-2) with Valkey, which meets the “at least two LiveKit nodes, no single point of failure” rule (9.5); recordings stay browser-made replay files in R2 (3.29); the storage allowances and the voice and audio cache use R2 (G.3, 8.14); the Exam & Admission Hub, Team Quest, Quiz Battle and the Live Game Show run on the App cluster, Centrifugo and LiveKit Egress; the sign-language avatar, AR Magic Book, Voice mode and Be the Teacher videos run in the browser; the teacher’s AI assistant, comic stories and Time-Travel answers use the launch GPU pool off-peak within caps; the study bots answer from the verified cache; the load test runs on the staging copy; the status page and synthetic checks run outside our servers. Enrolment is unlimited, so servers are added before classes start when the capacity plan says so (8.5).

Admission limits, the capacity guard and Nexa Co-Admin (38.19, 38.20) also run on existing capacity, Co-Admin on the local model at low priority.

- One Kubernetes cluster for all servers (16 at launch, App-3 from December 2026, the NexaLogicDev Heavy, Fast and sandbox pools from March 2027): the control plane runs on Redis1, Redis2 and App-1 on the OVHcloud private network in Virginia, and the six Dallas GPU servers join as worker nodes. Zone labels (Virginia/Dallas) and taints keep App pods in Virginia, GPU work in Dallas, and keep only GPU-serving pods on the Dallas GPU servers. Patroni, Valkey, LiveKit and HAProxy keep their own clusters on their pinned servers (8.13).
- Operations placement: Forgejo (code hosting and CI) runs on the App cluster with builds on Core-1 at night; an OpenTelemetry Collector runs on every node and sends metrics and logs to Prometheus and Loki on Redis1/Redis2; Alertmanager sends alerts; Harbor on Core-1 mirrors images and model weights.
- The final architecture's stated total is 16 servers: 14 core plus 2 dedicated recording nodes. Use the node table above as the canonical 16-row inventory; resolve any procurement discrepancies before deployment. App-3, the approved addition above, joins the App pool in December 2026; the full ecosystem register of 28 servers is in 7.2.

7.2 Ecosystem server register — all 28 servers

Rows 1–16 are the launch baseline above. The ecosystem adds the servers below; every one enters the Security & Control Centre inventory (D01) before it takes traffic. Pool owners, lending rules and the placement order for every server are in 8.17.

#	Node	Specification	Location	From	Funding	Role
17	App-3	OVHcloud ADVANCE-2: EPYC 4344P 8C/16T, 64 GB, 2 × 960 GB NVMe RAID 1, 1 Gbps	Vint Hill, Virginia	Dec 2026	A	App/API pool worker (no data)
18	DR replica	OVHcloud ADVANCE-2	Hillsboro, Oregon	Nov 2026	A	Disaster-recovery replica, second region
19–21	Dev-H1, Dev-H2, Dev-H3	4× RTX A6000 48 GB (2 NVLink pairs), 2× Xeon E5-2699 v4 (44 cores), 512 GB RAM, 240 GB SSD + 4 TB NVMe + 16 TB SATA, 1 Gbps; US$1,599 a month each	USA (Dallas zone where offered), one rack / private LAN	Jun 2027	B	NexaLogicDev Heavy Lane — MiniMax-M2.7, one copy per server (GLM-5.2 TP=4 × PP=3 as fallback); lent one server at a time in the 1–7 am window (8.17)

#	Node	Specification	Location	From	Fundin g	Role
22–25	Fast-1–Fast-4	Database Mart dedicated-GPU VPS: RTX PRO 6000 96 GB, 32 vCPU, 84 GB RAM, 400 GB SSD; US$749 a month each on the 3-month term, US$648 on the 24-month term after benchmarks	Dallas (with the launch GPU pool)	Sep 2027	B	NexaLogicDev Fast Lane — Qwen3-Coder-Next, one copy per GPU; more GPUs one at a time at >70% load; idle GPUs lent (8.17)
26–27	SBX-1, SBX-2	OVHcloud ADVANCE-2	Virginia	Mar 2027 / Sep 2027	B	Sandbox pool: AI loops, tests, scans, database backends (gVisor/Kata); previews run in the browser (32.14)
28	SBX-3	OVHcloud ADVANCE-2 (≈US$198 a month + ≈US$198 installation); ordered at ≥250 NexaLogicDev users in the IDE at once or 60% sandbox load for a week, in service at ≥300 or 70%	Virginia	At trigger (≈Mar–Jun 2028)	B	Third user-code sandbox server (8.17)

Totals at full build-out: 28 servers — 13 GPU servers (6 launch: 2 × RTX PRO 6000, 2 × RTX PRO 5000, 2 × RTX PRO 4000; 4 Fast-lane RTX PRO 6000; 3 Heavy-lane servers with 12 × RTX A6000) and 15 CPU servers. Count by date: 17 in November 2026 (16 + recovery replica) → 18 (December 2026, App-3) → 19 (March 2027, SBX-1) → 22 (June 2027, Heavy servers) → 27 (September 2027, Fast-1–Fast-4 and SBX-2) → 28 (SBX-3 at its trigger).

08. Hybrid Resource Pool, Scheduling and Failover

8.1 Hybrid Shared Resource Pool

- Dynamically route eligible inference and processing workloads according to live load, queue depth, GPU memory availability, task class and priority.
- Use admission control and bounded concurrency; do not promise that every request can run immediately.
- Keep product-level isolation at API, queue, cache, database access and billing layers while sharing eligible physical compute.
- Use backpressure to prevent queue growth from causing memory exhaustion or OOM.
- The capacity dashboard should report current local load, overflow/fallback load, GPU utilization, queue age, circuit-breaker state and estimated cost—but cost estimates must use measured/provider-specific rates.

8.2 X-Priority policy

Priority	Workload	Header	Behavior
1 — highest	NexaTute live-class speech / real-time classroom / supported Virtual Lab experience	X-Priority: 100	Protected low-latency lane; should not be silently starved by background video
2 — medium	Remon Motion Studio paid remake/production	X-Priority: 50	Standard paid lane with bounded concurrency; persist job/progress state when capped
3 — lowest	Motion Studio free/trial batch generation	X-Priority: 10	Background queue; throttle or process sequentially during peak demand

8.3 Three-tier resilience model

- Tier 1 — local execution: route to the local inference/compute pool when healthy and capacity is available.
- Tier 2 — cloud API fallback: if the local GPU cluster is unavailable, saturated or beyond defined latency/error thresholds, eligible AI requests go through the Portkey gateway to the fixed cloud fallback chain in 8.9. Each provider has its own timeout, retry limit and circuit breaker; a slow, failing or over-budget provider is skipped.
- Tier 3 — circuit breaker / controlled hold: if local and fallback paths fail or limits are exceeded, stop admitting the affected new workload, preserve durable job/session state, return a clear user message and allow controlled recovery.
- Use timeouts, bounded retries, exponential backoff, provider quotas, secret management and health checks. Avoid unbounded retry loops and retry storms.
- The example gateway snippets in source documents are design illustrations, not production-ready code; hard-coded global locks, placeholder API keys, unverified provider identifiers and incomplete payload mapping must be replaced and tested.

8.4 Circuit-breaker UX

NexaTute: explain that traffic pressure temporarily limits new lab work, preserve the current session where possible, and tell the learner when to retry; when live Nexa Teacher classes are paused by load, offer the learner's recorded class, mock tests and summaries with an ETA, or continue with the Nexa Co-Teacher in text, and credit the lost minutes (3.16). Motion Studio: explain that new rendering is temporarily paused, confirm that script/storyboard drafts are saved only if the durable write succeeded, and show queue status. Do not tell users that a session has been transferred to cloud backup unless the transfer and state continuity are actually confirmed.

All-capacity-down message: when the local GPU cluster and both cloud fallback providers are unavailable or over their spend cap (8.9 L4), every new request in both products shows "The server is very busy right now. Please try again after some time." The message is shown in the user's interface language, retries are automatic with back-off, and existing sessions, drafts and queued jobs are kept.

Edge fallback: the same message is also stored as a Cloudflare custom error page and Worker response, in every interface language, so users still see it when the origin is unreachable (both load balancers, app nodes or the whole data centre down) instead of a browser error; a Cloudflare-hosted status page shows current incidents.

8.5 Scale-out Hybrid Architecture for 16,000 CCU

The 16 servers in Section 07 are the launch fleet; App-3, the first App-pool step, is taken early in December 2026, before the NexaTute launch (Section 07). Because every server is rented, the architecture scales horizontally: rented nodes are added to product-specific pools when measured load reaches the triggers below, and hourly cloud GPUs absorb short peaks. The 16,000 CCU target (8,000 NexaTute + 8,000 Motion Studio) is served by the scale-out fleet plus burst capacity and remains a design target until the load tests in H.3 pass. At about 8% concurrency, 16,000 CCU corresponds to roughly 200,000 active users (the Year 3 target), so scale-out follows real demand and is funded from revenue at that stage.

Unlimited admission (decided 27 October 2026): there are no seat caps on enrolment, Crash Packs or crash programs. The capacity plan in platform operations turns enrolments and class start dates into a forecast of peak 1:1, batch and recording load and the GPU servers needed, and alerts with enough lead time to add servers before classes start; servers are always monitored, and the overload mix and API fallback stay as the safety net. Only the Free Beta keeps its ≈2,000 admission cap (11.11). Unlimited admission stays the default; from 29 October 2026 the admin may also set optional limits per class or course with a waiting list, and a capacity guard sends new sign-ups to the waiting list above 75% of the load-test-proven capacity without touching enrolled learners or live classes (38.19).

Ordering rule: because rented GPU servers can take days to weeks to deliver, a new server is ordered when its pool reaches 60% sustained load and is brought into service when the pool reaches the 80% trigger in the table below; hourly cloud GPUs cover the gap if demand arrives first.

Layer / pool	Launch fleet (16 rented; App-3 from December 2026)	16,000 CCU fleet (rented)	Scale-out trigger
Edge and load balancing	Cloudflare + LB-1/LB-2 active-active cluster (8.13)	LB-1 + LB-2 active-active; LB-3 only if ingress passes the trigger (uploads go straight to R2, 4.16)	Combined ingress >70% of 1 Gbps
Database connection pooling	PgBouncer sidecar on App-1/App-2, and on App-3 from December 2026 (OVHcloud private network)	PgBouncer sidecar on every App node, including added ones	With each App node
NexaTute API pool	App-1 + App-2 app cluster, plus App-3 from December 2026 (≥2 replicas) + overflow on spare Core-1/Core-2 cores (8.12)	3 App nodes (Kubernetes autoscaling)	CPU >60% or API p95 >300 ms for 10 min
Motion Studio API pool	App-1 + App-2 app cluster, plus App-3 from December 2026 (≥2 replicas) + overflow on spare Core-1/Core-2 cores (8.12)	2 App nodes (Kubernetes autoscaling)	CPU >60% or API p95 >300 ms for 10 min
Shared LLM pool (RTX PRO 6000, 96 GB)	Unified 4-GPU cluster (8.11)	12 GPU nodes	Priority-100 queue wait p95 >1.5 s or GPU >80% for 10 min

Layer / pool	Launch fleet (16 rented; App-3 from December 2026)	16,000 CCU fleet (rented)	Scale-out trigger
Speech pool (RTX PRO 5000, 48 GB)	Unified 4-GPU cluster, preferred on AI-2 Node A; the in-house VITS server voices run on CPU (8.12)	≈3 GPU nodes (speech-to-text fallback, GPU voices and batched class-summary transcription) plus 1 OVHcloud CPU server for the in-house VITS server voices (8.6)	Server-fallback sessions >80% of the pool limit, class-summary queue >10 minutes, or voice CPU >70% for 10 minutes
Video generation and assembly	Unified GPU cluster; assembly on the executing node's local SSD	AI-2 Node B + 2–3 RTX PRO 6000 nodes with LightX2V and FastWan (4 without them, 8.15) + hourly cloud GPU burst	Paid render queue ETA >15 min; from October 2027 also ≥60% video-pool busy time at the daily peak for a week → order a new GPU server, in service at 80% (8.17)
LiveKit SFU (human-led classes)	Core-1 (10 Gbps) + Core-2 (1 Gbps) cluster = 11 Gbps	Core-1 + Core-2; ≈4.2 Gbps at target fits Core-1 alone (≈42%). If more is needed, rent another unit of the Core-1 10 Gbps server type if the supplier can provide it, otherwise add 1 Gbps LiveKit nodes (≈600 Mbps usable each)	Core-1 egress >6 Gbps or Core-2 >600 Mbps
TURN relay	LiveKit embedded TURN on Core-1/Core-2; coturn fallback	Same; dedicated TURN node only if relayed traffic exceeds 30%	Relay share >30% or node CPU >60%
Recording / egress	Rec-1, Rec-2 in the GPU cluster (recording first); overflow to spare Core-1/Core-2 CPU for track-only recording, then to idle GPU nodes	3 Rec nodes with RTX PRO 4000 (lightweight track recording; download MP4s made in the browser, 9.5) — the benchmark sets the final number	>25 concurrent track recordings per node, until the benchmark sets the real limit
PostgreSQL	DB-1 + DB-2 Patroni cluster (synchronous standby) + live pgBackRest backup to R2	DB-1 + DB-2 + DB-3 (async, reporting)	DB-3 at 50% of target load
Valkey (Redis-compatible)	Redis1 + Redis2 cross-replicated cluster with Sentinel	Same nodes; add memory or nodes when needed	Memory >70%
NexaLogicDev Heavy pool	Dev-H1–Dev-H3 from June 2027 (Funding B), one MiniMax-M2.7 copy each	More 4× A6000 servers, one M2.7 copy each	Heavy queue wait p95 above the admin limit or >70% busy for a week, or Sonnet case-1 use rising (32.1)
NexaLogicDev Fast pool	Fast-1–Fast-4 (RTX PRO 6000) from September 2027 (Funding B)	More RTX PRO 6000 GPUs one at a time	Order at 60% load for a week; in service at 70% (32.1)

Layer / pool	Launch fleet (16 rented; App-3 from December 2026)	16,000 CCU fleet (rented)	Scale-out trigger
NexaLogicDev sandbox pool	SBX-1 from March 2027, SBX-2 from September 2027 (Funding B)	SBX-1–SBX-3, then more ADVANCE-2 servers as needed; never borrows other servers	Order at ≥250 users in the IDE at once or 60% load for a week; in service at ≥300 or 70% (7.2)
Operations	Monitoring on Redis1/Redis2; one Kubernetes cluster for all servers with its control plane on Redis1, Redis2 and App-1	Dedicated Ops node	At paid pilots
Total	16 servers; 17 from December 2026 with App-3	≈37–39 servers + hourly GPU burst (≈42 before the 17 October 2026 changes; confirmed by the benchmark)	Each step load-tested before use

8.6 Capacity Model (Planning Estimates)

All figures are planning estimates for sizing; they must be replaced by benchmark results (Gate 2) before any capacity claim is published.

Workload	Planning assumption	Demand at target	Pool sizing
NexaTute AI Tutor LLM	8,000 CCU; one turn every 45 s; ≈250 output tokens; every turn on Qwen3-30B-A3B with thinking off by default (3B active parameters, about as fast per token as Qwen3-8B, 8.15)	≈44,000 tokens/s	≈7.5 RTX PRO 6000 at ≈6,000 tokens/s blended
Motion Studio script / storyboard LLM	8,000 CCU; one 1,500-token generation per user every 10 min	≈20,000 tokens/s	≈4 RTX PRO 6000 at ≈5,000 tokens/s
Safety moderation (Qwen3Guard)	Every prompt and every generated answer or script is checked by the safety model; Motion Studio key images and sampled frames add ≈5–10% (4.15)	≈15% extra LLM GPU load	≈+1.7 GPUs at target
Photo questions and answer checking (3.10)	About 5% of AI Tutor turns include an image; a Qwen3-VL turn costs about 3× a text turn; printed text and formulas go to CPU OCR first	≈+10% LLM-pool load	Within the shared LLM pool and burst; a daily image limit per learner if needed
Pronunciation scoring	Short phoneme-model passes on recorded answers in English speaking practice	Small	Speech pool on AI-2 Node A
Class summaries (3.7)	Every live and group class summarised; ≈240 rooms at target, about 60 minutes of speech each; faster-whisper batched transcription; ≈1,500 summary tokens per class	≈240 audio-hours per hour at peak; LLM load negligible	≈+0.5–1 RTX PRO 5000 with batched transcription (8.15; ≈1.5–2 without it) at the 16,000 CCU target; a small share of AI-2 Node A at launch

Workload	Planning assumption	Demand at target	Pool sizing
Learning Insights (3.4)	Score update after every answer (≈1 ms CPU); step grading and mistake tagging of written answers by Qwen3-30B-A3B (thinking off); weekly trends and rankings in the batch window	Negligible CPU; ≈+3–5% LLM-pool load for step grading	App pool CPU and the shared LLM pool; no new servers
Audio lessons and guardian voice reports (3.10)	One audio summary per lesson, made once; one 1-minute Bangla voice report per guardian per week	TTS batch work only	Speech pool in the late-night batch window; files served from the CDN
Learning OS platform services (3.11)	Institution workspaces, messaging, xAPI records, public API traffic and agent routing	≈+10–15% App-tier CPU and memory	App pool; App-3 joins in December 2026 and App-4 follows the 8.5 trigger (CPU >60%)
Weekly national live quiz (3.10)	Up to ≈20,000 players at once; questions and answers over Centrifugo; host video as HLS through the CDN	≈20,000 WebSocket connections; one HLS stream	Centrifugo on the App cluster; one Rec node makes the HLS stream; time set by load test so it does not clash with the NexaTute evening peak
Explainer videos (3.10)	One 1–2 minute Quick-mode video per hard topic, made once	Small batch render work	Motion Studio pipeline at P10 in the batch window; served from the CDN
Series and promo videos (4.18)	Drafts after every finished episode (recap, teaser, three thumbnails); exported promos cut from rendered footage	≈1–2% of Motion Studio GPU time (moment selection and AI thumbnails); cuts on CPU and NVENC	Existing GPU pool and assembly node; no new servers
3D classrooms (3.15)	Room design (a few KB) loaded once per session; 3D assets from the CDN, cached in the browser	No server rendering or GPU work	App pool and Cloudflare CDN; no new servers
NexaTute-Bangla model training (2.5)	Periodic LoRA fine-tunes on curated data (2.5); occasional larger runs	Several GPU-hours per LoRA run	AI-1 nodes in the late-night batch window; larger runs on rented cloud GPU hours, not on the launch fleet
In-house voices and speech fine-tunes (4.2)	Six VITS voices and Bangla/Urdu Whisper-small fine-tunes before launch; retrained when quality data grows	A few GPU-days in total	AI-1/AI-2 nodes in the late-night batch window; model files served from the CDN
Skill courses (3.12)	AI-tutor turns as in other courses; one Office file check (a few CPU-seconds) per practice task; Penpot sessions for the design course; human-taught Quran and mentor sessions over LiveKit	Small next to the AI Tutor load; ≈+5% App-tier CPU at target	Shared LLM pool, App cluster workers and LiveKit; no new servers
Category knowledge bases and quality routing (3.13)	One retrieval per AI-tutor turn; new material embedded nightly; premium hard questions on the cloud only while a gate is not met	Retrieval ≈10–30 ms per turn; knowledge bases a few GB in PostgreSQL	App cluster, PostgreSQL and the BGE-M3 helper on the GPU pool; cloud spend inside the cloud budget (15.2); no new servers
AI content pipeline (3.13)	First build ≈100,000 questions and notes across categories, then a few thousand new or replaced items a week; ≈3,000 tokens per item for generation, independent solving and checks	≈300 million tokens for the first build ≈ 15 GPU-hours on one RTX PRO 6000; weekly runs well under one hour	AI-1/AI-2 nodes in the late-night batch window; first build from 5 November 2026, in the weeks before the NexaTute launch; no new servers

Workload	Planning assumption	Demand at target	Pool sizing
Translation of course content (3.13)	Only the course-and-language pairs offered in the 3.13 table; translation, back-translation and comparison ≈1,500 tokens per item per language	≈30–40 GPU-hours for the first build; later only new or changed items	AI-1 nodes in the late-night batch window from 5 November 2026, before the NexaTute launch; no new servers
Medical and engineering tools (3.13)	A few CPU-seconds per circuit diagram or simulation; notebooks and 3D anatomy in the browser; virtual-patient turns like normal AI-tutor turns	Small	App cluster sandbox workers, CDN and the shared LLM pool; no new servers
Practice lab sandboxes (3.6)	Docker/Kubernetes and cybersecurity labs at ≈0.5 vCPU and 512 MB each; browser labs need no server	50 sandboxes at launch ≈ 25 vCPU; ≈200 at target ≈ 100 vCPU	Spare Core-1/Core-2 cores and App workers (8.12); capped with a queue; extra OVHcloud CPU servers as use grows; no new servers at launch
Foreign-language speaking practice (11.2)	Server voices for Japanese, Korean, German and Chinese conversation lines; lesson audio cached	≈+1–2 RTX PRO 5000 per 1,000 learners practising speaking at the same time	Speech pool; beyond the limit, practice falls back to text with cached audio
National live model tests (3.10)	Up to ≈20,000 learners starting at the same minute; paper from the CDN; answers sent in batches	Burst of ≈20,000 WebSocket connections and a few thousand requests per second at start and end	CDN, Centrifugo and the App cluster; merit list in PostgreSQL; no new servers at launch
Phone tutor (IVR) (3.10)	Server speech recognition and voice for every call (no browser)	≈+1 RTX PRO 5000 per ≈400 calls at the same time	Speech pool and LiveKit cluster; calls queue with a busy message when full
Auto-dubbing (4.2)	Transcription, translation and new voice per added language	A few GPU-minutes per output minute per language	Speech and LLM pools at P50/P10; mostly in the daytime Motion Studio window
Own avatar and voice (3.10, 38.16)	Voice clones made once per person; per-person Bangla/Urdu VITS training ≈1–2 GPU-hours each; lessons in a teacher's voice cached; guardians' family voices, verified teachers and 18+ learners (one free each)	Small; training only in the batch window; NexaTute clone jobs off-peak at P10 within a daily GPU-minute cap (value set by the super admin, to confirm)	AI-2 Node A and AI-1 batch window; avatars in the browser; no new servers; avatar checks on Core-2 and lighter copies on the CPU worker pool
Shared LLM pool total	Product peaks do not fully overlap; includes the ≈15% safety-model load	≈13 GPUs if peaks coincide	12 GPUs + cloud burst (≈10% sharing gain)

Workload	Planning assumption	Demand at target	Pool sizing
Server speech fallback	20% of NexaTute CCU on weak devices (1,600), plus learners in any language whose browser voice has not yet passed its benchmark; 5 s speech in and 15 s voice out per turn	≈180 s STT + ≈530 s TTS audio per second	Speech-to-text on ≈1–2 RTX PRO 5000 (faster-whisper); voices on CPU — the in-house VITS voices through ONNX Runtime, roughly one core per 30–60 learners (≈30–55 cores at target: spare Core-1/Core-2 cores plus one OVHcloud CPU server beyond ≈8,000 NexaTute CCU, 8.12); Chatterbox and Indic Parler-TTS on GPU only for languages whose VITS voice has not passed; without the CPU voices this row would cap NexaTute at ≈3,000–4,000 CCU; beyond the limit new sessions degrade to text (8.9)
Human-led live classes	30% of NexaTute CCU (2,400 participants); speaker 720p ≈1.2 Mbps + 160p tiles	≈4.2 Gbps (≈2.4 Gbps when switched to 480p)	Core-1 (10 Gbps) + Core-2 (1 Gbps) = 11 Gbps total; ≈4.2 Gbps uses ≈38%; rooms placed by available bandwidth (most on Core-1)
Live-class recording	≈240 rooms, 20% recorded	≈50 concurrent room recordings	3 Rec nodes at ≈25 track recordings each (lightweight recording, 9.5); the benchmark sets the real per-node limit; overflow to spare Core-1/Core-2 CPU, then idle GPU nodes
Class recording (AI-tutor and batch classes)	Every class recorded as an event log with the learner's own audio (≈5 MB per 75-minute class)	No server encoding; ≈1.5 TB a month in R2 for 10,000 learners who use all their daily minutes	R2 storage and upload bandwidth only; replay voices made in the browser (11.3)
Batch classes (3.17)	One Nexa Teacher session per batch of up to 50; about one turn every 30 s while teaching, plus poll explanations; AI voice made in each browser; one server voice stream per batch for weak devices	≈5% of a 1:1 learner's LLM load per batch learner; poll bursts and events on Centrifugo for every learner	Shared LLM pool at P100 and the App cluster; the 7–9 am and 4–6:10 pm windows sit outside the NexaTute evening peak; no new servers
Nexa Co-Teacher (3.16)	Text only; up to 25 topics per learner a day; caches and knowledge-base retrieval first	Counted inside the 8,000 NexaTute CCU and the one-turn-every-45-s AI Tutor row, which stays the upper bound	Shared LLM pool; on overload the cloud API (Mistral Large 3, then GPT-6 Luna) within the spend cap; no new servers
AI Parent–Teacher Meeting (3.20)	10 minutes a month per learner, off-peak	≈9 concurrent sessions at 10,000 learners	Shared Nexa Teacher pool, outside the evening peak
Free-tier help minutes (3.22)	5 minutes a month per free user	≈12 concurrent sessions even if all 50,000 free users use them over 12 hours	≈1% of the fleet; behind plan holders in the queue
Nexa Co-Teacher Pass (3.22)	25 topics a day per Pass user; verified cache first	Text only	Local LLM pool only; no cloud API during overload
Script Checker (3.21)	Overnight batch of scanned answer scripts	Set by institution uploads	Qwen3-VL in the quietest GPU hours (for example 2–6 am)

Workload	Planning assumption	Demand at target	Pool sizing
Written checks and coding tests (3.27)	Up to 10 written checks a learner a day by plan (38.2); a three-page script ≈1–2 GPU-minutes of handwriting reading and marking (estimate, to benchmark); graded coding runs of a few CPU-seconds each with time and memory limits	Capped at 1,500 GPU-minutes a day for written checks (default, editable); results in 10–15 minutes, queued at the 4–9 pm peak	Launch GPU pool, AI-2 Node A first (8.17); practice-lab gVisor sandboxes on spare Core-1/Core-2 cores and App workers (8.12); no new servers
Engagement, teacher assistant and bots (3.32)	Team Quest, Quiz Battle and the Live Game Show for every learner; teacher-assistant drafts (20 a day per teacher, default); comic stories; Time-Travel and debate turns within their caps; bot answers from the cache	Per plan limits in 38.2; bots within each channel's budget (38.18)	Centrifugo, LiveKit Egress and the App cluster; LLM, FLUX and speech on the launch GPU pool off-peak at P10–P50; no new servers
Nexa Teacher Live (3.21)	One public session a week	One Nexa Teacher session and one LiveKit Egress stream	Negligible
Engagement models, link step and circular tracker (3.19, 3.20, 3.23)	Nightly retraining; one fetch and extraction per new link; daily circular collection	CPU only	App nodes and spare Core-2 cores; no GPU
Live Study Hall (3.20)	Presence, round timers, group sync (Yjs) and moderator help	Centrifugo messages; Nexa Co-Teacher LLM for short help; Qwen3Guard on reports	Shared LLM pool only for Help requests (short answers); no new GPU
Study Hall Q&A (31.3)	Up to 10 questions and 30 answers a learner a day; images resized in the browser	Image checks and OCR per post; AI verification of new answers at P10; Moderator answers only after 30 minutes without a reply	Spare CPU (Core-2) and the shared LLM pool at P10; no new GPU
MindForge (37)	Free: 3 puzzles a day; Pro: unlimited practice generated ahead of demand	Puzzle generation and solver checks (CPU); nightly IRT calibration; weekly forecasts; verbal items made in batches	CPU worker pool and the shared LLM pool at P10; no new GPU
Platform admin, accounting, notifications, staff chat and Growth Platform (38, 3.26)	Staff users; nightly export at about 2:00 am Dhaka; journey events and night-batch website reading; admin pushes and staff chat, staff calls in the platform admin panel; Marketing Hub imports, exports and channel posts	Small — a few LiveKit rooms for staff calls; one daily summary per chat (a few thousand tokens) at P10	App cluster (Centrifugo, notification service), LiveKit on Core-1/Core-2, CPU workers and Core-1 night batch; chat summaries on the shared LLM pool at P10; no new servers

Workload	Planning assumption	Demand at target	Pool sizing
Nexa Notes videos (3.25)	A few videos a week (the starting rhythm is two a week, 3.26); HLS at 720p, 480p and 240p; subtitles in six languages	≈10–20 CPU core-minutes of transcoding and under 1 GPU-minute of Whisper per 10-minute video, plus subtitle translation; ≈0.2–0.4 GB of R2 per 10-minute video (≈20–40 GB a year) — estimates to benchmark	CPU worker pool (App-1–App-3) at P10 in the night window; AI-2 Node A off-peak; R2 + CDN; no new servers
Websites, CMS and sales tools (3.25, 4.19)	Static pages on the CDN; a few CMS LLM calls a week; pre-sale chats cache-first	Small	Local LLM pool off-peak for the CMS; pre-sale chats like Nexa Co-Teacher text turns
AI Marketing Hub (3.26)	A few articles and videos a week, 1–2 Reels a day, about 30 emails a day per mailbox	Small	LLM off-peak; marketing videos at P10 in the batch window, never ahead of paying users
Motion Studio rendering	8,000 CCU = users active in the studio; renders are queued jobs	Set by GPU-minutes per output minute (benchmark)	5 local GPUs + hourly burst; queue shows ETA
API and database	≈0.2 requests/s per CCU	≈3,200 requests/s	5 App nodes; PostgreSQL primary + 2 replicas

Cache-hit sensitivity (NexaTute verified-answer cache, 8.14): every learner question answered from the cache skips LLM generation. The table shows how the NexaTute share of launch capacity and the LLM GPUs needed at 16,000 CCU change with the cache-hit rate (Motion Studio load, the ≈15% safety-model load and the ≈10% sharing gain unchanged).

NexaTute cache-hit rate	NexaTute CCU at launch (same 16 servers)	LLM GPUs needed at 16,000 CCU	Monthly saving vs no cache
0% (current plan)	≈2,000	12	—
20%	≈2,500	11	1 GPU server ≈ US$650 (≈৳79,000)
35%	≈3,100	10	2 GPU servers ≈ US$1,300 (≈৳1.6 lakh)
50%	≈4,000	8	4 GPU servers ≈ US$2,590 (≈৳3.2 lakh)

Planning rule: size the fleet for a 20–30% cache-hit rate and treat 50% as upside. Measure the real hit rate during the closed beta and paid pilots; if ≥50% is sustained for 30 days, lower the LLM GPU count and scale-out triggers in 8.5 accordingly. Cloud fallback cost falls in the same proportion as the cache-hit rate.

Single-product capacity of the 16 launch servers (planning estimates, to be load-tested):

Not yet counted in these figures (upside, to be confirmed by benchmark): lesson cache, meaning-matched cache, request coalescing, rule-based grading and the in-browser LLM may remove a further ≈20–35% of NexaTute LLM calls; RIFE frame interpolation and batched rendering may cut Motion Studio video GPU time per output minute by ≈40–60%. Capacity claims use only measured results.

Scenario	Without optimization	With 8.14/8.15 optimizations	Main limit
NexaTute only (CCU)	≈3,000	≈8,000–10,000 (25% cache); ≈11,000–15,000 (50% cache) on the GPUs; the App/API tier allows ≈6,000–7,500 until App-3 joins in December 2026	App/API tier until App-3, then LLM, then live-class recording
Motion Studio only — users in the studio	≈1,700	≈4,500–6,000 (one GPU for scripts, with suffix decoding and the compact storyboard schema)	Script LLM
Motion Studio only — video minutes per day (all Cinematic)	≈65	≈1,000–1,600 with LightX2V, FastWan and Wan2.2-S2V once benchmarked (≈650 realistic and ≈1,080 at full use without them)	Video GPUs
Motion Studio only — video minutes per day (Quick/Enhanced mix, 4.8)	—	≈3,000–4,500	Hero-shot GPU time

Both products at the same time — the limit of each layer (17 October 2026 review, planning estimates including all approved changes; the first limit sets the combined capacity):

Layer	Limit at the busiest hour (planning estimate)	Status
App/API tier — App-1/App-2 plus spare Core-1/Core-2 cores; ≈0.2 requests/s per CCU, ≈640 requests/s per App node	≈6,000–7,500 combined CCU; ≈7,000–9,000 with App-3 (≈1,920 requests/s)	First limit until App-3 joins in December 2026
NexaTute LLM GPUs	≈8,000–10,000 NexaTute CCU at a 25% cache-hit rate; ≈11,000–15,000 at 50%	Within limits
Server voices for weak devices (20% of NexaTute CCU)	≈10,000 NexaTute CCU with the CPU VITS voices; one OVHcloud CPU server beyond ≈8,000	Within limits only because of the CPU voices (≈3,000–4,000 without them)
Motion Studio script LLM	≈4,500–6,000 users in the studio (suffix decoding and the compact storyboard schema)	Within limits
Live-class bandwidth (Core-1 + Core-2, 11 Gbps)	≈16,000 NexaTute CCU	Within limits
Live-class recording (Rec-1/Rec-2)	≈8,000 NexaTute CCU, then overflow to spare CPU	Soft limit
Uploads	Straight from the browser to R2 (4.16); no load on the LB or App nodes	Fixed

Layer	Limit at the busiest hour (planning estimate)	Status
Batch-class windows (7–9 am and 4–6:10 pm, 3.17)	Each batch learner counts as one App/API CCU for polls and events; the 4–6:10 pm window overlaps the Motion Studio daytime load, so batch admission per window is capped from the measured App capacity	New; within limits at launch (BCS and SSC batches first)
Video GPUs	Cinematic ≈1,000–1,600 minutes a day; Quick/Enhanced ≈3,000–4,500 minutes a day	Within limits

Result: ≈6,000–7,500 combined CCU on the 16 launch servers (≈75,000–94,000 active users at 8% concurrency) and ≈7,000–9,000 with App-3 from December 2026 (≈87,000–112,000 active users); after that the GPUs are the next limit. The 16,000 CCU target needs ≈37–39 servers (≈US$15.5–16.5k a month) instead of ≈42 (8.5, 15.6). The Free Beta admission cap stays at ≈2,000 combined CCU until these figures are measured and is then raised to at most 70% of the measured capacity (11.11). The plan model of 18 October 2026 does not change these figures: Batch and Combo learners use fewer 1:1 minutes and a batch class needs one LLM session for up to 50 learners, so the one-turn-every-45-seconds assumption for every NexaTute CCU stays an upper bound (3.16, 3.17).

Production impact of the 4.10–4.16 additions (planning estimates before the launch benchmark; the figures above already include the earlier 8.14/8.15 optimizations):

Product / mode	Expected change in production	Main reason
NexaTute	≈0–2% lower	Almost all additions are Motion Studio; NexaTute peaks at night and Motion Studio in the day; NexaTute keeps its 60% LLM-pool floor and LiveKit always gets CPU first
Motion Studio — Quick mode	≈0–2% lower	Rendering stays in the creator's browser; approved animation library means few frame checks
Motion Studio — all modes combined	≈5–10% lower without the controls below; ≈3–5% with them	Video upscaler, frame safety checks, face restoration, music and quality-gate re-renders
Motion Studio — Cinematic Escalating tiers	≈10–20% lower without the controls below	Heaviest mode plus video upscaling and fallback re-renders

Extra GPU time added by each change (Motion Studio, per output minute):

Addition	Extra GPU time per output minute	Applies to
Video upscaler (RealBasicVSR/BasicVSR++) instead of Real-ESRGAN	+5–12%	Escalating tiers only
Quality-gate fallback to the next native tier	+2–5%	Failed scenes only
Qwen3-VL frame and gesture checks	+3–5%	Enhanced and Cinematic
NSFW classifier	<1%	All modes
GFPGAN close-up face restoration	+1–2%	Escalating tiers
ACE-Step music	+1–3%	Popular beds made in the night batch window

Addition	Extra GPU time per output minute	Applies to
BiRefNet and Demucs	<1%	Product ads and uploads only
VMAF, sharpness, Chromaprint, PySceneDetect, HMA	0% GPU	Chromaprint, PySceneDetect and HMA on spare Core-1/Core-2 CPU; VMAF and sharpness on the Dallas assembly-node CPU

- Offsets already decided: the Free Beta Cinematic pool starts at 10–15% of measured capacity instead of ≈400 minutes/day, dialogue scenes move from Cinematic to Quick/Enhanced until Cinematic lip-sync is licensed, and Cinematic 1080p Native is off at launch and paused automatically above 80% load (4.9, 4.14) — so first-month load stays about the same or lower; the difference appears as paid use grows.
- Controls that keep the loss at ≈3–5%: (1) Qwen3-VL runs only on flagged frames or scenes with people or characters (4.15); (2) the full video upscaler only for 1080p Escalating, a lighter model for 720p Escalating (4.13); (3) FP16, torch.compile and CUDA graphs for the upscaler and GFPGAN, as for Wan and FLUX — all open-source, no proprietary runtime; (4) music beds, template previews and popular scenes made in the night batch window after the NexaTute peak.
- Multiple aspect ratios: a crop of a 16:9 render to 9:16 loses resolution and needs upscaling or a new render, so the primary format is rendered natively and extra formats are cropped and lightly upscaled at the normal per-minute rate (4.10); without this rule Cinematic cost could rise by 10–30% for creators who order every format.
- The 8.15 production optimizations added on 17 October 2026 (thinking switch, output limits, Triton batching with MPS, batched Whisper, smaller frame-check images, A14B only for hero shots, INT8 embeddings) mainly protect capacity: the 8.6 sizing already assumes non-thinking answers of ≈250 tokens, so the thinking switch and output limits stop hard-maths thinking and long answers from eating into it rather than adding capacity. Triton batching with MPS, batched Whisper, smaller frame-check images, A14B only for hero shots and INT8 embeddings free GPU time, mostly for Motion Studio video (≈15–30%), more than offsetting the ≈3–5% cost of the 4.10–4.16 additions. The later 17 October 2026 changes (CPU server voices, LightX2V, FastWan, suffix decoding, Qwen3-8B routing and uploads straight to R2) bring the 16,000 CCU fleet to ≈37–39 servers instead of ≈42 (8.5).
- Final figures come from the benchmark after the servers arrive on 5 November 2026 (G.7, 13.2), measured with Locust, the LiveKit CLI load tester and vLLM's benchmark scripts (H.3); capacity claims use measured results only.

Recording-node sizing for NexaTute live classes (assumes 30% of learners in live classes of 10, 20% of classes recorded, lightweight track recording at ≈25 recordings per Rec node with download MP4s made in the browser, and 20% headroom): 3,000 CCU — 2 Rec nodes (minimum, for redundancy); 8,000 — 3; 12,000 — 4; 16,000 — 5. Rec nodes need CPU cores for track recording, not a large GPU (downloads no longer use them, 9.5); record classes on request rather than automatically to reduce the count.

8.7 Load Balancing

- Edge: Cloudflare (WAF, DDoS protection, CDN) routes to the active-active HAProxy pair LB-1/LB-2 on keepalived virtual IPs (LB-3 added at scale); least-connections balancing, active health checks and per-product rate limits; a failed LB is removed within seconds.
- API: stateless FastAPI pods per product behind Kubernetes Services, autoscaled on CPU and p95 latency; NexaTute and Motion Studio never share API pods.

- LLM: vLLM instances behind the open-source Portkey gateway using least-outstanding-requests and prefix-cache-aware routing; vLLM priority scheduling maps X-Priority 100/50/10 so NexaTute requests are served first on shared GPUs.
- Queues: separate Celery queues per product and priority (nexatute.speech, ms.paid, ms.free) on separate Valkey instances; KEDA scales workers on queue length.
- Real time: the LiveKit cluster places each room on the least-loaded SFU node; the TURN pair is load-balanced by DNS.
- Database: PgBouncer transaction pooling with separate per-product pools and connection caps; reporting and read-only queries go to replicas.

8.8 Bulkheads and Reserved Capacity

Each product has dedicated pools on its critical path, and shared GPU capacity is split into guaranteed floors, so overload in one product cannot starve the other.

Pool	NexaTute	Motion Studio	Borrowing rule
Shared LLM pool	Guaranteed 60% of capacity	Guaranteed 25% of capacity	15% floating. NexaTute (100) may pre-empt Motion Studio work above its 25% floor; Motion Studio may borrow idle NexaTute capacity while NexaTute load is below 50% (normally in the daytime) and returns it within one scheduling cycle; at night NexaTute may use idle Motion Studio capacity, returned as its requests finish.
Speech pool	First call for weak-device fallback	Voice-over TTS/STT at 50 (paid) or 10 (free/trial)	Motion Studio speech jobs use only capacity not needed by NexaTute fallback
Video generation pool	—	Dedicated	Paid (50) before free/trial (10); free/trial runs only on idle GPUs
LiveKit and TURN nodes	Dedicated	—	Not shared
Rec nodes (Rec-1/Rec-2)	Recording first (highest priority)	Light GPU work (post-processing and overflow) only while no recording runs	Pre-empted instantly when a recording starts; queued recordings overflow first to spare Core-1/Core-2 CPU (track-only), then to idle GPU nodes
API pools, Valkey and database pools	Dedicated	Dedicated	Not shared; separate connection caps

NexaLogicDev pools and lending between all three products (Heavy and Fast GPU pools, CPU workers, sandbox pool) follow the same owner-first rule and are set out in 8.17.

8.9 Priority Circuit Breakers and Degradation Ladder

Every dependency call (LLM, speech, video, database, R2 and cloud API) passes through a circuit breaker scoped per product and per pool. Closed → Open when the error rate exceeds 5% or p95 latency exceeds 3× its SLO over 30 s; Open → Half-open after 30 s, admitting 10% probe traffic; Half-open → Closed after three healthy intervals. Because breakers are per product, an open Motion Studio breaker never blocks NexaTute traffic, and vice versa.

Level	Trigger	NexaTute (X-Priority 100)	Motion Studio (50 paid / 10 free)
L0 Normal	All pools healthy	Full service	Full service
L1 Pressure	Pool utilisation >80% or queue wait rising	Keep thinking off for all turns and send small text tasks to Qwen3-4B; shorten maximum output; use floating LLM capacity	Pause free/trial (10) jobs; paid jobs continue
L2 High load	Priority-100 queue wait p95 >1.5 s	Pre-empt Motion Studio LLM work above its floor; load-aware routing offers recordings, mock tests and summaries first with an ETA (3.16); live Nexa Teacher classes are never sent to the cloud, while the Nexa Co-Teacher may use the cloud API within the spend cap; live video 720p → 480p	Paid jobs held to the reserved floor with ETA shown; hourly GPU burst for paid renders
L3 Overload	Local and burst capacity saturated	No new live Nexa Teacher classes: learners continue with the Nexa Co-Teacher in text and the lost minutes are credited back (3.16); new weak-device sessions switch from server voice to text; existing sessions protected; free-tier and Nexa Co-Teacher Pass users get cached answers and wait in the queue — the cloud API serves plan holders only (3.22)	New renders paused (breaker open); scripts, editing and drafts stay available
L4 Protect	Error budget breached, or the local GPU cluster and both cloud providers (GPT-6 Luna, Mistral Large 3) are unavailable or capped	New sessions see the message “The server is very busy right now. Please try again after some time.” and wait in a queue; active classes are never dropped	New requests see the message “The server is very busy right now. Please try again after some time.”; read-only studio; queued jobs persist and resume automatically

Cloud LLM use for NexaTute is limited to the Nexa Co-Teacher’s text answers and quality routing (3.13), and uses only providers with zero-retention and no-training terms and strips personal data before sending; for sessions of minors, if no compliant provider is available the ladder moves to L4 queueing instead of sending data to the cloud. Every level change is logged, alerted and reversed automatically when the trigger clears for 10 minutes. Free-tier and Nexa Co-Teacher Pass requests never go to the cloud (3.22).

Cloud fallback order (both products, text/LLM tasks only): GPT-6 Luna (OpenAI) first, then Mistral Large 3 (Mistral AI) as the last cloud option. Every hop keeps the product identity, priority, trace ID and idempotency key, strips personal data, and counts against the per-product monthly spend cap; when both providers are open-circuited or the cap is reached, the ladder moves to L4 queueing. Speech, video and recording never use cloud fallback. Free Beta users never use the cloud fallback (11.11). NexaTute exception (3.16): the Nexa Co-Teacher uses the cheaper order — verified-answer cache, then Mistral Large 3, then GPT-6 Luna — and live Nexa Teacher classes never use the cloud fallback.

For NexaTute text answers the order is the verified-answer cache first, then Mistral Large 3, then GPT-6 Luna (decided 23 October 2026, 3.16); the Nexa Teacher stops under overload while the Nexa Co-Teacher keeps running this way, and Free Beta and free-tier users never use the cloud APIs.

Quality routing (3.13): besides overload fallback, hard questions in a premium category that has not passed its gate on the local model are sent to GPT-6 Luna, with personal details removed; this stops automatically once the local model passes the gate.

8.10 Resource Efficiency

- Base load runs on monthly rented servers; peaks run on hourly cloud GPUs — spot/preemptible for Motion Studio renders and batch work, on-demand for NexaTute real-time traffic (8.15) — released automatically after 30 minutes idle.
- Daily load pattern (Bangladesh time, planning assumption to be confirmed from launch data): Motion Studio is busiest in the daytime (≈9 am–6 pm) and NexaTute in the evening and night (≈6 pm–1 am). Because the two peaks do not overlap, the shared GPU cluster serves each product with most of its capacity at that product's own peak instead of splitting it: Motion Studio borrows idle NexaTute GPUs in the daytime, and NexaTute uses idle Motion Studio capacity at night.
- Schedule-based pre-scaling before the daily peaks is decided from live load tests after launch. Motion Studio paid server renders use the delivery-speed tiers (G.8.3): Express (+20%) renders at high priority on whatever GPU capacity is free and shows its calculated delivery time; Normal (within 12 hours) and Slow (within 24 hours, 10% off) fill quieter hours, with Slow orders — and Normal orders whose window allows it — running in the 1–7 am window. NexaTute extra 1:1 minutes for adult courses are 20% cheaper from 1 am to 7 am (11.12); school and PlayBoom learners get no night discount.
- NexaTute batch classes (7–9 am and 4–6:10 pm) and the load-aware router (recordings, mock tests and summaries first when the GPU pool is busy) move learning away from the evening peak (3.16, 3.17).
- Free/trial work runs only on idle capacity; target steady GPU utilisation is 60–75% to keep headroom for priority-100 traffic.
- A mixture-of-experts model (Qwen3-30B-A3B, 3B active parameters), thinking off by default and prefix caching of curriculum context raise tokens per GPU.
- Browser-side voice, Nexa Teacher video, whiteboard, Virtual Lab, 3D classroom, AI-tutor recording and recording-download MP4s keep graphics, speech and encoding load off the servers.
- Escalating tiers (720p Escalating, 1080p Escalating) generate at the next lower native resolution and upscale, cutting video GPU time per output minute compared with native generation. 1080p Native (4.9) is the most GPU-heavy tier and is paused automatically for new orders when cluster load passes the admin threshold.
- API, worker and burst pools scale down automatically when load falls; every scale step is recorded with its cost.

8.11 Unified 6-Node GPU Cluster

From launch, all six GPU servers — AI-1 Node A and Node B (RTX PRO 6000, 96 GB), AI-2 Node A and Node B (RTX PRO 5000, 48 GB) and Rec-1 and Rec-2 (RTX PRO 4000, 24 GB) — work as one GPU pool at Database Mart in Dallas, Texas instead of six fixed-role servers; they are worker nodes of the single Kubernetes cluster that spans all servers, whose control plane is in Virginia (8.13, 8.16). Any idle GPU takes the highest-priority eligible work, so capacity left idle by one product is used by the other, and no single GPU failure stops a service. Kueue (Apache-2.0) queues GPU jobs with priority and pre-emption, vLLM priority scheduling orders LLM requests, and the NVIDIA GPU Operator (Apache-2.0) manages drivers and GPU health. Scale-out GPU nodes in 8.5 join the same cluster.

- Creator-tool models (4.10–4.12) share this pool: ACE-Step music (≈8–16 GB VRAM), BiRefNet background removal (≈2 GB) and Demucs music separation (≈2–4 GB) run at P50 for paid work and P10 for free work, prefer the Rec-1/Rec-2 and AI-2 nodes when free, and fall back to CPU for small jobs; popular music beds and template previews are made in the night batch window. They need no new servers; their GPU-minutes are measured in the launch benchmark.
- Quality and safety models also share this pool: RealBasicVSR/BasicVSR++ (≈4–8 GB) and GFPGAN (≈1–2 GB) for escalated final videos, and the NSFW classifier (<1 GB) for frame checks, at P50 paid / P10 free; Qwen3-VL frame checks use the existing Qwen3-VL deployment and run only on flagged frames or scenes with people or characters; production impact and efficiency controls are in 8.6. If the Dallas GPU site is unavailable, Motion Studio jobs fail over automatically to cloud GPUs (the existing burst path) within a monthly spending cap (4.14).

GPU memory (VRAM) placement — total working memory on every GPU stays under 90% of VRAM; content-generation models (Wan 2.2, FLUX.1-schnell, ACE-Step) run one at a time per GPU; small models (under ≈4 GB) may share a GPU through Triton and NVIDIA MPS; the per-GPU limits are set by vLLM's gpu_memory_utilization and NVIDIA MPS memory limits, while Kueue admits GPU jobs by priority and quota:

GPU node	VRAM	Models placed here	Rule
AI-1 Node A / Node B	96 GB each	Qwen3-30B-A3B with the thinking switch (prefill / decode), Qwen3Guard, EAGLE-3 draft; Wan 2.2 A14B (Cinematic hero shots) and Wan2.2-S2V-14B (Cinematic dialogue, after its gate) in the daytime on Node B; Qwen3-8B no longer runs here (8.15)	Qwen3 stays loaded on both nodes for hot standby: with Wan A14B on Node B the LLM keeps ≈46 GB and Wan ≈40 GB (FP8, one expert at a time, the other in the 84 GB RAM); Wan pauses after its current scene if Node A fails; vLLM sleep mode is used only in the night batch window
AI-2 Node A	48 GB	Speech (faster-whisper, Chatterbox, Indic Parler-TTS, wav2vec2; the in-house VITS server voices run on CPU, 8.12), Qwen3-VL-8B (photo questions, uploads, frame and gesture checks)	Qwen3-VL frame checks only on flagged frames or scenes with people (4.15)
AI-2 Node B	48 GB	Wan 2.2 TI2V-5B (the FastWan distilled version after its gate), RealBasicVSR/BasicVSR++, RIFE; GFPGAN, NSFW classifier and OpenCLIP/DINOv2 served by Triton with dynamic batching and NVIDIA MPS	Video node; final MP4 compile on its local SSD

GPU node	VRAM	Models placed here	Rule
Rec-1 / Rec-2	24 GB each	Recording first; when idle: ACE-Step, BiRefNet, Demucs, Real-ESRGAN previews, AV1 copies, 3D browser-render fallback (4.8), Qwen3-8B LLM overflow (the only place Qwen3-8B runs) and speech-to-text overflow	Recording pre-empts all other work

Workload roles by node — the work each GPU takes first, and what it can also run when idle:

Node	GPU	Preferred workloads	Can also run
AI-1 Node A	RTX PRO 6000, 96 GB	NexaTute LLM prompt processing (prefill) for Qwen3-30B-A3B and the Qwen3Guard safety model	Motion Studio LLM; Wan 2.2 A14B video when NexaTute load is low; in the late-night batch window: AI content pipeline generation (3.13), NexaTute-Bangla LoRA fine-tunes (2.5) and voice/speech model training (4.2)
AI-1 Node B	RTX PRO 6000, 96 GB	NexaTute LLM answer generation (decode) and hot standby; Motion Studio script LLM	Wan 2.2 A14B 720p video and Wan2.2-S2V-14B dialogue scenes; FLUX.1-schnell images (Nunchaku 4-bit after its gate); in the late-night batch window: Mistral Small 3.2 independent solving for the content pipeline (woken with vLLM sleep mode)
AI-2 Node A	RTX PRO 5000, 48 GB	Speech: server speech-to-text fallback and the GPU voices (Chatterbox, Indic Parler-TTS, cloned voices, and any language whose VITS voice has not yet passed its benchmark) — the in-house VITS server voices run on CPU (8.12), foreign-language practice voices, phone-tutor calls, auto-dubbing voices and Motion Studio voice-over; Qwen3-VL photo questions; class-summary transcription; pronunciation scoring	LLM overflow (Qwen3-30B-A3B FP8 when the node is free); overflow video upscaling and face restoration when AI-2 Node B is busy (4.13); in the late-night batch window: foreign-language lesson audio and audio lessons
AI-2 Node B	RTX PRO 5000, 48 GB	Batched Wan 2.2 TI2V-5B video, RIFE interpolation, scene quality checks, video upscaling and quality gate (4.13), FFmpeg/GStreamer assembly and promo cuts (4.18) on its local 400 GB SSD	Speech-to-text overflow; LLM overflow (Qwen3-30B-A3B FP8 when the node is free)
Rec-1	RTX PRO 4000, 24 GB	Lightweight track recording — always the top priority on this node (download MP4s are made in the browser, 9.5)	Only while no recording is running: preview upscaling, RIFE, AV1 viewing copies, scene quality checks, video upscaling (4.13), 3D browser-render fallback (4.8), Qwen3-8B LLM overflow, speech-to-text overflow
Rec-2	RTX PRO 4000, 24 GB	Lightweight track recording — always the top priority on this node (download MP4s are made in the browser, 9.5)	Only while no recording is running: preview upscaling, RIFE, AV1 viewing copies, scene quality checks, video upscaling (4.13), 3D browser-render fallback (4.8), Qwen3-8B LLM overflow, speech-to-text overflow

- Priority classes: REC — recording/egress, the highest priority on Rec-1/Rec-2; P100 — NexaTute LLM, server speech fallback and safety model; P70 — real-time work borrowed from another

product's pool, always below the owner's P100 (8.17); P50 — Motion Studio paid LLM, voice-over and renders; P10 — free/trial work.

- Batch-class AI sessions and Nexa Co-Teacher text run in the NexaTute LLM lane (P100) on AI-1 Node A/B, with overflow as for other NexaTute LLM work; they add no new models (3.16, 3.17).
- Guaranteed floors across the whole cluster: NexaTute 60%, Motion Studio 25%, 15% floating (8.8). When NexaTute load is below 50% (normally in the daytime) Motion Studio may use the idle GPUs and returns them within one scheduling cycle when learner demand rises. At night, when Motion Studio is quiet, NexaTute may also use Motion Studio's idle share; it is returned as NexaTute requests finish, without cutting any live session, when paid Motion Studio work arrives.
- Pre-emption order: P10 jobs first, then P50 jobs above their floor. Renders checkpoint after every scene, so a pre-empted job resumes without losing finished scenes; LLM answers are never cut mid-response — a GPU is handed over only after its in-flight requests finish.
- Hot standby: the NexaTute LLM always runs on at least two GPUs (AI-1 Node A and Node B), also while Wan 2.2 A14B uses Node B in the daytime — the LLM keeps ≈46 GB there (FP8 weights and a smaller KV cache) and Wan runs in FP8 in ≈40 GB with one expert on the GPU at a time (Wan2.2-S2V-14B for Cinematic dialogue uses the same budget); if Node A fails, Wan finishes its current scene and pauses so Node B serves the full LLM load. With prefill/decode disaggregation, if either node fails the other returns to normal mode and runs both steps, so one node failure does not stop the AI Tutor.
- Rec-1/Rec-2: when no recording is running, they take post-processing work (preview upscaling, RIFE, AV1 viewing copies, scene quality checks) and light overflow (Qwen3-8B LLM, speech-to-text) and 3D browser-render fallback chunks (4.8), capped at about 16 GB VRAM and part of the CPU, so recording always has room. When a recording starts, that work is pre-empted at once — NVENC is a separate encoder engine, so in-flight CUDA work drains while the recording begins.
- Recording overflow: if recordings are waiting because Rec-1/Rec-2 are full, track-only recordings start first on spare Core-1/Core-2 CPU (no encoding needed); only if that is also full do idle GPUs on the other cluster nodes take them using their own NVENC encoders. Overflow recording uses idle capacity first and may pre-empt only P10 free/trial work — never NexaTute P100 or Motion Studio paid P50 work — and moves back to Rec-1/Rec-2 for new recordings once they have free slots.
- Circuit breakers: one per product and workload class (LLM, speech, video) on the cluster, plus a GPU-node health breaker — a node with GPU errors, overheating or out-of-memory events is cordoned, its work is rescheduled to the other nodes, and it is probed back after recovery. When the whole cluster is saturated, the degradation ladder in 8.9 applies: free/trial is paused first and NexaTute is protected last.
- Data path: the GPU node that runs a render keeps its scene clips, intermediate files and final MP4 assembly on its own local SSD (AI-2 Node B preferred; AI-1 nodes have 400 GB, AI-2 Node A and Rec nodes 320 GB). If AI-2 Node B is busy or fails, other nodes continue assembly; a rescheduled job resumes from its last finished scene. Nothing heavy is written to Core-1.

Capacity effect (planning estimate): peak LLM throughput rises from ≈12,000 to ≈21,000 tokens/s when all four GPUs serve LLM (an RTX PRO 5000 delivers roughly 75% of an RTX PRO 6000), and Motion Studio can borrow both 96 GB GPUs in the daytime when learner load is low for Wan 2.2 A14B 720p video. After the ≈15% safety-model load (8.6), estimated launch capacity rises from ≈2,200 to ≈3,000 combined CCU (≈40% more) on the same 16 servers before the 8.14/8.15 optimizations (with them, see the combined table in 8.6); the 16,000 CCU target still needs the scale-out fleet in 8.5. Benchmarks must confirm these figures.

From June 2027 the NexaLogicDev Heavy pool (3 × 4× RTX A6000) and from September 2027 its Fast pool (4 × RTX PRO 6000, at this same Dallas site) join this cluster; the pools lend idle capacity to each other, owner first, by the placement order in 8.17.

8.12 CPU Work Placement

The six Database Mart GPU servers are dedicated-GPU VPS: the GPU is passed through to us in full, but the virtual CPUs may be shared with other customers. Their CPUs therefore run only the helpers that GPU serving needs, and all other CPU work runs on the ten dedicated CPU servers (eleven with App-3 from December 2026) — mainly the spare cores of Core-1 and Core-2 beside LiveKit, and the App nodes.

Node	Reserved for GPU serving	Spare CPU work (low priority)	Planning estimate
AI-1 Node A / Node B (2 × 32 cores)	8 cores each: vLLM scheduler, tokenizer, API server, safety and embedding helpers, Kubernetes agents	None planned — the virtual CPUs may be shared; light batch work only if the first-month benchmark shows no CPU steal	Not counted
AI-2 Node A (24 cores)	10 cores: audio decode/resample, voice activity detection, Bangla/Urdu text-to-phoneme, Opus encoding	None planned	Not counted
AI-2 Node B (24 cores)	16 cores during assembly: FFmpeg/GStreamer muxing, audio mix, subtitle burn-in, promo cuts and thumbnail frame scoring (4.18), quality-gate measurements (VMAF, sharpness, flicker at one frame per second, 4.13) and the Content Credentials manifest; 2 cores for the Triton Inference Server	None planned	Not counted
Rec-1 / Rec-2 (2 × 24 cores)	All cores: lightweight track recording (light, well under one core each); download MP4s are composited in the user's browser (9.5)	None — freed cores stay reserved for recording peaks	0 spare cores

Node	Reserved for GPU serving	Spare CPU work (low priority)	Planning estimate
Core-1 (48 cores, dedicated)	LiveKit SFU and TURN (≈16 cores at target)	Practice-lab sandboxes (gVisor), AI content pipeline parsing and checks (Docling, SymPy, Ragas, MinHash), small Motion Studio text tasks with llama.cpp and Qwen3-4B (CPU), free-trial scripts at P10, in-house VITS server voices through ONNX Runtime (P100 for NexaTute), 2D browser-render fallback in headless Chromium (4.8), load-test runners outside class hours, track-only recording overflow, PySceneDetect Shorts cutting, Chromaprint fingerprint checks and server-side auto-reframe (4.10–4.12) and Hasher-Matcher-Actioner hash matching of uploads (4.15)	≈28 spare cores
Core-2 (32 cores, dedicated)	LiveKit SFU and TURN (≈10 cores)	API overflow pods, Document-to-Video parsing, the Document Intake Service (Bijoy conversion, file parsing, upload OCR), WeasyPrint PDFs, OCR (Tesseract, PP-FormulaNet), Office file checks, schemdraw/ngspice jobs, analytics batch jobs, Yjs whiteboard sync, in-house VITS server voices, ClickHouse for Langfuse traces and Ralph LRS (capped pod)	≈20 spare cores
App-1 / App-2 / App-3 (16 threads each, dedicated; App-3 from December 2026)	NexaTute and Motion Studio API pods and platform services	Practice-lab sandboxes when Core-1/Core-2 are busy	Shared with API autoscaling

- CPU pinning: on the GPU servers the helper cores are pinned to the GPU-serving pods; on Core-1/Core-2, LiveKit and TURN get guaranteed CPU first, so batch work can never slow live classes.
- CPU budget for the creator tools: PySceneDetect, Chromaprint, hash matching, server auto-reframe, and the Document Intake Service share the spare Core-1/Core-2 cores after LiveKit (VMAF, sharpness and flicker measurements run on the CPU of the Dallas assembly node, one frame per second, so no video crosses sites); when spare-core use stays above 70% for a week, another OVHcloud CPU server is added (8.5). CI image builds (Forgejo Actions with Podman) run on Core-1 only in the night window.
- Spare-CPU pods on Core-1, Core-2 and the App nodes run in a low Kubernetes priority class with CPU and memory limits; they are evicted first when LiveKit, an API pod or a recording needs resources.
- Server voices for weak devices (8.6): the in-house VITS voices run on CPU through ONNX Runtime on spare Core-1/Core-2 cores — roughly one core per 30–60 learners on the server voice (estimate, measured in the benchmark) — so the GPUs keep only speech-to-text, Chatterbox, Indic Parler-TTS and cloned voices; one OVHcloud ADVANCE-2 CPU server is added beyond ≈8,000 NexaTute CCU or when spare-core use passes 70% (8.5).
- Batch classes and recordings (3.17, 11.3): the batch-class service, load-aware router and event-log recording writer run in the NexaTute API namespace on the App cluster; the one server VITS voice stream per batch for weak devices runs with the other CPU voices on spare Core-1/Core-2 cores; raised-hand audio is recorded track-only, like other overflow recordings.
- Engagement models, the link step and the circular tracker (3.19, 3.20, 3.23): model training and inference, the gVisor link fetcher with Docling, and Scrapy collection run in low-priority pods on the App nodes and spare Core-2 cores; the Script Checker uses the GPU only in the quietest night hours (3.21).
- Website builds and checks (3.25): Next.js page rebuilds, sharp image processing and Lighthouse CI checks run in low-priority pods on the App nodes; the CDN serves the pages.
- ClickHouse (Langfuse traces and Ralph LRS xAPI statements) runs on Core-2 in a low-priority pod with CPU and memory limits, so LiveKit always comes first, and is backed up every night with ClickHouse's native BACKUP to Cloudflare R2; PostgreSQL stays the source of truth, so losing ClickHouse loses only trace and analytics copies.
- API overflow: when the App nodes exceed 60% CPU or p95 300 ms, extra FastAPI pods for the affected product start on spare Core-1/Core-2 cores before any new App server is rented (8.5), keeping NexaTute and Motion Studio pods separate.
- Estimated effect: about 45–50 spare cores on Core-1 and Core-2 at launch, roughly three 16-thread App servers. When practice-lab sandboxes or batch work outgrow them, more OVHcloud ADVANCE-2 servers (≈US$198 a month each, ≈30 sandboxes each) are added before any CPU work returns to the GPU servers. Benchmarks must confirm these sizes.

8.13 Clustered Application, Data, Cache and Edge Tiers

From launch, every non-GPU server pair runs as a cluster, so the loss of any single server does not stop either product. Product isolation is kept inside each cluster through separate namespaces, databases, instances, credentials and quotas.

Tier	Nodes and clustering	Failover	Product isolation
Edge / load balancers	LB-1 + LB-2 active-active HAProxy with keepalived (two virtual IPs, one per node); Cloudflare load-balances across both; no control-plane or monitoring roles on the internet-facing LBs	A failed LB's virtual IP moves to the other node in ≈2–3 s; combined ingress 2 × 500 Mbps	Per-product frontends, rate limits and priority admission

Tier	Nodes and clustering	Failover	Product isolation
Database pooling	PgBouncer as a sidecar on every App node on the OVHcloud private network, always pointing at the current Patroni primary, with pool sizes divided across the App nodes so the total stays within the PostgreSQL connection cap — moved from LB-1/LB-2 because the LB pair is at RedSwitches while App and DB are at OVHcloud, so every query would otherwise cross providers twice	Survives the loss of either LB	Separate per-product pools and connection caps
Application	App-1 + App-2, plus App-3 from December 2026, as one Kubernetes app node pool; every NexaTute and Motion Studio API deployment runs ≥2 replicas spread across the nodes; internal tools (Chatwoot, Unleash, Langfuse and the shared AI CMS, 3.25) run as low-priority pods in their own namespace, and Learning OS platform services (Keycloak, APISIX, Centrifugo, Ralph LRS) run in a platform namespace with ≥2 replicas	If one node fails the others serve both products; autoscaling and spare Core-1/Core-2 cores (8.12) add capacity	Separate namespaces, resource quotas and network policies per product
Database	DB-1 + DB-2 Patroni PostgreSQL cluster: primary + synchronous standby (falls back to asynchronous if the standby is down)	Automatic failover in ≈30 s with no committed-data loss; DB-3 async replica added at scale (8.5)	Separate databases, roles and credentials per product
Live database backup	pgBackRest continuous WAL archiving to a dedicated Cloudflare R2 bucket (archive every ≤60 s) plus daily differential and weekly full backups, AES-256 encrypted	Point-in-time restore to any second; RPO ≤1 minute even if both DB nodes are lost; 14 days of point-in-time recovery and 8 weekly fulls retained	Write-only backup credentials; restore drill every month
Database archive	Monthly partitions; partitions older than 12 months exported to an encrypted R2 archive bucket (Parquet) and dropped from DB-1/DB-2	Keeps the 2 × 960 GB DB disks below 70%; archived data restorable on request	Separate archive prefixes and credentials per product
Cache / queue	Valkey on Redis1 + Redis2 with cross-replication and Sentinel: NexaTute instance (6379) primary on Redis1, replica on Redis2; Motion Studio instance (6380) primary on Redis2, replica on Redis1; AOF persistence; daily encrypted RDB snapshot to R2 kept 48 hours (see below)	Sentinel promotes the replica in ≈10 s; if both nodes are lost, restore the latest snapshot (≤24 h old) — PostgreSQL remains the source of truth for jobs and credits	Instances, ports, memory limits and credentials stay separate per product; snapshots use write-only credentials, and session tokens in them expire with the 48-hour retention

Tier	Nodes and clustering	Failover	Product isolation
Quorum and monitoring	One Kubernetes cluster for all servers: 3-member control plane and etcd on Redis1, Redis2 and App-1 (App-3 and later App nodes are workers only, keeping an odd quorum of three) (OVHcloud private network only; Patroni uses the Kubernetes API as its configuration store and never writes to etcd directly); the Dallas GPU servers are worker nodes; Valkey Sentinels on Redis1, Redis2 and App-1; monitoring on Redis1/Redis2 — Prometheus as two replicas, Grafana, Alertmanager, and Loki storing logs in Cloudflare R2	Quorum and monitoring survive the loss of any one member	—
Local mirror on Core-1	Harbor container registry (Apache-2.0) and an AI model-weight cache (Qwen3 family, Qwen3-VL, Qwen3Guard, Mistral Small 3.2, Wan 2.2, FLUX.1-schnell, Whisper, TTS voices, RealBasicVSR/BasicVSR++, GFPGAN, ACE-Step, Demucs, BiRefNet and the NSFW classifier) on Core-1's 4 TB NVMe for the Virginia nodes; the Dallas GPU nodes pull images and model weights from Cloudflare R2 and keep them on their local SSDs, so no large pulls cross between sites	If Core-1 is down, nodes pull images and models from Cloudflare R2 instead — the mirror is never a single point of failure	Low disk-I/O priority and a bandwidth cap so live-class media always comes first; large pulls only outside class peak hours

- Result: every tier from edge to database now survives a single-server failure at launch; the remaining single-node risks are only capacity limits, handled by the degradation ladder (8.9).
- Valkey snapshot: once a day an AES-256-encrypted RDB snapshot of each Valkey instance is uploaded to a separate Cloudflare R2 bucket and kept for 48 hours only (lifecycle rule deletes older copies), for fast warm restart. This is an optional convenience, not a live backup — sessions and cache can be rebuilt, and queued work is recovered from PostgreSQL job states.

8.14 Prompt Caching and Context Trimming

These rules reduce the tokens processed per request, so the same GPUs serve more learners and creators and the cloud fallback budget lasts longer. They apply to the local LLM pool and to the GPT-6 Luna → Mistral Large 3 cloud fallback (the Nexa Co-Teacher uses Mistral Large 3 first, 8.9).

- Prompt order for caching: every request is built as fixed system prompt → lesson or project context → conversation summary → recent turns → new question. The stable parts come first so they can be reused from cache.
- Local prefix caching: vLLM automatic prefix caching is enabled on all LLM replicas, and the gateway routes a learner's or creator's session to the same replica where possible (prefix-cache-aware routing, 8.7), so the shared system prompt and curriculum context are not recomputed on every turn.
- Cloud prompt caching: the same stable prefix is sent to GPT-6 Luna and Mistral Large 3 so their prompt-caching discounts apply where the provider supports them.
- Context trimming (NexaTute): keep only the last 6 turns word for word; older turns are folded into a running summary of at most 300 tokens, refreshed every few turns by Qwen3-30B-A3B

(thinking off) on the local cluster; retrieve only the 3–5 most relevant curriculum passages (BGE-M3 + reranker) instead of whole chapters.
- Context trimming (Motion Studio): script and storyboard requests send the project brief, the approved outline, a compact Character/World Bible summary and only the scene being worked on — never the full project history.
- Hard limits per request: at most 4,000 input tokens on the local cluster and 3,000 on cloud fallback, plus a maximum output length for each task type; personal data is stripped before any cloud call (8.9).
- Verified-answer cache: answers to frequently asked questions that have passed safety and quality checks are stored with their curriculum version and served directly, without calling an LLM, for matching questions — including the same question asked in other words or in Bangla, Banglish or English, matched by meaning with BGE-M3 at a high similarity threshold and only within the same class, subject and curriculum version; questions below the threshold go to the LLM. Entries expire when the curriculum or policy version changes.

Voice and audio cache (decided 29 October 2026): the TTS audio of repeated verified explanations is stored in R2 and replayed, keyed by the answer, language and voice or avatar — never audio that contains a learner's name or personal data. It is on, kept 90 days and limited to 50 GB (default, editable), purged automatically when the answer is corrected (the hub and correction history) and by hand with two-person approval; the cache-hit rate and the TTS minutes saved appear in platform operations (38.18).

- Lesson cache (generate once, reuse): the first time a lesson (class, subject, chapter, topic and language) is taught, the Nexa Teacher's lecture script, voice audio, lip-sync timing and whiteboard steps are checked for safety and quality and saved to R2 with the curriculum version; every later learner who opens the same lesson gets the saved lesson from the Cloudflare CDN, and the LLM only answers that learner's own questions. A saved lesson is regenerated when the curriculum or policy version changes.
- Offline lessons: a learner can download saved lessons (from the lesson cache) on Wi-Fi into encrypted browser storage (OPFS/IndexedDB, installable web app) and watch them later without internet, with no server load while offline. Asking the Nexa Teacher a question needs a connection. Offline viewing of saved lessons is included in an active plan; wallet-only learners follow the rule in 11.9.
- Learner feedback loop: an answer that many learners rate 👍 and that passes the safety check is added to the verified-answer cache; an answer rated 👎 goes to review, and reviewed corrections become training data for the Bangla fine-tune (8.15), so cache hit rate and answer quality rise over time.
- Monitoring: track prefix-cache hit rate (target ≥50%), average input tokens per turn (target ≤2,500), verified-answer cache hit rate (exact and meaning-matched), lesson cache hit rate and cost per learner-hour on the capacity dashboard.

Estimated effect (planning estimate, to be benchmarked): input-token cost on cloud fallback falls by roughly 50–70%, and local LLM throughput rises by about 10–20% because less context has to be processed on each turn.

8.15 Software Optimization Plan

These software optimizations raise output on the same hardware. Prompt caching, context trimming and the verified-answer cache are covered in 8.14 and are not repeated here. Gains are planning estimates from published results and must be confirmed in the Gate 2 benchmarks.

Area	Optimization	Expected gain (estimate)	When
LLM	FP8 KV cache in vLLM	≈2× concurrent sessions per GPU	Before launch

Area	Optimization	Expected gain (estimate)	When
LLM	EAGLE-3 speculative decoding with a small Qwen3 draft head	≈1.5–2.5× faster answers	Before launch
LLM	vLLM tuning: chunked prefill, batch size and sequence limits	≈10–20%	Before launch
LLM	Bangla fine-tune (LoRA, the NexaTute-Bangla model, 2.5) on verified Qwen3-30B-A3B answers, plus a difficulty router that turns thinking on only for hard questions	Better Bangla answers and fewer thinking-mode turns, measured by the 3.13 gate	After launch (Q2 2027)
LLM	NVFP4 quantization on the Blackwell GPUs	≈1.5–2× over FP8	After launch, only if Bangla quality holds
LLM	SGLang benchmarked against vLLM on the same workload	Keep whichever engine is faster	Gate 2 benchmark
LLM	Prefill/decode disaggregation (vLLM or SGLang): AI-1 Node A processes prompts (prefill) and AI-1 Node B generates answers (decode), passing the KV cache between them; needs a fast private link between the two nodes and returns to normal mode automatically if either node is lost	Peak-time slowest-response latency ≈20–40% lower; smoother answer streaming under load	Before launch; kept only if the benchmark beats normal mode
LLM	Subject-wise LoRA adapters on one Qwen3 base model (vLLM multi-LoRA serving) instead of separate models per subject	Fewer model copies in GPU memory, more concurrent sessions, better subject accuracy	After launch, with the Bangla fine-tune
LLM	LMCache (Apache-2.0): keep the KV cache of long conversations in CPU RAM or local SSD and reuse it on the next turn	≈2–3× throughput on long multi-turn sessions	Before launch (benchmark first)
LLM	vLLM sleep mode: when a GPU switches between LLM and Wan 2.2 work, the model weights are parked in CPU RAM and restored in seconds instead of being reloaded from disk; the hot-standby Qwen3 copies on AI-1 Node A and Node B are never put to sleep in the daytime (8.11)	GPU switch ≈5–10 seconds instead of ≈1–2 minutes, so idle GPUs can be lent more often	Before launch
LLM	Request coalescing (NexaTute): when many learners ask the same question at about the same time, one answer is generated and streamed to all of them	≈5–15% lower LLM load at peak; every learner gets the same checked answer	Launch
LLM	Grammar-constrained structured output (xgrammar in vLLM, Apache-2.0): quizzes, storyboards and lab diagnostics can only be written in the valid JSON format	No retries for broken output; ≈5–15% fewer LLM calls on structured tasks	Launch

Area	Optimization	Expected gain (estimate)	When
Browser	Small LLM in the learner's browser (WebLLM, Apache-2.0, with Qwen3-0.6B or 1.7B on WebGPU) on capable devices for light tasks: Banglish-to-Bangla conversion, spelling correction, quiz hints and question clean-up before the cache lookup; weak devices keep using the server	≈10–20% fewer server LLM calls	Before launch; benchmark on target phones and laptops
Browser	Prefetch: while a learner watches one part of a lesson, the browser downloads the next saved part from the Cloudflare CDN	No wait between lesson parts; no extra server load (served by the CDN)	Launch
Assessment	Rule-based checking of MCQ, numeric and formula answers (SymPy, BSD) without an LLM; the LLM only explains written answers and mistakes	Quiz-grading LLM calls close to zero	Launch
Video	Lightning distilled LoRA for Wan 2.2 (4–8 sampling steps)	≈5–10× faster renders	Before launch
Video	TeaCache, SageAttention, FP8 inference, torch.compile and CUDA graphs	≈1.5–2× further; torch.compile and CUDA graphs add ≈1.2–1.3×	Before launch
Video	Frame interpolation: Wan 2.2 A14B generates at 16 fps and RIFE (MIT) adds the in-between frames for smooth 24/30 fps output; TI2V-5B already renders at 24 fps, so RIFE is used there only for 30 fps output	≈1.5–2× more output per GPU-hour; no visible quality loss in normal scenes (fast-motion scenes checked)	Before launch
Video	Image-to-video from an approved key image: for Cinematic hero shots FLUX.1-schnell makes a key image, the creator approves it, and Wan 2.2 animates that image	Same character look in every scene; ≈30–50% fewer rejected renders	Launch
Video	Automatic per-scene quality check before delivery: blur and flicker checks, prompt match (OpenCLIP, MIT) and character consistency (DINOv2, Apache-2.0); only a failed scene is rendered again, never the whole video	Fewer bad videos reach creators, so fewer costly remakes	Launch
Video	Batched rendering: scenes with the same resolution and length, even from different creators, run together in one batch on the 96 GB GPUs	≈1.2–1.5× throughput, no quality change	Before launch, benchmark
Video	Fast preview upscaler: previews use the compact Real-ESRGAN model (realesr-general-x4v3); final videos use the video upscaler and quality gate (4.13)	Faster previews and less GPU time; final-video quality unchanged	Launch

Area	Optimization	Expected gain (estimate)	When
Video	Audio-first timing: the voice-over is generated and locked first, and each scene's length is set from the audio timing before rendering	No re-renders for picture–voice mismatch; more accurate lip-sync	Launch
Video	Character LoRA for series: for a creator who uses the same characters in every episode, a small Wan 2.2 / FLUX.1-schnell LoRA is trained once from approved images (≈15–30 GPU-minutes, in the late-night batch window) and reused in every episode	Same character look across episodes; far fewer rejected renders	After launch, for series creators
Video	Render modes and per-scene routing (4.8): browser render (including standard voices and the cinematic 3D look), scene library/cache, 2.5D, preview-first	GPU cost per output minute ≈3–5× lower	Cache and preview before launch; browser render at launch once the 13.2 device tests pass (until then the same renderer runs on the server)
Video	Spot / preemptible cloud GPUs for Motion Studio's daytime burst rendering and for batch work, with on-demand GPUs as fallback; scene checkpoints let an interrupted job resume on another GPU	Burst GPU cost ≈50–70% lower	Before launch (Motion Studio and batch only; NexaTute real-time burst stays on-demand)
Image	Image cache: lab and whiteboard images and diagrams made by FLUX.1-schnell are saved by prompt, style and model version and reused for identical requests	No GPU time for repeated images	Launch
Recording	Lightweight group-study recording: separate tracks and whiteboard event log; download MP4s made in the user's browser (9.5)	No download compositing or encoding on the Rec nodes; less R2 storage	Launch
Delivery	AV1 viewing copies: in-app previews and recording playback are encoded in AV1 with NVENC (H.264 for devices that cannot play AV1); downloaded files stay H.264 MP4	≈30–50% less preview and playback bandwidth and storage at the same quality	Launch
Speech	faster-whisper with Whisper large-v3-turbo; TTS phrase cache and request batching	≈4–8× faster STT; fewer TTS calls	Before launch
Speech	Sentence-by-sentence streaming: the Nexa Teacher starts speaking and lip-syncing (browser TTS or server fallback) as soon as the LLM finishes the first sentence	Time to the first spoken word ≈3–5 seconds → under 1 second; fewer retries and drop-offs	Launch
API and database	Async FastAPI with uvloop, Valkey response cache, Cloudflare compression; index review of hot queries	Lower latency and CPU per request	Before launch

Area	Optimization	Expected gain (estimate)	When
API	Cloudflare Workers edge cache for read-only APIs (course catalogue, lesson content, package and price lists), served from the Dhaka edge and purged on change	Fewer App and database requests; faster responses in Bangladesh	Before launch
Scheduling	Late-night batch window (≈1 am–8 am, after the NexaTute night peak and before the Motion Studio daytime peak): weekly progress and weakness reports, subject trends and improvement rankings (3.4), guardian voice reports and new audio lessons, analytics, verified-answer cache refresh and other non-real-time jobs run at P10 on idle GPUs and CPUs	GPUs stay free for NexaTute at night and for paid Motion Studio renders in the daytime	Before launch
LLM	One hybrid Qwen3-30B-A3B with a per-request thinking switch (enable_thinking): thinking off by default and on only for hard maths, the Examiner agent and premium courses, within a token budget; the separate 2507 Instruct/Thinking models are used only if the benchmark shows much better quality, because running both doubles VRAM	Protects the 8.6 sizing, which already assumes non-thinking answers of ≈250 tokens; avoids the VRAM of a second model	Before launch, after the 3.13 quality gate
LLM	Maximum output length per agent; spoken AI Tutor answers kept short, with a "more detail" option	Protects the 8.6 sizing (keeps answers near ≈250 tokens)	Launch
Vision	NVIDIA Triton Inference Server (BSD-3-Clause) with dynamic batching for the NSFW classifier, BiRefNet, OpenCLIP, DINOv2 and GFPGAN; NVIDIA MPS lets these models share one GPU (own models only, never user code or lab sandboxes); internal only, mTLS through SPIRE	≈50–70% less GPU time for these models, lower VRAM	Launch
Speech	faster-whisper BatchedInferencePipeline for non-live jobs (class summaries, dubbing, audio checks)	≈3–4× faster for these jobs	Launch
Safety	Qwen3-VL second-stage frame checks on 448–672 px frames sent in batches; the NSFW classifier and child-safety hash matching keep full-size images; precision/recall tested on Bangla and Banglish sets, with fallback to full size	≈50–60% cheaper frame checks	Launch, after the safety test

Area	Optimization	Expected gain (estimate)	When
Video	Wan 2.2 A14B only for Cinematic hero shots; TI2V-5B (native 720p, 24 fps) for every other scene; the model for the 480p tiers (A14B at 480p or TI2V-5B) is chosen by the benchmark	≈20–35% less Cinematic GPU time	Launch
Search	BGE-M3 as ONNX INT8 on spare Core-2 CPU, only if Ragas retrieval quality drops by no more than 2%	Frees GPU memory	After the benchmark
Video	LightX2V (Apache-2.0) as the Wan 2.2 inference engine: step-distilled models, FP8/INT8 kernels and CPU offloading; output quality checked against the standard pipeline	≈1.3–2× faster Wan renders on top of the Lightning LoRA path (estimate)	Before paid launch, after the benchmark
Video	FastWan distilled Wan 2.2 TI2V-5B (FastVideo, Apache-2.0; weights licence to verify) for the non-hero scenes	≈3–5× faster TI2V-5B scenes (estimate)	Before paid launch, after the quality gate
Video	Wan2.2-S2V-14B (speech-to-video; Apache-2.0 on its model page, to verify) for Cinematic dialogue close-ups with realistic lip-sync, replacing MuseTalk/LatentSync; runs like A14B on the 96 GB GPUs; until it passes, dialogue scenes use Quick or Enhanced (4.14)	Realistic Cinematic lip-sync from the same model family, with no separate lip-sync model	After its licence check and quality gate
Image	Nunchaku 4-bit (SVDQuant) FLUX.1-schnell (Apache-2.0; weights licence to verify) for key images and thumbnails	≈2–3× faster images with ≈3× less VRAM (estimate)	After the image-quality benchmark
LLM	LMCache CacheBlend for knowledge-base turns: reuses the stored KV cache of retrieved curriculum passages even when their order or neighbours change	≈2–3× faster prompt processing on knowledge-base turns (estimate), with answer quality checked	After the benchmark
LLM	Tasks that used Qwen3-8B (simple tutor turns, summaries, mistake tagging, explanations, context summaries) run on Qwen3-30B-A3B with thinking off, or on Qwen3-4B on CPU for small text tasks; Qwen3-8B stays only as LLM overflow on Rec-1/Rec-2, where Qwen3-30B-A3B does not fit	Frees the VRAM of a separate 8B model on AI-1 for KV cache; one main model to tune and check; similar speed because only 3B parameters are active	Before launch, after the 3.13 quality gate
LLM	n-gram/suffix speculative decoding for the Motion Studio script LLM (vLLM's built-in method, or Arctic Inference, Apache-2.0, if faster) plus a compact storyboard JSON schema	≈1.3–2× faster scripts and storyboards (estimate); ≈4,500–6,000 users in the studio (8.6)	Before paid launch, after the benchmark

Area	Optimization	Expected gain (estimate)	When
Speech	Server voices for weak devices on CPU: the in-house VITS voices through ONNX Runtime (MIT) on spare Core-1/Core-2 cores, plus one OVHcloud CPU server beyond ≈8,000 NexaTute CCU; the GPUs keep speech-to-text, Chatterbox, Indic Parler-TTS and cloned voices	Frees ≈3 RTX PRO 5000 at the 16,000 CCU target; without it the server voice would cap NexaTute at ≈3,000–4,000 CCU	Before launch
Scheduling	Kubernetes Dynamic Resource Allocation (DRA) with the NVIDIA DRA driver (Apache-2.0): GPU requests by memory and sharing mode instead of whole GPUs; optional	Finer GPU sharing; used only if stable on the chosen Kubernetes version	Evaluation after launch

Combined effect (estimate): LLM throughput ≈2–3× and Motion Studio render speed ≈5× or more on the same servers. Together with the cache-hit sensitivity in 8.6, NexaTute launch capacity could reach ≈4,000–6,000 CCU on the 16 launch servers while they also serve Motion Studio (NexaTute-only figures are in 8.6). Quality gate: every optimization must pass the same Bangla and curriculum test set before and after it is enabled; any measurable quality drop means it stays off until fixed.

8.16 Data Centre Regions (USA)

- Location: all 16 launch servers are in the USA, from three suppliers. Core site in Northern Virginia: OVHcloud US, Vint Hill (App-1, App-2, DB-1, DB-2, Redis1, Redis2; App-3 from December 2026) and RedSwitches, Washington DC (Core-1, Core-2 for LiveKit, LB-1, LB-2), about 1–3 ms apart. GPU site: Database Mart, Dallas, Texas (AI-1 A/B, AI-2 A/B, Rec-1/Rec-2), about 30 ms from Virginia. Hourly GPU burst and Motion Studio overflow run in the USA; the disaster-recovery replica runs in a second region (OVHcloud Hillsboro, Oregon). No servers are hosted in Bangladesh or Europe; Cloudflare's Dhaka edge serves cached content locally.

Alternatives checked: Hetzner GEX45 and GEX63 GPU servers were rejected (Europe-only, higher cost and no 10 Gbps unmetered bandwidth); servers in Bangladesh, India or Singapore cannot be had with more than 1 Gbps unmetered bandwidth (28 October 2026), so the launch stack stays in the USA and an edge server is added only if the Free Beta latency shows the need (9.5).

- Keep each cluster inside one region: Patroni synchronous replication, etcd quorum, Valkey Sentinel, the App and LB pairs and the LiveKit cluster need low latency between members, so each of these clusters runs entirely within one region; only asynchronous replicas, backups and burst capacity may cross regions.
- Site links: OVHcloud, RedSwitches and Database Mart are joined over the internet through encrypted WireGuard tunnel pairs (two tunnels per link, automatic failover) with mTLS between services. Patroni, etcd and Valkey Sentinel stay on the OVHcloud private network; GPU workers reach the queues and APIs in Virginia only through the tunnels, and nothing that needs a quorum spans two sites. Recording nodes in Dallas pull class media from LiveKit in Virginia (≈3 Mbps per recording); track-only overflow runs on spare Core-1/Core-2 CPU next to LiveKit. If the Virginia–Dallas link fails, the Dallas nodes keep their running pods (the node-eviction timeout is extended so a short outage does not reschedule GPU work), new AI requests use the cloud fallback (8.9) and Free Beta users wait in the queue.
- Latency to learners in Bangladesh: the US East coast is typically ≈220–250 ms round trip from Bangladeshi broadband users (≈110–125 ms one way). That is still within the usual limit for live voice and video (about 150 ms one way, ITU-T G.114), but with less margin than a European or Asian site, so live classes must be tested on Bangladeshi mobile networks before go-live. AI

requests travel App (Virginia) → GPU cluster (Dallas) and back, adding ≈30 ms per call — small next to the seconds an answer takes, and hidden further by sentence-by-sentence streaming. Saved lessons, browser models, AV1 viewing copies and read-only APIs come from the Cloudflare Dhaka edge, and voice and avatar run in the browser.

Region	Servers (proposed)	Why	Watch-outs
USA East — Northern Virginia (OVHcloud Vint Hill + RedSwitches Washington DC)	Ten servers (eleven with App-3 from December 2026): App, DB and Valkey clusters at OVHcloud; LiveKit cluster with TURN (Core-1, Core-2) and LB pair at RedSwitches; control plane on Redis1, Redis2 and App-1	Main US internet hub with good routes to Asia and Europe; the data tier shares one provider's private network	Confirm RedSwitches offers the Core-1 10 Gbps server; LB ↔ App traffic crosses providers within the same metro (≈1–3 ms)
USA Central — Dallas (Database Mart)	Six GPU servers as worker nodes of the single Kubernetes cluster (8.11); hourly GPU burst and Motion Studio overflow rendering nearby	Required RTX PRO GPUs at the quoted prices	Confirm which Database Mart site (Dallas or Kansas City) hosts each GPU server; ≈30 ms to Virginia
USA West — Oregon (OVHcloud Hillsboro)	Disaster-recovery replica (asynchronous)	A second region for full-site recovery	Asynchronous replication only
Cloudflare (Dhaka edge)	CDN, WAF, edge busy page and cached content (9.4)	Content served locally in Dhaka, cutting latency and international traffic	—

- International learners (Hindi, Urdu, Arabic and French markets) use the same LiveKit cluster in Virginia; Cloudflare serves static content, saved lessons and model files from their nearest edge.
- Disaster recovery across regions: the R2 live database backup plus an asynchronous replica in the other region allow a full-region restore within the RTO in 19.5.
- Data protection: learner data is processed and stored in the USA, so data-processing agreements are needed with every supplier, cross-border transfer rules for Bangladesh must be confirmed, and GDPR safeguards apply to any learners in the EU; children's data needs extra care. Legal review is required before launch (19.4).
- Growth options (2.5): telco zero-rating may need a dedicated learning domain and dedicated Cloudflare IP addresses; government (B2G) contracts may require hosting government school data in Bangladesh, which would run as a separate in-country tenant of the same platform.

8.17 Ecosystem Resource Sharing and Placement Priority

Rule: every product — NexaTute (with PlayBoom and the Live Study Hall), Remon Motion Studio and NexaLogicDev — may use any idle GPU or CPU capacity in the ecosystem through one scheduler. Each pool has an owner whose work always comes first; borrowed work runs only on idle capacity and is handed back within one scheduling cycle. A fixed placement order decides which server each workload tries first. This extends the two-product sharing of 8.8–8.11 to all 28 servers; the sandbox, data, edge and recovery servers are the only exceptions, for security and failover reasons.

Table 1 — Pools, owners and lending

Pool	Servers	Owner (first call)	Lends to	Lending condition
Launch GPU pool	AI-1 Node A/B, AI-2 Node A/B, Rec-1/Rec-2 (6 GPU servers, Dallas)	NexaTute (P100, 60% floor) and Motion Studio (25% floor); recording first on Rec-1/Rec-2 (8.8)	NexaLogicDev Fast-lane overflow (paid users) on AI-1/AI-2 nodes	NexaTute load below 50% and no Motion Studio Express backlog; Rec nodes are not lent to NexaLogicDev
Fast GPU pool	Fast-1–Fast-4 (4 × RTX PRO 6000, Dallas)	NexaLogicDev Fast lane (Qwen3-Coder-Next)	NexaTute LLM and speech-to-text overflow, Motion Studio script LLM and renders, ecosystem batch work	Fast-lane load below 50% for 10 minutes; at least 2 GPUs always keep Qwen3-Coder-Next loaded
Heavy GPU pool	Dev-H1–Dev-H3 (12 × RTX A6000)	NexaLogicDev Heavy lane (MiniMax-M2.7, one copy per server)	Motion Studio Normal/Slow renders; NexaTute and Motion Studio batch work (content pipeline, translation, LoRA training, class summaries, Script Checker, explainer and marketing videos)	Lending window 1–7 am Bangladesh time when the Heavy queue has been empty for 30 minutes, and June–September 2027 (before launch) outside benchmark slots; lent one server at a time while the others keep serving (the whole pool only with the GLM-5.2 fallback)
CPU worker pool	App-1–App-3; spare Core-1/Core-2 cores; Heavy-server CPUs (3 × 44 cores, 512 GB RAM)	Each product's API pods on the App nodes (separate namespaces and connection caps, 8.8); LiveKit on Core-1/Core-2	Trusted CPU work of every product: in-house VITS voices, OCR, Office checks, site builds, transcoding, analytics, nightly model retraining	App CPU below 60%; Core cores only while LiveKit has headroom; Dev CPUs only for trusted batch and near-real-time work, never API or database pods
Sandbox pool	SBX-1–SBX-3 (OVHcloud ADVANCE-2)	NexaLogicDev user code, builds and tests	Not lent, and does not borrow	One-way isolation: nothing else runs on these servers and user code runs nowhere else
Data, edge and recovery	DB-1/DB-2, Redis1/Redis2, LB-1/LB-2, the LiveKit role of Core-1/Core-2, recovery replica	Their pinned role	Not lent	Fixed roles keep data safe and failover predictable (8.13)

Table 2 — Placement order (which server is tried first)

Workload	Priority	1st	2nd	3rd	Last resort
NexaTute AI Tutor, Nexa Co-Teacher and batch-class LLM	P100	AI-1 Node A/B	AI-2 Node A/B, then Rec-1/Rec-2 while no recording runs	Idle Fast GPU at P70 (Qwen3-30B-A3B from local cache; text only)	Cloud API within the spend cap (8.9)

Workload	Priority	1st	2nd	3rd	Last resort
NexaTute server speech (weak devices) and class-summary transcription	P100 (summaries P10 → P100)	AI-2 Node A	Other launch GPUs	Idle Fast GPU — adult learners only; PlayBoom and school learners' audio stays on the launch pool	Text fallback (8.9)
NexaTute voice clones (Avatar & Voice Studio) and Nexa Notes subtitles	P10, off-peak; daily GPU-minute cap for clones	AI-2 Node A (speech pool)	Other launch GPUs in the night batch window (Bangla and Urdu VITS training)	Idle Fast GPU — adult voices only (guardians, teachers, 18+ learners)	Waits for the next off-peak window (no cloud)
Written-answer checking (Qwen3-VL-8B reading and Qwen3 marking, 3.27)	P50, queued (free-tier checks P10); daily GPU-minute cap	AI-2 Node A	Other launch GPUs outside the live-class peak	Dev-H servers in the night lending window	Waits in the queue with a notification — never an outside AI service
Motion Studio script and storyboard LLM	P50	AI-1 Node B	AI-2 nodes	Idle Fast GPU	Hourly cloud GPU
Motion Studio Express render	P50, first in queue	Any free launch GPU (AI-2 Node B, then AI-1 Node B in the daytime)	Idle Fast GPUs	Dev-H servers in its lending window	Hourly cloud GPU to keep the shown ETA
Motion Studio Normal and Slow renders	P50 / P10	Dev-H servers in the 1–7 am lending window	Idle launch GPUs	Idle Fast GPUs	Cloud only if the 12-hour or 24-hour delivery time is at risk
Ecosystem batch work (content pipeline, translation, LoRA, Script Checker, model evaluation, explainer and marketing videos, staff-chat summaries, Nexa Notes subtitle translation)	P10	Dev-H servers in the lending window	AI-1 nodes in the night batch window	Idle Fast GPUs	Waits for the next window (no cloud)
NexaLogicDev Fast lane	P100 paid / P10 free	Fast-1–Fast-4	Idle Heavy servers (M2.7 serves the request), then idle AI-1/AI-2 GPUs at P70 (Qwen3-Coder-Next from local cache)	An extra Fast GPU rented at the 70% trigger	MiniMax M3 API for paid users; free users wait (32.1)

Workload	Priority	1st	2nd	3rd	Last resort
NexaLogicDev Heavy lane	P100 paid	The Dev-H server with the shortest queue (one M2.7 copy each)	The other Dev-H servers	A lent-out server is reclaimed: guest work checkpoints and M2.7 reloads in ≈1–2 minutes, with the ETA shown	Case 1: the Fast Lane writes the plan and Claude Sonnet 5.5 reviews it; our servers build (Pro/Team, Auto-Architect only — Student/Hobby never use the Heavy Lane) (32.1)
CPU work (voices, OCR, builds, analytics)	By product	App-1–App-3	Spare Core-1/Core-2 cores	Heavy-server CPUs	Extra OVHcloud CPU server (8.5)
MindForge puzzle generation and calibration, Study Hall Q&A checks, Growth Platform website reading and accounting exports	P10	CPU worker pool (App-1–App-3)	Spare Core-1/Core-2 cores	Heavy-server CPUs (trusted batch only)	Waits for the night window
Graded coding-test runs (3.27)	P50 while a test runs	Practice-lab gVisor sandboxes on spare Core-1/Core-2 cores (LiveKit keeps its CPU first, 8.12)	App-1–App-3 workers	—	Queue with ETA; never the NexaLogicDev sandbox pool or the Heavy servers
Nexa Notes HLS transcoding and avatar LOD copies	P10	CPU worker pool (App-1–App-3) in the night window	Spare Core-2 cores (never Core-1, 9.3)	Heavy-server CPUs (trusted batch only)	Waits for the next night window
AI write → run → fix loops, tests, scans and database backends (NexaLogicDev; previews run in the browser, 32.14)	P100 paid / P10 free	SBX-1	SBX-2	SBX-3	Queue with ETA; never on another server

Priority and hand-back rules

- Who wins on a pool: REC (recording on Rec-1/Rec-2) → the owner's real-time work (P100) → real-time work borrowed from another product (P70, new) → paid renders and paid batch (P50) → free/trial and internal batch work (P10). NexaLogicDev paid interactive work is P100 on its own Dev pool and P70 when it borrows the launch pool; NexaTute real-time work is P100 on the launch pool and P70 on the Fast GPUs.
- Lend and reclaim: a pool lends when its owner's load stays below 50% for 10 minutes and takes capacity back above 70%. The borrower warms a model when its own pool passes 60% and sends traffic from 70%. On reclaim, LLM answers finish (never cut mid-answer), renders checkpoint after the current scene and batch jobs checkpoint; hand-back within one scheduling

cycle (≤60 seconds for LLM GPUs, one scene for renders, ≈1–2 minutes for a Heavy server while M2.7 reloads, ≈3–5 minutes for the whole pool with the GLM-5.2 fallback).

- Floors stay: on the launch pool NexaTute 60%, Motion Studio 25%, 15% floating (8.8); on the Dev pool the Heavy lane is never lent outside its window and at least 2 Fast GPUs always serve Qwen3-Coder-Next. Guests use only idle capacity above these floors.
- Models ready everywhere: every GPU server keeps the weights it may serve on its local NVMe (Heavy servers 4 TB; the launch and Fast GPUs hold both Qwen3-30B-A3B and a 4-bit Qwen3-Coder-Next copy of ≈45 GB), pulled from the Harbor mirror on Core-1, so a borrowed GPU starts serving in ≈30–90 seconds.
- How the choice is made: the RemonLogic Resource Broker ranks servers by owner, free VRAM, GPU busy % (DCGM Exporter), queue wait and network distance; Kueue cohorts enforce the lending and borrowing limits and pre-emption; the Inference Gateway sends each LLM request to the least-loaded allowed vLLM instance by priority; KEDA scales model replicas on queue depth.
- Safety and privacy: only RemonLogic-built, signed images run as guests; guest pods run in their own product namespace with Cilium network policy, per-product OpenBao secrets and no database credentials of the lending product; prefix and KV caches are never shared across products; raw audio, images and face data of minors (PlayBoom and school learners) are processed only on the launch pool; Dev-pool guests receive pseudonymous job payloads only; user code never leaves the sandbox pool.
- Accounting: every GPU-second and CPU-second is recorded per product (DCGM Exporter, Kueue and OpenCost); the monthly report shows borrowed hours at cost per server-hour between the Funding A and Funding B budgets.
- Control: lending windows, thresholds and a per-pool on/off switch are admin-panel settings; lending stops automatically when a pool's circuit breaker opens (8.9); the capacity dashboard (I.2.6) shows each pool's owner load and its lent and borrowed capacity.
- Expected effect (planning estimates, to benchmark): lending the Heavy pool (12 × A6000) for about 6 hours a night adds roughly 25–30% to daily Motion Studio video capacity for Normal and Slow orders and moves batch work off the launch GPUs; Fast-GPU overflow adds NexaTute LLM headroom at the evening peak when NexaLogicDev is quiet; the Heavy servers' 132 CPU cores can carry the in-house VITS voices and batch CPU work and may delay the extra OVHcloud CPU server (8.5). Pools that cannot be shared — sandbox, data and edge — are sized on their own (SBX-3, 7.2). Wan 2.2 on A6000 (no FP8) is benchmarked before render lending starts; the Fast GPUs run Wan 2.2 in FP8 like the launch pool.
- Video GPU trigger: from October 2027 the video pool's busy time is reviewed weekly; a new video GPU server (RTX PRO 6000 class, Dallas, 8.5) is ordered when the pool is ≥60% busy at its daily peak for a week and put into service at 80% or as soon as paid Express or Normal ETAs start to slip. Lent Heavy-pool hours count only inside the lending window. The capacity check projected 53–82% video-pool use in December 2027 and 65–100% in March 2028 under the 15.8 base case before lending; Normal and Slow delivery (G.8.3) move most renders out of the daytime peak and night lending adds roughly 25–30%, so the trigger is expected later — how much later depends on the share of Express orders, shown on the admin dashboard.

8.18 Browser-First Execution (all products)

Principle: NexaTute (with PlayBoom and the Live Study Hall), Remon Motion Studio and NexaLogicDev run in the user's web browser everything that a normal device can run. Our servers keep only five kinds of work: (1) large AI models (LLMs, video and image generation); (2) anything that needs secret keys (payments, external APIs, C2PA signing); (3) results that must be trusted (money, ranked-test grades, safety checks, certificates); (4) work that must continue after the tab closes; (5) the fallback for light or weak devices. This lowers server load, keeps personal material on the device and makes the products feel faster.

Product	Runs in the user's browser	Stays on our servers
NexaTute, PlayBoom, Live Study Hall	Voice (Whisper, in-house VITS); Nexa Teacher video and 3D avatar; Smart Whiteboard; Virtual Lab; 3D classroom and Classroom Builder; AI-tutor recording and download MP4s; on-screen OMR; face verification; PlayBoom animations and handwriting tracing; small WebLLM model. Added: OCR of printed text (Tesseract.js — only hard questions go to the GPU); pronunciation scoring (wav2vec2 through ONNX Runtime Web); MCQ and numeric checking and the Learning Insights view; search in the learner's own notes; lesson cache and prefetch; image resizing before upload; MindForge puzzles, practice scoring and offline daily puzzles; Study Hall Q&A image resizing, QR and phone-number blurring; calculator and dictionary; PlayBoom offline packs, read-aloud checks and music pads	Nexa Teacher and Nexa Co-Teacher LLM; Qwen3-VL for hard photo questions; safety models; live classes (LiveKit) and recording; ranked tests and the master learner record; payments; server voice and speech fallback; MindForge baseline, monthly check and calibration; Study Hall Q&A safety re-checks
Remon Motion Studio	Quick-mode render, 3D avatar and voice, Quick-mode lip-sync, timeline editor, uploads straight to R2. Added: Enhanced (2.5D) motion and render from server-made key images (Three.js); captions, logos, brand kit and thumbnail text; audio mixing and loudness (Web Audio); auto-reframe and Shorts scene detection; final MP4 assembly for Quick and Enhanced (WebCodecs + Mediabunny); preview upscaling and product background removal on strong devices	Cinematic video (Wan 2.2); FLUX images; script and storyboard LLM; voice cloning and dubbing; safety checks; C2PA signing; server takeover after 30 minutes offline (4.8)
NexaLogicDev	Editor, lint and format, previews (Sandpack, WebContainer, php-wasm), design canvas, voice to code, offline mode and Git (isomorphic-git) — 32.14	Heavy and Fast AI lanes; AI write → run → fix loop, tests, scans and database backends in the sandbox; deploys, domains and payments

Device tiers

Tier	Devices	What runs where
Full	Android 10 or later with at least 4 GB RAM, recent Chrome or Edge, laptops and desktops	Every browser feature, including in-browser models
Light	Android phones with 3 GB RAM; Safari and iPhone	Browser features without heavy models; voice, OCR, pronunciation, upscaling and Enhanced render use the server when needed
Server fallback	Below 3 GB RAM	Mainly on the server, at the same price; ETAs may be longer at peak

- Device Check: at first sign-in, and after a browser update, the app measures RAM, WebGPU and WebAssembly support, free storage and connection speed in a few seconds and sets the tier; users can see and change it in settings.
- Data Saver: on by default on mobile data — browser models (Whisper ≈75–150 MB, small LLMs, upscalers) download only on Wi-Fi or with the user's consent, are cached on the device and update only when they change; videos and previews use lower quality. On Wi-Fi the full tier returns.

- Trust rule: anything that decides money, ranked-test grades, safety or certificates is computed or verified on the server; the browser shows its result first and the server result is the record.
- Measured offload: the capacity dashboard (I.2.6) shows, per feature, the share run in the browser and the share that fell back to the server, by tier and device type, from the browser OpenTelemetry SDK (no personal content is sent). Once Free Beta data exists these shares replace the planning estimates in 8.6 and 32.13.
- Capacity effect: server CPU and GPU load falls for OCR, pronunciation scoring, Enhanced renders, MP4 assembly and IDE previews. The 8.6 figures stay as the upper bound until measured, and server fallback capacity is kept for the light and fallback tiers.

09. Data, Storage, Recording and Media Lifecycle

9.1 Data authority

- PostgreSQL is the durable source of truth for accounts, projects, scenes, jobs, credit reservations, usage ledger, settlement/refund events and other transactional state.
- Redis is for transient session/cache/presence, queue/broker, throttling and progress signaling; it must not be the sole record of financial state or durable job completion.
- Define primary/replica topology, replication lag policy, backup retention, restore tests and failover decision process.
- PostgreSQL runs as a Patroni cluster with a live pgBackRest backup to Cloudflare R2 (RPO ≤1 minute); configuration and recovery details are in 8.13.
- Learner passport and institution data (3.11): each institution's records are kept apart by tenant; learners can export their own record, and verifiable certificates are signed with keys held in OpenBao.
- Training and research data (2.5): only consented data is used, personal details are removed with Presidio before any model training or research export, minors' data needs guardian consent, and every export is logged.
- Archive policy: high-volume tables (lab and assessment events, usage ledgers, audit logs) use monthly PostgreSQL partitions; partitions older than 12 months are exported to an encrypted archive bucket in Cloudflare R2 as compressed Parquet files and dropped from the live database. Learning data stays restorable from the archive for 5 years and billing records for 7 years (11.9). An alert fires when DB disk use exceeds 70%.
- Staff chat (38.15): full messages are kept 90 days and the encrypted daily summaries 6 months (a compliance case until it is closed or a legal hold ends) — the lawyer confirms the periods; deletion runs as a nightly job and every archive access is logged.
- Tests (3.27): written-test images are kept under the same rules as other learner images (19.4, 36.1) and are seen only by the learner and the guardian; adult proctoring snapshots and the consent records are kept 6 months and then deleted (decided 25 October 2026); marking schemes keep every version.
- Use idempotency keys to prevent duplicate charges or duplicate rendering on retries.

9.2 Video job state machine

State	Required behavior
Requested / validated	Authenticate user, validate plan/content policy, permissions, quota and job payload
Credits reserved	Persist reservation atomically before paid GPU work
Queued	Persist priority, enqueue time, attempts, job ID and product ownership
Processing	Track step, progress, worker/node, heartbeat and bounded retry count

State	Required behavior
Compiled	Final MP4 created locally; verify format, size, duration and expected output
Uploaded	Transfer final artifact to R2; verify checksum/size and durable acknowledgement
Completed	Persist final artifact reference, usage settlement and access policy
Failed / retryable	Preserve draft and local output as applicable; release/refund credits according to explicit policy
Cleanup complete	Delete only exact task-owned file after upload verification; record cleanup event

9.3 Scratch-buffer and safe cleanup rules

- Keep heavy GStreamer writes, temporary chunk buffers, intermediate files and final MP4 compilation on the local storage of the GPU node executing the job (AI-2 Node B preferred), never over the network and never on Core-1.
- Rec-1/Rec-2 each have a 320 GB local SSD buffer in the v24.0 inventory; monitor free space and queue age, and apply backpressure before disks fill.
- Upload the completed final file, validate its checksum/size and receive durable acknowledgement before deleting local output.
- On upload or validation failure, retain the local file for retry/recovery.
- Cleanup must target only the exact task-owned file inside the designated scratch directory. Reject symlinks, non-regular files, path traversal and unexpected extensions. Do not use recursive deletion as a job cleanup shortcut.
- Use append-only usage/job events where possible, and record upload, validation and cleanup results for audit.

9.4 Cloudflare R2 and delivery

- Use R2 for final media and permitted archives; object-storage lifecycle/retention settings must match product policy.
- CDN delivery: everything that can be cached reaches users through Cloudflare's CDN (Dhaka edge for Bangladesh) — the web app, browser speech and LLM models, 3D assets, saved lessons, the image cache, audio lessons, explainer videos, the live-quiz HLS stream, AV1/H.264 viewing copies and finished Motion Studio videos, plus read-only APIs through Cloudflare Workers. Only public content is cached at the edge — Nexa Notes videos, images, pages and app files; private recordings are never edge-cached and are served through signed links valid 15 minutes (default, editable), so a copied link fails for anyone else; R2 has no egress fee. Live classes (LiveKit WebRTC), AI answers and write APIs cannot be cached and go to the servers in the USA.

Video delivery (decided 28–29 October 2026; free and open-source parts): Nexa Notes and group-class videos use FFmpeg (LGPL-only build, NVENC or OpenH264 instead of GPL x264) to make 720p, 480p and 240p, Shaka Packager for the HLS segments and playlists, R2 storage (no egress fee) behind the Cloudflare CDN on our own domain with the Dhaka (DAC) edge, the R2 location hint APAC and Tiered Cache, and hls.js with Video.js or our own light player for controls, captions and mistake markers (Safari and iPhone play HLS natively); browser uploads use Uppy with S3 multipart; Cloudflare Stream (paid per minute) is not needed. Class recordings are lightweight replay files and need only R2, the signed links and our replay player — no FFmpeg or HLS. The Free Beta measures which edge each operator's users reach (the cf-ray colo, for example DAC) and the cache-hit rate, shown as the Edge check in platform operations (38.18); a BDIX-connected local cache server is only a later fallback if some operators do not reach the Dhaka edge.

- Creator-tool libraries — music beds, the CC0 sound-effect library, templates and caption fonts — are stored in R2 (≈50–100 GB at launch) and served through the CDN; uploaded source videos for Shorts or translation follow the same storage, retention and 72-hour rules as finished videos, and a takedown (4.12) deletes them.
- Creator and learner uploads go straight from the browser to R2 in resumable parts (Uppy S3 multipart with short-lived presigned URLs from the App, each scoped to one object), within the per-package size and length limits (4.16); the App never receives the file, and the Document Intake Service (5.3) reads it from R2; project export bundles are built by an App-cluster worker and delivered as signed R2 links.
- NexaTute uploads (3.14) are stored privately in R2; homework images are deleted by an R2 lifecycle rule after 90 days; teacher question banks enter the shared knowledge base only after the 3.13 verification, in the late-night batch window.
- As soon as the final MP4 passes validation it is uploaded to Cloudflare R2 and the local copy is purged (9.3). If the client has storage available (an active package with free space under G.3), the file stays in their R2 storage. If the client has no storage available (no active package storage or the package's storage allowance is full, G.3), the file is placed in a temporary R2 delivery bucket, a signed download link valid for 72 hours is emailed to the client, and the file is deleted automatically after 72 hours by an R2 lifecycle rule.
- Notifications: ready-video emails and 72-hour download links are sent through a managed transactional email service with US and EU regions (for example Amazon SES), with SPF/DKIM/DMARC, bounce handling and automatic retry; the same notice is also shown in the studio and sent as a browser push notification, so delivery does not depend on email alone.
- Use scoped service credentials, private objects by default, short-lived signed URLs and authorization checks before delivery.
- Use retry-safe uploads and checksum/integrity verification. Never expose storage credentials in the browser.
- Set retention, deletion and legal-hold rules; define recovery if an object upload succeeds but the database update fails, or vice versa.
- Website media (3.25): the CMS media library is stored in R2 as WebP/AVIF images and served through the CDN; page versions and their approvals are kept in PostgreSQL. Nexa Notes videos are stored in R2 as HLS renditions (720p, 480p, 240p) with WebVTT subtitles and the master MP4 and served through the CDN; YouTube videos are only embedded click-to-load. Avatar GLB/VRM files and their lighter copies are stored in R2 and cached in the browser; voice samples and voice models are encrypted (38.16).

9.5 Recording and LiveKit

- Recording is by choice (3.29, 11.3): an AI-tutor class is recorded only when the learner chooses it, and a batch class once for the batch when at least one member asks. For an AI-tutor class the learner's browser writes an event log — the AI's text, whiteboard events and timing — with only the learner's own voice as Opus audio, and uploads it straight to R2 through a short-lived signed URL; the AI voice is not stored but re-created on replay by the in-browser VITS voice with the pinned voice version (server CPU voices on weak devices). A batch class is recorded once for the whole batch by the batch-class service (event log plus raised-hand audio, 3.17). The server verifies consent, size and checksum before accepting a file. Rec-1/Rec-2 handle recording/egress for human-led live and group classes and for server-fallback sessions; Core-1 and Core-2 form the LiveKit SFU cluster.
- Storage: an event-log recording is about 5 MB per 75-minute class (an MP4 would be about 280 MB), so 10,000 learners who use all their daily minutes add about 1.5 TB a month to R2; R2 has no egress fee, so unlimited replay adds no delivery cost.
- Learner-record growth (3.16, 3.19, 3.23): Nexa Co-Teacher summaries and handover notes are a few KB of text per learner a day in PostgreSQL; xAPI statements from both AI roles go to

ClickHouse; link results are kept only as short summaries and topic mappings, never as copies of the page.

- Download MP4 made in the browser: when a user downloads a recorded class, the user's browser builds the MP4 itself from the stored tracks and the whiteboard event log — the same replay the NexaTute player already shows. Frames (the video tiles and the Smart Whiteboard canvas) are drawn in a Web Worker with OffscreenCanvas and encoded to H.264 with WebCodecs, using the hardware encoder where available; the class audio tracks are mixed with Web Audio and encoded as Opus, which MP4 supports (I.2.4; AAC only if the player-compatibility test in 13.2 shows it is needed); Mediabunny (MPL-2.0) writes the MP4, which is saved straight to the device and never stored on the server. The job runs in 1-minute parts kept in the browser's private storage (OPFS), so it resumes after the tab is closed or the connection drops; progress, time left and a keep-this-tab-open note are shown, and the Screen Wake Lock keeps the PC awake (as in 4.8). Nothing is composited or encoded on the servers for downloads, so Rec-1/Rec-2 only record. On phones the MP4 takes longer to make; the recording can always be watched in the NexaTute player without making an MP4.
- Lightweight group-study recording: for group study and human-led classes, Rec-1/Rec-2 save the separate tracks without compositing — Opus audio, the speaker and listener video tracks at the recorded profiles, and the whiteboard as a Yjs event log; the NexaTute player replays them together in the browser. An MP4 is made only when a user asks to download the recording, by that user's browser (above), and it is not stored on the server. Expected effect: no compositing or encoding on the Rec nodes for downloads, and less R2 storage because the whiteboard is stored as events and download MP4s are not kept.
- Rec-1/Rec-2 are part of the unified GPU cluster (8.11): recording always has top priority on them, they take post-processing and light overflow work only when idle, and queued recordings overflow first to spare Core-1/Core-2 CPU, then to idle GPUs on the other cluster nodes.
- Recording layout: recordings use the same speaker/listener profiles and automatic switching as live video (3.2), for both browser and Rec-node recordings.
- LiveKit bandwidth: Core-1 (10 Gbps) + Core-2 (1 Gbps) = 11 Gbps unmetered. New rooms are placed by available bandwidth, so most traffic runs on Core-1. If Core-1 fails, Core-2 alone (1 Gbps) keeps classes running by switching every room to speaker 480p / listeners 160p and queueing new rooms (8.9 L2–L4); at about 0.7 Mbps per participant this protects roughly 1,400 participants.

LiveKit infrastructure (approved 28 October 2026): at least two LiveKit nodes with Valkey (Redis) for routing, so there is no single point of failure — met by Core-1 and Core-2; TURN on TLS port 443; Opus FEC always on and RED switched on automatically when packet loss appears (the rule is editable in platform operations); short-lived room tokens, role permissions and signed webhooks; per-class connection quality stored in ClickHouse and shown by area and operator; a monthly LiveKit version check, with updates through the software-updates panel (38.18). There is no edge server at launch: the Free Beta measures latency and a 1 Gbps Singapore or Mumbai edge is added only if needed; AI response time is cut first — the Nexa Teacher speaks from the first sentence and cached verified answers are used.

- Core-1 is the only server type available with 10 Gbps; other servers come with 1 Gbps or 500 Mbps. LiveKit therefore keeps most rooms on Core-1, and scale-out uses another Core-1-type server only if the supplier can provide one, otherwise several 1 Gbps LiveKit nodes.
- Recording permission, visible status, consent, plan allowance and retention must be checked before starting.
- Summary-only audio: when recording is off, class audio is captured only with the participants' consent to make the class summary; it is transcribed, then deleted as soon as the summary is saved, and is never replayable or billed as recording.
- Capacity depends on participant tracks, subscribed layers, bitrate, codec, room topology, recording layout and network conditions; validate using representative load tests.

- Test recording failure, reconnects, egress retries, audio/video synchronization, final container integrity and upload recovery.

10. Security, Privacy, Safety and Governance

Scope: Section 10 sets cross-platform principles; the detailed security architecture, authentication flow, roles, incident response and security tests are controlled by Section 19.

10.1 Cross-platform security controls

- Authenticate all users and services; use least-privilege service identities and scoped secrets.
- Separate tenant and product access; enforce authorization on every object, job, recording and signed URL.
- Use encryption in transit and supported encryption at rest; maintain auditable access and administrative logs.
- Staff chat (38.15) is encrypted in transit and at rest but not end-to-end, because every message is checked on send; files are virus-scanned; the staff policy states that work chats are monitored; push text never contains secrets.
- Rate-limit ingress, validate payloads, constrain upload types/sizes and protect internal service endpoints.
- Threat-model scratch cleanup, path traversal, symlink/TOCTOU races, object-storage callbacks, signed URL exposure and job replay.
- Keep API keys in a secrets manager/environment injection system; never commit real secrets into source/configuration examples.

10.2 NexaTute safety and learner privacy

- Separate student, teacher, parent/guardian, institution-admin and support permissions where those roles exist.
- For minors, apply guardian/consent review, recording status, retention and deletion controls appropriate to target jurisdictions.
- Learning OS (3.11): no private adult–minor messaging without guardian visibility; AI proctoring only with consent and only as flags for a human reviewer; outside plug-in apps get no learner data beyond their approved scopes; long-term AI memory is opt-in and can be cleared.
- Own avatar and voice (3.10): voices are cloned only for adults (teachers, parents and learners aged 18 or over) after the speaker reads a randomly generated consent sentence aloud, so another person’s recording cannot be used; cloned voices are used only inside NexaTute for learning, cannot be downloaded, carry an inaudible watermark, and can be deleted by their owner at any time; children choose cartoon-style avatars, never photo-real likenesses of themselves. Guardians clone only their own voice for PlayBoom stories, never a child’s; verified teachers may make an avatar from their own face photo after a liveness check and face match, used only in their own content (38.16).
- Tests (3.27): written scripts are checked by Nexa Guard, stripped of location data and never sent to an outside AI service; proctoring is limited to certificate and institution exams, with no video or snapshots of learners under 18 and no emotion or attention reading; employer tests are for adults only.
- Review curriculum alignment, scientific correctness, accessibility, language quality and model failure modes.
- Provide reporting and human review for high-risk cases. Dangerous real-world procedures should not be presented as AI-only instructions; the Smart Lab MVP is a virtual simulation environment.
- Explain that learning analytics are coaching signals, not guarantees of academic results.

- Wellbeing: if a learner shows signs of serious distress or a safety risk, the AI pauses teaching, responds with care, informs the guardian in an age-appropriate way and shows helpline information.
- Anti-bullying in batches (3.17): Nexa Guard (bad-language filter, report button and behaviour-change ladder) and nicknames; pair rooms for minors use text chat.
- Content safety, extending 4.15 to NexaTute: learners' spoken words in live classes are filtered in real time from the speech-to-text transcript; Nexa Teacher and Nexa Co-Teacher answers are checked by Qwen3Guard; teachers' lesson scripts and question banks are checked before publishing, under a teacher code of conduct; context rules follow the subject, so religion discussed in religious studies or wars in history are not treated as insults or violence.
- Behaviour-change ladder — education, not punishment, and always private: 1st offence, a gentle message explaining why; 2nd, a 5-minute "digital manners" mini-lesson (respectful language and respect for other people's religion and country) and a private note to the guardian; 3rd, chat and uploads off for 24–72 hours while study continues; serious cases (pornography, child abuse, threats), an immediate block, an alert to the NexaTute human controller, and the guardian or institution informed and the case reported as the law requires. Respect points and a good-conduct badge reward clean conduct, and a clean-conduct streak restores lost privileges. Teachers: the institution admin is informed and the account reviewed, with warning → suspension → ban and appeal, as in 4.15.
- Human controllers (19.3): their names, email addresses and mobile numbers are set in the admin panel; normal alerts go by push notification and email, very serious alerts by push notification, email and SMS. An alert that is not acknowledged within 15 minutes escalates to a second controller with another SMS; at least two controllers cover a 24/7 rota, and every alert and action is logged.
- Face data (3.16): face templates are biometric data — collected only with consent (guardian consent for minors), encrypted with OpenBao-managed keys, used only to confirm the account owner and presence, never to judge mood or attention, never sent to a cloud service, and deleted with the account.
- Nexa Co-Teacher data isolation (3.16): answers come only from the learner's own data and the shared course content, enforced with row-level security and tested in 13.2.
- Links (3.23): shared links pass a blocklist, a sandbox and the content checks first; serious cases reach the human controller.
- Purchases by minors (11.9): every purchase by an under-18 learner needs the guardian's approval, within a monthly spending limit the guardian sets.
- Engagement features (3.19): no camera or voice mood reading; no infinite scroll, autoplay or reward loops for children; guardian time limits and quiet hours; each feature has its own consent switch (19.4).

10.3 Motion Studio content, voice and rights safety

- Record consent and rights evidence for voice cloning, likeness/avatar assets, customer media and any imported copyrighted material; content safety, child safety and enforcement follow 4.15.
- Trace religious citations and attribution for Islamic content; mark uncertain/unverified sources rather than presenting them as verified.

11. Commercial Model & Approved Pricing

Authority: NexaTute Price Model v2.0, approved 18 October 2026; it replaces the wallet-only Final Price Model v1.0 of 15 October 2026 and the older BCS-package and general-wallet generations. NexaTute is sold as monthly plans — Batch, Combo and Personal (11.1); the prepaid wallet stays for

extra 1:1 minutes, add-ons and other per-minute services at the rates in 11.2. All prices include VAT. Planned launch: 1 January 2027.

11.1 NexaTute Plans — Batch, Combo and Personal

Three plans, each sold per month:

Plan	What the learner gets
Batch	Group classes with the Nexa Teacher (one AI for up to 50 learners, five days a week, 3.17) + 5 minutes a day of 1:1 with the Personal Nexa Teacher
Combo — most popular	Group classes + 30 minutes a day of 1:1 with the Personal Nexa Teacher, starting from the day's weak topics
Personal	75 minutes a day of 1:1 with the Personal Nexa Teacher for all subjects, or 40 minutes a day for a single subject; no group classes

1:1 minutes can be used at any time of day (an ETA is shown at peak, 3.16); unused daily minutes do not carry over. A batch starts on its scheduled date with any number of learners (3.17).

All subjects — price per month (VAT included):

Course and group-class subjects	Group classes a day	Batch	Combo	Personal (75 min/day)
Class 4–5	—	—	—	৳1,000
English medium Class 4–5 (Cambridge Primary Stage 4–6 / Edexcel iPrimary)	—	—	—	৳1,400
Class 6–8 (Mathematics, Science, English)	2 subjects × 40 min	৳500	৳750	৳1,000
English medium Class 6–8 (Cambridge Lower Secondary / Checkpoint / Edexcel iLowerSecondary — Mathematics, English, Science)	2 subjects × 40 min	৳900	৳1,200	৳1,500
Class 9–10 / SSC (Mathematics, Science, English)	3 subjects × 40 min	৳600	৳800	৳1,000
SSC Vocational (common subjects Bangla, English, Mathematics, Physics/Chemistry in the group class; trade subjects in the 1:1 time)	3 subjects × 40 min	৳600	৳800	৳1,000
HSC (Physics, Chemistry, Higher Mathematics/Biology, English)	3 subjects × 40 min	৳700	৳850	৳1,000
HSC Vocational (common subjects in the group class; trade subjects in the 1:1 time)	3 subjects × 40 min	৳700	৳850	৳1,000
Admission Test Preparation (subjects by university unit)	3 subjects × 40 min	৳700	৳850	৳1,000
BCS + Job Test Preparation (Bangla, English, Mathematics/Mental Ability, General Knowledge — Bangladesh and International)	3 subjects × 40 min	৳750	৳850	৳1,000

Course and group-class subjects	Group classes a day	Batch	Combo	Personal (75 min/day)
O-Level / IGCSE / IB / Edexcel (Mathematics, English, Physics, Chemistry, Biology in rotation, three classes per subject a week; batches per board)	3 subjects × 40 min	৳1,500	৳1,800	৳2,000
A-Level (the learner's three subjects)	3 subjects × 40 min	৳1,900	৳2,200	৳2,500
IELTS / TOEFL / PTE Academic (Listening, Reading, Writing, Speaking)	4 modules × 30 min	৳1,800	৳2,600	৳3,500
JLPT / TOPIK / EPS-TOPIK	3 subjects × 40 min	৳2,100	৳2,800	৳3,500
SAT / GRE / GMAT; PLAB / USMLE / OET	3 subjects × 40 min	৳2,400	৳3,200	৳4,000

Single subject — price per month (Combo is sold for the whole course only):

Course	Batch (per subject)	Personal single (40 min/day)
Class 4–10, SSC and HSC Vocational	—	৳500
English medium Class 4–5 / Class 6–8	— / ৳400	৳800 / ৳700
HSC	৳300	৳500
Admission Test Preparation	৳300	৳500
BCS + Job Test Preparation	৳400	৳500
O-Level / IGCSE / IB / Edexcel	৳600	৳1,000
A-Level	৳700	৳1,250
IELTS / TOEFL / PTE Academic (per module)	৳500	৳1,750
SAT / GRE / GMAT (per section)	৳1,300	৳2,000
JLPT / TOPIK / EPS-TOPIK; PLAB / USMLE / OET	Whole course only	Whole course only

Personal only — price per month:

Course	Single (40 min/day)	All (75 min/day)
University and BTEB diplomas (any subject the learner chooses)	৳750	৳1,500
ACCA / CA / CMA	৳2,000 per paper	—
PlayBoom (Play class–Class III; replaces PlayBoom, Section 30)	—	৳599 per child per 30 days; ৳5,750 per year; 30% off from the 2nd child

Course	Single (40 min/day)	All (75 min/day)
Foreign languages (Japanese, Korean, German, Arabic, Chinese, French)	৳1,250 per language	—
Skill courses (3.12)	৳800 per course	—
Other Popular & High-Demand Courses (11.2 catalogue)	৳1,250 per course	৳2,000 All-access (any catalogue course)
Any course not in the catalogue	Priced live from its valuation (11.13)	Priced live from its valuation (11.13)

Included in every plan:

- The Nexa Co-Teacher in text, 25 topics a day (3.16); recording of any class the learner chooses, with unlimited replay while it is kept (11.3, 3.29); storage by plan (G.3); unlimited mock tests and class summaries; the Virtual Lab (3.6); the AI daily study plan, Virtual Batch rank and the weekly guardian report (3.10, 3.17); one face-verified learner per account (3.16). Every feature of 3.10 is in the launch release; the 2027 engagement features switch on at the paid launch, with the free and paid limits of 3.32. True/false, fill-in-the-blank and matching tests are unlimited in every plan; written checks follow the plan limits in 38.2 (3.27).
- Extra 1:1 minutes: when the daily minutes are used up the learner can continue with the Personal Nexa Teacher; the wallet is charged at the course's per-minute rate (11.2), 20% less from 1 am to 7 am for adult courses (11.12).
- Scope: help is unlimited inside the purchased syllabus; a question outside it gets a short pointer and the offer of a Doubt Pass (11.14), the right course's Daily Pass or a subject upgrade (3.16).
- Renewal: at purchase the learner, guardian or teacher chooses auto-renew or manual renewal, and that choice stands until they change it (11.7).
- Prices include VAT. Add-ons, discounts and the price fine-tuning rules are in 11.12, USD prices in 11.2 and institution prices in 11.8.

English-medium Class 4–8 (approved 27 October 2026; Play class–Class III English medium stays in PlayBoom): the courses above open with the English-medium launch after the quality gate; Bangla and Bangladesh Studies support is included in the paid plans; the usual second-child, prepay and institution discounts apply; the Checkpoint Crash Pack is ৳900 and the Daily Pass ৳30 (11.12); USD prices are in 11.2. They are added to the revenue figures only after Free Beta data (15.8).

- Other Popular & High-Demand Courses (11.2) are Personal only: ৳1,250 a month per course for 40 minutes a day — the same 1,200 minutes would cost ৳3,000 at the ৳2.50 wallet rate — or All-access ৳2,000 for 75 minutes a day across any catalogue courses (calculated at ৳2.00 a minute). Catalogue courses 8 (Digital Marketing) and 9 (UX/UI) stay at the skill-course price of ৳800 because they are taught as skill courses (3.12); a batch can open later if a course gets enough learners. Prepay packages (11.12): 3 months ৳3,375 per course or ৳5,400 All-access; 6 months ৳6,250 or ৳10,000. "Any course" is priced live from its valuation (11.13); extra minutes cost ৳2.50 (৳2.00 from 1 am to 7 am).

Wallet top-ups — for extra 1:1 minutes, add-ons, extra images, Group Study and other per-minute services:

Item	৳700 Deposit	৳1,200 Deposit
Deposit amount	৳700	৳1,200
Wallet validity	45 days	75 days

Item	৳700 Deposit	৳1,200 Deposit
Billing basis	Learning minutes consumed	Learning minutes consumed
Renewal	Unlimited renewals under the approved price model	Unlimited renewals under the approved price model

- Mini top-up: ৳200 for 15 days, at the same per-minute rates, for learners who cannot deposit ৳700 at once. It can be repeated, and a later ৳700 or ৳1,200 deposit carries any remaining balance into the new validity.
- Image uploads (3.14): learners and guardians each get 10 free images a day for homework help, notes and answer checking; every extra image costs 1 learning minute at the course rate from the learner's wallet. Teacher lesson-script uploads and question-bank imports are free with no cap. The limits are set in the admin panel.

11.2 Wallet Rates, Course Categories & Subcategories

NOTE: AI Tutor can teach you in 6 Languages (Bangla, English, Hindi, Urdu, Arabic & French)

The per-minute rates below are wallet rates for extra 1:1 minutes beyond a plan's daily minutes, Group Study, the phone tutor and other per-minute services; plan prices are in 11.1. Recording is included at no extra charge (11.3), so the former "with recording" rates are withdrawn.

Category	Wallet rate per minute
Early classes — Class 1–3 (Bangla Medium and English Version), PlayBoom (plan only)	৳1.00
Bangla Medium — Class 4–12	৳1.00
English Version (Bangla Curriculum) — Class 4–12	৳1.00
BCS + Job Test Preparation (BCS, Bank and Government jobs)	৳1.00
Admission Test Preparation (University, Medical, Engineering)	৳1.00
SSC and HSC Vocational (BTEB)	৳1.00
Diploma in Engineering and other BTEB diplomas	৳1.50
Cambridge IGCSE	৳2.00
IB	৳2.00
Edexcel	৳2.00
University subjects (Engineering, Medical, Business, Science, Arts)	৳1.50
Cambridge A-Level	৳2.50
Professional Qualifications — ACCA, CA, CMA	৳4.00

Category	Wallet rate per minute
Medical Licensing Abroad — PLAB, USMLE, OET	৳4.00
Admission & Aptitude Tests — GRE, GMAT, SAT	৳4.00
English Proficiency Tests — IELTS, TOEFL, PTE Academic	৳3.50
Foreign Language Courses — Japanese, Korean, German, Arabic, Chinese, French	৳2.50
Language Tests for Work and Study Abroad — JLPT (Japanese), TOPIK and EPS-TOPIK (Korean)	৳3.50
Other Popular & High-Demand Courses (subject catalogue below)	৳2.50
AI can teach any Course as you required	৳2.50

Each category above is sold only after it passes its quality gate in 3.13; until then it is shown as "coming soon". Categories that meet the accuracy threshold also carry the "NexaTute Verified" badge with their measured accuracy (3.13). Each course is offered only in the teaching languages whose quality gate it has passed (3.13).

Other Popular & High-Demand Courses — Subject Catalogue

1. **Artificial Intelligence (AI) & Machine Learning** — AI concepts, model development and applied machine learning
2. **Data Science & Advanced Analytics** — Data analysis, statistical methods and advanced analytics
3. **Cloud Computing & DevOps** — Cloud platforms, deployment, automation and DevOps practices
4. **Cybersecurity** — Security principles, defensive practices and risk awareness
5. **Software & Systems Administration (IT Operations) — Operating systems, system administration and IT operations**
6. **Robotics & Industrial Automation** — Robotics fundamentals, control systems and industrial automation
7. **Human-Only Skill Assets (Soft Skills) — Communication, critical thinking, teamwork and leadership**
8. **Digital Marketing & Content Creation — Digital campaigns, audience development and content strategy (taught with hands-on practice as a skill course, 3.12)**
9. **UX/UI & Product Design — User research, interaction design and product experience (graphic-design practice in the skill course, 3.12)**
10. **Game Development & Virtual Environments** — Game design, development workflows and virtual environments
11. **Applied Finance & FinTech** — Financial analysis, digital finance and financial technology
12. **E-Commerce & Global Store Management** — Online retail operations, marketplace and store management
13. **International Standards & Auditing (ISO Certifications) — Standards awareness, audit concepts and certification preparation**
14. **Corporate Law, Compliance & Intellectual Property (IP) — Business law concepts, compliance and intellectual property**
15. **Agile Project Management & Product Leadership** — Agile methods, project delivery and product leadership

16. **Global Supply Chain & Logistics Management** — Supply chain operations, logistics and international movement of goods
17. **Renewable Energy & Sustainability** — Renewable energy systems and sustainability principles
18. **Healthcare Management & Health Tech** — Healthcare operations, management and health technology
19. **Creative Arts & Commercial Media Production** — Creative workflows, media production and commercial applications

The following approved catalogue expands the "Other Popular & High-Demand Courses" category. All subjects listed here use the single rate specified in the Category-Based Learning Rates table; rates are intentionally not repeated for each subject.

International prices (USD): learners whose billing country is outside Bangladesh (for example India, Pakistan, the Middle East and the diaspora, 2.5) pay the monthly plan prices below — the BDT price × 0.015, rounded to the nearest dollar (an exact 0.50 rounds down — the rule applies to every USD price with no exception, so ৳500 = $7; decided 29 October 2026) — through Paddle as Merchant of Record, billed by the USA LLC (0.2, 11.7). Extra 1:1 minutes use the USD wallet rates in the second table, with wallet top-ups of $10 (45 days) or $20 (75 days). Free Trial, Free Beta, refund and lifecycle rules are the same as in Bangladesh; international Group Study and institution prices are quoted case by case until usage data is available. USD prepay totals are rounded down to a whole dollar (for example Class 6–8 Batch at $7 a month: 3 months $18, 6 months $35). Paddle's automatic tax mode applies by the customer's country: tax is included in the price or added at checkout depending on the country, and the buyer sees the total before paying (11.7).

Course (USD per month)	Batch	Combo	Personal (all)	Personal (single)
Class 4–5	—	—	$15	$7
Class 6–8	$7	$11	$15	$7
English medium Class 4–5	—	—	$21	$12
English medium Class 6–8	$13	$18	$22	$6 (Batch per subject) / $10
Class 9–10 / SSC, SSC Vocational	$9	$12	$15	$7
HSC, HSC Vocational, Admission Test Preparation	$10	$13	$15	$7
BCS + Job Test Preparation	$11	$13	$15	$7
University, BTEB	—	—	$22	$11
O-Level / IGCSE / IB / Edexcel	$22	$27	$30	$15
A-Level	$28	$33	$37	$19
IELTS / TOEFL / PTE Academic	$27	$39	$52	$26 per module
JLPT / TOPIK / EPS-TOPIK	$31	$42	$52	—
SAT / GRE / GMAT	$36	$48	$60	$30 per section
PLAB / USMLE / OET	$36	$48	$60	—

Course (USD per month)	Batch	Combo	Personal (all)	Personal (single)
ACCA / CA / CMA	—	—	—	$30 per paper
PlayBoom (Play class–Class III)	—	—	$9 ($86 per year)	—
Foreign languages	—	—	—	$19 per language
Skill courses	—	—	—	$12 per course
Other Popular & High-Demand Courses	—	—	$30 (All-access)	$19 per course
Any course (valued live, 11.13)	—	—	BDT price × 0.015	BDT price × 0.015

USD wallet rates for extra 1:1 minutes and other per-minute services:

International category	Wallet rate per minute
School and general subjects (Bangladesh ৳1.00–৳1.50 categories)	$0.03
Cambridge IGCSE / IB / Edexcel / A-Level, foreign language courses, other popular courses and any course	$0.05
IELTS, TOEFL, PTE Academic, GRE, GMAT, SAT, ACCA, CA, CMA, PLAB, USMLE, OET, JLPT, TOPIK and EPS-TOPIK	$0.08

11.3 Built-In Recording Policy

Feature	Approved policy
Recording charge	None — recording is free and by choice: in the 1:1 class the learner chooses “Record this class” or “Don’t record”; in a batch class one shared recording is made when at least one learner wants it (3.29); it never starts automatically; the former +৳0.50 per minute premium is withdrawn
Storage	Counts toward the plan’s storage allowance (G.3)
Retention	12 months, then deleted automatically; the learner may keep a recording until the course ends (3.29)

Feature	Approved policy
Format	AI-tutor and batch classes are stored as an event log (AI text, whiteboard events and timing) with only the learner's own Opus audio — about 5 MB per 75-minute class; the AI voice is re-created on replay by the in-browser VITS voice with the pinned voice version (server CPU voices on weak devices). Live and group-class recordings are stored as separate tracks with the whiteboard event log and replayed in the NexaTute player. Any recording can be downloaded as an H.264 MP4 made in the user's browser (9.5).
Batch recordings	One shared recording per batch class when at least one learner wants it, for the approved members only, with consenting learners' audio and nicknames; it counts as each member's own class and shows all questions asked in class (3.17, 3.29)
Download	Free
Separate recording add-on	None

11.4 Group Study Pricing

Group size	Rate per student/min
2–3 students	৳0.75
4–5 students	৳0.60
6–10 students	৳0.50
11–20 students	৳0.40

Billing unit — per student per minute (৳/student/min). Each participant is charged the applicable per-student rate for each billable minute. Recording is included at no extra charge (11.3).

Guardian–counsellor call (3.10): ৳100 per 15-minute call with a human counsellor, booked from the guardian dashboard and paid from the wallet or through aamarPay (11.7); counsellors are contracted per call.

Learning guarantee (2.5; terms approved 28 October 2026, wording to be confirmed by the lawyer): it covers all paid plans — not the Free Beta, Free Trial, free tier or Daily Pass. It is measured on the Mastery Map for the topics studied in 4 weeks; an average gain under 5 points qualifies (default, editable). It depends on everyone taking part: the learner studies 4 weeks in a row with at least 3 sessions a week of at least 20 active minutes, completes at least 80% of due reviews, takes the start test and the end check and makes honest attempts; for learners under 18 the guardian has given consent, has not blocked study time, has opened the weekly report and confirmed the schedule at least once; an institution teacher assigns practice in the app and does not override the plan, and weeks blocked by the institution extend the window. The minutes used in those 4 weeks come back as wallet credit (not cash), valid 6 months, at most once per course every 3 months (defaults). The check runs automatically every 4 weeks with a "Claim" button; a person reviews

edge cases within 7 days with the evidence and an appeal; server outages extend the window. It is not a grade or result guarantee. All numbers are editable in the admin panel with two-person approval.

Skill courses (3.12): ৳800 per course a month for 40 minutes a day of 1:1 (11.1; $12 international); extra minutes at the "other/any course" wallet rate (৳2.50 a minute; $0.05 international, 11.2). The final-project mentor review and live Quran lessons with a human teacher are booked like the guardian–counsellor call, at ৳100 per 15 minutes (proposed).

11.5 Free Trial Package

Setting	Policy
Package	Free Trial Package
Package validity	7 days
AI Tutor limit	15 minutes/day
Total AI Tutor allowance	105 minutes
Cost / payment card	Free / no card required
Recording	Not included
Storage	1 GB (G.3) for the 7-day package validity — videos can be stored and watched (no recordings)

- Free Trial Package includes complete syllabus access, learning modules, class summaries, KPI tracking, full dashboard, weakness detection, unlimited mock tests and AI Tutor up to 15 minutes/day.
- The Free Trial Package excludes certificates, data export and offline lesson downloads; storage, video and recording are as shown in the table above.
- At the end of the trial the Nexa Teacher gives the learner and guardian an honest summary, from the trial data, of which plan — Batch, Combo or Personal — suits them best; the learner or guardian decides (3.16).
- Lifetime free tier (3.22): a learner who does not buy after the trial keeps the free tier for life.
- Second Chance (3.20): students who failed SSC or HSC, admission re-takers and BCS repeaters get a fresh 7-day Free Trial together with a free result analysis.

11.6 Refund Policy

Scenario	Window	Refund policy
Fresh purchase	0–30 days	100% of the unused portion (unused days of a monthly plan, or unused wallet balance)
Year Pass, commitment plan, Head Start, Olympiad	0–30 days	Unused part refunded, as for a fresh purchase
After 30 days	31+ days	No refund
Technical issue	Any time	Full refund for affected portion

Scenario	Window	Refund policy
Platform outage	Automatic	1:1 minutes lost during a platform outage, a failed session or a stopped Nexa Teacher class (3.16) are credited to the wallet automatically within 24 hours at the course's per-minute rate, and wallet validity is extended by the outage duration
Batch class stopped by a server problem	Automatic	Make-up class the same day or the next day, plus the recording (3.17)
Wrong charge	Any time	Immediate reversal
Duplicate charge	Any time	Immediate reversal
Shikkha Salami gift card	—	No cash refund; valid for 12 months and can be given to someone else
Teacher-marketplace item	7 days	Refunded if not opened

11.7 Payment Methods

Market	Methods	Payment type / notes
Bangladesh	aamarPay payment gateway: bKash, Nagad, Rocket, Visa/Mastercard and the other methods aamarPay supports	Plan purchases, renewals, wallet top-ups and add-ons; auto-renew only where aamarPay supports a tokenised recurring payment for the method, otherwise a renewal reminder with a payment link
International	Paddle (Merchant of Record), billed by the USA LLC (0.2); cards, PayPal and the local methods Paddle supports	USD plans, PlayBoom and wallet top-ups (11.2); Paddle's automatic tax mode by the customer's country — tax included in the price or added at checkout — and Paddle pays it; the LLC pays RemonLogic the monthly service fee

aamarPay merchant eligibility and recurring-payment support (Bangladesh) and the Paddle seller account of the USA LLC (international) must be confirmed before launch; listing a method does not establish that it is integrated or enabled. Bangladesh prices include VAT, and receipts show the VAT included. VAT 15/115 applies only to Bangladesh (aamarPay) revenue. Paddle revenue is booked as Paddle's payout — gross less the buyer-country tax Paddle pays and the Paddle fee, shown as its own expense line — at 0% Bangladesh VAT as an export of services (the chartered accountant confirms). The Paddle fee is a setting, not a fixed number: package, percentage, fixed fee and effective date are editable, changed with the finance lead's approval (the maker cannot approve) and logged (38.5).

App stores (decided 28 October 2026; replaces "no in-app purchase"): NexaTute is on Google Play, the Apple App Store and the Microsoft Store as well as the installable web app. A purchase inside a store app is paid through that store's billing; a purchase on the website goes through our own gateway (aamarPay in Bangladesh, Paddle abroad). Store purchases cost 30% more than the website price (default, editable), shown as three lines — "Plan price (incl. VAT) — if you buy on the website: ৳800", "App-store charge (30%): ৳240" and "Total — if you buy through Google Play / App Store / Microsoft Store: ৳1,040" — never folded into one price. A learner under 18 buys in a store app through the store's billing with the guardian's approval, never through aamarPay inside the app. The website gateway and the three stores are sales-channel rows of the admin price table (38.17), and store payouts are booked net of the store fee (38.8). Alternative billing can be switched on when Google's new rules reach Bangladesh (target September 2027).

11.8 B2B Pricing — Institution Discounts

Schools, coaching centres and other institutions buy Batch and Combo seats for their learners at a discount: minimum 20 learners; 30% off for 20–99 learners and 40% off for 100 or more, on the per-learner monthly prices in 11.1 (other courses get the same discounts). Institutions also get the institution dashboard, reports for all their learners and their own branding through the Classroom Builder (3.11, 3.15).

PlayBoom for institutions: preschools, kindergartens and schools buy PlayBoom seats at 25% off for 20–99 children (৳449 per child a month) and 35% off for 100 or more (৳389), on the ৳599 price; institution dashboard and reports included. NexaTute's own institution discounts above (30% / 40%) are unchanged.

Course (per learner per month)	Batch, 20–99	Batch, 100+	Combo, 20–99	Combo, 100+
Class 6–8	৳350	৳300	৳525	৳450
Class 9–10 / SSC, SSC Vocational	৳420	৳360	৳560	৳480
HSC, HSC Vocational, Admission Test Preparation	৳490	৳420	৳595	৳510
BCS + Job Test Preparation	৳525	৳450	৳595	৳510
O-Level / IGCSE / IB / Edexcel	৳1,050	৳900	৳1,260	৳1,080
A-Level	৳1,330	৳1,140	৳1,540	৳1,320

Contracts above 2,000 learners, government programmes and Personal-plan seats are quoted case by case — except Other Popular course seats (11.1), which get the same discounts: ৳875 (20–99 learners) and ৳750 (100+) per course, or ৳1,400 and ৳1,200 for All-access. Signed proposals must specify class limits, AI Tutor allowance, recording allowance, storage and overage rates; unlimited classes are not promised.

Script Checker for schools and coaching centres (3.21): 2 scripts a day free, then ৳5 a script; teachers approve the final marks.

11.9 Plan and Wallet Lifecycle & Retention

- Day 0: a monthly plan activates on payment and renews every month — automatically or manually, as chosen at purchase (11.1); the prepaid wallet activates after a ৳200 mini top-up, or a ৳700 or ৳1,200 deposit, and freezes at Day 15, 45 or 75. When a plan ends without renewal or a

wallet expires, media playback is locked until the learner renews; the included 10 GB storage follows the active plan or wallet validity.

- Plan changes: an upgrade (Batch → Combo → Personal) takes effect at once, paying the price difference for the remaining days — for example SSC Batch → Combo with 15 days left: (৳800 − ৳600) × 15/30 = ৳100; a downgrade takes effect from the next renewal.
- Failed auto-renewal: a 3-day grace period (as in Motion Studio) with push, email and SMS reminders; then the plan pauses and the learner's data is kept.
- Purchases by under-18 learners: a request goes to the guardian's phone, and payment happens only after the guardian approves; the guardian can set a monthly spending limit. A purchase above the limit cannot be approved until the guardian raises the limit — there is no one-time override.
- Academic Year Pass and commitment plan (11.14): the Year Pass covers the January–December school year; the commitment plan runs 12 months paid monthly, with the 12th month free — leaving midway has no penalty, only the free last month is lost.
- Free Trial Package (11.5): when the trial ends, the learner continues with a monthly plan (11.1) or a wallet deposit.
- Offline lessons: saved lessons can be downloaded and watched offline at no extra charge while a plan is active; with a wallet only, a lesson can be downloaded while the balance covers its length, and offline minutes are charged at the category rate when the device next goes online; downloaded lessons lock when the plan or wallet lapses, in line with the media lock above. Free Trial users cannot download lessons.
- Family, gift and referral: siblings in a family account spend from one guardian wallet, each with their own profile and progress; gift top-ups and referral bonus minutes are added to the recipient's wallet and follow its validity (3.10). The second child in a family gets 30% off a plan (11.12).
- Sponsor-a-Learner (3.10): sponsors buy a monthly plan or wallet minutes at the normal prices for a named learner or for learners selected by need through partner schools and NGOs; minutes go to the learner's wallet, sponsors see only consented and anonymised progress and certificates, receive a CSR receipt, and are named to the learner only if both sides agree. Rural girls are given priority (27.1).
- National live model tests (3.10) are free to take for learners with an active plan or wallet; the AI explanations after the test are billed at the learner's category rate. The phone tutor (IVR) is billed at the learner's category rate from the wallet linked to the phone number, or through the telco bill at a rate agreed with the telco (2.5).
- The source price model describes a 90-day post-expiry media grace period and a later deletion point; exact timestamps and wallet recovery/forfeiture rules must be confirmed before launch.
- Learning data: scheduled deletion after 5 years. Billing records: scheduled deletion after 7 years, subject to applicable legal requirements.
- Storage access matrix in the price model: video, PDF, image, audio and notes are unavailable while locked; metadata remains viewable; download is limited to the last 30 days; delete remains available.

11.10 Remon Motion Studio Pricing (Separate Product)

Motion Studio is priced separately from NexaTute; all its pricing is in Appendix G (G.8 packages, per-minute rates and delivery-speed tiers (G.8.3), USD prices (G.8.4), G.3 storage and voice entitlements, G.4 remakes, G.9 Islamic discount).

11.11 Free Beta

Free Beta is a free testing programme: 20 days for NexaTute (20 November–9 December 2026), 30 days for Motion Studio (from 20 December 2026) and 20 days for PlayBoom (1–20 January 2027).

Its purpose is to test the platform with real users while protecting service quality: admission is capped, daily use is limited, and free work never uses paid cloud capacity.

Setting	Free Beta policy
Programme name	Free Beta (shown with this name on every screen and in all marketing)
Duration	NexaTute 20 days (20 November–9 December 2026; paid plans from 1 January 2027); Motion Studio 30 days from 20 December 2026 (paid packages from a date to be set — to confirm); PlayBoom 20 days (1–20 January 2027; paid from 21 January 2027 only after its readiness checklist passes)
Products	NexaTute and Remon Motion Studio
Cost / payment card	Free / no card required
NexaTute AI Tutor	30 minutes/day (unused minutes do not carry over); video watching and storage within the free 1 GB allowance (G.3); no recording
NexaTute batch classes	Free group classes for BCS + Job Test Preparation and SSC batches, to measure class completion and the Batch → Combo/Personal upgrade rate before other courses open (3.17)
MindForge	MindForge Pro open to everyone to pilot items and calibrate finish-date estimates (37.6); afterwards a 7-day free trial (guardian permission under 18)
Motion Studio — Quick and Enhanced	1 minute of video per day
Motion Studio — Cinematic	1 minute of video per day, within the daily Cinematic pool below
Motion Studio output	All modes: 480p Native and 720p Escalating only (no 720p Native, 1080p Escalating or 1080p Native); all outputs watermarked (as in G.11)
Motion Studio video length	Maximum 15 seconds per video (up to 4 videos a day in Quick/Enhanced and 4 in Cinematic)
Concurrent users	Admission cap ≈2,000 combined CCU; new users wait in a queue; paid admission is unlimited (8.5)
Cloud LLM fallback	Off for Free Beta users
Priority	P10 free/trial lane (8.2)
Admin control	Daily minutes, video length, output tiers, end date and admission cap are separate admin-panel settings (4.9) and can be changed during the beta without a code change

- Admission cap: when ≈2,000 combined CCU are online (two-thirds of the ≈3,000 launch baseline), new users wait in a queue that shows their place in line (for example "You are #123 in line"), using the platform's own queue or Cloudflare Waiting Room (a paid Cloudflare feature).

Users already inside keep full service. The cap is raised only when load tests and the live dashboard show headroom, and is always kept at or below 70% of measured capacity.

- Daily Cinematic pool: Cinematic is the most GPU-heavy mode, so all Free Beta Cinematic minutes together are also limited by a daily pool set from the measured render capacity (starting point 10–15% of the measured Cinematic render capacity, raised only after the benchmark; 480p Native minutes count at their measured GPU time; 720p Escalating minutes count against the pool at their measured GPU time, including the upscaling pass; because 480p needs roughly half the GPU time of 720p, the pool can be raised once the render benchmark confirms it). When the pool is used up, new Cinematic requests are queued for the next day's pool and the user is told the expected time; Quick and Enhanced stay available.
- No cloud fallback: when the local GPU cluster is full, Free Beta requests wait in the queue or see the busy message (8.4); they are never sent to GPT-6 Luna or Mistral Large 3, so the cloud budget (15.2) is kept for paying users.
- Relationship to the Free Trial: during the Free Beta its allowances replace the 7-day Free Trial (11.5, G.11); after the Free Beta ends, new users receive the normal Free Trial.

11.12 Add-ons, Discounts and Price Rules

All prices include VAT.

Exam Crash Pack — per course, for the last 30 days before the exam, +45 minutes a day of 1:1 (about 60% of the course's Personal All price):

Course	Exam Crash Pack
Class 6–10 / SSC, HSC, Vocational, Admission Test Preparation, BCS + Job Test Preparation	৳600
University and BTEB	৳900
O-Level / IGCSE / IB / Edexcel; ACCA / CA / CMA (per paper)	৳1,200
English medium Checkpoint (Cambridge Lower Secondary)	৳900
Other Popular courses that end in an outside certification exam (for example ISO, cloud, cybersecurity, Agile/PMP)	৳1,200
A-Level	৳1,500
IELTS / TOEFL / PTE Academic; JLPT / TOPIK / EPS-TOPIK	৳2,100
SAT / GRE / GMAT; PLAB / USMLE / OET	৳2,400

There is no Crash Pack for PlayBoom, skill courses, foreign-language courses or Other Popular courses without an outside certification exam. There are no seat caps on Crash Packs or crash programs: admission is unlimited and servers are added before classes start (8.5; decided 27 October 2026).

Crash programs (approved 27 October 2026, for every Crash Pack): 1, 3 or 6 months, with extra 1:1 time that tapers — 1 month +45 minutes a day; 3 months +20 (months 1–2) then +45; 6 months +20 (months 1–4), +30 (month 5) and +45 (month 6). The price is the 1-month Crash Pack's per-minute rate × the minutes, less a small commitment discount; programs longer than the time left before the exam are not sold; USD = the Taka price × 0.015, nearest dollar with .5 rounding down. Every value — lengths, daily minutes per month, prices and discount, buffer % and phase % — is changeable in the admin panel (38.17).

Course	1 month	3 months	6 months
Class 6–10 / SSC, HSC, Admission, BCS, Vocational	৳600 ($9)	৳1,067 ($16)	৳1,933 ($29)
University; English medium Checkpoint	৳900 ($13)	৳1,600 ($24)	৳2,900 ($43)
O-Level / IGCSE; ACCA / CA / CMA (per paper)	৳1,200 ($18)	৳2,133 ($32)	৳3,867 ($58)
A-Level	৳1,500 ($22)	৳2,667 ($40)	৳4,833 ($72)
IELTS / TOEFL / PTE; JLPT / TOPIK / EPS-TOPIK	৳2,100 ($31)	৳3,733 ($56)	৳6,767 ($102)
SAT / GRE / GMAT; PLAB / USMLE / OET	৳2,400 ($36)	৳4,267 ($64)	৳7,733 ($116)

The Checkpoint prices (৳900 / ৳1,600 / ৳2,900) are approved; the other 3- and 6-month prices are computed by the same rule. Scheduling: a day-1 diagnostic per syllabus objective; the time budget is the days to the exam × the daily minutes, minus holidays, Ramadan and school exams, with a 10% buffer; the need is the remaining topics × the learner's own pace; an honest coverage percentage is shown before payment; topics are ordered by prerequisites, exam weight and weakness ("almost mastered" first in short programs). Phases: 1 month — targeted practice, two full mocks a week and a light rehearsal in the last days; 3 months — gap repair, finish the syllabus, mocks; 6 months — about 60% learning, 25% consolidation and 15% mocks with two buffer weeks. The plan is re-run weekly and after every mock (on track, tight or needs a plan, with levers); topics are dropped only with the learner's or guardian's agreement; weekly milestones appear in the guardian report; batch crash classes share one plan per series and 1:1 time fills personal gaps. There is no grade guarantee — the learning guarantee applies (11.4).

Daily Pass — 24 hours with 20 minutes of 1:1, the Nexa Co-Teacher and mock tests, paid through aamarPay (bKash and the other methods). The Daily Pass of another course also answers out-of-scope questions (3.16), and after repeat purchases the app shows how much a monthly plan would save:

Course	Daily Pass
School (Class 1–10), SSC, HSC, Vocational, Admission Test Preparation, BCS + Job Test Preparation, PlayBoom, skill courses	৳20
University and BTEB	৳30
English medium Class 4–8	৳30
O-Level / IGCSE / IB / Edexcel	৳40
A-Level; foreign languages; Other Popular & High-Demand Courses	৳50
IELTS / TOEFL / PTE Academic; JLPT / TOPIK / EPS-TOPIK	৳70
SAT / GRE / GMAT; PLAB / USMLE / OET; ACCA / CA / CMA	৳80
Any course (11.13)	Valuation rate × 20

Discounts and rewards:

- Prepay: 3 months at 10% off; 6 months for the price of 5 (one month free). The Academic Year Pass and the commitment plan are in 11.14.
- One discount at a time: a plan gets only the biggest of prepay (including the Year Pass), second child and B2B; the result reward, best-question reward, referral minutes and the night discount are given separately.
- Second child in a family: 30% off a plan.
- Result reward: GPA 5 in SSC or HSC, or passing the BCS Preliminary, earns one month free and, with consent, a success story.
- Best-question reward: +10 bonus minutes in the wallet when a learner's question or explanation is verified into the answer cache (8.14).
- Night discount: for adult courses — for example BCS + Job Test Preparation, University and IELTS-type tests — extra 1:1 minutes are 20% cheaper from 1 am to 7 am; school learners and PlayBoom get no night discount.
- Referral bonus minutes follow 3.10; sponsorship follows 11.9.

Price fine-tuning: prices are tuned from the Free Beta and the first paid month against these targets:

Target	If it is not met
Combo is 40–50% of subscribers	Personal above 40%: Personal +৳100. Batch above 60% and Combo below 25%: Combo –৳50, or a 3-day Combo trial for Batch learners
Batch → Combo upgrades of at least 15% a quarter	A 3-day Combo trial for Batch learners
Monthly churn of 10% or less	Find the cause and fix the features or quality
Gross margin of at least 60% per learner	That course's prices +10%
Personal learners use under 60 of their 75 daily minutes on average while the GPU pool stays under 70%	Above both: rely on the peak ETA (3.16) and add capacity (8.5)

A price change applies only to new subscribers — existing subscribers keep their price for three months — and prices change at most once a quarter. NexaTute prices, daily minutes and limits are admin-panel settings, changed only under these rules. After VAT, Personal (about 54%) and Institution (about 59.7%) margins sit below the 60% target, and no price or limit change is needed (decided 24 October 2026); counting written checks brings Combo to about 60% and Personal to about 50% (accepted, 38.2).

11.13 "Any Course" Valuation

A course that is not in the catalogue has no fixed price; the AI values it and shows the price live:

- The AI builds the course outline and places the course in an 11.2 rate band (৳1.00–৳4.00 a minute) by level, difficulty and market value, plus ৳0.50 a minute if a lab or sandbox is needed.
- Prices from that rate: Single (40 minutes a day) = rate × 500; All-access (75 minutes a day) = rate × 800; Daily Pass = rate × 20 — for example, Interior design at ৳2.00 → ৳1,000, ৳1,600 and ৳40.
- The learner sees the price with a "why this price" breakdown before buying; the price is locked for 3 months after purchase, and the admin can override it in the admin panel.
- Once a student agrees to a valued price, that price is final and the course is added to the price table at that price.
- Before valuing a new request the AI searches the price table for the same course by content, not just by name (for example "Interior design" = "Home interior decoration"); if it finds one, it

shows that price with no new valuation, so the table has no duplicates. A course already in the catalogue shows the catalogue price.
- An added course is labelled “AI-generated” until it passes the 3.13 quality gate, and then gets the “NexaTute Verified” badge.

11.14 Passes, Programmes and Commissions

Taka prices include VAT; USD prices follow BDT × 0.015 (11.2), with tax by Paddle’s automatic mode for the customer’s country (11.7).

Item	Price or terms
Nexa Co-Teacher Pass (3.22)	৳100 a month per course ($1); 25 topics a day; cached answers and the queue during overload
Doubt Pass (3.16)	৳50 for 20 questions outside the learner’s syllabus, valid 30 days ($1); paid from the wallet; under 18 needs the guardian’s approval; a separate product, price editable in the admin price table (38.17)
Head Start camp (3.20)	৳500 (batch class and 10 minutes a day of 1:1); free with a 6-month prepay bought in December
Olympiad track (3.20)	Batch ৳600 / Personal ৳1,250 a month
Study Abroad Application Coach (3.20)	৳2,000 one-time pack ($30)
Academic Year Pass	12 months for the price of 10 (January–December school year), sold in December with Head Start
Commitment plan	A 12-month commitment paid monthly, with the 12th month free; no penalty for leaving midway — only the free last month is lost
Guardian learning (3.20)	50% off Spoken English, computer, foreign-language and skill courses for guardians of active learners (for example ৳800 → ৳400)
Script Checker (3.21)	2 scripts a day free per institution, then ৳5 a script
Shikkha Salami gift cards (3.21)	Daily Pass or 1- or 3-month plans; VAT charged at sale; valid for 12 months; a unique code per card; about 5% retailer commission
Affiliate commission (3.21)	10% for the first 3 months on plans sold with the affiliate’s code; paid through aamarPay only after the 30-day refund window; self-referrals blocked; tax withheld (to confirm with an accountant)
Teacher marketplace (3.21)	NexaTute takes 20% commission; tax withheld on payouts (to confirm with an accountant)
Job pathway (3.21)	Employers and licensed agencies pay per hire or a monthly fee (amount to be set)
MindForge Free (37.8)	For everyone, including the free tier: 3 puzzles a day, the weekly challenge and simple progress
MindForge Pro (37.8)	৳199 a month ($3) as an add-on to any plan or bought alone — not included in Combo or Personal; prepay 3 months ৳537, 6 months ৳995; 30% off the second child; institutions ৳139 a seat (20–99) and ৳119 (100+); under-18 purchases need the guardian’s approval

Item	Price or terms
Recruitment tests add-on (3.27)	৳399 a month ($6); up to 3 recruitment or skill tests a day; adults (18+) only — learners under 18 cannot buy it; free for learners on BCS + Job Test Preparation and the other job-test courses; price, daily limit and the per-plan setting are edited in the admin price table with the normal approval (38.2)

These lines and the affiliate cost go into the investor revenue model when it is rebuilt (15.7).

12. Delivery Roadmap and Release Gates

Scope: Section 12 defines release gates; the dated product roadmap is controlled by Section 17.

Gate	Deliverable	Exit evidence
1. Architecture validation	Review v24.0 inventory, routing, node mappings, storage paths, ports, network and service isolation	Signed-off architecture and actual procurement/deployment inventory
2. Technical proof	Benchmark each AI model, TTS/STT, LiveKit, recording, GStreamer/FFmpeg, DB and R2 path independently	Dated benchmark reports with model version, concurrency, hardware, error rates and cost
3. Product MVP	Complete end-to-end NexaTute and Motion Studio user journeys	Functional tests for auth, entitlements, billing, job state, recording, delivery and cleanup
4. Safety/security review	Threat model, privacy review, voice rights, content-plan verification, child safety and data retention	Remediation register closed or risk accepted by authorized owner
5. Paid pilots	Run limited learner and creator cohorts	Retention, usage, quality, willingness-to-pay, support burden and per-job cost
6. Controlled scale	Increase concurrency gradually and test failover/circuit breaker	Load tests, recovery drills, saturation behavior and rollback plan
7. Expansion	Expand markets and languages only after product/operational gates	Curriculum/language validation, legal review, support readiness and unit economics

Timeline: for NexaTute, Gates 1–4 are completed in Months 1–3 (October–December 2026) — before the Free Beta on 20 November 2026 and the paid launch on 1 January 2027; for Motion Studio, Gates 1–4 are completed by its Free Beta on 20 December 2026 — the design, functional and safety work and the production benchmarks and checks on the rented fleet from 5 November; Gate 5 runs on each product's launch cohort (NexaTute in Months 4–6, Motion Studio from its paid start); Gate 6 controlled scale starts from Month 7; Gate 7 expansion follows 29.4 from 2028.

13. KPIs, Testing Plan, Risk Register and Investor Diligence

13.1 KPIs to collect

Scope: this table lists measurement areas; numeric targets are controlled by Section 22 (OKRs and KPI Dashboard).

Area	Metrics
NexaTute learning	Tutor latency; answer quality; assessment accuracy; repeated misconception resolution; D1/D7/D30 retention; lesson completion; learner/teacher satisfaction; Thinking Profile — unseen-problem performance, question level, transfer between subjects, confidence against accuracy and persistence (3.18)
North Star	Number of learners whose mastery rose each week; features, A/B tests and the team are measured against it (3.19)
Plans and batch classes	Plan mix (Batch, Combo, Personal); Batch → Combo/Personal upgrade rate; class completion and attendance; poll accuracy and confusion rate; recording-first deflection rate (help needs solved by recordings, mock tests, summaries and the Nexa Co-Teacher before a live class, 3.16); Nexa Co-Teacher topics per learner; 1:1 minutes used per plan
Free tier and growth	Free → paid %; help-button → purchase % (3.22)
MindForge	Pro attach rate among paying learners (target 15%); Pro churn; weekly active free-puzzle users; finish-date forecast accuracy; in the Free Beta, the share of learners doing puzzles on at least 3 days a week and Pro trial → purchase (37.10)
Learning proof	Learning gain per course; effect size of the partner-university study; accuracy of predicted rank, estimated IELTS band and GPA against real results (3.24)
Website and sales	Landing-page conversion; drop % at each funnel step; plan-finder completion; abandoned-checkout recovery; pre-sale chat → purchase; review ratings (3.25, 4.19)
Marketing	Posts and reach per channel; organic search clicks; leads by source; outreach reply and meeting rates; spam-report and unsubscribe rates; cost per lead and per customer (3.26)
Real-time classroom	Join success; connection time; audio/video latency and loss; reconnect rate; TURN relay share; recording completion rate
Virtual Lab / browser	Supported device/browser matrix; frame-time distribution; memory use; crash rate; synchronization conflicts; accessibility
Motion Studio production	Queue wait; script/storyboard acceptance; first-pass video approval; remake rate and completion; render success and duration; share of browser renders finished on the server (4.8); delivery time; retry rate; quality-gate failure rate (4.13); final MP4 validation; upload success; cost per output minute; trial-to-paid conversion; creator churn
Reliability	API error rate; p95/p99 latency; queue age; GPU VRAM/utilization; OOM incidents; database lag; cache saturation; fallback frequency; prefix-cache hit rate; input tokens per turn; verified-answer cache hit rate; cloud fallback cost per learner-hour
Commercial	Conversion; active paid accounts; churn/retention; revenue per account; support cost; gross margin; refund rate; CAC/payback once measured; price fine-tuning targets (11.12)
Security/safety	Unauthorized access tests; content-policy outcomes; consent coverage; deletion completion; incident rate; review queue SLA

Measurement: product events are stored in TimescaleDB and shown on Grafana dashboards (funnels, retention cohorts, D1/D7/D30); feature flags and A/B tests for onboarding, messages and prices run on Unleash (Apache-2.0, self-hosted), and experiment results are analysed in TimescaleDB and Grafana.

13.2 Required technical tests

Ecosystem additions are part of this test plan: PlayBoom readiness tests (30.7), NexaLogicDev tests and KPIs (32.11–32.13), the Day-1 must-have list (17.5) and the Security & Control Centre gates and test records (34.4, 34.7).

- Load-test NexaTute and Motion Studio separately, then together with the intended priority mix; do not infer CCU from hardware count alone.
- Run GPU memory and model concurrency tests, including simultaneous LLM, TTS/STT, vision and encoding loads.
- Test the unified GPU cluster: priority pre-emption (P10 → P50), scene-checkpoint resume after pre-emption, LLM hot-standby failover when one GPU node is lost, GPU health breaker cordon/recovery, and lending of idle NexaTute GPUs to Motion Studio and their return, instant pre-emption of light work on Rec-1/Rec-2 when a recording starts, and overflow of queued recordings to spare Core-1/Core-2 CPU and then to idle GPU nodes without touching P100/P50 work, and prefill/decode fallback to normal mode when one AI-1 node is lost.
- Test spare-CPU use on GPU nodes: CPU pinning of GPU-serving pods, eviction of low-priority CPU pods under pressure, API overflow pods starting on AI-1/AI-2 within the autoscaling window, and no CPU work scheduled on Rec-1/Rec-2.
- Test local inference failure, cloud fallback, provider rate limit, fallback failure, circuit breaker and recovery.
- Test the cloud fallback chain hop by hop: local cluster down → GPT-6 Luna; GPT-6 Luna failing or rate-limited → Mistral Large 3; both failing or the spend cap reached → L4 queueing; and confirm that personal data is stripped and minors' sessions are never sent to a non-compliant provider, and that the L4 message "The server is very busy right now. Please try again after some time." appears in both products when everything is unavailable.
- Test prompt caching and context trimming: prefix-cache hit rate with session-affinity routing, running-summary quality versus full history, input-token caps per request, verified-answer cache expiry on curriculum or policy change, and cost per learner-hour before and after.
- Test each 8.15 optimization against the baseline: throughput, latency, render time and the Bangla/curriculum quality test set, with a one-switch rollback for every optimization; including multi-LoRA adapter switching, LMCache hit rate on long sessions, Cloudflare Workers cache purge on content change, and that late-night batch jobs never delay real-time or paid work; and spot-GPU interruption with scene-checkpoint resume and on-demand fallback; prefill/decode disaggregation latency against normal mode and automatic return when one node is lost; and wrong-match rate of meaning-matched cached answers and lesson-cache reuse after a curriculum change; lightweight recording replay sync and the browser-made download MP4 (9.5); rule-based grading accuracy; GPU switch time with vLLM sleep mode; in-browser LLM quality and fallback on weak devices; time to first spoken word; and image-cache reuse; RIFE output on fast-motion scenes; character consistency of image-to-video; false-pass and false-fail rate of the per-scene quality check; batched-render throughput; AV1 playback with H.264 fallback on low-cost phones; and that only safety-checked 👍 answers enter the verified-answer cache; request coalescing for identical questions; JSON validity with xgrammar; picture–voice sync with audio-first timing; and character consistency across episodes with a character LoRA; offline lesson download, encrypted playback, minute deduction on reconnect and lock on wallet expiry; prefetch of the next lesson part on slow mobile networks; and preview-upscaler speed with unchanged final-video quality.
- Test the 4.8 render modes: browser render on the supported device matrix, the 3-second device test, rendering in a hidden tab (Web Worker), chunk resume after PC shutdown, power cut and internet loss, server takeover after 30 minutes for both a lost connection and a closed studio, the server rendering backwards while a returning browser continues forwards with no chunk kept twice, detection of a tampered chunk (changed frames or audio) before signing, 3D fallback chunks on the Rec-node GPU, browser voices against the server voices in a native-

speaker test, the cinematic 3D look on mid-range PCs, seamless chunk joins and audio, final-encode quality checks, ETA accuracy and notifications, and lip-sync quality in Bangla and Urdu.

- Test the 4.18 promo system: moment selection that never uses scenes marked as spoilers, frame-accurate cuts at scene boundaries, audio and video sync, EBU R128 loudness, Bangla title text on thumbnails, export sizes for each platform, and that autopilot drafts are never charged before export.
- Test the 3.15 classroom: load time on 4G and on low-end Android phones, frame rate in the Full, Light and 2D versions, the device check's choice, the same room design for everyone in a live class, the Smart Whiteboard on the room's board, the accessibility modes, the approval flow for posters shown to minors, and that nothing is rendered on the server.
- Test the Cloudflare edge busy page with the origin offline, the 60%-order / 80%-activate scale-out rule, automatic minute and credit restoration after an outage, email and push delivery of ready-video notices, and round-trip latency from major Bangladeshi broadband and mobile networks to the LiveKit cluster in Virginia, Virginia ↔ Dallas latency for AI calls and recording ingest, and WireGuard tunnel failover between the three providers; and the Free Beta controls (11.11): admission cap and waiting-room queue, daily limits (NexaTute 30 minutes, Motion Studio 1 minute Quick/Enhanced and 1 minute Cinematic), the 15-second maximum video length, the 480p Native / 720p Escalating tier limit, the daily Cinematic pool, admin-panel setting changes (4.9) taking effect without a restart, the 1080p Native load pause, the copyright controls for uploaded videos (4.12), escalating-tier quality gate and video-upscaler GPU time (4.13), the 8.6 production-impact estimates and the multi-format crop rule (4.10), Document Intake Service parsing and Bijoy conversion, daily image limits and the pre-production fee (3.14, 4.17), content-safety layers including bad language and obscene gestures (4.15), Cinematic GPU-minutes per output minute and the Free Beta Cinematic pool (4.14), Content Credentials on every export and AI-disclosure on publishing (4.10), Bangla caption rendering and music auto-ducking, and no cloud fallback for beta users.
- Test queue saturation and verify X-Priority 100 traffic remains protected without starving paid production unfairly.
- Benchmark the 8.15 additions before enabling them: answer quality with thinking on and off per course category (3.13 gate), Triton batching throughput and MPS isolation, batched Whisper speed, frame-check precision/recall at reduced image size on Bangla and Banglish sets, Cinematic quality with TI2V-5B versus A14B, and BGE-M3 INT8 retrieval quality with Ragas.
- Benchmark the 17 October 2026 additions before use: LightX2V and FastWan render speed and quality against the standard Wan 2.2 pipeline; Wan2.2-S2V-14B lip-sync on Bangla and Urdu dialogue; Nunchaku 4-bit FLUX.1-schnell image quality; CacheBlend speed and answer quality on knowledge-base turns; suffix decoding on scripts and storyboards; the former Qwen3-8B tasks on Qwen3-30B-A3B (thinking off) and Qwen3-4B against the 3.13 gate; the CPU server voices (learners per core, time to first audio, the same voice in the browser and on the server); Kubernetes DRA only if it is stable.
- Test the App tier and uploads: App-3 joining the pool (pods spread over three nodes, PgBouncer pools within the PostgreSQL connection cap, HAProxy health checks) with a Locust load test before it takes traffic by 15 December 2026; uploads straight to R2 (presigned-URL expiry, resume after a dropped connection, 100 creators uploading 2 GB files at once with no load on the LB or App nodes, and the Document Intake Service reading from R2); ClickHouse on Core-2 (its limits never slow LiveKit; nightly BACKUP and a restore drill).
- Run the capacity tests with Locust (API), the LiveKit CLI load tester (live classes) and vLLM's benchmark scripts (LLM), record the measured limit of every layer in the 8.6 combined table, and raise the Free Beta admission cap above ≈2,000 only up to 70% of the measured capacity (11.11).
- Test DB primary failure, replica lag, PgBouncer pools, Redis outage/recovery, duplicate task delivery and idempotent credit settlement.

- Test every 8.13 cluster: LB virtual-IP failover, App-node loss with both products still serving, Patroni switchover and failover, pgBackRest point-in-time restore from R2, Valkey Sentinel promotion for both instances, and quorum loss of one member, and restore of the daily encrypted Valkey RDB snapshot from R2 with the 48-hour lifecycle deletion verified.
- Test Motion Studio assembly on another GPU node when AI-2 Node B is lost, the 72-hour emailed download link and automatic deletion for clients without storage, control-plane quorum on Redis1/Redis2/App-1, Dallas worker-node behaviour during a Virginia–Dallas link outage, CDN cache hit rate and signed-token checks at the edge, monitoring survival when one Redis node fails, the safety-model load in LLM capacity tests, the cloud-fallback spend cap, and DB partition export to the R2 archive and restore, and the Core-1 local mirror (bandwidth cap during classes, fallback to R2 when Core-1 is down).
- Test large video jobs, scratch disk pressure, process crash, partial upload, checksum mismatch and exact-file cleanup.
- Test LiveKit at representative room/track/bitrate combinations, track recording on both Rec nodes (recordings per node, A/V sync, fallback if a node fails), and network/TURN edge cases.
- Test download MP4s made in the browser (9.5): the time to build a 60-minute class on a mid-range PC and on a low-cost Android phone, audio, video and whiteboard sync, H.264 + Opus playback in common players and phones (AAC only if this test needs it), resume after the tab is closed or the connection drops, and that no compositing or encoding runs on the servers.
- Test browser support for WebGL/WebGPU/Wasm, in-browser STT/TTS and recording, server fallback for weak devices under load, 720p/160p and 480p/160p profiles with automatic switching driven by server load and client internet speed (including switch-over time, no dropped session and recording continuity during a switch), accessibility, low-end device fallback and localization including Arabic RTL.
- Test the 3.10 engagement features: photo-question and answer-script accuracy on Bangla and English handwriting, leaderboard privacy for minors, flashcard review scheduling, Quiz Battle latency in Group Study, pronunciation-score agreement with human raters, and guardian report delivery by WhatsApp and SMS; Android app parity, certificate QR verification, referral fraud controls, forum moderation, uploaded-notes privacy, writing-feedback accuracy, and family-wallet sharing and gift top-ups.
- Test class summaries: time from class end to summary (AI-tutor ≤2 minutes, live and group ≤15 minutes at peak), Bangla and English transcription accuracy, summary quality against teacher-written summaries, P10-to-P100 promotion, and deletion of summary-only audio when recording is off.
- Test Learning Insights: mastery-score accuracy against later test results, correct roll-up from topic to chapter and subject, improvement-ranking sense checks by teachers, next-step plan generation, and teacher overrides; step-grading and mistake-tagging accuracy against teacher marking, hint and confidence weighting, misconception flags after repeated errors, lab process scores, and teacher adjustments taking precedence; rubric agreement between AI and teacher marking, IRT item-difficulty estimates, and — before any neural model is adopted — a held-out comparison against BKT/IRT for every major learner group.
- Test engagement and support: placement-test accuracy against later mastery, wrong-answer refund and cache correction, exam-countdown dates, re-engagement message timing and opt-out, share-card referral tracking, mini top-up carry-over, Chatwoot AI-to-agent handover, and Unleash experiment assignment; hint-ladder and confidence-check data in the Learning Insights, study-time controls, mastery gates, teach-back feedback quality, audio-lesson and Bangla voice-report quality, study-buddy matching with guardian approval, and counsellor-call booking, payment and refund; interest-based examples staying correct, puzzle-of-the-day sharing, mixed-practice selection, diagram accuracy, story-mission progress, explainer-video review before publishing, mini-game and simulation performance on low-end phones, AR fallback, and a 20,000-player live-quiz load test.

- Test the Learning OS (3.11): tenant isolation between institutions, Keycloak roles for guardians and minors, agent routing quality, curriculum-graph links, LTI 1.3 and xAPI conformance, SCORM import, API-gateway scopes and rate limits, messaging safety rules, proctoring consent, and verifiable-certificate signing and checking.
- Test the growth initiatives (2.5): single sign-on across both products without data mixing, anonymisation before training, Bangla benchmark gates for a new model, zero-rated domain coverage, NexaTute Box offline use and nightly sync, and outcome-guarantee eligibility and refunds.
- Test all six teaching languages (4.2, 29): per-language answer accuracy with native-speaker review, browser speech-recognition word error rate (especially Bangla and Urdu), voice listening tests, switch to the server voice when a browser model fails, right-to-left screens for Urdu and Arabic, Nastaliq and Bangla conjuncts in PDFs and certificates, and OCR of Hindi, Urdu, Arabic and French photo questions.
- Test the skill courses (3.12): Office file checks against hand-marked samples (formatting, formulas, charts), sandbox isolation of uploaded files, Penpot load and project storage, campaign-simulator labelling as practice, final-project mentor review flow, certificate issue to the learner passport, and Quran lessons kept teacher-led.
- Test course category readiness (3.13): per-category benchmark and gate results, cited sources in answers, refusal to guess on Bangladesh facts, the AI content pipeline (licence rejection, disagreement between the two solver models, copy check, automatic retirement of bad items), the weekly current-affairs update with two-source agreement, IRT calibration of question banks, IELTS/TOEFL band estimates against trainer marks, IB integrity rules, and that quality routing sends only hard premium questions to the cloud and stops after the gate passes, and that the NexaTute Verified badge appears, shows the right measured accuracy, and is removed and restored automatically; per-language gates, back-translation drift detection, language choices limited to passed languages, and badges shown per language.
- Test medical and engineering support (3.13): numeric answers re-checked by SymPy, circuit diagram and simulation results against known circuits, sandbox isolation for ngspice jobs, 3D anatomy loading on low-end phones, virtual-patient case marking, the source-and-year note on every dose or guideline answer, and that medical image questions use only labelled data.
- Test the new course categories and labs: PlayBoom with guardian start and child-safety checks, vocational and diploma question banks, SAT and PTE Academic mock scoring, PLAB/USMLE/OET gates and medical safeguards, foreign-language voices and pronunciation feedback, JLPT/TOPIK/EPS-TOPIK mock formats, CJK fonts and stroke order, sandbox isolation and 30-minute limits, Juice Shop challenges with no outside network access, and the sandbox queue when the cap is reached.
- Test the new features: board-book solutions against the textbook exercises, a national model test with ≈20,000 simulated takers (paper release, answer batching, merit-list speed and correctness), sponsor wallet credits and anonymised sponsor reports, phone-tutor calls over a test SIP trunk with keypad answers, auto-dubbing quality and timing per language, and scheduled publishing to test accounts on each platform.
- Benchmark the GPU servers in the first month of the 3-month term: CPU steal on the virtual CPUs, disk read speed for model loading, GPU throughput against the 8.6 assumptions, and that Core-1/Core-2 batch work never delays LiveKit.
- Test own avatar and voice: consent-sentence check rejects replayed recordings, no cloning for learners under 18, watermark detection on generated audio, no download of cloned voices, deletion on request, the G.3 allowances (in NexaTute one of each free, then the marketplace; XP only under 18), and linking a Motion Studio character or voice into NexaTute.
- Test the 18 October 2026 additions: batch classes with 50 simulated learners per batch across the 7–9 am and 4–6:10 pm windows (poll bursts, Centrifugo fan-out, one server voice stream per batch); load-aware routing from free to busy to overloaded, with ETAs and minute credits; the Nexa Co-Teacher’s 25-topic limit and cross-learner isolation (row-level security); the Nexa

Co-Teacher cloud order (Mistral Large 3 → GPT-6 Luna) with personal data removed; event-log recording and replay with the pinned voice version; face verification (liveness, OTP fallback, template deletion); aamarPay callbacks and auto and manual renewals; the live speech-to-text content filter; and the human-controller alert channels (push, email, SMS).

- Test link-step blocking with known malware, phishing and adult URLs, and the gVisor fetch limits (3.23); guardian purchase approval and spending limits (11.9); consent switches and the under-18 lock (19.4); Nexa Co-Teacher Pass limits and its cache-only path in overload (3.22); Script Checker marks against teachers' marks (3.21); affiliate self-referral blocking and payout after the refund window, and gift-card code uniqueness (11.14); unlimited admission with the capacity-plan alerts (8.5); and the engagement models for quality and bias across learner groups (3.19, 19.7).
- Test MindForge (37): every generated puzzle has exactly one solution, IRT calibration and DIF reports on the Free Beta data, range reporting, finish-date forecast accuracy, the consent switch and institution visibility, the 30-minute stop message for under-18 learners and offline puzzle sync; Study Hall Q&A (31.3): link and QR blocking, image checks, face blurring for under-18 posts, the verification flow and the daily limits; accounting (38.8–38.13): balanced journals, month lock, maker-cannot-approve, the approval thresholds, interest accrual for EMI and bullet loans with part-payments, partial payments, payroll, the share-transfer cap-table update, audit-log immutability, the cloud export with checksum and the monthly restore test; Growth Platform journeys (3.26): frequency caps, quiet hours, guardian-only messages for minors, the holdout group and one unsubscribe for all products.
- Test the 24–25 October additions: admin push routing, the 4-hour reminder and 24-hour hand-over, quiet hours that never hold back urgent or compliance alerts, and push text without secrets; the chat compliance check (blocked secrets, alerts per category), the virus scan of files, call permissions per role and no call recording, archive access with 2-step login and the 90-day and 6-month deletion jobs (38.15); avatar upload checks (glTF-Validator, lip-sync), the consent sentence and speaker verification for every clone, watermark and C2PA, the daily GPU-minute cap and no purchase button under 18 (38.16); Nexa Notes HLS playback on a 3G phone, reviewed subtitles in six languages and that a YouTube embed makes no request before the click (3.25); channel token refresh and expiry alerts, import undo, and the export watermark, OTP and 24-hour link (3.26).
- Test the tests of 3.27: handwriting-reading accuracy on Bangla and English scripts against human markers; that nothing is marked before the learner confirms unsure parts; part-wise marks against the board schemes; the fill-in tolerance rules; the recheck limits; the queue and notification at the 4–9 pm peak, the daily GPU cap with its 80% alert and that no script reaches an outside AI service; resume after a connection loss with the timer rules; shuffling and similarity flags that reach a person; the CAT stopping rule and level ranges; accent-bias checks; the coding runner's time, memory and no-network limits; no snapshots under 18; institution API scopes, rate limits and key revocation; and only error-notebook counts in the guardian report.
- Test the additions of 26–29 October 2026: microphones off by default, Raise hand, one follow-up, auto-off after every turn and the disturbance pause (3.28); audio-only at 15% packet loss, reconnect with minutes returned, Lite mode, data saver and the low-battery offer; recording by choice in 1:1 and batch classes, consent and exclusion of non-consenting learners, the SHA-256 check before Watch, 12-month deletion and signed 15-minute links that fail for others (3.29); the hub record statuses, masked numbers and no reminders to children after 9 pm (3.30); the sign-language caption rule and the 1.5× minute rate (3.31); free and paid limits of every 2027 feature, under-18 locks and the Beta-off switches (3.32); the feedback button, paused message and What's new sheet never interrupting a class or test (3.33); storage warnings at 80% and 95% (G.3); the three-line app-store price (11.7); OTP protection, passkeys for admin staff, the two-person approval and automatic rollback of every software update, and the staging load-test thresholds (38.18). Also test the admission limits and waiting-list order, the 48-hour offer hold, the capacity guard switching on at 75% and releasing below 60% without touching enrolled

learners or live classes, the emergency close reverting after 60 minutes without a second confirmation, the six-language admissions message, and Nexa Co-Admin refusing every request outside the asking admin's permissions and every item of the never-allowed list, never acting as the second approver and logging every request (38.19, 38.20).

- Test CMS approval, scheduling, expiry and rollback; prices on every page equal the price table; the claims check blocks unproven claims; axe-core and Lighthouse CI gates; the cookie consent manager and Turnstile; pre-sale AI handover to a person and to the guardian for minors; abandoned-checkout reminders only with consent; CRM lead capture; a Play Store build without in-app purchase (11.7); and prediction-accuracy reports against real results (3.24).
- Test that nothing new is published or sent without approval; approval routing, reminders and escalation; unsubscribe within 48 hours and the suppression list; the daily email cap; SPF, DKIM and DMARC on the outreach domain; ready-to-post packs while platform approvals are pending; report delivery (3.26).

13.3 Principal risks and mitigations

Scope: architecture-level risks only; the complete scored risk register is controlled by Section 23.

Risk	Mitigation / evidence required
16,000+ CCU target not achieved	Progressive load tests by product and combined mix; scale-out fleet, capacity model and degradation ladder in 8.5–8.9; publish only measured capacity
GPU memory starvation / OOM	Admission control, model sizing, priority-aware scheduler, monitoring, fallback and circuit breaker
Cross-product interference	Separate API pools, queues, Valkey instances and data scopes; guaranteed GPU floors and per-product circuit breakers (8.8–8.9)
Long-running render failure	Durable state machine, idempotency, bounded retries, checkpoints and user-visible status
Scratch disk fills or unsafe deletion	Local scratch ownership, disk alarms/backpressure, checksum-gated upload, exact-file safe purge
Model/provider/license changes	Pin versions where possible; verify license, availability, price, and fallback contract
Voice cloning misuse	Consent/rights records, policy checks, audit trail and abuse response
Religious/factual misinformation	Source tracing, uncertainty labels, policy versioning and human review
Minor privacy / recording risk	Consent, visible recording status, role-based access, retention and deletion
Unproven pricing/unit economics	Paid pilots, cost-per-job measurements, regional tax/payment review
Overstated investor claims	Separate shipped features, MVP scope, roadmap and hypotheses; attach evidence to metrics

13.4 Investor diligence inputs still missing

- Verified customer traction, paid users, retention cohorts and revenue data.
- Confirmed team biographies (the hiring plan is in 14.2).

- Signed term sheet; the funding ask, valuation, runway and use of funds in Section 15 are working assumptions.
- Supplier quotes and confirmed server SKUs, warranty and deployment status (data centres: OVHcloud US, RedSwitches and Database Mart in the USA, 8.16).
- Benchmark reports supporting capacity, latency, quality and cost claims.
- Legal review for target markets, privacy, child safety, IP, voice cloning and content licensing.
- Revenue model with the v2.0 plan prices (11.1): 15.7 uses planning assumptions until the release plan and Free Beta data are in.

14. Organization, Governance & Hiring Plan

Purpose: define ownership, staffing sequence and operating responsibilities for NexaTute (with PlayBoom), Remon Motion Studio, NexaLogicDev and the shared platform. Headcount and salary figures below are planning assumptions carried forward from the working proposal; they require reconciliation before approval.

14.1 Operating Structure

Function / role	Primary accountability
CEO / Founder	Company strategy, capital allocation, executive decisions, investor and partner relationships.
CTO / Engineering Lead	Architecture, engineering delivery, reliability, security engineering and technical hiring.
NexaTute team	Learning product, tutor experience, curriculum/content workflows, student and teacher experience.
Remon Motion Studio team	Creation workflow, script/storyboard, voice/media pipeline, rendering, remake policy and delivery.
NexaLogicDev team	AI Cloud IDE, builders, Auto-Architect, MiniMax-M2.7 / Qwen3-Coder-Next serving and the sandbox; billing, wallet and sandbox platform work by the shared team (Section 32)
Platform / SRE	Shared infrastructure, observability, deployment, backups, incident response and capacity testing.
Security / Governance	Access control, privacy, AI safety, content verification, licensing and audit evidence.
Business & Operations	Sales, customer support, finance operations, vendor management and administration.

14.2 Hiring Plan (Proposed)

Role	Count	Target timing	Indicative monthly salary (BDT)
CEO / Founder	1	Current	Not specified

Role	Count	Target timing	Indicative monthly salary (BDT)
CTO	1	Month 1	৳150,000
Backend Engineer	2	Month 1	৳80,000 each
AI/ML Engineer	1	Month 1	৳100,000
Content Lead	1	Month 2	৳60,000
Full-stack Engineer	1	Month 1	৳80,000
Content Writers	2	Month 3	৳35,000 each
Marketing Manager	1	Month 3	৳70,000
Customer Support	2	Month 3	৳30,000 each
DevOps / SRE Engineer	2	Month 1	৳90,000 each
UX/UI Designer	1	Month 2	৳60,000
B2B Sales	1 (2nd after a B2B revenue threshold)	Year 2 (Month 13)	৳60,000
Partnership Manager	1	Month 3–4	Not specified
AI/ML Engineer — NexaLogicDev (Heavy and Fast model serving)	1	Month 6 (March 2027)	৳100,000
Full-stack Engineer — NexaLogicDev (IDE, builders)	1	Month 6 (March 2027)	৳80,000
NexaLogicDev revenue-gated roles: 2nd Full-stack (MRR ≥ ৳2 lakh), dedicated Support (≥150 paid users), Developer Advocate (MRR ≥ ৳4 lakh)	3	When the trigger is met	৳80,000 / ৳35,000 / ৳50,000 — paid from NexaLogicDev revenue
NexaLogicDev pre-launch QA (contract)	1	Months 9–12	৳50,000 — Funding B
Subject reviewers (part-time, contract)	As needed	Before each course opens	Not specified
MindForge psychometrician (part-time, contract)	1	Month 2 (November 2026)	≈৳35,000 (estimate, to be confirmed)
MindForge item reviewer (part-time, contract)	1	Month 3 (December 2026)	≈৳20,000 (estimate, to be confirmed)
Exam & Admission Hub circular reviewer (part-time, contract)	1	Month 2 (November 2026)	To confirm

Role	Count	Target timing	Indicative monthly salary (BDT)
English-medium content reviewer (part-time, contract)	1	Before the English-medium launch	To confirm

All launch-critical roles join in Months 1–3 (October–December 2026): the DevOps and support roles before the NexaTute Free Beta on 20 November 2026, the Motion Studio roles before its Free Beta on 20 December 2026, and the rest before the 1 January 2027 paid launch. The partnership manager (Month 3–4) covers schools and institutions, affiliates, gift-card retailers, employers and agencies and the teacher marketplace (3.21); part-time subject reviewers check content before each course opens (3.13). On payroll (26 October 2026): Md Lablu Mia (operations and SRE owner), Nazia Rahman (Data Protection Officer), Tanzila Akter (partnership manager), the two Human Controllers and the security owner (Mst Sonia Sabuz); how their salaries sit in the Funding A salary line (0.5) is to be confirmed.

MindForge (37.5): a part-time psychometrician prepares the pilot protocol, the ethics application and the calibration plan before the Free Beta and owns validity, reliability and fairness evidence; a part-time item reviewer checks samples and bias. MindForge has no dedicated content writers because its items are generated and checked by code and AI.

14.3 People & Culture

- Remote-first operating model with Bangladesh as the initial hiring and operating focus.
- Use clear role ownership, documented runbooks, code review, onboarding and recurring training.
- Promote AI-first problem solving, customer empathy, responsible AI use and security-by-design.
- Equity or other long-term incentives may be considered subject to legal, tax and board approval.
- Any three-month trial/probation arrangement must be reflected in compliant employment contracts.

14.4 Hiring Budget Reconciliation Note

Reconciled from the role-by-role plan in 14.2 (CEO compensation not specified and excluded): all 14 launch-critical hires — including two DevOps/SRE engineers for the clustered infrastructure — join in Months 1–3, so Year 1 has 15 staff including the CEO and a base-salary run-rate of ৳9.9 lakh/month from Month 3; Year 2 — 17 staff after the two B2B Sales hires, run-rate ৳11.1 lakh/month; Year 3 — 25 staff / ৳18 lakh monthly remains a planning target. Cumulative base salaries over the 18-month runway total ৳1.80 crore; with an assumed 10% employer on-cost (festival bonus, benefits and statutory costs) the requirement is about ৳1.98 crore, so the NexaTute and Motion Studio base was ৳2.00 crore; in the ecosystem budget the Salaries line in 15.2 is ৳2.30 crore for all three products, including the MindForge part-time roles (reconciliation below). Finance must confirm start dates, the on-cost rate and CEO compensation.

Ecosystem reconciliation: salaries for all three SaaS products are budgeted together inside the unchanged ৳5 crore Funding A. The NexaTute and Motion Studio base requirement stays ≈৳1.98 crore; one B2B Sales hire is deferred until B2B revenue justifies it (−≈৳4 lakh); the NexaLogicDev AI/ML and Full-stack engineers from Month 6 add ≈৳25.7 lakh including the 10% on-cost. This brought the Salaries line to ৳2.20 crore; with the MindForge part-time roles below it is ৳2.30 crore (15.2). NexaLogicDev's other roles are revenue-gated and its pre-launch QA contract is paid from Funding B (0.5).

MindForge reconciliation: the part-time psychometrician (≈৳35,000 a month, Months 2–18) and item reviewer (≈৳20,000 a month, Months 3–18) cost ≈৳9.2 lakh over the window as contract fees with no

employer on-cost — estimates to be confirmed. They are paid inside the unchanged ৳5 crore Funding A by moving ৳10 lakh from the Content line (৳25 → ৳15 lakh) to Salaries (৳2.20 → ৳2.30 crore); the ≈৳0.8 lakh left over stays in Salaries as headroom. The Content line can give this up because MindForge items are generated by code and AI rather than written by hand (to be confirmed with the founders).

15. Funding Ask, Use of Funds & Milestones

This section records the working investor discussion assumptions. They are not an offer of securities, a guaranteed return or a verified forecast.

15.1 Funding Assumptions

Item	Working assumption	Status
Funding ask	৳5 crore	Proposed; not confirmed
Runway	18 months	Target; depends on actual burn
Pre-money valuation	৳30 crore	Negotiation assumption
Post-money valuation	৳35 crore	৳30 crore pre-money + ৳5 crore new funding
Equity offered	14.29%	৳5 crore ÷ ৳35 crore post-money; requires legal/financial validation

15.2 Use of Funds

Category	Amount	Share
Salaries	৳2.30 crore	46%
Infrastructure	৳1.65 crore	33%
Marketing	৳55 lakh (NexaTute + Motion Studio ৳52 lakh; NexaLogicDev launch ৳3 lakh)	11%
Content	৳15 lakh	3%
Legal	৳10 lakh	2%
R&D	৳5 lakh	1%
Working capital	৳10 lakh	2%
Contingency	৳10 lakh	2%
Total	৳5.00 crore	100%

Ecosystem allocation: Funding A stays final at ৳5.00 crore and now carries salaries and marketing for all three SaaS products. Sources inside the ask: Infrastructure ৳1.80 → ৳1.65 crore (+৳15 lakh; the 17-month rent window to March 2028 brings the ≈৳1.71 crore estimate to ≈৳1.645 crore, 15.6), R&D

৳10 → ৳5 lakh (+৳5 lakh; NexaLogicDev model benchmarking runs in Funding B's first three months), salary-line rounding (+৳2 lakh), one deferred B2B Sales hire (+≈৳4 lakh) and ৳3 lakh moved within Marketing. Uses: the NexaLogicDev AI/ML and Full-stack engineers from Month 6 (≈৳25.7 lakh) and the NexaLogicDev launch campaign (৳3 lakh). NexaLogicDev infrastructure is Funding B (৳1.50 crore, 0.5).

MindForge allocation: ৳10 lakh moves from Content (৳25 → ৳15 lakh) to Salaries (৳2.20 → ৳2.30 crore) for the MindForge part-time psychometrician and item reviewer (≈৳9.2 lakh, 14.4); the Funding A total stays ৳5.00 crore.

Infrastructure is rented, not purchased: estimated ≈US$5,530 (≈৳6.75 lakh) per month for the 16-server fleet and the disaster-recovery replica (15.6) — 2× RTX PRO 6000, 2× RTX PRO 5000 and 2× RTX PRO 4000 GPU servers, Core-1 at 10 Gbps and Core-2 at 1 Gbps and 8 CPU servers — × 18 months ≈ ৳1.21 crore, plus ≈৳1.5 lakh one-time setup, plus ≈৳13 lakh for Cloudflare R2/CDN and other services, plus a dedicated ৳30 lakh cloud-LLM fallback budget for the GPT-6 Luna (OpenAI) → Mistral Large 3 (Mistral AI) chain (≈৳1.7 lakh per month, enforced by the gateway spend cap in 8.9) plus ≈৳1.5 lakh for the 3-month starting term on the GPU servers, ≈ ৳1.67 crore; App-3 from December 2026 adds ≈৳4.1 lakh (≈৳1.71 crore in total), within the ৳1.65 crore Infrastructure line set from the 17-month rent window (15.6). Rates are October 2026 listed prices at ≈৳122/US$ (15.6) and must be confirmed by supplier quotes. To fund rental, the cloud fallback budget and pre-launch hiring (including a second DevOps engineer) within the ৳5 crore ask, Marketing, Content, Legal, R&D, working capital and contingency were reduced; restoring the earlier Marketing, Legal and contingency budgets would have required about ৳6 crore, but Funding A stays final at ৳5 crore. Scale-out beyond the launch fleet (8.5) is added only when its triggers are reached and is funded from revenue at that stage; at the full 16,000 CCU fleet the estimated rent is ≈US$15.5–16.5k (≈ ৳18.9–20.1 lakh) per month after the 17 October 2026 changes (≈US$17,670 before them), plus hourly burst (15.6).

15.3 Proposed Milestones

Target date	Milestone	Qualification
Month 2 (Nov 2026)	NexaTute Free Beta	Servers rented from 5 November; NexaTute Free Beta 20 November–9 December 2026
Month 3 (Dec 2026)	NexaTute feature set complete	NexaTute launch on 1 January 2027 after Gates 1–4
Month 4 (Jan 2027)	≈300 paying NexaTute learners; ≈240 paying Motion Studio creators	Base case 15.8; define active-user and verified-account rules
Month 5 (Feb 2027)	≈৳13 lakh net revenue a month (all products)	Base case 15.8; revenue recognition and refunds must be defined
Month 9 (Jun 2027)	≈1,600 paying NexaTute learners and ≈530 PlayBoom children	Base case 15.8; monthly spend covered from July 2027
Month 12 (Sep 2027)	≈2,400 paying NexaTute learners; NexaTute MRR ≈৳21.7 lakh; NexaLogicDev launch	Base case 15.8 at ≈৳900 ARPU (VAT included); ecosystem net revenue ≈৳48.1 lakh a month

Target date	Milestone	Qualification
Month 18 (Mar 2028)	Cumulative result positive (from February 2028); ≈৳83 lakh net revenue a month	Base case 15.8; re-run monthly from actual data

Timeline anchor: Month 1 = October 2026. Servers are rented from 5 November 2026; the NexaTute Free Beta runs 20 November–9 December 2026 (Month 2) and the Motion Studio Free Beta starts on 20 December 2026 (Month 3); the NexaTute feature set is built and acceptance-tested in Months 1–3 for its 1 January 2027 paid launch; Year 1 OKR targets marked "Month 12" refer to September 2027 and the 18-month runway ends in March 2028.

15.4 Illustrative Investor Scenarios

Horizon	Illustrative company valuation	Interpretation
Year 2	৳100 crore	Scenario only; not a forecast
Year 3	৳300 crore	Scenario only; depends on growth and market conditions
Year 5	৳1,000 crore	Scenario only; no guaranteed return

Prior draft exit examples included strategic acquisition in 5–7 years, PE buyout in 7–10 years and IPO after 10+ years. These are possible scenarios, not commitments. Dilution, liquidation preferences, taxes, transaction costs and the actual exit value affect investor proceeds.

15.5 Funding Reconciliation Required

The ৳5 crore figure is treated in this master as the initial 18-month funding ask. The later Risk Register's ৳10 crore base, ৳12 crore buffered and ৳15 crore worst-case figures are total programme contingency scenarios, not additional funding asks. Finance must replace this planning classification if the scenarios represent future rounds or another time horizon.

15.6 Infrastructure Cost Breakdown

Monthly rent for the launch fleet from the suppliers' listed prices in October 2026 (US$, excluding taxes; ≈৳122/US$). GPU servers use 24-month reserved rates; signed quotes replace these figures before ordering. The Database Mart GPU servers are dedicated-GPU VPS, so they start on a 3-month term (≈US$400 more a month: 2 × 749 + 2 × 418 + 2 × 230) while CPU steal, disk speed and GPU throughput are benchmarked in the first month; the 24-month term is signed only after the benchmarks pass, and all other CPU work runs on the dedicated CPU servers (8.12).

Servers	Supplier and plan	US$ per month	Notes
AI-1 Node A/B	Database Mart, Dallas — RTX PRO 6000 96 GB, 32 vCPU, 84 GB RAM	2 × 648 = 1,296	12-month term: 2 × 698
AI-2 Node A/B	Database Mart, Dallas — RTX PRO 5000 48 GB, 24 vCPU, 56 GB RAM	2 × 360 = 720	12-month term: 2 × 389
Rec-1/Rec-2	Database Mart, Dallas — RTX PRO 4000 24 GB, 24 vCPU, 56 GB RAM	2 × 187 = 374	12-month term: 2 × 209; 1-month: 2 × 252

Servers	Supplier and plan	US$ per month	Notes
App-1/2, DB-1/2, Redis1/2	OVHcloud US, Vint Hill — ADVANCE-2: AMD EPYC 4344P 8 cores/16 threads, 64 GB DDR5, 2 × 960 GB NVMe	6 × 198 = 1,188	One-time installation equal to one month (≈US$1,188)
App-3 (from December 2026)	OVHcloud US, Vint Hill — ADVANCE-2, the same as App-1/App-2	198	One-time installation ≈US$198; ordered 1 December 2026 (Section 07, 8.5)
Core-1	RedSwitches US — AMD EPYC 9555P 64 cores, 384 GB, 2 × 3.84 TB NVMe, 10 Gbps unmetered (unlimited traffic)	1,315	Unlimited traffic, so LiveKit growth needs no traffic add-on
Core-2	RedSwitches US — 2 × Xeon Gold 6130 (32 cores/64 threads), 128 GB, 1 Gbps unmetered (unlimited traffic)	259	Unlimited traffic
LB-1/LB-2	RedSwitches US — entry server (about 6 cores, 32 GB)	2 × ≈90 = ≈180	Estimate; US listing to be confirmed
Disaster-recovery replica	OVHcloud US, Hillsboro — ADVANCE-2	198	Second region (8.16)
Total servers	18 servers from December 2026 (16 + App-3 + recovery replica); 17 in November	≈5,728 (≈৳6.99 lakh); ≈5,530 in November	Previous estimate ≈US$6,200
Cloudflare and other services	R2 storage, CDN, Workers, transactional email	≈৳72,000	To re-quote: advanced WAF rules and Waiting Room need a Cloudflare Business or higher plan, and the C2PA signing certificate from a CA on the C2PA trust list is an annual cost (G.7)
Cloud LLM budget	GPT-6 Luna → Mistral Large 3, capped by the gateway (8.9)	≈৳1.67 lakh	Unchanged (৳30 lakh over 18 months)
Platform total per month		≈৳9.38 lakh (≈৳9.14 lakh in November)	

Notes: the inventory's "16 cores" for App, DB and Valkey servers is read as 16 threads, which ADVANCE-2 provides; the Core-1 listing has more cores and memory than the 48-core, 256 GB inventory specification because it is the US 10 Gbps server currently listed; RedSwitches currently lists its US servers in New York — if Washington DC is not offered at order time, New York (a few milliseconds from Virginia) is used with no design change. Over 18 months, including the 3-month starting term, the infrastructure costs ≈৳1.67 crore, or ≈৳1.71 crore with App-3 from December 2026 (≈৳4.1 lakh including installation), against the ৳1.65 crore Infrastructure line in 15.2. Because rent starts on 5 November 2026, the window to March 2028 holds about 17 months of rent, about ৳6–7 lakh less than this figure. At 16,000 CCU the earlier 8.5 plan (16 × RTX PRO 6000, 7 × RTX PRO 5000, 3 × RTX PRO 4000 and 16 CPU servers, plus the recovery replica) costs ≈US$17,670 (≈৳21.6 lakh) a month before hourly burst; with the 17 October 2026 changes (CPU server voices, LightX2V and FastWan, uploads straight to R2) the plan is ≈37–39 servers at ≈US$15.5–16.5k (≈৳18.9–20.1 lakh) a month, to be confirmed by the benchmark.

Running costs added on 26–29 October 2026 (samples — to confirm): SMS to guardians is free for families and on by default, so NexaTute pays the licensed SMS provider (about ৳0.35 an SMS); outbound guardian voice reminders about ৳0.80 a minute (fallback ৳0.95) and WhatsApp voice messages about ৳0.50 each; the IPTSP phone-tutor line from launch (per-minute cost to confirm);

WhatsApp bot replies are charged after 1,000 free service messages a month per number. Each line is capped by a monthly service budget (38.18). LiveKit needs no new server — Core-1 and Core-2 already form the two-node cluster (9.5) — and the storage allowances stay within the R2 budget.

15.7 Unit Economics

Planning assumptions (to be replaced with measured data after launch): the plan prices in 11.1 with a plan mix of 40% Batch, 40% Combo and 20% Personal (the 11.12 targets); 85% of learners in the ৳1,000-group courses and 15% in higher-priced courses; about ৳50 a month of add-ons (extra minutes, Crash Packs, Daily Passes) and about 8% of discounts (prepay, sibling) — a blended ARPU of about ৳900 including VAT, as in 15.3. Prices include VAT: at the standard 15% rate (to be confirmed in the legal and tax review) and after 2% payment fees and 1% refunds, net revenue is about ৳759 per paying learner a month. NexaTute carries 60% of the platform cost in 15.6 (≈৳5.48 lakh a month), in line with its LLM-pool share.

Paying learners	Net revenue per month	Infrastructure cost per paying learner	Result after NexaTute's infrastructure share, per month
500	৳3.80 lakh	৳1,096	−৳1.68 lakh
1,000	৳7.59 lakh	৳548	+৳2.11 lakh
2,000	৳15.18 lakh	৳274	+৳9.70 lakh
5,000	৳37.95 lakh	৳110	+৳32.47 lakh
10,000	৳75.90 lakh	৳55	+৳70.42 lakh

What this means for the plan prices: the infrastructure cost per paying learner falls from about ৳548 at 1,000 learners to about ৳55 at 10,000, so the plans carry a wide margin at scale; a Personal learner who uses all 75 minutes every day costs only about ৳45 a month in compute when the launch fleet is full (≈৳0.02 a minute). The risk is volume, not price: about 720 paying learners cover NexaTute's share of the infrastructure, and about 3,660 paying learners (on NexaTute alone) cover the whole monthly spend of ≈৳27.8 lakh (৳5 crore over 18 months); Motion Studio revenue lowers that number. These figures are higher than the ≈620 and ≈3,150 of the wallet model only because prices now include VAT. Below about 720 paying learners each month runs at a loss on infrastructure, so the Free Beta and Free Trial must stay inside their caps.

The income lines approved on 18 October 2026 (11.14), PlayBoom, Motion Studio, NexaLogicDev and the affiliate cost are now combined in the ecosystem revenue model (15.8).

Motion Studio: the rent of one GPU-minute is about ৳1.83 on an RTX PRO 6000, ৳1.02 on an RTX PRO 5000 and ৳0.53 on an RTX PRO 4000 at full use. Each per-minute rate in G.8 therefore pays for a limited number of GPU-minutes per finished minute of video; the benchmarked GPU-minutes per output minute (8.6) must stay below the 70%-use column. Quick mode renders in the browser and uses almost no GPU, so it carries the widest margin; monthly package fees cover storage and platform costs, and the pre-production fee (৳20 per 2 minutes, 4.17) covers the script, key-image and preview compute of about ৳3–10 per 2 minutes, including projects that never reach a final video.

Mode and tier (G.8)	Rate per minute	RTX PRO 6000 GPU-minutes covered (full use)	At 70% average use
3D Animation, Premium Professional, 480p Native	৳25	≈14	≈10

Mode and tier (G.8)	Rate per minute	RTX PRO 6000 GPU-minutes covered (full use)	At 70% average use
Cinematic, 480p Native (Freelancer packages)	৳41	≈22	≈16
Cinematic, 720p Native (Freelancer packages)	৳61	≈33	≈23
Cinematic, 1080p Escalating (Freelancer packages)	৳74	≈40	≈28
Cinematic, 1080p Native (all paid packages; after benchmark)	৳162	≈89	≈62

Customer acquisition cost (CAC) and lifetime value (LTV): with ≈৳783 ARPU after VAT (≈৳900 including VAT) and ≈90% gross margin at scale, LTV depends mainly on how long learners stay. The rule is to keep LTV at least three times CAC and recover CAC within six months; CAC is measured per channel every month, and churn from the first paying cohorts replaces the assumptions below.

Monthly churn	Average lifetime	LTV	Maximum CAC (LTV ÷ 3)
5%	20 months	৳14,090	৳4,700
8%	12.5 months	৳8,810	৳2,940
12%	8.3 months	৳5,870	৳1,960

With the ৳55 lakh marketing budget (15.2), a CAC of ৳1,000 would bring about 5,500 paying learners and a CAC of ৳2,000 about 2,750 — enough in both cases to pass the ≈720-learner infrastructure break-even, and in the first case to approach the ≈3,660-learner company break-even together with Motion Studio revenue.

15.8 Ecosystem Revenue Model

Revenue lines added on 26–29 October 2026 and not yet in the base case: the white-label licence fee and the NexaTute for Partners prices (sample ৳150 a seat a month or ৳1.50 a minute) are samples — to confirm; the Doubt Pass (৳50), storage add-ons, crash programs and English-medium Class 4–8 courses join the model after Free Beta data.

This model replaces the single-product ARPU view for investor use. It covers the 18 months of Funding A (October 2026–March 2028) for all four paid products, the 11.14 income lines and the affiliate cost. All inputs are planning assumptions to be replaced with Free Beta and first-cohort data; prices are VAT-inclusive and are converted to net revenue after 15% VAT, 2% payment fees, 1% refunds and the reseller discount.

Assumptions

Line	Assumption
NexaTute	Paying learners 300 in January 2027 rising evenly to 4,000 by March 2028; blended ARPU ৳900 incl. VAT (11.1 mix 40% Batch / 40% Combo / 20% Personal, add-ons and discounts, 15.7)
PlayBoom	Paying children 100 in February 2027 rising to 1,500 by March 2028; ARPU ≈৳540 incl. VAT (৳599 per child less sibling 30% off, annual plans and Daily Passes)
Remon Motion Studio	Paying creators 150 in December 2026 rising to 1,500 by March 2028; ARPU ≈৳3,000 incl. VAT (package fee + per-minute usage + pre-production)

Line	Assumption
NexaLogicDev	Paid users 40 at the September 2027 launch rising to 375 by March 2028 (on the way to ≈500 by May 2028); net ≈৳1,800 per user (32.13)
MindForge Pro	Add-on subscribers = 15% of paying NexaTute learners (the attach-rate target, 37.10), from January 2027 (45 → 600 by March 2028); ARPU ≈৳180 incl. VAT (৳199 less prepay, second-child and institution discounts); learners who buy Pro without a plan are not counted
Reseller sales	15% of sales through resellers at an average 18% discount (all products, 33)
11.14 lines	Nexa Co-Teacher Pass 2% of free users (2,000 → 20,000); Olympiad 2% of learners at ≈৳800; Study Abroad Coach 20 packs a month; Script Checker ≈2,000 paid scripts a month; Head Start 10% of learners in December 2027; gift cards 2% of NexaTute sales less 5% retailer commission; teacher marketplace 20% of ≈৳1.5 lakh sales a month from January 2028; job pathway ≈10 placements × ৳5,000 from July 2027 (fee to be set); Academic Year Pass and commitment plan change cash timing, not the totals
Affiliate cost	15% of new NexaTute learners through affiliates, 10% of their first 3 months
Spend	Funding A ৳5 crore spread evenly (≈৳27.8 lakh a month) plus Funding B spend from March 2027 (32.13); market entry (India Q2 2028) is outside this window

Month by month (net revenue vs. spend; paying users per product; net revenue includes MindForge Pro)

Month	NexaTute	PlayBoom	Motion	NexaLogic Dev	Net revenue	Spend	Result	Cumulative
Oct 26	0	0	0	0	৳0.0 L	৳27.8 L	−৳27.8 L	−৳27.8 L
Nov 26	0	0	0	0	৳0.0 L	৳27.8 L	−৳27.8 L	−৳55.6 L
Dec 26	0	0	150	0	৳3.7 L	৳27.8 L	−৳24.1 L	−৳79.6 L
Jan 27	300	0	240	0	৳8.2 L	৳27.8 L	−৳19.6 L	−৳99.2 L
Feb 27	564	100	330	0	৳13.0 L	৳27.8 L	−৳14.8 L	−৳114.0 L
Mar 27	829	208	420	0	৳17.8 L	৳28.2 L	−৳10.4 L	−৳124.4 L
Apr 27	1,093	315	510	0	৳23.0 L	৳28.2 L	−৳5.2 L	−৳129.7 L
May 27	1,357	423	600	0	৳27.8 L	৳28.2 L	−৳0.4 L	−৳130.1 L
Jun 27	1,621	531	690	0	৳32.6 L	৳34.1 L	−৳1.5 L	−৳131.5 L
Jul 27	1,886	638	780	0	৳37.8 L	৳34.1 L	৳3.8 L	−৳127.8 L
Aug 27	2,150	746	870	0	৳42.6 L	৳34.1 L	৳8.6 L	−৳119.2 L
Sep 27	2,414	854	960	40	৳48.1 L	৳38.1 L	৳10.0 L	−৳109.2 L
Oct 27	2,679	962	1,050	96	৳54.0 L	৳38.1 L	৳15.8 L	−৳93.4 L
Nov 27	2,943	1,069	1,140	152	৳59.8 L	৳38.6 L	৳21.1 L	−৳72.3 L

Month	NexaTute	PlayBoom	Motion	NexaLogic Dev	Net revenue	Spend	Result	Cumulative
Dec 27	3,207	1,177	1,230	208	৳66.9 L	৳38.2 L	৳28.8 L	−৳43.6 L
Jan 28	3,471	1,285	1,320	263	৳71.6 L	৳38.2 L	৳33.5 L	−৳10.1 L
Feb 28	3,736	1,392	1,410	319	৳77.4 L	৳38.2 L	৳39.3 L	৳29.1 L
Mar 28	4,000	1,500	1,500	375	৳83.2 L	৳37.7 L	৳45.6 L	৳74.7 L

18-month totals by product

Revenue line	Net revenue, 18 months
Remon Motion Studio	৳325.0 L
NexaTute subscriptions and add-ons	৳238.2 L
PlayBoom	৳49.6 L
MindForge Pro add-on (from January 2027)	৳7.1 L
NexaLogicDev (from September 2027)	৳26.2 L
11.14 income lines	৳22.8 L
Affiliate commission (cost)	−৳1.4 L
Total net revenue	৳667.5 L
Total spend in the window (Funding A ৳500 L + Funding B to March 2028)	৳592.8 L

- Base case: the ecosystem covers its monthly spend from July 2027 and the 18-month cumulative result turns positive in February 2028; March 2028 net revenue is ≈৳83 lakh a month against ≈৳38 lakh of spend. MindForge Pro adds ≈৳7 lakh of net revenue over the window and does not change the break-even months.
- Conservative case (half the paying users on every line): monthly spend is covered from about February 2028 and the window ends ≈৳2.5 crore below spend — still inside the ৳6.5 crore raised (Funding A + B), so the plan stays funded.
- Motion Studio carries the most revenue in the window because it launches first and has the highest ARPU; NexaTute overtakes it as learners grow. The 3-year targets in 15.3/15.4 stay as targets; this model is the investor base case until measured data replaces it.
- Risks: volume (CAC and conversion), Motion Studio ARPU, and Free Beta conversion. Each assumption is an admin-reviewed input; the model is re-run monthly from actual data.

16. Competitive Landscape & Positioning

16.1 Referenced Alternatives

Category	Examples cited in working notes	Competitive dimension
Bangladesh learning platforms	10 Minute School	Local curriculum, brand, content and distribution
Free learning resources	Khan Academy	Free access, breadth and self-paced learning
International education products	Coursera, Udemy	Course catalogue, brand, pricing and structured learning
General AI assistants	ChatGPT, Google Gemini	Flexible Q&A, general reasoning and broad availability
Language learning	Duolingo	Habit loops, gamification and language-focused learning
Offline alternatives	Local coaching and private tutors	Human guidance, local trust and established relationships

16.2 Proposed Differentiation

- NexaTute: an AI-Powered Learning Operating System for Bangladesh — Bengali-first personalized learning, curriculum-aligned experiences, AI tutor workflows, browser-side 3D/virtual lab and collaborative learning tools.
- NexaTute plan ladder (11.1): AI batch classes at coaching-centre prices, Combo with daily 1:1 time and Personal 1:1 teaching, plus a Nexa Co-Teacher that teaches in text (learners can speak a question and listen to the answer) and measured Thinking Profile gains (3.16–3.18).
- Remon Motion Studio: an AI-assisted script-to-video workflow with structured editing, voice/media processing, remake controls and validated final delivery.
- Shared infrastructure can lower duplicated platform work, while separate product runtimes, permissions, queues and accounting preserve product isolation.
- Initial niche focus proposed in earlier planning: Cambridge-related learning and BCS preparation, subject to validated demand and approved pricing.

16.3 Claims That Require Evidence

- Claims such as “50× cheaper than coaching,” “14,000+ topics,” or superiority over named competitors must not be published until independently substantiated and precisely scoped.
- Competitor capabilities, current pricing, market share and user counts require dated source checks before external use.
- Positioning charts are hypotheses for strategy discussion, not objective market measurements.
- Hardware VRAM capacity or model count alone does not prove a sustainable competitive moat.
- Learning claims on the website, in sales and in investor material use only the published Learning Outcomes Report and prediction-accuracy results (3.24); the Landing Page AI cannot publish a claim without them (3.25).

17. Product & Platform Roadmap

Launch rule: the servers are rented from 5 November 2026; NexaTute opens its 20-day Free Beta on 20 November 2026 and its paid plans on 1 January 2027 with every feature in Sections 03 and 11 — including every feature once dated Q2 or Q4 2027 (decided 26 October 2026) and the 2027 engagement features, which switch on at the paid launch (3.32); Remon Motion Studio opens its 30-day Free Beta on 20 December 2026 with every feature in Section 04 and Appendices F–G except those marked for a later quarter, and its paid packages start on a date to be set (to confirm). Later quarters cover optimization, scale, new offerings and market expansion only. Dates are targets, not evidence that a feature has shipped. Each release must pass security, quality, licensing, cost and operational acceptance gates.

17.1 NexaTute

Target period	Planned focus
Q4 2026 (Months 1–3)	Build, integrate and acceptance-test the full feature set; closed beta; from 5 November 2026 the AI content pipeline first build, voice training and GPU benchmarks run on the rented fleet in the late-night window while Motion Studio serves its users
Q1 2027 — Launch (1 Jan 2027)	The launch release holds every approved NexaTute feature in Sections 03 and 11 — including those earlier planned for Q2–Q4 2027, Q1 2028 or the release plan — plus PlayBoom (paid from 21 January 2027, Section 30) and the Study Hall groups and moderator (31). Any feature that has not passed its tests by launch is shown as “coming soon” and switched on from the admin panel when it passes.
Q2 2027	Quality, latency and cost optimization from launch-cohort data; any launch feature still failing its gate switches on as it passes
Q3 2027	Controlled scale-up with load-tested capacity increases
Q4 2027	Curriculum depth and content expansion
Q1 2028	Teacher/creator marketplace (a launch-scope feature) at scale; new course categories
Q2 2028	India (Bengali-speaking) market entry per 29.4
Q3 2028	Pakistan market preparation per 29.4
Q4 2028	Continuous improvement and new course categories

Launch scope: all approved NexaTute and PlayBoom features are in the launch scope and target launch; labels such as “Release plan” or a later quarter elsewhere in this document now mean “in the launch scope, switched on when its tests pass”. Market entries (India Q2 2028, Pakistan Q3 2028) keep their dates. Risk: the launch scope is very large for the planned team, so the go-live gates (34) and the “coming soon” rule decide what is visible on 1 January 2027. Alternative billing in

the Play Store app is switched on when Google's new billing rules reach Bangladesh (target September 2027, 11.7).

17.2 Remon Motion Studio

Target period	Planned focus
Q4 2026 — Free Beta from 20 Dec 2026 (paid date to confirm)	Servers rented from 5 November 2026 and production checks on the rented fleet; the minimum launch feature list (4.14) goes live on 20 December 2026 in the 30-day Free Beta (11.11) and as paid packages from a date to be set (to confirm), and any feature that has not yet passed its tests is shown as "coming soon" and switched on from the admin panel (4.9) when it passes. Planned for launch: script/storyboard, cinematic and animated video, drama series, Reels & Shorts, Brand Kit, consent-gated voice cloning and multilingual voice, advanced editing, Islamic-content verification, avatar animation, Document-to-Video, analytics, multi-client review, API and webhooks; music and sound effects, all aspect ratios, basic timeline editor, AI content labels, styled Bangla captions (4.10); template library, product ad maker, hook/title/thumbnail suggestions, long video → Shorts, referral programme (4.11); copyright controls for uploaded videos (4.12); series and promo videos — intro, outro, recap, next-episode preview, thumbnails, teaser, trailer and highlights (4.18)
Q1 2027	Optimization from the first creator cohort; shares the fleet with the NexaTute launch on 1 January 2027 (daytime Motion Studio, night-time NexaTute)
Q2 2027	Render quality, render-time and cost optimization; auto-dubbing and direct publishing to YouTube, Facebook and TikTok (4.2, 4.6); video translate, AI presenter, podcast/audio → video, team workspace and the mobile app (4.11)
Q3 2027	Controlled scale-up with load-tested capacity increases
Q4 2027	Template library expansion; creator marketplace and client service marketplace at 20% commission (4.11)
Q1 2028	Marketplace expansion: more asset types and international sellers (new offering)
Q2 2028	White-label options for Enterprise
Q3 2028	Interactive Branching Drama general availability (Enterprise beta at launch)
Q4 2028	Evaluate global expansion

17.3 Shared Platform

Target period	Planned focus
Q4 2026 — Fleet from 5 Nov 2026	Rented server fleet from 5 November 2026 (GPU servers on a 3-month term with first-month benchmarks), secrets management (OpenBao), monitoring and alerting, AI gateway and cloud fallback, analytics — ready for the NexaTute Free Beta on 20 November 2026 and the Motion Studio Free Beta on 20 December 2026; App-3 joins the App pool in December 2026 after its load test (Locust, LiveKit CLI and vLLM benchmarks, H.3), before the NexaTute launch
Q1 2027 — NexaTute launch (1 Jan 2027)	Production operations, on-call rota and incident response for both products (Motion Studio in its Free Beta since 20 December 2026)
Q2 2027	Capacity and cost optimization
Q3 2027	Disaster-recovery drills and resilience improvements
Q4 2027	Analytics and reporting enhancements
Q1 2028	Evaluate multi-region design
Q2 2028	SDK and plugin interfaces
Q3 2028	Marketplace foundations
Q4 2028	Ecosystem expansion

17.4 Release Gates

- Do not announce a release as complete until functional, security, accessibility and regression tests pass.
- Load testing must report achieved CCU, p95/p99 latency, error rate, GPU utilization, queue delay and cost.
- Verify third-party/model licensing, voice consent, content safety and data-retention requirements before release.
- Use staged rollout, monitoring, rollback criteria and named operational ownership.

17.5 Day-1 must-have list (launch release)

Every approved feature is in the launch scope, but these must be live and gate-passed on launch day; everything else switches on from the admin panel as it passes its tests (“coming soon” until then). Motion Studio keeps its 4.14 minimum launch list.

#	NexaTute — must be live on 1 January 2027
1	Sign-up with Nexa ID, face verification with OTP fallback, age check, guardian consent and per-feature consent switches
2	Batch, Combo and Personal plans; aamarPay (Bangladesh) and Paddle (international); auto or manual renewal; guardian purchase approval; wallet top-ups and Daily Pass

#	NexaTute — must be live on 1 January 2027
3	Nexa Teacher 1:1 with voice in Bangla and English and the Smart Whiteboard
4	Batch classes for BCS + Job Test Preparation and SSC with polls, confusion meter and question wall
5	Nexa Co-Teacher (text, 25 topics a day) on the shared learner record
6	Automatic class recording, replay and class summaries
7	Mock tests and on-screen OMR practice for BCS and SSC
8	AI study plan, Learning Insights basics and the weekly guardian report
9	BCS and SSC course content that has passed the quality gate (3.13)
10	Load-aware routing, ETA, busy page and automatic minute restoration
11	Safety: Qwen3Guard, child-safety hash matching, behaviour ladder and human-controller alerts
12	Privacy settings, data download and deletion
13	Free Trial, free tier basics and the silent Live Study Hall
14	Installable web app (PWA) and notifications (push, email, SMS)
15	Support inbox (AI first, then a person) and the admin panel for plans, prices, controllers and reviewers
16	Monitoring, backups, restore test and go-live gates G01–G10 (34)

#	PlayBoom — must be live on 21 January 2027
1	Guardian account with child profiles, guardian PIN and OTP; no face templates
2	Per-child plan ৳599 with sibling discount, annual plan and ৳20 Daily Pass
3	Play/Nursery/KG voice-first core: letters, numbers, rhymes, phonics, colours and shapes
4	Class I–III core lessons in Math, Bangla and English (NCTB)
5	AI Play Buddy (voice + animated avatar) within the 30-minute daily AI limit
6	Digital handwriting tracing (paper scan may follow as it passes)
7	Screen-time rules: 25-minute sessions, up to 3 with a 2-hour gap, guardian note
8	Parent dashboard basics and the child-safety rules of 30.5
9	Readiness checklist passed (30.7)

Early follow-ups (switch on as they pass): Study Hall groups and the Nexa Study Hall Moderator, labs, Classroom Builder, other courses and batches, Olympiad and Head Start, marketplace, job pathway, phone tutor and the remaining 3.10 features.

17.6 Roadmap 2027–2028 (approved future features)

For investors: every approved feature that is not live on the first day of the Free Beta, with the point at which it switches on. Dates after launch follow measured demand and gates (17.4); the public "Coming in 2027" page on the landing page lists the learner-facing items (3.32).

When	What switches on
Free Beta, 20 November–9 December 2026	The core learning product (Sections 03 and 11) with the Free Beta limits (11.11); the source line, the 20-20-20 eye break and the Beta poll; the Free-Beta safety items (3.33); everything else in 3.32 shown as "Coming soon"
Paid launch, 1 January 2027	The 2027 features of 3.32 — Team Quest and Quiz Battle, the Shorts question, AI Debate Chamber, family quiz cross-rewards, Live Game Show, Be the Teacher, AR Magic Book, Time-Travel role-play, Nexa Radio, mystery missions, group mind maps, self check-in and breaks, Learn → Earn, the teacher's AI assistant, Helper rewards, the Parent–Teacher Meeting by phone, the job board, Nexa Swap, the Study Buddy companion, white-label portals, AI comic stories, yearly sponsorship packages, the what-if simulator, Error Autopsy, Voice mode and the study bots — and every feature earlier dated Q2 or Q4 2027 (3.10, 3.12)
21 January 2027	PlayBoom paid, after its readiness checklist (30.7)
After the Motion Studio Free Beta (from 20 December 2026; paid start to confirm)	Motion Studio paid packages and delivery-speed tiers (G.8.3); Cinematic 720p Native and higher only after the video optimisations pass (35.5)
After the paid launch, phased	NexaTute for Partners (API and LTI tool); white-label partners' own store apps; English-medium Class 4–8 courses with the English-medium launch after the quality gate; new countries' curricula as each passes the quality check (30.9)
From the Free Beta measurements	A LiveKit edge server in Singapore or Mumbai (1 Gbps) only if latency needs it; a BDIX-connected local cache only if some operators do not reach the Dhaka edge (9.4, 9.5); hub ideas beyond the 54 (3.30)
March and September 2027	NexaLogicDev development (March) and launch (September 2027, Section 32)
By September 2027	Alternative app-store billing when Google's new rules reach Bangladesh (11.7)
1 November 2027	Reminder to buy the separate outreach domains (3.26)
Before 2 December 2027	EU AI Act compliance for MindForge assessment (37.9); until then assessment stays off in the EU
2028	NexaLogicDev later updates (32.8) and SBX-3 at its trigger (7.2); MindForge IQ-style scores only after norms, a technical manual and external validation (37.9)
After the lawyer and DPO check	Nexa Co-Admin on WhatsApp and Messenger for staff; imo once business API access is confirmed (38.20)

18. Technology Stack, Model Registry & Interfaces

This is a proposed stack inventory, not a statement that every component is deployed. Licence policy: every self-hosted component must be free, open source and permitted for commercial use (OSI-approved licence such as MIT, Apache-2.0, BSD, PostgreSQL, MPL-2.0, LGPL or AGPL), recorded with its SPDX identifier; only the open-source editions of open-core products are used.

Paid or non-open-source services are listed as deliberate exceptions in group 12 of the register (18.1). Pin exact versions and validate compatibility, support status and resource requirements before implementation.

18.1 Unified Software Register (RemonLogic Ecosystem)

Unified Software Register — the single list of every software component, model and service used by NexaTute and Remon Motion Studio. Product: N = NexaTute only (including PlayBoom and Live Study Hall), M = Motion Studio only, B = NexaTute and Motion Studio, L = NexaLogicDev only, E = every product of the ecosystem; a Motion Studio component that NexaTute reaches only through the Motion Studio API, such as the server renders of explainer videos, stays M (6.4). Licence: SPDX identifier. Status: Approved; Verify licence (blocked from production until confirmed, 18.4); Condition (open source with an obligation stated in the row); Evaluation only; Blocked until approved; Exception (deliberate paid or non-open-source service). Sections 4.7, 5.1 and 5.2 refer to this register and do not repeat it.

Totals: 244 register rows. Groups 1–12 hold 192 rows — 94 shared by NexaTute and Motion Studio (B), 58 NexaTute-only (N), 21 Motion Studio-only (M) and 19 used by every product (E); group 13 holds the 52 ecosystem additions — 25 NexaLogicDev (L), 14 used by every product (E), 11 NexaTute-only (N) and 2 shared by NexaTute and Motion Studio (B). Each component appears in one row only (names checked after normalising case, punctuation, brackets, vendor prefixes, “API” suffixes and version numbers), and the shared platform (servers, data, AI serving, safety and uploads) means no component runs twice.

Software	Licence (SPDX)	Product	Used for	Runs on	Status
1. Web apps and browser					
Next.js 16, React 19, Tailwind CSS 4	MIT	B	Web apps of both products; landing pages and AI CMS sites of both products (3.25, 4.19); Nexa Notes, the NexaTute landing-page blog with articles and videos, and the country and syllabus pages (3.25)	Browser; static files via Cloudflare CDN	Approved
sharp	Apache-2.0 (bundled libvips LGPL-3.0-or-later)	B	Image resizing and WebP/AVIF for the CMS media library (3.25); strips location data and resizes Study Hall Q&A images on the server (31.3)	App cluster	Condition: libvips used as a dynamic library (18.4)
next-intl	MIT	B	Interface translations (29.2); CMS page languages and hreflang (3.25); one URL per syllabus country and board, for example nexatute.com/in/cbse, with hreflang (3.25)	Browser	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
vanilla-cookieconsent	MIT	B	Cookie consent manager on both sites (3.25, 19.4); the Marketing switch is off by default, opt-in only and never offered to learners under 18; no advertising on our own sites (3.25)	Browser	Approved
web-vitals	Apache-2.0	B	Real-user page speed (Core Web Vitals) for the drop-off analysis (3.25)	Browser	Approved
Noto fonts (Bengali, Devanagari, Nastaliq Urdu, Naskh Arabic, CJK)	OFL-1.1	B	Every script, right-to-left layout, PDFs	Browser, CDN, WeasyPrint	Approved
Capacitor	MIT	B	Android apps of both products; the Android admin app of the platform and Institute admin panels, with push through FCM (38.15)	Device	Approved
Three.js	MIT	B	NexaTute 3D teacher, labs and classrooms (3.15); Motion Studio Quick-mode characters, AI presenter, the cinematic 3D look and Enhanced (2.5D) motion in the browser (WebGPU with WebGL2 fallback, 4.8, 8.18); GLB/VRM avatars of the Avatar & Voice Studio (38.16); the Bangla Sign Language avatar in every class (3.31)	Browser; headless Chromium fallback (4.8)	Approved
React Three Fiber	MIT	N	3D lab and classroom interface (3.15)	Browser	Approved
Ammo.js	Zlib	N	Virtual Lab physics	Browser	Approved
@pixiv/three-vrm (VRM avatars)	MIT	B	Own avatars (3.10, G.3); the VRM avatar library of the Avatar & Voice Studio — official Nexa Teacher avatars, Premium avatars and verified teachers' photo avatars (38.16)	Browser	Approved
Rhubarb Lip Sync	MIT	B	Lip-sync for the 3D teacher and Quick-mode characters; lip-sync check of every uploaded avatar before review (38.16)	Browser; server fallback	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
MediaPipe (Face Detection, Face Landmarker, proctoring signals)	Apache-2.0	B	NexaTute proctoring and the face-verified single user (liveness check and presence, 3.16); Motion Studio auto-reframe and thumbnail face detection (4.18); blurs faces in Study Hall Q&A images posted by under-18 learners (31.3); liveness check before a verified teacher's photo avatar is made (38.16); limited proctoring of certificate and institution exams — face check and random presence checks, no emotion or attention reading (3.27)	Browser; spare Core-1 cores; assembly-node CPU for thumbnail frames	Approved
SFace face-recognition model (OpenCV Zoo)	Apache-2.0	N	Face match for the face-verified single user (3.16); match of a teacher's avatar photo with the verified face (38.16)	Browser (ONNX Runtime Web); only an encrypted template on the server	Verify licence (model weights)
Uppy with the AWS S3 multipart plugin	MIT	B	Resumable uploads straight to R2 with presigned URLs (3.14, 4.16) and browser-render chunks (4.8)	Browser → Cloudflare R2 (presigned URLs from the App)	Approved; replaces tus/tusd so uploads do not pass through the LB or App nodes
Revideo	MIT	B	Browser rendering, timeline editor and templates; intro, outro and title cards (4.18); Motion Canvas dropped; NexaTute Wrapped and personal mistake videos in the learner's browser (3.20)	Browser; headless Chromium fallback through Puppeteer (4.8)	Approved
WebCodecs, OPFS/IndexedDB, Screen Wake Lock, Web Workers/OffscreenCanvas, Beacon API	Web standard	B	Browser encoding, offline chunks, keep-awake, background rendering and the closing signal (4.8); download MP4s of class recordings (9.5); test answers autosaved in IndexedDB and synced every 10 seconds (3.27)	Browser	Approved
Mediabunny	MPL-2.0	B	MP4 files in the browser: Motion Studio render chunks (4.8) and download MP4s of class recordings (9.5)	Browser	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Web Audio API	Web standard	B	Voice playback, audio mixing preview, and mixing class audio for download MP4s (9.5); PlayBoom tabla, dhol and harmonium pads (30.8)	Browser	Approved
Yjs	MIT	N	Whiteboard sync	Browser; Core-2	Approved
y-websocket	MIT	N	Yjs whiteboard sync server	Core-2	Approved
Excalidraw	MIT	N	Smart Whiteboard, always in the browser (also on the classroom board, 3.15)	Browser	Approved
WebLLM	Apache-2.0	E	Small in-browser LLM (NexaTute); optional offline autocomplete in NexaLogicDev (32.14)	Browser	Approved
transformers.js, ONNX Runtime Web	Apache-2.0, MIT	E	Browser Whisper and in-house VITS voices (NexaTute, Motion Studio 4.8, NexaLogicDev voice to code 32.14); SFace face match (3.16); pronunciation scoring, preview upscaling and background removal in the browser on capable devices (8.18)	Browser	Approved
KaTeX, Markmap, Mermaid	MIT	N	Maths, mind maps, diagrams	Browser	Approved
Apache ECharts	Apache-2.0	E	Dashboards and charts; the target-clients map of the AI Marketing Hub, drawn from boundary shapes served from our own CDN with no external map tiles, and its import and export reports (3.26)	Browser	Approved
FullCalendar (standard)	MIT	N	AI daily study plan (3.16)	Browser	Approved
ts-fsrs	MIT	N	Spaced-repetition flashcards; the FSRS review scheduler and the overdue-review KPI on the Mastery Map (3.29)	Browser / App cluster	Approved
Pyodide	MPL-2.0	N	Python practice	Browser	Approved
CodeMirror 6	MIT	N	Code editor; the coding-test editor (3.27)	Browser	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Blockly	Apache-2.0	N	Block programming; PlayBoom arrow-path coding puzzles (30.8)	Browser	Approved
v86	BSD-2-Clause	N	Browser Linux practice	Browser	Condition: GPL components shipped unmodified with source
JupyterLite	BSD-3-Clause	N	Browser notebooks	Browser	Approved
Phaser	MIT	N	Mini-games	Browser	Approved
hanzi-writer	MIT	N	Chinese stroke order	Browser	Approved
KanjiVG	CC-BY-SA-3.0	N	Japanese stroke order	Browser	Condition: attribution and share-alike
OpenDyslexic	OFL-1.1	N	Accessibility font	Browser	Approved
Avro phonetic rules (avro-py)	MIT	N	Bangla phonetic typing	Browser	Approved
WebXR	Web standard	N	AR 3D models	Browser	Approved
MindAR, model-viewer	MIT; Apache-2.0	N	AR Magic Book: image tracking of an NCTB book page and the 3D model shown on it; iPhone through Quick Look (3.32)	Browser	Approved
axe-core	MPL-2.0	B	Accessibility testing; check of every CMS change (3.25)	CI	Approved
Lighthouse CI	Apache-2.0	B	Page-speed and SEO check before CMS publishing (3.25)	CI runners and App cluster	Approved
2. Backend, platform and integrations					
Python 3.12+	PSF-2.0	B	Backend runtime	App cluster, workers	Approved
FastAPI, uvloop	MIT; MIT OR Apache-2.0	B	APIs in separate product namespaces	App nodes (App-1/App-2; App-3 from December 2026)	Approved
Celery 5.5	BSD-3-Clause	B	Asynchronous jobs	App-cluster workers	Approved
Instructor, Pydantic	MIT	B	Structured AI output	App cluster	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Keycloak	Apache-2.0	B	Identity, SSO, roles, team workspaces; 2-step login for every admin and finance user and a step-up OTP before money, role or export actions (38.12); staff suspend and reactivate with sessions ended at once (38.12); 2-step login before the staff-chat summary archive opens (38.15); OAuth client-credential keys with scopes for the institution roster and results API (18.3); passkeys and hardware security keys (WebAuthn/FIDO2) for admin staff, required for the super admin, money approvers and the security owner, with an authenticator-app fallback and 10 backup codes; learner and guardian login security — OTP resend wait, daily maximum, lockout, session length, device limit and sign-out of all devices (38.12, 38.17); passkey-linked staff chat-app accounts for Nexa Co-Admin and the secure confirmation link (38.20)	App cluster	Approved
argon2-cffi (Argon2id)	MIT	B	Password hashing	App cluster	Approved
Apache APISIX, Redoc	Apache-2.0, MIT	B	Public developer API and docs (Learning OS, Agency API); the institution roster and results API with scopes and rate limits (18.3)	App cluster (APISIX in standalone mode, no separate etcd)	Approved
Portkey AI Gateway	MIT	B	Cloud-fallback routing and spend caps; Nexa Co-Teacher order Mistral Large 3 → GPT-6 Luna (8.9)	App cluster	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
WeasyPrint, HarfBuzz	BSD-3-Clause, MIT	E	PDFs: reports, certificates, script and storyboard downloads; PDF exports of target-client lists from the AI Marketing Hub for all three products (3.26); printed papers from the paper builder, with the answer key kept restricted (3.27)	App cluster	Approved
In-house notification service with pywebpush (Web Push, VAPID)	In-house; MPL-2.0 (pywebpush); Web standard (Web Push)	E	Notifications by email, Web Push, WhatsApp and SMS; human-controller safety alerts (10.2); guardian purchase-approval requests, smart reminders and circular alerts (3.19, 3.20, 11.9); abandoned-checkout reminders (3.25); daily, weekly and monthly marketing reports (3.26); journey messages for all three products with frequency caps and quiet hours (3.26); accounting due-date and loan reminders (38.8, 38.10); admin push notifications in the platform and Institute admin panels — the bell with the unread count, info, important and urgent levels, routing by role or named user from the control panel, a reminder after 4 hours and the next approver after 24 hours, quiet hours and digests, staff-chat and call pushes; Web Push with VAPID keys to browsers and the installable admin app, the Android admin app through FCM, optional email, SMS and WhatsApp (38.15); “result ready” notifications for queued written checks (3.27)	App-cluster Celery workers; browser (Web Push)	Approved; replaces Novu, which needs MongoDB (SSPL, not open source)

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Chatwoot	MIT	B	Support inbox; pre-sale AI on Facebook Messenger, WhatsApp and the website chat, connected as an in-house agent bot through webhooks, not Chatwoot's paid Captain AI (3.25); Instagram DMs for the AI Marketing Hub (3.26); the Messenger and WhatsApp study bots for registered users (3.32)	App cluster	Condition: non-Enterprise code only
Unleash (open-source edition)	Apache-2.0	B	Feature flags, gradual rollouts, A/B variants and package switches (4.9); results analysed in TimescaleDB and Grafana; A/B tests of engagement features (3.19); landing-page A/B tests (3.25); platform-wide module switches (38.2); the platform-wide written-checking switch (38.2); Free Beta off / paid launch on switches of the 2027 features and the paused-feature message (3.32, 3.33)	App cluster (PostgreSQL)	Approved; replaces GrowthBook, which needs MongoDB (SSPL, not open source)
Centrifugo OSS	Apache-2.0	B	Real-time notices; Motion Studio render connection signal (4.8); batch-class polls, confusion meter, question wall and the AI text stream (3.17); Live Study Hall presence and round timers (3.20); internal staff chat of the platform and Institute admin panels — messages, presence and read state — and the live notification bell (38.15); Team Quest, Quiz Battle and the Live Game Show (3.32); the Beta community rooms (3.33)	App cluster	Condition: OSS edition only
LangGraph	MIT	N	AI agent layer (3.11); one memory for the Nexa Teacher and the Nexa Co-Teacher (3.16)	App cluster	Approved
SymPy	BSD-3-Clause	N	Rule-based answer checking; equivalent mathematical answers in fill-in-the-blank tests (3.27)	App cluster	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
scikit-learn	BSD-3-Clause	N	Analytics experiments; learner-state detection (3.19)	App cluster	Approved
LightGBM	MIT	N	Learner-state, drop-out and reminder-timing models (3.19)	App cluster (CPU)	Approved
Vowpal Wabbit	BSD-3-Clause	N	Next-activity contextual bandit for lessons, quizzes, Shorts and challenges (3.19)	App cluster (CPU)	Approved
pyBKT, girth	MIT	N	Mastery tracing and IRT; item analysis of question banks (19.7); IRT calibration and DIF bias checks of MindForge items (37.5); IRT calibration for the adaptive placement test and smart practice (CAT; item selection in-house, 3.27); the Mastery Map with a confidence range and per-class improvement (3.29)	App cluster	Approved
LanguageTool	LGPL-2.1-or-later	N	English grammar feedback	Separate internal service	Condition: separate service
segno	BSD-3-Clause	N	Certificate QR codes	App cluster	Approved
Ralph LRS, PyLTI1p3, scorm-again, @digitalbazaar/vc	MIT, MIT, MIT, BSD-3-Clause	N	xAPI, LTI 1.3, SCORM, verifiable credentials; xAPI statements from both AI roles (3.16); MindForge skill badges as Open Badges 3.0 (37.9); LTI 1.3 beside the institution roster and results API (3.27)	App cluster (Ralph on ClickHouse) / browser	Approved
LiveKit Agents	Apache-2.0	N	The Nexa Teacher as a LiveKit agent with turn detection, so learners can interrupt it; the microphone turn rules of the 1:1 and 1:50 classes (3.28). LiveKit SIP is not used — phone calls go through a licensed provider (3.28)	Core-1/Core-2	Approved
google-api-python-client	Apache-2.0	B	YouTube publishing (4.6); YouTube link details in the link step (3.23)	App-cluster workers	Approved
Penpot	MPL-2.0	N	Graphic-design course	App cluster	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
LibreOffice (headless)	MPL-2.0	N	Skill-course file checks	Sandboxed App workers	Approved
gVisor	Apache-2.0	E	Practice-lab sandbox isolation; sandboxed fetching of shared links (3.23); NexaLogicDev user-code sandboxes with Kata Containers (32.2); graded coding tests in skill courses — our own runner with time and memory limits and no network, not Judge0 (3.27)	Core-1 / App cluster; NexaLogicDev sandbox pool	Approved
Docker Engine (Moby)	Apache-2.0	N	Container practice inside gVisor lab sandboxes (course content only; not used to run the platform)	Lab sandboxes	Condition: lab use only
OWASP Juice Shop	MIT	N	Security practice lab	Sandbox	Approved
schemdraw, ngspice	MIT, BSD-3-Clause	N	Circuit diagrams and simulation	Core-2 sandbox (App cluster when busy)	Approved
3. Data					
PostgreSQL 18	PostgreSQL	B	Main database and usage ledger; row-level security for the Nexa Co-Teacher's per-learner data (3.16); shared learner record of the Nexa Teacher and Nexa Co-Teacher (3.16); voucher, affiliate and marketplace ledgers (11.14); CMS page blocks with versions and the B2B CRM (3.25); marketing leads with consent and opt-out records (3.26); staff-chat messages (kept 90 days), the encrypted chat-summary archive (6 months), notification routing and history (38.15); avatar and voice library versions and clone consent records (38.16)	DB-1/DB-2	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
pgvector	PostgreSQL	B	Vector search and knowledge bases; learner interest map (3.19) and content-based course matching for "any course" (11.13); similar Study Hall Q&A questions (31.3); look-alike organisations (3.26)	DB cluster	Approved
TimescaleDB (Apache-2.0 edition)	Apache-2.0	B	Product analytics; funnel events for the drop-off analysis (3.25); marketing attribution and reports (3.26)	DB cluster	Condition: Timescale-Licence features excluded
Apache AGE	Apache-2.0	N	Curriculum knowledge graph	DB cluster	Approved
Patroni, pgBackRest	MIT	B	PostgreSQL HA (Kubernetes API as configuration store) and backups to R2	DB-1/DB-2	Approved
PgBouncer	ISC	B	Connection pooling	Sidecar on every App node	Approved
Valkey + Sentinel	BSD-3-Clause	B	Cache, queues, sessions (6379 NexaTute, 6380 Motion Studio); context card for the shared learner record (3.16); verified Study Hall Q&A answers (31.3); Valkey streams for journey events (3.26)	Redis1/Redis2	Approved
boto3	Apache-2.0	B	Cloudflare R2 access	App cluster, workers	Approved
4. Text AI and evaluation					
vLLM (Router, LMCache, multi-LoRA, sleep mode)	Apache-2.0	E	LLM serving; LMCache CacheBlend for knowledge-base passages; n-gram/suffix speculative decoding for the script LLM (8.15); pipeline parallelism splits a model across GPUs inside a NexaLogicDev Heavy server, and across the three Heavy servers only for the GLM-5.2 fallback (32.1)	GPU cluster; Heavy pool	Approved; pipeline parallelism benchmarked before use
Arctic Inference	Apache-2.0	M	Suffix speculative decoding for the Motion Studio script LLM, used only if faster than vLLM's built-in method (8.15)	GPU cluster (test)	Evaluation only

Software	Licence (SPDX)	Product	Used for	Runs on	Status
NVIDIA Triton Inference Server	BSD-3-Clause	B	Dynamic batching of small vision and safety models (8.15)	AI-2 Node B	Approved: internal only, mTLS
SGLang	Apache-2.0	B	Alternative engine, benchmark only	GPU cluster (test)	Evaluation only
llama.cpp	MIT	B	Qwen3-4B text tasks on CPU; small offline model in the NexaTute Box	Core-1 spare cores; NexaTute Box	Approved
xgrammar	Apache-2.0	B	Constrained JSON output	GPU cluster	Approved
Qwen3-30B-A3B (FP8)	Apache-2.0	E	Main tutor and script LLM, also the former Qwen3-8B tasks; hybrid thinking switch, off by default; landing-page and CMS copy, the weekly drop-off analyst and the pre-sale AI (3.25); marketing articles, video scripts, email replies and reports (3.26); encrypted daily summaries of staff chats at P10 (38.15); subtitle translation for Nexa Notes videos (3.25); AI column mapping of bulk imports and organisation details for AI target finding, for all three products (3.26); marking of short answers and transcribed written answers against the approved marking scheme (3.27); Nexa Co-Admin answers and drafts on our own servers at low priority, and the AI translation of the admissions message (38.19, 38.20)	AI-1 Node A/B	Approved
Qwen3-8B	Apache-2.0	B	LLM overflow only (its former tasks run on Qwen3-30B-A3B or Qwen3-4B, 8.15)	Rec-1/Rec-2 when idle	Approved
Qwen3-4B	Apache-2.0	B	Small text tasks; NexaTute Box offline model	Core-1 (CPU); NexaTute Box	Approved
Qwen3-0.6B / 1.7B	Apache-2.0	N	In-browser model	Browser	Approved
Qwen3 EAGLE-3 draft head	To confirm	B	Speculative decoding	AI-1	Verify licence (specific draft-head weights)

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Subject LoRA adapters	In-house	N	Subject tuning on licensed curriculum data	AI-1	Approved
Mistral Small 3.2 (24B)	Apache-2.0	N	Independent solver (3.13)	GPU cluster, night window	Approved
Qwen3Guard-Gen-8B	Apache-2.0	B	Text safety moderation (4.15); safety check of shared-link text (3.23); Study Hall Q&A posts (31.3); compliance check of every staff-chat message on send, with rules — passwords, keys and bank details blocked, other categories alerted to the concerned admin (38.15); Nexa Notes post and video text (3.25)	AI-1 Node A	Approved
Qwen3-VL-8B	Apache-2.0	B	Photo questions, uploads, frame and gesture checks; Script Checker marking of scanned answer scripts (3.21); handwriting reading of written tests, with unsure parts confirmed by the learner before marking (3.27)	AI-2 Node A; other launch GPUs for queued written checks within the daily cap (8.17)	Approved
BGE-M3, BGE-reranker-v2-m3	MIT, Apache-2.0	N	Retrieval and meaning-matched cache	GPU cluster (BGE-M3 ONNX INT8 on Core-2 CPU if quality holds)	Approved
LLaMA-Factory	Apache-2.0	N	Fine-tuning (uses TRL internally)	GPU cluster, night window	Approved
lm-evaluation-harness, promptfoo	MIT	N	Model and category gates	CI, night window	Approved
Ragas, sacrebleu	Apache-2.0	N	Grounding and back-translation checks	Night window	Approved
Presidio	MIT	N	Anonymising training and research data	Night window	Approved
Scrapy, feedparser	BSD-3-Clause, BSD-2-Clause	E	Licensed source collection; circular and deadline tracker (3.20); organisations' public contact pages for the target-client database, read in the night batch and respecting robots.txt and site terms (3.26)	Core-1	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
datasketch (MinHash)	MIT	N	Copy checks for content and teacher uploads; code-similarity check of coding tests, flagged for human review (3.27)	Core-1	Approved
Langfuse	MIT	B	LLM tracing; prompt versions with rollback on the operations page (38.3)	App cluster; data in ClickHouse, PostgreSQL and R2	Condition: MIT core only
ClickHouse	Apache-2.0	B	Event store for Langfuse LLM traces (30-day retention) and Ralph LRS xAPI statements; PostgreSQL stays the source of truth; event data for the engagement models (3.19); MindForge events (37.11); journey reports with holdout groups (3.26); per-class LiveKit connection quality by area and operator (9.5); the Free Beta dashboard (3.33); recording and report events (3.29)	Core-2 (CPU and memory capped; nightly native BACKUP to R2)	Approved
5. Speech and voice					
faster-whisper with Whisper large-v3-turbo	MIT	B	Server speech-to-text, dubbing, summaries, audio checks (batched for non-live jobs); subtitles of Nexa Notes videos — a transcript in the spoken language, translated by Qwen3 into the other five languages, each language checked by a reviewer — off-peak at P10 (3.25)	AI-2 Node A	Approved
Whisper small (browser) with Bangla and Urdu fine-tunes	MIT	N	In-browser speech-to-text	Browser	Approved
Common Voice data	CC0-1.0	N	Fine-tune data	—	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Chatterbox Multilingual V3	MIT	B	Voices and voice cloning (not Bengali or Urdu); NexaTute voice clones from the Avatar & Voice Studio — guardians' family voices for PlayBoom Play Buddy stories (never a child's voice), verified teachers and learners aged 18 or over — the same engine and consent rules as Motion Studio voice cloning (35.2, 38.16)	AI-2 Node A; NexaTute clone jobs off-peak at P10 within a daily GPU-minute cap (8.17)	Approved
Indic Parler-TTS	Apache-2.0	B	Bengali and Urdu voices	AI-2 Node A	Approved
In-house VITS voices	MIT (code); voice data owned	B	Browser voices (both products, 4.8); server voices for weak devices on CPU; Bangla fallback for Motion Studio; batch-class AI voice made in each browser from the AI's text (3.17); replay of event-log recordings with the pinned voice version (11.3); the six-language voice library of the Avatar & Voice Studio, and per-person Bangla and Urdu voice models for NexaTute clones, trained in the late-night batch window (38.16)	Browser; Core-1/Core-2 CPU through ONNX Runtime (one OVHcloud CPU server beyond ≈8,000 NexaTute CCU)	Approved
ONNX Runtime (server)	MIT	B	In-house VITS server voices on CPU, and the BGE-M3 INT8 option (8.12, 8.15)	Core-1/Core-2 spare cores; OVHcloud CPU server when needed	Approved
wav2vec2-lv-60-espeak-cv-ft, g2p_en, CMU Pronouncing Dictionary	Apache-2.0, Apache-2.0, BSD-2-Clause	N	Pronunciation scoring — in the browser (ONNX Runtime Web) on capable devices, on the server for light and weak devices (8.18); PlayBoom letter-sound and read-aloud checks (30.8); dictionary pronunciations (3.20); speaking marks never depend on accent, with regular accent-bias checks (3.27)	Browser; AI-2 Node A fallback	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
AudioSeal	MIT (verify weights)	B	Inaudible watermark on cloned voices in both products, including cloned-voice audio in Motion Studio (35.2); NexaTute clones from the Avatar & Voice Studio (38.16)	AI-2 Node A (speech pool)	Verify licence (weights)
6. Document intake (5.3)					
Docling	MIT	E	Document parsing; text of shared web pages (3.23); bulk import of organisation lists from PDF files in the AI Marketing Hub for all three products (3.26)	Core-1 (content pipeline), Core-2 (uploads)	Approved
pypdf, python-docx, python-pptx	BSD-3-Clause, MIT, MIT	E	PDF, Word and PowerPoint uploads; pypdf stamps the exporter's watermark on every page of PDF exports from the AI Marketing Hub (3.26)	Core-2; App cluster (exports)	Approved
openpyxl	MIT	E	Excel question banks and Excel course; Excel exports of accounting reports (38.13); Excel and CSV import and export of target-client lists in the AI Marketing Hub (3.26)	Core-2; App cluster	Approved
Tesseract OCR	Apache-2.0	B	Printed text in 6 languages — in the learner's browser through Tesseract.js first, server CPU for light and weak devices (8.18); Study Hall Q&A images — search, duplicates and hidden phone numbers or links (31.3)	Browser (Tesseract.js); Core-2 fallback	Approved
PaddleOCR PP-FormulaNet	Apache-2.0	B	Formula reading	Core-2	Approved
pillow-heif (libheif)	BSD-3-Clause (decode-only build; libheif and libde265 LGPL-3.0; no x265)	B	iPhone HEIC photos	Core-2	Verify licence
Bijoy → Unicode converter	To confirm	B	Bijoy-encoded Bangla files	Core-2	Verify licence
Fountain screenplay parser	To confirm	M	Screenplay scripts	Core-2	Verify licence

Software	Licence (SPDX)	Product	Used for	Runs on	Status
7. Video, image and audio generation					
Wan 2.2 (TI2V-5B / A14B)	Apache-2.0	M	Video generation: A14B for Cinematic hero shots only, TI2V-5B for other scenes	AI-1 Node B (daytime), AI-2 Node B	Approved after model-page check
Lightning LoRA, TeaCache, SageAttention	To confirm	M	Faster Wan 2.2	GPU cluster	Verify licence
LightX2V	Apache-2.0	M	Inference engine for Wan 2.2 (8.15)	GPU cluster	Verify licence; benchmark first
FastWan2.2-TI2V-5B (FastVideo)	Apache-2.0	M	Distilled TI2V-5B for faster non-hero scenes (8.15)	AI-2 Node B	Verify licence (weights); quality gate
FLUX.1-schnell	Apache-2.0	B	Key images and thumbnails; NexaTute answer pictures and diagrams (3.10) and CMS banners for both products (3.25); blog and social images (3.26); AI comic stories per chapter after expert review (3.32)	GPU cluster	Approved after model-page check
Nunchaku (4-bit SVDQuant FLUX.1-schnell)	Apache-2.0	B	Faster FLUX.1-schnell images for both products with less VRAM (8.15)	GPU cluster	Verify licence (weights); quality gate
Depth-Anything V2 Small	Apache-2.0	M	2.5D depth maps (Small model only)	GPU cluster	Approved
Character LoRA	In-house, per creator	M	Character consistency in series	GPU cluster	Approved
RIFE (Practical-RIFE)	MIT	M	Frame interpolation for Wan 2.2 A14B (16 → 24/30 fps), and 24 → 30 fps where needed	GPU cluster	Approved
Real-ESRGAN	BSD-3-Clause	M	Preview and backup upscaling; a compact model upscales previews in the browser on strong devices (8.18)	Rec-1/Rec-2, GPU cluster	Approved
RealBasicVSR / BasicVSR++ (MMagic)	Apache-2.0	M	Final escalated video upscaling (4.13)	AI-2 Node B (overflow: AI-2 Node A, Rec-1/Rec-2)	Verify licence (weights)
SeedVR2	Apache-2.0	M	1080p Escalating candidate	GPU cluster (test)	Evaluation only

Software	Licence (SPDX)	Product	Used for	Runs on	Status
GFPGAN	Apache-2.0	M	Close-up face restoration	AI-2 Node B	Verify licence (face-detector weights)
OpenCLIP, DINOv2	MIT, Apache-2.0	M	Scene quality and character checks; thumbnail frame scoring (4.18)	GPU cluster	Approved
Wan2.2-S2V-14B (speech-to-video)	Apache-2.0	M	Cinematic dialogue lip-sync close-ups; replaces MuseTalk/LatentSync (8.15)	AI-1 Node B (daytime), like A14B	Verify licence; quality gate before use
ACE-Step	Apache-2.0	M	AI background music, including trailer and teaser music (4.18)	Rec-1/Rec-2 when idle	Verify licence
Demucs (htdemucs)	MIT	M	Voice and music separation	Rec-1/Rec-2 or CPU	Verify licence (weights)
rembg with BiRefNet	MIT	M	Background removal for product ads	Rec-1/Rec-2	Approved
PySceneDetect	BSD-3-Clause	M	Shorts and promo cutting (4.11, 4.18)	Core-1	Approved
Puppeteer, Chromium (headless)	Apache-2.0, BSD-3-Clause	B	Server fallback renderer for browser renders (Revideo renderVideo, 4.8); JavaScript and HTML/CSS answers of NexaTute coding tests checked inside the gVisor sandbox (3.27)	Spare Core-1/Core-2 CPU (2D); Rec-1/Rec-2 GPU when idle (3D); practice-lab gVisor sandboxes for coding tests	Approved
libass	ISC	M	Styled captions	Assembly node	Approved
c2pa-rs / c2patool	MIT OR Apache-2.0	B	Content Credentials; Content Credentials on NexaTute cloned-voice audio (38.16)	Assembly node; remote signing on the App cluster, key in OpenBao; App cluster for NexaTute audio	Approved
Chromaprint	LGPL-2.1-or-later	B	Audio fingerprint check; copyright check of the audio of Nexa Notes videos (3.25)	Core-1	Condition: separate process
libvmaf (VMAF filter in FFmpeg), OpenCV	BSD-2-Clause-Patent, Apache-2.0	B	Quality gate (VMAF on 1 frame per second); thumbnail frame sharpness (4.18); OpenCV QR-code detection re-checks Study Hall Q&A images on the server (31.3)	Dallas assembly-node CPU; Core-2 (uploads)	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
CC0 sound-effect library	CC0-1.0	M	Sound effects	R2 + CDN	Approved
8. Safety					
NSFW image classifier (e.g. Falconsai ViT)	Apache-2.0	B	First-pass image and frame screening; images on shared links (3.23); Study Hall Q&A images (31.3); Nexa Notes video frames (3.25); written-test pages (3.27)	AI-2 Node B / CPU	Verify licence
Meta Hasher-Matcher-Actioner with PDQ and TMK+PDQF	BSD-3-Clause	B	Child-safety hash matching of uploads; images on shared links (3.23); Study Hall Q&A images (31.3); Nexa Notes videos (TMK+PDQF, 3.25); written-test pages (3.27)	Core-1/Core-2	Approved; hash lists by application
URLhaus feed (abuse.ch)	CC0-1.0	N	Blocklist of malware URLs for links shared by users (3.23)	App cluster (link step of the Document Intake Service)	Approved
StevenBlack hosts (unified hosts with the pornography and gambling extensions)	MIT	N	Blocklist of phishing, malware, pornography and gambling domains for shared links (3.23)	App cluster (link step of the Document Intake Service)	Verify the licence of each source list in the chosen variant
9. Live classes and media					

Software	Licence (SPDX)	Product	Used for	Runs on	Status
LiveKit, LiveKit Egress, LiveKit Ingress	Apache-2.0	N	Live classes and recording egress; batch-class raised-hand audio, adult pair rooms and one server voice stream per batch (3.17); weekly Nexa Teacher Live stream to Facebook and YouTube (3.21); staff voice and video calls in the platform admin panel, allowed per role and never recorded (38.15); a live human invigilator for institution exams, not recorded (3.27); simulcast and dynacast with audio-only fallback, reconnect, raise-hand speaking permissions, join/leave webhooks, TURN on 443, FEC and RED, short-lived tokens (3.28, 9.5); Ingress (OBS, RTMP, WHIP) for the human-teacher revision class; the Live Game Show HLS stream (3.32)	Core-1/Core-2; Rec-1/Rec-2	Approved
coturn	BSD-3-Clause	N	TURN fallback	Core-1/Core-2	Approved
GStreamer	LGPL-2.1-or-later	B	Recording and media pipelines	Rec and GPU nodes	Condition: dynamic libraries
FFmpeg (LGPL build; NVENC and OpenH264 encoders), Shaka Packager	LGPL-2.1-or-later; OpenH264 BSD-2-Clause; Shaka Packager BSD-3-Clause	B	Encoding and assembly; Nexa Notes videos transcoded to HLS at 720p, 480p and 240p on CPU with the OpenH264 encoder, so the build stays free of GPL x264 (3.25); Shaka Packager makes the HLS segments and playlists of Nexa Notes and group-class videos (9.4); class recordings are replay files and need neither	GPU nodes; CPU worker pool (App-1–App-3) at P10 for Nexa Notes HLS	Condition: no GPL or non-free build flags
10. Infrastructure, operations and compliance					
Kubernetes, etcd, containerd	Apache-2.0	B	One cluster for all servers (16 at launch, App-3 from December 2026); rolling node-by-node software updates in the maintenance window (38.18)	Control plane on Redis1/Redis2/App-1	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Argo Rollouts	Apache-2.0	E	Rolling updates with health checks and automatic rollback to the previous version when a check fails; every update starts only after two-person approval in the software-updates panel, never automatically (38.18)	Kubernetes control plane (Redis1/Redis2/App-1)	Approved
Podman	Apache-2.0	B	Image builds (Docker dropped)	CI	Approved
Harbor	Apache-2.0	B	Container registry and model-weight mirror	Core-1	Approved
Forgejo, Forgejo Actions	GPL-3.0-or-later	B	Code hosting and CI	App cluster; builds on Core-1 at night	Condition: used unmodified internally
Syft, Grype, Trivy, ScanCode Toolkit	Apache-2.0	E	SBOM, vulnerability and licence scans (18.4); ScanCode also scans the licences of NexaLogicDev-generated code (32.11); the daily new-version check of all server software in the software-updates panel (38.18)	CI; NexaLogicDev sandbox pool	Approved
OWASP ZAP	Apache-2.0	B	Dynamic security testing of every web app and API	CI, staging	Approved
Locust, LiveKit CLI (load test), vLLM benchmark scripts	MIT, Apache-2.0, Apache-2.0	B	Load and capacity tests before any capacity claim (8.6, 13.2, H.3); the staging load-test panel with pass thresholds before the Free Beta, before the paid launch and after every major update (38.18); the proven capacity behind the capacity guard (38.19)	Test runners on spare Core-1 cores outside class hours	Approved
Semgrep Community Edition	LGPL-2.1	E	Static code security analysis of our own code; AI security audit of NexaLogicDev-generated code before deploy (SQL injection, XSS and similar, 32)	CI; NexaLogicDev sandbox pool, run as a separate tool	Condition: separate tool; registry rules used internally under the Semgrep Rules License, or own rules

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Falco	Apache-2.0	B	Runtime threat detection on Kubernetes	Every node	Approved
Cilium with WireGuard	Apache-2.0	B	Networking and encrypted site links	All nodes	Approved
SPIRE	Apache-2.0	B	Service identities for mTLS with Cilium mutual authentication	Every node	Approved
cert-manager	Apache-2.0	B	TLS certificates (Cloudflare Origin CA and Let's Encrypt)	Cluster	Approved
OpenBao	MPL-2.0	B	Secrets and signing keys; OAuth tokens of the AI Marketing Hub channel connections, encrypted and refreshed automatically with expiry alerts (3.26); keys of the encrypted staff-chat summary archive (38.15)	3-node Raft cluster on Redis1, Redis2 and App-1; daily encrypted snapshot to R2	Approved
KEDA, Kueue	Apache-2.0	B	Autoscaling and GPU job queue	Cluster	Approved
NVIDIA GPU Operator	Apache-2.0	B	GPU software on Kubernetes	GPU nodes	Approved
NVIDIA DRA Driver for GPUs	Apache-2.0	B	Optional Kubernetes Dynamic Resource Allocation for finer GPU sharing (8.15)	GPU nodes (test)	Evaluation only
HAProxy, keepalived	GPL-2.0-or-later	B	Load balancing and failover	LB-1/LB-2	Condition: used unmodified
Prometheus, Alertmanager	Apache-2.0	B	Metrics and alerts; synthetic-check results and alerts (38.18)	Redis1/Redis2	Approved
Prometheus Blackbox Exporter	Apache-2.0	N	Synthetic checks every 5 minutes — page load, class join and delay — from small probes on Grameenphone, Robi, Banglalink and BDIX broadband connections (probe set-up to confirm), feeding the Edge check (38.18)	Probes in Bangladesh; results to Prometheus on Redis1/Redis2	Approved
Grafana, Loki	AGPL-3.0-only	B	Dashboards and logs (stored in R2); funnel and drop-off dashboards behind the admin analysis page (3.25)	Redis1/Redis2	Condition: used unmodified
OpenTelemetry Collector	Apache-2.0	B	Telemetry from every node	Every node (DaemonSet)	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
k3s, Nginx	Apache-2.0, BSD-2-Clause	N	NexaTute Box offline device (rclone has its own row in group 13)	On-site box	Approved
11. Content and datasets					
Z-Anatomy 3D models	CC-BY-SA-4.0	N	Anatomy	CDN	Condition: attribution and share-alike
OpenStax Anatomy & Physiology, PTB-XL	CC-BY-4.0	N	Medical content and ECG questions	Knowledge base	Condition: attribution
IfcOpenShell	LGPL-3.0-or-later	N	Selected BIM assets (Bonsai, GPL-3.0, internal tool only)	Asset preparation	Condition
CC0 3D model and texture libraries (Poly Haven, Kenney, Quaternius) and commissioned models	CC0-1.0 (commissioned models owned by RemonLogic)	B	Classroom rooms and props (3.15); Motion Studio Quick-mode 3D scenes	R2 + Cloudflare CDN; cached in the browser	Approved; each asset reviewed before it enters the library
12. Deliberate exceptions (paid or not open-source)					
Cloudflare R2, CDN, Workers, WAF, Waiting Room, Turnstile, Cloudflare for SaaS, Email Routing, Web Analytics	Proprietary (paid and managed services)	E	Storage, edge cache, protection, waiting room; cache purge after CMS publishing and Turnstile spam protection on site forms (3.25); Nexa Notes HLS videos and subtitles in R2 behind the CDN and Turnstile on the sign-up and Nexa Notes email forms (3.25); NexaLogicDev free sites at username-site.nexalogicdev.app on Workers + R2, custom domains with automatic SSL (Cloudflare for SaaS), email forwarding and site analytics (32.7, 32.10); the Dhaka (DAC) edge, R2 location hint APAC and Tiered Cache, only public content cached (9.4); Turnstile before every OTP is sent (38.18); the public status page status.nexatute.com (3.33); white-label custom domains (3.32)	Cloudflare	Exception
Amazon SES (or equivalent)	Proprietary (paid service)	B	Transactional email; opt-in newsletters, never cold outreach (3.26)	US/EU regions	Exception

Software	Licence (SPDX)	Product	Used for	Runs on	Status
WhatsApp Business Platform, Bangladeshi SMS gateway, licensed voice/IVR provider and IPTSP hotline, imo business messaging	Proprietary (paid service)	E	Guardian reports and reminders; SMS for serious human-controller alerts (10.2) and the OTP fallback of face verification (3.16); pre-sale chat and abandoned-checkout reminders for both products (3.25); journey messages and B2B follow-ups for all three products with per-channel consent (3.26); Institute Admin reminders (38.7); optional SMS and WhatsApp copies of urgent admin notifications (38.15); OTP and notifications in NexaLogicDev-generated apps (32); sign-up OTP by SMS or WhatsApp; guardian SMS free and on; outbound guardian voice reminders through a BTRC-licensed voice/IVR provider and the inbound phone-tutor line on a licensed IPTSP number (3.28); WhatsApp and SMS study bots (3.32); waiting-list offers by free SMS (38.19); Nexa Co-Admin for staff on WhatsApp and imo through a separate “NexaTute Staff” account — off until the lawyer and DPO check, imo locked off until business API access is confirmed (38.20)	Managed	Exception
aamarPay payment gateway (bKash, Nagad, Rocket, cards)	Proprietary (payment service)	B	Payments: NexaTute plans, renewals, wallet top-ups and add-ons (11.7); Motion Studio package payments through the same gateway; affiliate payouts and gift-card sales (11.14); VAT 15/115 applies only to these Bangladesh sales (11.7, 38.5)	Managed	Exception

Software	Licence (SPDX)	Product	Used for	Runs on	Status
YouTube Data API, Meta Graph API (Facebook, Instagram and Pages), TikTok Content Posting API	Proprietary (platform APIs)	E	Direct publishing; title, description and chapters of YouTube links shared by learners (3.23); auto-published NexaTute Shorts (3.21); Facebook Messenger for the pre-sale AI (3.25); marketing posts, Reels, Instagram publishing and comment replies (3.26); the Channels page of the AI Marketing Hub — each product's official YouTube, Facebook and Instagram accounts connected by OAuth (we never see or keep a password), the super admin connects and the security owner approves, tokens in OpenBao (3.26); click-to-load youtube-nocookie embeds of our own YouTube videos in Nexa Notes, with no request to YouTube before the click (3.25); NexaLogicDev Facebook Page → Website import with the page owner's consent (32.6); the Messenger Platform for Nexa Co-Admin staff chats, off until the lawyer and DPO check (38.20)	Managed	Exception; YouTube API audit and Meta App Review
Google Search Console, Bing Webmaster Tools, Google Business Profile	Proprietary (free services)	B	Search indexing and search-query data for SEO and the Landing Page AI's signals; business reviews (3.25); Bing keyword research data (3.26)	Managed	Exception
LinkedIn Community Management API	Proprietary (platform API)	E	Company-page posts, videos and comment replies (3.26); the LinkedIn channel connection on the Channels page for each product (OAuth, 3.26); Motion Studio direct publishing to LinkedIn after launch (4.6)	Managed	Exception; LinkedIn approval needed

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Google Trends API, Google Ads API (Keyword Planner)	Proprietary (free services)	B	Keyword and trend research for SEO and the content plan (3.26)	Managed	Exception; Trends API in alpha, Keyword Planner needs Basic access
Outreach mailbox (Google Workspace or Zoho Mail)	Proprietary (paid service)	B	B2B outreach email sent and read over SMTP/IMAP on a separate outreach domain (3.26)	Managed	Exception
GPT-6 Luna, then Mistral Large 3	Proprietary (paid APIs)	B	Cloud LLM fallback; the NexaTute Nexa Co-Teacher uses Mistral Large 3 first, then GPT-6 Luna, and live Nexa Teacher classes never use the cloud (8.9)	Via Portkey	Exception
NVIDIA driver, CUDA, NVENC, MPS	Proprietary (free)	B	GPU runtime, hardware encoding and GPU sharing for own models (MPS)	GPU nodes	Exception
13. RemonLogic Ecosystem additions					
MiniMax-M2.7 (4-bit AWQ)	MiniMax licence (modified MIT); commercial use needs MiniMax's written authorization	L	NexaLogicDev Heavy Lane: architecture, Auto-Architect plans, design fixes and final review (32.1)	Heavy pool — one copy per Dev-H server (vLLM, TP=4 or TP=2 × PP=2 inside the server)	Approved; licence request and benchmark before launch
GLM-5.2 (4-bit AWQ/GPTQ) — fallback	MIT (verify)	L	Heavy Lane fallback if the M2.7 licence is refused (32.1)	Heavy pool — vLLM TP=4 × PP=3 across the three Dev-H servers	Fallback; benchmarked in June 2027
Qwen3-Coder-Next (FP8 or AWQ 4-bit)	Apache-2.0 (verify)	L	NexaLogicDev Fast Lane: everyday coding, builders, files from the Heavy plan, small fixes (32.1)	Fast pool — Fast-1–Fast-4 (RTX PRO 6000), one copy per GPU; NexaTute pool overflow off-peak	Approved; licence and benchmark check
Monaco Editor, xterm.js	MIT	L	Browser code editor and terminal of the NexaLogicDev IDE (32.2)	Browser	Approved
Sandpack	Apache-2.0	L	In-browser previews of generated apps	Browser	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
WebContainer API (StackBlitz)	Proprietary (commercial licence)	L	Power mode: full Node.js previews in the browser for Team plan users, agencies and Auto-Architect module buyers (32.2)	Browser	Exception; commercial licence requested from StackBlitz
WordPress Playground	GPL-2.0-or-later	L	Browser testing of generated WordPress themes and plugins (later update)	Browser	Approved; GPL code kept separate from RemonLogic code
Kata Containers	Apache-2.0	L	Isolated sandbox runtime for user code, builds and tests, alongside gVisor (32.2)	Sandbox pool (OVHcloud ADVANCE-2)	Approved
Jest, PHPUnit	MIT, BSD-3-Clause	L	Generated unit tests and the write → run → fix loop	Sandbox pool	Approved
OSV-Scanner	Apache-2.0	L	Dependency vulnerability scan of generated code (32.11); licence scan with ScanCode Toolkit (group 10)	Sandbox pool	Approved
Name.com Reseller API (Openprovider later)	Proprietary (managed)	L	International domain registration for Go Live (32.10)	Managed	Exception
GitHub, Vercel and Netlify APIs	Proprietary (platform APIs)	L	GitHub sync; one-click deploy to the user's own Vercel/Netlify account (32.10)	Managed	Exception
bKash, Nagad, Rocket and SSLCommerz APIs	Proprietary (payment services)	L	Payment modules inserted into generated apps, using the customer's own merchant account (32.4)	Customer's merchant	Exception
Pathao, Steadfast and RedX courier APIs	Proprietary (services)	L	F-commerce order manager in generated apps (later update); OTP and SMS use the Bangladeshi SMS gateway row of group 12	Managed	Exception

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Paddle	Proprietary (Merchant of Record)	E	International payments for every product, billed by the USA LLC (0.2, 11.7); automatic tax by the customer's country — included in the price or added at checkout; revenue booked as Paddle's payout (gross less the tax Paddle pays and the Paddle fee, an editable setting approved by the finance lead) at 0% Bangladesh VAT as an export of services, to be confirmed by the accountant (11.7, 38.5)	Managed	Exception
SpeechBrain ECAPA-TDNN	Apache-2.0 (verify model)	B	Speaker verification for voice-cloning consent (35.2); NexaTute voice clones of guardians, verified teachers and 18+ learners (38.16)	Speech pool (GPU or CPU)	Verify licence
OMR practice engine (in-house)	Proprietary (RemonLogic)	N	On-screen OMR sheets for SSC, HSC, Admission and BCS practice; rule-based scoring, Nexa Co-Teacher explanations (3.10)	Browser + App cluster	Approved
MiniMax API (MiniMax M3)	Proprietary (managed API)	L	Fast-lane fallback when the Fast Lane and idle Heavy servers are unavailable; paid users only; monthly spend cap (32.1)	Managed (MiniMax cloud) through the Portkey gateway	Exception
Anthropic API (Claude Sonnet 5.5)	Proprietary (managed API)	L	Heavy-lane plan review (Heavy unavailable, two failures, every Complex Module) and completion after a third failure; Pro/Team plans and Auto-Architect only; monthly spend cap (32.1)	Managed (Anthropic cloud) through the Portkey gateway	Exception
Gitleaks	MIT	L	Finds and removes keys and secrets from code and prompts before any external API call (32.1)	App cluster	Approved
ESLint, Stylelint	MIT	L	Lint in the browser (Web Workers) while the user types (32.14)	Browser	Approved
Prettier (with plugin-php)	MIT	L	Formatting in the browser (Web Workers) for JS/TS, CSS, HTML and PHP (32.14)	Browser	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
php-wasm (WordPress Playground runtime)	GPL-2.0-or-later (verify); PHP under the PHP License	L	PHP, Laravel (without a database) and WordPress previews in the browser (32.14)	Browser, unmodified separate component	Approved; GPL code kept separate from RemonLogic code
GrapesJS	BSD-3-Clause	L	Website Builder and Wireframe Generator canvas in the browser (32.2, 32.14)	Browser	Approved
dnd-kit	MIT	L	Drag and drop in the builders and the IDE (32.14)	Browser	Approved
Workbox	MIT	E	Service worker for the offline PWA (32.14); PlayBoom offline packs (30.8) and MindForge daily puzzles offline (37.6); Wi-Fi prefetch of the next day's lessons and Nexa Radio (3.28, 3.32)	Browser	Approved
y-indexeddb (with Yjs)	MIT	L	Offline project storage and conflict-free sync when back online (32.14)	Browser	Approved
Web Speech API	Browser built-in (vendor cloud in Chrome)	L	Voice fallback on weak devices only, with a notice to the user (32.14)	Browser	Approved; fallback only
isomorphic-git	MIT	L	Git clone, commit and branch in the browser; push and pull through the GitHub API with the user's token (32.14)	Browser	Approved
OpenTelemetry JS (browser SDK)	Apache-2.0	E	Browser-versus-server share per feature, tier and device for the capacity dashboard; no personal content sent (8.18)	Browser → OpenTelemetry Collector	Approved
Device Check and Data Saver (in-house)	Proprietary (RemonLogic)	E	Measures RAM, WebGPU/WebAssembly, storage and connection at first sign-in to set the device tier; Wi-Fi-only model downloads and lower video quality on mobile data (8.18)	Browser	Approved
lottie-web	MIT	N	Light animations for PlayBoom Play/Nursery/KG lessons (30.1)	Browser	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
RemonLogic Resource Broker (in-house)	Proprietary (RemonLogic)	E	Placement order, lend and reclaim thresholds, lending windows and model warm-up across all pools; drives Kueue cohorts (lending and borrowing limits) and the Inference Gateway (8.17)	App cluster	Approved
Gateway API Inference Extension (Inference Gateway) with Envoy Gateway	Apache-2.0	E	Routes every LLM request to the least-loaded allowed vLLM instance in any GPU pool, by product and priority (8.17)	App cluster (gateway) → all GPU pools	Approved; benchmark
NVIDIA DCGM Exporter	Apache-2.0	E	Per-GPU busy %, memory and health for the placement score, lending thresholds and the capacity dashboard (8.17)	Every GPU server (with the GPU Operator)	Approved
OpenCost	Apache-2.0	E	GPU and CPU usage and cost per product and per funding budget (8.17); daily cost per learner on the operations page (38.3)	App cluster	Approved
rclone (with crypt)	MIT	E	Nightly lesson sync from R2 to the NexaTute Box (2.5); encrypted export of the accounting database and reports to Google Drive, OneDrive and Dropbox (38.13)	NexaTute Box; nightly job on the CPU worker pool	Approved
pgaudit	PostgreSQL	E	Database audit logging of reads and changes of accounting, payroll, share and access tables, next to the in-house append-only action log (38.12)	DB-1/DB-2 (PostgreSQL 18 extension)	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
Casbin (PyCasbin) with the SQLAlchemy adapter	Apache-2.0	E	Role and scope permissions — view, edit, approve and export per module and sub-module — masking rules and maker-checker approvals in the admin panel and accounting (38.12); Keycloak stays the identity provider; people management in Roles and access — role changes (old → new role with a required reason) and suspend or reactivate, approved by the security owner and never by the requester (38.12); notification routing and call permissions per role (38.15); every Nexa Co-Admin request checked against the asking admin's own permissions (38.20)	App cluster (library in the admin API; policies in PostgreSQL)	Approved
OR-Tools (CP-SAT solver)	Apache-2.0	N	Proves that every generated MindForge logic-grid, number and pattern puzzle has exactly one solution (37.5)	CPU worker pool at P10	Approved
jsQR	Apache-2.0	N	Finds QR codes in Study Hall Q&A images so that they are blurred before upload (31.3); the server re-checks with OpenCV	Browser	Approved
math.js	Apache-2.0	N	Scientific, graphing and unit-conversion calculator, with the per-exam calculator rules in mock tests (3.20); instant check of equivalent mathematical answers in fill-in-the-blank tests, re-checked on the server with SymPy (3.27)	Browser	Approved
Open English WordNet	CC-BY-4.0	N	English meanings, synonyms and antonyms in the dictionary (3.20); Bangla ↔ English subject glossaries, idioms and phrases are made by the AI content pipeline and reviewed (3.13)	Browser (cached word data); App cluster	Condition: attribution

Software	Licence (SPDX)	Product	Used for	Runs on	Status
email-validator, dnspython	Unlicense, ISC	E	Format and domain (MX) checks of organisations' official email addresses in the target-client database; no probing of other mail servers (3.26); format and domain checks of every row of a bulk import (3.26)	App-cluster workers	Approved
Organisation open data: Wikidata, OpenStreetMap, government lists (for example Bangladesh school and education-board lists)	CC0-1.0, ODbL-1.0; per source	E	Organisations for the target-client database, each field with its source and lawful basis (3.26); OpenStreetMap-derived boundary shapes for the target-clients map, served from our own CDN (3.26)	Night batch on Core-1 (with Scrapy)	Condition: ODbL attribution and share-alike for derived OpenStreet Map data; each government list's terms checked
Ahrefs or Semrush API (optional)	Proprietary (paid service)	B	Optional paid keyword data for the Landing Page AI keyword-research module, connected only if the admin switches it on (3.25)	Managed	Exception; off by default
hls.js, Video.js	Apache-2.0	N	Adaptive video player of Nexa Notes — Auto, 720p, 480p and 240p for weak internet, WebVTT subtitles in six languages, chapters and no autoplay; Safari plays HLS natively (3.25); Video.js (or our own light player) for controls, captions and mistake markers on group-class videos (9.4)	Browser; HLS files from R2 through the CDN	Approved
glTF-Validator (Khronos)	Apache-2.0	N	Automatic check of every uploaded GLB or VRM avatar — file structure, size, textures and bones — before the cultural review and two-person approval (38.16)	Browser pre-check; re-checked on Core-2 at upload (trust rule, 8.18)	Approved
glTF-Transform	MIT	N	Lighter copies (LOD) and compression of approved avatars — meshopt geometry and WebP textures through sharp — so that weak devices load them quickly (38.16)	CPU worker pool at P10	Approved

Software	Licence (SPDX)	Product	Used for	Runs on	Status
ClamAV (clamd)	GPL-2.0-only	N	Virus scan of files and images sent in staff chat, of avatar uploads and of Nexa Notes video uploads before anyone can open them (38.15, 38.16, 3.25)	Core-2 (uploads), next to the Document Intake Service	Condition: separate service, used unmodified
Firebase Cloud Messaging (FCM)	Proprietary (free Google service); Firebase Android SDK Apache-2.0	N	Push notifications to the Android admin app of the platform and Institute admin panels, sent by the in-house notification service; the text never shows secrets, salaries, student names or amounts (38.15)	Managed (Google)	Exception (Google service)
X API (formerly Twitter)	Proprietary (paid platform API)	E	A possible X channel on the AI Marketing Hub Channels page — not connected: the API is paid and the owner decides after checking the price (3.26)	Managed	Blocked until approved

18.2 Model Registry (Indicative)

Detail view of the model rows in the register (18.1): working VRAM and cautions. Licence, product and placement are the same as in 18.1; if they differ, 18.1 is the authority.

Model / component	Product	Working VRAM estimate	Status / caution
Qwen3-30B-A3B (FP8)	B	~31 GB	Apache-2.0; MoE with 3B active parameters for high throughput; one hybrid model with a per-request thinking switch; also serves the former Qwen3-8B tasks (8.15)
Wan 2.2 (TI2V-5B / A14B)	M	~24–80 GB	Apache-2.0 (verify); video generation — benchmark GPU-minutes per output minute; generates up to 720p in tested use; 1080p Native for Cinematic only after its benchmark and quality gate (4.9)
LightX2V (Wan 2.2 inference engine)	M	Same as the Wan model it runs	Apache-2.0 (verify); step-distilled and quantized Wan inference (8.15); output checked against the standard pipeline

Model / component	Product	Working VRAM estimate	Status / caution
FastWan2.2-TI2V-5B	M	~24 GB	Apache-2.0 (verify weights); distilled TI2V-5B for faster non-hero scenes (8.15)
FLUX.1-schnell	B	~24 GB	Apache-2.0 (verify); images and thumbnails for both products (18.1)
Nunchaku 4-bit FLUX.1-schnell (SVDQuant)	B	~8 GB	Apache-2.0 (verify weights); faster key images with less VRAM; quality checked against the FP8 model (8.15)
Depth-Anything V2 Small	M	~1 GB	Apache-2.0 (only the Small model; larger variants are non-commercial and excluded); depth maps for 2.5D scenes
Real-ESRGAN	M	~2 GB	BSD-3-Clause; preview and backup upscaler; final Escalating videos use RealBasicVSR/BasicVSR++ (4.13)
RIFE (Practical-RIFE)	M	<1 GB	MIT; frame interpolation 16 → 24/30 fps
ACE-Step	M	~8–16 GB	Apache-2.0 (verify code and weights); AI background music for Motion Studio (4.10); MusicGen and Stable Audio excluded
BiRefNet	M	~2 GB	MIT; background removal for the product ad maker, used through rembg (MIT) (4.11); a smaller variant runs in the browser on strong devices (8.18)
Demucs (htdemucs)	M	~2–4 GB (CPU possible)	MIT code (verify weights); separates voice from music in uploaded videos for music replacement (4.12)
MediaPipe Face Detection / Face Landmarker	B	<100 MB (CPU / browser)	Apache-2.0; auto-reframe for 16:9, 9:16, 1:1 and 4:5 (4.10); liveness check for face verification (3.16)
SFace (OpenCV Zoo)	N	<40 MB download, cached on the device; runs in the browser	Apache-2.0 (verify model weights); face match only; the template is encrypted and deleted with the account (3.16)
RealBasicVSR / BasicVSR++	M	~4–8 GB	Apache-2.0 (MMagic; verify weights); video super-resolution for final escalated videos (4.13)

Model / component	Product	Working VRAM estimate	Status / caution
SeedVR2	M	~16–40 GB	Apache-2.0 (verify); evaluated later for 1080p Escalating only (4.13)
GFPGAN	M	~1–2 GB	Apache-2.0 (verify face-detector weights); light close-up face restoration (4.13); CodeFormer excluded
NSFW image classifier (e.g. Falconsai ViT)	B	<1 GB (CPU possible)	Apache-2.0 (verify); fast first-pass frame and image safety check (4.15)
OpenCLIP / DINOv2	M	~1–2 GB each	OpenCLIP MIT, DINOv2 Apache-2.0; per-scene quality and character-consistency checks
Qwen3-8B	B	~6 GB (4-bit)	Apache-2.0; LLM overflow on Rec-1/Rec-2 only (it fits in 24 GB, where Qwen3-30B-A3B does not); its former tasks run on Qwen3-30B-A3B or Qwen3-4B (8.15)
Mistral Small 3.2 (24B)	N	~14 GB (4-bit)	Apache-2.0; independent solver in the AI content pipeline (3.13), loaded only in the late-night batch window; also the AI verifier of MindForge verbal-reasoning and real-world items (37.5)
Subject LoRA adapters (Qwen3 base)	N	~0.1–0.5 GB each	Trained in-house on licensed curriculum data; loaded on demand by vLLM multi-LoRA
Character LoRA (Wan 2.2 / FLUX.1-schnell)	M	~0.1–0.5 GB each	Trained per creator from their own approved images; stored with the project and deleted with it
Qwen3-4B (CPU, llama.cpp)	B	CPU RAM ~3 GB (4-bit)	Apache-2.0; small Motion Studio text tasks on spare CPU
Qwen3-0.6B / Qwen3-1.7B (browser, WebLLM)	N	~0.4–1.2 GB download (4-bit), cached on the device	Apache-2.0; small in-browser model on capable devices
Qwen3 EAGLE-3 draft head	B	~1–2 GB	Apache-2.0; speculative-decoding draft for faster LLM answers (8.15)
Qwen3-VL-8B	B	~8 GB (4-bit)	Apache-2.0; photo questions, formula reading and answer-script checking (3.10); handwriting reading of written tests (3.27)

Model / component	Product	Working VRAM estimate	Status / caution
PaddleOCR PP-FormulaNet	B	CPU or ~1 GB	Apache-2.0 (code and model weights); lightweight formula recognition on spare CPU
Tesseract OCR (Bengali, English, Hindi, Urdu, Arabic, French)	B	CPU only	Apache-2.0; printed text in photo questions
Whisper large-v3-turbo (faster-whisper)	B	~1.5 GB	MIT; ≈4–8× faster than the original large-v3 with similar accuracy
Whisper small (browser), with in-house Bangla and Urdu fine-tunes	N	~250 MB download (8-bit), cached on the device	MIT weights; fine-tunes use Common Voice (CC0) and consented recordings; word-error-rate benchmark gate per language
In-house VITS voices (6 languages, browser)	B	~30–60 MB each, cached on the device	MIT code; trained on commissioned voice-artist recordings owned by RemonLogic; character-based, no GPL phonemiser; listening-test gate per language; the same voices run on server CPU through ONNX Runtime for weak devices (8.12)
wav2vec2-lv-60-espeak-cv-ft	N	~1.5 GB	Apache-2.0 (model weights); phoneme recognition for pronunciation scoring
Chatterbox Multilingual V3	B	~6 GB	MIT; covers English, Hindi, Arabic and French (also Japanese, Korean, German and Chinese for language courses) — not Bengali or Urdu; also the NexaTute voice clones (38.16), off-peak within a daily GPU-minute cap
Indic Parler-TTS	B	~4 GB	Apache-2.0; Bengali and Urdu officially supported; validate Urdu voice quality
BGE-M3	N	~2 GB	MIT; approximate
BGE-reranker-v2-m3	N	~2 GB	Apache-2.0; approximate
Qwen3Guard-Gen-8B	B	~6 GB (4-bit)	Apache-2.0; multilingual safety moderation

Model / component	Product	Working VRAM estimate	Status / caution
Wan2.2-S2V-14B (speech-to-video)	M	~40 GB (FP8 with CPU offloading on the 96 GB GPUs; to be measured)	Apache-2.0 (verify on the model page); Cinematic dialogue close-ups only; replaces MuseTalk/LatentSync; blocked from production until its licence check and quality gate pass (8.15)
Engagement models (LightGBM, Vowpal Wabbit)	N	CPU only (no VRAM)	MIT and BSD-3-Clause; trained in-house on NexaTute's own event data; used only where the consent switches are on (3.19, 19.4)
MiniMax-M2.7 (4-bit AWQ)	L	≈140 GB per copy on one Dev-H server (4 × 48 GB) incl. KV cache	MiniMax licence (commercial authorization needed); ≈229B total / ≈10B active; one copy per server; MiniMax recommends temperature 1.0; the official FP8 weights cannot run on A6000
GLM-5.2 (4-bit) — fallback	L	≈372–475 GB across 12 RTX A6000 (3 servers) incl. KV cache	MIT (verify); ≈744B total / ≈40B active; pipeline-parallel over the private LAN; used only if the M2.7 licence is refused
MiniMax M3 (API)	L	None — managed API	Fast-lane fallback only; not self-hosted
Claude Sonnet 5.5 (API)	L	None — managed API	Heavy-lane plan review and rescue only; not self-hosted
Qwen3-Coder-Next	L	≈80 GB (FP8) or ≈45 GB (4-bit) on one RTX PRO 6000 96 GB	Apache-2.0 (verify); ≈80B total / ≈3B active; 256K context; one copy per GPU
SpeechBrain ECAPA-TDNN	B	<1 GB (CPU possible)	Apache-2.0 (verify model); consent speaker verification only, in both products
AudioSeal	B	<1 GB (CPU possible)	MIT (verify weights); audio watermark only

18.3 Proposed API Surface

- Identity: register, login, refresh, logout, profile and MFA/session management.
- NexaTute: tutor query, Nexa Co-Teacher query, assessment submission, progress/KPI, plan purchase and renewal, wallet balance/top-up, batch enrolment and batch-class events (polls, questions), face-verification result, recording start/stop and LiveKit token issuance. Tests (3.27): answer autosave and resume, written-check upload, learner confirmation and result, recheck request and coding-test submission.

Institution roster and results API (3.27): OAuth 2.1 client-credential keys from Keycloak with scopes — roster read and write, assign tests, results read — through the APISIX gateway with rate limits; webhooks when results are ready; each key approved by the institution head and the NexaTute

security owner, logged, rotated and revocable; LTI 1.3 stays available (3.11). NexaTute for Partners (3.32) uses the same keys, scopes, rate limits and LTI 1.3 for partner schools and companies, with spend caps and a DPA per partner. Study bots receive WhatsApp Cloud API and Messenger webhooks for registered, linked accounts only and reply from the verified cache.

- Motion Studio: project create/update, job submit/status/cancel, render request and final-artifact delivery status.
- Administration: usage and job dashboards, moderation/review queues, support and audit views, and Motion Studio Package Settings (limits, video length, tiers, modes, fees and rates; 4.9).
- API counts (including “120+ endpoints”) remain planning estimates until an OpenAPI specification is generated, reviewed and versioned.

18.4 Licence & Compatibility Gate

Maintain a software bill of materials and model registry with exact package/model IDs, versions, licence text, commercial-use permissions, attribution obligations, model weights/source hashes and approval owner. Unknown or restrictive licences must block production use until resolved.

International-standard practice: every release produces a machine-readable SBOM in CycloneDX or SPDX format (Syft), is scanned for known vulnerabilities (Grype, Trivy) and for licences (ScanCode Toolkit) in Forgejo CI, and is blocked if any component has an unknown, restrictive or changed licence. Licences are recorded as SPDX identifiers in the register (18.1), and the licence-compliance process follows OpenChain (ISO/IEC 5230), so investors, B2B and government partners can audit it. Model weights are recorded with their source, hash and licence the same way.

Dependency rule: every component's own runtime dependencies — databases, message brokers, bundled libraries and model weights — are checked against the same licence policy before approval. Under this rule GrowthBook and Novu were replaced (both need MongoDB, licensed under SSPL, which is not open source), ClickHouse was added for Langfuse and Ralph LRS, APISIX runs without a separate etcd, and pillow-heif is built without the GPL x265 encoder.

Licences to confirm before the creator tools go live: ACE-Step (code and weights), Demucs weights, RealBasicVSR/BasicVSR++ weights, GFPGAN face-detector weights, SeedVR2, the NSFW classifier, the Fountain parser, the Bijoy converter, pillow-heif (decode-only build without x265, which is GPL), the EAGLE-3 draft-head weights and the 17 October 2026 additions (LightX2V, the FastWan TI2V-5B weights, Wan2.2-S2V-14B and the Nunchaku 4-bit FLUX.1-schnell weights); Semgrep registry rules are used only internally under the Semgrep Rules License; Chromaprint stays a separate process so its LGPL-2.1 licence does not extend to RemonLogic code; music and sound-effect library items must be CC0 or owned by RemonLogic. For NexaTute face verification: the SFace model weights (3.16). For shared links: the licence of each source list in the chosen StevenBlack hosts variant and the abuse.ch URLhaus terms (3.23). For the CMS: sharp’s bundled libvips (LGPL-3.0-or-later) is used as a dynamic library (3.25).

19. Security, Privacy, Safety & Incident Response

19.1 Security Architecture

Control area	Proposed control
Edge	Cloudflare WAF/DDoS controls and TLS; strict origin protection
Identity	Short-lived access JWTs, refresh rotation/reuse detection, MFA for privileged roles and session revocation

Control area	Proposed control
Password/session	Argon2id or an equivalently reviewed password hash; Secure/HttpOnly/SameSite cookies where applicable
Authorization	RBAC plus tenant, entitlement and resource-ownership checks on every protected operation
Secrets	OpenBao-backed secret storage, scoped credentials, rotation and no secret logging; OpenBao runs as a 3-node Raft cluster (Redis1, Redis2, App-1) with daily encrypted snapshots to R2, Shamir 3-of-5 unseal keys held by named officers and a written break-glass procedure
Data	Encryption in transit and at rest; PostgreSQL RLS where appropriate; audited privileged access
AI safety	Prompt-injection resistance, content moderation, PII handling, human review and traceable policy versions
Media	Private R2 objects, scoped signed URLs, consent/rights workflow and verified artifact delivery
Payments	Server-side verification, idempotent callbacks, atomic ledger updates and reconciliation; never trust browser redirect alone

19.2 Authentication Flow

Browser/mobile → Cloudflare WAF and TLS → LB-1/LB-2 (HAProxy) → authentication service (Keycloak) → credential and account-status checks → MFA/step-up when required → short-lived access token plus rotated refresh session → authorization checks (role, tenant, entitlement and resource ownership) → product API → audit and monitoring.

Sign-in additions (28–29 October 2026): at sign-up the user chooses where the OTP goes — SMS, email or WhatsApp — with Cloudflare Turnstile checked before any OTP is sent; learner and guardian login security follows the admin settings (OTP resend after 60 seconds, 5 a day, 5 wrong codes → 15-minute lock, 30-day sessions, 3 devices, sign out of all devices); admin staff sign in with passkeys or hardware security keys (WebAuthn/FIDO2), required for the super admin, money approvers and the security owner, optional for other staff, with an authenticator-app code as fallback, 10 one-time backup codes, lost-key recovery only with two-person approval and an identity check, and every key added or removed logged and alerted; institution head admins may require passkeys for their staff (38.12, 38.17).

19.3 Roles & Governance

Role	Minimum principle
Student	Own learning records and permitted personal resources only

Role	Minimum principle
Parent	Only explicitly linked child accounts and consent-authorized information; approves purchases by under-18 learners and decides their consent switches (19.4)
Teacher	Assigned classes, learners and permitted classroom resources
Creator	Owned/authorized projects, assets and render jobs
Admin	Scoped administrative permissions with audit trails; custom roles per module and sub-module, scope limits, masking and maker-checker approvals (38.12)
Human controller	Receives safety alerts — push and email, plus SMS for serious cases — and acts on serious cases; named in the admin panel with name, email and mobile number (10.2); an unacknowledged alert escalates to a second controller after 15 minutes; at least two controllers on a 24/7 rota (10.2)
Data Protection Officer	Owns consent records, data-subject requests and the lawyer's review under the Bangladesh Personal Data Protection Act 2026 (19.4)
Superadmin	Break-glass access, MFA, approval and enhanced logging
Institute Admin	Own institution only; no details of a student's mental distress or safety incidents; cannot change prices or publish content without the checks (38.7)
Finance roles	Finance lead, accounts officer, payroll officer, founders and board, department manager (team totals only), external auditor or chartered accountant (read-only and export, about 30 days) and investor (investor report and own shares) (38.12)

19.4 Privacy, Retention & Compliance

- Publish clear privacy notices, lawful basis/consent flows, age-appropriate controls and data-subject request processes for applicable jurisdictions.
- Because servers are hosted in the USA, sign data-processing agreements with every hosting, storage, email and cloud-LLM provider, confirm Bangladesh cross-border transfer requirements, and apply GDPR safeguards for learners in the EU (8.16). Under the Bangladesh Personal Data Protection Act 2026, transfers outside Bangladesh need equivalent protection, so the lawyer's review covers the US servers specifically; if needed, children's data is kept in another location.
- Retention periods must be approved by data category and legal requirements; avoid claiming GDPR/COPPA/FERPA or other compliance until counsel and evidence support it.

- Use data minimization, deletion verification, access logging, backup-retention rules and controlled exports.
- Keep separate product authorization and usage boundaries even when databases or physical infrastructure are shared.
- Biometric and children's data: face templates for the face-verified single user are biometric data — collected only with consent (guardian consent for minors), encrypted, used only for account and presence checks, never sent to a cloud service, and deleted with the account; minors get COPPA-style and UK Children's Code-style protection (guardian consent, minimal data, no advertising or tracking, safest defaults), and EU learners get their GDPR rights to download and delete their data (19.7).
- Bangladesh Personal Data Protection Act 2026: a Bangladeshi lawyer reviews NexaTute before launch and a Data Protection Officer is appointed (19.3). The engagement features and personalised offers (3.19, 3.22) are explained in the Terms & Conditions, and each has its own on/off switch, chosen at purchase — or at sign-up for free users — and changeable at any time in Settings. For under-18 learners the guardian decides (a teacher only for adult learners, or where an institution holds guardian consent); those switches stay locked off for under-18 learners until the lawyer confirms (decided 26 October 2026).
- Retention values approved (26 October 2026): compliance case records kept 6 months after the case closes; mentor notes deleted 1 year after the student leaves; sessions end after 30 minutes idle; 2 OTP exceptions a month; account and learning records deleted 5 years after last use. A student leaving an institution keeps the account (deleted 5 years after last use); the institution can view it for 90 more days and the family can download it at any time.
- Backups on the privacy page: daily copies are kept 30 days and month-end and year-end copies longer (changeable); deleted personal data leaves each backup copy when that copy expires — the lawyer confirms the wording.
- Recordings (3.29): every recording has a consent record and an access log; notifications keep the service categories and marketing separate (3.30).
- Age check at sign-up: date of birth, and an OTP to the guardian's phone for under-18 learners; every consent choice is recorded.
- Language: the Terms & Conditions, Privacy Policy and all legal documents are in English because international students may enrol; the consent screen also shows a 3–4-line plain-language summary in the learner's own language (for example Bangla), so consent is informed and clearly communicated. The Terms and this summary also state that NexaTute practice exams are not recognised by any education board or exam authority (3.27).
- Website cookies (3.25, 4.19): only essential cookies load until the visitor agrees in the cookie consent manager; analytics are aggregated, with no session recording or tracking-based personalisation for children; every Terms & Conditions and Privacy Policy version keeps an acceptance log.
- Marketing data (3.26): leads keep their source and lawful basis or consent; an opt-out goes to a suppression list kept only to stop future emails; organisations that never reply are deleted after 12 months; no lead records or profiles of minors are built from marketing.

19.5 Incident Response & Recovery Targets

Stage	Indicative target	Expected action
Detect	0–5 minutes	Alert, triage and establish incident owner
Assess	5–30 minutes	Scope impact, severity, data exposure and service blast radius

Stage	Indicative target	Expected action
Contain	30–60 minutes	Disable compromised credentials/routes, isolate workloads, preserve evidence
Communicate	1–4 hours	Notify leadership and affected parties as required by law/policy
Eradicate	4–24 hours	Patch root cause, rotate secrets and validate clean state
Recover	24–72 hours	Restore safely, monitor and verify service integrity
Postmortem	Within 72 hours after stabilization	Document timeline, root cause, corrective actions and owners

These are planning targets, not demonstrated response performance. With the 8.13 clusters, the targets are: single-node failure — RTO ≈30 s (database) and ≈10 s (Valkey) with no committed-data loss; full restore from the R2 live backup — RTO ≤4 hours and RPO ≤1 minute. These require architecture-specific backup, replication and restore tests before they can be treated as service commitments.

19.6 Security Acceptance Tests

- Test refresh-token rotation/reuse detection, MFA recovery, privilege escalation, tenant isolation and session revocation.
- Run backup restore, DB failover, R2 recovery, incident tabletop and secret-rotation drills.
- Test payment duplicate callbacks, delayed callbacks, failed settlement and atomic credit reservation.
- Test prompt injection, PII leakage, unsafe content, Islamic-source attribution and human-review escalation.
- Log security events without recording passwords, tokens, payment secrets or unnecessary sensitive prompt content.
- Run Semgrep (static analysis) and OWASP ZAP (dynamic testing) on every release, Falco runtime detection on every node, and SBOM vulnerability scans (Grype, Trivy); a critical finding blocks the release.
- Test face-template deletion, Nexa Co-Teacher cross-learner isolation (row-level security) and breach notification within 72 hours (tabletop), and commission a yearly external penetration test. The first external penetration test runs before the 1 January 2027 paid launch, with high findings fixed and retested, and repeats every 12 months (default, editable) and after major changes; a “Report a security problem” page and security.txt invite responsible disclosure (36.6).

19.7 International Standards Alignment

RemonLogic follows these international standards and frameworks. "Aligned" means the controls are designed to the standard; no certification or compliance is claimed until an audit or counsel confirms it (19.4).

Area	Standard / framework	How it is applied
Information security	ISO/IEC 27001 (aligned controls), CIS Benchmarks	Risk register, access control, logging, backup and incident response (19.1–19.6); hardened Linux and Kubernetes baselines; breach notice to affected users and guardians within 72 hours; yearly external penetration test; ISO/IEC 27001 certification path after launch (3.24)
Application security	OWASP ASVS Level 2, OWASP Top 10, OWASP Top 10 for LLM Applications	Security requirements per feature; Semgrep and OWASP ZAP in CI; prompt-injection and data-leak tests (19.6)
AI governance	ISO/IEC 42001 (aligned), NIST AI RMF	Model registry, evaluation gates, human review, content safety and audit trails (3.13, 4.15, 18.2); model-behaviour changes such as the thinking switch, smaller frame-check images, INT8 embeddings, distilled and 4-bit video and image models (LightX2V, FastWan, Nunchaku), CacheBlend, suffix decoding and the Qwen3-8B routing change are logged and switched on only after measured quality checks; ISO/IEC 42001 certification path after launch (3.24)
AI transparency in education	EU AI Act (education uses are high-risk); UNESCO guidance on AI in education	The Nexa Teacher and Nexa Co-Teacher are always labelled as AI; major assessment and mastery decisions can be reviewed by a person; AI scoring is explained in plain language; reported wrong answers are reviewed by people; regular AI quality and bias tests (3.4, 3.10, 3.16); speaking tests never marked on accent, with regular accent-bias checks (3.27)
Children’s data and privacy	COPPA, UK Age Appropriate Design Code, GDPR (EU learners)	Guardian consent under 18, minimal data, no advertising or tracking, safest defaults (private profile, chat only inside the batch); guardians can view and delete all data; GDPR data download, deletion and consent records (19.4); a cookie consent manager on both sites (3.25)
Bangladesh data protection	Personal Data Protection Act 2026	Guardian consent for children’s data; a consent switch per feature at purchase and sign-up; the guardian decides for under-18 learners; profiling switches locked off for minors if the lawyer requires it; review of transfers to the US servers; Data Protection Officer (19.4)
Education quality	ISO 21001 (educational organizations)	Written learning outcomes per course, a defined feedback and complaint process with response times, and regular course quality reviews linked to the 3.13 quality gate
Assessment integrity	Classical item analysis, IRT and computerised adaptive testing	Question-bank item analysis (difficulty and discrimination, girth); shuffled questions and options and answer-pattern similarity flags in ranked, batch, live, certificate and institution tests for human review, never an automatic penalty; adaptive tests only for placement and practice; limited proctoring only in certificate and institution exams (3.27); QR-verified certificates (3.10)

Area	Standard / framework	How it is applied
Thinking-skills assessment (MindForge)	AERA/APA/NCME Standards; ITC/ATP Guidelines; EU AI Act high-risk (Annex III); ISO/IEC 23894 and 27701; QTI 3.0, CASE, Open Badges 3.0; OECD PISA and ISCED; ESSA tiers; ISO 10667; EN 301 549	Practice, not an IQ test; calibrated items, DIF checks and range reporting; no IQ-style score until validated and normed; EU assessment part off until compliant (37.9)
Workplace assessment	ISO 10667 (assessment service delivery)	Employer and organisation tests for adults only, with candidate consent, explained scores and the statement that results are never the only basis for hiring (3.27)
Software supply chain	SPDX, CycloneDX, OpenChain ISO/IEC 5230	SBOM and licence scanning on every release; SPDX licence IDs in the register (18.1, 18.4)
Content provenance	C2PA Content Credentials	Every Motion Studio export signed (4.10); browser-made chunks are checked like uploads before signing (4.8)
Accessibility	WCAG 2.2 Level AA	Screen-reader labels and alt text, full keyboard use, colour-blind mode, large font and high contrast, captions on recordings and AI speech, extendable time limits; axe-core tests in CI; dyslexia and ADHD options (3.10); a third-party audit with an Accessibility Conformance Report (VPAT) (3.24)
Inclusive design	Universal Design for Learning (UDL)	Inclusion features organised under the three UDL principles — engagement, representation, and action and expression (3.24)
Language levels	CEFR (A1–C2)	English and foreign-language progress and certificates show the CEFR level (3.24)
Learning interoperability	LTI 1.3, xAPI, SCORM, W3C Verifiable Credentials 2.0	Learning OS integrations and certificates (3.11); every learning activity is recorded as an xAPI statement
APIs	OpenAPI 3.1, OAuth 2.1 / OpenID Connect	Public API specification and Keycloak sign-in (18.3); client-credential keys with scopes for the institution roster and results API (3.27)
Media quality	EBU R128 loudness, ITU-T G.114 latency	Loudness normalisation of every export; live-class latency targets (8.16)
Testing and capacity	ISO/IEC/IEEE 29119 (test process), ISO/IEC 25010 (performance efficiency)	Pre-agreed pass/fail criteria and measured capacity reports from Locust, the LiveKit CLI and vLLM benchmarks before any capacity claim (H.3)
Payments	PCI DSS scope minimisation	Card data entered only on the payment provider's pages; RemonLogic never stores card numbers
Localisation	Unicode, CLDR, ISO 8601	Bangla and right-to-left scripts, local number, date and currency formats (29)

20. Core Architecture Flows & System Diagrams

The following flow descriptions provide the text equivalent of the four proposed architecture diagrams. They are logical flows; actual implementation must be verified against deployed configuration.

20.1 NexaTute Learning Flow

Learner browser/mobile → Cloudflare (WAF, Dhaka edge cache) → LB-1/LB-2 (HAProxy) → NexaTute API pods on the App cluster → authentication, entitlement and face-verification checks → load-aware routing (3.16): the live Nexa Teacher when the GPU pool is free, recordings, mock tests and summaries first when it is busy, the text Nexa Co-Teacher when it is overloaded → lesson and verified-answer caches → AI gateway → GPU cluster (Nexa Co-Teacher overflow to Mistral Large 3, then GPT-6 Luna) → answer streamed sentence by sentence → progress/KPI event → PostgreSQL and the NexaTute Valkey instance. Batch classes: 3.17; details: 6.2; browser rendering: 3.2.

20.2 Live Class & Recording Flow

Teacher/student clients → LiveKit token endpoint with authorization → LiveKit SFU cluster on Core-1 and Core-2 → media routing; LiveKit embedded TURN (coturn as fallback) relays media when direct connectivity fails → Rec-1/Rec-2 record classes as separate tracks with the whiteboard event log → private R2 → replay in the NexaTute player; a download MP4 is made in the user's browser (9.5). Consent, retention and access rules: 9.5. AI-tutor and batch classes are recorded as event logs with the learner's own audio and replayed in the browser (11.3).

20.3 Remon Motion Studio Video Pipeline

Creator request or uploaded script → job record and idempotency key → Document Intake Service, copyright and content-safety checks → package settings, spending cap and price confirmation → pre-production fee and credit reservation → script and storyboard → voice-over first and key-image approval → queue with product priority → render on the GPU cluster → per-scene quality and frame-safety checks → video upscaling and quality gate for escalated tiers → music, captions and aspect ratios → final MP4 validated and signed with Content Credentials on the executing node → upload to private R2 → verify → clean up → settle credits. Full flow: 6.3; scratch and cleanup rules: 9.3.

20.4 Payment & Credit Flow

Client initiates checkout → backend creates a pending transaction and idempotency key → aamarPay processes the payment → provider callback/webhook is verified server-side → transaction state is reconciled → atomic ledger entry activates or renews the plan, or credits the correct product wallet → user receives confirmation. Browser redirects are informational only and cannot establish payment success. Duplicate callbacks must not double-credit; refunds and charge reversals require auditable compensating ledger entries. Keep NexaTute and Motion Studio product ledgers logically separate. Auto-renew uses a tokenised recurring payment only where aamarPay supports it for the method; otherwise the learner is reminded before the renewal date (11.7). The Play Store app has no purchase flow at launch; plans are bought on the website (11.7).

20.5 Shared Infrastructure & Isolation

LB-1/LB-2 route application traffic with HAProxy; PgBouncer runs as a sidecar on the App nodes next to the database network (8.13). The App nodes and shared workers must preserve product-specific runtime contexts, authorization, queues, cache namespaces and job accounting. Valkey keeps a separate NexaTute instance (6379) and Motion Studio instance (6380), each with a primary and a replica across Redis1/Redis2 (8.13). GPU sharing is permitted only through controlled, priority-aware scheduling.

20.6 Priority & Failover Rules

Priorities follow the X-Priority policy in 8.2. The header alone does not enforce priority. Queue routing, worker scheduling, admission control, back-pressure, retry limits and starvation protection must implement the policy. Failover must respect product isolation and must not move heavy GStreamer scratch writes onto Core-1.

21. User Personas

21.1 Primary Personas

Persona 1 — BCS Aspirant

Attribute	Value
Name	Rahim
Age	26
Location	Dhaka
Education	Bachelor's
Goal	Pass BCS
Pain points	Expensive coaching (৳5,000–10,000/month), limited time
Willingness to pay	৳500–2,000/month
Expected AI Tutor use	30–60 min/day
Implied monthly spend	Combo ৳850 a month for BCS + Job Test Preparation (Batch ৳750, Personal ৳1,000) — within willingness to pay
Preferred device	Android phone
Total study time (all sources)	4–6 hours/day

Persona 2 — Class 9–10 Student

Attribute	Value
Name	Fatima
Age	15
Location	Chittagong
Education	Class 10
Goal	Good grades in SSC
Pain points	Physics/Math difficulty, no tutor
Willingness to pay	৳300–800/month

Attribute	Value
Expected AI Tutor use	10–25 min/day
Implied monthly spend	Batch ৳600 or Combo ৳800 a month (Personal single subject ৳500) — within willingness to pay
Preferred device	Smartphone
Total study time (all sources)	2–3 hours/day

Persona 3 — Parent

Attribute	Value
Name	Mr. Ahmed
Age	42
Location	Dhaka
Occupation	Businessman
Goal	Child's success
Pain points	Limited visibility into child's progress
Willingness to pay	৳500–2,000/month
Child’s expected AI Tutor use	20–60 min/day
Implied monthly spend	Combo ৳800 or Personal ৳1,000 a month per child; a second child 30% off — within willingness to pay
Preferred device	Smartphone

Persona 4 — University Student

Attribute	Value
Name	Karim
Age	21
Location	Dhaka
Education	Engineering
Goal	CGPA + skills
Pain points	Complex topics, no guidance
Willingness to pay	৳500–1,500/month
Expected AI Tutor use	10–30 min/day

Attribute	Value
Implied monthly spend	Personal ৳750 (single subject) or ৳1,500 (all subjects) a month; University is Personal only — mostly within willingness to pay
Preferred device	Laptop
Total study time (all sources)	3–4 hours/day

Persona 5 — Content Creator

Attribute	Value
Name	Sajid
Age	28
Location	Dhaka
Occupation	YouTuber
Goal	Faster video production
Pain points	Expensive editing tools, time
Willingness to pay	৳3,000–8,000/month
Likely package	Premium Professional (৳3,999/month) plus G.8 per-minute charges — e.g., 80 minutes of 1080p Escalating at ৳48/min = ৳3,840, total ৳7,839/month, within the ৳3,000–8,000 willingness to pay. Premium Agency (৳7,999) suits agency teams with larger budgets.
Preferred device	Desktop

Persona 6 — PlayBoom Guardian

Attribute	Value
Name	Nasrin
Age	32
Location	Narayanganj
Occupation	Bank officer; two children (KG and Class II)
Goal	A safe, playful start in letters, numbers, Bangla and English without too much screen time
Pain points	No time to teach every day; worried about YouTube content and screen time
Willingness to pay	৳500–1,200/month per child
Expected use	KG child 1 × 25 min a day; Class II child 45 min a day

Attribute	Value
Implied monthly spend	PlayBoom ৳599 + ৳419 (second child 30% off) = ৳1,018 a month — within willingness to pay
Preferred device	Tablet or the guardian's smartphone

Persona 7 — Freelancer

Attribute	Value
Name	Tanvir
Age	24
Location	Rajshahi
Occupation	Freelance web developer (Fiverr, Upwork)
Goal	Deliver more client websites and apps faster, at a higher margin
Pain points	Repetitive work, foreign AI tools priced in dollars, slow PCs
Willingness to pay	৳1,000–3,000/month
Expected use	Daily coding and builders; 2–4 client projects a month
Implied monthly spend	NexaLogicDev Pro ৳2,499 + occasional top-ups; Freelancer Toolkit and Go Live bundles resold to clients
Preferred device	Laptop

Persona 8 — Small Business Owner (F-commerce)

Attribute	Value
Name	Rupa
Age	35
Location	Sylhet
Occupation	Runs a clothing shop on a Facebook Page
Goal	A real website with bKash payment and courier booking, without hiring a developer
Pain points	Orders lost in Messenger, no technical skills, worried about cost
Willingness to pay	৳500–2,500/month
Expected use	Facebook Page → Website once, then weekly updates and order management
Implied monthly spend	Hobby ৳499 + Shop Go Live bundle (৳799 first year) + Care Plan ৳499; later the AI Shop Assistant ৳299–699
Preferred device	Smartphone

Persona 9 — Digital Agency

Attribute	Value
Name	Pixel Bangla (agency of 8)
Age	Owner 34
Location	Dhaka
Occupation	Web, app and marketing agency
Goal	Ship client ERP modules and websites faster under its own brand
Pain points	Developer cost, delivery delays, client handover
Willingness to pay	৳8,000–20,000/month
Expected use	Team plan with Power mode (WebContainer API), Auto-Architect modules, client preview links, white-label
Implied monthly spend	Team ৳9,999 + one Medium App (৳4,999) or Complex Module (৳9,999) a month; Motion Studio promo videos
Preferred device	Desktop

Persona 10 — Startup Development Team

Attribute	Value
Name	ShopKoro (startup, 5 developers)
Age	Team 23–30
Location	Dhaka
Occupation	Early-stage e-commerce startup
Goal	Build and iterate the product with a small team and a small budget
Pain points	Foreign AI-tool subscriptions per seat in dollars, code quality, security
Willingness to pay	৳8,000–15,000/month
Expected use	Team plan (5 seats, pooled credits), GitHub sync, auto tests and security audit, Power mode
Implied monthly spend	Team ৳9,999 + top-ups ≈৳1,000–2,500
Preferred device	Laptop

21.2 Persona Distribution

Persona	% of users	Revenue %
BCS Aspirant	30%	21%
Class 9–10	20%	6%
Parent	12%	8%
University	12%	7%
Content Creator	8%	25%
PlayBoom Guardian	8%	4%
Freelancer	4%	4%
Small Business (F-commerce)	3%	3%
Digital Agency	1.5%	12%
Startup Development Team	1.5%	10%
Total	100%	100%

Note: Revenue % is the share of combined ecosystem revenue, estimated as user share × midpoint willingness-to-pay (BCS ৳1,250; Class 9–10 ৳550; Parent ৳1,250; University ৳1,000; Content Creator ৳5,500; PlayBoom Guardian ৳850; Freelancer ৳2,000; Small Business ৳1,500; Digital Agency ৳14,000; Startup Development Team ৳11,500). Content creators, agencies and development teams are a small share of users but over 45% of revenue, so Motion Studio and NexaLogicDev retention matter as much as NexaTute volume.

22. Success Metrics & OKRs

22.1 Year 1 OKRs

Objective 1 — Achieve Product-Market Fit

Key Result	Target	Timeline
D30 Retention	>40%	Month 12
NPS	>50	Month 12
User Satisfaction	>4.5/5	Month 12
Feature Adoption	>60%	Month 12

Objective 2 — Grow User Base

Key Result	Target	Timeline
Total Users	≈15,500 (≈12,300 free)	Month 12
Paying Users	≈3,270 NexaTute incl. PlayBoom	Month 12

Key Result	Target	Timeline
Monthly Growth	>20%	Ongoing
Referral Rate	>30%	Month 12

Objective 3 — Build Sustainable Revenue

Key Result	Target	Timeline
NexaTute MRR	৳21.7 lakh (+ PlayBoom ≈৳4.6 lakh)	Month 12
NexaTute ARPU	৳900	Month 12
Motion Studio MRR	৳28.8 lakh (15.8 base case)	Month 12
Gross Margin	>70%	Month 12
CAC Payback	<6 months	Month 12

Objective 4 — Ensure Reliability

Key Result	Target	Timeline
Uptime	>99%	Ongoing
API Latency (P95)	<3 seconds	Ongoing
Error Rate	<0.5%	Ongoing
Support Response	<4 hours	Ongoing

22.2 Year 2 OKRs

Objective	Key Result	Target
Scale Users	Total Users	50,000
Paid Conversion	Paying Users	10,000 (20% of users) at ৳880 ARPU
Revenue Growth	NexaTute MRR	৳88 lakh
Revenue Growth	Motion Studio MRR (illustrative)	৳55 lakh
Profitability	Break-even	Monthly from July 2027; cumulative from February 2028 (15.8)
Team Growth	Team Size	17
Product	Mobile Apps	Live since launch; maintain feature parity

22.3 Year 3 OKRs

Objective	Key Result	Target
Market Leadership	Users	200,000
Paid Conversion	Paying Users	50,000 (25% of users) at ৳910 ARPU
Revenue	NexaTute ARR	৳54.6 crore
Revenue	Motion Studio ARR (illustrative)	৳33 crore
Profitability	Net Margin	>20% (must remain below gross margin)
Expansion	Regional Markets	3+
Ecosystem	Marketplace GMV	৳10 crore

22.4 KPI Dashboard

Category	Metrics
Learning	Tutor latency, answer quality, assessment accuracy, retention; North Star: learners whose mastery rose this week (3.19); learning gain and prediction accuracy (3.24)
Engagement	Daily active users, session length, frequency
Revenue	MRR, ARR, ARPU, LTV, CAC; landing-page conversion and funnel drop by step (3.25, 4.19)
Retention	D1, D7, D30, churn rate
Quality	NPS, CSAT, error rate
Operations	Uptime, latency, cost per session
Security	Incidents, compliance, audit

Consistency note: the investor base case is the 15.8 revenue model. Month 12 (September 2027): ≈2,400 paying NexaTute learners × ৳900 ARPU = ≈৳21.7 lakh NexaTute MRR, ≈850 PlayBoom children, ≈360 MindForge Pro subscribers (≈৳0.65 lakh MRR) and ≈960 Motion Studio creators (≈ ৳28.8 lakh MRR); end of 2027: ≈3,200 paying learners. Year 2 (end 2028) keeps the target of 10,000 paying learners × ৳880 ARPU = ৳88 lakh MRR and Year 3 the target of 50,000 × ৳910 = ৳4.55 crore MRR (৳54.6 crore ARR). The rising paid-conversion rate is the key assumption and must be validated with cohort data; all figures are unverified forecasts.

23. Risk Register (Complete)

23.1 Risk Matrix

#	Risk	Probability	Impact	Severity	Mitigation
1	16,000 CCU not achieved	High	High	Critical	Early load testing
2	Model quality poor	Medium	High	High	Multi-model fallback and evaluation
3	Content copyright issue	Medium	High	High	Legal review and licensing
4	Voice cloning abuse	Medium	Critical	Critical	Consent framework and monitoring
5	Payment gateway delay	High	Medium	High	Evaluate multiple providers
6	Team hiring delay	High	Medium	High	Remote hiring and contractors
7	Competition from big tech	High	High	Critical	Niche focus
8	Cash flow crisis	Medium	Critical	Critical	Reserve and staged spending
9	Regulatory changes	Low	High	Medium	Legal monitoring
10	Data breach	Low	Critical	High	Security audit and incident response
11	Server hardware failure	Medium	High	High	Redundancy and recovery tests
12	API cost spike	Medium	Medium	Medium	Budget limits
13	User acquisition cost high	High	Medium	High	Organic acquisition and CAC tracking
14	Churn rate high	Medium	High	High	Retention strategy
15	AI hallucination	High	Medium	High	RAG and verification
16	Content generation quality	Medium	High	High	Human review
17	Cyber attack	Medium	Critical	Critical	WAF and monitoring
18	Natural disaster	Low	Critical	High	Multi-region recovery
19	Key person dependency	High	High	Critical	Documentation and cross-training

#	Risk	Probability	Impact	Severity	Mitigation
20	Market timing wrong	Medium	High	High	Paid pilots and pivot ability
21	Deepfakes and misuse of real people's face or voice	Medium	Critical	Critical	Random-sentence consent check, public-figure block, AudioSeal watermark, C2PA labels, report and takedown (4.15, 10.3)
22	Child-safety violation in uploads or generated content	Low	Critical	High	Zero-tolerance hash matching (PDQ/TMK), NSFW and Qwen3-VL checks, account ban and reporting to the authorities (4.15)
23	Illegal content and takedown demands	Medium	High	High	Content policy reviewed by a Bangladeshi lawyer, takedown and repeat-infringer process, audit log (4.12, 4.15)
24	Platform API dependency (YouTube, Facebook, TikTok)	Medium	Medium	Medium	Download-and-upload fallback, early API audits, no feature that works only through one platform (4.6, 4.14)
25	Licence or dependency change in open-source software	Medium	High	High	SBOM and licence scans on every release, pinned versions and a replacement plan per component (18.4)
26	AI batch-class teaching quality not yet proven	Medium	High	High	Interactive class format, segment reports, A/B-tested explanations and weekly human review; batch classes open first for BCS + Job Test Preparation and SSC with Free Beta measurement (3.17)

#	Risk	Probability	Impact	Severity	Mitigation
27	Plan mix hurts margins (learners move between plans)	Medium	Medium	Medium	Clear price gaps between plans and the fine-tuning rules, at most one change a quarter (11.12)
28	One learner's data shown to another through the Nexa Co-Teacher or batch recordings	Low	Critical	High	Per-learner retrieval with row-level security, isolation tests and batch recordings limited to batch members (3.16, 3.17, 13.2)
29	Misuse or breach of face (biometric) data	Low	Critical	High	Consent (guardian consent for minors), browser-side matching, encrypted templates, deletion with the account and legal review (10.2, 19.4)
30	Profiling minors breaks the Bangladesh data law	Medium	High	High	A consent switch per feature, the guardian decides for under-18 learners, switches locked off if the lawyer requires it, Data Protection Officer (19.4)
31	Transfer of learners' data to the US servers is challenged	Medium	High	High	Lawyer's review of the US servers; children's data moved to another location if needed (19.4)
32	Affiliate, gift-card or marketplace fraud	Medium	Medium	Medium	Commission only after the 30-day refund window, self-referral blocking, unique card codes, tax withheld and reviewed payouts (11.14)

#	Risk	Probability	Impact	Severity	Mitigation
33	Harmful or pirated content arrives through shared links	Medium	High	High	Blocklists, sandboxed fetching, Qwen3Guard and image checks, no links from minors in batch chat, question leaks reported to the human controller (3.23)
34	Exam-season demand for 1:1 time overloads the GPU pool	High	Medium	High	Unlimited admission with the capacity plan — servers are added before classes start, with alerts and lead time; load-aware routing and the overload mix stay as the safety net (8.5, 3.16)
35	The free tier costs more than it converts	Medium	Medium	Medium	Free features run on the CDN, in the browser and from the question bank; GPU only for the 5 help minutes; free → paid % measured in the Free Beta (3.22)
36	App-store payment rules block or tax mobile sales	Medium	Medium	Medium	Purchases inside store apps use the store's billing with the 30% app-store charge on its own line; website purchases use our gateway; alternative billing when Google's rules reach Bangladesh (11.7)
37	The AI CMS publishes a wrong price or an unproven claim	Low	High	Medium	Prices only from the price table, claims check, admin approval and one-click rollback (3.25)

#	Risk	Probability	Impact	Severity	Mitigation
38	The pre-sale AI gives wrong advice or sells to a child directly	Low	High	Medium	Answers from the price table and verified content; B2B and complex cases to a person; purchases by minors only through the guardian (3.25)
39	Outreach email is reported as spam or breaks the data law	Medium	High	High	Organisations only, low daily cap, one-click unsubscribe, separate domain, lawyer review (3.26)
40	A platform restricts or bans an account for automated posting	Low	Medium	Medium	Official APIs only, platform approvals, posting limits, a person approves new content (3.26)
41	NexaLogicDev used to build phishing or scam sites	Medium	Critical	Critical	Acceptable Use Policy, prompt and pre-deploy scans, 24-hour takedown, DNS disable, human decisions (32.3, 36.5)
42	MiniMax-M2.7 commercial licence refused or delayed, or its 4-bit build too weak	Medium	High	High	Licence request sent now; GLM-5.2 fallback on the same three servers; June 2027 benchmark against the official FP8 weights and GLM-5.2 on our own tasks (32.1)
43	WebContainer licence too costly or refused	Medium	Medium	Medium	Power mode only for Team, agencies and module buyers; Sandpack and own sandbox cover the launch (32.10)
44	USA LLC compliance missed (Form 5472, Delaware tax)	Low	High	Medium	doola filing and bookkeeping, related-party log, yearly reminders (0.2)

#	Risk	Probability	Impact	Severity	Mitigation
45	Paddle onboarding delayed or refused	Medium	Medium	Medium	Apply as soon as the LLC opens; Bangladesh sales unaffected; international sales wait (11.7)
46	Child-safety or privacy failure in PlayBoom	Low	Critical	High	Guardian accounts, no face templates, no chat, local AI only, readiness checklist and lawyer review (30.5, 36.1)
47	Launch scope too large for the launch dates	High	High	Critical	Day-1 must-have list, “coming soon” switch, go-live gates (17.5, 34)
48	Revenue below the 15.8 base case	Medium	High	High	Model re-run monthly from actual data; staged spending; marketing tied to measured CAC (15.8)
49	Cinematic margins if the video optimizations fail	Medium	Medium	Medium	Native Cinematic tiers stay “coming soon” until the optimizations pass (35.1)
50	Dev GPU supplier changes price or availability	Low	Medium	Low	Month-to-month terms, phased renting and alternative GPU suppliers (32.13)
51	Shared-pool contention slows a product’s real-time work, or a guest workload lands on the wrong server	Medium	High	High	Owner-first P100, floors and 60/70% lend–reclaim thresholds, one-cycle hand-back, per-pool on/off switch, signed images and namespace isolation; sandbox, data and edge never lent (8.17)

#	Risk	Probability	Impact	Severity	Mitigation
52	External AI API dependency: price change, outage, policy change or customer concern about code leaving our servers	Medium	Medium	Medium	Local servers first; monthly spend caps; secret removal; “RemonLogic servers only” switch; alternative APIs (GLM-5.2 via Fireworks, DeepSeek V4 Pro) behind the same gateway (32.1)
53	Browser-side features fail on weak devices or some browsers, or model downloads cost users too much mobile data	Medium	Medium	Medium	Device Check tiers, Data Saver (Wi-Fi-only model downloads), feature detection with server fallback, tests on low-end Android and Safari before launch (8.18, 32.14)

Severity rule (applied consistently to every row): Low × High or Medium × Medium = Medium; Low × Critical, Medium × High or High × Medium = High; Medium × Critical, High × High or High × Critical = Critical.

23.2 Risk Response Plan

Risk	Response
CCU not achieved	Scale servers and optimize models based on evidence
Model quality	Fine-tune where justified and use evaluated fallback models
Copyright	Legal agreements, rights records and original content
Voice abuse	Consent, watermarking where appropriate and monitoring
Payment delay	Multiple eligible gateways and reconciliation
Hiring delay	Remote recruitment and contractors
Competition	Niche focus and product quality
Cash flow	Reserve, staged commitments and revenue validation
Regulatory	Legal advisor and compliance reviews
Data breach	Security audit, containment and recovery drills

23.3 Contingency Budget

Scenario	Budget
Base case	৳10 crore
With 20% buffer	৳12 crore
Worst case	৳15 crore

Reconciliation applied: the ৳5 crore ask is the initial 18-month round; the ৳10/12/15 crore figures are total programme contingency scenarios. Confirm the classification, time horizon and gross-margin assumptions with finance before investor circulation.

24. Exit Strategy

24.1 Exit Options

Option	Timeline	Valuation	Multiple vs ৳35 crore post-money*
Strategic Acquisition	5–7 years	৳500–1,000 crore	14–29×
IPO	10+ years	৳2,000 crore+	57×+
PE Buyout	7–10 years	৳400–800 crore	11–23×
Merger	3–5 years	৳200–400 crore	6–11×
Secondary Sale	3–5 years	৳100–200 crore	3–6×

*Multiple = scenario company valuation ÷ ৳35 crore post-money entry valuation, before any further dilution. For example, at a 14.29% stake a ৳500 crore exit would be worth about ৳71 crore to the investor before dilution, liquidation preferences, taxes and transaction costs. These are illustrative scenarios, not offers or guaranteed returns.

24.2 Potential Acquirers

Category	Potential organizations
EdTech	Unacademy, Coursera, 10 Minute School
Technology	Google, Microsoft, Meta, Amazon
Local	Grameenphone, Robi, bKash
Investment firms	Tiger Global, Sequoia, a16z
Strategic	Publishing houses and testing companies

Listing an organization does not imply interest, contact or commitment.

24.3 IPO Path

Period	Milestone
Years 1–3	Growth and profitability discipline
Years 4–5	Scale and strengthen market position
Years 6–7	IPO readiness preparation
Years 8–10	IPO execution if commercially and legally appropriate

24.4 Exit Readiness Checklist

- Audited financials for three years, when available and applicable
- Clean capitalization table
- IP protection and licensing records
- Team retention and succession planning
- Customer concentration analysis
- Compliance and governance
- Scalable systems and documented operations
- Evidence supporting market leadership claims

25. Advisory Board

25.1 Board Composition

Role	Expertise	Commitment	Compensation
Education Expert	Curriculum, pedagogy	Monthly	Equity + stipend
AI/ML Expert	Models and infrastructure	Monthly	Equity
Legal Advisor	IP and compliance	Quarterly	Retainer
Finance Advisor	Fundraising and M&A	Quarterly	Equity
Islamic Scholar	Religious accuracy and ethics	As needed	Honorarium
Industry Expert	EdTech and SaaS	Monthly	Equity

25.2 Advisory Roles

Advisor area	Contribution
Education	Curriculum validation and pedagogy
AI/ML	Model selection and optimization
Legal	Compliance, contracts and IP
Finance	Fundraising and financial planning
Islamic	Content accuracy and ethics

Advisor area	Contribution
Industry	Market insights and networking

25.3 Meeting Cadence

Type	Frequency
Board Meeting	Quarterly
Advisor Call	Monthly
Strategic Review	Semi-annual
Annual Retreat	Yearly

26. Partnerships Strategy

26.1 Partnership Types

Type	Partner	Purpose
Distribution	Schools, coaching centers	User acquisition
Campus ambassadors	School, college and university students	Peer-to-peer acquisition: each ambassador gets a referral code and earns 10% commission for the first 3 months on plans sold with it, or free minutes (affiliate program, 3.21; referral programme, 3.10)
Affiliates	Coaching centres and YouTubers	Plan sales with their own code at 10% commission for the first 3 months (3.21)
Content	Curriculum providers and publishers	Authorized content and licensing
Technology	Cloudflare and eligible AI/cloud providers	Infrastructure
Payment	aamarPay (bKash, Nagad, Rocket and cards)	Transactions
Research	Universities	Innovation; the learning-efficacy comparison study (3.24)
Corporate	Companies	B2B training
Retail	bKash agents, mobile-recharge shops and bookshops	Shikkha Salami gift cards (3.21)
Employment	Employers and licensed agencies	Job pathway for adult learners (3.21)

26.2 Target Partnerships

Partner target	Type	Timeline
100+ schools	Distribution	Year 1–2
500+ coaching centers	Distribution	Year 2–3
500+ campus ambassadors	Campus ambassadors	Year 1
Grameenphone, Robi or Banglalink	Telco (zero-rating / data bundles)	Year 1–2
a2i and Ministry of Education	Government (B2G)	Year 2
UNICEF, World Bank and education NGOs	Development partners (B2G)	Year 2–3
Cambridge	Content	Year 1
NCTB	Content	Year 1
Cloudflare	Technology	Now / ongoing evaluation
bKash	Payment	Year 1
aamarPay	Payment gateway	From launch
5+ universities	Research	Year 2

All entries are targets, not confirmed partnerships. Availability, eligibility, agreements and licensing must be verified.

26.3 Partnership Benefits

Partner	Potential benefit
Schools	Free or discounted access under agreed terms
Coaching centers	Revenue share
Content rights holders	Licensing fees
Technology providers	Credits and support where available
Payment providers	Transaction processing
Research partners	Grants and talent development

27. Sustainability & Social Impact

27.1 Social Impact

Area	Commitment / target
Free Access	10% free access for eligible low-income users

Area	Commitment / target
Scholarships	1,000+ free accounts
Teacher Training	Free workshops
Digital Literacy	Community programs
Rural Access	Offline or low-bandwidth mode (saved lessons can be downloaded for offline viewing, 8.14); audio-only low-internet mode (3.2); Sponsor-a-Learner with priority for rural girls (3.10)
Inclusion	Support for learners with dyslexia, ADHD or slower learning (3.10); wellbeing and anti-bullying safeguards (10.2)
Women in Tech	30% female engineers

27.2 Environmental Impact

Area	Action
Green Computing	Efficient models
Server Efficiency	Modern hardware and utilization monitoring
Paperless	Digital-first operations
Remote Work	Reduce commuting
Carbon Offset	Evaluate tree planting / credible offset programs

27.3 Ethical AI

Principle	Implementation
Transparency	Explain AI limitations
Fairness	Bias testing
Privacy	Data minimization
Safety	Child protection
Accountability	Human oversight

27.4 Community Programs

Program	Description
Code for Kids	Free coding classes
BCS Study Group	Free community support

Program	Description
Teacher Fellowship	Educator support
Content Contributors	Open-source content initiatives

Targets should be supported by eligibility rules, funding, delivery plans and measurable outcome reporting.

28. Customer Support Plan

28.1 Support Tiers

Tier	Target response time	Channels
Free (Free Trial Packages)	72 hours	Email
Basic (NexaTute plans and wallet; Freelancer Mini/Starter)	24 hours	Email + Chat
Premium (Premium Starter/Professional/Agency)	4 hours	Email + Chat + Phone
Enterprise (B2B and Motion Studio Enterprise)	1 hour	Dedicated support + SLA

28.2 Support Channels

Channel	Availability
Email	24/7 intake
In-app Chat	9 AM–9 PM; pre-sale AI on the website chat 24/7 (3.25)
Phone	9 AM–6 PM
WhatsApp	9 AM–9 PM; pre-sale AI 24/7 (3.25)
Facebook Messenger	Pre-sale AI 24/7; a person 9 AM–9 PM (3.25)
Social media comments	AI replies 24/7 from verified content; complaints to a person (3.26)
Community Forum	24/7 access

Support tooling: Chatwoot (MIT, self-hosted) gives one inbox for in-app chat, email, WhatsApp and phone notes; an AI assistant answers first in Bangla or English from the knowledge base (28.4) and hands over to an agent when needed; payment questions (bKash, Nagad, cards) are checked automatically against the payment ledger.

28.3 Support Metrics

Metric	Target
First Response Time	<4 hours overall target
Resolution Time	<24 hours target
CSAT	>4.5/5
Ticket Volume	<5% of users
Self-Service Rate	>60%

28.4 Knowledge Base

Content	Target volume
FAQ	100+ articles
Video Tutorials	50+ videos
User Guides	20+ guides
Community Posts	1,000+

28.5 Escalation Matrix

Level	Owner	Timeline
L1	Support Agent	0–4 hours
L2	Senior Support	4–24 hours
L3	Engineering	24–72 hours
L4	CTO/CEO	72+ hours

Critical security, payment and outage incidents should follow a severity-based escalation path rather than waiting for the standard window.

29. Localization Strategy

29.1 Language Priority

Language	Priority	Timeline	Market
Bengali	P0	Launch	Bangladesh
English	P0	Launch	Bangladesh + Global
Hindi	P1	Launch (AI Tutor teaching); market entry per 29.4	India

Language	Priority	Timeline	Market
Urdu	P1	Launch (AI Tutor teaching); market entry per 29.4	Pakistan
Arabic	P2	Launch (AI Tutor teaching); market entry per 29.4	Middle East
French	P2	Launch (AI Tutor teaching); market entry per 29.4	Africa + EU

29.2 Localization Scope

Digits: the interface uses Latin digits in every language; the exception is PlayBoom's child views, which show Bangla digits (১, ২, ৩) to Bangla-medium children and Arabic-Indic digits to Arabic children (decided 24 October 2026). Urdu uses Noto Nastaliq Urdu with a taller line height and Arabic Noto Naskh Arabic.

Element	Bengali	English	Hindi	Urdu	Arabic	French
UI	Launch	Launch	Launch (NexaTute); Year 2 (Motion Studio)	Launch (NexaTute); Year 2 (Motion Studio)	Launch (NexaTute); Year 3 (Motion Studio)	Launch (NexaTute); Year 3 (Motion Studio)
Content	Full curriculum at launch	Full curriculum at launch	Each subject follows the learner's own country curriculum; a country opens when its content passes the quality gate (29.4)	Each subject follows the learner's own country curriculum; a country opens when its content passes the quality gate (29.4)	Each subject follows the learner's own country curriculum; a country opens when its content passes the quality gate (29.4)	Each subject follows the learner's own country curriculum; a country opens when its content passes the quality gate (29.4)
AI Tutor teaching	Launch	Launch	Launch	Launch	Launch	Launch
Voice	Launch	Launch	Launch	Launch	Launch	Launch
Currency	৳	$	₹	Rs (PKR)	Local (SAR, AED, etc.)	€ / local
Payment	Eligible local providers	Eligible international providers	UPI / local	Local providers	Local providers	Local providers
Curriculum	NCTB (Bangla medium and English Version); Cambridge or Edexcel for English medium	Cambridge / relevant	CBSE / relevant	Pakistan national / relevant	Local curriculum	French / Francophone

NexaTute's app is in all six languages from launch (Urdu and Arabic right-to-left): the chosen language changes the app text, the Nexa Teacher and Nexa Co-Teacher teaching language, and notifications, SMS, email and reports, and a guardian can choose a different language from the learner. Remon Motion Studio keeps its own interface schedule (Bangla and English at launch, Hindi

and Urdu in Year 2, Arabic and French in Year 3); until then its creators use the English or Bangla interface. Standing rule: every subject follows each learner's own country curriculum — in NexaTute, PlayBoom (30.9), MindForge (37.3) and every future feature; a country's curriculum opens only after its content passes the quality gate (3.13), following the NexaTute launch order. Full local curricula (CBSE, Pakistan national, Middle East and Francophone) are added with market entry in 29.4.

29.3 Cultural Adaptation

Aspect	Action
Islamic Content	Qualified scholarly review
Local Examples	Regional context
RTL Support	Arabic and Urdu right-to-left interface (Urdu in Nastaliq); formulas and code stay left-to-right
Fonts and text shaping	Noto fonts for every script; WeasyPrint PDFs render conjuncts and joined letters correctly (4.2)
Language quality	Answer-accuracy, speech-recognition and voice listening benchmarks gate each language; every course is also gated per language and offered only in the languages it has passed (3.13)
Date Format	Locale-aware formatting
Currency	Local display and clear pricing
Holidays	Regional calendars

29.4 Regional Expansion Plan

Order	Region	Timeline
1	Bangladesh	2027
2	India (Bengali-speaking)	2028
3	Pakistan	2028–29
4	Middle East	2029–30
5	Global	2030+

30. NexaTute PlayBoom — Play Class to Class III

PlayBoom is the NexaTute module that teaches children from Play class to Class III through games, an AI Play Buddy, handwriting practice and a parent dashboard. It replaces Kids Mode: the Kids Mode prices (৳400/৳700), its ৳20 Daily Pass and its USD prices are withdrawn, Kids Mode rooms become PlayBoom rooms (3.15) and the weekly guardian + child quiz runs in PlayBoom. Source:

"NexaTute Live Study Hall & PlayBoom" scope document (20 October 2026), reconciled with this master.

30.1 Learning experience

- **Play, Nursery and KG:** fully voice-first with pictures and animation — letter and number recognition, rhymes, phonics, colours and shapes; no reading is needed to use the app.
- **Class I–III: the child's own country curriculum (30.9) — in Bangladesh the NCTB textbooks chapter by chapter: Bangla, English and Mathematics, with Science, Bangladesh and Global Studies and religion in Class III.**
- Mini-games, puzzles, adventures, stories, word building, number practice, spelling and learning missions.
- AI Play Buddy with voice and an animated avatar: adaptive explanations, questions and missions; always says it is an AI and asks for clarification when unsure.
- Adaptive learning by age, class, skill level, performance and mistake pattern; skill mastery, diagnostic checks, mistake notebook, smart revision and a learning-goal tracker.
- Rewards for effort, consistency and personal improvement — not only speed; no money-bought rewards for children.

30.2 Handwriting Studio

- Tracing first, then letter writing: digital practice on the Smart Whiteboard, touchscreen or stylus.
- Paper practice: the guardian photographs or uploads the notebook page; handwriting recognition and answer feedback run on the existing Qwen3-VL deployment (photo questions, 3.10).
- Image quality check: blurry, cut-off, tilted or unreliable photos are not guessed — the app asks for a clearer photo.
- Answer correctness and handwriting readability/neatness are scored separately.
- 10 successful scans a day; failed or unclear scans do not count (to be verified in testing).

30.3 Price and daily limits

Item	Limit / price	Notes
Monthly	৳599 per child per 30 days ($9)	One price for every class; USD by the NexaTute rule (BDT × 0.015, rounded), paid through Paddle abroad
Siblings	30% off from the 2nd child	Same sibling rule as NexaTute; only one discount at a time (the larger)
Daily Pass	৳20 for 24 hours	Within the same daily limits
Institutions	25% off for 20–99 children (৳449); 35% off for 100+ (৳389)	Preschools, kindergartens and schools; dashboard and reports included
Annual	৳5,750 ($86)	20% off ৳7,188, rounded to a round figure
Play / Nursery / KG	25 minutes a day by default; the guardian can allow up to 3 sessions of 25 minutes	At least 2 hours between sessions; when the 2nd or 3rd session is turned on, a one-line note says that under 1 hour of daily screen time is recommended at this age — the guardian decides
Class I–III	45 minutes a day by default; up to 75 minutes	Guardian-controlled

Item	Limit / price	Notes
AI Voice + Video	30 minutes a day combined	Voice and video at the same time are not double-counted; other activities continue within the daily limit
Handwriting analysis	10 successful scans a day	Failed or unclear uploads do not count
Connection	Online only	Safe pause/resume and progress reconciliation after reconnect; no duplicate charge or progress loss; paper writing away from the app is not app time

30.4 Parent dashboard and support features

- Progress, goals, daily screen-time and session-length controls, subject controls and explainable progress reports (one integrated feature, not duplicates).
- Guardian verification and consent, separate profiles for each child, account recovery and data-deletion requests.
- AI confidence and correction, AI quality monitoring (Bangla/English speech, handwriting and curriculum benchmarks), incident and complaint workflow with human support.
- Low-bandwidth mode, inclusive modes (audio instructions, simplified UI, adjustable difficulty); WCAG 2.2 AA is the accessibility target — no compliance claim without an audit.

30.5 Child safety — NexaTute minors rules applied

Area	PlayBoom rule
Account	Account in the guardian's name with a profile per child; switching profiles needs the guardian PIN
Consent	Guardian OTP at sign-up; a separate on/off switch per feature (19.4); profiling switches stay locked off if the lawyer finds minors' profiling not allowed
Face verification	Off — no biometric templates of young children; guardian PIN and OTP instead
Purchases	Only the guardian buys, within the guardian's monthly limit (11.9)
AI	Runs only on RemonLogic servers — children's sessions are never sent to a cloud API; the AI says it is an AI, never encourages secrecy or excessive use
Voice, images, handwriting	The child's private data; microphone/camera indicator; the guardian can view and delete everything; retention and deletion as in 19.4
Other children	No chat between children; leaderboards by nickname only and only with guardian permission
Messages and marketing	Everything goes to the guardian; no direct messages or ads to children
AI training	Only with guardian consent and with personal data removed (Presidio)
Safety	Report/block, human escalation, child-safety hash matching and the 4.15 checks on every image

30.6 Architecture

- Same shared system as NexaTute, Motion Studio and NexaLogicDev: voice (VITS), avatar and animation run in the browser; the LLM runs on the shared Qwen3 pool; handwriting on Qwen3-VL. No new servers; capacity is confirmed in the Free Beta. The additions in 30.8 run mostly in the browser (offline packs with Workbox, read-aloud checks, music pads); Play Buddy voice and video stay within 30 minutes a day.
- Screens: welcome and guardian consent; child profiles and selector; child home; Math/Bangla/English areas; games, stories and missions; AI Play Buddy; digital Handwriting Studio; paper scan and quality check; handwriting feedback; mastery and diagnostics; mistake notebook and revision; goals and rewards; parent dashboard and screen-time controls; subscription, payment, renewal and receipt; recovery, privacy, consent withdrawal and deletion; voice/video permissions; support and incidents; admin AI-quality monitoring; loading, empty, error, reconnect, quota-exhausted and AI-uncertain states.

30.7 Launch plan and readiness checklist

- In-house testing → Free Beta 1–20 January 2027 (quotas, fair use and cost controls apply; guardian verification and age-appropriate safety for every age in the beta) → feedback and fixes → paid from 21 January 2027 only when the checklist passes. The beta never converts to paid automatically; a critical safety or privacy blocker postpones launch.
- Checklist: child-safety and guardian-consent flow; privacy, deletion and access-control tests; age-appropriate AI, escalation and human support; WCAG 2.2 AA audit; Math, Bangla, English and handwriting benchmarks; voice/video quality and scan accuracy; quota, billing, reconnect and duplicate-charge tests; load, backup and disaster-recovery tests; parent/child usability and complaint workflow; Bangladesh legal review (and each new market's review before entering it).

30.8 Approved additions

Price, daily limits and the rules of 30.5 stay as they are: no ads, no chat between children, no links and no purchases by children. Most additions run in the child's browser; Play Buddy voice and video stay inside the 30-minute daily limit, and no servers are added.

Area	What the child or guardian gets
Story time	Picture stories read by Play Buddy; the child picks what happens next and taps a word to hear it; a bedtime audio-only mode
Letter and sound journey	Bangla vowels and consonants and English phonics with mouth animation; gentle pronunciation help in the browser that never says "wrong"
Reading levels (Class I–III)	Letter → word → sentence → story, with read-aloud checks in the browser
Dictation and conjunct letters (Class II–III)	Dictation and যুক্তবর্ণ (conjunct letters) checked in the Handwriting Studio (30.2)
Rhymes and sing-along	Own or copyright-free rhymes only, animated with Motion Studio
Hands-on mathematics	Counters, number line, tangram and a "grocery shop" game with real Bangladeshi taka coins and notes
World around me	Animals, fruits, the six seasons, body parts, hand-washing, road safety and the festivals of all religions
Small home science	Float or sink, growing a seed and similar experiments done with the guardian; photos stay private

Area	What the child or guardian gets
“Why?” box	The child asks “why?” aloud and Play Buddy answers simply, within the 30-minute voice limit; the guardian sees the questions
Little thinking games	Matching, sorting, patterns, memory cards and mazes — not MindForge and with no measuring (37)
Arrow-path coding puzzles	From age 5, in the browser (Blockly)
Bangladesh stories	Copyright-free folk tales such as ঠাকুরমার ঝুলি (Thakurmar Jhuli); age-appropriate stories for 21 February, 26 March and 16 December; a map puzzle, rivers, the doel and the tiger
Music	Tabla, dhol and harmonium pads and a clap-along rhythm game (Web Audio in the browser)
Drawing and colouring	Saved only to the guardian’s private gallery
Feelings and manners	Short stories about feelings and manners; there is no emotion detection of the child
Safety stories	Water and drowning safety, fire and road safety, and “if I get lost”, including learning the guardian’s phone number; age-appropriate body safety appears only if the guardian turns it on
Healthy habits	A 2-minute tooth-brushing song, vegetables and a sleep routine; a bedtime lock set by the guardian; 2-minute movement breaks and eye rest; one bedtime lock for the whole family (8 pm, default, editable), a movement break after every 2 games and one flower in “my garden” for every 10 stars (decided 24 October 2026)
Rewards	A sticker book and “my garden” that grows with learning stars; Play Buddy dress-up unlocked by learning — no purchases, loot boxes or streak pressure
Family	Parents abroad or grandparents record stories and cheers the child hears (guardian-approved family only, private); “play at home today” screen-free family cards; siblings take turns on one phone in cooperative games — never online play with strangers; birthday greetings and festival themes
Guardian support	A “ready for Class I” checklist (pre-reading, pre-maths, fine motor) and a weekly tip; a monthly private “my work” album as a PDF
Access and inclusion	The Sensory-friendly mode for children with autism or speech delay — the same mode as for every NexaTute learner (3.31) — (soft colours, no sudden sounds, a picture routine; the guardian turns it on); Bangla sign-language videos for key words, captions and high contrast
Kindergartens	Classroom mode for kindergarten teachers on a TV or projector and printable worksheets; institution discounts as in 30.3
Offline packs	Lesson packs downloaded on Wi-Fi and played offline (Workbox service worker); progress syncs when the device is back online
Class III graduation	A “graduation” certificate into NexaTute Class 4, where MindForge starts (37)

30.9 Curricula and NCTB books

Standing rule: every subject follows each child’s own country curriculum. At setup the guardian picks the child’s curriculum — the app suggests one from the country and the guardian confirms it. Lessons are written by the AI in its own words; no textbook text or images are copied (3.13).

Path	Curriculum and books
Bangladesh — NCTB Bangla medium	Class I and II follow their full NCTB textbooks by name, chapter by chapter, so that parents can follow school: আমার বাংলা বই (Amar Bangla Boi), English for Today and গণিত (Mathematics). Class III adds প্রাথমিক বিজ্ঞান (Primary Science), বাংলাদেশ ও বিশ্বপরিচয় (Bangladesh and Global Studies) and religion — Islam, Hindu, Buddhist or Christian studies, chosen by the guardian; Class III also adds the religious-language book of the chosen religion, on by default and switchable off by the guardian — for example Islam → the Arabic book আদদুরুসুল আরাবিয়্যাহ (decided 24 October 2026)
Bangladesh — NCTB English Version	The same NCTB books in English (for example EV Mathematics), plus Bangla; Class III uses the NCTB EV books such as "EV Primary Science" and "EV Bangladesh and Global Studies" (names checked against NCTB's official list)
Bangladesh — English medium	Cambridge Primary or Pearson Edexcel iPrimary, or "my school's own book list", which the AI maps to lessons and a content reviewer checks
Other countries	Their own curriculum, in line with the six languages: India CBSE or state boards; Pakistan national or provincial curriculum; Arab countries their ministry curriculum; France Éducation nationale; Cambridge or IB for expatriate families. A country opens only after its content passes the quality gate, following the NexaTute launch order
Bangladeshi children abroad	The host country's curriculum for school subjects, plus a separate "learn Bangla" part to keep the mother tongue
Play, Nursery and KG	Skills-based (letters, numbers, colours), aligned with each country's pre-primary guidance; the teaching language is chosen by the guardian

- Country curriculum quality check (decided 24 October 2026, instead of a part-time teacher per country): the AI checks first against the official curriculum, facts, language and age fit; the platform's own verified teachers in that country act as Teacher Reviewers (institution teachers, free teacher-tool users and verified marketplace teachers), and every chapter needs two verified teachers' approval before it opens, rewarded with a free plan, a small per-chapter fee or a "Verified Reviewer" badge; each country opens first as a labelled Beta for volunteer families with consent and is fixed from their feedback before it opens to everyone; after launch learners and guardians give feedback on every lesson (for example "doesn't match my school book") into a fix queue. No child sees content that has not passed the check, and the book list is checked against NCTB's official list before any content is made (G.7).

31. Live Study Hall — Groups, Shared Lab, Moderator and Q&A

The Live Study Hall (3.20) stays free as a silent, timer-based study room. From this edition, paid users can also study in class-wise groups with a shared Laboratory and Whiteboard and call the Nexa Study Hall Moderator for short help.

Feature	Free	Paid
Silent Study Hall (25-minute study + 5-minute break, one question per round, XP, streak)	Yes	Yes
Class-wise groups of 2–10 (same class or batch)	—	Yes

Feature	Free	Paid
Shared Laboratory and Whiteboard in the group	—	Yes
Help button — Nexa Study Hall Moderator answers by text and voice	—	Yes
Study Hall Q&A — ask, answer and correct, with images (31.3)	Yes	Yes

31.1 Groups

- Learners of the same class or batch form a group of 2–10; minors need guardian approval (as for study buddies).
- Talking in the group: text chat only for minors; voice allowed for adults (as in batch pair rooms); minors may still ask the Nexa Study Hall Moderator by voice.
- The Laboratory and Whiteboard run in the browser and sync through Yjs — no GPU.

31.2 Nexa Study Hall Moderator

- **Always on, rule-based (CPU):** term-list checks on nicknames and avatars, the Report button, and XP/streak abuse detection (hidden tab, no answer to the round question, several accounts on one device earn no XP).
- **Needs-help detection (CPU signals):** repeated wrong steps in the lab, stuck on one problem for long, “Bujhini” taps, wrong round answers — the moderator then offers help (“Looks like you are stuck here — want help?”).
- **Help button (paid):** the moderator answers by text and voice. Answers come from the existing Nexa Co-Teacher LLM (Qwen3-30B-A3B); the voice is played in the learner’s browser (VITS) and spoken questions are understood with in-browser Whisper; a group question is answered once for the whole group.
- **Short help only:** hints and pointing out where the mistake is, about 2–3 minutes. Teaching a full topic is handed to the Nexa Co-Teacher (25 topics a day) or the Nexa Teacher (plan 1:1 minutes or wallet).
- **Reports:** reported names or text are checked with the existing Qwen3Guard; serious or unclear cases go to the NexaTute human controller; consequences follow the behaviour ladder; every action is audit-logged and can be appealed. The moderator never switches chat or voice on by itself.
- **Cost:** no new model, GPU or server — short help uses the shared LLM pool at the Nexa Co-Teacher priority.

Screens: hall home and directory; join or create private hall; study room and timer; break screen; question and result; XP and streak history; group room with Lab and Whiteboard; Help and moderator answer; moderator notice and report incident; incident status; guardian approval and safety settings; admin incident queue and human review; loading, reconnect, empty and error states.

31.3 Study Hall Q&A

Free for everyone. Anyone can ask any question in text with images, anyone can answer, and anyone can answer again or correct an answer in the comments under it. Boards are tagged by subject and course; answers collect “helpful” votes and the asker’s “this helped”. Before a question is posted, a matching verified answer is shown first. The Q&A boards are the community forum of 3.10.

Details (approved 24 October 2026 with the X4 prototype): 3 “Helpful” votes or the asker’s “This helped me” start the AI check, and an answer becomes “NexaTute verified” only when the AI check

also agrees; a Nexa Study Hall Moderator answer always stays “AI hint”, never verified; an answer verified into the cache earns paid users +10 bonus minutes and free users a Helper badge and 50 XP; a reported post stays visible until it is reviewed; learners aged 18 or over may post a visible face photo with a “share with care” note (under-18 faces are blurred); 9 or more digits count as a phone number and are masked, as are social links (wa.me, t.me, facebook.com); Q&A uses the group-chat behaviour ladder, with posting paused for 24 hours at step 3.

Step	What happens
1. Ask	The learner writes the question and may add images; a matching verified answer (pgvector similarity over BGE-M3 embeddings) is shown before posting
2. Answer and correct	Others answer or correct in the comments; if nobody answers within 30 minutes, the Nexa Study Hall Moderator posts a short answer or hint labelled “AI”
3. Check	Correctness is never decided by votes alone: community signals are combined with the existing AI verification against the verified knowledge base (3.13)
4. Verified	When both agree, the answer is marked “NexaTute verified”; a disagreement or a new topic goes to the human reviewer queue (38.5)
5. Wrong answers	Not deleted, but marked “not verified yet — see the corrections below”
6. Cache	Verified answers are cached in Valkey, so the next learner gets the answer at once and the Nexa Co-Teacher checks it first; a permanent copy is kept in PostgreSQL with pgvector for similar questions
7. Question bank	A verified question enters the question bank only after a content reviewer approves it — anonymous, tagged by syllabus topic and difficulty, with no verbatim guidebook text and no leaked exam questions. The Terms & Conditions say posts may be used anonymously in the question bank

Images yes, links never

- No link works for anyone: URLs are not clickable, and disguised links (for example “dot com” written out) and QR codes inside images are caught.
- Every image is checked by Nexa Guard before it is posted — the NSFW classifier and child-safety hash matching (4.15). Location data is stripped, phone numbers and QR codes are blurred, and the image text is read by OCR for search and duplicate detection.
- The browser resizes the image, finds QR codes (jsQR) and phone numbers (Tesseract.js) and blurs them before upload to Cloudflare R2; the server repeats every check (OpenCV QR detection, Tesseract, the NSFW classifier, PDQ hash matching), because safety results are always decided on the server (8.18).

Safety for minors, rewards and limits

- Nicknames only — no profile photo or real name; no private messages; phone numbers, email addresses and social-media IDs are masked; under-18 learners cannot post face photos (faces are blurred, MediaPipe face detection); the report button and the behaviour-change ladder apply (10.2).
- Rewards: an answer verified into the cache earns +10 bonus minutes for paid users and a Helper badge with XP for free users.
- Limits: 10 questions and 30 answers or comments a day; half of that for new accounts in their first 3 days.
- Architecture: no new servers — Valkey, PostgreSQL with pgvector, Cloudflare R2 for images and the AI checks at low priority (P10) on the shared pool; Qwen3Guard checks the text of every post.

32. NexaLogicDev — AI Cloud IDE, No-Code Builders and App Factory

NexaLogicDev (formerly NexaPrompt) is the "Build" product of the ecosystem: an AI-powered cloud IDE, visual builders and the Nexa Auto-Architect app factory, priced for Bangladesh and run on RemonLogic's own servers. Source: NexaLogicDev Master Blueprint v7.2 (21 October 2026) with all later decisions of 21 October 2026. Development starts in March 2027 (Month 6) and the product launches in September 2027; its infrastructure is Funding B (৳1.50 crore) and its core team is in Funding A.

32.1 AI architecture

Lane	Servers	Model	Setup	Work
Heavy	3 × (4× RTX A6000 48 GB)	MiniMax-M2.7 (≈229B total / ≈10B active), 4-bit AWQ; GLM-5.2 (≈744B / ≈40B, MIT) as fallback if the licence is refused	One copy per server (TP=4, or TP=2 × PP=2 inside the server — whichever is faster in the benchmark); GLM-5.2 fallback: TP=4 × PP=3 across the servers	Architecture, Auto-Architect plans, design decisions, final review
Fast	4 × RTX PRO 6000 96 GB (Database Mart Dallas) from launch; more GPUs one at a time at >70% busy	Qwen3-Coder-Next (≈80B total / ≈3B active, 256K context), FP8 or AWQ 4-bit	One copy per GPU	Everyday coding, Website Builder, writing files from the Heavy plan, small fixes

- Why MiniMax-M2.7: the model fits inside one server, so no model traffic crosses the network; three independent copies give ≈3–5× the Heavy-lane throughput of GLM-5.2, answers come ≈3–4× faster (≈10B instead of ≈40B active parameters), and one server failure leaves two-thirds of the lane running. Its published SWE-bench Pro score is 56.2% against 62.1% for GLM-5.2 (≈10% lower). Commercial use needs MiniMax's written authorization, requested now; GLM-5.2 on the same three servers (pipeline-parallel, ≈12 KB per token across the 1 Gbps LAN) is the fallback.
- The Heavy servers (A6000) have no FP8, so M2.7 runs as a 4-bit AWQ build that is checked against the official FP8 weights on our own tasks; MiniMax recommends temperature 1.0, which is tested in the write → run → fix loop. The Fast GPUs (RTX PRO 6000) run Qwen3-Coder-Next in FP8 if the KV cache leaves enough room for concurrent users, otherwise in 4-bit AWQ (≈45 GB) — decided in the benchmark. Model choice is the best open model now and can be updated before launch; the whole ecosystem can move to a newer model family together.
- Complex-reasoning pipeline: Router (simple → Fast) → clarifying questions and Wireframe approval → Architect (MiniMax-M2.7: schema, APIs, modules, file list, tests) → Builder (Fast lane writes files in parallel) → Verify loop (sandbox build, tests and security scan; small errors fixed by Fast, design errors or 2+ failures go back to M2.7) → final M2.7 review → delivery.
- System design closes the gap to frontier models: write → run → fix until build and tests pass; official docs (Laravel, React, WordPress, bKash) and RemonLogic templates indexed for retrieval; Project Memory; speculative decoding, prefix and semantic caches (tenant-, project- and permission-aware).
- Honesty rule: published SWE-bench Pro scores are 56.2% for MiniMax-M2.7, 62.1% for GLM-5.2 and 44.3% for Qwen3-Coder-Next against 89.9% for Claude Opus 5.5 (vendor and third-party figures, to be re-benchmarked on our servers). Claude Sonnet 5.5 (81.3%) is used only for plan review and rescue on Pro/Team plans and Auto-Architect (below), never presented as our

own model. Marketing says “the best AI app factory at Bangladesh prices”, never “equal to Opus”.

- The old Qwen2.5-Coder-32B / Qwen3-Coder-480B plan and its “8–15 concurrent Heavy jobs” target are withdrawn; the new target is set from benchmarks.

Lane access and API fallback

Plan	Lanes	External API	After two failed attempts
Free	Fast Lane only	None — waits in the queue with an ETA	Task stops; credits for the failed attempts are refunded
Student, Hobby	Fast Lane only (an Auto-Architect module they buy is built like a Pro task)	MiniMax M3 API when the Fast Lane and the idle Heavy servers are unavailable	Task stops; credits refunded; a Pro upgrade is offered to complete it
Pro, Team, Auto-Architect buyers	Fast Lane and Heavy Lane	MiniMax M3 API for the Fast Lane; Claude Sonnet 5.5 API for Heavy-lane plan review and rescue	Sonnet corrects the plan (case 2); on a third failure Sonnet completes the task (case 3)

Heavy-lane case (Pro, Team, Auto-Architect)	Trigger	What happens	API cost per task
Normal	Heavy Lane available	MiniMax-M2.7 on our servers writes the plan; the Fast Lane builds; sandbox tests	None (own GPU ≈৳0.67 per Heavy credit)
Case 1	Heavy Lane unavailable (busy beyond the admin limit, or down)	The Fast Lane writes the plan; Claude Sonnet 5.5 reviews and corrects only the plan (database schema, APIs, modules, file list); our servers do the rest	≈৳4–6
Case 2	Heavy Lane fails twice in a row	Sonnet reviews and corrects only the plan; our Heavy Lane does the rest	≈৳4–6
Case 3	Heavy Lane fails again with the corrected plan	Sonnet completes the task	≈৳60–170
Complex Module	Every Complex Module	Sonnet reviews the plan once before the build	≈৳4–6

- Fast Lane fallback: Fast-lane work first overflows to idle Heavy servers (M2.7 serves the request) and idle launch GPUs (8.17); only when these are also unavailable do paid users go to the MiniMax M3 API — the Fast Lane never uses Sonnet; free users wait.
- Routing: every step goes through the Portkey AI gateway with circuit breakers; the customer sees the same credit price whichever path runs the task, and only the ETA can change.
- Hand-back: when our servers free up (queue below the limit for 2 minutes), the next step of the task returns to them; a call in progress is never cut. A task that reached Sonnet in cases 2–3 finishes its Sonnet step on Sonnet, then the next task starts on our servers again.
- Spend caps: each API has its own monthly cap in the admin panel (starting values ৳20,000 for MiniMax M3 and ৳50,000 for Claude Sonnet 5.5, paid from NexaLogicDev revenue and changed from measured use); at the Sonnet cap, case-1 work waits for the Heavy Lane with an ETA.
- Privacy: before any API call, keys and secrets are removed from the code and prompt (Gitleaks), environment files and customer data samples are never sent, and only the files the task needs

leave our servers; both providers are used under API terms that do not train on the data; the Terms and Privacy Policy name them (36.2).
- "RemonLogic servers only": agencies, Team accounts and any customer can switch this on per project; the API paths are then off and the task waits for our servers.
- Quality loop: every Sonnet plan correction and completion is logged with the failing M2.7 attempt, so recurring failure types feed the templates, retrieval and prompts that keep our own lanes improving.

32.2 Core product

- **NexaBrain Chat:** the orchestrator on the right of the screen; routes work to the right module; chat in Bangla, English, Hindi, Urdu and Arabic, answering in the user's language; code identifiers always in professional English; voice-to-code in Bangla, Hindi, Urdu and Arabic (in-browser Whisper, 32.14).
- **Developer IDE:** Monaco editor, xterm.js terminal (hidden in beginner mode), AI Auto-Fix in the sandbox, GitHub sync (clone, push, pull requests), legacy converter (jQuery/Core PHP → React/Laravel), auto unit tests (Jest/PHPUnit), mobile app generator (React Native/Flutter), AI security audit before deploy, offline PWA editing. Browser-first (32.14): the editor, lint and format, previews (Sandpack for everyone, WebContainer API "Power mode" for Team plan users, agencies and Auto-Architect module buyers — for everyone if the StackBlitz price allows — and php-wasm for PHP and WordPress), the design canvas, voice and offline mode run in the user's browser; the RemonLogic sandbox runs only the AI write → run → fix loop, tests, security scans and database-backed backends.
- **Auto Website Builder:** drag-and-drop canvas in the browser (GrapesJS, 32.14), prompt-to-site (five pages with text and images), auto SEO, export to React/Next.js or a WordPress theme ZIP, one-click deploy.
- **WordPress Builder Mode** (later update): form-based plugin, theme and template generators, WordPress Playground testing.
- **Nexa Wireframe Generator:** wireframe first, drag-and-drop, preview and edit, approve → full build on the Heavy lane.
- **Nexa Auto-Architect (premium):** consultant chat, modular checklist (Accounting, HR, Inventory…), read-only build log, database + backend + frontend ZIP; Basic App ৳1,499, Medium App ৳4,999, Complex Module ৳9,999 per module (Complex in a later update); no full-ERP bundle.
- **Billing safety:** 75% (yellow) and 90% (red) credit alerts, hard lock at zero, graceful pause with durable checkpoints, auto-resume after recharge.
- **Enterprise and security:** Yjs collaboration, RBAC, 2FA and audit logs, anti-fraud, Bangla README and voice-over client handover, user owns generated code (licences in the Terms), isolated sandbox with quotas, network policy and secret isolation.

32.3 Mandatory before launch

- **Abuse prevention:** blocks phishing, fake bank/bKash/government login pages, scam shops and malware at prompt time; automatic content and link scan before deploy; report and takedown; incidents to the Security & Control Centre (AI detects and recommends, people decide).
- **Published credit table** with separate Fast and Heavy rates and a credit estimate before every task (32.9).
- **Free Tier:** Fast lane only, 3 credits a day and 1 new static site a day (32.10).
- **Version history and rollback:** a snapshot before every AI change; diff view; one-click rollback.
- **Project Memory:** each project's plan, decisions, file list and history kept per tenant and project.

32.4 Bangladesh market features

- Ready payment modules for bKash, Nagad, Rocket and SSLCommerz, added to any generated app in one click (signed webhooks, idempotent handling).
- Bangladeshi templates — e-commerce, coaching centre, school management, pharmacy POS, restaurant, real estate — with correct Bangla fonts and Unicode.
- Mushak 6.3 VAT invoice and BDT format in the Auto-Architect Accounting module (format checked in legal review).
- Bangladeshi SMS gateway integration for OTP and notifications; low-bandwidth IDE mode.

32.5 Ecosystem links

- Learn → Build: NexaTute coding learners continue in NexaLogicDev; Explain mode explains each step (on by default for Free and Student); Student Plan for verified NexaTute learners.
- Motion Studio: one-click promo videos and handover videos; credits deducted at the Motion Studio price, ৳5 = 1 credit (e.g. a 1-minute 720p promo at ৳45 = 9 credits).
- Nexa ID and one wallet; NexaCoins and NexaTute XP in one reward system; Reseller Program coverage (33).

32.6 Engagement

Feature	What it does	When
Facebook Page → Website	Builds a site from a business's Facebook Page through Meta's official API with the owner's consent, with a WhatsApp order button	Launch
Site analytics and lead alerts	Visitors and form leads on a dashboard; new leads by WhatsApp or SMS (Cloudflare analytics)	Launch
"Made with NexaLogicDev" badge	On free sites, with a referral link that earns the owner bonus credits; paid plans can remove it	Launch
Daily Build Challenge and streaks	A small daily task; a 7-day streak earns bonus credits	Launch
Unfinished-project reminders	"Your site is 80% ready" — resumes from the checkpoint	Launch
Remix Gallery	Remix public sites/templates; the creator earns NexaCoins per remix	Later update
Freelancer Toolkit	Client proposals and quotes, Fiverr/Upwork portfolio export, handover documents	Later update
Festival templates	Eid sale, Pohela Boishakh, Victory Day, Black Friday — pushed from the season calendar	Later update
Weekly AI site review	SEO, speed and accessibility suggestions each week; leads to the Care Plan	Later update
Two-sided referral	Both people get bonus credits, alongside the 10% lifetime referral commission	Later update
Certified Builder	Verified certificate for NexaTute course + real NexaLogicDev project; Job Board priority; "Add to LinkedIn Profile" button (0.7)	Later update

Feature	What it does	When
Weekly Bangla Live Build Session	Building a real Bangladeshi business site live on LiveKit; recordings become tutorials	After the Developer Advocate hire

32.7 Free sites

- All free sites are served from one wildcard Cloudflare Worker + R2 at username-site.nexalogicdev.app (Cloudflare Pages' per-account project limit is avoided; near-zero cost).
- 1 new site a day, at most 5 live at once (older sites stay in the project, offline); a site with no visitors for 90 days goes offline automatically.
- Stock or template images on free sites (optionally one free AI image a day).

32.8 Later updates (approved)

- AI Shop Assistant: a Bangla chatbot on the user's site that answers from the site's content and sends orders to WhatsApp — ৳299–699 a month with a monthly conversation cap (uses Fast-lane capacity).
- F-commerce Order Manager: order dashboard, one-click Pathao/Steadfast/RedX booking and tracking, cash-on-delivery reconciliation.
- Expert Help: hire a verified Certified Builder from the Job Board when the AI cannot finish; 15% commission.
- Play Store publishing: PWA/Android (TWA) build and publishing help, ≈৳999 one-time.
- University and polytechnic lab licence (classroom mode, teacher dashboard, assignments, linked to NexaTute) and SME-digitalisation bulk licences (SME Foundation, a2i, NGOs) through the Partnership Manager.
- Local SEO and Google listing (Search Console, sitemap, Bangla keywords); one-click translation Bangla ↔ English ↔ Arabic; international domain sales (32.10).

32.9 Credits, plans and prices

1 credit = 20,000 Fast-lane tokens or 1,000 Heavy-lane tokens (input tokens count at one-fifth). A small edit costs about 0.1–0.3 credit; a full architecture plan about 15–30 credits. Sandbox build/test runs are free within fair use. Each task shows its estimate first; the charge never exceeds the estimate by more than 20%, and credits are refunded if the AI cannot complete the task. Tokens per credit are a config value re-tuned after benchmarks without changing prices.

Plan	Price / month	Credits / month	Conditions
Free	৳0	3 a day	Fast lane only; 1 new static site a day (5 live)
Student	৳299	250	Verified NexaTute learner; Fast lane only (MiniMax M3 API fallback); Explain mode on
Hobby	৳499	400	Fast lane only (MiniMax M3 API fallback)
Pro	৳2,499	1,500	Fast and Heavy lanes within credits; Sonnet plan review and rescue (32.1)
Team	৳9,999	6,000	5 seats, pooled credits, RBAC; Fast and Heavy lanes; Sonnet plan review and rescue

Top-up (valid 180 days)	৳199	৳499	৳999	৳2,499	৳4,999
Credits	35	100	210	550	1,150
Per credit	৳5.69	৳4.99	৳4.76	৳4.54	৳4.35

Other revenue	Price
Pay-Per-Deploy (username.nexalogicdev.app or one custom domain)	৳199 a month
Go Live Bundles (domain + static hosting + SSL + email forwarding + Google submission)	Starter (.xyz) ৳499 first year, ৳3,199 renewal; Business (.com, recommended) ৳2,199 first year, ৳3,199 renewal; Shop (.shop) ৳799 first year, ৳8,999 renewal — renewal shown prominently at purchase; .store, .online and .site not offered (very high renewals)
Care Plan	Static ৳299; WordPress ৳499; WooCommerce/dynamic ৳999 a month (uptime, updates, monthly backup)
Managed app hosting (later)	৳799 per app a month, when ≥100 paying users ask for two months
Marketplace commission	30%; 20% for Verified Creators (20+ sales, rating ≥ 4.5)
Referral	10% of the referred user's payments for life, plus two-sided bonus credits
White-label	Licence for agencies under their own brand
AI Shop Assistant / Play Store publishing / Expert Help	৳299–699 a month / ≈৳999 one-time / 15% commission

Unit check (22 October 2026 hardware: Fast lane 4 × RTX PRO 6000 at US$648–749, Heavy lane MiniMax-M2.7 on 3 servers at US$1,599): GPU cost per credit ≈৳0.73–0.84 (Fast) and ≈৳0.67 (Heavy) at planned capacity, down from ≈৳1.12 and ≈৳2.02 with the earlier A6000/GLM-5.2 plan; top-ups leave ৳3.57–4.67 per credit after VAT, gateway fee and reseller discount (≈76–86% margin), so credit prices stay. API fallback costs: MiniMax M3 ≈৳0.18 per credit (Fast-lane fallback); Claude Sonnet 5.5 plan review ≈৳4–6 per task and completion after a third failure ≈৳60–170 per task, only on Pro/Team plans and Auto-Architect modules (a Complex Module nets ≈৳8,214). Auto-Architect GPU cost ≈৳70 / ৳211 / ৳423 against ৳1,231 / ৳4,107 / ৳8,214 net. Go Live bundles use Name.com list prices (21 October 2026: .com $12.99 / renewal $19.99; .xyz $1.99 / $20.99; .shop $2.99 / $59.99) at ৳122 = US$1; reseller-account prices are expected to be lower. Storage by plan (Free 1 GB, Student 3 GB, Hobby 5 GB, Pro 25 GB, Team 100 GB pooled) and the storage add-ons are in G.3.

International (USD) prices (BDT + 35% at ৳122 = US$1, whole dollars; billed by the USA LLC through Paddle):

Item	BDT	USD
Student / Hobby / Pro / Team (per month)	৳299 / ৳499 / ৳2,499 / ৳9,999	$3 / $6 / $28 / $111
Top-ups	৳199 / ৳499 / ৳999 / ৳2,499 / ৳4,999	$2 / $6 / $11 / $28 / $55
Auto-Architect Basic / Medium / Complex	৳1,499 / ৳4,999 / ৳9,999	$17 / $55 / $111

Item	BDT	USD
Pay-Per-Deploy / Care Plan	৳199 / ৳299–999	$2 / $3–11
Go Live first year: Starter / Shop / Business	৳499 / ৳799 / ৳2,199	$6 / $9 / $24
Go Live renewal: .xyz and .com / .shop	৳3,199 / ৳8,999	$35 / $100

32.10 Hosting, domains and Go Live

- Now: Next.js/Node apps deploy in one click to the user's own Vercel or Netlify account; Laravel, PHP and WordPress projects export as a ZIP with a Bangla guide for cPanel hosting (passwords never stored); static sites on Cloudflare with automatic SSL.
- Custom domains through Cloudflare for SaaS with automatic SSL issue and renewal.
- Later: RemonLogic managed hosting on separate hosting servers (≈60 apps per ADVANCE-2, ≈ ৳403 cost per app) once ≥100 paying users ask for two months in a row.
- Domains: international TLDs only, fully automated through a reseller API — Name.com first (no deposit), Openprovider when volume grows; no .com.bd/.bd (the registry needs a trade licence and TIN and manual steps). Domains are registered in the customer's name and can be transferred out.
- Go Live: buy → DNS records and www redirect → SSL → site connected (static on Cloudflare; dynamic to Vercel/Netlify via API) → free email forwarding (Cloudflare Email Routing) with SPF, DKIM and DMARC → Search Console and sitemap — live in 1–2 minutes.
- Governance: the AI only proposes DNS changes; a tested deterministic workflow applies them after the user previews and approves, with an audit log and one-click rollback (Security & Control Centre rule, 34).

32.11 Marketplace, hackathons, onboarding, support and KPIs

- **Marketplace:** templates ৳99–4,999; seller KYC through the reseller KYC flow; monthly payout (minimum ৳500) to bKash, Nagad or bank after a 14-day hold; refund within 7 days if a template does not build or is not as described; code fingerprinting, licence scan and build test; a person reviews each new seller's first three items.
- **Hackathons:** weekly Prompt Battle with credit prizes (500 / 300 / 150); monthly hackathon with cash ৳10,000 / ৳5,000 / ৳3,000; free entry, skill-judged on a published rubric (working build 40%, problem solved 30%, code quality 20%, presentation 10%); cash prizes for 18+ or with guardian consent; withholding tax confirmed with a tax adviser.
- **Onboarding:** five-step Bangla tutorial — a first landing page in 10 minutes — with 10 bonus credits.
- **Support:** AI first, then a person. Free: community and 72 hours; Student/Hobby: 48 hours; Pro: 24 hours; Team: 8 business hours plus WhatsApp; payment problems 24 hours on any plan; site down or double charge 4 hours.
- **KPIs:** build success after the verify loop ≥90%; free → paid ≥4%; monthly churn ≤8%; refunds ≤3%; abuse takedown <24 hours; NPS ≥40; paid users 100 / 250 / 500 at Funding-B months 9 / 12 / 15; Sonnet cases 1–3 below 5% of Heavy tasks and API spend inside both caps (32.1).

32.12 Team, timeline and first launch scope

- Funding A pays one AI/ML Engineer (Heavy and Fast model serving) and one Full-stack Engineer from Month 6 (March 2027) plus a ৳3 lakh launch campaign; the shared team (CTO, Backend, DevOps, UX) builds billing, wallet and sandbox platform pieces.

- Revenue-gated hires: 2nd Full-stack at MRR ≥ ৳2 lakh; dedicated Support at ≥150 paid users (shared support until then); Developer Advocate at MRR ≥ ৳4 lakh; performance ads at 20% of NexaLogicDev revenue. Pre-launch QA contract (Months 9–12) from Funding B.
- Launch scope (September 2027): IDE, Website Builder, Wireframe Generator, Auto-Architect Basic, the five mandatory items (32.3) and five engagement features (32.6). WordPress Mode and the Complex Module follow in a later update. Risk: with two dedicated engineers the date depends on the shared team's time after the NexaTute launch.
- Development phases: (1) shared Kubernetes, PostgreSQL and Valkey namespace and sandbox pool; (2) user/credit/expiry schema and OAuth; (3) vLLM Fast (Qwen3-Coder-Next) and Heavy (MiniMax-M2.7; GLM-5.2 fallback); (4) Next.js IDE, NexaBrain Chat, Website Builder; (5) Wireframe Generator and Auto-Architect; (6) payments, billing middleware, alerts, pause/resume and public beta; (7) abuse prevention, credit table, Free/Student plans, version history, Project Memory, Bangladesh modules and ecosystem integration.

32.13 Funding B and unit economics

Item	Value
Funding B	৳1.50 crore for 18 months from March 2027 (separate from the ৳5 crore)
GPU rate	Heavy servers US$1,599 per server per month on every term; Fast RTX PRO 6000 US$749 per GPU on the 3-month term, then US$648 on the 24-month term once the benchmarks pass
GPU phasing	Months 1–3: no GPU servers (AI tested on the NexaTute pool off-peak); Months 4–18: 3 Heavy servers (45 server-months); Months 7–18: 4 Fast RTX PRO 6000 GPUs (48 GPU-months); more Fast GPUs one at a time at >70% load for a week
18-month cost	≈৳1.44 crore (GPU ≈৳1.27 crore, sandbox incl. SBX-3 ≈৳9.4 lakh, cloud/domains/email ≈৳3.6 lakh, hackathon ≈৳2.2 lakh, QA ≈৳2 lakh); contingency ≈ ৳5.6 lakh
Run rate after launch	≈৳10.2 lakh a month after the 3-month term (3 Heavy servers, 4 Fast GPUs, 2 sandbox servers); ≈৳10.5 lakh with SBX-3
Break-even	≈570 paid users (≈585 with SBX-3) with the mix 75 Student, 200 Hobby, 175 Pro, 30 Team plus top-ups, Auto-Architect, deploy and Care Plan (net ≈৳1,800 per user a month after VAT, gateway fee and reseller discount); the earlier ≈500 assumed a single A6000 Fast server at launch
Capacity	Fast Lane (4 × RTX PRO 6000, Qwen3-Coder-Next 4-bit): ≈250–500 tasks at once, ≈1,100–1,900 users in the IDE. Heavy Lane (three M2.7 copies): ≈30–60 Heavy tasks at once, ≈90–225 Pro/Team users in the IDE — now the first limit. Sandbox pool (SBX-1–SBX-3): ≈150–180 running sandboxes, used only for AI loops, tests, scans and database backends because previews run in the browser (32.14), so it serves roughly 1.5–2× more IDE users than before (≈700–1,000). Paid users served: ≈750–1,800 if 28% are online at peak (conservative), ≈2,100–4,500 at a typical 10%. 5,000 paid users need ≈4–6 sandbox servers, 4–6 Heavy servers and 4–6 Fast GPUs (+≈US$1,800–6,600 a month against ≈৳90 lakh net revenue a month), added step by step at the 60/70% triggers. Planning estimates to benchmark.
SBX-3 (third sandbox server)	≈US$198 a month + ≈US$198 installation from its trigger (≈March–June 2028): ≈ ৳1.0–1.7 lakh inside Funding B's window, included in the 18-month cost above
External API fallback	MiniMax M3 API (Fast-lane fallback) and Claude Sonnet 5.5 API (Heavy-lane plan review and rescue, 32.1) are paid from NexaLogicDev revenue within monthly caps (starting ৳20,000 and ৳50,000), not from Funding B

Item	Value
Resource sharing	The Heavy servers and Fast GPUs lend idle GPU (and Heavy-server CPU) capacity to NexaTute and Motion Studio, and paid Fast-lane overflow may use idle NexaTute GPUs, owner first (8.17); borrowed hours are shown at cost per server-hour in the monthly report
Revenue	≈৳75 lakh net over Funding B's 18 months if paid users reach 500 by Month 15; ≈৳15 lakh falls inside Funding A's window

32.14 Browser-first IDE

Everything a user needs while writing and previewing code runs in the user's web browser; the servers keep only the work that must run on our side. This keeps the IDE fast on a normal internet connection, keeps code on the user's device while it is edited and previewed, and leaves the sandbox servers free for the AI.

Feature	Runs in	Software	Server role
Code editor and terminal	Browser	Monaco Editor, xterm.js	None
Lint and format	Browser (Web Workers)	ESLint, Prettier with plugin-php, Stylelint; Monaco's built-in TypeScript/JavaScript workers	None
Frontend and Node.js preview	Browser	Sandpack (everyone); WebContainer API "Power mode" (Team, agencies, Auto-Architect buyers; everyone if the StackBlitz price allows)	None
PHP, Laravel (without a database) and WordPress preview	Browser	php-wasm; WordPress Playground	None
Wireframe, website and design editor	Browser	GrapesJS canvas; dnd-kit drag and drop	Receives only the approved wireframe for the Heavy Lane
Voice to code (Bangla, Hindi, Urdu, Arabic, English)	Browser first	In-browser Whisper with the same Bangla/Urdu fine-tunes as NexaTute (WebGPU/WASM); Web Speech API only as a weak-device fallback	Server speech-to-text (speech pool) as the last fallback
Offline mode	Browser	PWA with a Workbox service worker; project files in IndexedDB/OPFS; Yjs + y-indexeddb for conflict-free sync; Local Storage only for small settings	Syncs when back online; AI tasks wait in a queue
Git (clone, commit, branch)	Browser	isomorphic-git	Push and pull through the GitHub API with the user's token
Offline autocomplete (optional)	Browser, capable devices	WebLLM with a small Apache-2.0 coding model	None

Feature	Runs in	Software	Server role
AI write → run → fix loop, tests, security scans, database-backed backends	Server sandbox (SBX)	gVisor/Kata, Jest/PHPUnit, Semgrep, OSV-Scanner; MySQL/PostgreSQL inside the sandbox	Yes — only these run on our servers

- Fallbacks: features are detected per browser; if WebContainer, WebGPU or php-wasm is not available (older or low-end devices, some Safari versions), the preview falls back to Sandpack or the server sandbox within fair use, and voice falls back as in the table; the low-bandwidth IDE mode (32.4) uses the same rules.
- Security: previews run in a sandboxed, cross-origin-isolated frame; no server secrets ever reach the browser; deploys still pass the server-side security scan (32.3).
- Privacy: in-browser Whisper keeps the voice on the device; when the Web Speech API fallback is used, the user is told that the browser vendor processes the audio.
- Capacity: because previews run in the browser, the sandbox servers serve roughly 1.5–2× more IDE users (32.13); the AI lanes are unchanged.

33. RemonLogic Ecosystem Reseller Program

Applies to NexaTute (including PlayBoom), Remon Motion Studio and NexaLogicDev (named "NexaPrompt" in the 20 October 2026 policy draft). Policy draft v1.0; law, tax, payment-gateway and final business terms are checked before publication.

33.1 Packs, discounts and margin

Pack	Price	Usable wallet	Validity	First-sale discount	Renewal discount (proposed)
Starter	৳10,000	৳10,000	180 days	15%	12%
Extreme	৳30,000	৳30,000	360 days	22%	17%

- Example: a ৳1,000 customer price costs a Starter reseller ৳850 (margin ৳150) and an Extreme reseller ৳780 (margin ৳220), before taxes, gateway fees, refunds or chargebacks.
- The discount applies to every product and every paid item: NexaTute plans, add-ons and passes; PlayBoom; Motion Studio package fees, per-minute rates and the Express surcharge; NexaLogicDev plans, top-ups, Auto-Architect and other paid items.

33.2 Operating rules

- **Registration and KYC:** correct name, email, mobile and business details; both email OTP and mobile OTP before activation; risk-based extra checks; fake identities and multiple fraudulent accounts are banned.
- **Pack purchase and wallet:** a pack activates only after a verified payment; a unique transaction/idempotency key prevents double credit or debit; every credit and debit goes to the ledger; what happens to unused balance at expiry is written in the published terms.
- **Customer sales:** the reseller states the correct product, plan, price, validity, limits and refund terms; order, reseller ID, product, price, discount, wallet debit and result are recorded together; nothing activates without a verified payment or approved sale; a failed sale reverses the wallet debit and retries never create a duplicate sale.

- **Refunds and chargebacks:** request with order ID, reason and evidence; the system checks eligibility; eligible refunds may be automatic, exceptions and high-value cases go to an admin; order, wallet ledger and margin are adjusted together; chargebacks put the transaction under review with evidence kept.
- **Suspension and reinstatement:** for fraud, fake KYC, abnormal sales, payment fraud, mistreating customers or policy breaches; limits on sales or new orders by risk; the reason and appeal route are given unless law forbids; the decision is recorded.
- **Dashboards:** resellers see pack status, wallet, validity, sales, failed orders, refunds and history; admins see KYC, payments, sales, ledger, refunds, chargebacks, risk signals, suspensions and the audit log under RBAC.
- **Security and fraud:** no plaintext OTPs, passwords, tokens or keys in logs; MFA for admins; rate limits, device/session risk checks and duplicate-request protection; audit trail with actor, time, reason and correlation ID for every money change; tested backups and access revocation.
- **Privacy, disputes and records:** collect only what is needed; deletion/export requests balanced with financial-record duty; local law checked before UAE or other markets; ticket numbers and a support → finance/reseller operations → admin review path; entities Reseller, KYCVerification, ResellerPackPurchase, WalletAccount, WalletLedgerEntry, Customer, Order, PaymentAttempt, Refund, Chargeback, ResellerAttribution, SuspensionCase, Appeal and AuditEvent, with database transactions, unique constraints, signed webhooks and state machines.

33.3 Decisions still needed before launch

Settlement and minimum withdrawal of margin; unused balance at pack expiry; renewal attribution when a customer or reseller changes; refund time limits and partial refunds; pack upgrade/downgrade; tax, VAT, gateway fee, withholding and currency conversion; whether the platform or the reseller collects customer payments (and its compliance model).

34. RemonLogic Security & Control Centre

Consolidated control baseline v1.6 (21 October 2026) for the whole ecosystem: an AI-assisted infrastructure-reliability and cybersecurity operations platform. Governance rule: AI observes, analyses, correlates, recommends and notifies; humans investigate, authorise and remediate; independent monitoring and alerting keep working when AI is unavailable. This is a design baseline, not proof that controls are implemented, tested or certified.

34.1 Scope and principles

- In scope: asset and service inventory and health; infrastructure, network, Kubernetes, application, database, GPU, storage, recording/media and external dependencies; identity, privileged access, credentials, vulnerabilities, attack surface, API abuse, supply chain, DNS/TLS; data integrity, transaction reconciliation, privacy, tenant isolation, audit evidence, incident response and forensics; backup, ransomware recovery, high availability, failure domains, continuity and crisis communication; AI safety, model quality, untrusted-content defence and cost controls; control ownership, risk register, compliance mapping, test evidence and release gates.
- Out of scope unless separately authorised: autonomous remediation; claims of 100% detection, guaranteed availability or certification without evidence; unapproved collection of customer content; inventing server counts before the asset register is reconciled.
- Principles: AI/human boundary; independent monitoring proven by dependency analysis; evidence-first findings; least privilege with short-lived credentials; safe degraded mode (Unknown/Degraded/Monitoring Gap is never shown as healthy); no single point of truth in AI; tenant and data protection; change traceability.

- Ecosystem inventory for D01: the 24–28 servers of 0.6 (launch fleet, App-3, recovery replica, Heavy, Fast and sandbox pools), Cloudflare, R2, payment gateways, Paddle, Name.com, the MiniMax and Anthropic API accounts and the LLC's accounts.

34.2 Control domains

ID	Domain	Minimum coverage
D01	Asset inventory, service health and telemetry coverage	Canonical inventory, owners, criticality, health, collector status, freshness, coverage gaps, unknown-state handling
D02	Monitoring integrity, independent watchdog and alert delivery	Tamper detection, watchdog, alert-rule integrity, end-to-end delivery acknowledgement, fallback communications
D03	Capacity, performance and cost governance	CPU/RAM/disk/GPU/VRAM, queues, DB connections, recording buffers, latency, forecasts, telemetry and AI cost
D04	Identity, privileged access and credential lifecycle	Admin anomalies, MFA, privilege escalation, service accounts, key rotation/expiry/revocation, emergency access
D05	Attack surface, vulnerability and API abuse protection	Exposed services, patch deadlines, DDoS/exhaustion signals, rate-limit tests, approved response boundaries
D06	Software supply chain and artifact integrity	SBOM, scanning, build provenance, signing of software and model artifacts, release holds
D07	Domain, DNS, TLS and external dependency security	Domain expiry, registrar access, DNS changes, certificates, provider outages, quotas and fallback tests
D08	AI safety, prompt-injection defence and model assurance	Untrusted-content isolation, approved data access, versioning, evaluation, drift, AI-off fallback, read-only boundary
D09	Data integrity, transaction consistency and reconciliation	Replication lag, duplicate/incomplete transactions, wallet/billing/subscription reconciliation, idempotency, ledger consistency
D10	Privacy, tenant isolation, retention and deletion	Minimisation, tenant boundaries, redaction, access logs, retention, deletion, backup lifecycle, legal hold
D11	Protected audit evidence, trusted time and forensics	Tamper-evident logs, trusted timestamps, clock drift, evidence access, chain of custody
D12	Dependency mapping and common-mode failure protection	Service graph, shared failure domains, blast radius, independent path validation
D13	Backup integrity, ransomware recovery and clean-room restore	Protected backups, separate credentials, restore verification, clean-room recovery, RTO/RPO
D14	High availability, network partition and split-brain protection	Quorum, fencing, leader election, failover, conflicting-write prevention, recovery validation
D15	Incident response, emergency communications and business continuity	Severity, ownership, escalation, crisis communications, manual procedures, drills
D16	Risk register, compliance mapping and control effectiveness	Owner, severity, mitigation, residual risk, NIST CSF 2.0 mapping, exceptions, management review
D17	Human approval, change management and production readiness	Maker-checker, preview, approval, rollback, emergency change review, release gates

ID	Domain	Minimum coverage
D18	Security simulation, resilience testing and continuous improvement	Alert-delivery, monitoring-shutdown and credential-compromise exercises, load/failover/restore tests, retests

34.3 AI permission boundary and severity

Unless a separately approved automation design says otherwise, AI cannot directly: run shell commands or scripts in production; change firewall, WAF, DNS, network, identity, Kubernetes, application or monitoring configuration; block accounts, revoke credentials, isolate hosts or quarantine files; modify databases, customer records, wallets, billing, subscriptions or ledgers; restart, deploy, scale, fail over, restore or delete evidence; suppress critical alerts or change alert rules. AI recommendations are labelled and include evidence, uncertainty, impact, verification steps and the owner. Customer-domain DNS for NexaLogicDev Go Live is applied only by an approved deterministic workflow after the customer's approval (32.10).

Severity	Indicative criteria	Minimum response
SEV-1 Critical	Material compromise, critical outage, loss of control-plane integrity, serious data/financial integrity issue, broad recovery event	Immediate on-call, incident commander, evidence preserved, approved plan invoked
SEV-2 High	Significant degradation, credible compromise indicators, important dependency failure, likely SLO breach	Urgent owner notification, time-bound investigation
SEV-3 Moderate	Limited impact, contained or recurring issue, incomplete suspicious evidence	Assigned owner and response target; trend and correlate
SEV-4 Low	Low-impact, expected change, advisory or informational	Record, deduplicate, review trends

34.4 Control records, go-live gates and metrics

- Every control has one record: ID and domain, objective and risk, scope and assets, accountable owner, data sources and freshness, detection logic and escalation, test procedure and evidence, status (Planned, Implemented, Passed, Failed, Blocked, Not Tested, Not Applicable, Degraded/Unknown), exceptions and remediation.
- Go-live gates for every paid launch (Motion Studio, NexaTute, PlayBoom, NexaLogicDev): G01 inventory and dependency graph approved; G02 critical telemetry verified; G03 watchdog and fallback alerts tested; G04 privileged access, secrets, tenant isolation and AI read-only boundary tested; G05 restore and clean-room recovery meet RTO/RPO; G06 load, failover and partition tests; G07 incident response exercised; G08 supply-chain verification and rollback tested; G09 critical risks fixed or formally accepted; G10 named owner approves release. A failed critical gate blocks release.
- Metrics: asset coverage, telemetry freshness, alert delivery success, detection and response latency, control effectiveness counts, recovery readiness, security posture, AI quality and cost — segmented by criticality, never one composite score.
- Legacy requirements 1–51 (v1.1–v1.5) are mapped to D01–D18 with traceability; duplicates are retired only after mapping and verification.
- Open decisions: canonical inventory, service criticality and RTO/RPO, data policy, on-call and alert targets, thresholds, compliance applicability, exception governance. Phases: 0 Reconcile →

1 Visibility & guardrails → 2 Detection & integrity → 3 Recovery & continuity → 4 Assurance & go-live. Owners (service, security, operations/SRE) are still to be assigned.

34.5 Duplicate-control consolidation rules

- Each requirement has one canonical Control ID and one authoritative record.
- One control may satisfy several risks, but its separate test cases and evidence obligations stay explicit.
- Overlapping legacy sections map to the canonical control instead of being copied as additional active controls.
- Controls are not merged just because their names sound similar when their failure modes or acceptance tests differ.
- A legacy-to-canonical traceability table is kept during migration; duplicates are retired only after every requirement is mapped and verified.
- Shared ownership never means shared credentials or unrestricted access — least privilege applies to every integration.

34.6 Metrics and formulas

Metric	Formula / measure
Asset coverage	Critical assets with valid telemetry ÷ inventoried critical assets
Telemetry freshness	Critical signals within their freshness threshold ÷ all critical signals
Alert delivery success	End-to-end delivery acknowledgements ÷ alerts sent (test and production)
Detection latency	Detection time − event time, with source timestamps and known clock uncertainty
Response latency	First owner action − detection time (measured separately from detection)
Control effectiveness	Counts of Passed, Failed, Not Tested, Blocked and Not Applicable, with evidence links
Recovery readiness	Restore tests passed ÷ restore tests run; achieved RTO/RPO vs target; open recovery blockers
Security posture	Overdue critical vulnerabilities, privileged-access exceptions, unreviewed changes, unresolved high risks
AI quality	Evaluation results, false positives/negatives where measurable, calibration, fallback activations
Cost posture	Telemetry and AI-analysis spend vs budget, without silently weakening required coverage

Every metric is segmented by criticality and service with its denominator and measurement window shown; no single composite score is presented as proof that the environment is secure.

34.7 Minimum control test record (template)

Field	Content
Control ID	Dxx-xxx
Control objective	What risk or failure mode is being controlled

Field	Content
Scope / asset(s)	Named service, environment and criticality
Preconditions	Approved test environment, permissions, safety controls
Test steps	Reproducible procedure
Expected result	Observable pass/fail criteria
Actual result	Measured outcome
Evidence	Timestamped logs, screenshots, test output or immutable evidence reference
Status	Passed / Failed / Not Tested / Blocked / Not Applicable
Owner / tester	Named accountable owner and test executor
Remediation / retest	Action, due date and retest evidence

34.8 Review and approval

Role	Name	Approval / date
Accountable service owner	Md Samsuzzaman	Pending
Security owner	Mst Sonia Sabuz	Pending
Operations / SRE owner	Md Lablu Mia	Pending

The baseline is ready for implementation planning once these owners approve the open decisions (34.4). It is not production-ready until the required controls are implemented, their tests pass with evidence and gates G01–G10 are approved for each paid launch.

35. Remon Motion Studio — Rights, Safety and International Pricing

35.1 Pricing changes

- Delivery-speed tiers replace the time-of-day rates: Express +20% (always offered, high priority, calculated ETA, surcharge refunded if more than 20% late), Normal within 12 hours (“usually much sooner”), Slow within 24 hours at 10% off (G.8.3).
- International prices in USD = BDT price + 35% at ৳122 = US$1 (G.8.4).
- Reseller discount applies to the package fee, per-minute rates and the Express surcharge (33).
- Videos ordered from NexaLogicDev deduct NexaLogicDev credits at the Motion Studio price, ৳5 = 1 credit.
- B2B and Enterprise: “Contact Sales Team”.
- Cinematic (+35%) and 1080p Native (৳120/৳162) rates stay at the present prices. Cost check: the video optimizations already in this blueprint (8.15/8.16 — Lightning 4–8-step LoRA, LightX2V, FastWan for non-hero scenes, TeaCache, SageAttention, FP8, torch.compile/CUDA graphs, RIFE, A14B only for hero shots, AI-Director routing) were counted at only half their expected gain: ≈11 GPU-minutes per output minute for Cinematic 720p Native, ≈৳16 at 70% GPU use — gross margin after VAT and gateway fees ≈65–77% across all Cinematic tiers and ≈65% for

1080p Native. These optimizations must be live before paid packages start; the Free Beta benchmark confirms the figures and rates remain admin-editable. If Lightning LoRA, LightX2V and FastWan have not passed before paid packages start (date to confirm), Cinematic 720p Native and higher tiers stay “coming soon” while Cinematic 480p Native and 720p Escalating stay on.

35.2 Voice-cloning consent and rights policy (approved)

- Who: one’s own voice, or another person’s voice with that person’s direct consent.
- Consent: the voice owner reads a freshly generated consent sentence (date, account name, “I consent”) live in the app — pre-recorded audio is not accepted; speaker verification (SpeechBrain ECAPA) matches the consent voice to the cloning sample; a third party confirms by OTP on their own phone.
- Not allowed: voices of under-18s; politicians, celebrities or religious figures without written-permission proof; deceased persons without the heirs’ permission.
- Use: the cloned voice works only in that account — it cannot be transferred or sold — and never for political speech, money requests, fraud or obscene content; 4.15 applies.
- Labels: every output with a cloned voice carries C2PA metadata and an inaudible watermark (AudioSeal), and an “AI voice” label when published directly. Licences of SpeechBrain and AudioSeal are verified before use.
- Withdrawal: consent can be withdrawn at any time; the voice model is deleted within 72 hours and no new renders are made. Consent records (recording, time, device, verification score) are kept for 3 years after deletion for disputes.
- Complaints: a claim that a voice was cloned without consent leads to takedown within 24 hours and account suspension while it is reviewed.

35.3 Content policy additions to 4.15 (approved)

- Deepfakes and impersonation: showing a real person saying or doing what they did not, presented as news or evidence, is blocked; stricter checks for election candidates during election periods.
- Fake news: copying a real news outlet’s logo or format is blocked.
- Scam advertising: fake investments, fake jobs or overseas-job lures and fake lotteries are blocked.
- Promotion of gambling/betting apps, drugs or weapons is blocked; health misinformation (e.g. miracle cures) is blocked.
- Islamic content: Quran and hadith quotations are checked against a verified text; mismatches go to human review.
- Every output carries C2PA metadata and a “Made with AI” label on direct publishing.
- The user has approved these rules as final; a Bangladeshi lawyer’s review of terms such as “anti-state”, “deepfake” and voice rights of deceased persons is still recommended.

35.4 Islamic content verification (approved)

- The AI checks first — is it Islamic content, are quotations correct, does it meet policy — and records a confidence score.
- Confident → the 25% Islamic discount (G.9) applies automatically. Unsure → human review queue; reviewers are chosen and assigned in the admin panel. Named reviewers (set up before the Free Beta): Qari Md Shahadat Hossain and Md Serajul Islam.
- Audit trail: AI score, reviewer, time, decision and reason; the creator can appeal.

35.5 Launch actions

- Apply now for the YouTube Data API audit, the TikTok Content Posting API review and Meta Business Verification with App Review for Facebook Pages and Instagram Reels, and request NCMEC hash-sharing access (0.7); until each approval arrives, creators download and upload themselves.
- Before submitting: publish the Privacy Policy, Terms and Data Deletion pages on remonlogicstudio.com, set up support@remonlogicstudio.com and record the demo screencasts of the real publishing flow.
- Check TikTok's branding rules: if watermarked Free Trial and Free Beta videos cannot be posted through the API, direct TikTok posting is offered on paid packages only.

36. Legal and Compliance Drafts (for lawyer review)

Drafted on 21 October 2026 so the product can be built to them now; a Bangladeshi lawyer reviews and corrects them before the NexaTute Free Beta (20 November 2026), following the draft legal opinion of 36.6. These are working drafts, not legal advice.

36.1 Children's data and profiling (NexaTute and PlayBoom)

- Scope: every learner under 18, and every PlayBoom child. The guardian is the decision-maker; the child's own account cannot change consent.
- Collection: only what each feature needs — no advertising IDs, no third-party trackers, no ads; birth date and guardian phone are collected for the age check and OTP.
- Consent: a separate on/off switch for each feature at sign-up and in Settings (19.4), with a 3–4 line plain-language summary in Bangla; the guardian can withdraw any consent at any time with immediate effect.
- Profiling: engagement intelligence, personalised offers and learning-style detection run only with the guardian's switch on; if the lawyer finds minors' profiling not permitted even with consent, these switches stay locked off for under-18s and only non-profiling adaptive teaching (answer correctness, mastery) is used.
- Biometrics: NexaTute face templates only with guardian consent, encrypted, used only to confirm the account owner, never for mood or attention; PlayBoom uses no face templates.
- Rights: the guardian can view, download and delete all of the child's data; deletion within 30 days except records the law requires (payments, consent logs).
- Retention: learning records while the account is active plus 12 months; voice, images and handwriting deleted 90 days after the plan ends unless the guardian keeps them; consent records 3 years.
- Safety: no private adult–child chat without guardian visibility; children's sessions never sent to cloud AI; AI training only with guardian consent and with personal data removed.
- MindForge (37.7): thinking-skill results are learning data under these rules, with a separate switch for sharing them with teachers and institutions. Open questions for the lawyer: whether the Personal Data Protection Act 2026 allows cognitive profiling of minors with guardian consent (if not, the MindForge profiling switches stay locked off for under-18 learners), and whether the MindForge assessment part must stay blocked for EU learners until the EU AI Act high-risk obligations are met, while practice runs.

36.2 Cross-border data transfer (servers in the USA)

- Position: all servers are in the USA (Dallas, Virginia, Washington DC, Hillsboro); the Personal Data Protection Act 2026 allows transfer outside Bangladesh only with equivalent protection, which this draft provides through the safeguards below.

- Safeguards: encryption in transit (TLS/mTLS, WireGuard) and at rest (OpenBao-managed keys); access limited to named RemonLogic staff with MFA and audit logs; data-processing agreements with every hosting provider (OVHcloud, RedSwitches, Database Mart, Cloudflare) that forbid use for their own purposes; breach notice within 72 hours; yearly penetration test.
- Transparency: the Privacy Policy names the countries and providers; the consent screen states that data is stored in the USA.
- NexaLogicDev AI providers: when our servers are busy or a hard task needs a plan review or rescue, the code needed for that task may be processed by the MiniMax or Anthropic API (32.1) after keys and secrets are removed; the Terms and Privacy Policy name both, and a "RemonLogic servers only" switch turns this off. NexaTute and PlayBoom learner data is never sent to these APIs.
- Fallback: if the lawyer or the National Data Governance Authority requires local storage for children's data, children's personal data moves to a Bangladesh or approved-region store while learning content stays on the main fleet.
- Registrations: the data-law registrations, the Data Protection Officer appointment and the DPO contact in the Privacy Policy before launch.

36.3 App-store payments

- Position (decided 28 October 2026): NexaTute is on Google Play, the Apple App Store and the Microsoft Store as well as the installable web app; purchases inside a store app use that store's billing and cost 30% more, shown as a separate app-store charge line; purchases on the website use aamarPay (Bangladesh) or Paddle (international) (11.7).
- The app does not link or steer users to outside payment in a way Google's policy forbids; the lawyer confirms the exact wording allowed in Bangladesh.
- When Google's alternative billing reaches Bangladesh (target September 2027), user-choice billing is switched on in the app.

36.5 NexaLogicDev Acceptable Use Policy (draft)

- Prohibited: phishing or fake login pages of any bank, mobile wallet (bKash, Nagad), government body or other organisation; scam shops, fake investment or job sites; malware or code that attacks other systems; sale of gambling, drugs or weapons; child-abuse or obscene content; theft of others' brands, trademarks or copyright; collecting personal data without consent; spam or mass-messaging tools.
- Checks: prompts and every deploy are scanned automatically for content and links; anything suspicious is held for human review before it goes live.
- Reports and takedown: every site carries a Report link; verified complaints are taken down within 24 hours, and serious cases (phishing, child abuse) immediately, with reporting as the law requires.
- Enforcement: warning → temporary suspension → permanent closure; serious breaches close the account at once; the reason is always given and the customer can appeal within 7 days.
- Domains: the customer owns the domain, but on abuse RemonLogic may disable its DNS on our Cloudflare account; the customer can still transfer the domain away.
- Decisions: AI detects and recommends; a person makes every takedown and account decision (Security & Control Centre, 34).
- Credits: unused credits are not refunded when an account is closed for a policy breach, unless the decision is overturned on appeal.

36.4 Other documents to draft from these positions

Terms & Conditions and Privacy Policy (English, with plain-language summaries), Guardian Consent form, Refund Policy (11.6), NexaLogicDev Acceptable Use Policy (36.5), Reseller Agreement (33), Marketplace Seller Terms (32.11), Voice-Cloning Consent terms (35.2), Content Policy (4.15, 35.3),

intercompany licence and service agreement between RemonLogic Private Limited and the USA LLC (0.2). Also: MindForge consent and child-assent forms and the DPIA (37.9); the Study Hall Q&A posting rules, including anonymous use in the question bank (31.3); the accounting approval matrix and access policy (38.12); and the country rules for marketing outreach (3.26).

36.6 Draft legal opinion, lawyer-list decisions and the legal go-live checklist

Decided 28 October 2026: follow NexaTute_Draft_Legal_Opinion_Bangladesh.docx for now (sent to the lawyer, not yet signed). It replaces the earlier law names in this document.

Topic	Position
Laws named	Personal Data Protection Act 2026 (Act 63; section 29 on cross-border transfers; NexaTute is the data fiduciary and providers are data processors), Cyber Security Act 2026 (Act 81), National Data Management Act 2026 (Act 80), Children Act 2013, Trademark Act 2009 and Copyright Act 2023 — on every legal page, consent screen and legal register
Cross-border transfers	A consent checkbox alone does not make foreign transfers lawful: a data processing agreement (DPA) with every provider — hosting, email, SMS, voice/IVR, IPTSP, analytics, AI APIs, Paddle, aamarPay and the app stores; US servers stay for now (36.2)
Hub records	The five record statuses, source fields and correction history of 3.30; sample dates never shown as official; never presented as an official board, university, Cambridge or Pearson service or with their logos; no logins to their accounts and no scraping of restricted systems
Notifications	Exam and deadline, admit card and seat plan, result and shortlist, mock exam and marketing kept separate, with consent where required and marketing opt-out (3.30)
Other systems	Never collect the passwords or OTPs of other systems
Lawyer-list decisions (28 October 2026)	Approved positions: minors' profiling switches stay off until the lawyer confirms; US servers for now; batch-recording consent, 12-month retention and guardian access (3.29); COPPA and GDPR; proctoring consent; the "not board-recognised" note; recruitment and job-pathway terms; learning-guarantee terms (11.4); the new legal pages and voice, SMS and IPTSP provider contracts; backup deletion follows the privacy page; MindForge assessment off in the EU and switchable per country; log-based "stuck / didn't understand" signals (not emotion reading); landing-page guarantee claims follow the guarantee terms. Every position is controllable from the admin panel (38.4).
Liability and privacy wording	NexaTute takes no liability for centre-uploaded Cambridge, Pearson or IB papers (the school's decision and responsibility, 3.30) or for teacher-, guardian- or school-approved content and uploads, which are the responsibility of whoever approved or uploaded them; the legal pages state that NexaTute does not interfere with the privacy of students, parents, teachers or schools and only filters bad language, pornography and similar harmful content (Nexa Guard) — the lawyer finalises the wording

Topic	Position
Legal pages	The six landing-page legal pages (3.25) plus: child and guardian consent policy, recording and class-report policy, AI transparency notice, community rules, the learning-guarantee terms, subscription terms (auto-renew, Crash Packs and crash programs, Daily Pass, gift cards, app-store charge), the Exam & Admission Hub disclaimer, the institution agreement with DPA (including centre-uploaded exam materials), the copyright and takedown policy, the data retention and deletion policy (merged with data deletion), the security incident response policy, the notification and communication policy, the user complaint and data-rights procedure, and the provider DPA template; later with their features: the marketplace seller agreement, affiliate terms, Sponsor-a-Learner and Sign Language Fund terms, Most Improved scholarship rules, proctoring consent and recruitment-test and job-pathway terms. All in six languages, the English version prevailing; after launch they show "In force from 1 January 2027" with the lawyer-approval note
Report a security problem	A responsible-disclosure page on the landing page (security-disclosure): the email security@nexatute.com, what to include, good-faith research safe harbour, rules (no access to other people's data, no denial of service, no social engineering, no physical attacks), a reply within 3 working days (default, editable), no paid bug bounty at launch but a public thank-you list, and /.well-known/security.txt; reports land in the admin security-reports queue (38.4)
New lawyer questions	Meta's 2026 AI Providers policy for the WhatsApp and Messenger study bots (3.32); the licence question for Cambridge and Pearson past papers; whether teacher or guardian approval changes NexaTute's liability for content; Nexa Co-Admin for staff on WhatsApp, Messenger and imo (all off until the lawyer and DPO check, 38.20)

Legal go-live checklist (before the 1 January 2027 paid launch; shown complete with its dates in the admin panel when done): the current law text checked; privacy policy and terms approved; consent flows and guardian consent; the source-rights check; the correction and deletion workflow; provider contracts and DPAs; notification opt-out; incident response and security testing, including an external penetration test with high findings fixed and retested (19.6); and the lawyer's final signed review. Later: official data partnerships, verification automation, the DPIA, regular reviews and a per-country analysis before each expansion.

37. NexaTute MindForge — Thinking-Skills Practice

MindForge is the NexaTute module for thinking-skills practice and development, for learners from Class 4 to adults. It is practice, not an IQ test. It covers eight skills — logical and abstract reasoning, working memory, visual-spatial thinking, numerical reasoning, verbal reasoning, attention, creative problem-solving and metacognition — and shares the one learner record with the Nexa Teacher, the Nexa Co-Teacher, the Nexa Study Hall Moderator and the mock tests. Source: "NexaTute MindForge — Product Blueprint", reconciled with this master. Clickable prototype: X18 (0.8).

37.1 Purpose and principles

- No IQ-style score until expert validation and norming (37.9); no "brain growth" or guaranteed-IQ claims. Results are shown as measured thinking-skill practice gains alongside the Thinking Profile (3.18).
- Supportive language only: no "weak" or "smart" labels — "needs more practice"; no MindForge leaderboards.
- MindForge results are never the basis for admission, grades or discipline.
- Who: Class 4 to adults. PlayBoom children do not get MindForge at launch; the Class III graduation certificate leads into Class 4, where MindForge starts (30.8).

- Scope control: after the additions in 37.6 no more MindForge features are added before the Free Beta; Free Beta data decides what is actually useful.

37.2 Feature groups

All twelve feature groups of the MindForge product blueprint are included.

Feature group	What it does
1. Cognitive assessment	Practice, the starting baseline and the monthly check across the eight skills, kept clearly apart from any formally validated assessment; results shown as a range
2. AI Cognitive Gym	Daily puzzles, patterns, memory, logic and attention tasks and multi-step reasoning, with adaptive difficulty, hints, worked explanations and retries
3. Development plans	Personal goals, recommended activities and checkpoints, changed only when there is enough evidence (37.3)
4. Metacognition Coach	The learner explains the approach and the clues used; confidence is compared with accuracy
5. Creative Thinking Studio	Open-ended ideas, alternative uses, story design and multiple-solution tasks; “make your own puzzle” (37.6)
6. Real-World Problem Lab	Planning, everyday numeracy and decisions in Bangladeshi contexts, with a fairness review (37.6)
7. Parent and teacher dashboard	Skill trends, practice consistency and reflections; supportive recommendations, never diagnostic labels; teachers see it only with consent (37.7)
8. Research and validation console	Item bank with versions, item difficulty, bias checks, reliability, pilot and norming studies and expert approval — in the admin panel (38.5)
9. Localisation	App text and language-free puzzles in all six languages from launch; verbal-reasoning items in Bangla and English first, other languages only after native review
10. Child safety and privacy	Guardian consent, minimal data, healthy-use limits and accessibility (37.7)
11. Institution portal	Class summaries in the Institute Admin with consent, assignments and aggregate reports (38.7)
12. Validated assessment service	Formal score reporting stays switched off until validation, norming, fairness, security and legal reviews are approved (37.9)

37.3 One learner record and syllabus–skill mapping

The Nexa Teacher, the Nexa Co-Teacher, the Nexa Study Hall Moderator, the mock tests and MindForge all write to the one shared learner record (3.16):

Source	What it writes to the learner record
Nexa Teacher	Topic mastery, poll results, the explanation type that works, the first wrong whiteboard step, the level of the learner’s own questions
Nexa Co-Teacher	Topics asked and question depth (help-assisted answers weigh less)
Nexa Study Hall Moderator	Where the learner got stuck, hints given, “Bujhini” taps and lab mistakes
Mock tests	Score, time per question and error type

Source	What it writes to the learner record
MindForge	Practice results per skill and confidence compared with accuracy

- Syllabus–skill mapping: each syllabus topic is tagged with the thinking skills it needs (geometry → visual-spatial; word problems → verbal and numerical reasoning; BCS mental ability → logic). It works both ways: a topic where the learner is stuck brings targeted MindForge practice, and strong reasoning brings harder problems from the Nexa Teacher. Reading, labs and the beyond-the-syllabus activities are mapped too; the mapping is kept in the CASE format (37.9).
- Standing rule: every subject follows each learner's own country curriculum, so the mapping is made per curriculum (29.2).
- Strengths and practice areas are flagged only with enough evidence — for example at least 20 attempts over at least 3 days.

37.4 Expected syllabus completion

- Method: remaining topics × the learner's own pace per topic type ÷ the available study time (plan minutes, batch routine, holidays and Ramadan) → a date range compared with the exam date, shown as "on track" or "at risk" with the levers that would change it.
- Recalculated every week and after every mock test; in the first 2 weeks it is a rough estimate from similar learners; its accuracy is tracked and recalibrated.
- Shown to the learner, in the guardian's weekly report, to the Nexa Teacher and in the institution's class summary.

Institute class summary (defaults approved 24 October 2026): a privacy threshold of 5 consented learners applies to sections, single skills, subjects, practice-set completion and exports; "tight" means the finish-date range straddles the exam date and "needs a plan" means even the earliest finish is after it; a half-yearly exam covers 55% of the syllabus and a model test 85%, other tests and finals all of it. On the guardian side the MindForge data-sharing switch starts off, Pro needs the guardian's approval again every month and revision time in the forecast is rounded to half a week.

37.5 Items, measurement and calibration

- Practice is unlimited and freshly generated. Pattern, number, logic-grid, memory and attention puzzles are generated by code, and a solver (OR-Tools CP-SAT) proves that each one has exactly one correct answer — on CPU, with no GPU. Verbal-reasoning and real-world problems are generated by the LLM and checked by the AI verifier of the content pipeline (the independent solver, 3.13), with human sample review.
- Measurement — the baseline, the monthly check, progress and the finish-date estimates — uses only AI-made items that enough learners have answered (for example 200 or more), so that their difficulty is calibrated automatically (item response theory with girth). Bias is checked with differential item functioning (DIF) by gender, urban or rural, Bangla or English medium and language; results are shown as a range (for example 62–68), never as one number.
- The Free Beta starts with unlimited practice and is itself the pilot that builds the calibrated set.
- Age bands: Class 4–5, Class 6–8, Class 9–12 and adults. There is no hand-written item bank and there are no dedicated content writers; the team adds one part-time item reviewer (sample checks and bias) and one part-time psychometrician (14.2).
- Items are stored in the QTI 3.0 format with versions; expert approval and retirement of items happen in the validation console (38.5).

37.6 Approved additions

Twenty-one additions are part of the module:

#	Addition
1	Practice tied to real exam sections — BCS mental ability, bank-job and admission analytical sections — shown next to the learner's mock-test score for that section
2	A starting baseline and a monthly check in the same format, shown honestly as practice gains
3	App text and language-free puzzles (patterns, shapes, numbers) in all six languages from launch; verbal-reasoning items in Bangla and English first, other languages only after native review; the weekly word challenge in Hindi, Urdu, Arabic and French stays in English until native review (decided 24 October 2026)
4	A free 1-minute thinking warm-up at the start of Nexa Teacher batch and 1:1 classes
5	Careless-mistake coach: many "careless" errors in mock tests lead to short attention and self-checking practice and a "check once at the end" habit
6	Healthy use for under-18 learners: a gentle stop message after 30 minutes of practice a day; no streak pressure (one rest day a week keeps the streak)
7	A MindForge puzzle in the weekly guardian + child quiz
8	MindForge Pro open to everyone during the Free Beta (20 November–9 December 2026) to pilot items and calibrate finish-date estimates; afterwards a 7-day free trial (guardian permission under 18)
9	Real-World Problem Lab with Bangladeshi contexts — a bKash monthly budget, bus fares, cricket run rate, market prices — and a fairness review (urban and rural, girls and boys)
10	Admin item-bank and validation console in the content-quality area (item difficulty, bias checks, expert approval, 38.5), and a part-time psychometrician on the team — required before any IQ-style score
11	Skill refresh: a puzzle type comes back after 7 and 30 days
12	"Why this puzzle": a one-line reason beside each recommendation (AI transparency)
13	Exam-eve mode: only a 5-minute light warm-up, no new hard puzzles
14	Accessibility: with the dyslexia or ADHD settings, untimed mode by default and read-aloud puzzles; no puzzle relies on colour alone; screen-reader alternatives for picture puzzles (WCAG 2.2 AA)
15	Low internet: daily puzzles are downloaded on Wi-Fi, playable offline and synced later (Workbox)
16	"Make your own puzzle" in the Creative Thinking Studio: reviewed puzzles enter the puzzle bank anonymously; reward +10 bonus minutes (paid) or a Helper badge (free)
17	Cooperative group puzzles in paid Study Hall groups (31.1), with no competition
18	A skill-based tip added to the weekly guardian coaching tip
19	Pro prepay follows the normal rule: 3 months at 10% off (৳537) and 6 months for the price of 5 (৳995) (37.8)
20	No hand-written item bank: practice is generated and solver-checked, and measurement uses only calibrated items (37.5); age bands Class 4–5, 6–8, 9–12 and adults
21	MindForge KPIs in the KPI panel of the content-quality area (37.10, 38.5)

37.7 Consent, age, safety and accessibility

- A separate consent switch decides whether MindForge data is shared with teachers; the guardian decides for under-18 learners. Institutions see MindForge data only with that consent, and never use it for admission, grades or discipline.
- Profiling rules of 19.4 and 36.1 apply: if the lawyer finds that cognitive profiling of minors is not allowed even with consent, the MindForge profiling switches stay locked off for under-18 learners (G.7).
- Under-18 learners need the guardian's approval to buy MindForge Pro (11.9). Emotion recognition is never used. Accessibility follows WCAG 2.2 AA (37.6, item 14).
- Where MindForge appears: the learner's MindForge tab and Thinking Profile; the guardian weekly report (expected completion date); the Institute Admin class summary (with consent); and the platform admin (module switch, item-bank review).

37.8 Price and plans

Item	Price or terms (VAT included)
MindForge Free	For everyone, including the free tier: 3 puzzles a day, the weekly challenge and simple progress
MindForge Pro	৳199 a month as an add-on to any plan or bought alone; not included free in Combo or Personal, to protect the 60% margin target (11.12); $3 for learners abroad (BDT × 0.015)
Prepay	3 months ৳537 (10% off); 6 months ৳995 (one month free)
Second child	30% off
Institutions	৳139 a seat for 20–99 learners; ৳119 a seat for 100 or more
Under-18 purchase	Needs the guardian's approval (11.9)
Free Beta and trial	Pro open to everyone during the NexaTute Free Beta (20 November–9 December 2026); afterwards a 7-day free trial (guardian permission under 18)

MindForge is its own module, with sub-modules, in the plans and modules table (38.2); which sub-modules sit in Free and which in Pro is set there.

37.9 International standards plan

Area	Standard	How it is applied	When
EU AI Act	High-risk education AI (Annex III); obligations from 2 December 2027 after the AI Omnibus	Risk management, data governance, technical documentation and logs, human oversight, accuracy evidence and EU database registration; emotion recognition is banned and not used; the assessment part stays off for EU learners until compliant (lawyer to confirm), while practice can run	Compliant before 2 December 2027

Area	Standard	How it is applied	When
Psychometrics	AERA/APA/NCME Standards for Educational and Psychological Testing; ITC/ATP Guidelines for Technology-Based Assessment, including translation and adaptation	Validity, reliability and fairness evidence and a technical manual; the six-language rule documented; IRT calibration, DIF checks and range reporting (37.5)	DIF checks, range reporting and the translation guideline before launch; technical manual before any IQ-style score
Norms and research ethics	Bangladeshi norm sample stratified by division, urban or rural, gender, school type and medium; separate norms per country	Pilot and norming studies approved by a partner-university ethics committee, with guardian consent and child assent, de-identified data and independent external validation	Ethics approval and pilot before launch; norms and external validation before any IQ-style score
AI and privacy management	ISO/IEC 23894 alongside ISO/IEC 42001; ISO/IEC 27701 on top of ISO/IEC 27001; data protection impact assessment (DPIA)	AI risk register and privacy controls for MindForge (19.7)	DPIA before launch
Children	UNICEF Policy Guidance on AI for Children; UK Age Appropriate Design Code; US COPPA (under 13)	Applied per market	Before each market opens
Interoperability	1EdTech QTI 3.0 (item bank); CASE (syllabus–skill mapping); Open Badges 3.0	Item, mapping and badge formats (verifiable credentials, 3.11)	QTI and CASE formats before launch
Frameworks	OECD PISA frameworks (creative thinking, problem solving); ISCED levels	The eight skills aligned to PISA; age bands aligned to ISCED	Planned
Evidence	Efficacy RCT with the partner university; ESSA evidence tiers for the US market; Digital Promise Research-Based Design certification	Claims only from evidence (16.3)	Planned
Adult aptitude practice	ISO 10667	Applies if MindForge is used for adult job or aptitude practice; never the sole basis for hiring	When offered
Accessibility	EN 301 549 alongside WCAG 2.2 AA; a written accommodations policy	Needed for EU public procurement	Planned

37.10 KPIs and Free Beta measures

- KPIs, shown in the KPI panel of the admin content-quality area (38.5): Pro attach rate among paying learners (target 15%), Pro churn, weekly active free-puzzle users and finish-date forecast accuracy.
- Free Beta measures: the share of learners doing puzzles on at least 3 days a week, finish-date forecast accuracy, and Pro trial → purchase.
- The revenue model counts MindForge Pro at the 15% attach-rate target (15.8).

37.11 Architecture

Where it runs	MindForge work
Learner's browser	Puzzle play, timers and practice scoring; offline daily puzzles (Workbox); read-aloud puzzles (in-house VITS voices); the MindForge tab and charts
CPU worker pool (App-1–App-3, spare Core-1 cores)	Puzzle generation with OR-Tools uniqueness checks at P10; nightly IRT calibration and DIF checks (girth); weekly finish-date forecasts; scoring of the baseline and monthly check, which are decided on the server (trust rule, 8.18)
Shared GPU pool, low priority	Generation of verbal and real-world items with their explanations in the batch window, and the few live explanations, at P10, so live classes are always protected
Data	The shared learner record and the versioned item bank in PostgreSQL; events in ClickHouse; nothing new outside the existing clusters
Hardware	Unchanged — no new servers or GPUs

38. Platform Administration, Institute Admin and Company Accounting

One platform admin panel runs NexaTute — plans and modules, operations, accounts, content quality, money, growth institutions, tests and marking, and every user-visible value (38.17) — and the company accounting of RemonLogic Private Limited for all products. Every area is role-based, every money action needs an approver other than the person who made it, and every action is logged. Clickable prototypes: X10, X12–X19, X21 and X22 (0.8). No new servers are needed (38.14).

38.1 Platform admin panel — areas and dashboards

Area	What the admin sees and does	Prototype
Overview and capacity	Totals of institutions, subscriptions and active courses; capacity and queues. Learners and usage: free and paid learners by class, learners by subject, live counts right now (1:1 with the Nexa Teacher, live batch classes, Study Hall, Nexa Co-Teacher), learners by country, and today's new free and new paid users	X10
Plans, prices and modules	Price table, the plans and modules table and the platform-wide module switches (38.2)	X10
Institutions	Onboarding — verify and approve new institutions (for example by EIIN or trade licence), branches, suspend and reactivate; the institution billing ledger (subscriptions, seat limits, dues, invoices); the institution support queue with SLA and onboarding progress	X10
Events, certificates and jobs	Create, publish and revoke events and certificates; the job-pathway listings manager (3.21)	X10
Classes and schedule	Language batches, timetable, holidays and Eid dates, Ramadan timetable, season calendar and make-up classes	X10
Platform operations	AI models and prompts, API spend caps, escalations, cost per learner, GPU load, feature flags, maintenance, backups and incidents (38.3)	X14

Area	What the admin sees and does	Prototype
Accounts, support and legal	Account search, support view, consent, data requests, terms versions, breach register and link blocklist (38.4)	X15
Content quality and money	Quality gate, answer and translation reviews, MindForge item bank, discounts, refunds, payouts, VAT and the price-rule KPI panel (38.5)	X16
Tests and marking	Written-check limits per plan and the platform switch (38.2); the written-check GPU queue and daily cap (38.3); marking schemes, question types per exam, the recheck queue, the AI-marking audit, and the fill-in and error-notebook rules (38.5)	X10, X14, X16
Every value and the classroom	The full Plans & prices editor, sales channels, product switches by country and age, classroom and microphone rules, login security, wellbeing, learning limits, money routing and AI settings — every user-visible value with two-person approval (38.17)	X10
Updates, load tests and delivery	Software updates with two-person approval and automatic rollback, the staging load test, the LiveKit panel, video and recording delivery with the Edge check, synthetic checks, the status page, OTP protection, service budgets, the Beta dashboard and the Security & controls tab (38.18)	X14
Admissions and capacity guard	Admission limits per class or course, the waiting list, the capacity guard, the master “All new admissions” switch with emergency close and the six-language admissions message (38.19)	X25, X14
Nexa Co-Admin	The AI assistant that acts only within the asking admin’s permissions, its activity log and settings, and the chat-app channels (38.20)	X27
Safety	Nexa Guard, the Human Controller rota, helplines set country-wise and Sponsor-a-Learner (0.8, 10.2)	X10
Growth	AI CMS, drop-off analysis, AI Marketing Hub and Growth Platform, affiliates, B2B CRM, reviews, founder and investor reports; Landing Page AI (38.6)	X13, X17, X19
Accounting and finance	Books, receivables and payables, payroll, rent and vendors, loans, investors and shares (38.8–38.11)	X10
Access and reports	Roles, 2-step login, the full action log and approvals — two-person approval for price changes, large refunds, data deletion and institution discounts (38.12); founder reports; people management — search staff, change a role with a reason, suspend and reactivate, approved by the security owner (38.12)	X10
Notifications and staff chat	The bell, the notification control panel and the internal staff chat with calls (38.15); the Institute side sits in the Institute Admin	X21, X12
Avatar & Voice Studio	Avatar and voice libraries, clone requests, consent and complaints, the marketplace and prices (38.16); the user side for learners, guardians and teachers	X22, X23

38.2 Plans and modules table and module switches

- A table of modules × plans — free tier, Daily Pass, Batch, Combo, Personal, PlayBoom and institution — where the admin ticks which modules each plan includes and sets their limits (for example 1:1 minutes a day). Institutions’ optional modules (Script Checker, mentors, placement, PlayBoom, MindForge) are set the same way. Every module and sub-module can be set per plan and platform-wide.

- Safety modules cannot be removed from any plan: Nexa Guard, guardian consent and purchase approval, data download and deletion rights, and accessibility.
- Changes apply to new subscribers; existing subscribers keep their plan for the 3-month lock and get advance notice in their own language before a module is removed.
- Before saving, the table shows a live AI-cost and margin check against the 60% target and dependency warnings (for example, removing the Nexa Co-Teacher breaks “Ask Co-Teacher” in batch classes). Changes need two-person approval and keep a version history with one-click rollback. The plan comparison on the website, in the app and in the Landing Page AI updates from the same table.
- Platform-wide module switches — for example turning PlayBoom off for everyone during maintenance — are kept separate from the per-plan table (Unleash flags).

Tests in the table (starting settings, decided 25 October 2026; in X10 the “Tests & marking” module group, which also holds the error notebook):

Module	Starting setting
Written checks — written tests and short answers (3.27)	Free tier 1 a week; Daily Pass 2 a day; Batch 4 a day; Combo 7 a day; Personal 10 a day; institutions 4 a day per learner
True/false, fill in the blank, matching and ordering	Unlimited in every plan, including the free tier
Human recheck	Free once per test, then up to 2 more a day
Board-style mixed tests	In every paid plan; not on the free tier
Written checking on or off	A platform-wide switch beside the per-plan limits (Unleash flag)
Employer and organisation assessments	Per plan: free (the starting setting for job-test courses, 11.14), a price, or off; the Recruitment tests add-on is a row of the price table (11.14); changes follow the normal two-person approval

Margin check: the cost of written checks is counted in the 60% margin check — at these limits Combo comes to about 60% and Personal to about 50%, which the owner accepted (25 October 2026); written checks run after live classes and the Nexa Co-Teacher on the GPU, with free-tier checks last, and the limits stay editable in this table.

The plans table also holds the 2027 features with their free and paid limits (3.32), the storage allowances and add-ons (G.3), the Doubt Pass and the crash programs (11.12, 11.14) — every one a (default, editable) row with two-person approval.

MindForge in the table (starting settings):

Module	Free tier and Daily Pass	Batch, Combo and Personal	PlayBoom	Institution
MindForge Free — 3 puzzles a day, weekly challenge, simple progress	Included	Included	Not offered (Class 4 and above)	Included
MindForge Pro — the full module	Add-on ৳199 a month or bought alone	Add-on ৳199 a month (not included in Combo or Personal)	Not offered	৳139 a seat (20–99); ৳119 (100+)

38.3 Platform operations

Page	What it shows and controls
AI models and prompts	Model and prompt versions with one-click rollback (Langfuse prompt management; vLLM model versions) — every change is logged as a model-behaviour change (19.7)
API fallback spend	Monthly spend caps per external API, current use and alerts (Portkey, 8.9)
Wrong-answer escalations	Counts of answers escalated from the local model to the cloud fallback and to human reviewers, by course
Cost per learner	Daily AI and infrastructure cost per learner, by plan and product (OpenCost and ClickHouse)
GPU load	Per-GPU busy share and memory (DCGM Exporter) and the placement order of 8.17
Written-check GPU queue	Queued written checks with waiting times; the evening peak (4–9 pm), when checks wait and the learner is notified when the result is ready; the daily GPU cap for written checks — 1,500 GPU-minutes (default, editable) with an alert at 80%. When the cap or the GPUs are full, checks wait — they are never sent to an outside AI service (8.17)
Feature flags and A/B tests	Gradual rollouts and experiments (Unleash)
Maintenance and busy page	Switches the Cloudflare maintenance or busy page for one product or all
Backup and restore	Status of database, cache and object-storage backups and the result of the monthly restore test
Incident log	Every incident with severity, owner, timeline and postmortem (19.5)
Updates, load tests and delivery	Software updates, the staging load test, the LiveKit panel, video and recording delivery, synthetic checks, the status page, OTP protection and service budgets (38.18)

38.4 Accounts, support and legal

- Account search and status for learners, guardians, teachers and institutions; a read-only support view that is always logged and needs the guardian’s permission for a child’s account.
- Face-check exceptions: an OTP exception when the face check fails (3.16); free-tier multi-account abuse flags.
- Consent records; data download and deletion requests under the Bangladesh Personal Data Protection Act 2026, handled by the Data Protection Officer (19.3, 36.1); Terms & Conditions versions with the acceptance log; the data-breach and incident register (72-hour notice, 19.5).
- Link blocklist: the URLhaus and StevenBlack lists plus the platform’s own entries (3.23).

Support queues (28–29 October 2026): Free Beta feedback tickets with screenshots and SLA timers (3.33); the security-reports queue fed by the “Report a security problem” page, with severity, status and a 3-working-day reply (36.6); the takedown log of centre-uploaded papers (3.30); forced sign-out of all devices (38.17).

Legal and compliance controls (28 October 2026), each logged with two-person approval: the minors’ profiling switches (locked off until the lawyer confirms), MindForge assessment per country (EU off by default), the data-region note (US servers, for now), the backup-deletion text taken from the privacy page, the centre-upload feature, the learning guarantee and its numbers, and the provider DPA register (one DPA row per provider, 36.6).

38.5 Content quality and money

Area	What the admin does
Quality	Course quality gate (3.13); “any course” price review and override (11.13); teacher-content check
Answers	The answer-cache and wrong-answer review queue, including the Study Hall Q&A reviewer queue (31.3); the circular tracker and post-exam solution review (3.20)
Languages and tools	The six-language translation review queue with native reviewers; dictionary and glossary entries and the calculator rules for each exam (3.20)
MindForge	Item bank and validation console — item difficulty, bias (DIF) checks, expert approval and retirement of items (37.5)
Sign language and student videos	The BdSL sign-check queue (a BdSL expert and a Deaf reviewer) with coverage per course and subject (3.31); review of Be the Teacher videos (Nexa Guard, then a subject expert), Time-Travel answers, AR models and comic stories (3.32)
Hub circulars	The circular-verification queue: AI-read dates, units, eligibility, fees and seats → the reviewer approves, edits or rejects, with the source PDF, the last-verified time and change detection (old → new) and who is notified (3.30)
Tests and marking	Marking schemes per board and exam with versions and a second person’s approval; the question types allowed for each exam; the human recheck queue (answer within 24 hours); the AI-marking audit — a sample of AI marks compared with human marks; fill-in-the-blank rules (spelling tolerance, digits, units, equivalent answers); error-notebook rules (retest days, removal after 2 correct in a row, once-a-day practice) (3.18, 3.27)
Money	Price table, discounts, Crash Pack and crash-program sales (no seat caps), gift cards, refunds, aamarPay and Paddle reconciliation, affiliate and marketplace payouts with tax withheld, and the VAT report in the Mushak 9.1 return format for the chartered accountant; VAT counted only on Bangladesh sales, and a VAT and Paddle view with Paddle payouts, buyer-country tax and the editable Paddle fee (11.7)
KPI panel	Warns when the price-rule targets (Combo 40–50% of subscribers, churn of 10% or less, margin of at least 60%) are missed and proposes the agreed fix (11.12); MindForge KPIs (37.10)

38.6 Growth

The growth area holds the AI CMS and the drop-off analysis (3.25), the AI Marketing Hub and RemonLogic Growth Platform with its CRM, journeys and target-client database (3.26; prototype X19), affiliates and the teacher marketplace (3.21), the B2B CRM, review collection, Sponsor-a-Learner (0.8) and the founder and investor reports built from the same accounting and KPI data. The Landing Page AI lives here too (3.25; prototype X13).

38.7 Institute Admin

Schools, colleges, coaching centres and universities run their own workspace (3.11). The Institute Admin sees only its own institution.

Area	What the institution admin does
Setup	Profile, logo and colours, academic year, campuses or branches and departments, classes, sections and shifts, subjects and syllabus mapping

Area	What the institution admin does
People	Bulk import from Excel or CSV, roll numbers, guardians linked by OTP with consent, staff roles (head teacher, coordinator, class teacher, subject teacher, accountant) and student status (active, paused, left, transferred)
Seats	Seats bought, assigned and reassigned when a student leaves
Classes	Batches by section or language, the institution's own routine, holidays and exam dates that update the AI study plan, and a calendar view
Academics	Homework and practice, class and model tests, the Script Checker (3.21) and the institution's own question bank after checks; the paper builder with sets A/B/C and approval by the head teacher or coordinator, proctoring of the institution's own exams under the 3.27 rules, written checks within the institution limit (38.2) and the roster and results API (18.3)
Reports	Attendance and participation (polls, questions, "didn't understand" taps), mastery by class, section and subject, a weak-topic heatmap, an at-risk early warning, section comparison, syllabus completion and student feedback; PDF and Excel export; the MindForge class summary only with consent (37.7)
Communication	Notices that each guardian and student receives in their own language, guardian-meeting scheduling, and automatic reminders by email and WhatsApp
Billing	Invoices, VAT receipts, seat changes and Sponsor-a-Learner applications
Mentors and counsellors	Institutions may add their own human mentors or counsellors; confidentiality rules protect students' mental-health notes
Placement	Only for institutions with learners aged 18 or over (BTEB, polytechnics, universities, skill centres), following the job-pathway rules (3.21)
Classroom, past papers and tools	The classroom speaking view with the microphone rules (3.28); "Our centre's past papers", a private area for approved Cambridge or Pearson centres and IB-authorised schools (3.30); the teacher's AI assistant, Team Quest launches, mystery missions and group mind maps for a class, and Nexa Swap supervision (3.32); passkeys for staff; the at-risk rule within the platform's limits; maintenance and What's new notices (3.33)

- By type: schools — NCTB Class 6–10, PlayBoom for the primary section (25% / 35% off, 30.3) and more guardian consent and contact; colleges — HSC groups and electives, admission preparation and the circular tracker, and 18+ students who consent for themselves; coaching centres — enquiries and free trials, frequent batch changes, model-test ranks, their own mock tests on the marketplace (20%), an affiliate code and a result showcase with consent; universities — 18+ learners and placement.
- Limits: the Institute Admin sees no details of a student's mental distress or safety incidents (these go to the guardian and the Human Controller) and cannot change prices or publish content without the checks. For a serious safety incident the institution receives a notice without details, for example "A safety matter for a student in 9-A is being handled by NexaTute; the guardian has been informed" (decided 26 October 2026). When a student leaves, the account is kept (deleted 5 years after last use), the institution can view it for 90 more days and the family can download it at any time.

38.8 Company accounting — books, receivables and payables

- Full double-entry books. Chart of accounts — assets, liabilities, equity, income and expenses; the accountant can add accounts. Journal — date and the month the entry is for, balanced debit and credit, description, proof, made by and approved by; most entries are made automatically

from payments, invoices, bills, payroll, loans and shares; manual entries only by the finance role, with approval; nothing is deleted, only reversed; closed months are locked.
- Ledger for every account with a running balance, and sub-ledgers for each employee, vendor, institution, lender and shareholder. Trial balance, profit and loss, balance sheet and cash flow are produced automatically.
- Receivables and payables with due-date reminders (for example which server bill is due on which date); loans and interest; investor details; profit and loss.
- Export to Tally, QuickBooks and Excel. The VAT return, income tax and the audit stay with a chartered accountant; tax and VAT rates are set by the accountant in settings, never hard-coded.

Partial payments, both ways

- "Record payment" with amount, date, method and reference; invoices and bills move from unpaid to partly paid to paid, and reminders continue for the balance.
- Receipts are applied to the oldest invoice first (the admin can pick another); ageing and the 60-day suspension rule follow the oldest unpaid invoice; overdue institutions can get an instalment plan, with no suspension while they pay on time.
- An overpayment is kept as an advance (a refund needs two approvals); gateway shortfalls are recorded as fees or chargebacks for reconciliation; unidentified bank receipts wait in a queue; small write-offs need one approval and bad-debt write-offs two.
- Two-person approval is judged on the bill's total, not on each instalment; USD bills paid at different rates record the exchange gain or loss; vendor advances are adjusted against later bills; a short loan instalment shows as overdue, with penalty interest kept separate.
- Partial receipts do not change profit and loss under accrual accounting; the accountant decides accrual or cash basis.

38.9 Payroll, rent and vendors

- New-employee form: join date, salary parts (basic, house rent, medical, conveyance), bank or bKash account and TIN; mid-month joiners are prorated.
- "Run payroll" for a chosen month with deductions (tax at source, advances, unpaid leave) and the festival bonus; two-person approval; a payslip for each employee. Individual salaries are visible only to finance and the founders; employee loans are deducted from salary automatically.
- Rent: the landlord is a vendor with the agreement (monthly rent, due day, deposit); a "Rent — <month>" bill is made automatically each month; the expense is booked to the month it is for, prepaid rent is spread month by month and the deposit is kept as an asset.

38.10 Loans taken and given

- New-loan form with a direction — loan taken or loan given: lender or borrower (bank, person or director, contact, agreement copy), amount, date, receiving or paying account, rate, tenure and repayment type (EMI, interest monthly with principal at the end, bullet or flexible). EMI schedules split principal and interest automatically.
- Loans given are assets (a loan receivable per borrower); their interest is "other income" below operating profit and accrues automatically at each month end — monthly (rate ÷ 12) or daily (days ÷ 365), as the agreement says.
- Receipts go to accrued interest first, then principal (or principal only if the agreement allows), with a preview of the split before saving; a borrower statement as a PDF and the payoff amount for today; reminders 3 days before and on the due date; penalty interest only if the agreement says so.
- After 90 days overdue, interest stops being booked as income and the loan is flagged for provision (the accountant decides). Lending always needs two approvers; a loan to a director cannot proceed without the general-meeting or board approval document.

- Bullet loans with partial repayments, taken or given: interest is charged on the outstanding principal for the days it was outstanding. A part-payment closes the interest to that day on the old balance, and the new, lower balance earns interest from then. Simple interest by default (unpaid interest does not earn interest); compounding or capitalisation only if the agreement says so; a rate change applies from its date; each interest entry shows its from–to dates, days, balance and rate. Example: ৳10,00,000 at 12% with ৳4,00,000 repaid on day 90 → interest to day 90 is 10,00,000 × 12% × 90/365 = ৳29,589, and from then the ৳6,00,000 balance earns about ৳197 a day.

38.11 Investors, shares and share transfers

- The investors and shares register shows how many shares each investor owns and how much they paid; the cap table updates automatically. A share issue splits the face value to share capital and the rest to share premium; partly paid shares carry a call schedule with reminders (voting and ownership follow the Articles; the lawyer confirms).
- Share transfer (secondary sale): a request with seller, buyer, number of shares and price; checks — it cannot exceed the seller's holding, lock-ins apply, ESOP-pool shares cannot be sold and the share class carries over; a pre-emption offer to the existing shareholders pro rata within the days set in the Articles, and tag-along if the shareholders' agreement says so.
- Transfer checklist: board resolution, RJSC Form 117, stamp duty, the seller's capital-gains tax challan, the old certificate returned and a new certificate issued. Status: requested → offered → board approved → RJSC filing → done or cancelled. Total shares do not change; the price is shown as the "last transfer price" without changing the valuation.
- The sale money is not the company's, so no journal entry is made for it; only stamp duty and legal costs paid by the company are expenses. Only the founders and the board can process transfers.

38.12 Access, approvals and the audit log

Group	Roles
Platform admin	Super admin (founders), security owner, operations, Nexa Guard controller, institution success (own institutions only), content and programmes, marketing (aggregates only), support (limited, logged look-ups) and analyst
Accounting	Finance lead, accounts officer, payroll officer, founders and board, department manager (team totals only), external auditor or chartered accountant (read-only and export, time-limited to about 30 days) and investor (investor report and own shares)

- No access by default. Custom roles are made by ticking view, edit, approve and export for each module and sub-module; scope limits (institutions, country, department) and masking of sensitive fields by role (individual salaries, only the last 4 digits of bank and bKash accounts, TIN, shareholder details).
- The maker can never approve their own work, including their own salary or expense. Every transaction needs approval, and who approves is set from the user roles.
- Role assignment is proposed by the super admin and approved by the security owner (Mst Sonia Sabuz); 2-step login for everyone; a fresh OTP before approving money, changing roles or exporting; a quarterly access review; accounts unused for 90 days are suspended automatically; access is removed the day an employee is marked as left; break-glass access is time-limited, needs a reason and is reviewed afterwards. Admin staff sign in with passkeys or hardware security keys — required for the super admin, money approvers and the security owner (19.2).

- People management (decided 24 October 2026; prototype X10): staff are searched by name or email and filtered by role and status (active, suspended, access ended); “Change role” shows the old → new role with a required reason and is approved by the security owner (a founder when it concerns the security owner), never by the requester; “Suspend” and “Reactivate” work without removing the person and carry a reason and the same approval (approval of Reactivate to confirm); every step is logged. Learner, guardian, teacher and institution search stays in Accounts (38.4). Approval exceptions and defaults (24 October 2026): in platform operations, feature flags, adding servers, restore tests and incident status changes need no approval and are only logged, and when the super admin asks for something himself the operations/SRE owner approves; in accounts, support and legal the defaults are a 30-day data-request deadline, a 72-hour breach notice to the authority and affected people and 7-year financial records (defaults, not legal claims); in content quality and money, refunds over ৳5,000 need two approvals; in growth the super admin cannot approve legal pages, contracts or payouts, affiliate codes with commission need approval, and a B2B deal at the contract stage stays blocked from onboarding until the lawyer or DPO approves.

Transaction	Approval (editable table by transaction type and amount)
Up to ৳1 lakh	One approver other than the maker (accounts officer or finance lead)
৳1–10 lakh	Finance lead
Over ৳10 lakh; any loan; share issue or transfer; month close	Finance lead and a founder
Payroll	Prepared by the payroll officer, approved by the finance lead
System entries	Batch-approved daily
Loans given	Always two approvers; a loan to a director also needs the general-meeting or board approval document
Write-offs and refunds of advances	Small write-offs one approval; bad-debt write-offs and refunds of overpayments two

- A rejection needs a reason, and nothing counts in the reports until it is approved.
- Audit log: who (name and role), when, where (device, browser, IP address, approximate city) and what (before → after) for every action, including views of salaries and shares and every export; the log cannot be edited or deleted.

38.13 Cloud export and backup

Item	Rule
Destinations	Google Drive, OneDrive and Dropbox on company accounts, each with a chosen folder — any one or all of them
Contents	Encrypted full backup of the accounting database, Excel and PDF reports, and Tally and QuickBooks files, in year and month folders
Schedule	Nightly at about 2:00 am Dhaka time, a final copy at each month close, and “export now”
Retention	The last 30 daily copies, every month-end and every year-end (changeable)
Keys	The encryption key is held by the founders and the finance lead

Item	Rule
Changes	A destination is changed by the super admin or the finance lead with the security owner's approval
Checks	Each export is logged with a checksum; a failure raises an alert and retries; a restore test runs every month

38.14 Architecture and software

Part	Where it runs and with what
Admin panel and Institute Admin	Separate admin namespaces on the App cluster (App-1–App-3); Keycloak for 2-step login and the step-up OTP; Casbin policies for roles, scopes, masking and approvals (stored in PostgreSQL)
Accounting data	Its own PostgreSQL schema on DB-1/DB-2 with row-level security, backed up by pgBackRest like all data; pgaudit records database reads and changes of accounting, payroll, share and access tables; the in-house action log is append-only
Reports and exports	WeasyPrint PDFs (payslips, statements), openpyxl Excel files, Tally and QuickBooks files made in-house; rclone (with encryption) runs as a nightly job on the CPU worker pool; the key and the cloud-drive tokens are kept in OpenBao
Operations pages	Read from Prometheus, DCGM Exporter, OpenCost, ClickHouse, Langfuse, Portkey and Unleash
Notifications and staff chat	In-house notification service on App-cluster Celery workers with pywebpush (web push) and FCM (Android admin app); Centrifugo for chat and the bell; PostgreSQL for messages, routing, history and the encrypted summary archive; LiveKit (Core-1/Core-2) for staff calls; Qwen3Guard and Qwen3 summaries on the shared GPU pool at P10; ClamAV on Core-2 (38.15)
Avatar & Voice Studio	Avatar files in R2 behind the CDN, checked by glTF-Validator (browser and Core-2) and slimmed by glTF-Transform on the CPU worker pool; voice clones on AI-2 Node A off-peak within the daily GPU-minute cap; consent records, voice samples and models encrypted in PostgreSQL and R2 (38.16)
Tests and marking	Written checks queued on the Celery workers to the launch GPU pool — Qwen3-VL-8B on AI-2 Node A reads the handwriting and the local Qwen3 marks — within the daily GPU cap (8.17); fill-in checks with math.js and SymPy; graded coding runs in the practice-lab gVisor sandboxes; marking schemes, rechecks and audits in PostgreSQL; scripts and snapshots in R2, private to the learner (3.27)
Isolation	Accounting and admin data are never reachable from NexaLogicDev sandboxes, learner apps or other institutions; support look-ups are read-only and logged
Hardware	Unchanged — no new servers; the load is small (staff users only); voice clones and chat summaries use off-peak time on the existing GPU pool (8.17); written checks and coding tests also run on existing capacity (8.17); the 26–29 October 2026 controls and panels also run on existing capacity (7.1)

38.15 Admin notifications and internal staff chat

Decided 24 October 2026 for the platform admin panel and every school and college admin panel (Institute Admin). Prototypes: X21 for the platform side and inside X12 for the Institute side; a bell sits in the header of X10, X13–X17 and X19, and X21 is linked from their sidebars.

Part	Rule
Bell and levels	A bell with the unread count on every admin page; one click opens the item. Info — in the app or the daily digest; important — push; urgent — push, email and SMS, even in quiet hours
Platform events	Approvals (money, refunds, payroll, loans, shares, prices and plans, roles and permissions, data deletion, terms, model rollback, API caps, maintenance, ad budget, campaigns) and important changes (a new price live, approve or reject results to the maker, break-glass, new-device login, exports, a failed backup, incidents, load over 60%, API spend at 80%, a bill or EMI due)
Institute events	Seat changes, staff roles, notices, exam-date changes, discount requests, pending guardian consent, at-risk counts, overdue bills, ticket SLA and onboarding steps
Control panel	Per event: the recipients by role or named user, the channel and the level. Approvals go to the approver from the approval rules (38.12), with CC allowed; a reminder after 4 hours and, after 24 hours, the next approver. Each user sets quiet hours and digests and can mute ordinary events, never their own approvals or security alerts; the Institute head admin routes events among its own staff only; changing who gets security alerts needs the security owner's approval; every notification and routing change is logged
Channels	Web push to the browser and the installable admin app (pywebpush), the Android admin app through Firebase Cloud Messaging (the listed exception) and, optionally, email, SMS and WhatsApp through the existing notification service. Push text never shows secrets — no salaries, student names or amounts
Staff chat	Staff only; each institute's chat is separate and the platform talks to an institute only through a support channel. One-to-one chat with anyone in one's own panel, with images and files virus-scanned. No free group creation: a group request (name, purpose, members) is approved by a role-based approver (department head or founder; the Institute head admin), plus automatic department and class groups by role; only the group owner adds or removes members, and this is logged
Calls	Voice and video only in the platform admin panel, allowed per role from the control panel (voice only, voice and video, group calls); calls are never recorded, and AI call notes are made only if everyone agrees
Chat pushes	A one-to-one message pushes; in a group only an @mention pushes (the rest is counted on the bell, with an optional all-messages push per group); an incoming call rings; a group request goes to the approver and the result to the requester; a compliance alert is urgent, ignores quiet hours and cannot be muted; the lock screen shows only the sender (a preview is allowed per role); no second push to a device already showing the chat
Compliance check on send	Our own Qwen3Guard plus rules check every message: passwords, API keys and bank details are blocked with a warning; confidential company information (prices, salaries, contracts, investment, code), learner or guardian personal data, harassment or attempts to move data out are sent but alert the concerned admin at once, chosen per category from the user roles (for example the security owner, DPO, finance lead or founder; the Institute head admin). A person decides — the AI never punishes — and the staff policy states that work chats are monitored
Summaries and retention	A daily AI summary of every chat (local Qwen3 at P10) goes to an encrypted archive that each person opens with their own password and 2-step login (no shared archive password, decided 26 October 2026); every access is logged; confidential and compliance chats are summarised in full detail, and the super admin chooses from the user roles who may see them. Full messages are kept 90 days and summaries 6 months (a compliance case until it is closed or a legal hold ends) — the lawyer confirms the periods (G.7). Not end-to-end encrypted, because every message is checked on send; encrypted in transit and at rest

Part	Rule
Systems	No new servers: the in-house notification service on the App-cluster Celery workers with pywebpush and FCM; Centrifugo for messages and the bell; PostgreSQL for messages, routing, history and the archive; LiveKit on Core-1/Core-2 for calls; Qwen3Guard and Qwen3 on the shared GPU pool; ClamAV on Core-2; archive keys in OpenBao (18.1, 38.14)

38.16 Avatar & Voice Studio

Decided 24 October 2026: a platform-admin module (prototype X22) with a user side for learners, guardians and teachers (prototype X23). It uses the character and voice-cloning engine and the consent and safety rules of Motion Studio (35.2), and its entitlements follow G.3.

Part	Rule
Avatar library	Upload of GLB or VRM avatars → automatic checks (glTF-Validator, size and textures, the lip-sync check) → cultural review → two-person approval → live. Each avatar has an age group and an unlock type (free, XP or Premium), versions and one-click rollback. Approved avatars get lighter copies for weak devices (glTF-Transform) and are rendered in the browser with Three.js and three-vrm
Voice library	Our own voices in six languages, each checked by a native reviewer; the admin adds the official Nexa Teacher avatars and voices
Who adds what	Learners choose avatars and voices from the library — no clones or photo avatars under 18. A guardian may clone their own voice for PlayBoom Play Buddy stories and invite family members aged 18 or over, never the child's voice. Verified teachers may clone their own voice and make an avatar from their own face photo after a liveness check and face match (MediaPipe, SFace), used only in their own content. Learners aged 18 or over may clone their own voice for pronunciation practice
Consent and safety	The Motion Studio voice-cloning rules apply (35.2): a freshly generated consent sentence (speaker, date, account and use) read live — on camera as well for a photo avatar; SpeechBrain ECAPA speaker verification; AudioSeal watermark, C2PA and an "AI voice" label. Clone requests, consent and complaints are approved by the Nexa Guard controller, never by the person who asked; a withdrawn voice stops at once and is deleted within 72 hours; consent records are kept 3 years; a complaint leads to removal and account suspension within 24 hours
Server work and GPU	Clones use the same server engine as Motion Studio: Chatterbox on AI-2 Node A (speech pool) for English, Hindi, Arabic and French, and per-person Bangla and Urdu VITS models trained in the late-night batch window (6.2). NexaTute clone jobs run off-peak at P10 within a daily GPU-minute cap set by the super admin (value to confirm); extra requests wait for the next window with the ETA shown (8.6, 8.17). Avatar checks run on Core-2 and the lighter copies on the CPU worker pool; files sit in R2 behind the CDN. No new servers or GPUs
Marketplace and pricing	Premium avatars, a guardian's family voice, a verified teacher's voice and photo avatar, and an 18+ learner's own voice: one of each is free and more are bought in the NexaTute marketplace (aamarPay in Taka, Paddle in USD); learners under 18 get one Premium avatar free and unlock others only with XP, never with money. Proposed prices: ৳149 ($2) per extra Premium or photo avatar and ৳299 ($4) per extra voice clone, changed in the admin with two approvals (to confirm); avatars by outside designers (20% commission) stay off for now (to confirm). Premium avatar and voice-clone rows sit in the plans and modules table (38.2)

38.17 Every user-visible value in the admin panel

Approved 28 October 2026: every remaining user-visible value is editable in the admin panel — the default is the value users see today, every change needs two-person approval and is logged, and changes apply to new subscribers or new events (existing subscribers keep the 3-month lock where it applies), in six languages.

Area	Controls
Plans & prices editor	The full price table — every course, add-on and pass (including the Doubt Pass), per-minute rates and the USD ×0.015 rule, discounts and rewards, trials, the billing lifecycle (failed renewal, grace, upgrades and downgrades), refund, gift-card, marketplace and affiliate rates; propose → two-person approval → effective date → version history; the Landing Page AI and growth pages link to it. Crash-program and English-medium prices use the same flow.
Sales channels	Website gateway, Google Play, Apple App Store and Microsoft Store — each on or off with its app-store charge (30%, default, editable) (11.7)
Product switches	On or off, scoped by country and age group, for Study Hall Q&A, the teacher marketplace, the Doubt Pass, gift cards, Sponsor-a-Learner, Head Start and the Year Pass; PlayBoom per-feature switches and content defaults; the master "All new admissions" switch with emergency close and the admission limits (38.19)
Classroom	Batch maximum (50), mic starts off, turn length (60 s), follow-ups (1), hand-raise limit (unlimited), the disturbance rule (3 in 10 minutes → 2-minute pause) (3.28); weak-internet audio-only threshold (15% packet loss) and noise suppression (on); 1:1 lesson-flow steps and exam-eve mode; batch teaching rules (adaptive pace, Peer Instruction, class format); the normal timetable; the weekly human revision-class slot and the mock-test tie-break rule
Login security	For learners and guardians: OTP resend wait (60 s) and daily maximum (5), wrong-code lockout (5 wrong → 15 minutes), session length (30 days), maximum devices per account (3) and "Sign out of all devices" (forced from 38.4 when needed); sign-up OTP channels — SMS, email or WhatsApp — each on or off with the SMS and WhatsApp sender numbers, API settings and the sending email address; face verification and OTP-exception rules
Safety and wellbeing	PlayBoom daily limits, the family bedtime lock and session rules; Nexa Guard filter categories, strictness by age and word lists per language; the behaviour ladder and Human Controller escalation rules; guardian SMS free and on (switch); the under-18 break reminder (45 minutes) and guardian daily limit; streak rules and the leaderboard (off by default; nicknames under 18); the default guardian monthly spending limit; sign-up by country and age (US under-13 waiting list, country waiting lists); the data retention schedule
Learning limits	Free-tier limits; Study Hall Q&A limits, auto-hint and contact masking; Study Hall rounds and the XP economy; MindForge healthy use and free limits; link-explainer limits; 1:1 minute and free-photo rules; the minute-wallet rules; teaching languages per course; smart-reminder caps and the engagement-feature switches; the Calm-mode suggestion before exams
Money routing	Payment gateways and methods (bKash, Nagad, card) per country; institution seat and billing rules; per-institution module and limit overrides (the Institute Admin shows what applies, 38.7); the at-risk early-warning rule
AI and content	AI answer routing and the fallback order; the voice and audio cache (on, kept 90 days, at most 50 GB, purged when an answer is corrected — default, editable, 8.14); PlayBoom chapter review and country-opening stages; support response-time promises and auto-replies; marketing message caps and quiet hours; the phone-tutor hotline minutes

Area	Controls
Operations	Auto-scaling and alert thresholds, the backup schedule, staff access rules and the software-update settings (38.18)
Doubt Pass	A separate top-up product for questions outside the learner's syllabus: ৳50 for 20 questions, valid 30 days, $1; paid from the wallet; under 18 needs the guardian's approval; price, size, validity and on/off are rows of the price table (11.14)

38.18 Software updates, load tests, delivery monitoring and service budgets

Approved 28–29 October 2026 for platform operations (prototype X14). Every change below needs two-person approval and is logged.

Panel	What it does
Software updates	An inventory of all server software — operating system and packages, LiveKit, Valkey (Redis), PostgreSQL, Centrifugo, the AI model servers, the web server and the Node and Python libraries — with the current and latest version, checked automatically every day from the SBOM (18.4). The admin is notified of every new version (a normal update as an important notice; a security patch as urgent push, email and SMS that overrides quiet hours). Every update, small or large, needs two-person approval — there are no automatic updates and the AI never updates anything by itself. Flow: release notes → the staging copy with automatic tests → two-person approval → rolling node-by-node update with health checks → automatic rollback to the previous version if a health check fails. The maintenance window is set in the admin panel (for example 02:00–05:00, editable) and never falls during live classes. Version control: the full history per program (who, when, from → to, result) and one-click rollback to any earlier version with approval. Customer notices per update: SMS, email, WhatsApp, a maintenance banner on the landing page and a notice on learner, guardian and institution dashboards, with timing and text in six languages; the "What's new" notes for each release are written here (3.33).
Admissions and capacity guard	Guard status, current load against the threshold, the waiting list per class or course and the link to the load test (38.19)
Load test (staging)	Runs only on the staging copy, never on live: a scenario editor (concurrent 1:50 classes, learners per class, 1:1 classes, ramp-up and duration) and a result table per component — LiveKit, Valkey, Centrifugo, PostgreSQL, the AI model servers and the web/API tier — with CPU, memory, p95 class-join time, p95 audio delay, error rate and pass or fail. Pass thresholds (default, editable): class join ≤3 s p95, audio delay ≤300 ms p95, errors below 1%, CPU below 75%. Run before the Free Beta (20 November 2026), before the paid launch (1 January 2027) and after every major update; starting a run needs two-person approval; Locust and the LiveKit CLI load tester (18.1); no new servers.
LiveKit	Rooms, participants, packet loss, delay and reconnects by area and operator, node health, TURN on 443, the version check and update checklist, the RED auto rule and FEC setting, and the Free Beta latency measurement with the "add an edge server" decision (9.5)

Panel	What it does
Video and recording delivery	Storage use by type (class recordings as replay files, Nexa Notes videos, group-class videos, uploads) with monthly cost and a 6-month forecast; the 12-month recording lifecycle with “keep until my course ends”; signed links for private recordings (15 minutes, default, editable); the HLS ladder 720p / 480p / 240p and the start quality on weak networks and data saver (240p, default, editable); CDN cache time for public Nexa Notes (7 days, default, editable — private recordings never cached) and purge; buffering, upload failures, recordings stuck in “Verifying” and playback errors; the software chain shown read-only (9.4); guardian access locked on; and the Edge check — the share of each operator’s users (Grameenphone, Robi, Banglalink, Teletalk, BDIX broadband) served from the Cloudflare Dhaka edge and the cache-hit rate, with an alert when an operator’s Dhaka-edge share falls below 70% (default, editable)
Synthetic checks	Every 5 minutes from Grameenphone, Robi, Banglalink and BDIX broadband: page load, class join and delay, with alerts; they feed the Edge check (9.4)
Status page and incidents	Incidents can be published to status.nexatute.com; the public text needs two-person approval (3.33)
OTP protection	Cloudflare Turnstile before an OTP is sent (on), per-IP (10 an hour) and per-number (5 a day) limits, SMS allowed only to expected country prefixes (Bangladesh +880 by default; others use email or WhatsApp), a daily SMS spend cap (৳2,000) and a spike alert at three times the hourly average (all default, editable)
Service budgets	A monthly budget for SMS, voice, AI APIs, R2 and WhatsApp, with an alert at 80% and a set action at 100% — slow down or switch channel (for example SMS → WhatsApp or email for OTPs, voice → push, AI API → queue and cached answers and cached audio first, R2 → alert only)
Area and operator notices	Notices to users of a weak-network area or operator (3.28)
Beta dashboard and security tab	The Free Beta live dashboard (3.33) and the “Security & controls” tab — control IDs, test records, go-live gates G01–G10, security metrics, the yearly penetration test, the DPIA and the vendor key register (34)

38.19 Admission limits, waiting list and capacity guard

Approved 29 October 2026 (“very important, otherwise server management will have problems”; prototypes X25 Plans & modules › Platform switches, the X14 capacity view, X01, X03, X08, X12, X18 and X20). Unlimited admission (8.5) stays the default; these controls protect live classes when demand runs ahead of proven capacity. Every change needs two-person approval (the asker never approves) and is logged; values marked (default, editable) are starting settings.

Control	Rule
Admission limit per class or course	No limit (default, editable): a new 1:50 batch opens automatically every 50 learners (the batch maximum of 38.17). Optional limit rows by class or course, country and teaching language, each with its limit, enrolled count and status Open / Full → waiting list / Closed; the existing sign-up-by-country Open / Waiting list / Closed setting stays and is linked.

Control	Rule
Waiting list	When a limit is reached, new learners join a waiting list and see their position; places are offered in order when someone leaves or the limit is raised; an offered place is held 48 hours (default, editable; 24 / 48 / 72) and then passes to the next person. The learner — and for a learner under 18 the guardian, who accepts or declines — is told by SMS (free), push and email (all on by default). No payment is taken until a place is offered and accepted; names on the admin list are masked (nickname or initial under 18).
Capacity guard	On (default): when load passes 75% (default, editable; 60 / 70 / 75 / 80) of the capacity proven by the last passing staging load test (38.18) — for example 2,000 proven concurrent learners → guard at 1,500 — new sign-ups and enrolments go to the waiting list automatically; it releases below 60% (default, editable). An important push goes to the Ops owner and the super admin when it switches on. It never blocks paying learners already enrolled and never interrupts live classes; the capacity plan (8.5) still adds servers before classes start.
All new admissions (master switch)	One switch for all new admissions, On by default; turning it off or on needs two-person approval. Existing learners keep studying, renewals continue and live classes are never interrupted. Institution contract seats are not affected unless “also pause new institution student activations” is ticked (unticked by default).
Emergency close	The super admin or the Ops owner can close admissions at once; a second admin must confirm within 60 minutes (default, editable; 30 / 60 / 120) or admissions reopen automatically; logged, with an important push and email to the super admin and the Ops owner.
Admissions message	The admin writes the message once in any one language with a reason (capacity, maintenance, exam period or other) and an optional reopening date; the AI translates it into the other five languages, each translation can be edited (marked “Edited by admin”), and the second approver checks the translations before all six are published together. It shows on the landing page and the sign-up screens (on by default), with the waiting list open while admissions are closed (on by default).
Where users see it	Sign-up and checkout show a kind “Join the waiting list” state with the position, what happens next and the notification channels, then “A place is ready — held for 48 hours” with Accept / Decline; the learner dashboard has a waiting-list card; the guardian app shows the child’s status and the offer; the Institute Admin notes that contract seats are not affected but the capacity guard can delay new student activations; landing-page course and price buttons switch to “Join the waiting list” with a one-line explanation and an FAQ entry “What is the waiting list?”. No seat counts, “only N left” or countdown pressure anywhere.
Operations view	Platform operations shows the guard status, the current load against the threshold, the waiting list per class or course and a link to the load-test card (38.18).

38.20 Nexa Co-Admin — AI assistant for admin staff

Approved 29 October 2026 (prototype X27, with a “Nexa Co-Admin” button in the header of the admin boards). Nexa Co-Admin is an AI inside the platform admin panel that serves each admin user only within that user’s own role and permissions — for example “show me the current price model” (shown only if the user may view prices) or “today Sonia was paid ৳5,000 salary — record it in accounting” (done only if the user may record payments). It is always labelled: “Nexa Co-Admin is an AI. It acts only within your own permissions.”

Rule	How it works
Permissions	Every request is checked against the asking admin's role, scope and masking rules (38.12); a request outside them is refused politely with "Request access" or "Send to Finance"-style options. A "what can you do for me" chip lists what the current role may read and do.
Actions	Co-Admin asks only for what is missing (for example which of two masked staff matches, the account paid from and the month), shows a preview of the exact entry (for the salary example: Dr Salary expense ৳5,000 / Cr Bank ৳5,000) and acts only after Confirm. Money, payroll and other two-person actions are submitted as "Waiting for second approval" and the approver is notified; Co-Admin can never be the second approver. An undo is a reversal entry that needs approval.
Levels per role (default, editable)	Super admin, Finance and Ops: do with confirmation (Ops: non-destructive operations only); Support, Content and Marketing: draft only. "Always ask before acting" is on and locked for money, people, legal and data deletion.
Never allowed (fixed list of 6)	Delete or change logs; show secrets, keys or passwords; change roles or permissions, including its own; approve as the second approver; export personal data without approval; act outside the asking admin's own permissions.
Activity log	Every request and action — time, admin (masked), role, what was asked, data read, action taken, approval status, model and result — in an append-only log kept 7 years (default, to confirm); export needs two people.
Model and limits	The own-server model first, so admin data stays on NexaTute servers; an outside API only for non-personal questions (off by default); 50 actions a day per admin (default, editable); voice input off by default; Co-Admin itself on by default; every setting change needs two-person approval.
Read-only answers	Counts with sources, for example Free Beta sign-ups from the Beta dashboard, with cells under 20 hidden.

Co-Admin on chat apps — WhatsApp, Messenger and imo (approved 29 October 2026; the same rules on each)

Rule	How it works
Channels	A separate "NexaTute Staff" account on WhatsApp, Messenger and imo, each switched on or off in the admin panel; all three are off until the lawyer and DPO check (36.6), and imo stays locked off until imo business API access is confirmed (to confirm). Once allowed: on for Super admin, Finance, Ops and Support, off for Content and Marketing (default, editable).
Linked accounts	Staff link their own number or account with a passkey (allow-list); access is removed at once when someone leaves or is suspended; one admin can turn off all linked accounts in an emergency, and turning them back on needs two people.
Answers	Personal data is masked; sensitive data such as salaries and identities appears only through an "open in app" link.
Actions	Off by default (read only). When switched on, an action is previewed in the chat and confirmed through a secure link with a passkey, Face ID or fingerprint, valid 15 minutes (default, editable); money still needs the second approval. Never through a chat app: role or permission changes, personal-data export, data deletion and API keys.

Rule	How it works
Limits and log	A 15-minute session timeout and 20 actions a day per person (defaults, editable); optional night quiet hours (off by default); every message is logged “via WhatsApp”, “via Messenger” or “via imo”.

Systems: no new servers — Co-Admin runs on the App cluster with the local Qwen3 on the launch GPU pool at low priority, permission checks through Keycloak and Casbin (38.14), the activity log in PostgreSQL, the existing notification service for approvals, and the chat channels through the existing WhatsApp, Meta and messaging rows of the register (18.1).

Appendix A. Canonical Architecture Decisions

Decision	Canonical rule
Authority	16-Oct-26_Final_Hybrid_Architecture_v24.0.docx is the infrastructure baseline and overrides conflicting earlier architecture documents; this master blueprint is the consolidated proposal.
Server count	16 rented servers at launch (14 core + 2 recording) and App-3 from December 2026; 28 servers in the full ecosystem register (7.2); elastic scale-out to ≈37–39 rented servers (≈42 before the 17 October 2026 changes) plus hourly GPU burst for 16,000 CCU (8.5).
Capacity	16,000+ combined CCU is a target; split target is 8,000 NexaTute + 8,000 Motion Studio; requires proof by load testing.
Product separation	Distinct product API/runtime, queues, cache namespaces, permissions, usage accounting and durable job records.
Hybrid pool	All six GPU servers run as one unified Kubernetes GPU cluster (8.11), with recording as the top priority on Rec-1/Rec-2 and overflow of queued recordings to spare Core-1/Core-2 CPU and then idle GPU nodes, with guaranteed per-product floors (NexaTute 60%, Motion Studio 25%, 15% floating), priority pre-emption (P100/P50/P10), LLM hot standby and per-product circuit breakers with a degradation ladder (8.8–8.11).
Ecosystem resource sharing	Every product may use any idle GPU or CPU capacity of the ecosystem through one scheduler, in the placement order of 8.17: the owner’s real-time work comes first, borrowed work runs at P70 or below on idle capacity only and is handed back within one scheduling cycle; the sandbox, data, edge and recovery servers are never lent; minors’ raw media stays on the launch pool.
NexaLogicDev AI lanes	Heavy Lane MiniMax-M2.7 on our own servers (GLM-5.2 if the licence is refused); Fast Lane Qwen3-Coder-Next on 4 × RTX PRO 6000; Free, Student and Hobby use the Fast Lane only; external APIs only as fallback through Portkey with spend caps — MiniMax M3 for the Fast Lane, Claude Sonnet 5.5 plan review and rescue for Pro/Team and Auto-Architect (32.1).
Browser-first execution	All three products run in the user’s browser whatever a normal device can run; our servers keep large AI models, secrets, trusted results (money, ranked grades, safety, certificates), work that must outlive the tab and the fallback for light or weak devices; Device Check tiers and Data Saver (8.18).
Administration and accounting	One platform admin panel with a plans and modules table, platform-wide module switches and role-based access; company accounting with double-entry books; maker-checker approvals by amount; an append-only audit log with pgaudit; a nightly encrypted export to company cloud drives and a monthly restore test (38).

Decision	Canonical rule
MindForge measurement	Practice items are generated and solver-checked; only calibrated items (IRT) are used for measurement; results are ranges; no IQ-style score until validated and normed (37).
Shared growth platform	One AI Marketing Hub, CRM, journey engine and target-client database for all three products, with separate brands, data and approvers; organisations only; no marketing to minors (3.26).
Admin notifications and staff chat	One notification service for every admin push (web push with pywebpush; the Android admin app through FCM as the listed exception) with role-based routing, escalation and quiet hours; staff chat on Centrifugo, PostgreSQL and LiveKit with a compliance check on send, virus-scanned files, unrecorded calls and encrypted daily summaries; no new servers (38.15).
Avatars and voice clones	One avatar and voice engine shared with Motion Studio: GLB/VRM avatars checked with glTF-Validator and rendered in the browser; voice clones only with the live consent sentence, speaker verification, AudioSeal and C2PA, off-peak on the speech GPU within a daily cap; no clones or photo avatars under 18 (38.16, G.3).
Website video	Nexa Notes videos are HLS files in R2 behind the CDN, transcoded on the CPU worker pool (never Core-1); YouTube only click-to-load; no ads or tracking on our sites (3.25).
Tests and written-answer checking	Written checks are queued jobs on our own GPUs within a daily cap — never an outside AI service; nothing is marked until the learner confirms unsure parts, and a person rechecks on request; adaptive tests only for placement and practice; proctoring only in certificate and institution exams, never in practice tests and with no video of children; coding tests in the existing gVisor sandbox (3.27).
Priority	NexaTute real-time workload X-Priority 100; NexaLogicDev paid interactive work X-Priority 100 on its own Dev pool; real-time work borrowed from another pool X-Priority 70; Motion Studio paid X-Priority 50; free/trial and internal batch X-Priority 10 (8.17).
Client-side rendering	NexaTute voice, Nexa Teacher video, whiteboard, lab, the 3D classroom (3.15), AI-tutor recording and download MP4s of class recordings (9.5) run in supported browsers, with a server speech fallback for weak devices (3.2) and no server rendering of the classroom; Motion Studio Quick-mode animation, 3D avatars and standard voices also render in the creator's browser, with chunk resume and a server takeover after 30 minutes without a connection (4.8).
Motion Studio scratch	Heavy render work and final MP4 compilation on the local SSD of the GPU node executing the job; never on Core-1 (9.3).
Upload safety	Validate final artifact and durable upload before deleting only the exact task-owned file; preserve local file on failure.
Storage	Cloudflare R2 for final validated outputs and permitted archives; scoped credentials, signed URLs and integrity verification; user uploads go straight from the browser to R2 with presigned multipart URLs (4.16).
Redis	Valkey cluster on Redis1/Redis2 with Sentinel: NexaTute instance 6379 (primary Redis1, replica Redis2) and Motion Studio instance 6380 (primary Redis2, replica Redis1); instances stay separate per product. Daily encrypted RDB snapshot to R2, kept 48 hours.
Database HA and backup	DB-1/DB-2 Patroni cluster with synchronous standby; live pgBackRest WAL backup to Cloudflare R2 (RPO ≤1 minute); the App nodes and LB-1/LB-2 run as clusters (8.13).
Content verification	Common safety verification on every prompt; Islamic plan adds source/attribution verification; plan mismatch guard and human review.
Commercial claims	Pricing, model licensing, capacity, costs and product outcomes require validation before public/investor claims.

Decision	Canonical rule
Duplicate information	Maintain one canonical description per feature/software/server/policy; cross-reference rather than duplicating detail.

Appendix B. Operational Acceptance Checklist

☐ Canonical v24.0 server inventory reconciled with actual supplier/deployed hardware.

☐ All 16 nodes have recorded IPs, ports, OS, drivers, GPU model, storage mounts, firewall rules and monitoring agents.

☐ App-1 and App-2 deploy as separate product runtimes with independent auth and rate limits.

☐ Redis1:6379 and Redis2:6380 are isolated and tested for restart/recovery.

☐ DB backup/restore, replication lag, failover and PgBouncer pooling are tested, including Patroni automatic failover, pgBackRest WAL archiving to R2 and point-in-time restore.

☐ Portkey provider/model identifiers, credentials, timeout/retry policies and fallback behavior verified in staging for the GPT-6 Luna (OpenAI) → Mistral Large 3 (Mistral AI) chain.

☐ X-Priority routing and queue starvation protections are load-tested.

☐ NexaTute browser rendering and accessibility tested across target devices/browsers.

☐ LiveKit media, TURN and Rec-1/Rec-2 recording tested at representative room topologies.

☐ Motion Studio script verification occurs before storyboard/video generation.

☐ Credits are reserved atomically; retries cannot double-charge or duplicate paid rendering.

☐ Every GPU node running a render keeps all heavy video scratch/intermediate work on its own local SSD (AI-2 Node B preferred).

☐ R2 upload checksum/size/durable acknowledgement gates cleanup.

☐ Upload failure retains recoverable local output; cleanup refuses symlinks and paths outside the assigned directory.

☐ Voice consent/rights workflow, Islamic source verification, plan mismatch guard and human-review queue tested.

☐ Pricing, taxes, language support, licenses and public claims approved by owners.

☐ Load-test report documents actual achieved CCU, p95/p99 latency, error rates, fallback usage and cost.

☐ Incident response, rollback, backup restore and circuit-breaker recovery drills completed.

Document Status Note

This final corporate edition is the single working proposal and architecture index for RemonLogic. The v24.0 hybrid architecture remains authoritative for locked infrastructure decisions. Product descriptions and commercial inputs are consolidated planning material; they do not establish that every feature is implemented or production-ready.

Appendix C. Glossary

Term	Definition
CCU	Concurrent Users
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
RLS	Row-Level Security
SLA	Service Level Agreement
SLO	Service Level Objective
RTO	Recovery Time Objective
RPO	Recovery Point Objective
NVENC	NVIDIA hardware video encoder
CRDT	Conflict-free Replicated Data Type
MoE	Mixture of Experts
AWQ	Activation-aware Weight Quantization
KV Cache	Key-Value Cache
HNSW	Hierarchical Navigable Small World
RRF	Reciprocal Rank Fusion
HyDE	Hypothetical Document Embeddings
VAD	Voice Activity Detection
TTS	Text-to-Speech
STT	Speech-to-Text
MTTR	Mean Time to Recovery

Appendix D. Change Log

Version	Date	Changes
v19.0	14 October 2026	Feature catalogue
v24.0	15 October 2026	Hybrid architecture baseline
v25.0	15 October 2026	Consolidated master source document
v26.0	15 October 2026	Added Sections 21–31 and Appendices A–D; retained existing source content

Version	Date	Changes
v26.3	15 October 2026	Group Study billing unit clarified (৳/student/min); baseline used for the Appendix E feature reconciliation audit
v28.4	16 October 2026	Remon Motion Studio Complete Features Reference (Appendix F); technical review against v24.0
v28.5	16 October 2026	Technical Reconciliation & Delta Register (Appendix I) operational controls
v28.9	16 October 2026	Corporate final: styling, authority hierarchy, feature consolidation and release-control appendices; verification corrections to equity, exit multiples, OKR assumptions, Motion Studio per-minute billing, Group Study recording, Free Trial Package, personas, risk matrix, H.2 roles and salary budget; 16,000 CCU scale-out hybrid architecture with load balancing, reserved capacity and priority circuit breakers (8.5–8.10)
v29.0	17 October 2026	Consolidated corrections (file revisions v89–v97): Motion Studio 1080p Native tier and admin package settings; creator tools, growth features, copyright and content safety; escalating-tier quality; launch risk controls; creator account, spending cap and pre-production fee; uploads for both products through the Document Intake Service; production-impact estimates; Unified Software Register with SPDX licences, SBOM and OpenChain process; architecture placement fixes

Version	Date	Changes
v29.1	17 October 2026	Architecture review fixes: Motion Studio request flow reordered (job record first, signing before upload); quality-gate measurements and Content Credentials on the Dallas assembly node; PgBouncer moved from LB-1/LB-2 to App-node sidecars (change from the v24.0 node assignment, approved 17 October 2026); HAProxy and Keycloak in the authentication flow; register corrections and additions (Docker Engine for labs, y-websocket, Web Audio, SPIRE, cert-manager, OWASP ZAP, Semgrep, Falco); Redis-node storage; International Standards Alignment (19.7)
v29.2	17 October 2026	Production optimizations (8.15): hybrid Qwen3 thinking switch, per-agent output limits, NVIDIA Triton batching with MPS, batched Whisper, smaller Qwen3-VL frame-check images, Wan 2.2 A14B only for Cinematic hero shots, BGE-M3 INT8 option; capacity note (8.6), VRAM table (8.11), register (Triton, MPS), tests (13.2) and AI-governance conditions (19.7)
v29.3	17 October 2026	Final dependency and consistency pass: GrowthBook and Novu replaced by Unleash and an in-house notification service (MongoDB/SSPL removed); ClickHouse added for Langfuse and Ralph LRS; APISIX standalone; OpenBao as a 3-node Raft cluster with snapshots and Shamir unseal; LLM hot standby kept while Wan A14B runs on AI-1 Node B; corrected VRAM rule; TI2V-5B native 24 fps; Triton on AI-2 Node B only; Patroni on the Kubernetes API; tusd; admin-panel settings completed; licence notes for pillow-heif, Semgrep and EAGLE-3; C2PA certificate and Cloudflare plan costs; list of items that need real-world evidence (G.7)

Version	Date	Changes
v29.4	18 October 2026	Approved changes of 17 October 2026: App-3 (OVHcloud ADVANCE-2) from December 2026, before the NexaTute launch; uploads straight from the browser to R2 (Uppy S3 multipart; tusd removed); capacity review with a per-layer combined table (8.6) and a 16,000 CCU fleet of ≈37–39 servers; thinking switch and output limits recorded as capacity protection; Motion Studio browser rendering extended (4.8): browser voices, cinematic 3D look at Quick rates, server takeover after 30 minutes for a lost connection or a closed studio, offline rendering, backwards server rendering, checks on browser-made chunks, 3D fallback on Rec-node GPUs, Mediabunny and Puppeteer/Chromium; CPU server voices (in-house VITS on ONNX Runtime); Qwen3-8B tasks moved to Qwen3-30B-A3B and Qwen3-4B; LightX2V, FastWan, Wan2.2-S2V-14B (replacing MuseTalk/LatentSync), Nunchaku 4-bit FLUX.1-schnell, LMCache CacheBlend, suffix decoding with Arctic Inference and optional Kubernetes DRA; Locust, LiveKit CLI and vLLM load tests; review fixes: launch row follows the minimum launch list (17.2), image-upload charge (11.1), per-GPU memory limits (8.11), ClickHouse moved to Core-2 with nightly backup, five new risks (23.1), review-queue staffing gap (G.7), and the two 8.11 GPU tables separated again

Version	Date	Changes
v29.5	18 October 2026	NexaTute Classroom Builder and Redesign (3.15): templates, editor, AI room designer, own posters with safety checks, institution branding and Kids Mode rooms, always rendered in the user's browser (Full, Light or 2D version; never on the server), with the Smart Whiteboard in the browser on the room's board, free for every user, from launch; Motion Studio Series and Promo Videos (4.18): intro, outro, recap, next-episode preview, thumbnails, teaser, trailer and highlights cut from rendered episodes, series autopilot drafts, promos billed at the Quick-mode rate with no pre-production fee, intros and outros billed once; CC0 3D asset libraries added to the register; flows (6.2, 6.3), capacity (8.6), placement (8.11, 8.12), admin settings (4.9), tests (13.2) and roadmap (17.1, 17.2) updated
v29.6	18 October 2026	Download MP4s of recorded live and group classes are now made in the user's browser (WebCodecs, Mediabunny, Web Audio, OffscreenCanvas; 1-minute parts with resume; H.264 + Opus) instead of GStreamer NVENC compositing on Rec-1/Rec-2, so the Rec nodes only record and download MP4s are not stored on the server (9.5, 11.3, 20.2); placement, capacity and tests updated (6.1, 07, 8.5, 8.6, 8.11, 8.12, 8.15, 13.2, H.2, I.2.4); register: Mediabunny, WebCodecs and Web Audio now used by both products; review fixes: the Appendix F heading now uses the Heading 1 style, so it appears in the contents and the navigation pane, contents entries added for Appendix F and 19.7, "headless Chromium" used throughout, and stale v28.9 labels removed from Appendix I

Version	Date	Changes
v29.7	18 October 2026	NexaTute Price Model v2.0 (approved 18 October 2026): monthly Batch, Combo and Personal plans with group classes, daily 1:1 minutes and wallet top-ups for extra minutes (11.1); USD plan prices (recalculated with the 0.50-rounds-up rule), institution discounts, aamarPay payments, auto or manual renewal and VAT-inclusive prices (11.2, 11.7, 11.8); recording included free everywhere and stored as event logs of about 5 MB per class (11.3, 9.5); Exam Crash Pack, Daily Pass, discounts, rewards and price fine-tuning rules (11.12); new sections 3.16 (AI Teacher and text-only AI Co-teacher, load-aware routing, syllabus scope, face-verified single user), 3.17 (batch classes, Combo and Virtual Batch with quality system and KPIs) and 3.18 (learning science, 1:1 methods, brain grooming and the Thinking Profile); NexaTute Laboratory highlighted (3.6); new engagement and beyond-syllabus features marked “Release plan” (3.10); mastery-based progression from launch in the Personal Plan; wellbeing, behaviour ladder, human-controller alerts, face-data and inclusion safeguards (10.2); international standards extended — children’s data, ISO 21001, AI transparency, assessment integrity, 72-hour breach notice and penetration test (19.4, 19.7); Motion Studio Express and Night render rates (4.8, 4.9, G.8.3); architecture, capacity, degradation ladder and placement updated (6.1, 6.2, 7.1, 8.4, 8.6, 8.9–8.12, 8.14, 20.1–20.4); register: SFace added, aamarPay replaces bKash/Nagad/Stripe, 178 rows; unit economics recalculated with VAT-inclusive plan prices (15.7); KPIs, tests and risks 26–29 added (13.1, 13.2, 23.1)

Version	Date	Changes
v29.8	18 October 2026	Decisions approved after v29.7: AI Teacher and AI Co-teacher share one learner record with handover notes, combined evaluation and Co-teacher class summaries (3.16); holiday, Ramadan, make-up and free-guest rules for batches (3.17); new sections 3.19 (engagement intelligence with consent switches and child safeguards), 3.20 (learning programmes: Parent–Teacher Meeting, Live Study Hall, Wrapped, mistake videos, Choose your AI Teacher, Head Start, Olympiad, Smart Suggestion, post-exam sprint, Second Chance, circular tracker, Study Abroad Coach, guardian learning, season calendar), 3.21 (free content funnel, affiliates, teacher marketplace, Script Checker, job pathway, gift cards), 3.22 (lifetime free tier, Help button, Co-teacher Pass) and 3.23 (links shared by users); Other Popular course plans and USD prices (11.1, 11.2), refunds, B2B, plan changes, grace period and guardian purchase approval (11.6, 11.8, 11.9), Crash Pack seat limit and one-discount rule (11.12), new 11.13 (“any course” valuation) and 11.14 (passes, programmes and commissions); Bangladesh Personal Data Protection Ordinance 2025 controls, English legal documents with a local-language consent summary, Data Protection Officer and the review of the US servers (19.3, 19.4, 19.7); human-controller escalation and link safety (10.2); architecture, capacity and storage (5.1, 6.2, 8.6, 8.9, 8.12, 9.5); register 182 rows with LightGBM, Vowpal Wabbit, URLhaus and StevenBlack hosts (18.1, 18.2, 18.4); partnership manager and subject reviewers (14.2); North Star metric, tests, risks 30–35 and partnerships (13.1, 13.2, 22.4, 23.1, 26.1)

Version	Date	Changes
v29.9	19 October 2026	New 3.24 (learning-efficacy reports, prediction accuracy, CEFR, UDL, external audits and the AI lesson rubric), 3.25 (Landing Page AI and AI CMS for both products: signals, approvals, guardrails, professional landing-page standard, CMS options, drop-off analysis page and sales tools — pre-sale AI, plan finder, checkout recovery, B2B CRM, reviews) and 4.19 (Motion Studio landing page, CMS and drop-off funnel); web app as the main mobile channel and no in-app purchase in the Play Store app at launch (3.10, 11.7); maps, stacks, flows, capacity, placement and storage (3.9, 4.7, 5.1–5.3, 6.2, 6.3, 8.6, 8.12, 8.13, 9.4, 20.4); register 187 rows with sharp, vanilla-cookieconsent, web-vitals, Lighthouse CI and a search and reviews services row, Turnstile added to the Cloudflare row, and product column corrected to shared for FLUX.1-schnell, Nunchaku, Revideo, google-api-python-client, the platform APIs and the WhatsApp Business Platform (18.1, 18.2, 18.4); A/B-test results in TimescaleDB everywhere (3.9, 3.19); standards, privacy, KPIs, tests, claims, risks 36–38, partnerships and support channels (19.4, 19.7, 13.1, 13.2, 16.3, 22.4, 23.1, 26.1, 28.2); company registration and app-store checks (Appendix G)
v30.0	19 October 2026	New 3.26 AI Marketing Hub for both products: articles, videos and Reels with approval, publishing to the blogs, YouTube, Facebook, Instagram, TikTok and LinkedIn, SEO, demos, query answers, B2B market research and outreach with approval routing for replies, lead data and daily, weekly and monthly reports; maps, stacks, flow and capacity (3.9, 4.7, 4.19, 5.1, 5.2, 6.1, 6.2, 7.1, 8.6); register 190 rows with the LinkedIn API, Google Trends and Keyword Planner APIs and the outreach mailbox (18.1); privacy, KPIs, tests, risks 39–40, support and launch controls (19.4, 13.1, 13.2, 23.1, 28.2, 4.14); lawyer check of outreach (Appendix G)

Version	Date	Changes
v31.0	21 October 2026	RemonLogic Ecosystem edition: new Section 00 (ecosystem vision, corporate structure with the Delaware LLC, shared platform, launch timeline, unified funding, salaries and marketing, 24–25-server architecture); new Sections 30 PlayBoom (Play class–Class III, replaces Kids Mode), 31 Live Study Hall groups and AI moderator, 32 NexaLogicDev (GLM-5.2 Heavy and Qwen3-Coder Fast lanes, credits and plans, hosting and domains, engagement, Funding B), 33 Reseller Program, 34 Security & Control Centre v1.6, 35 Motion Studio rights, safety and international pricing; G.8.3 delivery-speed tiers replace the time-of-day rates; G.8.4 USD prices; Free Beta dates (NexaTute 12–31 December 2026, PlayBoom 1–20 January 2027); Funding A reallocated inside ৳5 crore (14.2, 15.2)
v31.1	21 October 2026	A6000 Dev GPU servers confirmed at US$1,599 per server per month on every term; Funding B re-phased (Heavy from June 2027, one Fast server at launch, second at ≥450 paid users; ≈৳1.36 crore with ≈৳14 lakh contingency; break-even ≈500 paid users); Go Live bundles set (Starter .xyz ৳499, Business .com ৳2,199, Shop .shop ৳799, with renewal prices); Cinematic and 1080p Native rates kept at present prices on the strength of the existing video optimizations (35.1)
v31.2	21 October 2026	International payments for NexaTute and PlayBoom move to Paddle through the USA LLC (11.2, 11.7); PlayBoom priced per child with 30% off from the 2nd child, a ৳20 Daily Pass and institution discounts of 25% (20–99) and 35% (100+); reseller discount on every product and paid item (33); ecosystem revenue model rebuilt with all products and the 11.14 lines (15.8); the launch scope now holds every approved NexaTute feature (17.1); legal and compliance drafts for lawyer review (36); Security & Control Centre completed with consolidation rules, metric formulas, test-record template and named owners (34.5–34.8)

Version	Date	Changes
v31.3	21 October 2026	Software register for the whole ecosystem (18.1: 22 new rows, products L and E), model registry (GLM-5.2, Qwen3-Coder-30B-A3B, SpeechBrain, AudioSeal), NexaLogicDev build layer in 6.1, 25-server register (7.2), Day-1 must-have list (17.5), on-screen OMR practice, Motion Studio optimization fallback and named Islamic reviewers (35), NexaLogicDev USD prices (32.9) and Acceptable Use Policy (36.5)
v31.4	21 October 2026	Deep-review corrections: 15.8 optimistic model confirmed as the investor base case with milestones (15.3), OKRs (22) and the consistency note aligned; Executive Summary, Product Portfolio and customer groups updated for the ecosystem; five new personas (PlayBoom guardian, freelancer, small business, digital agency, startup team) and a new persona distribution (21); ten new risks 41–50 (23); WebContainer Power mode for Team, agencies and module buyers; stale salary and infrastructure figures corrected (14.4, 15.2, 15.6); test-plan pointers (13.2)
v31.5	21 October 2026	Ecosystem resource sharing (new 8.17): every product may use any idle GPU or CPU capacity, owner first, with pool owners, lending conditions, a placement order per workload, the new P70 class for borrowed real-time work, hand-back, privacy and accounting rules; Dev Heavy servers lent whole in the 1–7 am window. SBX-3 added (26 servers: 7.2, 0.6, 8.5, 32.13). Video GPU order trigger at 60% (approved). Software register +4 rows (Resource Broker, Inference Gateway with Envoy Gateway, DCGM Exporter, OpenCost); architecture layer, Appendix A and risk 51 updated. Corrected risks 41, 42, 48 and 50, whose Probability and Severity values had been swapped in v31.4.

Version	Date	Changes
v31.6	22 October 2026	NexaLogicDev models and hardware: Heavy Lane MiniMax-M2.7 (one copy per Dev-H server; licence requested; GLM-5.2 fallback), Fast Lane Qwen3-Coder-Next on 4 × RTX PRO 6000 at Database Mart Dallas replacing the A6000 Fast servers; 28 servers (7.2, 0.6); software register (217 rows) and model registry, 32.1 architecture and honesty rule (Opus 5.5), 8.17 pools, Funding B (≈৳1.44 crore, contingency ≈৳5.6 lakh, break-even ≈570 paid users), 15.8 spend and risk 42 updated.
v31.7	22 October 2026	NexaLogicDev Heavy-lane order and API fallback (32.1): own servers first, MiniMax M3 API on local failover or when busy or down (paid users), Claude Sonnet 5.5 API after two failed attempts or for Complex Modules (Pro/Team, Auto-Architect); spend caps, secret removal (Gitleaks), “RemonLogic servers only” switch, privacy text (36.2), credit cost check (32.9), Funding B note (32.13), 8.17 placement, register 220 rows, model registry, architecture layer, D01 inventory and risk 52.
v31.8	22 October 2026	Lane access and API fallback redesigned (32.1): Free, Student and Hobby on the Fast Lane only; the MiniMax M3 API is the Fast-lane fallback (paid users); for Pro/Team and Auto-Architect, Claude Sonnet 5.5 reviews the plan when the Heavy Lane is unavailable (case 1), after two failures (case 2) and for every Complex Module, and completes the task after a third failure (case 3); hand-back, spend caps, privacy; plan conditions (32.9), unit check, 8.17 placement, register, model registry, architecture layer, 0.3 and 36.2 updated; Fast-lane FP8/4-bit choice left to the benchmark. Review fixes: 8.5 scale-out rows for the Heavy and Fast pools, Appendix A lane rule, Sonnet-use and API-spend KPIs (32.11).

Version	Date	Changes
v32.0	22 October 2026	Corporate Edition: designed cover, Document Control and Approvals, Executive Dashboard, Document Map and full two-level contents; approval-date and version tags moved out of the body into this log; NexaLogicDev capacity statement (32.13) and video-trigger note (8.17) updated.
v32.1	22 October 2026	Browser-first NexaLogicDev IDE (new 32.14): code editor, lint and format (Web Workers), previews (Sandpack, WebContainer, php-wasm), GrapesJS design canvas, voice to code (in-browser Whisper first) and offline mode (PWA, IndexedDB/OPFS, Yjs) run in the user's browser; the sandbox keeps AI loops, tests, scans and database backends; register +10 rows (230), architecture layer, 0.6, 7.2, 8.17, 32.2, 32.13 capacity and risk 53 updated.
v32.2	22 October 2026	Browser-first execution for all products (new 8.18): NexaTute adds OCR, pronunciation scoring, MCQ/numeric checking, Learning Insights view, own-notes search, lesson cache and image resizing in the browser; Motion Studio adds Enhanced (2.5D) render, captions and brand overlays, audio mixing, auto-reframe, Shorts detection, Quick/Enhanced MP4 assembly and on-device upscaling and background removal; NexaLogicDev adds Git (isomorphic-git); device tiers, Device Check, Data Saver, trust rule and measured browser offload; register 231 rows (two duplicate rows merged into the existing transformers.js/ONNX Runtime Web and WebLLM rows); 0.3, layer table, Appendix A, Executive Dashboard and risk 53 updated; owner roles confirmed.

Version	Date	Changes
v32.3	22 October 2026	Domains, brands, official channels and platform approvals (new 0.7): company and product domains (all owned) with nexalogicdev.app for customer sites on the Public Suffix List; official handles; YouTube, TikTok, Meta (Facebook and Instagram), WhatsApp, LinkedIn and NCMEC approvals with timing; Instagram Reels added to Motion Studio direct publishing (LinkedIn after launch); “Add to LinkedIn Profile” on certificates; 35.5 launch actions, platform-publishing register and architecture rows, Executive Dashboard and Document Map updated.
v32.4	23 October 2026	Product names: AI Teacher → Nexa Teacher (always shown with an “AI” label), AI Co-teacher → Nexa Co-Teacher, Study Hall AI moderator → Nexa Study Hall Moderator, renamed throughout the body; Nexa Co-Teacher gains spoken questions and read-aloud answers in the browser (no voice conversation); minors may ask the Moderator by voice; “Add to LinkedIn Profile” only on 18+ accounts; helpline numbers, Sponsor-a-Learner sponsors and Eid holiday dates managed in the admin panel; NexaTute hi-fi mockup (75 mobile and 10 desktop screens) referenced; new 0.8, 3.16, 31.1, 0.7 and Document Map updated.
v32.5	24 October 2026	Decisions of 23–24 October 2026. New 37 MindForge: thinking-skills practice for Class 4 to adults (not an IQ test), 12 feature groups, one learner record with the Nexa Teacher, Nexa Co-Teacher, Moderator and mock tests, syllabus–skill mapping, evidence rule, expected syllabus completion, consent and age rules, the 21 approved additions, AI-generated and solver-checked items with an IRT-calibrated measurement set (Free Beta = pilot), the international standards plan (EU AI Act high-risk from 2 December 2027, AERA/APA/NCME, ITC/ATP, IRT/DIF, norms and ethics, ISO/IEC 23894/27701, DPIA, UNICEF, UK AADC, COPPA, QTI 3.0, CASE, Open Badges 3.0, OECD PISA/ISCED, RCT/ESSA/Digital Promise, ISO 10667, EN 301 549) and KPIs;

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		MindForge Free and Pro (৳199 a month, $3, prepay ৳537/৳995, second child 30% off, institutions ৳139/৳119) in 11.11, 11.14 and the plans and modules table. New 38 Platform Administration, Institute Admin and Company Accounting: admin areas and the learners-and-usage dashboard; plans and modules table and platform-wide module switches; platform operations (X14); accounts, support and legal (X15); content quality and money incl. the Mushak 9.1 VAT report and the price-rule KPI panel (X16); growth (X17); Institute Admin with mentors, placement (18+) and limits; double-entry books, receivables and payables, partial payments both ways, payroll, rent and vendors, loans taken and given (automatic interest, bullet loans with part-payments), investors, share issue and share transfer; roles, approval rules, maker-cannot-approve and the full audit log; cloud export to Google Drive, OneDrive and Dropbox. Also: 31.3 Study Hall Q&A (free; images yes, links never; verification flow; Valkey cache with a PostgreSQL/pgvector copy; question bank after review; rewards and limits; Moderator answer after 30 minutes; safety for minors) replacing the community-forum row of 3.10; 30.8 PlayBoom additions (both rounds) and 30.9 curricula with the NCTB books for Class I–III; 3.26 renamed and extended into the RemonLogic Growth Platform for all three products (one CRM record per organisation, event-driven journeys with a 10% holdout, the target-client database of organisations, the country rules engine; 5.3, 0.3); Landing Page AI additions (3.25); calculator and dictionary (3.20); batches separated by teaching language (3.17); Nexa Guard named (0.8, 3.17, 10.2); no one-time override above the guardian's spending limit (11.9); NexaTute app in six languages from launch and the standing rule that every subject follows the learner's own country curriculum (29.2, 0.8); helplines set country-wise; 0.8 mockup table with clickable prototypes X1–X19 (incl. X19 AI Marketing Hub) and tooltips. Software register 231 → 239 rows: new rclone (moved out of the

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		NexaTute Box row; also the accounting export), pgaudit, Casbin, OR-Tools, jsQR, math.js, Open English WordNet, email-validator with dnspython, organisation open data (Wikidata, OpenStreetMap, government lists) and the optional Ahrefs or Semrush API (exception); Workbox, openpyxl, Scrapy, the notification service and the WhatsApp Business Platform become ecosystem-wide (E) and OpenCV shared (B); duplicates carried over from earlier editions merged so that each component has one row (AudioSeal and Semgrep rows merged; gVisor and ScanCode Toolkit listed once, now E); totals recounted from the table (the previous B/N counts had not been updated when two rows became E in v32.2); architecture rows in 3.9, 6.1, 8.6, 8.17, 8.18 and Appendix A; hardware unchanged (no new servers). Cost and revenue: a part-time MindForge psychometrician (≈ ৳35,000 a month from Month 2) and item reviewer (≈৳20,000 a month from Month 3), ≈৳9.2 lakh in all, paid inside the unchanged ৳5 crore Funding A by moving ৳10 lakh from Content (৳25 → ৳15 lakh) to Salaries (৳2.20 → ৳2.30 crore) — fees and the move to be confirmed; MindForge Pro in the 15.8 base case at a 15% attach rate (ARPU ≈৳180 incl. VAT): 18-month net revenue ≈ ৳6.60 → ≈৳6.68 crore, March 2028 ≈ ৳83 lakh a month, Month 12 ≈৳48.1 lakh a month, break-even months unchanged (14.2, 14.4, 15.2, 15.3, 15.8, 0.5, Executive Dashboard). Legal open items in 36.1, 36.4 and G.7 (EU assessment block under the AI Act; minors' cognitive profiling under the 2025 data ordinance; accounting and outreach questions). Fixes: the old name "AI Study Hall Moderator" left in 3.20 and in the 31 and 31.2 headings, and "an Nexa", corrected; the numbered-phase labels inherited from v31.2 reworded by meaning as the launch scope or launch release (17.1 roadmap rows and launch-scope note, the 17.5 heading and text, risk register 23, Executive Dashboard and this log) — no decision changed.

Version	Date	Changes
v32.6	25 October 2026	Decisions of 24–25 October 2026 that touch software, architecture and hardware placement. New 38.15 Admin notifications and internal staff chat (prototype X21; Institute side in X12): the bell with info, important and urgent levels, platform and Institute events, the notification control panel with routing by role or named user, a reminder after 4 hours and the next approver after 24 hours, quiet hours and digests; web push (pywebpush), the Android admin app through Firebase Cloud Messaging (exception) and optional email, SMS and WhatsApp through the existing notification service; staff-only chat with approved groups, voice and video calls only in the platform admin panel and never recorded, a compliance check on send (Qwen3Guard plus rules), virus-scanned files (ClamAV), encrypted daily AI summaries, messages kept 90 days and summaries 6 months (lawyer to confirm). New 38.16 Avatar & Voice Studio (X22; user side X23): GLB/VRM library with automatic checks (glTF-Validator), cultural review, two-person approval and lighter copies (glTF-Transform); six-language voice library; guardians' family voices for PlayBoom, verified teachers' voices and photo avatars (liveness and face match) and 18+ learners' own voices on the same Chatterbox/VITS engine and consent rules as Motion Studio, with SpeechBrain verification, AudioSeal and C2PA, off-peak on AI-2 Node A within a daily GPU-minute cap; one free of each and more from the marketplace, XP only under 18 (G.3; prices and the cap to confirm). People management in Roles and access (38.12). 3.25 Nexa Notes with videos (FFmpeg HLS at 720p/480p/240p on the CPU worker pool, Whisper and Qwen3 subtitles in six languages checked by reviewers, R2 + CDN, hls.js, click-to-load youtube-nocookie embeds, schema.org Article and VideoObject), the country and syllabus picker, six legal pages, University, Polytechnic and Vocational courses, sign-up and the Marketing cookie switch; 3.26 Channels (OAuth, tokens in

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		OpenBao; X left out until the owner decides), Import, AI target finding, Export with the exporter's watermark and the Target-clients list and map without external map tiles; Paddle's automatic tax by country, an editable Paddle fee and VAT 15/115 only on Bangladesh sales (11.2, 11.7, 11.14, 38.5; accountant to confirm). Software register 239 → 241 rows: new hls.js, glTF-Validator, glTF-Transform, ClamAV, Firebase Cloud Messaging (exception) and the X API (blocked until approved); 44 rows updated, product codes following their new users (Apache ECharts N → E; Qwen3-30B-A3B, vLLM, Docling, pypdf, WeasyPrint, the Cloudflare row, the platform-API row and the LinkedIn API B → E; c2pa-rs, Chromaprint and SpeechBrain M → B); true duplicates merged so that each component has one row — Web Push (VAPID) into the notification service, vLLM pipeline parallelism into vLLM, the NexaLogicDev Cloudflare Workers + R2 row into the Cloudflare row, Meta Graph API (Pages) into the platform-API row, and the second mentions of the LinkedIn API and the Bangladeshi SMS gateway moved into their own rows; libvmaf and FFmpeg rows renamed; 18.2 product codes aligned with 18.1; totals recounted from the table. Also 0.8 (X20–X23; 23 clickable prototypes), 00, 3.9, 3.10, 4.7, 6.1, 6.2, 8.6, 8.17 (two placement rows), 9.1, 9.4, 10.1, 10.2, 13.2, G.7, Appendix A, the Executive Dashboard and the Document Map. Hardware unchanged — 16 launch servers, App-3 and the 28-server register; every new workload is placed on an existing node (7.1): HLS transcoding on App-1–App-3 at P10 in the night window (never Core-1), voice clones and subtitles on AI-2 Node A off-peak, Nexa Notes video growth in R2 of ≈20–40 GB a year (estimate).
v32.7	25 October 2026	Owner decisions of 25 October 2026 made after v32.6. New 3.27 Tests, Written-Answer Checking and Exam Integrity (prototype X7; admin in X10, X14 and X16): true/false with an optional "why" and exam-rule negative marking, fill in the blank checked automatically (spelling

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		tolerance with the correct spelling shown, Bangla and English digits, units, equivalent mathematical answers with math.js and SymPy), matching and ordering, short answers and written tests by photo or scan of several pages or by typing; handwriting read by Qwen3-VL-8B on our own GPUs, unsure parts confirmed by the learner before marking, part-wise marks by the board scheme, a model full-mark answer, mistakes into the error notebook and a human recheck (free once per test, then 2 more a day); results in 10–15 minutes, queued with a notification at the 4–9 pm peak, scripts never sent to an outside AI service; board-style mixed tests per country curriculum. Exam recovery (IndexedDB autosave, 10-second sync, timer rules), shuffled questions and options with sets A/B/C and an answer-similarity check for human review, adaptive tests only for placement and smart practice (IRT items with 200+ answers, about 10–20 questions), the practice-exam “not recognised by any education board” note (also in the Terms, 19.4), accent-fair speaking marks, coding tests on our own gVisor runner (not Judge0), the paper and blueprint builder, certificate levels and the retake rule, limited live proctoring (certificate and institution exams only; no video or snapshots under 18; no emotion or attention reading), employer and organisation assessments (18+, ISO 10667, never the only basis for hiring; free for BCS + Job Test Preparation and other job-test learners, otherwise the Recruitment tests add-on at ৳399 a month ($6) for up to 3 tests a day, 18+ only, 11.14) and the institution roster and results API with OAuth scopes and webhooks beside LTI 1.3 (18.3). Error notebook in the 1:1 class (3.18; X2, X8): toolbar button and side panel, the after-class count, the full notebook page, Practise now, the home card and only mistake counts in the guardian report. Admin “Tests & marking”: written-check limits per plan — free tier 1 a week, Daily Pass 2, Batch 4, Combo 7 and Personal 10 a day, institutions 4 (to confirm) — with

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		true/false, fill-in and matching unlimited and a platform switch (38.2); the written-check GPU queue with a daily cap of 1,500 GPU-minutes and an 80% alert (default, editable, 38.3); marking schemes with versions and approval, the 24-hour recheck queue (to confirm), the AI-marking audit, fill-in and error-notebook rules (38.5); the Institute Admin academics row (38.7); 38.1 and 38.14. Values marked "(default, editable)" are starting settings in the admin panel. Software register unchanged at 241 rows — no new component was needed; 21 rows extended (Qwen3-VL-8B, Qwen3-30B-A3B, math.js, SymPy, pyBKT/girth, gVisor, Puppeteer/Chromium M → B, CodeMirror 6, datasketch, MediaPipe, LiveKit, the IndexedDB row, the NSFW classifier, HMA, wav2vec2, Keycloak, APISIX, PyLTI1p3, WeasyPrint, the notification service, Unleash) and the 18.2 Qwen3-VL row; totals recounted. Also 0.8, 3.4, 3.9, 3.25 (the recruitment-test offer on the landing page: plan add-ons, the BCS + Job course card, an employer block and the FAQ), 3.10 (national live model tests now use shuffling and similarity review instead of proctoring; answer-script checking is a queued job, no longer P100), 3.11, 3.12, 3.21, 6.1, 6.2, 8.6, 8.17 (two placement rows), 9.1, 10.2, 11.1, 11.14 (Recruitment tests add-on row), 13.2, 19.4, 19.7 (assessment-integrity row, new workplace-assessment row), G.7 and Appendix A. Hardware unchanged — no new servers: written checks run on the launch GPU pool with AI-2 Node A first, and graded coding runs in the existing practice-lab gVisor sandboxes (7.1).
v32.8	29 October 2026	Owner decisions of 26–29 October 2026 and an A-to-Z check of every memory line. New dates: NexaTute Free Beta 20 November–9 December 2026 (paid plans from 1 January 2027); Remon Motion Studio Free Beta from 20 December 2026 (replaces 10 November; paid start to confirm). New 3.28 ("1:50 AI class", microphones off with Raise hand, one follow-up, auto-off after

Version	Date	Changes
		every turn, the disturbance rule, LiveKit class features, weak-network and weak-phone help, voice channels without SIP, guardian SMS free and on), 3.29 (recording by choice, library, class reports, mistake timeline, per-class improvement, Main Student Dashboard, Mastery Map, All Exams), 3.30 (Exam & Admission Hub ideas 1–54, record statuses, past-question and centre-upload rules), 3.31 (Bangla Sign Language avatar, Sensory-friendly mode, Calm mode, Voice mode, 20-20-20), 3.32 (Team Quest, Quiz Battle, Live Game Show Saturday 7:00 pm, the 2027 engagement, strategic and trust features with the free and paid placement), 3.33 (Free-Beta safety, What's new sheet, status page), 17.6 (roadmap 2027–2028), 36.6 (draft legal opinion, lawyer-list decisions, legal pages, security-disclosure page, legal go-live checklist), 38.17 (every user-visible value, the Plans & prices editor and the 22 + 17 admin gaps) and 38.18 (software updates with two-person approval and automatic rollback, staging load test, LiveKit and delivery panels, synthetic checks, OTP protection, service budgets). Updated in place: every NexaTute feature once dated Q2 or Q4 2027 is in the launch release (3.10, 3.12, 3.18, 17); unlimited admission (8.5, 11.12, 13.2, 23); storage by plan with add-ons for all three products (G.3; 11.3, 11.5, 11.11, 32.9); app-store billing with a separate 30% charge line (11.7, 36.3); English-medium Class 4–8 prices, crash programs and the Doubt Pass (11.1, 11.2, 11.12, 11.14); USD prices with .5 rounding down (11.2); the learning-guarantee terms (11.4); the Personal Data Protection Act 2026 (19, 36, 38); the CDN rule for private recordings and the video delivery chain (9.4); LiveKit infrastructure (9.5); the 25 October test values confirmed (3.27, 38.2, 38.5, 9.1); and the gap-check fixes in 0.8, 3.25, 3.26, 8.9, 8.16, 19.2, 19.4, 19.6, 29.2, 30.8, 31.3, 37.4 and 38.4–38.15. Software register: 3 new rows (MindAR and model-viewer, Prometheus Blackbox Exporter, Argo Rollouts), 5 rows renamed or extended by merging (FFmpeg +

Version	Date	Changes
		Shaka Packager, hls.js + Video.js, LiveKit + Ingress, LiveKit Agents without SIP, the SMS/WhatsApp row + licensed voice provider and IPTSP line) and 15 more rows extended. Hardware unchanged — no new servers: LiveKit already runs on two nodes (Core-1, Core-2), and every new feature uses existing capacity (7.1).

Version	Date	Changes
v32.9	29 October 2026	Owner decisions of 29 October 2026 made after v32.8. New 38.19 Admission limits, waiting list and capacity guard (X25, X14; learner, guardian, institution and landing-page states): no limit by default with a new 1:50 batch every 50 learners; optional limits by class or course, country and teaching language; a waiting list with a 48-hour offer hold, free SMS, push and email, the guardian deciding under 18 and no payment until a place is accepted; a capacity guard at 75% of the load-test-proven capacity (release below 60%) that never affects enrolled learners or live classes; one master “All new admissions” switch with an emergency close that reverts unless a second admin confirms within 60 minutes; institution contract seats unaffected unless ticked; an admissions message written once and AI-translated into all six languages, checked by the second approver; no scarcity wording. New 38.20 Nexa Co-Admin (X27): an AI assistant acting only within the asking admin’s own role and permissions, with preview and confirmation, the normal second approval for money (never itself the second approver), per-role levels, a fixed never-allowed list of 6, an append-only activity log kept 7 years (to confirm), the own-server model first and daily caps; Co-Admin for staff on WhatsApp, Messenger and imo — each switchable, all off until the lawyer and DPO check, imo locked off until business API access is confirmed — with passkey-linked accounts, masked answers, secure-link confirmation, a 15-minute timeout, 20 actions a day and every message logged by channel. Also 0.8 (prototype structure: legal pages X24, admin X10/X25/X26, Co-Admin X27, every board ≤ 1.8 MB), 3.25, 7.1, 8.5, 11.2 (USD rounding to confirm: ৳500 = $8 in the approved list vs $7 by the .5-down rule), 13.2, 17.6, 36.6, 38.1, 38.17 and 38.18. Software register unchanged at 244 rows — imo business messaging joins the WhatsApp/SMS row and the Messenger Platform the Meta Graph API row; Keycloak, Casbin, Qwen3-30B-A3B and Locust rows extended. Hardware unchanged —

Version	Date	Changes
v32.10	29 October 2026	Owner decision of 29 October 2026 (15:53): ৳500 = $7 — the USD rule (Taka × 0.015, nearest dollar, an exact .5 rounds down) applies to every USD price with no exception; the "(to confirm)" note of v32.9 is removed and the rule is stated as final (11.2). Every NexaTute USD figure was checked against the rule. Changed: Class 6–8 Batch $8 → $7; Personal single $8 → $7 for Class 4–5, Class 6–8, Class 9–10/SSC and SSC Vocational, HSC, HSC Vocational and Admission Test Preparation, and BCS + Job Test Preparation (11.2); Nexa Co-Teacher Pass $2 → $1 (৳100 × 0.015 = $1.50; 3.22, 11.14); the USD prepay example now reads Class 6–8 Batch 3 months $18, 6 months $35 (11.2). All other USD prices already follow the rule; the international per-minute wallet rates ($0.03 / $0.05 / $0.08) and wallet top-ups are separately approved rates, not conversions; Motion Studio and NexaLogicDev keep their +35% USD rule. Software register and hardware unchanged.

Appendix E. NexaTute Feature Reconciliation Audit

Integrated from Audit Report v1.0 | 15 October 2026

Purpose of this appendix: provide a consolidated, preliminary feature-reconciliation audit alongside the RemonLogic Master Project Proposal. The companion Excel matrix is the row-level working register; this appendix is the narrative audit summary.

Compared sources: Master v26.3, NexaTute Complete Master Document v19.0, and NexaTute Architecture v19 Reconciled v24.0.

Purpose

This report provides a preliminary feature inventory to compare legacy NexaTute feature terminology with the consolidated Master document. It is not a claim that every feature has been implemented or fully specified.

Audit method

A working inventory was created across learner experience, AI tutor/live learning, assessment, content/curriculum, interactive learning, analytics, administration, billing, and infrastructure/security. Each item was checked for textual mention in the legacy and current documents. Exact-term matching is useful for triage but cannot prove full semantic coverage.

Preliminary findings

The working matrix contains 116 feature/capability rows. Rows are marked as potentially missing when a term is found in the legacy source but not exact-matched in Master v26.3; rows appearing in

both are marked partial/needs verification because a name match does not prove a complete workflow.

Key review areas: Learning Pattern Analysis, Teacher Suggestions, Adaptive Class Modification, Per-Class Evaluation, KPI definitions, stakeholder dashboards, curriculum/topic mapping, content QA, assessment-to-remediation loop, and operational acceptance criteria.

Known reconciliation issue

The legacy documents mention differing feature/KPI totals in different contexts (including 272 features / 816 mode combinations and 42 or 46 KPIs). These counts must be reconciled against a canonical register; they should not be presented as confirmed until definitions and duplicate handling are settled.

Group Study billing unit

The pricing table defines Group Study rates per student per minute (৳/student/min). This unit is explicitly clarified in Master v26.3; the obsolete per-group-versus-per-student clarification line has been removed. This item is therefore treated as resolved in v26.3.

Recommended next step

Treat the spreadsheet as a triage matrix. Review each row against source sections, assign a stable feature ID, define acceptance criteria, and record evidence (section/page) before declaring any feature complete or missing.

Audit Control and Use

This is a preliminary reconciliation, not a certification that features are implemented or production-ready. Exact-term matching can miss synonyms and context. Use the companion spreadsheet “NexaTute_Feature_by_Feature_Reconciliation_Matrix_v1.0.xlsx” as a triage register; confirm every row against authoritative source sections, record evidence and acceptance criteria, and update statuses only after review. Do not present the 116 working rows as the complete NexaTute feature count.

Appendix F. Remon Motion Studio — Complete Features Reference

This appendix contains the Motion Studio feature scope only. Package fees, per-minute rates, recording, storage entitlements and metering are governed by G.3, G.4, G.8 and G.9.

F.1 Executive Summary

Remon Motion Studio is an AI-powered video production platform covering idea, script, storyboard, cinematic and animated production, drama series, editing, safety verification, rendering and secure delivery. This consolidated document includes the final package fees and per-minute rates, core feature catalogue, highlighted content formats and billing rules.

Billing, packages, per-minute rates, recording and metering are defined in G.8 (remakes in G.4); this feature appendix does not override the commercial authority.

F.2 Feature Access by Package

F.2 is derived from the package baseline in G.8. The package names, monthly fees, supported output tiers and maximum video lengths below mirror G.8; G.8 remains the sole authority for per-minute rates, recording, metering and storage policy. F.2 records feature access only and does not create a separate price or overage rule. Packages have no included minute quota; generated minutes are billed at the G.8 per-minute rates, and each project that uses AI pre-production carries the pre-production fee (4.17). Each package is one subscription covering 3D Animation Video, Cinematic Video and Islamic Content; only the per-minute rate changes with the render mode (G.8.1 for Quick/Enhanced, G.8.2 for Cinematic, less 25% for eligible Islamic Content under G.9).

Package	G.8 package anchor	Feature-access implication
Free Trial Package	৳0; 7 days; 1 min/day; maximum 15 sec per video; 480p Native and 720p Escalating only (G.11)	Freelancer Starter features in Quick and Enhanced modes; watermarked output; no recording
Freelancer Mini	৳499/month; 480p Native, 720p Escalating, 720p Native and 1080p Native; 5 sec–45 sec	Basic script/storyboard/video workflow; Reels & Shorts (up to 45 sec); limited Brand Kit and remake workflow.
Freelancer Starter	৳999/month; 480p Native, 720p Escalating, 720p Native, 1080p Escalating and 1080p Native; 5 sec–2 min	Basic video generation/editing, script/storyboard, Reels & Shorts (up to 2 min), standard remake workflow and project management.
Premium Starter	৳1,999/month; 480p Native, 720p Escalating, 720p Native, 1080p Escalating and 1080p Native; 5 sec–5 min	Premium short-form, Brand Kit, drama-series recap/preview/teaser/trailer/highlights and repurposing.
Premium Professional	৳3,999/month; all G.8 output tiers; 5 sec–10 min	Document-to-Video, Natural-Language Editing, Advanced Audio Direction and analytics where supported.
Premium Agency	৳7,999/month; 720p Native, 1080p Escalating and 1080p Native; 5 sec–20 min	Multi-client review/approval, advanced publishing/reporting, plan-limited API and webhooks.
Islamic Content (within any package)	Same fee as the chosen package; per-minute rates −25% (G.9)	All features of the chosen package plus Islamic source verification (Quran ayah/translation, Hadith reference/authenticity, sensitive-depiction review)

Package	G.8 package anchor	Feature-access implication
Enterprise	Custom fee, per-minute rates and scope under G.8 contract	Custom features, integrations, API/webhooks and approved Interactive Branching Drama scope.

Feature availability by package:

Feature	Freelancer Mini	Freelancer Starter	Premium Starter	Premium Professional	Premium Agency	Enterprise
Core script/storyboard/video workflow	Included	Included	Included	Included	Included	Custom
Free optional recording	Included	Included	Included	Included	Included	Included
Islamic content workflow	Included	Included	Included	Included	Included	Included
Cinematic/animated drama	Included	Core workflow	Core workflow	Core workflow	Core workflow	Custom
Drama-series repurposing	Basic	Basic workflow	Included	Included	Included	Custom
Reels & Shorts Generator	Included	Included	Included	Included	Included	Included
Brand Kit, intro/outro & thumbnails	Limited	Basic export only	Included	Included	Included	Custom
Document-to-Video	—	—	—	Included	Included	Custom
Natural-Language Editing	—	Basic workflow	Basic workflow	Included	Included	Custom
Advanced Audio Direction	—	—	—	Included	Included	Custom
Content Performance Analytics	—	—	—	Where supported	Where supported	Custom
Multi-client review/approval	—	—	—	—	Included	Custom
Advanced publishing/reporting	—	—	—	—	Included	Custom
API & webhooks	—	—	—	—	Plan-limited	Custom
Interactive Branching Drama	—	—	—	—	—	Beta/approved scope

F.2.1 Freelancer Mini

- Basic editing and limited standard remake workflow.
- Islamic Content Creation subject to required verification and applicable 25% discount policy.
- Reels & Shorts Generator (up to 45 seconds), basic intro-outro and thumbnail workflow.
- Basic AI video generation, script/storyboard, basic cinematic and animated video workflow.
- Each generated video follows the G.8 Freelancer Mini length range of 5 seconds to 45 seconds; longer requests must be rejected by validation.

F.2.2 Freelancer Starter

- AI script/storyboard, basic video generation and basic editing.
- Standard remake workflow and project management.
- Reels & Shorts Generator (up to 2 minutes).
- Islamic Content Creation subject to required verification.

F.2.3 Premium Starter

- All Freelancer Starter capabilities.
- Reels & Shorts Generator; Brand Kit and Visual Identity.
- Drama-series recap, next-episode preview, teaser, trailer and highlights.

F.2.4 Premium Professional

- All Premium Starter capabilities.
- Document-to-Video for supported PDF, DOCX, PowerPoint and URL inputs.
- Natural-Language Editing Assistant and Advanced Audio Direction.
- Content Performance Analytics where platform data is available.

F.2.5 Premium Agency

- All Premium Professional capabilities.
- Multi-client workflow, client review/approval, advanced publishing and reporting.
- Plan-limited API and webhooks subject to security and integration permissions.

F.2.6 Enterprise

- Custom per-minute rates, features and integrations.
- Enterprise API/webhooks and custom workflows subject to agreement.
- Interactive Branching Drama according to approved release scope.
- Contact the Sales Team for a custom quotation.

F.3 Highlighted Content Creation Features

F.3.1 Cinematic Drama Creation

- Cinematic storytelling, dramatic scenes, camera framing/movement, lighting and composition.
- Dialogue, emotional beats, scene-by-scene generation and short-film workflows.
- Visual style options subject to the selected model, package and output validation.

F.3.2 Animated Drama Creation

- 2D/3D animation-style stories and expressive animated characters.
- Animated environments, sequencing scenes, dialogue/narration and visual continuity.
- Fantasy, adventure, family, educational and other suitable story formats.

F.3.3 Drama Series Production

- Episode-based scripts, storyboards and production workflows.
- Recurring character and world references to support continuity across episodes.
- Opening recap, next-episode preview, trailer, teaser, highlights and best moments (made and charged as in 4.18).
- Season and episode content repurpose according to plan entitlements.

F.3.4 Reels & Shorts Generator

- Vertical 9:16 short-form video workflows, with 16:9, 1:1 and 4:5 versions from the same project (4.10).
- Configurable 15–180 second target duration, capped by the package’s maximum video length in G.8 (45 seconds on Freelancer Mini; 2 minutes on Freelancer Starter).
- Hooks, captions/subtitles, highlights and multiple short-form versions.
- Repurpose suitable scenes from longer dramas or videos, including the creator's own uploaded or imported videos (4.11) under the copyright controls in 4.12.

F.3.5 Cinematic Intro & Outro Creator

- Animated logo reveals, cinematic opening titles and branded opening sequences.
- Outro/end screens and follow/subscribe calls to action.
- Transitions, AI music and the sound-effect library with auto-ducking (4.10).
- Reusable templates are integrated with Brand Kit (4.18).

F.3.6 AI Thumbnail Creator

- Thumbnail concepts from a video, episode, script or topic (4.18).
- Character close-ups, dramatic composition, title placement and brand styling.
- Episode-specific variants and platform-appropriate export dimensions.
- Review text rendering and typography before publishing.

F.4 Core Production Workflow

1. Idea, prompt, brief or supported document intake.
2. AI script generation and review.
3. Scene-by-scene storyboard, camera/visual direction and planning.
4. Character & World Bible references for continuity.
5. Common Safety Verification and Islamic-specific verification when selected.
6. AI Director orchestration and scene-by-scene video generation.
7. Voice, narration, multilingual subtitles, music and sound effects.
8. Editing, scene reorder, targeted regeneration and remake.
9. Rendering, final MP4 compilation, output validation and secure delivery.
10. Usage ledger, minute-usage reconciliation, job status and project history.

F.5 Consolidated Feature Catalogue

Feature group	Consolidated features
Idea & pre-production	Prompt/brief intake; script and dialogue generation; storyboard; scene breakdown; camera/lighting direction; approval and revision history.
Character & world continuity	Character references; appearance/costume consistency; locations; props; world references; reusable assets.
Video production	Cinematic drama; animated drama; short-film workflow; drama-series episodes; scene-by-scene generation; targeted scene regeneration.
Short-form & repurposing	Reels/Shorts; recap; next-episode preview; teaser; trailer; highlights; best moments; multiple short-form variants.
Branding & packaging	Brand Kit; logo/colour/font/watermark; cinematic intro; outro/end screen; thumbnail concepts; reusable templates.
Voice & audio	Text-to-speech; permitted voice cloning; transcription; subtitles/captions; emotional audio direction; music/SFX balancing.
Editing & remake	Natural-language editing; script/dialogue changes; scene reorder; clip extraction; supported voice/language changes; preview and approval.

Feature group	Consolidated features
Documents & inputs	Supported PDF, DOCX, PowerPoint and URL to script/storyboard/video workflow.
Publishing & analytics	Scheduling; publishing status; retry where APIs allow; views/watch time/retention/engagement insights where data is available.
Collaboration & developer tools	Multi-client workspace; client review/approval; plan-limited API; webhooks; enterprise integrations.
Safety & verification	Common safety verification; Islamic source tracing; rights/consent checks; plan mismatch guard; verification cache; human-review queue.
Rendering & delivery	Durable jobs; retry/resume; FFmpeg/GStreamer; NVENC where supported; final MP4 validation; Cloudflare R2; secure signed delivery.
Billing & reliability	Plan entitlements; minute-usage ledger; atomic reservations; idempotency; usage estimates; failed-job restoration policy; monitoring.

F.6 Technical Architecture & Delivery

- Dashboard/API for projects, plans, scripts, reviews and approvals.
- Core application for authentication, billing, minute-usage ledger and policy guards.
- AI orchestration and durable job queue for scripts, storyboards, verification and production.
- Rendering, assembly, upload and cleanup follow 6.3 and 9.3.
- Do not send heavy scratch/intermediate file writes over the network to Core-1.
- After final MP4 compilation and validation, a background task transfers only the finished file to Cloudflare R2; delivery and the 72-hour link rule follow 9.4.
- Use accepted output duration for video-minute accounting, not GPU processing time.
- Use durable job state, retries/resumption, idempotency, observability and atomic credit reservations.
- Benchmark real output resolution, throughput, GPU cost and verify model/software licensing before production claims.
- Delivery-speed tiers (G.8.3): Express (+20%, high priority with a calculated ETA), Normal (within 12 hours) and Slow (within 24 hours, 10% off) apply to paid server renders; Quick mode renders in the browser and is exempt; the usage ledger records the tier.

F.7 Content Plan Safety Rules

- User selects Islamic Content Creation or General Content Creation at the start because verification requirements differ.
- Every prompt passes Common Safety Verification; generated frames, audio and uploads are also checked (4.15).
- Islamic plan adds religious source tracing and claim verification.
- If General plan is selected for clearly Islamic content, explain the difference and offer a plan switch or proceeding with stated verification scope; never silently switch.
- Run script-level verification after script generation and before storyboard/video generation.
- Cache verification results by prompt/content and policy version where appropriate.
- Record selected plan and billing entitlement per job.
- Maintain a human-review queue for uncertain or sensitive cases.

Appendix G. Final Cross-Product Completeness & Release Checklist

G.1 Cross-Product Completeness Checklist

- NexaTute sections, architecture, monthly Batch, Combo and Personal plans with wallet top-ups, the v2.0 price model, popular courses, group-study billing unit, B2B terms, and launch policies are present and reviewed.
- Capacity figures and feature availability are labelled as design targets/planned scope unless benchmarked or production-verified.

G.2 Remon Motion Studio Pricing Authority

Motion Studio is priced separately from NexaTute. Appendix F keeps the feature catalogue only. Appendix G.3 governs shared entitlements, G.4 governs remakes and Motion Studio credits, G.8 is the single package/per-minute rate/recording authority, and G.9 governs the Islamic-content discount.

G.3 Cross-Platform Storage, Character & Voice Entitlements

This section is the single canonical policy for NexaTute, Remon Motion Studio and NexaLogicDev storage, and for NexaTute and Motion Studio characters and voices. Product sections must reference G.3 rather than duplicate storage, character or voice-entitlement rules.

Included storage

Included storage (decided 29 October 2026; replaces the flat 10 GB per package): storage is given by plan or package size, plus paid storage add-ons, on all three products. It keeps users from needless uploads and recordings. Every value is an admin-panel setting with two-person approval (38.17).

NexaTute plan	Storage
Free user, Free Trial, Daily Pass (and Free Beta users)	1 GB
Nexa Co-Teacher Pass; MindForge Pro	2 GB
Batch; PlayBoom	6 GB
Combo	8 GB
Personal	10 GB (the same for every course; storage follows the plan type)
Active Crash Pack or crash program	+2 GB
Several plans	The largest allowance counts (plus a Crash Pack)
Institution	A pooled 50 GB + 1 GB per seat

NexaTute rules: warnings at 80% and 95%; when the storage is full, before a new recording or upload the learner chooses “delete old items” or “buy an add-on” (nothing is deleted automatically beyond the 12-month recording rule, 3.29); files are kept 30 days after a plan ends, with download allowed and a notice before deletion; a storage meter for the learner, the guardian and the institution; add-ons under 18 need the guardian’s approval; a store purchase shows the website and app-store lines (11.7).

Storage add-on (per month, all products)	Price
+20 GB	৳80 ($1)
+50 GB	৳180 (NexaTute $3; Motion Studio and NexaLogicDev $2)
+100 GB	৳300 (NexaTute $4; Motion Studio and NexaLogicDev $3)

NexaTute USD prices follow BDT × 0.015 (11.2); Motion Studio and NexaLogicDev follow their +35% USD rule, as approved. Remon Motion Studio (same model, approved 29 October 2026): Free and Free Trial 1 GB, Freelancer Mini 5 GB, Freelancer Starter 10 GB, Premium Starter 20 GB, Premium Professional 40 GB, Agency 80 GB; no-storage delivery keeps the 72-hour emailed download link (below). NexaLogicDev (same model, approved 29 October 2026): Free 1 GB, Student 3 GB, Hobby 5 GB, Pro 25 GB, Team 100 GB pooled for 5 seats (32.9). The included quota is tied to the active plan or package and is not a promise of permanent storage.

No-storage delivery: clients without available storage (no active package storage or the allowance is full) receive a 72-hour emailed download link and the file is then deleted (9.4).

Lifetime character creation allowance and add-on

For each platform, the first 2 character creations per user account are free for lifetime use. The 3rd and every subsequent character creation costs ৳150 per character as a one-time lifetime charge. This explicitly supersedes the earlier draft that proposed 3 free character creations; the approved allowance is 2 free characters, not 3.

Lifetime voice-cloning allowance and add-on

For each platform, the first 2 voice-cloning creations per user account are free for lifetime use. The 3rd and every subsequent voice-cloning creation costs ৳150 per voice clone as a one-time lifetime charge. Voice cloning still requires documented speaker consent, source-rights evidence, intended-use records and applicable abuse/safety checks.

NexaTute uses the same character-creation and voice-cloning engine as Motion Studio. A character or voice already created on one platform can be linked to the other with the same account without a new charge (proposed); creating a new one uses that platform's allowance above.

NexaTute allowances (Avatar & Voice Studio, decided 24 October 2026): in NexaTute the two-free / ৳150 rule above is replaced — Premium avatars, a guardian's family voice, a verified teacher's voice and photo avatar, and an 18+ learner's own voice each have one free, and more are bought in the NexaTute marketplace (prices to confirm, 38.16); learners under 18 get one Premium avatar free and unlock others only with XP, never with money. Remon Motion Studio keeps the rule above.

Checkout, entitlement tracking and anti-duplication controls

Track lifetime free-use counts and paid add-on purchases durably per user account and platform. Before creating a character or voice clone beyond the free allowance, show the count, ৳150 one-time charge and lifetime entitlement terms, then require explicit confirmation. Reserve/record the entitlement atomically, prevent retries or duplicate requests from charging or consuming allowance twice, and provide a clear transaction history. Do not create duplicate copies of this policy in other pricing tables; use a reference to G.3.

G.4 Hybrid Billing and Operating Rules

- Hybrid billing: apply eligible free allowances and available purchased Motion Studio credits first. If no eligible allowance or credit balance is available, show the applicable per-minute remake

rate and estimated total at checkout before charging. No monthly remake package, package renewal, or unlimited package is offered.

- R1 is 30%, R2 is 50%, R3 is 75%, and R4 is 100% of the original production tier's base rate. R5 is 91% of base (130% former rate reduced by 30%); no separate R5 surcharge applies.
- Remake levels (proposed definitions; commercial sign-off required): R1 Basic edit — text/script, caption, subtitle or scene-order changes with light re-render; R2 Partial regeneration — selected scenes, music or voice replaced at the same tier; R3 Major regeneration — most scenes regenerated at the same tier; R4 Full remake — whole video regenerated at the original tier; R5 Multilingual remake — a new language/voice version of the full video.
- The R1 basic-edit allowance is limited to 3 per video and R2 to 2 per video; R3 is limited to 1 per video. Paid Motion Studio credits have 30-day validity from purchase and do not renew; unused credits expire at expiry.
- R2 and R3 same-tier eligibility/discount applies only when the remake is based on the same production tier as the original video.
- R4 uses the original tier's base rate.
- R5 is calculated as base rate × 0.91. Apply the 25% Islamic discount to that exact amount when the Islamic verification requirements are met, then round to the nearest whole taka; 0.50 rounds upward. No additional surcharge is added.
- Islamic discount eligibility requires both common safety verification and Islamic religious verification; the selected content plan must not be silently switched.
- Before any paid remake starts, perform entitlement checks and reserve credits atomically. Release reservations on pre-processing failure; settle only the measured/authorized work and prevent duplicate charging.
- Free remake is limited to low-compute text/script and scene/layout edits that do not trigger heavy GPU rendering, paid voice cloning, or new 3D asset generation.
- Show the estimated charge and obtain explicit confirmation before queueing paid GPU/TTS work.
- Purchased Motion Studio credits are not transferable unless a signed contract explicitly states otherwise. Show credit expiry and the applicable refund treatment (G.10) before checkout; do not refer to package renewal or package credits.

G.5 Cost and Billing Examples

Example	Calculation	Charge
2-minute 3D Animation Freelancer Starter 720p Native reference, no recording	2 × ৳45	৳90
2-minute 3D Animation Freelancer Starter 720p Native reference, recording enabled	2 × ৳45	৳90
2-minute 3D Animation Islamic Freelancer Starter 720p Native reference, no recording	2 × rounded ৳34	৳68
2-minute 3D Animation Islamic Freelancer Starter 720p Native reference, recording enabled	2 × rounded ৳34	৳68
2-minute Cinematic Freelancer Starter 720p Native reference	2 × ৳61	৳122
2-minute Islamic Cinematic Freelancer Starter 720p Native reference	2 × rounded ৳46 (Islamic 3D Animation ৳34 × 1.35 = 45.9)	৳92

Examples use the G.8 base-rate references, the Freelancer Starter maximum video length of 2 minutes and whole-taka rounding. They are illustrative billing examples, not additional overage charges.

G.6 Unit Economics Checklist

- Model cost by task type, prompt/context size, GPU time, fallback provider and retries.
- Measure storage, recording egress, bandwidth, CDN, payment fees, taxes, refunds and support.
- Separate free/trial cost from paid-production cost.
- Calculate gross margin using measured utilization and actual provider prices.
- Do not claim savings or margins until measured.

G.7 Open Decisions Before Launch

- Publish the exact 30-day expiry timestamp and timezone at checkout and in receipts.
- Confirm applicable legal, tax, payment-provider, privacy and data-retention requirements.
- VAT (Mushak) invoices for Motion Studio business clients are decided with the deferred Bangladesh legal and tax items (4.16).
- Confirm aamarPay merchant eligibility and recurring-payment support for each method in Bangladesh and the USA LLC's Paddle seller account for international payments (11.7); where auto-debit is not available, use renewal reminders, a 3-day grace period and wallet top-up (4.14).
- Confirm with a Bangladeshi lawyer: profiling of minors, the consent switches and the transfer of learners' data to the US servers under the Personal Data Protection Act 2026 (19.4).
- Ask the lawyer (MindForge, 37.9): whether MindForge's assessment part must stay blocked for EU learners until the EU AI Act high-risk obligations are met (from 2 December 2027), while practice runs; and whether the Personal Data Protection Act 2026 allows cognitive profiling of minors with guardian consent.
- Confirm the MindForge part-time psychometrician and item reviewer fees (estimates ≈৳35,000 and ≈৳20,000 a month) and the ৳10 lakh move from Content to Salaries (14.4, 15.2).
- Check the PlayBoom book list against NCTB's official list before content is made, and decide whether to add the Class III Arabic book (30.9).
- Accounting (38): the chartered accountant sets tax and VAT rates, chooses accrual or cash basis and confirms the provision rule for loans 90 days overdue; the lawyer confirms voting and ownership of partly paid shares under the Articles and the share-transfer checklist (RJSC Form 117, stamp duty).
- Growth Platform (3.26): the lawyer confirms Bangladesh's rules for outreach to organisations and the lawful basis recorded for each target-client field.
- Staff chat (38.15): the lawyer confirms the monitoring notice under the Personal Data Protection Act 2026, the 90-day message and 6-month summary retention, and whether an institute's chat summaries stay with RemonLogic or in the institute's own archive.
- VAT (11.7, 38.5): the chartered accountant confirms that Paddle revenue is at 0% as an export of services and that VAT 15/115 applies only to Bangladesh (aamarPay) sales.
- Tests (3.27): the lawyer confirms the consent text and 30-day retention of adult webcam snapshots, the guardian-consent wording for presence checks under 18, the practice-exam note and the terms of employer assessments (18+, candidate consent, ISO 10667, never the only basis for hiring).
- Confirm with an accountant: tax withheld on affiliate and marketplace payouts and VAT on gift cards (11.14).
- Company registration checklist for the accountant and lawyer: trade licence, TIN, VAT registration (BIN) so that VAT-inclusive prices can be charged, aamarPay merchant KYC, digital business registration if needed, and any registration the National Data Governance Authority requires under the new data law.

- Confirm with the lawyer that a Play Store app without in-app purchase, with buying on the website, meets Google's payments policy in Bangladesh (11.7).
- Confirm with the lawyer that B2B outreach to organisations' official addresses meets the data law, and set up the outreach domain with SPF, DKIM and DMARC before the first campaign (3.26).
- Confirm the Express surcharge (+20%), its ETA accuracy and the late-delivery refund rate (G.8.3) from the launch load data; the Night-render seat limit was removed with the time-of-day rates on 21 October 2026.
- Document the Islamic-content verification workflow and audit trail for discount eligibility.
- Define voice-cloning consent and rights workflow.
- Set B2B class, AI, recording, storage, overage rates and any contracted exceptions.
- Confirm the Cinematic Video per-minute rates (+35%, G.8.2) against the benchmarked GPU minutes per output minute before launch; server fallback for an unfinished browser render is charged at the same price.
- Benchmark Cinematic 1080p Native (Wan 2.2 run at 1080p) for GPU-minutes per output minute and visual quality, and confirm the ৳120 / ৳162 1080p Native rates against the measured cost, before switching Cinematic 1080p Native on in the admin panel (4.9).
- Apply for YouTube Data API and TikTok Content Posting API audits and for free child-safety hash-list access now (4.14, 4.15); have a Bangladeshi lawyer finalise the content-policy definitions (4.15).
- Benchmark the escalating-tier video upscaler (4.13) and Wan 2.2 GPU-minutes per output minute before paid packages start (date to confirm); set quality-gate thresholds, bitrates and the Free Beta Cinematic pool from the results (4.14).
- Items that only real-world evidence can close (the design is otherwise complete): the November benchmarks in 13.2 (GPU-minutes, VRAM fits, quality gates, capacity), the licence confirmations in 18.4, YouTube and TikTok API audits, child-safety hash-list access, the C2PA certificate and Cloudflare plan, a Bangladeshi lawyer's review of the content policy (4.15), and the deferred legal, tax, team, budget and go-to-market items.
- Buy a C2PA signing certificate from a certificate authority on the C2PA trust list (a self-signed certificate shows as an unknown signer on platforms), and confirm the Cloudflare plan needed for advanced WAF rules and Waiting Room; update 15.6 with both costs.
- Motion Studio refund treatment is defined in G.10 and the Motion Studio Free Trial Package in G.11; confirm both with legal and payment providers before launch.
- Staffing gap: the human review queues — Islamic content review within 3–7 days (4.5), content-safety appeals and unclear cases (4.15), reported wrong answers (3.10) and copyright takedowns (4.12) — need named reviewers and on-call cover that are not yet in the hiring plan or budget. They are decided with the deferred team and budget items (14.2, 15.2); until then the admin panel shows the age of each queue so the review times can be watched.

G.8 Subscription Packages, Per-Minute Rates & Recording Rules

One subscription for all video types: each Motion Studio package (one monthly fee) lets the creator make 3D Animation Video (Quick and Enhanced modes), Cinematic Video and Islamic Content in any mix. Only the per-minute rate changes with the render mode — G.8.1 rates for Quick and Enhanced, G.8.2 rates for Cinematic — and the 25% Islamic Content discount (G.9) applies to whichever rate is used. The usage ledger records the mode and the Islamic eligibility for every generated minute.

G.8.1 Packages and 3D Animation Video rates — monthly fees for every Motion Studio package, and per-minute rates for the Quick and Enhanced render modes (4.8):

Package	Monthly fee	Storage policy (G.3)	480p Native	720p Escalating	720p Native	1080p Escalating	1080p Native	Video length	Target user
Free Trial Package	৳0 for 7 days	See G.3	Free	Free (highest trial tier)	N/A	N/A	N/A	1 min/day (7 min in total); max 15 sec per video	New creators; Quick and Enhanced modes only, watermarked (G.11)
Freelancer Mini	৳499	See G.3	৳30/min	৳35/min	৳45/min	N/A	৳120/min	5 sec to 45 sec	Reels, Shorts, promo ads
Freelancer Starter	৳999	See G.3	৳30/min	৳35/min	৳45/min	৳55/min	৳120/min	5 sec to 2 min	Reels, Shorts, promo, trailer, teaser, short story
Premium Starter	৳1,999	See G.3	৳30/min	৳35/min	৳45/min	৳55/min	৳120/min	5 sec to 5 min	Cinematic/ animation short stories and cartoons
Premium Professional	৳3,999	See G.3	৳25/min	৳30/min	৳39/min	৳48/min	৳120/min	5 sec to 10 min	Cinematic/ animation series, drama, cartoon
Premium Agency	৳7,999	See G.3	N/A	N/A	৳35/min	৳44/min	৳120/min	5 sec to 20 min	Cinematic/ animation short film, series, drama, cartoon
Islamic Content (within any package)	Same as the chosen package	See G.3	−25% of that package's rate (G.9)	−25% (G.9)	−25% (G.9)	−25% (G.9)	−25% (G.9)	Same as the chosen package	Islamic creators; requires Common Safety and Islamic religious verification
Enterprise	Custom	See G.3; custom add-ons by contract	Custom	Custom	Custom	Custom	Custom	Custom	Custom scope

G.8.2 Cinematic Video rates — per-minute rates for the Cinematic render mode (4.8) within the same G.8.1 subscription; each rate is the 3D Animation Video rate plus 35%, rounded to the nearest whole taka (0.50 rounds up); 1080p Native is ৳120 × 1.35 = ৳162 in every paid package. There is no separate Cinematic monthly fee.

Package	Monthly fee	Storage policy (G.3)	480p Native	720p Escalating	720p Native	1080p Escalating	1080p Native	Video length	Target user
Free Trial Package	Not available	—	—	—	—	—	---	—	Cinematic mode is not part of the free trial
Freelancer Mini	Same subscription (G.8.1)	See G.3	৳41/min	৳47/min	৳61/min	N/A	৳162/min	5 sec to 45 sec	Reels, Shorts, promo ads
Freelancer Starter	Same subscription (G.8.1)	See G.3	৳41/min	৳47/min	৳61/min	৳74/min	৳162/min	5 sec to 2 min	Reels, Shorts, promo, trailer, teaser, short story
Premium Starter	Same subscription (G.8.1)	See G.3	৳41/min	৳47/min	৳61/min	৳74/min	৳162/min	5 sec to 5 min	Cinematic/ animation short stories and cartoons
Premium Professional	Same subscription (G.8.1)	See G.3	৳34/min	৳41/min	৳53/min	৳65/min	৳162/min	5 sec to 10 min	Cinematic/ animation series, drama, cartoon
Premium Agency	Same subscription (G.8.1)	See G.3	N/A	N/A	৳47/min	৳59/min	৳162/min	5 sec to 20 min	Cinematic/ animation short film, series, drama, cartoon

Package	Monthly fee	Storage policy (G.3)	480p Native	720p Escalating	720p Native	1080p Escalating	1080p Native	Video length	Target user
Islamic Content (within any package)	Same subscription (G.8.1)	See G.3	Islamic 3D rate +35% (G.9)	Islamic 3D rate +35% (G.9)	Islamic 3D rate +35% (G.9)	Islamic 3D rate +35% (G.9)	Islamic 3D rate +35% (G.9)	Same as the chosen package	Islamic creators; same verification rules
Enterprise	Same contract (G.8.1)	See G.3; custom add-ons by contract	Custom	Custom	Custom	Custom	Custom	Custom	Custom scope

Output tier ladder (lowest to highest): (1) 480p Native — generated directly at 480p; (2) 720p Escalating — the 480p native output AI-upscaled to 720p; (3) 720p Native — generated directly at 720p; (4) 1080p Escalating — the 720p native output AI-upscaled to 1080p; (5) 1080p Native — made directly at 1080p, the highest tier, at ৳120 per minute (3D Animation) in every paid package and managed from the admin panel (4.9). Quick mode draws its frames at 1080p in the browser and Enhanced mode builds its key image, depth map and parallax at 1080p, so both give true 1080p Native output; Cinematic mode needs Wan 2.2 to run above its tested 720p range, so Cinematic 1080p Native is switched on only after its GPU benchmark and quality gate pass (G.7). Escalating tiers reuse the lower native generation plus video upscaling (RealBasicVSR/BasicVSR++, with the 4.13 quality gate; Real-ESRGAN for previews), so they cost less GPU time and 720p Escalating is priced between 480p Native and 720p Native. 1080p Native is not part of the Free Trial or the Free Beta, which use 480p Native and 720p Escalating only. A package can only use the tiers marked with a rate in the table above, subject to the live admin settings (4.9).

G.8.3 Delivery-speed tiers (approved 21 October 2026; replaces the time-of-day render rates of 18 October 2026) — for paid server renders only; Quick mode renders in the creator's browser and has no delivery tier:

Tier	Price	Delivery	If late
Express	G.8.1 / G.8.2 rate + 20%	Always offered, no daily limit; rendered at high priority on whatever GPU capacity is free in the shared cluster; the system calculates and shows the expected delivery time at order	The 20% surcharge is refunded automatically if delivery is more than 20% later than the ETA shown (e.g. ETA 40 min → refund after 48 min)
Normal	G.8.1 / G.8.2 rate	Within 12 hours — shown as "within 12 hours, usually much sooner"	Motion Studio credit if later than 12 hours
Slow	G.8.1 / G.8.2 rate – 10%	Within 24 hours; scheduled into quiet hours and the 1–7 am window	Motion Studio credit if later than 24 hours

The tier adjustment applies to the final per-minute rate after the G.9 Islamic discount, rounded to the nearest whole taka (0.50 rounds up); the usage ledger records the tier. Free Trial and Free Beta renders are free and have no delivery tier. Scheduling: Express takes free GPU capacity anywhere in the shared cluster at high priority; Slow orders, and Normal orders whose 12-hour window allows it, run in quieter hours and the former night window (1–7 am), so the late-night batch capacity stays in use without a separate night rate.

G.8.4 International prices (USD, approved 21 October 2026) — BDT price + 35%, converted at ৳122 = US$1; monthly fees rounded to the nearest dollar, per-minute rates to the cent; billed through

Paddle by the USA LLC (0.2). The G.9 Islamic discount, the Express (+20%) and Slow (−10%) tiers apply as in BDT.

Package	Monthly	480p Native	720p Esc.	720p Native	1080p Esc.	1080p Native
Freelancer Mini	$6	$0.33	$0.39	$0.50	—	$1.33
Freelancer Starter	$11	$0.33	$0.39	$0.50	$0.61	$1.33
Premium Starter	$22	$0.33	$0.39	$0.50	$0.61	$1.33
Premium Professional	$44	$0.28	$0.33	$0.43	$0.53	$1.33
Premium Agency	$89	—	—	$0.39	$0.49	$1.33
Cinematic (any package)	Same fee	3D rate + 35%	+35%	e.g. $0.67	+35%	$1.79

Pre-production fee: US$0.22 per started 2 minutes of planned video (US$0.17 for Islamic content). Enterprise: Contact Sales Team.

Authority split: G.3 remains the single source for included storage and lifetime character/voice entitlements. G.8 is the final source for package fees, per-minute rates, delivery-speed tiers (G.8.3), USD prices (G.8.4), recording inclusion and minute metering for one subscription per package, with 3D Animation Video rates (G.8.1) and Cinematic Video rates (G.8.2, +35%). F.2 is a feature-access view derived from this G.8 package baseline; if package values change, update F.2 from G.8. Packages do not include a minute quota: every generated minute is billed at the package's per-minute rate for the selected output tier.

- The monthly package fee covers package access and features; generated video minutes are billed separately at the per-minute rates above.
- Auto-dubbing (4.2): each extra language is billed per output minute at 50% of the package's 3D Animation rate for the chosen tier (proposed); direct publishing (4.6) is included in every paid package.
- Creator tools and growth features (4.10–4.11) and the series and promo tools (4.18) are included in every package fee; promos cut from the creator's own rendered episodes are billed at the Quick-mode rate of the chosen tier (4.18). Motion Studio pricing is package fee + pre-production fee + per-minute video rate: the pre-production fee is ৳20 per started 2 minutes of planned video (৳15 for Islamic content), covering script, storyboard, scene list, key images and the 480p preview (4.17); videos are billed at the per-minute rates above. The creator marketplace and the client service marketplace take a 20% commission on each sale; the seller receives 80%.
- Standard video generation and eligible Islamic-content usage are billed per minute at the selected package's rates; optional recording is included at no extra charge. Paid remakes and Motion Studio credits follow G.4.
- Usage is metered from the first second of video generation using the approved seconds-to-minutes conversion; there is no bonus allowance.
- There is no included minute quota and therefore no quota-exhaustion pause; show the estimated per-minute charge before generation and require confirmation. No separate overage fee applies.
- Show the selected plan, per-minute rate, minutes generated this billing cycle and usage estimate clearly; never deduct the same usage twice. The creator's monthly spending cap and its 50/80/100% alerts follow 4.16; paid-package videos carry full commercial rights (4.16).
- Publish policies for failed jobs, partial outputs, retry handling, credit restoration, retention and secure downloads.

G.9 Islamic Content Discount

Apply a 25% discount only to eligible underlying usage rates for Islamic Content; it does not reduce the monthly subscription fee. Islamic content is not a separate package; it is made within the same subscription as 3D Animation and Cinematic videos. Eligibility requires Common Safety Verification plus religious source verification, including Quran/translation attribution, Hadith collection/reference/authenticity checks and sensitive-depiction review. The discount applies in every render mode. Islamic 3D Animation Video rates are the G.8.1 rates less 25%; Islamic Cinematic Video rates are those Islamic 3D Animation rates plus 35%, matching the Cinematic uplift in G.8.2. All rates are rounded to the nearest whole taka (0.50 rounds up); for 1080p Native this gives ৳90 (Islamic 3D Animation, ৳120 less 25%) and ৳122 (Islamic Cinematic, ৳90 × 1.35 = ৳121.50 rounded up) in every paid package.

Islamic Content rates after the 25% discount (the monthly package fee is not discounted):

Package family and package	480p Native	720p Escalating	720p Native	1080p Escalating	1080p Native
3D Animation — Freelancer Mini / Starter / Premium Starter	৳23/min	৳26/min	৳34/min	৳41/min (not Mini)	৳90/min
3D Animation — Premium Professional	৳19/min	৳23/min	৳29/min	৳36/min	৳90/min
3D Animation — Premium Agency	N/A	N/A	৳26/min	৳33/min	৳90/min
Cinematic — Freelancer Mini / Starter / Premium Starter	৳31/min	৳35/min	৳46/min	৳55/min (not Mini)	৳122/min
Cinematic — Premium Professional	৳26/min	৳31/min	৳39/min	৳49/min	৳122/min
Cinematic — Premium Agency	N/A	N/A	৳35/min	৳45/min	৳122/min

G.10 Motion Studio Refund Policy

Motion Studio package fees and purchased Motion Studio credits follow the same windows as the NexaTute refund policy (11.6):

- Fresh purchase, 0–30 days: 100% refund of unused credits; the monthly package fee is refundable in full only if no video was generated in that billing cycle.
- 31+ days: no refund.
- Failed render or technical issue: minutes or credits consumed by the affected job are restored or refunded.
- Wrong or duplicate charge: immediate reversal.
- Platform outage: minutes or credits consumed by jobs that failed or stalled during a platform outage are restored automatically within 24 hours, and expiring credits are extended by the outage duration.
- Expired credits (30-day validity) are not refundable.

G.11 Motion Studio Free Trial Package

- Validity: 7 days; free, no payment card required; one trial per user account with duplicate-claim controls (4.6).

- Generation allowance: 1 minute of video per day for the 7-day validity (7 minutes in total; unused time does not carry over to the next day), with a maximum of 15 seconds per video, at 480p Native or 720p Escalating only (no 720p Native, 1080p Escalating or 1080p Native), with Freelancer Starter features; outputs are watermarked.
- Processed in the free/trial batch lane at X-Priority 10 in the Quick and Enhanced render modes only (no Cinematic); recording is not included.
- Voice cloning and character creation only within the G.3 lifetime free allowance.
- The daily allowance, maximum video length, output tiers and validity are admin-panel settings (4.9) and can be changed without a code change.

Appendix H. Implementation Readiness & Final Acceptance Controls

Purpose: provide a controlled evidence framework for implementation readiness, server acceptance, capacity testing, recovery readiness and final document integrity. These controls do not replace the canonical v24.0 architecture baseline.

Evidence rule: a checklist item is not complete merely because it appears in this proposal. Mark an item Verified only when a dated artifact, test result, configuration export, signed review or other reproducible evidence is linked in the project record.

H.1 Implementation Readiness Matrix

Use one row per deployable feature, service, server role and operational control. Keep planning status separate from verified deployment status.

Status	Meaning	Minimum evidence required
Planned	Approved in design, not yet configured.	Approved requirement, owner and acceptance criteria.
Configured	Configuration exists in a non-production or production environment.	Versioned configuration, environment identifier and reviewer.
Tested	A defined test has passed in a named environment.	Test case ID, date, tester, result, logs/metrics and defects.
Production Verified	Production operation has been observed and accepted.	Production evidence window, alert/dashboard evidence, rollback path and approver.
Blocked / Exception	Not met, unavailable or intentionally deferred.	Reason, risk owner, mitigation, due date and explicit acceptance/waiver.

H.1.1 Required register fields

- Unique control ID and requirement/source reference (including v24.0 section or decision where applicable).
- Product scope: NexaTute, Remon Motion Studio, shared platform, or corporate operations.
- Assigned owner and independent reviewer; target date and current status.
- Environment and software/model version; exact test procedure and pass/fail threshold.
- Evidence URI/path, date captured, result, defect/ticket link and retest result.
- Risk classification, dependency, exception approval and next review date.

H.1.2 Release gate

A release cannot be labelled “Production Verified” while a critical security, data-loss, cross-product isolation, unsafe scratch-file deletion or unresolved P0 defect remains open. Capacity claims must remain “design target” until the relevant load-test gate is passed.

H.2 Server-by-Server Acceptance Checklist

Reconcile all 16 server records against the canonical v24.0 inventory before procurement, provisioning and acceptance. The inventory below is a checklist template, not a statement that the servers have been deployed or tested.

Node / group	Required checks	Acceptance evidence
AI-1 Node A / Node B	CPU, RAM, SSD, GPU model/VRAM, driver/runtime compatibility, inference service health, GPU memory headroom, temperature/power and telemetry.	Supplier/SKU record; OS and driver output; model smoke test; GPU metrics; failure/restart test.
AI-2 Node A	CPU/RAM/storage/network; STT/TTS dependencies; concurrent workload limits; consent and access controls for voice features.	Hardware inventory; dependency/license list; TTS/STT test corpus results; audit logs.
AI-2 Node B and any GPU node running assembly	Canonical local 400 GB SSD role; GStreamer/FFmpeg build; Opus preservation; local chunk/intermediate/final MP4 workflow; no heavy scratch writes to Core-1.	Local disk/path test; recording test; codec probe; checksum and R2 acknowledgement test; safe-cleanup test.
Rec-1 / Rec-2	Dedicated recording role, ingress/network, lightweight track recording and whiteboard event log, replay sync in the NexaTute player, recording isolation, disk alarms and retries; no download compositing on the node (download MP4s are made in the browser, 9.5); GPU driver version and NVENC for AV1 viewing copies.	Parallel recording tests; sample playback; failure/recovery test; upload logs and checksums. NVENC encode benchmark (sessions per node, quality, CPU/GPU use).
Core-1 / Core-2	Core-1 and Core-2 as one LiveKit SFU cluster: room placement across both nodes, node-failure failover, capacity (rooms, tracks, subscribed layers, bitrate), 10 Gbps (Core-1) and 1 Gbps (Core-2) network with bandwidth-weighted room placement, embedded TURN/TLS on 443 and coturn fallback, RAID 1 NVMe health, access restrictions and disk/IO monitoring.	LiveKit cluster load test at representative room topologies; single-node failure drill; TURN relay/connectivity test; network and storage benchmarks; configuration export. Core-1 failure drill confirming Core-2 (1 Gbps) switches all rooms to 480p/160p and applies admission control.
App-1 / App-2 / App-3	API routing, authentication, rate limits, product isolation, health probes, circuit breakers, graceful overload response and rollback.	API contract tests; authz tests; load/soak logs; fault injection; user-visible error validation.
DB-1 / DB-2, Redis1 / Redis2, LB-1 / LB-2 (HAProxy), App-node PgBouncer sidecars and monitoring nodes	Canonical node assignment; PostgreSQL primary/replica health, replication lag and failover; PgBouncer pooling; Redis persistence/restart where intended; retention, access controls, alert delivery, capacity and recovery.	Config exports; DB health and failover tests; backup/restore and replication tests; alert test; retention and access review.
All remaining nodes in canonical inventory	Confirm exact node name, role, hardware, network port, storage, dependencies, monitoring and failover relation directly from v24.0.	Signed 16-row inventory reconciliation; no undocumented role substitution.

H.2.1 Fleet-wide acceptance checks

- Record exact vendor, model/SKU, warranty, datacenter/location, operating system, firmware, driver and software versions.
- Verify network paths, firewall rules, DNS, TLS, service identity, secrets handling and least-privilege access.
- Test monitoring/alerting, log correlation, clock synchronization, disk-full behavior, process restart and node unavailability.
- Verify resource reservations and isolation between NexaTute and Motion Studio; shared physical capacity must not merge permissions, queues, billing or durable records.
- Record deviations from v24.0 in a signed exception/change register; do not silently “correct” locked hardware specifications.

H.3 Capacity and Load-Test Evidence Plan

The 16,000+ combined CCU figure remains a design target, not a benchmark-verified capacity claim. Use representative workload mixes and publish measured results, limitations and test environment details before making public performance claims.

Load-test tools: Locust (MIT) for API and web traffic, the LiveKit CLI load tester (Apache-2.0) for live classes and vLLM's benchmark scripts for LLM throughput (18.1); the results replace the 8.6 planning figures layer by layer.

Test stage	Required workload / checks	Required report outputs
Baseline smoke	One user per key journey; API, AI request, live class, recording, video generation, playback and payment/usage ledger in staging.	Pass/fail per journey, latency, errors, resource use and build/version.
Per-product load	NexaTute and Motion Studio tested independently at increasing concurrency; distinguish interactive, live, batch and rendering workloads.	Achieved concurrency, throughput, p50/p95/p99 latency, error rate, queue age, GPU/RAM/CPU/network/disk use.
Combined target ramp	Ramp toward the stated 8,000 NexaTute + 8,000 Motion Studio target using realistic traffic mix and declared test duration.	Maximum sustainable CCU by product, bottleneck, queue behavior, cost metrics and clear target pass/fail.
Soak / peak / recovery	Long-duration steady load, burst, queue backlog, model/provider slowdown, node loss, GPU OOM, DB/Redis interruption and recovery.	Stability over time, memory/disk growth, recovery time, duplicate/lost job count and backlog drain rate.
Recording / media stress	Concurrent rooms/tracks, recording on both Rec nodes, local scratch pressure, interrupted upload, checksum mismatch and R2 retry.	Recording success rate, audio/video integrity, scratch peak, upload latency and exact-file cleanup proof.
Client compatibility	Supported browsers/devices, WebGL/WebGPU/WASM capability, low-end fallback, accessibility and Arabic/Urdu right-to-left layout.	Browser/device matrix, failures, performance observations and approved fallbacks.

H.3.1 Pre-agreed pass/fail criteria

Before testing, engineering must document numeric SLOs for latency, error rate, availability, queue wait, recording success, render completion, data integrity and recovery. Do not invent thresholds after seeing the results. Define ramp steps, duration, user think-time, request mix, test data, environment parity and abort criteria in the test plan.

H.3.2 Reporting and claim controls

- Attach raw test output, dashboards, run scripts, workload profile, software/config versions and known limitations.
- Report both achieved and failed target levels; do not extrapolate beyond tested workload without clearly labelling the estimate.
- Separate CCU from requests per second, live media participants, queued jobs and active GPU render jobs.
- Repeat tests after material changes to hardware, drivers, models, scheduler, database, media pipeline or network.

H.4 Security, Backup, Disaster Recovery and Incident Acceptance

Control area	Required validation	Evidence / exit condition
Identity and access	MFA for privileged users where supported, least privilege, service identities, key rotation, offboarding and audited admin actions.	Access review, test account results, secret-rotation record and audit-log sample.
Product/data isolation	Attempt cross-product access to APIs, queues, caches, files, recordings, credit ledgers and admin functions.	Negative authorization tests pass; no cross-tenant/product data exposure.
Data protection	TLS in transit, encryption and key management at rest where applicable, signed URLs, scoped R2 credentials and retention enforcement.	Config review, access tests, key/credential inventory and expiry/deletion evidence.
Backup and restore	Restore databases, configuration and required durable job/usage records into an isolated environment. Validate media object recovery strategy.	Dated restore drill, integrity checks, measured restore duration and approver. pgBackRest WAL archive in R2 verified by a monthly point-in-time restore.
RPO / RTO	Owners set business-approved Recovery Point Objective and Recovery Time Objective for each critical service.	Approved target table plus measured drill results; gaps have remediation owners.
Incident response	Runbook for security incident, data exposure, provider outage, GPU/DB failure, recording failure and R2 upload failure.	Named on-call owner, escalation tree, communication templates and tabletop/drill result.
Data lifecycle	Retention, user deletion, legal hold, recording consent/status and safe media cleanup.	Test proving exact object/job deletion only after durable completion/acknowledgement rules are met.
AI/content safety	Abuse reporting, voice-cloning consent/rights, content policy checks, religious/factual review where applicable and human-review queue.	Policy version, audit trail, test cases, escalation and reviewer sign-off.

H.4.1 Recovery acceptance rules

- Do not set final RPO/RTO values without business-owner approval and a successful timed recovery drill.
- Do not delete local scratch or source artifacts until the documented validation, checksum, durable upload acknowledgement and persisted verified-state gates have passed.
- Keep backups protected from ordinary application credentials; periodically test restore rather than relying on successful backup job status alone.
- Document any third-party provider dependency and manual fallback, including which functions degrade or pause during provider outage.

H.5 Final Document Integrity & Word Acceptance Checklist

This checklist governs the final editable Word master. The document must be opened in Microsoft Word (or a compatible renderer for preliminary checks), fields refreshed, and pagination inspected after the final content change.

Check	Required action	Completion evidence
Native Table of Contents	Use Word heading styles and an actual TOC field; select Update entire table after all edits.	TOC field exists; heading titles and page numbers match the final layout.

Check	Required action	Completion evidence
Heading numbering	Normalize heading syntax and hierarchy; ensure unique, sequential chapter numbering without renumbering canonical quoted material incorrectly.	Automated heading audit plus visual inspection of every major section.
Cross-references	Update page/section references, figure/table references, Appendix references and document-control/version labels.	No stale or broken references in final Word file.
Duplicate content	Check repeated Roadmap/KPI material, pricing, feature lists and inherited audit notes; preserve distinct required detail while identifying the controlling section.	Duplicate audit log and approved decision for each apparent repetition.
Pricing exclusions	Confirm previously removed 12.1, 12.2 and 12.3 pricing sections have not returned under duplicate or alternate headings.	Search report for exact headings and semantic duplicate review.
Page layout	Inspect page-by-page for orphan headings, split rows, blank pages, clipped tables, header/footer collisions and unreadable code blocks.	Rendered PDF page count and page-review log; issues fixed and re-rendered.
Tables and appendices	Repeat table headers across pages; verify each Appendix title, order, TOC entry and internal links.	Table/Appendix checklist signed off.
Version and status	Ensure cover, header/footer, change log, document-control table and final version note consistently say v29.6; retain the v24.0 authority rule.	Global search for stale version labels and sign-off.
Claims and evidence	Check that “planned”, “target”, “configured”, “tested” and “production verified” are used accurately.	Claims review with evidence links or explicit “requires validation” label.
Final handoff	Save editable DOCX and final PDF from the same last-saved Word revision.	Both files have matching version, date, page count and content.

H.5.1 Word refresh procedure

- Open the DOCX in Microsoft Word and allow fields to update if prompted.
- Press Ctrl+A, then F9 to update fields; if Word asks, choose “Update entire table” for the Table of Contents.
- Inspect the entire TOC and all section/Appendix page starts after pagination stabilizes.
- Review the document in Print Layout, including tables spanning pages, headers/footers and final pages.
- Save the DOCX, export a fresh PDF, then compare page count and final version label.

H.5.2 Final sign-off record

Role	Name / date	Decision / evidence reference
Architecture owner	_____	v24.0 reconciliation and deviations register: _____
Infrastructure / SRE reviewer	_____	Server acceptance and load/DR evidence: _____
Security / privacy reviewer	_____	Security and data-lifecycle acceptance: _____

Role	Name / date	Decision / evidence reference
Document control reviewer	_____	TOC, numbering, page layout and duplicate audit: _____
Release approver	_____	Approved / Approved with exceptions / Rejected: _____

H.6 Status and Limitations

The controls in Appendix H are acceptance criteria and templates—not evidence that hardware has been purchased, services deployed, security audited, recovery targets achieved or 16,000+ CCU demonstrated. Populate the registers with real evidence before marking any item complete. Canonical v24.0 remains authoritative for locked server inventory, node assignments, client-side rendering, priority policy, failover design and scratch-file safety policy.

Final-edition note: this appendix is retained as the release-control framework for the corporate final product.

Appendix I. Technical Reconciliation & Delta Register

This register records implementation clarifications and source conflicts that require evidence or an explicit owner decision; it does not duplicate the canonical architecture, pricing or feature catalogues.

I.1 Review method and scope

The master (reviewed at v28.4 and carried into v28.9) was reviewed against the supplied canonical Hybrid Architecture v24.0, including its software inventories, 16-server inventory, data-flow diagram, failover/circuit-breaker material, scratch-cleanup example, UI overload messages, Portkey configuration, GStreamer pipeline, PgBouncer configuration, X-Priority matrix and capacity-estimator material. Existing reviewed content is retained in v28.9. This section adds only material that was absent or needed an explicit reconciliation; it does not repeat the full feature catalogue, server inventory, pricing tables or security checklist.

Authority remains unchanged: v24.0 controls locked physical inventory, node assignments, client-side rendering, priority policy, failover intent and scratch-file safety. Appendix F remains the authority for the Motion Studio feature reference only; Appendix G.8 is the sole authority for package fees, per-minute rates, recording and metering. All capacity and feature claims remain targets until measured and accepted.

I.2 Operational clarifications

I.2.1 User-facing overload and queue messages

NexaTute classroom banner (overload, 3.16): “প্রিয় শিক্ষার্থী, এই মুহূর্তে সার্ভারে অতিরিক্ত চাপ থাকায় AI টিচারের লাইভ ক্লাস সাময়িকভাবে বন্ধ আছে। AI কো-টিচার লেখার মাধ্যমে আপনার পড়া চালিয়ে যাবে, আর হারানো মিনিট স্বয়ংক্রিয়ভাবে ফেরত দেওয়া হবে। AI টিচার আনুমানিক {ETA} মিনিট পর আবার পাওয়া যাবে।”

NexaTute busy notice (load-aware routing, 3.16): “AI টিচার আনুমানিক {ETA} মিনিট পর ফ্রি হবে। ততক্ষণ আপনার আগের রেকর্ড করা ক্লাসটি দেখুন — লাইনে আপনার জায়গা রাখা আছে।”

Motion Studio rendering notice: “আমাদের ভিডিও রেন্ডারিং ইঞ্জিনটি ওভারলোড সুরক্ষার জন্য সাময়িকভাবে নতুন রেন্ডারিং রিকোয়েস্ট নেওয়া স্থগিত করেছে। আপনার তৈরি করা স্ক্রিপ্ট ও স্টোরিবোর্ড ড্রাফট হিসেবে নিরাপদে সেভ করা হয়েছে। পূর্বের কিউ (Queue) খালি হওয়া মাত্রই আপনার রেন্ডারিং প্যানেলটি আবার সচল হবে।”

Implementation note: overload wording follows the I.3 row “Overload UI wording”; preserve draft/job IDs and never claim a session was transferred when the operation failed.

I.2.2 Explicit routing and fallback configuration

Portkey gateway routing order: local vLLM on the GPU cluster first, then the cloud fallback chain in 8.9, with retries only for transient status codes. The master treats this as a required deployment configuration record rather than a promise that fallback is already operational. NexaTute exception: the AI Co-teacher routes verified-answer cache → Mistral Large 3 → GPT-6 Luna, and live AI-teacher classes are not routed to the cloud (8.9).

Before release, record exact provider/model identifiers, API base URLs, secret injection method, request timeout, per-provider retry policy, retryable status codes, circuit-breaker thresholds, maximum concurrent cloud calls, budget ceilings, and a test showing recovery after the cloud limit is reached. Keep provider credentials out of source files and examples.

A fallback request must preserve product identity, authorization, request/trace ID, idempotency key where applicable, usage accounting, and the product priority. Do not send a request to an unapproved provider or exceed configured spend limits.

I.2.3 PgBouncer deployment specifics

The v24.0 appendix contains a base PgBouncer configuration with transaction pooling and separate read/write database aliases. The implementation register requires a concrete, environment-specific PgBouncer configuration and a tested mapping to DB-1 primary and DB-2 replica.

Validate authentication method and secret storage, pool_mode, pool sizes, reserve pool, server/client connection limits, query/session compatibility with transaction pooling, health checks, primary/replica routing, replication lag handling, and failover behavior. The sample is a starting point only; it is not a production-ready credential or topology declaration.

I.2.4 Recording compositor and audio preservation

LiveKit Egress remains assigned to Rec-1/Rec-2; render assembly follows the local-SSD scratch rules in 9.3. For Motion Studio renders, only a single final, validated MP4 may be uploaded to Cloudflare R2 after compilation; class recordings are uploaded as separate tracks plus the whiteboard event log, and their download MP4 is made in the user’s browser (9.5).

Audio-path rule: preserve encoded Opus where the selected container/muxer and pipeline support it. Do not introduce an automatic Opus-to-AAC/faac transcode merely because an older sample pipeline contains voaacenc. Decode Opus only where required by mixing/compositing; validate audio/video synchronization, stream/container compatibility and final playback in acceptance tests.

The older v24.0 GStreamer example includes an AAC encoder stage; that example is not copied as-is because it conflicts with the approved Opus-preservation rule. Select the final pipeline based on the actual incoming stream, deployed plugins and desired output container.

I.2.5 Scratch cleanup: mandatory safety gates

The v24.0 example describes a checksum-gated purge, but its illustrative code does not itself prove that checksum verification succeeded before deletion. The implementation must explicitly enforce all of the following gates: task-owned path only; reject symlinks and non-regular files; resolve and enforce the approved local scratch root; verify the final file exists and is non-empty; compare expected size/checksum; receive durable object-storage upload acknowledgement; then mark the upload state verified before cleanup.

On upload, checksum, storage acknowledgement or database-state failure, retain the local final file for recovery and raise an observable alert. Cleanup must be idempotent, race-aware and unable to delete paths outside the task's assigned directory. Never send heavy scratch/intermediate writes to

Core-1. Use the relevant local worker SSD: the executing GPU node for Motion Studio render scratch; Rec-1/Rec-2 local buffers for recording jobs.

I.2.6 Capacity estimator and dashboard semantics

The v24.0 estimator appendix uses an illustrative 12,000 local / 16,000 total model and sample cost constants. These values conflict with the canonical 16,000+ combined target split as 8,000 NexaTute + 8,000 Motion Studio and are not benchmark evidence. The updated master does not adopt the 12,000 local threshold or sample cost rates as approved production limits.

Dashboard labels must distinguish measured live values, configured limits, simulation assumptions and target values. Before publishing cost/CCU estimates, define a versioned benchmark-derived capacity model with workload mix, room/track topology, model/runtime settings, bitrate, p95/p99 latency, failure rate, fallback ratio and cost inputs. Label simulated estimates as estimates, not achieved capacity.

I.2.7 Hardware inventory and node-role verification

This master already carries the 16-node baseline, so the inventory is not duplicated here. During implementation, verify each actual node against v24.0 before deployment: AI-1 A/B (32c/84 GB, 400 GB SSD, RTX PRO 6000 96 GB each); AI-2 A (24c/56 GB, 320 GB SSD, RTX PRO 5000 48 GB); AI-2 B (24c/56 GB, 400 GB HP SSD, RTX PRO 5000 48 GB); Rec-1/Rec-2 (24c/56 GB, 320 GB SSD, RTX PRO 4000 24 GB each); Core-1 (48c/256 GB, 2×4 TB NVMe R1, 10 Gbps); Core-2 (32c/128 GB, 2×960 GB NVMe R1, 1 Gbps); App-1/App-2, DB-1/DB-2, Redis1/Redis2 and LB-1/LB-2 must match the full inventory in Section 07.

Record actual SKU, serial/asset ID, OS/kernel, GPU driver, CUDA/runtime, storage mount and filesystem, network rate, firewall rules, monitoring agent and verification date. This is a verification register, not a second hardware inventory.

I.3 Reconciliation register

Source item	Issue found	Canonical handling (current edition)
v24.0 Appendix E capacity estimator	Illustrative 12,000 local + 4,000 cloud overflow at 16,000	Canonical target remains 8,000 NexaTute + 8,000 Motion Studio; no threshold is treated as production-approved until load-tested.
v24.0 locked 16-server inventory	The 16 servers alone cannot carry the 16,000 CCU target	The 16 servers are the rented launch fleet; the scale-out fleet in 8.5 is the approved architecture change for 16,000 CCU
v24.0 node assignment (App tier)	Two App nodes are the first capacity limit when both products run together (8.6)	App-3 (OVHcloud ADVANCE-2) joins the App pool in December 2026 (approved 17 October 2026); the 16-row inventory stays the launch baseline (Section 07)
v24.0 GStreamer example	Contains voaacenc after Opus parsing/decoding	Preserve encoded Opus where supported; transcode only if an explicitly approved output requirement demands it.
v24.0 scratch purge example	Comment references checksum confirmation but code does not perform/verify the upload receipt and checksum in the shown function	Require explicit checksum + durable upload acknowledgement + persisted verified state before deletion.

Source item	Issue found	Canonical handling (current edition)
Overload UI wording	Sample says session is being transferred to cloud backup	Display this only after confirmed transfer; otherwise show a truthful queued/temporarily unavailable state.
Fallback code sample	Illustrative sample has simplified payloads and a global in-memory lock	Production implementation must use provider-compatible request schemas, shared circuit-breaker state, bounded retries, timeouts, concurrency/spend limits, observability and recovery tests.
Capacity and feature claims	Targets and sample calculations are not benchmark evidence	Keep 16,000+ CCU as a design target; publish achieved capacity only with reproducible test evidence.

I.4 Final deduplication and release acceptance

- Keep one canonical 16-server hardware inventory (Section 07); this review does not add a duplicate inventory table.
- Keep one NexaTute price model and one Motion Studio package table; the two billing systems remain separate.
- Keep one Motion Studio feature authority in Appendix F; this Appendix I delta register adds only operational controls missing from the master.
- Keep the priority policy at X-Priority 100 for NexaTute real-time learning workloads, 50 for Motion Studio paid production, and 10 for free/trial batch work; implement bounded queues and starvation protection.
- Keep client-side processing for NexaTute's voice (STT/TTS), AI-teacher video, 3D avatar/lip-sync display, whiteboard UI, virtual-lab visuals, the 3D classroom, AI-tutor session recording and download MP4s of class recordings; do not shift these graphics workloads to server-side rendering by default.
- Confirm that render scratch, assembly, upload and cleanup follow 9.3.
- Treat 16,000+ combined CCU as a design target, not a verified outcome. Require reproducible load tests, security review, license review and deployment evidence before launch claims.
- Do not consider this Word document evidence of deployment. Track actual owners, versions, licensing, test results, rollback and operational acceptance separately.

I.5 Final version note

Final corporate pass: Appendix G.3 remains the authority for cross-platform storage, character and voice entitlements. Appendix G.8 is the single Motion Studio authority for package fees, per-minute rates, recording inclusion and metering. Duplicate feature, pricing and hardware descriptions were consolidated; unresolved values remain labelled for owner validation.